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Mathematical Tools for the Study of the Incompressible Navier-Stokes Equations and Related Models

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Preface

This monograph is a revised and augmented version of a previous book [24] written in French and published in 2006. Our objective is twofold. First, we want to introduce the reader to the modelling and the mathematical analysis in fluid mechanics. The central models we deal with are the incompressible Stokes and Navier–Stokes equations, whose derivation is exposed in the first chapter. Second, we introduce mathematical tools in a sufficiently general way so that the reader should be able to use them for studying many other kinds of nonlinear evolution partial differential equations.

In this spirit, we tried to make the book as self-contained as possible. The only prerequisites that are needed to begin reading are basic results in calculus, integration, and functional analysis. The second chapter is a recap of more or less standard analysis results that are used in this book. This includes for instance: the definitions and properties of weak and weak-$\star$ convergences in a Banach space, a short introduction to distribution theory, basic compactness results, the Bochner integral and related spaces, basic results in spectral theory, and so on.

In Chapter III, we introduce and study Sobolev spaces on Lipschitz domains of $\mathbb{R}^d$. The first section contains main definitions and properties of such domains. We describe, in particular, how to define the integral of functions defined on the boundary of such domains and we prove the fundamental Stokes formula. Sobolev spaces are then introduced and thoroughly studied in Section 2. An important role is played (in this chapter but also in the next ones) by suitable mollifying operators that permit us to prove the density of sets of smooth functions in various functional spaces. We also study embedding theorems, extension operators, trace and trace lifting operators, duality theory for Sobolev spaces, and the very important Poincaré and Hardy inequalities. The third section of the chapter is devoted to the introduction of suitable normal/tangential coordinates near the boundary of a sufficiently smooth domain. This framework is useful for the study of tangential Sobolev spaces and thus for the proof of regularity properties, up to the boundary, to solutions of elliptic problems (like the Stokes problem, for instance). All these notions are illustrated in the last section of the chapter where a complete study of the Laplace problem with Dirichlet or Neumann boundary conditions is given.

Chapter IV concerns the steady (incompressible) Stokes equations. This is a central topic for the study of any model in incompressible fluid mechanics. A basic tool in this theory is the Nečas inequality, a complete proof of which is given in Section 1. This inequality is used in the next section to study relations between the gradient fields and the divergence-free vector fields. This is fundamental in order to build the pressure in the Stokes system. The next two sections are dedicated to particular properties of the divergence and curl operators as well as some related function spaces. The analysis of the Stokes problem with Dirichlet boundary conditions is undertaken in Section 5, which includes the definition of the Stokes operator and application to the unsteady Stokes problem through the semigroup approach. We postponed to Section 6 the complete proof of elliptic regularity properties of the Stokes equations. The last three sections are concerned with some nonstandard variants of the Stokes problem, considering different kind of boundary or interface conditions.

The steady and unsteady Navier–Stokes equations are studied in Chapter V. For the unsteady problem, we prove existence and uniqueness (in 2D) of global weak solutions, we also investigate smoother solutions and discuss the parabolic regularisation properties of the system. Then, for the steady problem, we discuss existence and uniqueness issues in the case of homogeneous and nonhomogeneous boundary data and we deal with very basic stability issues for these solutions.

In Chapter VI, we study the most complex model considered in this book which is the one of the (unsteady) incompressible Navier–Stokes equations for a nonhomogeneous fluid. The first ingredient that is useful in the analysis of this problem is the study of weak solutions for the transport equation. We propose here, in the first section of this chapter, a self-contained exposition of the Di Perna–Lions theory of renormalized solutions for such equations including more recent developments related to the associated trace problem. Those results are then used to prove global existence of weak solutions for the initial- and boundary-value problem associated with the complete Navier–Stokes system in the second section of the chapter.

The final chapter is concerned with two different issues related to boundary conditions arising in numerical simulations of incompressible flows. The first section deals with the problem of outflow boundary conditions that one needs to choose in the case where the computational domain is, for practical reasons, smaller than the real physical domain. We analyse here a nonlinear boundary condition relating normal stress and momentum flux at the outflow boundary. In the second section of the chapter, we study the penalty method which can be used, for instance, to impose a homogeneous Dirichlet boundary condition on the boundary of an obstacle located in a computational domain whose geometry is simple. We prove convergence of the penalised solution towards the exact solution when the penalisation parameter goes to 0, through the description of a suitable boundary layer. Additionally to their practical interest, the analysis of these two problems gives the opportunity to introduce mathematical methods that the reader may find useful in other contexts.

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Contents

Chapter I
The equations of fluid mechanics

This first chapter introduces the equations of fluid mechanics and the various models which we study later. Among the many books published on fluid mechanics, and more generally on the mechanics of continuous media, we recommend that readers consult, for example, references [90, 66, 102, 127, 126, 123] if they seek a more detailed description of the various concepts. Moreover, some of the thermodynamics concepts we use here are very briefly described in Appendix B.

1 Continuous description of a fluid

1.1 The continuous medium assumption. Density

A fluid consists in a large number of molecules in motion without a precise shape at rest (contrary to a solid). A first approach to studying a fluid might involve writing down the equations of motion for each of the particles by considering their interactions (collisions, characterised by the *mean free path*, but also long-range interaction). One such, so-called *statistical*, approach led to the kinetic theory of fluids and to statistical mechanics more generally. It can be traced back to the work of Maxwell in the middle of the nineteenth century.

In a large number of physical situations, if the mean density of the fluid studied is not too low (i.e., if the characteristic lengths of the problem are large compared to the mean free path of the particles), then the fluid can be considered as a continuous medium. This means that the movement of the particles can be considered as a whole and not independently for each particle. Hence, we can define *macroscopic* quantities that characterise the system: density, velocity, and so on.

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An intuitive description of a continuum medium is the following. Let δV be an elementary volume of space around a point M at a given time t in the considered fluid. Here, the volume δV is supposed to be small compared to the size of the problem studied but, nevertheless, sufficiently large compared to the mean free path of the molecules. This implies that this volume δV contains a large number of molecules. If we now denote δm as the mass contained in this elementary volume, then the density of the fluid at the point M at time t can be intuitively defined as the ratio

$$\rho \approx \frac{\delta m}{\delta V}.$$

If it can be established by measurements that this quantity is essentially independent of the elementary volume δV selected around M at time t, provided that its size remains within the limits described above, then it is said that the continuous medium assumption holds. In that case, an intuitive statistical definition of the velocity of the flow at point M and at time t is defined as the velocity of the centre of mass of the particles contained in the elementary volume δV

$$v \approx \frac{\sum_{i=1}^{N} m_i v_i}{\sum_{i=1}^{N} m_i},$$

where m_i and v_i represent the mass and velocity of each of the molecules in δV and N is the number of these molecules. Here also, this ratio is expected to be essentially independent of the size of the volume δV. An interactive illustration of the validity of the continuous medium assumption is available in [125]. A more formal definition of the density is the following.

Definition I.1.1. *We say that the continuous medium description is valid, if there exists a nonnegative function $(t, X) \mapsto \rho(t, X)$ such that the mass contained in any fluid volume ω at time t can be expressed as follows*

$$\textit{Mass in } \omega \textit{ at time } t = \int_{\omega} \rho(t, X)\, dX. \tag{I.1}$$

This function ρ is called the density field of the fluid under study.

From the previous formal discussion, it should be clear that formula (I.1) can only be expected to be valid for fluid volumes ω whose size is sufficiently larger than the mean free path of the molecules in the fluid.

Under normal conditions of temperature and pressure, the mean free path in a conventional gas (oxygen, hydrogen, nitrogen, etc.) is on the order of 100 nm $= 10^{-7}$ m, whereas in water this characteristic length is around 10^{-10}m. This is, of course, very small compared with the characteristic size of problems studied in the majority of actual situations. However, in more extreme situations, for example, at very high altitudes where the air is rarefied, the mean free path in the medium may be very much longer and its order of magnitude can become comparable to the dimensions of the real problems being investigated (e.g. the flow of air around a space shuttle in the upper layers of the atmosphere). In this case, the fluid can no longer be assumed to be a continuous medium and it becomes necessary to use a microscopic statistical description of the medium. Throughout the sequel of this book, we assume that we are dealing with a continuous medium.

Notation: The main notation and properties used here concerning common differential operators are reviewed in appendix A.

1.2 Lagrangian and Eulerian coordinates

In fluid mechanics there are two "canonical" coordinate systems in which the various equations of motion can be written. *Lagrangian* coordinates are associated with a fluid particle (or a fluid volume element) as described below and follow it throughout its evolution. By contrast, *Eulerian* coordinates are the coordinates of the fixed reference frame associated with the experiment.

In some circumstances, such as for studying free surface flows or fluid–solid interactions, *Lagrangian* coordinates are actually quite useful. However, since the work of Euler in the eighteenth century, it has been more usual and convenient to write flow equations using *Eulerian* coordinates, and it is these that we will use here.

The continuous medium assumption allows us to consider the movement of the fluid molecules as a whole and not individually. Let $\omega \subset \mathbb{R}^3$ be the volume of space occupied by the fluid at the initial time. The flow is described by a family of bijective maps $(\varphi_t)_t$ defined on ω, such that for any $\Omega_0 \subset \omega$, the set $\Omega_t = \varphi_t(\Omega_0)$ exactly contains, at time t, the same fluid molecules that were present in Ω_0 at the initial time (we always assume in this book that there is no matter source or sink in the system, otherwise the equations should be adapted accordingly).

Definition I.1.2. *Such a family of sets $(\Omega_t)_t$, with $\Omega_t = \varphi_t(\Omega_0)$, indexed by the time variable t, is called a* fluid element*.*

In the case where Ω_0 is a single point $\Omega_0 = \{x_0\}$, we call it a fluid particle*.*

This is a purely mathematical concept, that is not related to any precise molecule of the fluid. In this last case, for any t we have $\Omega_t = \{\varphi_t(x_0)\}$. As a consequence, $\varphi_t(x_0)$ is the position in a fixed reference frame at time t of the fluid particle that occupied the position x_0 at the initial time 0.

We may thus introduce for any t, t_0, and x_0 the point $X(t, t_0, x_0) = \varphi_t(\varphi_{t_0}^{-1}(x_0))$ which is the position at time t of the fluid particle whose position was x_0 at time t_0. The map $t \mapsto X(t, t_0, x_0)$ is called the *trajectory* of this fluid particle.

Knowing X or $(\varphi_t)_t$ is equivalent and allows the evolution of the system to be perfectly determined. Nevertheless, we are not generally interested in identifying the positions of each of the fluid particles of the system at a given time; rather, it is preferable to work with the macroscopic velocity field v defined by the kinematic relation

$$v(t,x) = \left(\frac{\partial X(s,t,x)}{\partial s}\right)_{|s\,=\,t}, \tag{I.2}$$

which means that $v(t,x)$ is the velocity of the fluid particle passing through x at time t. Equivalently, for any t and any x_0, this definition reads

$$v(t,\varphi_t(x_0)) = \frac{\partial \varphi_s(x_0)}{\partial s}_{|s\,=\,t}. \tag{I.3}$$

Remark I.1.1. From (I.3), we see that, provided the velocity field v is known and is smooth enough, it is theoretically possible to determine the position field X by resolving the following differential equation, sometimes known as the *characteristic equation*,

$$\frac{\partial X(t,t_0,x_0)}{\partial t} = v(t,X(t,t_0,x_0)), \quad \text{and} \quad X(t_0,t_0,x_0) = x_0. \tag{I.4}$$

The coordinates (t,X) above, constitute the so-called Eulerian coordinate system. Working in Eulerian coordinates involves fixing a position X and writing the mechanical and thermodynamical balance equations at this fixed point.

Let $f(t,X)$ be any quantity related to the description of the fluid (density, temperature, etc.). The price to pay for the use of Eulerian coordinates is that the partial derivative of f with respect to time $\partial f/\partial t$ does not correctly describe the variation of the quantity f with time for the particle passing through the fixed position X at time t. This is because it does not account for the motion of the particle. In this case, the correct concept for a time derivative of f in the Eulerian description is that of the material derivative defined by

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + v\cdot\nabla f. \tag{I.5}$$

This definition comes from the following calculation of the variation with respect to time of the quantity f associated with a fluid particle.

$$\begin{aligned}
&\frac{d}{dt}\big(f(t,X(t,t_0,x_0))\big)\\
=&\frac{\partial f}{\partial t}(t,X(t,t_0,x_0)) + \frac{\partial X}{\partial t}(t,t_0,x_0)\cdot\nabla f(t,X(t,t_0,x_0))\\
=&\frac{\partial f}{\partial t}(t,X(t,t_0,x_0)) + v(t,X(t,t_0,x_0))\cdot\nabla f(t,X(t,t_0,x_0))\\
=&\frac{Df}{Dt}(t,X(t,t_0,x_0)).
\end{aligned}$$

For simplicity, when stating and describing the results of this chapter, a number of assumptions are made regarding the regularity of the variables used to characterise the flow (density, velocity, etc.). Note, however, that it is possible to weaken those assumptions in some cases.

2 The transport theorem

We begin by describing a fundamental result of differentiation, known in fluid mechanics as the transport theorem. It shows us how to differentiate the value of a scalar quantity integrated over a fluid element $(\Omega_t)_t$.

We assume that, for each t, the map φ_t is a smooth diffeomorphism and that $t \mapsto \varphi_t$ is smooth. To simplify the notation a little we also set $\varphi(t,x) = \varphi_t(x) = X(t,0,x)$.

Theorem I.2.1. *For any function f of class $\mathcal{C}^1$ with respect to the variables $(t,X) \in \mathbb{R} \times \mathbb{R}^3$, we have*

$$\frac{d}{dt}\int_{\Omega_t} f(t,X)dX = \int_{\Omega_t}\left(\frac{\partial f}{\partial t} + \operatorname{div}(fv)\right)dX,$$

for any time t, where v is the vector field in $\mathbb{R}^3$ defined in (I.2).

Proof.

At a fixed t, the change of variables $x \in \Omega_0 \mapsto X = \varphi(t,x) \in \Omega_t$ ensures that

$$\int_{\Omega_t} f(t,X)dX = \int_{\Omega_0} f(t,\varphi(t,x))|J(t,x)|dx,$$

where J is the Jacobian determinant of the map $x \mapsto \varphi(t,x)$.

Moreover the determinant is a trilinear function with respect to its three rows. From (I.3) we deduce the identity

$$\frac{\partial}{\partial t}\frac{\partial \varphi_i}{\partial x_j}(t,x) = \frac{\partial}{\partial x_j}\frac{\partial \varphi_i}{\partial t}(t,x) = \frac{\partial}{\partial x_j}\left(v_i(t,\varphi(t,x))\right).$$

Omitting to write the points where the derivatives are evaluated, we obtain

$$\frac{d}{dt}J(t,x) = \begin{vmatrix} \frac{\partial v_1}{\partial x_1} & \frac{\partial v_1}{\partial x_2} & \frac{\partial v_1}{\partial x_3} \\ \frac{\partial \varphi_2}{\partial x_1} & \frac{\partial \varphi_2}{\partial x_2} & \frac{\partial \varphi_2}{\partial x_3} \\ \frac{\partial \varphi_3}{\partial x_1} & \frac{\partial \varphi_3}{\partial x_2} & \frac{\partial \varphi_3}{\partial x_3} \end{vmatrix} + \begin{vmatrix} \frac{\partial \varphi_1}{\partial x_1} & \frac{\partial \varphi_1}{\partial x_2} & \frac{\partial \varphi_1}{\partial x_3} \\ \frac{\partial v_2}{\partial x_1} & \frac{\partial v_2}{\partial x_2} & \frac{\partial v_2}{\partial x_3} \\ \frac{\partial \varphi_3}{\partial x_1} & \frac{\partial \varphi_3}{\partial x_2} & \frac{\partial \varphi_3}{\partial x_3} \end{vmatrix} + \begin{vmatrix} \frac{\partial \varphi_1}{\partial x_1} & \frac{\partial \varphi_1}{\partial x_2} & \frac{\partial \varphi_1}{\partial x_3} \\ \frac{\partial \varphi_2}{\partial x_1} & \frac{\partial \varphi_2}{\partial x_2} & \frac{\partial \varphi_2}{\partial x_3} \\ \frac{\partial v_3}{\partial x_1} & \frac{\partial v_3}{\partial x_2} & \frac{\partial v_3}{\partial x_3} \end{vmatrix}.$$

Applying the chain rule then gives

$$\frac{\partial}{\partial x_j}\left(v_k(t,\varphi(t,x))\right) = \sum_{l=1}^{3} \frac{\partial v_k}{\partial X_l}(t,\varphi(t,x))\frac{\partial \varphi_l}{\partial x_j}(t,x). \tag{I.6}$$

We define the row vectors L_k and M_k as

$$L_k = \left(\frac{\partial \varphi_k}{\partial x_1}(t,x), \frac{\partial \varphi_k}{\partial x_2}(t,x), \frac{\partial \varphi_k}{\partial x_3}(t,x)\right),$$
$$M_k = \left(\frac{\partial}{\partial x_1}\left(v_k(t,\varphi(t,x))\right), \frac{\partial}{\partial x_2}\left(v_k(t,\varphi(t,x))\right), \frac{\partial}{\partial x_3}\left(v_k(t,\varphi(t,x))\right)\right).$$

Equation (I.6) shows that

$$M_k = \sum_{l=1}^{3} \frac{\partial v_k}{\partial X_l}(t,\varphi(t,x))\, L_l,$$

hence, this gives

$$\det(M_1,L_2,L_3) = \det\left(\sum_{l=1}^{3}\frac{\partial v_1}{\partial X_l}L_l, L_2, L_3\right) = \sum_{l=1}^{3}\frac{\partial v_1}{\partial X_l}\det(L_l,L_2,L_3)$$
$$= \frac{\partial v_1}{\partial X_1}\det(L_1,L_2,L_3) = \frac{\partial v_1}{\partial X_1}(t,\varphi(t,x))\, J,$$

because the determinant is trilinear and alternating.

Similar calculations yield

$$\det(L_1,M_2,L_3) = \frac{\partial v_2}{\partial X_2}(t,\varphi(t,x))\, J,\ \det(L_1,L_2,M_3) = \frac{\partial v_3}{\partial X_3}(t,\varphi(t,x))\, J.$$

However, we have seen that we can write

$$\frac{\partial J}{\partial t} = \det(M_1,L_2,L_3) + \det(L_1,M_2,L_3) + \det(L_1,L_2,M_3).$$

This results in

$$\frac{\partial J(t,x)}{\partial t} = (\operatorname{div} v)(t,\varphi(t,x))\, J(t,x).$$

We have defined $\varphi(t)$ to be a $\mathcal{C}^1$-diffeomorphism that depends continuously on t, so its Jacobian J is not zero and it takes all its values in $\mathbb{R}_*^+$ or in $\mathbb{R}_*^-$, domains over which the "absolute value" function is $\mathcal{C}^\infty$ and has the "sign" function as its derivative. Therefore, the function $|J|$ is differentiable, and

$$\frac{\partial |J|}{\partial t} = \operatorname{sgn}(J)\frac{\partial J}{\partial t} = \operatorname{sgn}(J)(\operatorname{div} v)(t,\varphi(t,x))\, J = (\operatorname{div} v)(t,\varphi(t,x))\, |J|.$$

By applying the theory for differentiation under the integral sign, we obtain

$$\begin{aligned}\frac{d}{dt}\int_{\Omega_t} f(t,X)\,dX &= \frac{d}{dt}\int_{\Omega_0} f(t,\varphi(t,x))|J(t,x)|\,dx\\ &= \int_{\Omega_0} \frac{\partial}{\partial t}\big(f(t,\varphi(t,x))|J(t,x)|\big)\,dx.\end{aligned}$$

Furthermore,

$$\begin{aligned}\frac{\partial}{\partial t}\big(f(t,\varphi(t,x))|J(t,x)|\big) =& \frac{\partial f}{\partial t}(t,\varphi)\,|J| + \left(\frac{\partial \varphi}{\partial t}\right)\cdot\nabla f\,|J| + f(t,\varphi)\frac{\partial}{\partial t}|J|\\ =& \left(\frac{\partial f}{\partial t}(t,\varphi) + v\cdot\nabla f + f(t,\varphi)(\operatorname{div} v)(t,\varphi(t,x))\right)|J|.\end{aligned}$$

An inverse change of variables in the above, gives the result that

$$\begin{aligned}\frac{d}{dt}\int_{\Omega_t} f(t,X)\,dX &= \int_{\Omega_t}\left(\frac{\partial f}{\partial t} + v\cdot\nabla f + f\ \operatorname{div} v\right)dX\\ &= \int_{\Omega_t}\left(\frac{\partial f}{\partial t} + \operatorname{div}(fv)\right)dX.\end{aligned}$$

□

Remark I.2.1. This result for differentiation of a scalar equation can be generalised to vector equations. For convenience, we give this result below in condensed form, in which $F(t,X)$ is a vector quantity:

$$\frac{d}{dt}\int_{\Omega_t} F(t,X)\,dX = \int_{\Omega_t}\left(\frac{\partial F}{\partial t} + \operatorname{div}(F\otimes v)\right)dX,$$

with the same vector field v as in the preceding theorem. See Appendix A for the definition of $\operatorname{div}(F\otimes v)$.

3 Evolution equations

In this section, we introduce the set of equations that describes the motion of a fluid. These are the equations for the conservation of mass, for the conservation of momentum, and finally for the conservation of energy. We use a Eulerian description of motion and limit ourselves to the case of Newtonian flows (see Section 4).

In all this section, $(\Omega_t)_t$ is an arbitrary fluid element as in Definition I.1.2.

3.1 Balance equations

3.1.1 Mass conservation

We recall that we assumed that there is neither a sink nor a source of matter within the flow considered. Therefore, the mass conservation property can be stated as follows: for any given initial volume Ω_0 the total mass of the matter contained in $\Omega_t = \varphi_t(\Omega_0)$ at time t is the same as the one contained in Ω_0 at the initial time.

Using the notion of density introduced in Definition I.1.1, this property can be expressed as

$$\frac{d}{dt}\int_{\Omega_t} \rho\, dX = 0.$$

Note that to avoid the equations appearing too cumbersome, we no longer express the dependency on (t, X) for the various unknowns in the problem, unless it is really necessary.

Using the transport theorem (Theorem I.2.1), this can be equivalently written as

$$\int_{\Omega_t} \left(\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho v)\right) dX = 0,$$

which has to hold for all time t and for any choice of the fluid element $(\Omega_t)_t$ evolving with the flow. Assuming that ρ and v are regular functions, this clearly leads to the following partial differential equation, known as the *conservation of mass equation* or the *continuity equation* or the *mass balance* equation

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho v) = 0. \tag{I.7}$$

3.1.2 Linear momentum equation

In this section we apply Newton's second law of motion in order to express the evolution of linear momentum for a fluid element.

The total linear momentum contained within a fluid element $(\Omega_t)_t$, at time t, is the quantity

$$\int_{\Omega_t} \rho v\, dX.$$

Newton's second law of motion states that the rate of change of this total momentum is equal to the sum of the external forces applied to this fluid element. In general, these forces are of two types:

- Body forces,

$$\int_{\Omega_t} \rho f\, dX,$$

where, in many situations, the mass density of forces f (which represents the external forces experienced by the fluid) is reduced to the acceleration due to gravity.

- Surface forces,

$$\int_{\partial\Omega_t} \tau(\nu)\,dy,$$

where ν is the unit normal outward from the surface $\partial\Omega_t$ of Ω_t. These forces are exerted on the investigated fluid element, $(\Omega_t)_t$, by the rest of the flow; they are called the *stress forces*. Of course, the vector $\tau(\nu)$ also depends on (t, y) but we have not expressed this explicitly in the equations.

Hence, in general, the equation for the linear momentum is written using the transport theorem for vector fields (Remark I.2.1) in the following way,

$$\int_{\Omega_t} \left(\frac{\partial \rho v}{\partial t} + \operatorname{div}(\rho v \otimes v)\right) dX = \int_{\Omega_t} \rho f\,dX + \int_{\partial\Omega_t} \tau(\nu)\,dy, \tag{I.8}$$

and has to be satisfied for any choice of the fluid element $(\Omega_t)_t$.

3.1.3 Angular momentum equation

We now apply the theory of angular momentum to the fluid element $(\Omega_t)_t$. This states that the rate of change of the total angular momentum of the fluid element, calculated with respect to a given fixed point, is equal to the sum of the moments of the forces exerted on the element calculated at the same point.

This can be written by considering a fixed point O and by defining r as the position vector $\overrightarrow{OX}$

$$\frac{d}{dt}\int_{\Omega_t} r \wedge (\rho v)\,dX = \int_{\Omega_t} r \wedge (\rho f)\,dX + \int_{\partial\Omega_t} r \wedge \tau(\nu)\,dy.$$

Since in Eulerian coordinates the position vector r does not depend on time, the transport theorem (Theorem I.2.1) tells us that the preceding equation can be written

$$\int_{\Omega_t} \left(r \wedge \left(\frac{\partial \rho v}{\partial t}\right) + \operatorname{div}\big((r \wedge \rho v) \otimes v\big)\right) dX = \int_{\Omega_t} r \wedge (\rho f) dX + \int_{\partial\Omega_t} r \wedge \tau(\nu)\,dy.$$

Furthermore, using Formula (A.10) from Appendix A, we have

$$\begin{aligned}
\operatorname{div}\big((r\wedge\rho v)\otimes v\big) &= (\operatorname{div} v)(r\wedge\rho v) + (v\cdot\nabla)(r\wedge\rho v)\\
&= r\wedge\big((\operatorname{div} v)\rho v\big) + \sum_{i=1}^{3} v_i\partial_{x_i}(r\wedge\rho v)\\
&= r\wedge\big((\operatorname{div} v)\rho v\big) + \sum_{i=1}^{3} v_i(r\wedge\partial_{x_i}\rho v) + \sum_{i=1}^{3} v_i\frac{\partial r}{\partial x_i}\wedge(\rho v)\\
&= r\wedge\big((\operatorname{div} v)\rho v\big) + r\wedge\big(v\cdot\nabla(\rho v)\big) + \sum_{i=1}^{3} v_i\frac{\partial r}{\partial x_i}\wedge(\rho v).
\end{aligned}$$

However, since the point O is fixed, we have $\partial r/\partial x_i = e_i$, the ith vector of the canonical basis of $\mathbb{R}^3$. Hence the final term of the preceding equation can be expressed by

$$\sum_{i=1}^{3} v_i\frac{\partial r}{\partial x_i}\wedge(\rho v) = \sum_{i=1}^{3} v_i e_i\wedge(\rho v) = v\wedge(\rho v) = 0.$$

By applying Formula (A.10) once more, we obtain

$$\operatorname{div}\big((r\wedge\rho v)\otimes v\big) = r\wedge(\operatorname{div}(\rho v\otimes v)).$$

In conclusion, the equation for conservation of angular momentum can be written in the form

$$\int_{\Omega_t} r\wedge\left(\frac{\partial\rho v}{\partial t} + \operatorname{div}(\rho v\otimes v)\right) dX = \int_{\Omega_t} r\wedge(\rho f)\, dX + \int_{\partial\Omega_t} r\wedge\tau(\nu)\, dy. \quad \text{(I.9)}$$

3.1.4 Evolution of energy

We start by recalling the first law of thermodynamics which states that the rate of change of the total energy of a system is the sum of the power of mechanical forces and of the heat exchange rate with the exterior.

The total energy per unit volume is given by the quantity $\rho\,\mathscr{E} \equiv \rho\,(e+\frac{1}{2}|v|^2)$ where e is the specific internal energy (*i.e.* per unit mass) related to the thermodynamic state of the fluid and $\frac{1}{2}\rho|v|^2$ represents the kinetic energy per unit volume.

The power of external forces can be broken down into two terms, one associated with the body forces and the other with the stresses applied at the surface

$$\dot{W} = \int_{\Omega_t} \rho\, f\cdot v\, dX + \int_{\partial\Omega_t} \tau(\nu)\cdot v\, dy.$$

We assume that there is no heat source (by radiation, for example) inside the volume; then the heat input rate in the volume Ω_t takes the form

$$\dot{Q} = -\int_{\partial\Omega_t} \Phi(t, y, \nu)\, dy,$$

where the assumption is made that the term Φ for heat transfer across the surface of Ω_t depends only on the position, time, and the outward normal of Ω_t at the point considered. As with the stress τ, we only express the dependency on (t, y) when necessary.

The energy balance can now be written as

$$\frac{d}{dt}\int_{\Omega_t} \rho\mathscr{E}\, dX = \int_{\Omega_t} \rho f \cdot v\, dX + \int_{\partial\Omega_t} \tau(\nu)\cdot v\, dy - \int_{\partial\Omega_t} \Phi(\nu)\, dy.$$

By applying the transport theorem to the derivatives with respect to time, the following conservation equation is obtained.

$$\begin{aligned} &\int_{\Omega_t} \left(\frac{\partial\rho\mathscr{E}}{\partial t} + \operatorname{div}(\rho\mathscr{E} v)\right) dX \\ &= \int_{\Omega_t} \rho f \cdot v\, dX + \int_{\partial\Omega_t} \tau(\nu)\cdot v\, dy - \int_{\partial\Omega_t} \Phi(\nu)\, dy. \end{aligned} \tag{I.10}$$

3.2 Cauchy's stress theorem

In this section, we study the general form of stresses $\tau(\nu)$ and, more precisely, we demonstrate the fundamental theorem below. This stipulates that stress is linear with respect to the normal to the surface of the volume element considered.

In this section, we explicitly express the dependence of stresses on (t, y) which are, therefore, denoted by $\tau(t, y, \nu)$.

Theorem I.3.1 (Cauchy's stress). *Assume that the density field ρ, the velocity field v, and the body force density f are regular. Let also assume that, the vector ν being fixed, the function $(t, y) \mapsto \tau(t, y, \nu)$ is continuous.*

Then, there exists a tensor-valued function $(t, y) \mapsto \sigma(t, y)$ such that for all (t, y) and for all unit vectors ν we have

$$\tau(t, y, \nu) = \sigma(t, y).\nu.$$

Proof.

The key point of the proof is an homogeneity argument comparing the sizes of surface and volume integrals in linear momentum balance equations on an elementary control volume whose size tends to 0. Let us give the details of these computations.

- We first show that $\tau(t, y, \nu) = -\tau(t, y, -\nu)$ for all (t, y, ν). Let us set

$$h = \frac{\partial \rho v}{\partial t} + \operatorname{div}(\rho v) - \rho f,$$

such that the equation for conservation of momentum (I.8) is written for any domain ω as

$$\int_\omega h\, dX = \int_{\partial\omega} \tau(t, y, \nu)\, dy. \tag{I.11}$$

Let y_0 be an unspecified point and ν_0 a unit vector. We consider a sphere ω centred on y_0 and divided into two hemispheres ω_1 and ω_2 by the plane Σ passing through y_0 and orthogonal to ν_0 (it is assumed that ν_0 is directed from ω_1 towards ω_2). By applying the conservation equation above to ω, to ω_1 and to ω_2, we obtain, respectively,

$$\int_\omega h\, dX = \int_{\partial\omega} \tau(t, y, \nu)\, dy,$$
$$\int_{\omega_1} h\, dX = \int_{\partial\omega_1\cap\Sigma} \tau(t, y, \nu_0)\, dy + \int_{\partial\omega\cap\partial\omega_1} \tau(t, y, \nu)\, dy,$$

and

$$\int_{\omega_2} h\, dX = \int_{\partial\omega_2\cap\Sigma} \tau(t, y, -\nu_0)\, dy + \int_{\partial\omega\cap\partial\omega_2} \tau(t, y, \nu)\, dy.$$

Inasmuch as $\omega = \omega_1 \cup \omega_2$ and $\partial\omega = (\partial\omega \cap \partial\omega_1) \cup (\partial\omega \cap \partial\omega_2)$, we can combine these three equations to obtain

$$\int_{\partial\omega_1\cap\Sigma} \big(\tau(t, y, \nu_0) + \tau(t, y, -\nu_0)\big)\, dy = 0.$$

This is valid for any sphere ω centred on y_0, which can be as small as desired, thus by continuity of the function $y \mapsto \tau(t, y, \nu_0) + \tau(t, y, -\nu_0)$ we obtain

$$\tau(t, y_0, -\nu_0) = -\tau(t, y_0, \nu_0).$$

- A priori, the function $\tau(t, y, \nu)$ is only defined for unit vectors ν. Because of the property above, we can extend this function to all vectors ξ of $\mathbb{R}^3$ by

$$\tau(t, y, \xi) = |\xi|\tau\left(t, y, \frac{\xi}{|\xi|}\right),$$

and

$$\tau(t, y, 0) = 0.$$

From this, it follows that this function is homogeneous of order 1 with respect to ξ. Indeed, we have

$$\tau(t,y,\lambda\xi) = |\lambda||\xi|\tau\left(t,y,\frac{\lambda\xi}{|\lambda\xi|}\right) = |\lambda||\xi|\tau\left(t,y,\operatorname{sgn}(\lambda)\frac{\xi}{|\xi|}\right)$$
$$= |\lambda||\xi|\operatorname{sgn}(\lambda)\tau\left(t,y,\frac{\xi}{|\xi|}\right) = \lambda\tau(t,y,\xi).$$

- We can now demonstrate the additive nature of the function $\xi \mapsto \tau(t,y,\xi)$. Let ξ_1 and ξ_2 be two vectors from $\mathbb{R}^3$. If these two vectors are colinear, then the homogeneity property above is sufficient to make the conclusion. Suppose, now, that ξ_1 is not colinear with ξ_2. Let y_0 be a fixed point. There is a plane $\mathscr{P}$ generated by the point y_0 and the vectors ξ_1, ξ_2 with an orientation, which is equivalent to choosing a unit normal ν_0 to this plane. We denote $\xi_i^\perp$ as the vector obtained by the rotation of the vector ξ_i through an angle of $\pi/2$ in the positive sense.

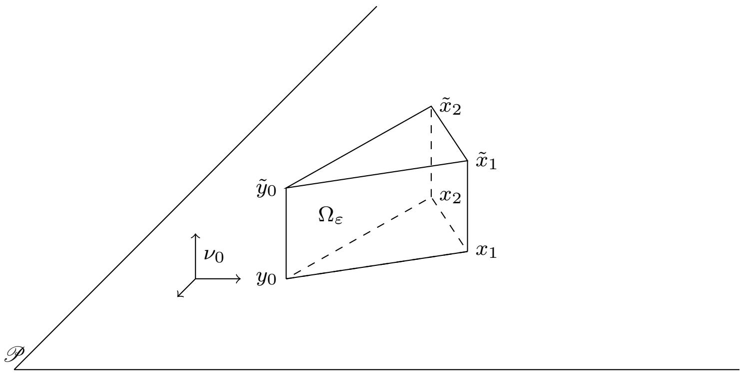


Fig. I.1 The prism Ω_ε

We now introduce the points

$$x_1 = y_0 + \varepsilon\xi_1^\perp, x_2 = y_0 - \varepsilon\xi_2^\perp,$$

and

$$\tilde{y}_0 = y_0 + \varepsilon\nu_0, \tilde{x}_1 = x_1 + \varepsilon\nu_0, \tilde{x}_2 = x_2 + \varepsilon\nu_0,$$

such that $y_0x_1x_2\tilde{y}_0\tilde{x}_1\tilde{x}_2$ form a right prism (denoted Ω_ε) of height ε and having the triangle $y_0x_1x_2$ as its base (see Figures I.1 and I.2).
If we now apply Equation (I.11) to this prism Ω_ε, then after division by ε^2, we obtain

$$\frac{1}{\varepsilon^2}\int_{\Omega_\varepsilon} h\,dX = \frac{1}{\varepsilon^2}\int_{\partial\Omega_\varepsilon} \tau(t,y,\nu)\,dy. \tag{I.12}$$

The surface terms can be broken into five components, three referring to the lateral faces of the prism and two concerning the bases of the prism. On the two bases, the outward unit normal is given by $\pm\nu_0$, and the outward unit normals of the lateral faces are exactly identical to the outward unit

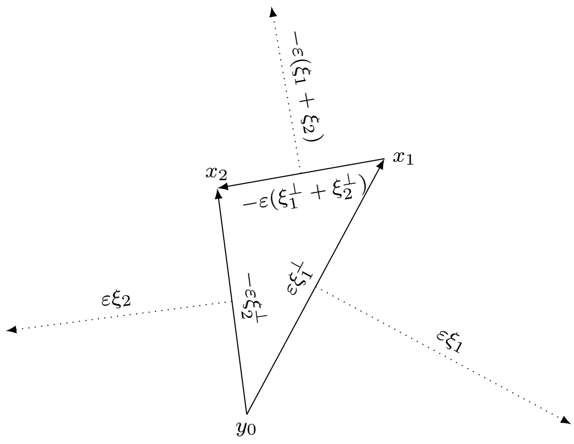


Fig. I.2 The triangle $y_0x_1x_2$ in the plane $\mathscr{P}$

normals of triangle $y_0x_1x_2$ in the plane $\mathscr{P}$ of the triangle. However, we have

$$x_1 - y_0 = \varepsilon\xi_1^\perp, x_2 - y_0 = -\varepsilon\xi_2^\perp, \text{and hence} x_2 - x_1 = -\varepsilon(\xi_1^\perp + \xi_2^\perp),$$

and, therefore, there exists a value $\eta \in \{-1, 1\}$ that does not depend on ε (it depends only on the orientation selected) such that the outward unit normals of the lateral faces of the prism are given by

$$\begin{aligned}
\nu(y_0x_1\tilde{x}_1\tilde{y}_0) &= \eta\frac{\xi_1}{|\xi_1|},\\
\nu(y_0x_2\tilde{x}_2\tilde{y}_0) &= \eta\frac{\xi_2}{|\xi_2|},\\
\nu(x_1x_2\tilde{x}_2\tilde{x}_1) &= -\eta\frac{\xi_1+\xi_2}{|\xi_1+\xi_2|}.
\end{aligned}$$

The surface term in (I.12), is then expressed by using the homogeneity of the function τ with respect to ξ, as

$$\begin{aligned}
&\frac{1}{\varepsilon^2}\int_{\partial\Omega_\varepsilon}\tau(t,y,\nu)\,dy\\
&= -\frac{\eta}{\varepsilon^2|\xi_1+\xi_2|}\int_{x_1x_2\tilde{x}_2\tilde{x}_1}\tau(t,y,\xi_1+\xi_2)\,dy\\
&\quad+\frac{\eta}{\varepsilon^2|\xi_1|}\int_{y_0x_1\tilde{x}_1\tilde{y}_0}\tau(t,y,\xi_1)\,dy+\frac{\eta}{\varepsilon^2|\xi_2|}\int_{y_0x_2\tilde{x}_2\tilde{y}_0}\tau(t,y,\xi_2)\,dy\\
&\quad+\frac{1}{\varepsilon^2}\int_{y_0x_1x_2}\tau(t,y,-\nu_0)\,dy+\frac{1}{\varepsilon^2}\int_{\tilde{y}_0\tilde{x}_1\tilde{x}_2}\tau(t,y,\nu_0)\,dy.
\end{aligned}$$

However, we have already seen that $\tau(t,y,-\nu_0) = -\tau(t,y,\nu_0)$ and, furthermore, the area of the triangle $y_0x_1x_2$ is equal to the area of the triangle $\tilde{y}_0\tilde{x}_2\tilde{x}_2$ and has the value $\varepsilon^2|\xi_1 \wedge \xi_2|/2$. It is therefore proportional to ε^2. Since we assume that $y \mapsto \tau(t,y,\nu_0)$ is continuous, we obtain, using the mean value theorem,

$$\begin{aligned}
&\frac{1}{\varepsilon^2}\int_{y_0x_1x_2} \tau(t,y,-\nu_0)\,dy + \frac{1}{\varepsilon^2}\int_{\tilde{y}_0\tilde{x}_1\tilde{x}_2} \tau(t,y,\nu_0)\,dy \\
=&\frac{1}{\varepsilon^2}\int_{y_0x_1x_2} (\tau(t,y,-\nu_0) + \tau(t,y+\varepsilon\nu_0,\nu_0))\,dy \\
=&\frac{C}{\mathrm{area}(y_0x_1x_2)}\int_{y_0x_1x_2} (\tau(t,y,-\nu_0) + \tau(t,y+\varepsilon\nu_0,\nu_0))\,dy \xrightarrow[\varepsilon\to 0]{} 0.
\end{aligned}$$

Let us now consider one of the terms on a lateral face, for example, $y_0x_1\tilde{x}_1\tilde{y}_0$. The area of this rectangular face is $\varepsilon|x_1 - y_0| = \varepsilon^2|\xi_1|$. The term corresponding to this face can therefore be written as

$$\begin{aligned}
\frac{\eta}{\varepsilon^2|\xi_1|}\int_{y_0x_1\tilde{x}_1\tilde{y}_0} \tau(t,y,\xi_1)\,dy =& \frac{\eta}{\mathrm{area}(y_0x_1\tilde{x}_1\tilde{y}_0)}\int_{y_0x_1\tilde{x}_1\tilde{y}_0} \tau(t,y,\xi_1)\,dy \\
&\xrightarrow[\varepsilon\to 0]{} \eta\tau(t,y_0,\xi_1).
\end{aligned}$$

Passing to the limits is justified by continuity (and thus uniform continuity in the neighborhood of y_0) of the function $y \mapsto \tau(t,y,\xi_1)$. The two remaining surface terms are treated in the same way.
It just remains to investigate the volume term in Equation (I.12). Remember that it was assumed that ρ and v are regular fields and therefore the function h is locally bounded. The volume of the prism Ω_ε is clearly proportional to ε^3, therefore we immediately obtain

$$\left|\frac{1}{\varepsilon^2}\int_{\Omega_\varepsilon} h\,dX\right| \leq C\varepsilon \xrightarrow[\varepsilon\to 0]{} 0.$$

Finally, we can pass to the limit in (I.12), when ε tends to 0, and we obtain

$$0 = \eta\tau(t,y_0,\xi_1) + \eta\tau(t,y_0,\xi_2) - \eta\tau(t,y_0,\xi_1+\xi_2).$$

After simplifying this equation by η, this can be expressed as

$$\tau(t,y_0,\xi_1+\xi_2) = \tau(t,y_0,\xi_1) + \tau(t,y_0,\xi_2).$$

All of the above demonstrate the linearity of the stress $\tau(t,y,\xi)$ with respect to ξ and therefore prove the existence of the tensors $\sigma(t,y)$ such that $\tau(t,y,\nu) = \sigma(t,y).\nu$, for each unit vector ν.

□

Definition I.3.2. *For all points (t,y), the tensor $\sigma(t,y)$ defined by the relation*

$$\tau(t,y,\nu) = \sigma(t,y).\nu, \textit{for all unit vectors } \nu,$$

is known as the stress tensor *of the flow.*

Again, to avoid cumbersome equations, we no longer indicate the dependence of the stress tensor on (t,y). Hence, we write the stress in the form $\tau(\nu) = \sigma.\nu$. The surface term in the energy conservation equation (I.10) is now expressed by using the divergence theorem (i.e., the Stokes formula; see Theorem III.1.8)

$$\int_{\partial\Omega_t} (\sigma.\nu)\cdot v\, dy = \int_{\partial\Omega_t} \nu\cdot({}^t\sigma.v)\, dy = \int_{\Omega_t} \operatorname{div}({}^t\sigma.v)\, dX.$$

Hence, the energy evolution equation becomes

$$\begin{aligned}&\int_{\Omega_t}\left(\frac{\partial\rho\mathscr{E}}{\partial t} + \operatorname{div}(\rho\mathscr{E}v)\right) dX\\ &= \int_{\Omega_t}\rho f\cdot v\, dX + \int_{\Omega_t}\operatorname{div}({}^t\sigma.v)\, dX - \int_{\partial\Omega_t}\Phi(\nu)\, dy.\end{aligned} \tag{I.13}$$

It is now possible to establish a result, similar to Cauchy's theorem, for the heat transfer terms Φ, which is stated in the following form.

Theorem I.3.3. *Let us assume that ρ, $\mathscr{E}$, v and f are regular functions. Let us further assume that for all ν the function $(t,y)\mapsto\Phi(t,y,\nu)$ is continuous; then there exists a continuous vector field φ, such that*

$$\Phi(t,y,\nu) = \varphi(t,y)\cdot\nu.$$

This vector field is called the heat flux in the flow.

The proof of this result is formally identical to that for Cauchy's theorem using the conservation of energy equation in the form (I.13). We can then apply the divergence theorem again to obtain a new form of the energy evolution equation

$$\begin{aligned}&\int_{\Omega_t}\left(\frac{\partial\rho\mathscr{E}}{\partial t} + \operatorname{div}(\rho\mathscr{E}v)\right) dX\\ &= \int_{\Omega_t}\rho f\cdot v\, dX + \int_{\Omega_t}\operatorname{div}({}^t\sigma.v)\, dX - \int_{\Omega_t}\operatorname{div}\varphi\, dX.\end{aligned}$$

Since this holds for any fluid element $(\Omega_t)_t$ evolving with the flow, we obtain the local form of the equation

$$\frac{\partial\rho\mathscr{E}}{\partial t} + \operatorname{div}(\rho\mathscr{E}v) - \operatorname{div}({}^t\sigma.v) + \operatorname{div}\varphi = \rho f\cdot v. \tag{I.14}$$

3.3 Evolution equations revisited

Now that we know, from Cauchy's theorem, that the stress forces have the tensor form $\tau(\nu) = \sigma.\nu$, we can deduce a simpler form for the conservation of momentum equation.

Indeed, the divergence theorem (Theorem III.1.8) tells us that for any domain Ω, we have

$$\int_{\partial\Omega} \sigma.\nu \, dy = \int_{\Omega} \operatorname{div} \sigma \, dX,$$

hence, the conservation equation (I.8) can be written as

$$\int_{\Omega_t} \left(\frac{\partial \rho v}{\partial t} + \operatorname{div}(\rho v \otimes v) \right) dX = \int_{\Omega_t} \rho f \, dX + \int_{\Omega_t} \operatorname{div} \sigma \, dX.$$

This relation having to be valid for any fluid element $(\Omega_t)_t$, we obtain *the linear momentum equation* in its local form

$$\frac{\partial(\rho v)}{\partial t} + \operatorname{div}(\rho v \otimes v) - \operatorname{div} \sigma = \rho f. \tag{I.15}$$

What will the angular momentum conservation relation now become? First we need to calculate the surface term and try to express it as a volume integral. To do this we attempt to write the term $r \wedge (\sigma.\nu)$ in the form $T.\nu$ where T is a new tensor to be determined. We therefore write

$$\begin{aligned} r \wedge (\sigma.e_i) = r \wedge \left(\sum_{j=1}^{3} \sigma_{ji} e_j \right) &= \left(\sum_{j=1}^{3} r_j e_j \right) \wedge \left(\sum_{j=1}^{3} \sigma_{ji} e_j \right) \\ &= (r_2\sigma_{3i} - r_3\sigma_{2i})e_1 + (r_3\sigma_{1i} - r_1\sigma_{3i})e_2 + (r_1\sigma_{2i} - r_2\sigma_{1i})e_3, \end{aligned}$$

which demonstrates that the ith column of the tensor matrix T in the canonical basis is simply

$$\begin{pmatrix} r_2\sigma_{3i} - r_3\sigma_{2i} \\ r_3\sigma_{1i} - r_1\sigma_{3i} \\ r_1\sigma_{2i} - r_2\sigma_{1i} \end{pmatrix}.$$

The surface term of (I.9) is expressed, by applying the divergence theorem to tensor T, as

$$\int_{\partial\Omega_t} r \wedge (\sigma.\nu) \, dy = \int_{\partial\Omega_t} T.\nu \, dy = \int_{\Omega_t} \operatorname{div} T \, dX.$$

Hence, Equation (I.9) becomes

$$\int_{\Omega_t} r \wedge \left(\frac{\partial \rho v}{\partial t} + \operatorname{div}(\rho v \otimes v) \right) dX = \int_{\Omega_t} r \wedge (\rho f) \, dX + \int_{\Omega_t} \operatorname{div} T \, dX.$$

Inasmuch as this relation must be true for any time t and any fluid element $(\Omega_t)_t$, this provides us with the equation for the angular momentum in its local form

$$r \wedge \left(\frac{\partial \rho v}{\partial t} + \operatorname{div}(\rho v \otimes v) - \rho f \right) - \operatorname{div} T = 0. \tag{I.16}$$

We now need to calculate the divergence of T. Let us begin with the first component, and note that the position vector, r, satisfies the relation $\partial r_i/\partial x_j = \delta_{ij}$ (the Kronecker delta):

$$\begin{aligned}
(\operatorname{div} T)_1 &= \sum_{i=1}^{3} \frac{\partial}{\partial x_i}(r_2 \sigma_{3i} - r_3 \sigma_{2i}) \\
&= \sum_{i=1}^{3} \left(r_2 \frac{\partial \sigma_{3i}}{\partial x_i} - r_3 \frac{\partial \sigma_{2i}}{\partial x_i} \right) + \sum_{i=1}^{3} \frac{\partial r_2}{\partial x_i} \sigma_{3i} - \frac{\partial r_3}{\partial x_i} \sigma_{2i} \\
&= \big(r \wedge \operatorname{div} \sigma \big)_1 + (\sigma_{32} - \sigma_{23}).
\end{aligned}$$

A similar calculation for the other two components gives

$$\operatorname{div} T = r \wedge \operatorname{div} \sigma + \begin{pmatrix} \sigma_{32} - \sigma_{23} \\ \sigma_{13} - \sigma_{31} \\ \sigma_{21} - \sigma_{12} \end{pmatrix}.$$

Hence, Equation (I.16) can now be written as

$$r \wedge \left(\frac{\partial \rho v}{\partial t} + \operatorname{div}(\rho v \otimes v) - \rho f - \operatorname{div} \sigma \right) - \begin{pmatrix} \sigma_{32} - \sigma_{23} \\ \sigma_{13} - \sigma_{31} \\ \sigma_{21} - \sigma_{12} \end{pmatrix} = 0.$$

However, Equation (I.15) demonstrates that the vector product which appears in this equation is identically zero (because the second factor is zero).

It has hence been shown that

$$\begin{pmatrix} \sigma_{32} - \sigma_{23} \\ \sigma_{13} - \sigma_{31} \\ \sigma_{21} - \sigma_{12} \end{pmatrix} = 0.$$

This fundamental property can be stated in the following way.

Theorem I.3.4. *Let us assume that the density field ρ, the velocity field v, and the body forces field f are smooth; then the stress tensor acting on the fluid in the flow is symmetric.*

This result has a number of fundamental consequences, as we show below. One of these is the existence, for all points (t, y), of an orthonormal basis of $\mathbb{R}^3$ that diagonalises the tensor σ. These basis vectors are known as the *principal axes* of the tensor and the corresponding eigenvalues are termed the *principal stresses* associated with the tensor σ.

Remark I.3.1. Now that we have established that the stress tensor is symmetric, it is clear that the evolution equation for the linear momentum implies the one for the angular momentum. Hence, from this point onwards, it is not needed to add this last equation in the system governing the motion of a fluid.

Using the symmetry of the stress tensor, *the evolution equation for the energy* (I.14) can be written in the form

$$\frac{\partial(\rho\mathscr{E})}{\partial t} + \operatorname{div}(\rho\mathscr{E}v) - \operatorname{div}(\sigma.v) + \operatorname{div}\varphi = \rho f \cdot v. \tag{I.17}$$

Remark I.3.2. The equations (I.15) and (I.17) obtained so far, are said to be *in the conservative form*, because they are expressed in the form of a divergence in (t, x) of a given quantity equal to a source term.

It is sometimes more relevant, for formal calculations or physical interpretations, to write the preceding equations in the *nonconservative* form, as follows,

- Linear momentum equation:

$$\underbrace{\rho\left(\frac{\partial v}{\partial t} + (v\cdot\nabla)v\right)}_{=\frac{Dv}{Dt}} - \operatorname{div}\sigma = \rho f. \tag{I.18}$$

- Energy equation:

$$\underbrace{\rho\left(\frac{\partial \mathscr{E}}{\partial t} + (v\cdot\nabla)\mathscr{E}\right)}_{=\frac{D\mathscr{E}}{Dt}} - \operatorname{div}(\sigma.v) + \operatorname{div}\varphi = \rho f \cdot v. \tag{I.19}$$

These equations are obtained algebraically, starting from our earlier equations and using the mass balance equation and (A.4), (A.10). From a formal point of view they are completely equivalent. From a mathematical point of view, when working with *weak* solutions (i.e., solutions that are not very regular), it is not always clear whether these two formulations are equivalent or even if we can attach a meaning to the nonconservative formulations.

4 Fundamental laws: Newtonian fluids and thermodynamics laws

In the preceding section, we provided the general form of the equations for the conservation of mass and the evolution of linear momentum and energy, describing the flow of a fluid that is subject to body forces and does not contain a source of matter or heat within the domain being studied.

These equations give rise to several, and yet undetermined, quantities, namely, the stress tensor σ (so far we only know that it is symmetric), the heat flux φ, and the specific internal energy e. These last two quantities are related to the other fields describing the flow via the laws of thermodynamics; we return to them in Section 4.4.

4.1 Fluids at rest

We consider initially the case of the stress tensor, σ. We start by remembering one of the fundamental properties of fluids.

Definition I.4.1. *In a fluid at rest, the stress acting on a surface element of a fluid element acts in the direction opposite to the one of the outward normal of the surface. Moreover, the modulus of this stress is independent of direction. It is denoted as p and referred to as the* hydrostatic pressure of the fluid*.*

Hence, from this definition of the pressure, for a fluid at rest, the stress at all points is $\tau(\nu) = -p\,\nu$, with $p \geq 0$. This means that the stress tensor can, in this case, be written as

$$\sigma = -p\,\mathrm{Id}.$$

When the fluid is in motion, the effects due to pressure and to the motion are separated by expressing the stress tensor in the form

$$\sigma = \mathscr{T} - p\,\mathrm{Id},$$

where the new tensor $\mathscr{T}$ (which is clearly symmetric) is called *the viscous stress tensor.*

Notice that it is postulated that the hydrostatic pressure p exactly coincides with the thermodynamic pressure P which is related to the state of the material at the molecular level as defined in Appendix B.

4.2 Newton's hypothesis

Let us consider a planar Couette flow for a viscous fluid (see also Section 7.2). This involves a fluid flow between two parallel plates that are assumed to be infinite (e.g., parallel to the plane generated by the first two coordinate vectors e_1 and e_2). We assume that under the action of a stress (i.e., of a force per unit surface) of constant magnitude and aligned along the direction of e_1 (this stress is denoted $F\,e_1$), the upper plate is driven at a velocity equal to $U_0 e_1$ whereas the lower plate is held at rest.

It can be observed experimentally for "classic" fluids, such as water for example, that the viscous stress F required to move the plate at velocity U_0 is given by

$$F = \mu \frac{U_0}{d}, \tag{I.20}$$

where μ is a constant that depends only on the fluid being considered. Put another way, the stress F is proportional to the velocity gradient U_0/d in the flow.

This stress effect is a form of resistance of the fluid to shearing. Indeed, if we cease to apply the driving force F at a given instant, then after a certain time the fluid will return to the rest state. However, we know that in the absence of a driving force and frictional forces, a point mass having a certain initial non-zero velocity maintains this velocity over time. In the case of water, for instance, we observe that the velocity is damped over time when the driving force is stopped. Such a fluid is said to be viscous.

We observed above that viscous effects appear once the fluid is subject to deformation. In a simplified way, in the case of the Couette flow, one can say that particles at height $x_3 = C + h$ have a horizontal velocity that is greater than that of particles at height $x_3 = C$, which creates the deformation of the fluid.

These ideas can be stated formally in the following way.

Definition I.4.2. *Consider a flow for which the velocity field is v. Let us consider three arbitrarily close points x_0, $x_1 = x_0 + (\delta x)\xi_1$, and $x_2 = x_0 + (\delta x)\xi_2$. We denote as $\tilde{x}_0$, $\tilde{x}_1$, and $\tilde{x}_2$ the positions at time $t + \delta t$ of the fluid particles that were situated respectively at x_0, x_1, and x_2 at time t.*

We say that the fluid behaves as a rigid body at the point x_0 at time t, if the triangles $x_0 x_1 x_2$ and $\tilde{x}_0 \tilde{x}_1 \tilde{x}_2$ can be superimposed up to a first-order correction with respect to δt and δx, for any choice of the directions ξ_1 and ξ_2.

More precisely, this means that for all i, j, k, we have

$$(\overrightarrow{\tilde{x}_i \tilde{x}_j} \cdot \overrightarrow{\tilde{x}_i \tilde{x}_k}) = (\overrightarrow{x_i x_j} \cdot \overrightarrow{x_i x_k}) \Big(1 + O\big(\delta t^2 + \delta x^2\big)\Big). \tag{I.21}$$

Definition and Proposition I.4.3. *A fluid behaves like a rigid body at (t, x) if, and only if, the tensor defined by*

$$\mathrm{D}(v) = \frac{1}{2}(\nabla v + {}^t\nabla v) = \left(\frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right)\right)_{1 \le i,j \le 3}$$

is zero at (t, x).

The tensor $\mathrm{D}(v)$ *is known as the* strain rate tensor *for the flow.*

Proof.

We denote as $s \mapsto x_0(s), x_1(s)$, and $x_2(s)$, the respective trajectories starting from x_0, x_1, and x_2 at time t, such that $\tilde{x}_i = x_i(t + \delta t)$. From the characteristic equation (I.4), we have

$$x_i'(s) = v(s, x_i(s)),$$

so that

$$x_i''(s) = \left(\frac{\partial v}{\partial t} + (v \cdot \nabla)v\right)(s, x_i(s)).$$

Let us carry out a Taylor expansion in the following way for each $j \in \{0, 1, 2\}$,

$$\tilde{x}_j = x_j + \delta t\, v(t, x_j) + \delta t^2 \int_0^1 (1-s)\left(\frac{\partial v}{\partial t} + (v \cdot \nabla)v\right)(t + s\,\delta t, x_j(t + s\,\delta t))\, ds.$$

However, since for $i \neq j$, $|x_i - x_j| \leq C\delta x$, and using the continuous dependence of the solutions of the characteristic equation with respect to the initial data, we get

$$|x_i(s) - x_j(s)| \leq C\delta x, \forall s \in [t, t + \delta t].$$

Hence, using the mean value theorem and assuming that the velocity field is sufficiently regular, it follows

$$\tilde{x}_j - \tilde{x}_i = x_j - x_i + \delta t(v(t, x_j) - v(t, x_i)) + O(\delta t^2 \delta x).$$

By performing a first-order expansion of the term $v(t, x_j) - v(t, x_i)$, we obtain

$$\overrightarrow{\tilde{x}_0\tilde{x}_1} = \tilde{x}_1 - \tilde{x}_0 = x_1 - x_0 + \delta t\delta x \nabla v(t, x_0).\xi_1 + O(\delta t \delta x^2) + O(\delta t^2 \delta x),$$

and

$$\overrightarrow{\tilde{x}_0\tilde{x}_2} = \tilde{x}_2 - \tilde{x}_0 = x_2 - x_0 + \delta t\delta x \nabla v(t, x_0).\xi_2 + O(\delta t \delta x^2) + O(\delta t^2 \delta x).$$

Next, we take the scalar product to obtain

$$\begin{aligned}\overrightarrow{\tilde{x}_0\tilde{x}_1} \cdot \overrightarrow{\tilde{x}_0\tilde{x}_2} &= \overrightarrow{x_0x_1} \cdot \overrightarrow{x_0x_2} + \delta t\delta x^2\big((\nabla v(t, x_0).\xi_1) \cdot \xi_2 + (\nabla v(t, x_0).\xi_2) \cdot \xi_1\big)\\ &\quad + O(\delta t \delta x^3) + O(\delta t^2 \delta x^2).\end{aligned}$$

By definition, we have $\overrightarrow{x_0x_1} \cdot \overrightarrow{x_0x_2} = (\delta x^2)(\xi_1 \cdot \xi_2) = C\delta x^2$, from which we can deduce

$$\begin{aligned}(\overrightarrow{\tilde{x}_0\tilde{x}_1} \cdot \overrightarrow{\tilde{x}_0\tilde{x}_2}) &= (\overrightarrow{x_0x_1} \cdot \overrightarrow{x_0x_2})\\ &\quad \times \left(1 + \delta t\big((\nabla v(t, x_0).\xi_1) \cdot \xi_2 + (\nabla v(t, x_0).\xi_2) \cdot \xi_1\big) + O(\delta t^2 + \delta x^2)\right).\end{aligned}$$

It can then be seen that condition (I.21) is thus equivalent to

$$(\nabla v(t, x_0).\xi_1) \cdot \xi_2 + (\nabla v(t, x_0).\xi_2) \cdot \xi_1 = 0,$$

for all vectors ξ_1 and ξ_2; this indicates exactly that the tensor $\nabla v(t, x_0)$ is antisymmetric or, equivalently, that $\mathrm{D}(v)$ is zero at the point (t, x_0).

$\square$

Remark I.4.1. It can be shown (see Lemma IV.7.5) that a velocity field v obeying $\mathrm{D}(v) = 0$ for all t, x is necessarily of the form

$$v(t, x) = v_0(t) + \omega(t) \wedge r,$$

where r is the position vector in a fixed reference frame. Alternatively we could express this by saying that, under these conditions, the motion is the superposition of a translational motion at uniform velocity (in x) and a rotational motion around a fixed axis, which corresponds to the possible motions of a rigid body.

Remark I.4.2. Elementary computations lead to the formulas

$$\begin{aligned}
\mathrm{Tr}(\mathrm{D}v) &= \mathrm{Tr}\,\nabla v = \operatorname{div} v,\\
\operatorname{div}({}^t\nabla v) &= \nabla(\operatorname{div} v),\\
\operatorname{div}(2\mathrm{D}v) &= \operatorname{div}(\nabla v + {}^t\nabla v) = \Delta v + \nabla(\operatorname{div} v),\\
\operatorname{div}((\operatorname{div} v)\,\mathrm{Id}) &= \nabla(\operatorname{div} v),
\end{aligned}$$

which are useful in the sequel.

The discussion above shows that all the information on the rate of deformation of the fluid in the flow is contained in the symmetric part $\mathrm{D}(v)$ of the velocity gradient tensor ∇v. Under these conditions, the *standard* viscous behavior of a fluid can be summed up as follows.

1. The viscous stress tensor $\mathscr{T}$ in a flow depends only on the strain rate tensor $\mathrm{D}(v)$.
2. The dependence of $\mathscr{T}$ on $\mathrm{D}(v)$ is linear.
3. The relation linking $\mathscr{T}$ and $\mathrm{D}(v)$ is isotropic.

Of course, not all fluids will conform to these properties. A fluid that satisfies these properties experimentally is known as a *Newtonian fluid.* A Newtonian fluid may also display non-Newtonian behavior in extreme conditions of temperature, pressure, or velocity.

Some comments are required on Newton's three assumptions. The first comes directly from the preceding considerations: the viscous stress in a fluid results directly from the action of the strain rate to which the fluid is subjected. It is therefore natural that the tensor $\mathscr{T}$ only depends on $\mathrm{D}(v)$. The assumption of linearity is a simple generalisation of (I.20) which shows, in the simple case of planar Couette flow (see also Section 7.2), that the stress F is linear with respect to the deformation U/d. We can therefore write $\mathscr{T} = L(\mathrm{D}(v))$, where L is a linear map defined from the set of all symmetric tensors into itself. Finally, the third proposition is very natural in physics and expresses the invariance of the rheological properties of the fluid studied when changing the orthonormal reference frame. The exact significance of this proposition is as follows. For all changes to the orthonormal reference frame (i.e., for all orthogonal matrices P) and for all symmetric tensors d, we have

$$L({}^tPdP) = {}^tPL(d)P. \tag{I.22}$$

Proposition I.4.4. *For a Newtonian fluid, the law giving the viscous stress tensor as a function of the strain rate tensor is necessarily in the form*

$$\mathscr{T} = 2\mu \mathrm{D}(v) + \lambda(\operatorname{div} v)\,\mathrm{Id},$$

where λ and μ are both real coefficients.

Proof.

Let d be a symmetric tensor and x an eigenvector of d for an eigenvalue Λ. We consider the matrix P of the symmetry with respect to the hyperplane orthogonal to x. That is to say $Px = -x$ and the restriction of P to the orthogonal of x is the identity.

Since x is an eigenvector of d, and by definition of P, we have ${}^tPdP = d$ such that property (I.22) gives

$${}^tPL(d)P = L(d),$$

or even, inasmuch as P is orthogonal,

$$L(d)P = PL(d).$$

If we apply this to x, we get $PL(d)x = L(d)Px = -L(d)x$ and hence, from the definition of P, $L(d)x$ is necessarily colinear to x, which proves that x is an eigenvector of $L(d)$. We denote the eigenvalue of $L(d)$ associated with x as $f(\Lambda)$.

Note that $f(\Lambda)$ does not depend on the eigenvector x for the eigenvalue Λ of d. Indeed, if y is another such eigenvector, then the preceding calculation shows that $tx + (1-t)y$ is also an eigenvector of $L(d)$ for all $t \in \mathbb{R}$. This proves that x and y are eigenvectors of $L(d)$ for the same eigenvalue. We can now calculate $L(d)$ when d is diagonal. Let E_i denote the matrix for which all the coefficients are zero except the coefficient (i,i) which has a value of 1. Following our discussion above, there exist values λ and μ such that

$$L(E_1) = 2\mu E_1 + \lambda\,\mathrm{Id}.$$

Now, let Q_i be the permutation matrix that exchanges the indices 1 and i, such that we have

$$E_i = {}^tQ_iE_1Q_i,$$

and hence

$$L(E_i) = L({}^tQ_iE_1Q_i) = {}^tQ_iL(E_1)Q_i = 2\mu{}^tQ_iE_1Q_i + \lambda\,\mathrm{Id} = 2\mu E_i + \lambda\,\mathrm{Id}.$$

This is true for all i, thus we have established that for all diagonal tensors d we have

$$L(d) = 2\mu d + \lambda(\mathrm{Tr}\, d)\,\mathrm{Id}.$$

Now, let d be any symmetric tensor. There exists an orthogonal matrix Q such that ${}^tQdQ$ is diagonal, giving

$$\begin{aligned} L(d) &= QL({}^tQdQ){}^tQ = Q\Big(2\mu{}^tQdQ + \lambda(\mathrm{Tr}({}^tQdQ))\,\mathrm{Id}\Big){}^tQ \\ &= 2\mu\, d + \lambda(\mathrm{Tr}\ d)\,\mathrm{Id}. \end{aligned}$$

As a consequence, the relation giving the viscous stress tensor can be expressed as

$$\mathscr{T} = L(\mathrm{D}(v)) = 2\mu\mathrm{D}(v) + \lambda(\mathrm{Tr}\,\mathrm{D}(v))\,\mathrm{Id} = 2\mu\mathrm{D}(v) + \lambda(\mathrm{div}\, v)\,\mathrm{Id},$$

because the trace of the strain rate tensor is simply the divergence of v.

□

We can now write the general form of the stress tensor for a Newtonian fluid

$$\sigma = \mathscr{T} - p\,\mathrm{Id} = 2\mu\mathrm{D}(v) + (\lambda\ \mathrm{div}\, v - p)\ \mathrm{Id}.$$

4.3 Consequences of the second law of thermodynamics

The second law of thermodynamics implies that the viscous stresses are necessarily dissipative, that is, that the power of these forces must be nonnegative. In order to describe this more clearly, it is useful to write the entropy evolution equation. This is done in detail below by using the results revised in Appendix B. In particular, we show that the local specific entropy equation can be expressed in the following conservative form,

$$\left(\frac{\partial\rho s}{\partial t} + \mathrm{div}(\rho s v)\right) + \frac{\mathrm{div}\,\varphi}{T} - \frac{\mathrm{Tr}(\mathscr{T}\mathrm{D}(v))}{T} = 0,$$

where s is the specific entropy (i.e., per unit mass) in the flow, and T is the temperature.

If we consider any fluid element $(\Omega_t)_t$ evolving with the flow under study, the total entropy evolution can be expressed, using the transport theorem, as

$$\begin{aligned}\frac{d}{dt}\int_{\Omega_t}\rho s\,dX &= -\int_{\Omega_t}\operatorname{div}\left(\frac{\varphi}{T}\right)dX - \int_{\Omega_t}\frac{\varphi\cdot\nabla T}{T^2}\,dX + \int_{\Omega_t}\frac{\operatorname{Tr}(\mathscr{T}\mathrm{D}(v))}{T}\,dX\\ &= -\int_{\partial\Omega_t}\frac{\varphi\cdot\nu}{T}dy - \int_{\Omega_t}\frac{\varphi\cdot\nabla T}{T^2}\,dX + \int_{\Omega_t}\frac{\operatorname{Tr}(\mathscr{T}\mathrm{D}(v))}{T}\,dX.\end{aligned}$$

If we integrate this equation between times t and $t+\delta t$, and if we call $S(t) = \int_{\Omega_t}\rho s\,dX$ the total entropy contained in the fluid element Ω_t, it follows that

$$\begin{aligned}&S(t+\delta t) - S(t) - \int_t^{t+\delta t}\int_{\partial\Omega_s}\frac{-\varphi\cdot\nu}{T}\,dy\,ds\\ &= \int_t^{t+\delta t}\int_{\Omega_s}\frac{-\varphi\cdot\nabla T}{T^2}\,dX\,ds + \int_t^{t+\delta t}\int_{\Omega_s}\frac{\operatorname{Tr}(\mathscr{T}\mathrm{D}(v))}{T}\,dX\,ds.\end{aligned}\tag{I.23}$$

The left-hand side of this equation can be written in the form $\Delta S - \Delta Q/T$ where ΔQ if the quantity of heat acquired by the fluid element through its boundary between the two times t and $t+\delta t$ (don't forget that ν is the *outward* normal to Ω_t). The second law of thermodynamics in the form (B.1) tells us that this quantity should be nonnegative, or even positive if the transformation between these two times is irreversible. In this form, the second law of thermodynamics is known as the *Clausius theorem*.

It can also be written in the local form:

$$\left(\frac{\partial\rho s}{\partial t} + \operatorname{div}(\rho s v)\right) + \operatorname{div}\left(\frac{\varphi}{T}\right) \geq 0.$$

We deduce from the Clausius theorem and from (I.23) that we must have

$$\varphi\cdot\nabla T \leq 0, \text{for any possible } \nabla T, \tag{I.24}$$

and

$$\operatorname{Tr}(\mathscr{T}\mathrm{D}(v)) \geq 0, \text{for any possible } \mathrm{D}(v). \tag{I.25}$$

By using the notation from the preceding paragraph, the relation (I.25) expresses that the quadratic form $d \mapsto \mathscr{P}(d) \equiv L(d) : d = \operatorname{Tr}(L(d)\,.\,d)$ defined over the set of all symmetric tensors, must be positive-semidefinite.

From the isotropy property (I.22), we know that $\mathscr{P}({}^tQdQ) = \mathscr{P}(d)$ for all orthogonal matrix Q. Hence, it is sufficient for us to study the quadratic form defined in $\mathbb{R}^3$ by

$$\mathscr{P}(d_1, d_2, d_3) = \mathscr{P}\left(\begin{pmatrix} d_1 & 0 & 0\\ 0 & d_2 & 0\\ 0 & 0 & d_3\end{pmatrix}\right).$$

Since the matrix of this quadratic form in the basis of $\mathbb{R}^3$ formed by the vectors

$$\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \text{ and } \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

is simply

$$\begin{pmatrix} 4\mu & 2\mu & 0 \\ 2\mu & 4\mu & 0 \\ 0 & 0 & 6\mu + 9\lambda \end{pmatrix},$$

we deduce that $\mathscr{P}$ is positive-semidefinite if and only if the following conditions hold,

$$\mu \geq 0 \text{ and } 2\mu + 3\lambda \geq 0. \tag{I.26}$$

The coefficient μ is known as the *dynamic viscosity* of the flow and the coefficient $2\mu/3 + \lambda$ (which is then nonnegative) is known as the *bulk viscosity* of the flow.

In many cases, the bulk viscosity is very low and can be neglected (Stokes' assumption); that is,

$$\lambda = -\frac{2}{3}\mu. \tag{I.27}$$

This finally gives rise to the following expression for the viscous stress tensor,

$$\mathscr{T} = 2\mu \left(\mathrm{D}(v) - \frac{1}{3}(\operatorname{div} v)\,\mathrm{Id} \right),$$

which, in this case, has a zero trace. This implies that all of the isotropic components of the stresses are contained in the pressure p.

In the case where the dynamic viscosity μ is small and can be neglected, we set $\mu = 0$, and the fluid is said to be *perfect* or *inviscid.*

To finish, we can now give a general expression for the heat flux, φ. Indeed, if we assume that the flux φ only depends on the temperature and its gradient, and that the dependence of φ on ∇T is invariant when changing between inertial reference frames, then we obtain Fourier's law which says that φ is proportional to the temperature gradient. Furthermore, the relation (I.24) tells us that the flux's direction is opposite to the one of the temperature gradient. We can therefore write

$$\varphi = -k\nabla T, \tag{I.28}$$

where the coefficient k is nonnegative and called the *thermal conductivity* of the fluid. This coefficient may depend on various characteristics of the fluid such as temperature or pressure.

4.4 Equation for the specific internal energy

Now that we have described the law which gives the stress tensor in a flow of a Newtonian fluid, we can concentrate on the various possible equivalent forms for the evolution equation for the energy, depending on the thermodynamic variables that we select to describe the system.

We start by establishing the equation obeyed by the specific internal energy e which is linked to the total energy per unit mass $\mathscr{E}$ by the equation

$$\mathscr{E} = e + \frac{1}{2}|v|^2,$$

which shows the separation of the total energy into the internal and kinetic energy terms. To do this, it is useful first to write the evolution equation for kinetic energy in the nonconservative form. This is obtained as a consequence of the conservation of momentum equation by taking the scalar product of (I.18) by v. This gives

$$\frac{\rho}{2}\left(\frac{\partial |v|^2}{\partial t} + v \cdot \nabla |v|^2\right) - (\operatorname{div} \sigma) \cdot v - \rho\, f \cdot v = 0.$$

Let us subtract this equation from the energy equation (I.19). We obtain

$$\rho\left(\frac{\partial e}{\partial t} + v \cdot \nabla e\right) + \operatorname{div}\, \varphi + (\operatorname{div} \sigma) \cdot v - \operatorname{div}(\sigma.v) = 0.$$

However, the tensor σ being symmetric, we have

$$\begin{aligned} v \cdot \operatorname{div} \sigma - \operatorname{div}(\sigma.v) &= \sum_{i,j=1}^{3} v_j \frac{\partial}{\partial x_i} \sigma_{ji} - \sum_{i,j=1}^{3} \frac{\partial}{\partial x_i}(\sigma_{ij} v_j) \\ &= -\sum_{i,j=1}^{3} \sigma_{ij} \frac{\partial}{\partial x_i} v_j = -\operatorname{Tr}(\sigma \nabla v). \end{aligned}$$

By breaking down σ into $\sigma = \mathscr{T} - p\,\mathrm{Id}$ where $\mathscr{T}$ is the viscous stress tensor, we obtain

$$\rho\left(\frac{\partial e}{\partial t} + v \cdot \nabla e\right) + \operatorname{div}\, \varphi + p\,(\operatorname{div} v) - \operatorname{Tr}(\mathscr{T}\, \nabla v) = 0.$$

For a Newtonian fluid, we saw that we have

$$\mathscr{T} = 2\mu \mathrm{D}(v) + \lambda\,(\operatorname{div} v)\ \mathrm{Id},$$

and hence we obtain

$$\begin{aligned}
\mathrm{Tr}(\mathscr{T}\nabla v) &= \mathrm{Tr}(\mathscr{T}\mathrm{D}(v)) \\
&= \mathrm{Tr}\big(\lambda(\operatorname{div} v)\mathrm{D}(v)\big) + 2\mu\,\mathrm{Tr}\big(\mathrm{D}(v)\mathrm{D}(v)\big), \\
&= \lambda(\operatorname{div} v)\,\mathrm{Tr}(\nabla v) + 2\mu\,\mathrm{Tr}\big(\mathrm{D}(v)\mathrm{D}(v)\big), \\
&= \lambda(\operatorname{div} v)^2 + 2\mu\mathrm{D}(v) : \mathrm{D}(v) = \lambda(\operatorname{div} v)^2 + 2\mu|\mathrm{D}(v)|^2,
\end{aligned}$$

because we have $\operatorname{div} v = \mathrm{Tr}(\mathrm{D}(v))$. As we have seen above, under the conditions of Equation (I.26) this quantity is always nonnegative in accordance with the second law of thermodynamics.

By applying Fourier's law (I.28), the equation expressing the change in specific internal energy with time can be written as

$$\rho\left(\frac{\partial e}{\partial t} + v\cdot\nabla e\right) - \operatorname{div}(k\,\nabla T) + p\,(\operatorname{div} v) - \lambda(\operatorname{div} v)^2 - 2\mu|\mathrm{D}(v)|^2 = 0. \quad \text{(I.29)}$$

In the conservative form, Equation (I.29) has the form:

$$\frac{\partial(\rho e)}{\partial t} + \operatorname{div}(\rho e v) - \operatorname{div}(k\nabla T) + p\,(\operatorname{div} v) - \lambda(\operatorname{div} v)^2 - 2\mu|\mathrm{D}(v)|^2 = 0.$$

4.5 Formulation in entropy and temperature

We can now use the thermodynamic concepts recalled in Appendix B to deduce the equivalent expression for the conservation equation using the specific entropy and temperature as variables. These concepts, even though they concern a priori the equilibrium thermodynamics for homogeneous systems, can be applied to any elementary fluid particle inasmuch as it is reasonable to assume that the relaxation time for the internal structure of the fluid particle is very much shorter than the characteristic time for the overall movement of the fluid. As a consequence, we assume that the fluid is everywhere and at any time at the thermodynamic equilibirum.

First let us write the differential of $(s,\rho)\mapsto e(s,\rho)$ in the form

$$de = T ds - p\,d\left(\frac{1}{\rho}\right),$$

where $1/\rho$ is the specific volume of the flow (refered to as v in the appendix). At the Lagrangian level, that is, using the material derivative D/Dt, we then deduce that

$$\begin{aligned}\frac{De}{Dt} &= T\frac{Ds}{Dt} - p\frac{D\left(\frac{1}{\rho}\right)}{Dt}\\ &= T\frac{Ds}{Dt} + \frac{p}{\rho^2}\frac{D\rho}{Dt}\\ &= T\frac{Ds}{Dt} - \frac{p}{\rho}(\operatorname{div} v),\end{aligned}$$

by using the mass conservation equation (I.7).

Equation (I.29) therefore becomes an equation for the specific entropy s

$$\rho T\frac{Ds}{Dt} - \operatorname{div}(k\,\nabla T) - \lambda(\operatorname{div} v)^2 - 2\mu\,|\mathrm{D}(v)|^2 = 0.$$

Let us now try to obtain the equation in terms of temperature T. To do this, we use the differential expression for the specific internal energy e as a function of the temperature and the specific volume (B.4). We write this as

$$de = c_v dT + \left(T\left(\frac{\partial p}{\partial T}\right)_\rho - p\right) d\left(\frac{1}{\rho}\right),$$

which using the conservation of mass equation (I.7) gives

$$\begin{aligned}\frac{De}{Dt} &= c_v\frac{DT}{Dt} - \frac{1}{\rho^2}\left(T\left(\frac{\partial p}{\partial T}\right)_\rho - p\right)\frac{D\rho}{Dt}\\ &= c_v\frac{DT}{Dt} + \frac{1}{\rho}\left(T\left(\frac{\partial p}{\partial T}\right)_\rho - p\right)(\operatorname{div} v).\end{aligned}$$

Hence, Equation (I.29) can be expressed in terms of temperature in the form

$$\rho c_v\frac{DT}{Dt} + T\left(\frac{\partial p}{\partial T}\right)_\rho (\operatorname{div} v) - \operatorname{div}(k\,\nabla T) - \lambda(\operatorname{div} v)^2 - 2\mu\,|\mathrm{D}(v)|^2 = 0,$$

or again in the conservative form, assuming that c_v is a constant

$$\begin{aligned}&\frac{\partial(\rho c_v T)}{\partial t} + \operatorname{div}(\rho c_v T v) + T\left(\frac{\partial p}{\partial T}\right)_\rho (\operatorname{div} v)\\ &- \operatorname{div}(k\,\nabla T) - \lambda(\operatorname{div} v)^2 - 2\mu\,|\mathrm{D}(v)|^2 = 0,\end{aligned}$$

5 Summary of the equations

If we consider a Newtonian fluid for which the viscosity coefficients (λ and μ) are assumed to be constant (in general they may depend on density and temperature), then the viscous stress tensor can be written as

$$\mathscr{T} = 2\mu \mathrm{D}(v) + \lambda(\operatorname{div} v)\,\mathrm{Id},$$

where λ and μ obey the conditions (I.26). Under these conditions, using Remark I.4.2, the flow of the fluid is governed by the following equations.

- Mass conservation:
$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho v) = 0.$$
- Linear momentum evolution:
$$\frac{\partial(\rho v)}{\partial t} + \operatorname{div}(\rho\, v \otimes v) - \mu\Delta v + \nabla\big(p - (\lambda+\mu)(\operatorname{div} v)\big) - \rho\, f = 0.$$
Remember that the equation governing kinetic energy can be deduced from the two preceding equations and therefore does not need to be added to the equations modelling the evolution of the flow. This property underlies all the mathematical techniques used in the rest of this book to study the equations of fluid mechanics. They are often called the *energy methods*.
- Evolution of the total energy of the system:
This last equation can take many formally equivalent forms, as a function of the thermodynamic unknowns that are considered. Each possesses both a conservative and a nonconservative form.
  - Using the total energy per unit mass $\mathscr{E} = e + \frac{1}{2}|v|^2$
$$\frac{\partial(\rho\mathscr{E})}{\partial t} + \operatorname{div}(\rho\mathscr{E}\, v) - \operatorname{div}(k\,\nabla T) - \operatorname{div}(\mathscr{T}.v) + \operatorname{div}(p\, v) - \rho\, f \cdot v = 0.$$
  - Using the internal energy:
$$\frac{\partial(\rho e)}{\partial t} + \operatorname{div}(\rho e\, v) - \operatorname{div}(k\nabla T) + p\,(\operatorname{div} v) - \lambda(\operatorname{div} v)^2 - 2\mu\,|\mathrm{D}(v)|^2 = 0.$$
  - Using the specific entropy:
$$\rho T\left(\frac{\partial s}{\partial t} + v\cdot\nabla s\right) - \operatorname{div}(k\,\nabla T) - \lambda(\operatorname{div} v)^2 - 2\mu\,|\mathrm{D}(v)|^2 = 0.$$
  - Using temperature, assuming a constant specific heat capacity at constant volume, c_v:
$$\frac{\partial(\rho c_v T)}{\partial t} + \operatorname{div}(\rho c_v T\, v) - \operatorname{div}(k\nabla T) + T\left(\frac{\partial p}{\partial T}\right)_\rho (\operatorname{div} v)$$
$$- \lambda(\operatorname{div} v)^2 - 2\mu\,|\mathrm{D}(v)|^2 = 0.$$

The motion is, therefore, perfectly defined by three evolution equations. Of course, the system must be closed by adding an equation of state which relates the various thermodynamic variables of the system to one another (the ideal gas law $p = k\rho T$, the Van der Waals law $(p + a\rho^2)(1 - b\rho) = k\rho T$, or many others). This set of equations is called the full Navier–Stokes–Fourier system. It has to be supplemented by initial and boundary conditions that we discuss briefly in Section 6.2.5.

Finally, we mention that the flow is said to be steady if all the fields associated with the flow (density, pressure, velocity, energy) are independent of the time t. However, this does not mean at all that the fluid is at rest (which would imply that $v = 0$).

6 Incompressible models

In many situations, it is useless or at least undesirable to solve the full Navier–Stokes–Fourier equations because some physical phenomena can be neglected in the considered applications. In this book we consider the case of *incompressible models*, that we describe in this section. Very briefly, we show that this kind of simplified models is relevant for liquids in standard conditions and for gases for low-speed flows. In other cases, general compressible models have to be considered, even though other simplifications are possible.

We refer, for instance, to [127] for a description of many different asymptotic regimes and simplified models derived from the full Navier–Stokes–Fourier system.

6.1 The incompressibility assumption

Definition and Proposition I.6.1. *The flow of a fluid is said to be* incompressible *if one of the following equivalent properties is satisfied.*

1. *The volume of any fluid element is constant along the time.*
2. *The velocity field v is divergence-free (it is also said to be* solenoidal*); that is*
$$(\operatorname{div} v)(t, x) = 0, \quad \forall (t, x).$$
3. *The density ρ is constant along the trajectories associated with the velocity field v.*

We show below that the term *incompressible* can lead to confusion with the notions of *compressibility* and *thermal expansion* of the fluid under study. More appropriate terms should be *isovolume flow* or *isochoric flow* to describe the fact that volumes are preserved in the evolution. However, the term *incompressible flow* is widely used in the literature and we follow this convention in this book.

Proof.

$1 \Leftrightarrow 2$: Let $(\Omega_t)_t$ be any fluid element. Applying the transport theorem to $f = 1$, we get

$$\frac{d}{dt}|\Omega_t| = \frac{d}{dt}\left(\int_{\Omega_t} 1\, dX\right) = \int_{\Omega_t} (\operatorname{div} v)\, dx.$$

Therefore, the conservation of the volume of $(\Omega_t)_t$, for any choice of fluid element is equivalent to the divergence-free condition $\operatorname{div} v = 0$.
$2 \Leftrightarrow 3$: We write the mass balance equation

$$\frac{\partial \rho}{\partial t} + v \cdot \nabla \rho + \rho(\operatorname{div} v) = 0;$$

that is,

$$\frac{D\rho}{Dt} + \rho(\operatorname{div} v) = 0,$$

if one uses the notation $D\rho/Dt$ for the material derivative of ρ as defined in (I.5).
The density ρ cannot vanish (if it becomes very small the continuous medium assumption is no longer valid) therefore we deduce that

$$\operatorname{div} v = 0 \quad \Longleftrightarrow \quad \frac{D\rho}{Dt} = 0,$$

which exactly express that the divergence-free condition on v is equivalent to the fact that the density ρ is constant along the characteristic curves of v.

□

Remark I.6.1. A flow can be incompressible even if the density is not constant. It is only required that the density of a particle of fluid remain constant during the evolution.

As an example of a nonhomogeneous incompressible flow we can consider water in the ocean whose density depends on the salinity but which is nevertheless incompressible.

Remark I.6.2. For an incompressible flow, if the density ρ is homogeneous at the initial time (that is to say that $x \mapsto \rho(0, x) = \rho_0$ is a constant), then the density remains constant

$$\rho(t, x) = \rho_0, \forall (t, x).$$

Indeed, we have seen that incompressibility implies that ρ is constant along trajectories of fluid particles

$$\rho(t, X(t, 0, x_0)) = \rho(0, x_0) = \rho_0, \forall t, \forall x_0.$$

Since, for a fixed time t the map $x_0 \mapsto X(t, 0, x_0) = \varphi_t(x_0)$ is a bijection, we deduce that $\rho(t, X)$ is indeed equal to ρ_0 for any t and any X and the flow is globally homogeneous.

Let us now investigate in which circumstances a flow can be supposed to be incompressible.

We start from the equation of state which gives a relation of the form $\rho = \rho(p, T)$. Therefore, we can compute the divergence of v through the material derivative of the pressure and the temperature as follows:

$$\operatorname{div} v = -\frac{1}{\rho}\frac{D\rho}{Dt} = \underbrace{-\,\frac{1}{\rho}\left(\frac{\partial \rho}{\partial p}\right)_T \frac{Dp}{Dt}}_{\text{compression}} \underbrace{-\,\frac{1}{\rho}\left(\frac{\partial \rho}{\partial T}\right)_p \frac{DT}{Dt}}_{\text{thermal expansion}}. \qquad \text{(I.30)}$$

The incompressibility condition for the flow, $\operatorname{div} v = 0$, is thus equivalent to assuming that the compression term and the thermal expansion term are both zero (or negligible) because it is not reasonable to think that the two terms, describing two different phenomena may compensate each other.

In the sequel of this chapter (and in the rest of the book), we assume that

- Either the coefficient of thermal expansion of the fluid $(1/\rho)(\partial\rho/\partial T)_p$ is negligible in the flow under study. This is, for instance, the case for barotropic fluids (those fluids for which the pressure depends only on the density at least in the conditions of the experiment).
- Or that the temperature is almost constant along trajectories of fluid particles (this occurs for instance in isothermal flows).

We now investigate the compression term in (I.30). This term can be supposed to be small enough in two different cases.

- Case 1: The isothermal compressibility coefficient of the fluid is zero (or is negligible). This means that

$$\beta = -\frac{1}{\rho}\left(\frac{\partial \rho}{\partial p}\right)_T \approx 0.$$

  This assumption is natural for liquids in standard temperature and pressure conditions. For instance, for liquid water the coefficient β is around 10^{-10}Pa^{-1} and the influence of the compressibility can be neglected for reasonable pressure variations in the flow.
- Case 2: The compressibility coefficient β is not small enough.
  In the case of gases (the air, for instance) this coefficient is around 10^{-5}Pa^{-1} which is no more negligible (recall that $10^5\text{Pa} = 1\text{bar}$).
  However, we can still make the incompressibility assumption for such flows as soon as the typical velocity is small compared to the isothermal sound speed. Let us illustrate this point through a quite formal dimensional analysis.

The isothermal sound speed c in such a fluid is defined through the formula

$$\frac{1}{c^2} = \sqrt{\left(\frac{\partial \rho}{\partial p}\right)_T}.$$

It can be shown, indeed, that pressure/density variations in the fluid travel at speed c. Let us introduce the Reynolds dimensionless number

$$\mathcal{R}e = \frac{\rho v_0 l_0}{\mu},$$

where l_0 and v_0 are typical length and velocity scales in the flow, chosen in such a way that

$$|\nabla v| \approx \frac{v_0}{l_0}, |\Delta v| \approx \frac{v_0}{l_0^2}.$$

As we show in Section 6.2.3, the Reynolds number accounts for the ratio between inertia effects and viscous effects in the system. The above formal approximations can be justified by using a suitable scaling of the equations. With this definition and (I.30) (without thermal expansion effects) we have

$$\operatorname{div} v = -\frac{1}{\rho c^2}\frac{Dp}{Dt}. \tag{I.31}$$

In order to simplify this formal analysis a little, we consider the steady case for which the material derivative of the pressure simply reads $Dp/Dt = v \cdot \nabla p$.

– Subcase 1: We assume that viscous effects are negligible (high Reynolds number) so that the momentum equation (without source terms) reduces to

$$-\frac{1}{\rho}\nabla p \approx (v \cdot \nabla)v.$$

It follows from (I.31) that

$$|\operatorname{div} v| = \frac{1}{\rho c^2}|v \cdot \nabla p| \approx \frac{1}{c^2}|v \cdot ((v \cdot \nabla)v)| \approx \frac{v_0^2}{c^2}|\nabla v|.$$

As a consequence, if the *Mach number* defined as the ratio $\mathrm{Ma} = v_0/c$ satisfies

$$\mathrm{Ma} \ll 1, \tag{I.32}$$

then the divergence of v can be assumed to be negligible compared to typical velocity gradients, and it can reasonably be assumed that the flow is incompressible.
It is commonly admitted that the condition $\mathrm{Ma} < 0.3$ is a good criterion for deciding whether compressibility effects have to be taken into account.

Note that Case 1 above is nothing but the assumption that c is infinite (or at least very large) which can also be understood as low Mach number asymptotics.

– Subcase 2: We assume that nonlinear inertia terms are negligible in front of viscous effects (low Reynolds number). In that case (using Stokes assumption (I.27)) we have

$$\nabla p \approx \mu \Delta v + \frac{\mu}{3}\nabla(\operatorname{div} v),$$

and we deduce that

$$\begin{aligned} |\operatorname{div} v| &= \frac{1}{\rho c^2}|v \cdot \nabla p| \approx \frac{\mu}{\rho c^2}|v \cdot \Delta v| \approx \frac{\mu}{\rho c^2 l_0^2} v_0^2 \\ &\approx \frac{1}{c^2}\frac{\mu v_0}{\rho l_0}|\nabla v| = \frac{v_0^2}{c^2}\underbrace{\frac{\mu}{\rho v_0 L}}_{=1/\mathcal{R}e}|\nabla v|. \end{aligned}$$

Therefore, in that case the flow will be considered to be incompressible as soon as

$$\mathrm{Ma} \ll \sqrt{\mathcal{R}e}. \tag{I.33}$$

This condition can be much more restrictive than (I.32).

In each of the two cases described above the sound speed c is assumed to be large (infinite in Case 1, finite but large enough in Case 2). This corresponds to the limit where $(\partial\rho/\partial p)_T \approx 0$. In particular, in this asymptotic regime, it is not theoretically possible to use the equation of state $\rho = \rho(p, T)$ to obtain an expression of the pressure as a function of the density $p = p(\rho, T)$. Indeed, when the above derivative is zero it means that this reciprocal function may not exist (at least in the asymptotic regime $\mathrm{Ma} \to 0$).

Therefore, it seems that for incompressible models the pressure is no longer related to the other thermodynamical variables and becomes an independent variable. This is consistent with the fact that we add a new equation to the whole system of evolution equations that v has to satisfy and which is the divergence-free condition.

From a mathematical point of view, the pressure gradient term in the momentum equation has now become a Lagrange multiplier related to the divergence-free constraint (see Remark IV.5.1). In particular, we observe that one can add a constant to the pressure without changing the equations so that only pressure gradients have a physical meaning.

This apparent ambiguity on the definition of the pressure can be analysed as follows. In the formal low-Mach number limit $\mathrm{Ma} \to 0$ it can be shown, in a suitable framework (see [127] for instance), that the thermodynamical pressure p_{Ma} (for which we explicitly show the dependence on the Mach number here) actually converges towards a constant p_0 and that there exists a function π such that

$$p_{\mathrm{Ma}} - p_0 \underset{\mathrm{Ma}\to 0}{\sim} \mathrm{Ma}^2\,\pi,$$
$$\nabla(p_{\mathrm{Ma}}) \underset{\mathrm{Ma}\to 0}{\sim} \mathrm{Ma}^2\,\nabla\pi.$$

Therefore, in this asymptotic incompressible regime, the pressure gradient term appearing in the momentum equation is in fact in the form $\mathrm{Ma}^2\nabla\pi$. This is exactly the Lagrange multiplier pressure term in the incompressible Navier–Stokes equations. It has to be interpreted as a pressure fluctuation and not as the absolute pressure; in particular this "incompressible" pressure has no particular sign.

Even though it should be less ambiguous to refer to this incompressible pressure as π instead of p, for instance, using the letter p to represent this pressure fluctuation is still widely practiced in the literature.

Rigorous mathematical justifications of the low-mach number limit (and thus of the incompressible limit for the isentropic or barotropic flows) are known in some cases. This goes beyond the scope of this book and we refer to the review papers [4, 46, 58, 4] for more details on this topic.

6.2 *Overview of the incompressible models*

Let us recall that, from now on, we neglect (the influence of) temperature variations in the flow under study and that we concentrate on the case of an incompressible flow. We are in position to describe the various models that we study in the next chapters.

Our assumption on temperature variations and incompressibility implies that there is no more energy evolution equation in the system and that the pressure p is not defined through an equation of state but rather in an indirect process associated with the divergence-free constraint. We recall that we showed in the previous section that this pressure term p has to be understood as a fluctuation of the actual thermodynamic pressure around its mean value.

6.2.1 The nonhomogeneous incompressible Navier–Stokes equations

The more general and more complex system that we study in this book (Chapter VI), is the following one:

$$\begin{cases} \dfrac{\partial \rho}{\partial t} + \mathrm{div}(\rho v) = 0, \\ \dfrac{\partial(\rho v)}{\partial t} + \mathrm{div}(\rho v \otimes v) - 2\,\mathrm{div}(\mu(\rho)\mathrm{D}(v)) + \nabla p = \rho f, \\ \mathrm{div}\, v = 0, \end{cases}$$

At least formally, the first two equations can be written in nonconservative form but we will see that, for the mathematical definition of a weak solution (ρ, v, p) to this system, the conservative form is more suitable (and not necessarily equivalent to the nonconservative one).

6.2.2 The homogeneous incompressible Navier–Stokes equations

Due to the fact that the density ρ is not constant, the previous system has many nonlinearities. That's the reason why, before studying these equations, we first concentrate our attention on the simpler and more standard case where the density is homogeneous (see Remark I.6.2). Notice that the mass balance equation and the divergence-free constraint are now strictly identical so that only the following system remains,

$$\begin{cases} \rho\left(\dfrac{\partial v}{\partial t} + (v\cdot\nabla)v\right) - \mu\Delta v + \nabla p = \rho f, \\ \operatorname{div} v = 0, \end{cases}$$

where ρ and μ are now given positive constants. This system is analysed in Chapter V as well as its steady version

$$\begin{cases} -\mu\Delta v + \rho(v\cdot\nabla)v + \nabla p = \rho f, \\ \quad \operatorname{div} v = 0. \end{cases}$$

6.2.3 Dimensionless equations

The homogeneous incompressible Navier–Stokes equations given above explicitly depend on two (constant) parameters: the density and the viscosity of the fluid. Because of the nonlinear term, the behavior of the solutions will also clearly depend on the magnitude of the velocity field and in the time/space scales at which the system is observed.

To make this point appear in the equations, we need to perform a dimensional analysis, that is, to put the equations into *dimensionless* form. This involves selecting, depending on the problem being studied, a timescale t_0 (the characteristic time of observation of the phenomenon) and a spatial scale l_0 (the characteristic size of the flow considered). From these two fixed scales, we can then deduce a characteristic velocity for the problem

$$v_0 = \frac{l_0}{t_0},$$

which is simply the mean velocity required to cross the distance l_0 in a time t_0. Of course, it is equivalent (and sometimes more natural) to determine l_0 and

v_0 from the geometry of the container and the inflow boundary conditions, and then deduce the time scale $t_0 = l_0/v_0$ at which interesting phenomena will certainly occur. Finally, we can set a characteristic size f_0 for the body forces applied to the flow (typically buoyancy forces).

We introduce new variables which are said to be *dimensionless* or *adimensional*, defined by

$$t' = \frac{t}{t_0}, x' = \frac{x}{l_0}, v' = \frac{v}{v_0}, f' = \frac{f}{f_0},$$

so that the equations above become

$$\begin{cases} \rho \left(\dfrac{v_0}{t_0} \dfrac{\partial v'}{\partial t'} + \dfrac{v_0^2}{l_0} (v' \cdot \nabla_{x'}) v' \right) - \mu \dfrac{v_0}{l_0^2} \Delta_{x'} v' + \dfrac{1}{l_0} \nabla_{x'} p = \rho f_0 f', \\ \operatorname{div}_{x'} v' = 0, \end{cases}$$

where $\operatorname{div}_{x'}$, $\nabla_{x'}$, and $\Delta_{x'}$ designate the usual differential operators acting on the new space variables x'.

We notice that through the relation among l_0, t_0, and v_0, the coefficients which appear in front of $\partial v'/\partial t'$ and $(v' \cdot \nabla_{x'})v'$ are the same. This is not coincidence and corresponds to the fact that the material derivative operator D/Dt is invariant, apart from a multiplicative constant, through the scaling of the system we proposed.

Hence, by setting $p' = p/\rho v_0^2$, we obtain

$$\begin{cases} \left(\dfrac{\partial v'}{\partial t'} + (v' \cdot \nabla_{x'}) v' \right) - \dfrac{\mu}{\rho v_0 l_0} \Delta_{x'} v' + \nabla_{x'} p' = \dfrac{f_0}{v_0/t_0} f', \\ \operatorname{div}_{x'} v' = 0. \end{cases}$$

It can be seen that the new system depends on two parameters only, the Reynolds number $\mathcal{R}e$ and the Froude number $\mathcal{F}r$, defined by

$$\mathcal{R}e = \frac{\rho v_0 l_0}{\mu}, \text{ and } \mathcal{F}r = \frac{v_0/l_0}{f_0}.$$

The Reynolds number measures the ratio between the order of magnitude of the inertial terms $\rho(v \cdot \nabla)v$ and the order of magnitude of the viscous terms. When the Reynolds number is small, viscous effects dominate; we say that the flow is *laminar*. When the Reynolds number is high, inertial effects dominate and we say that the flow is *turbulent*.

The transition between these two behaviors is, of course, rather fuzzy and depends largely on the physical case being investigated. We do not go further into such considerations here, although they are fundamental in many real applications; see for instance [59].

In the most common case, where the body forces reduce to gravity, the Froude number measures the ratio between the mean acceleration in the flow v_0/t_0 and the acceleration due to buoyancy f_0.

In the following, we assume that the Froude number is equal to 1; by contrast, we attempt, as far as possible, to observe the influence of the Reynolds number in the mathematical results presented in this book, at least when studying nonlinear models. Again, to prevent notation becoming too unwieldy, we make no further reference to the "primes" on the variables and the unknowns, so that the dimensionless equations we study in Chapter V are

$$\begin{cases} \dfrac{\partial v}{\partial t} - \dfrac{1}{\mathcal{R}e}\Delta v + (v \cdot \nabla)v + \nabla p = f, \\ \operatorname{div} v = 0. \end{cases} \tag{I.34}$$

6.2.4 The homogeneous Stokes problem

In the case where non-linear effects can be neglected in the homogeneous Navier–Stokes equations (i.e., when the Reynolds number $\mathcal{R}e$ is sufficiently small) then System (I.34) formally reduces to the Stokes equation

$$\begin{cases} \dfrac{\partial v}{\partial t} - \dfrac{1}{\mathcal{R}e}\Delta v + \nabla p = f, \\ \operatorname{div} v = 0. \end{cases}$$

This problem is linear with constant coefficients. Therefore, it is clear that most of the mathematical difficulties for its study are contained in the steady Stokes problem

$$\begin{cases} -\dfrac{1}{\mathcal{R}e}\Delta v + \nabla p = f, \\ \operatorname{div} v = 0, \end{cases}$$

in particular concerning the existence and properties of the pressure.

The study of these equations in various contexts (in particular for different kinds of boundary conditions) is the topic of Chapter IV. It gives us the opportunity to introduce many mathematical tools which are useful in the study of more general fluid mechanics models.

6.2.5 Initial and boundary data

For unsteady models, it is clearly required to impose initial conditions in order to define the evolution of the system. We show that we only need to impose the initial value of the density (in the nonhomogeneous case) and of the velocity. It has no mathematical meaning to impose an initial value for the pressure because this unknown has the role of a Lagrange multiplier associated with the incompressibility condition and thus, is defined in some indirect way.

Concerning boundary conditions, the most widely used condition is a Dirichlet boundary condition for the velocity

$$v = v_b, \quad \text{on the boundary.}$$

When $v_b = 0$, it is called an homogeneous Dirichlet boundary condition or a no-slip condition. No boundary condition is necessary for the pressure.

We study these boundary conditions for the homogeneous steady and unsteady Stokes and Navier–Stokes equations.

As far as the nonhomogenous fluids are concerned, we have to discuss possible boundary conditions associated with the density ρ. We have seen that, in incompressible models, the material derivative of ρ is zero, which means that ρ is constant along trajectories associated with the flow, that is, along the solutions of the characteristic equation (I.4). Two cases have to be considered and are precisely discussed in Chapter VI.

- If v is tangent to the boundary of the domain under study, then the solutions of the characteristic equation never cross the boundary of the domain. Therefore, it is very likely that no boundary condition is necessary for the density.
- If v is not tangent to the boundary of the domain, then the characteristic curves will certainly cross the boundary. In that case, boundary data for ρ are necessary on the inflow points of the boundary.

For the steady Stokes problem we study in Chapter IV more general boundary conditions: Neumann-like boundary conditions, normal stress boundary conditions, and vorticity/pressure conditions.

We also investigate, in Chapter VII, outflow nonphysical boundary conditions for the Navier–Stokes problem which are necessary to solve the problem in the case of a computational domain obtained after truncation of a physical unbounded domain.

6.2.6 Additional comments

For very high values of the Reynolds number, the incompressible Navier–Stokes system (I.34) formally becomes the incompressible Euler system

$$\begin{cases} \dfrac{\partial v}{\partial t} + (v \cdot \nabla)v + \nabla p = f, \\ \operatorname{div} v = 0, \end{cases}$$

which accounts for the flow of perfect (or inviscid) fluids. From a mathematical point of view, this new system is hyperbolic and thus of a very different nature than the Navier–Stokes system which is parabolic. We do not study the

Euler equations in the present book, and we refer, for instance, to [88, 86, 41] for questions related to this system.

Similarly, the compressible Navier–Stokes equations and the full Navier–Stokes–Fourier system are not analysed in this book. The interested reader can find many results in [87, 57, 93]. Notice, however, that many mathematical tools presented in this book are useful in the study of compressible flows.

7 Some exact steady solutions

Before attacking a precise analysis of the preceding equations in the following chapters, we would like to present quickly some examples of steady situations for which explicit formulas exist. These give the velocity field and the pressure field in very specific geometries.

In this section we discuss steady solutions of the homogeneous incompressible Navier–Stokes equations (I.34). We are therefore seeking a pair (v, p) which is a solution of

$$\begin{cases} -\dfrac{1}{\mathcal{R}e}\Delta v + (v\cdot\nabla)v + \nabla p = g, & \text{in } \Omega \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ v = v_b, & \text{on } \partial\Omega, \end{cases} \tag{I.35}$$

where g represents the buoyancy forces exerted on the fluid and v_b the Dirichlet boundary data. Since these gravitational forces derive from a potential, we can introduce π_0 such that $\nabla\pi_0 = g$. Then by setting $\pi = p - \pi_0$, we can always return to the case where the source term is zero. For example, if we assume that gravity is constant and directed along e_3 on the experimental scale, then $\pi_0 = -g\, x_3$.

If we hope to find exact analytical solutions, it is of course necessary that the domain Ω exhibit a relatively simple geometry, for which it is possible to perform all the calculations. We note that such simple geometries are, nevertheless, representative of significant physical situations.

Once these analytical expressions of the solutions are obtained, we can then ask if they correspond effectively to the observed physical solutions. Unfortunately, as soon as the Reynolds number $\mathcal{R}e$ is large, we note that this is not always the case. We should then study the stability of the solutions presented. These ideas, which are of great practical importance, are not covered in this text. Among the very abundant literature on the subject, we refer, for instance, to [40, 53, 127] or to more specialised texts such as [43].

7.1 Poiseuille flow in a pipe

Let us consider an infinite tube with axis e_3 and an ellipse-shaped cross section. The cross-section of the tube, denoted as E, is defined by the equation

$$\left(\frac{x_1}{R_1}\right)^2 + \left(\frac{x_2}{R_2}\right)^2 \leq 1. \tag{I.36}$$

From the symmetry of the domain considered, we can look for a velocity field solution of the equations, which depends on x_1 and x_2 only. Very generally, such a velocity field can be expressed as

$$v = (u_1(x_1, x_2), u_2(x_1, x_2), u_3(x_1, x_2)).$$

Since u_3 does not depend on x_3, the divergence-free condition gives

$$\frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} = 0.$$

Hence, by setting $\tilde{v} = (u_1, u_2)$, we see that the two-dimensional velocity field $\tilde{v}$ satisfies the following two-dimensional steady Navier–Stokes equations, obtained by considering the two main components of the system of equation (I.35)

$$\begin{cases} -\dfrac{1}{\mathcal{R}e}\Delta\tilde{v} + (\tilde{v}\cdot\nabla)\tilde{v} + \nabla\pi = 0, & \text{in } E, \\ \operatorname{div}\tilde{v} = 0, & \text{in } E. \end{cases}$$

Furthermore, the homogeneous boundary conditions for the velocity at the wall give us $\tilde{v} = 0$ at the boundary of the elliptic section E.

Now we multiply the equation by $\tilde{v}$ and we integrate it over E. After integration by parts (formally justified because of the boundary conditions), using the divergence-free condition we obtain

$$\frac{1}{\mathcal{R}e}\int_E (|\nabla u_1|^2 + |\nabla u_2|^2)\, dx_1 dx_2 = 0.$$

We often make use of this type of computation in the rest of this text. It shows that u_1 and u_2 are constants but, since they are zero at the boundary of E, we obtain $u_1 = u_2 = 0$. In consequence, the pressure π satisfies the relations

$$\frac{\partial \pi}{\partial x_1} = \frac{\partial \pi}{\partial x_2} = 0.$$

We thus see that the velocity field that we are seeking is necessarily of the form

$$v = (0, 0, u(x_1, x_2)),$$

where $u = u_3$ is a scalar function. It can easily be shown that such a velocity field does indeed have zero divergence and moreover that we have

$$(v \cdot \nabla)v = u\frac{\partial u}{\partial x_3} = 0.$$

Hence (v, π) is a solution of the steady Navier–Stokes equations if the scalar functions u and π satisfy

$$\begin{cases} -\dfrac{1}{\mathcal{R}e}\Delta u + \dfrac{\partial \pi}{\partial x_3} = 0, \\ \dfrac{\partial \pi}{\partial x_1} = \dfrac{\partial \pi}{\partial x_2} = 0. \end{cases}$$

Since u only depends on x_1 and x_2, by differentiating the first equation with respect to x_3, we obtain

$$\frac{\partial^2 \pi}{\partial x_3^2} = 0.$$

We denote the constant equal to $\partial\pi/\partial x_3$ as p_1, such that the equation for u becomes

$$-\frac{1}{\mathcal{R}e}\Delta u + p_1 = 0,$$

in the elliptic equation (I.36).

This equation is associated with the zero velocity boundary conditions at the wall $u = 0$ over the boundary of the ellipse. The unique solution of this equation is obtained in an analytical manner

$$u = p_1 \frac{-\mathcal{R}e}{\frac{2}{R_1^2} + \frac{2}{R_2^2}} \left(1 - \left(\frac{x_1^2}{R_1^2} + \frac{x_2^2}{R_2^2}\right)\right).$$

The pressure field associated with this flow is therefore given (apart from one constant) by

$$\pi = p_1\, x_3.$$

Hence, this is the pressure gradient which drives the flow.

We observe that the actual pressure of the fluid is in fact given by $p = \pi + \pi_0$ where π_0 is the potential of the gravitational force

7.2 Planar shear flow

We now consider the two-dimensional planar flow of a viscous Newtonian fluid between two horizontal plates. This flow has already served as an example to illustrate Newton's theory in Section 4.

The upper plate is driven at a velocity $U \geq 0$, and the lower plate at a velocity $-V \leq 0$. The open set Ω delimiting the fluid domain is $\mathbb{R}\times]-h,h[$. Guided once again by the geometry of the domain, we look for a velocity field which only depends on the transverse variable x_2. However, the divergence-free condition requires that the second component of the velocity field be a constant. The condition $v \cdot \nu = 0$ at the boundary of the domain shows us that v is necessarily of the form

$$v = (u(x_2), 0).$$

As we saw previously, such a velocity field automatically cancels the inertia term $(v \cdot \nabla)v$. By introducing the total pressure, the equation satisfied by the couple (u, π) is written as

$$\begin{cases} -\dfrac{1}{\mathcal{R}e}\Delta u + \dfrac{\partial \pi}{\partial x_1} = 0, \\ \dfrac{\partial \pi}{\partial x_2} = 0. \end{cases}$$

Since u only depends on x_2, by differentiating the first equation with respect to x_1 we obtain

$$\frac{\partial^2 \pi}{\partial x_1^2} = 0.$$

Hence by denoting the constant equal to $\partial\pi/\partial x_1$ as p_1, we immediately obtain the general solution of the ordinary differential equation for u:

$$u = \frac{\mathcal{R}e\, p_1}{2} x_2^2 + b\, x_2 + c.$$

The boundary conditions

$$u(h) = U, u(-h) = -V,$$

then allow the flow to be completely determined. Finally, we obtain

$$u = \frac{\mathcal{R}e\, p_1}{2}(h^2 - x_2^2) + \frac{U+V}{2h} x_2 + \frac{U-V}{2}.$$

The pressure field associated with the flow is again linear and given by

$$\pi = p_1\, x_1.$$

The actual pressure in the flow is again given by $p = \pi + \pi_0$. This time, the driving elements of the flow are both the pressure gradient and, of course, the boundary conditions.

7.3 Couette flow between two cylinders

Let us consider a domain limited by two coaxial cylinders, with axes directed along e_3 and having infinite height. The external cylinder is fixed and the internal cylinder is driven in a uniform rotation. Let the rotation vector $\mathbf{\Omega}$ be colinear with e_3.

The velocity field of a rigid solid which is submitted to a uniform rotation is given by

$$v = \mathbf{\Omega} \wedge \mathbf{r},$$

where $\mathbf{r}$ is the position vector, which is also written as

$$v = \mathbf{\Omega} \wedge (re_r),$$

where $r = |\mathbf{r}|$ and e_r is the unit rotation vector related to the description of the domain in cylindrical coordinates.

Inspired by this formula, we thus look for a solution of the Navier–Stokes equations between the two cylinders in the form

$$v = f(r)\mathbf{\Omega} \wedge e_r,$$

where f has to be determined. We note that, according to Equation (A.1), such a vector field is necessarily divergence-free.

We show that in this particular case, the inertia term can be written in the form of a gradient. Indeed, according to Equation (A.6), we have

$$(v \cdot \nabla)v = \frac{1}{2}\nabla|v|^2 - v \wedge \operatorname{curl} v,$$

and it is sufficient then to show that the last term is a gradient. For this we use the particular form chosen for the velocity field v. By using Equation (A.8), this becomes

$$\operatorname{curl} v = \operatorname{div}(f(r)e_r)\,\mathbf{\Omega} - (\mathbf{\Omega} \cdot \nabla)(f(r)e_r),$$

however, the last term is zero because $\mathbf{\Omega}$ is directed along $e_z = e_3$ and $f(r)e_r$ does not depend on z. By use of Equation (A.1), the first term becomes

$$\operatorname{curl} v = \operatorname{div}(f(r)e_r)\,\mathbf{\Omega} = \frac{1}{r}\frac{\partial}{\partial r}\left(rf(r)\right)\mathbf{\Omega}. \tag{I.37}$$

Hence, the double vector product identity (A.3) gives us

$$\begin{aligned} v \wedge \operatorname{curl} v &= \frac{1}{r}\frac{\partial}{\partial r}\left(rf(r)\right) f(r)(\mathbf{\Omega} \wedge e_r) \wedge \mathbf{\Omega} \\ &= |\mathbf{\Omega}|^2 \frac{1}{r}\frac{\partial}{\partial r}\left(rf(r)\right) f(r) e_r. \end{aligned}$$

Therefore, if $r \mapsto g(r)$ designates an indefinite integral of the function

$$r \mapsto \frac{1}{r}\frac{\partial}{\partial r}\left(rf(r)\right) f(r),$$

then

$$v \wedge \operatorname{curl} v = |\mathbf{\Omega}|^2 g'(r) e_r = |\mathbf{\Omega}|^2 \nabla g(r),$$

which shows that the inertia term can now be written as a gradient that is denoted by $(v \cdot \nabla)v = \nabla\widetilde{\pi}(r)$. We note, furthermore, that following the preceding calculations, the function $\widetilde{\pi}(r)$ is defined by

$$\widetilde{\pi}'(r) = -\frac{|\mathbf{\Omega}|^2}{r} f^2(r) = -\frac{|v|^2}{r}.$$

The steady Navier–Stokes equation can now be written as

$$-\frac{1}{\mathcal{R}e}\Delta(\mathbf{\Omega} \wedge f(r)e_r) + \nabla(\pi + \widetilde{\pi}) = 0.$$

We look for particular solutions which satisfy the relations

$$\Delta(\mathbf{\Omega} \wedge f(r)e_r) = 0, \text{ and } \nabla(\pi + \widetilde{\pi}) = 0.$$

This equation gives us $\pi = -\widetilde{\pi}$ because the pressure is defined apart from a constant. If we then introduce an indefinite integral F of the function f, the first equation can be written as

$$\Delta(\mathbf{\Omega} \wedge \nabla F(r)) = 0,$$

or even

$$\mathbf{\Omega} \wedge \Delta(\nabla F(r)) = 0.$$

However, since $F(r)$ only depends on r, its gradient lies in the plane e_r, e_θ and therefore this equation is equivalent to

$$0 = \Delta(\nabla F(r)) = \nabla(\Delta F(r)).$$

Hence, the function $\Delta F(r)$ is constant and we must now solve $\Delta F(r) = a$, which by using Equation (A.2) can be written as

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial F}{\partial r}\right) = a.$$

We immediately obtain

$$F(r) = a\frac{r^2}{4} + b \ln r + c.$$

We find the function f by differentiation

$$f(r) = F'(r) = a\frac{r}{2} + \frac{b}{r}.$$

Let R_1 be the radius of the inner cylinder and R_2 that of the outer cylinder; then the boundary conditions are written as

$$v(R_1) = \mathbf{\Omega} \wedge (R_1 e_r) \text{ and } v(R_2) = 0,$$

which gives, for the function f:

$$f(R_1) = R_1, \text{ and } f(R_2) = 0.$$

We can now determine the constants a and b and obtain

$$f(r) = \frac{R_1^2 R_2^2}{R_2^2 - R_1^2}\left(\frac{1}{r} - \frac{r}{R_2^2}\right),$$

which gives us the velocity field through the equation

$$v = f(r)(\mathbf{\Omega} \wedge e_r) = f(r)|\mathbf{\Omega}|e_\theta.$$

We should note that the function $f(r)$ is decreasing and convex. It is now possible to calculate the pressure in this flow by using the fact that $\pi = -\widetilde{\pi}$ and that $\widetilde{\pi}$ is the indefinite integral of the function $-|\mathbf{\Omega}|^2 f^2(r)/r$. After all the calculations, we obtain

$$\pi(r) = |\mathbf{\Omega}|^2 \left(\frac{R_1^2 R_2^2}{R_2^2 - R_1^2}\right)^2 \left(\frac{r^2}{2R_2^4} - \frac{1}{2r^2} - \frac{2}{R_2^2}\ln r\right).$$

This function is increasing with respect to r, which proves that the pressure is higher in the neighborhood of the outer cylinder than in the neighborhood of the inner cylinder. Furthermore, the actual pressure in the flow is obtained by adding the gravitational potential $p = \pi + \pi_0$.

We also note that, contrary to the preceding examples, this analytical solution of the Navier–Stokes equations does not depend on the Reynolds number for the flow. Indeed, we note that in this particular flow the inertia term is exactly compensated by the pressure term, as well as the fact that the viscous term is identically zero.

Chapter II
Analysis tools

The goal of this chapter is to describe the analysis tools that we use in later chapters. We have gathered together fundamental concepts required to study many linear or nonlinear evolution partial differential equations coming from many areas of physics and biology, for instance.

We start in Section 2 by presenting, without proof, some classic results of functional analysis such as the open mapping theorem, the Banach–Steinhaus theorem and the Lax–Milgram theorem, proofs of which can easily be found in the literature (see, e.g., [27, 104, 105]). We also give definitions of weak and weak-$\star$ convergence, which are frequently used in the analysis of partial differential equations. We pay particular attention to the expression of these results into some fundamental spaces, namely the Lebesgue spaces. The section is completed by a short introduction to distribution theory and by the description of some basic properties of Lipschitz continuous functions.

Section 3 aims at describing some tools around the notion of compactness which is fundamental when one deals with nonlinear terms in partial differential equations. We recall in particular the Schauder fixed-point theorem.

In the analysis of evolution problems, one of the usual ways for establishing existence theorems is first to obtain energy estimates. In general, these are deduced from elementary differential inequalities involving real functions of a single real variable (the time variable t, in the problems which concern us). In Section 4 of this chapter therefore, we describe the links between the concepts of weak differentiation and standard differentiation as applied to numerical functions of one single real variable. Finally, we prove the various Gronwall type inequalities, which allows us to obtain the desired estimates in most cases.

Section 5 is dedicated to the introduction and study of the spaces of functions integrable on an interval of $\mathbb{R}$ with values in a Banach space. This is also known as the Bochner integral theory. In particular, we prove the Aubin–Lions–Simon compactness theorem [14, 84, 109], a fundamental result for the study of nonlinear problems. The main ingredient of this proof is the Ascoli

F. Boyer and P. Fabrie, *Mathematical Tools for the Study of the Incompressible Navier-Stokes Equations and Related Models*, Applied Mathematical Sciences 183,
DOI 10.1007/978-1-4614-5975-0_2,

theorem, reviewed at the start of this chapter. We also present the definition and main properties of the Fourier transform for this class of functions.

We conclude the chapter, in Section 6, with a very short introduction to the spectral theory of self-adjoint unbounded operators with compact resolvent.

1 Main notation

Throughout this book, the space dimension is denoted by $d \in \mathbb{N}^*$ (typically $d = 2$ or $d = 3$ in the case of fluid mechanics applications). The Euclidean norm on $\mathbb{R}^d$ is denoted by $x \mapsto |x|$ and the associated inner product by $(x, y) \mapsto x \cdot y$.

For all multi-index $\alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{N}^d$ we denote its length by $|\alpha| = \alpha_1 + \cdots + \alpha_d$. For any function f we define $\partial^\alpha f = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_d}^{\alpha_d} f$ as soon as this partial derivative exists (in a classic or in a weak sense).

For any open set $\Omega \subset \mathbb{R}^d$ we use the following standard functional spaces.

- The set $\mathcal{C}^k(\Omega)$, $k \geq 0$, of functions with continuous partial derivatives up to order k.
- The subset $\mathcal{C}_b^k(\Omega) \subset \mathcal{C}^k(\Omega)$ of functions such that all partial derivatives up to order k are bounded.
- The set $\mathcal{C}^{0,\alpha}(\Omega)$, $\alpha \in]0, 1]$ of α-Hölder continuous functions. In the case $\alpha = 1$, $\mathcal{C}^{0,1}(\Omega)$ is the set of Lipschitz continuous functions. The Lipschitz seminorm of such a function is defined by

$$\mathrm{Lip}(f) = \sup_{\substack{x,y\in\Omega \\ x\neq y}} \frac{|f(x) - f(y)|}{|x - y|} < +\infty.$$

- The set $\mathcal{C}^{k,\alpha}(\Omega)$, $k \geq 0$, $\alpha \in]0, 1]$ of functions in $\mathcal{C}^k(\Omega)$ whose partial derivatives of order k are α-Hölder continuous.
- The set $\mathcal{C}_c^\infty(\Omega)$ of functions in $\mathcal{C}^\infty(\Omega)$ which are compactly supported in Ω. Another usual notation for this space, in particular in the theory of distributions, is $\mathcal{D}(\Omega)$.
- The set $\mathcal{C}_c^\infty(\overline{\Omega})$ of the restrictions to Ω of functions in $\mathcal{C}_c^\infty(\mathbb{R}^d)$.

Moreover, for any function u defined on Ω we denote as $\bar{u}$ its extension by 0 on the whole space defined by

$$\bar{u}(x) = \begin{cases} u(x), & \text{for } x \in \Omega, \\ 0, & \text{for } x \notin \Omega. \end{cases}$$

For any $x \in \Omega$, we define $\delta(x)$ to be the signed distance from x to the boundary, which is defined by

$$\delta(x) = \begin{cases} d(x, \partial\Omega) & \text{for } x \in \overline{\Omega}, \\ -d(x, \partial\Omega) & \text{for } x \notin \Omega. \end{cases} \tag{II.1}$$

By using the triangle inequality, it is obvious to check that δ is Lipschitz continuous on $\mathbb{R}^d$ and that $\mathrm{Lip}(\delta) \leq 1$.

2 Fundamental results from functional analysis

2.1 Banach spaces

In this section we recall essential results of functional analysis. We do not provide proofs; the reader can find these in the classic monographs on the subject such as [27], [104], and [105].

For any normed vector space E, we denote its topological dual as E', that is, the space of continuous linear functionals on E. For $f \in E'$ and $x \in E$, we introduce the duality bracket

$$\langle f, x \rangle_{E',E} = f(x).$$

We reserve the notation $(\cdot, \cdot)_H$ for a scalar product in a Hilbert space, H.

Let E and F be two normed vector spaces and $S : E \mapsto F$ be a continuous linear function. We define the *adjoint* or *transposed* function, denoted ${}^tS : F' \mapsto E'$, by

$$\langle {}^tSf, x \rangle_{E',E} = \langle f, Sx \rangle_{F',F}, \forall f \in F', \forall x \in E.$$

From the Hahn–Banach theorem, we can express the norm of an element from a normed vector space E by duality as follows.

Proposition II.2.1. *Let E be a normed vector space. Then for all $x \in E$, we have*

$$\|x\|_E = \sup_{f \in E', f \neq 0} \frac{|\langle f, x \rangle_{E',E}|}{\|f\|_{E'}} = \sup_{\|f\|_{E'} \leq 1} |\langle f, x \rangle_{E',E}|.$$

Another consequence of the Hahn–Banach theorem is the following useful density criterion for a subspace of a given normed space.

Proposition II.2.2. *Let E be a normed vector space and F be a vector subspace of E. We assume that any continuous linear functional on E which vanishes on F is identically zero. Then, F is a dense subspace of E.*

The following result (due to Banach), gives a characterisation of the isomorphisms between Banach spaces.

Theorem II.2.3 (Open mapping). *Let E and F be two Banach spaces. If u is a surjective, continuous linear function from E into F, then u is an open* *map, which means that the image under u of all open sets of E is an open set of F. In particular, if u is bijective, its reciprocal function is continuous and, consequently, spaces E and F are algebraically and topologically isomorphic.*

Finally, the last result that we recall in this section, which is sometimes called the "uniform boundedness principle" shows that if a family of continuous linear functions defined on a Banach space is pointwise bounded, then it is uniformly bounded.

Theorem II.2.4 (Banach–Steinhaus). *Let $(u_i)_{i\in I}$ be a family of continuous linear functions of a Banach space E within a normed vector space F, indexed by a set I. We assume that for all $x \in E$, the family*

$$(u_i(x))_{i\in I},$$

is bounded in F. Then, the family $(u_i)_{i\in I}$ is uniformly bounded in the sense of the norm of the operators; that is,

$$\sup_{i\in I} \|u_i\|_{L(E,F)} < +\infty;$$

or equivalently,

$$\exists C > 0, \textit{such that } \|u_i(x)\|_F \leq C\|x\|_E, \forall i \in I, \forall x \in E.$$

We conclude this section by introducing the Lax–Milgram theorem, which is an important tool in the study of linear partial differential problems in variational formulation.

Theorem II.2.5 (Lax–Milgram). *Let V be a Hilbert space, $a : V \times V \to \mathbb{R}$ a bilinear form, and $L : V \to \mathbb{R}$ a linear form.*

Assume that a and L are continuous and that a is coercive, that is,

$$\exists \alpha > 0,\ a(v, v) \geq \alpha\|v\|_V^2,\ \forall v \in V;$$

then there exists a unique solution $v \in V$ to the problem

$$a(v, w) = L(w), \forall w \in V. \tag{II.2}$$

Moreover, this solution satisfies

$$\|v\|_V \leq \frac{\|L\|_{V'}}{\alpha}. \tag{II.3}$$

2.2 Weak and weak-$\star$ convergences

We do not go into the details here of the general theory of weak and weak-$\star$ topologies (see [27] for a more complete study). Rather, we simply recall the sequential properties of these topologies, which are essential later. In this book, these notions are mainly used in the framework of Lebesgue and Sobolev spaces (see Section 2.3.4).

Definition II.2.6. *Let E be a Banach space and E' its dual space.*

- *We say that a sequence $(u_n)_n$ of elements of E weakly converges towards $u \in E$, if for any $f \in E'$ we have*

$$f(u_n) = \langle f, u_n \rangle_{E',E} \xrightarrow[n\to\infty]{} \langle f, u \rangle_{E',E} = f(u).$$

- *We say that a sequence $(f_n)_n$ of elements of E' weakly-$\star$ converges towards $f \in E'$, if for any $u \in E$, we have*

$$f_n(u) = \langle f_n, u \rangle_{E',E} \xrightarrow[n\to\infty]{} \langle f, u \rangle_{E',E} = f(u).$$

Of course, as soon as the space on which we are working is infinite-dimensional (functional spaces of type $L^p(\Omega)$, for example), the closed bounded subsets of this space are not necessarily compact for the topology of the norm on E. Nevertheless, the following result establishes the property of weak compactness of closed bounded sets.

Theorem II.2.7 (Weak and Weak-$\star$ compactness).

- *Let E be a reflexive Banach space (i.e., E is isomorphic with E'' via the natural embedding). Then, from any* bounded *sequence of elements of E, we can extract a subsequence which weakly converges in E.*
- *Let E be a separable Banach space (i.e., one which contains a dense countable subset). Then, from any* bounded *sequence of elements of E', we can extract a subsequence which weakly-$\star$ converges in E'.*

One of the important consequences of the Banach–Steinhaus theorem (Theorem II.2.4) is the property of lower semicontinuity of the norm for weak and weak-$\star$ topologies on a Banach space.

Corollary II.2.8. *Let E be a Banach space, and $(u_n)_n$ be a sequence of elements of E (or E', respectively) which weakly converges (or weakly-$\star$, respectively) towards $u \in E$ (or $u \in E'$, respectively). Then the sequence $(u_n)_n$ is bounded in E (or in E', respectively) and we have*

$$\|u\|_E \leq \liminf_{n\to\infty} \|u_n\|_E, \text{ (resp., } \|u\|_{E'} \leq \liminf_{n\to\infty} \|u_n\|_{E'}\text{)}.$$

The following proposition is often used to prove the weak (resp., weak-$\star$) convergence of a whole sequence.

Proposition II.2.9. *Let E be a reflexive Banach space (resp., the dual of a separable Banach space) and $(x_n)_n$ a bounded sequence in E.*

We assume that there exists $x \in E$ such that every weakly convergent (resp., weakly-$\star$ convergent) subsequence of $(x_n)_n$ has a limit equal to x; then the whole sequence $(x_n)_n$ weakly converges (resp., weakly-$\star$ converges) to x.

Proof.

We only give the proof in the reflexive case, the other case being similar. Assume that $(x_n)_n$ does not weakly converge to x. This means that there exists $f \in E'$ such that $(\langle f, x_n\rangle_{E',E})_n$ does not converge to $\langle f, x\rangle_{E',E}$. Hence, there exists $\varepsilon > 0$ and a subsequence $(x_{\varphi(n)})_n$ such that

$$|\langle f, x_{\varphi(n)} - x\rangle_{E',E}| \geq \varepsilon, \forall n \geq 0. \tag{II.4}$$

Since $(x_{\varphi(n)})_n$ is bounded in E which is reflexive, Theorem II.2.7 shows that there exists a new subsequence $(x_{\varphi(\psi(n))})_n$ that weakly converges in E. By assumption its weak limit is necessarily equal to x which implies that

$$\langle f, x_{\varphi(\psi(n))} - x\rangle_{E',E} \xrightarrow[n\to\infty]{} 0.$$

This is a contradiction with (II.4) and the claim is proved.

□

A consequence of this result is that a bound in a "small" space and a weak convergence in a "large" space implies the weak convergence in the "small" space. The precise statement is the following.

Proposition II.2.10. *Let E, F, G be three Banach spaces such that $E \subset G$, $F \subset G$ with continuous embeddings. We assume that F is reflexive.*

Let $(x_n)_n$ be a sequence in $E \cap F$ such that there exists $x \in E$ satisfying

$$(x_n)_n \text{ is bounded in } F,$$

$$(x_n)_n \text{ weakly converges towards } x \text{ in } E.$$

Then,

$$(x_n)_n \text{ weakly converges towards } x \text{ in } F.$$

Proof.

From Proposition II.2.9, the claim will be proved if we show that x is the unique possible weak limit in F of subsequences of $(x_n)_n$.

Let $(x_{\varphi(n)})_n$ a subsequence which weakly converges in F towards some limit $y \in F$. The embedding $F \subset G$ is continuous, therefore we know that $(x_{\varphi(n)})_n$ weakly converges to y in G.

On the other hand, we know by assumption that $(x_{\varphi(n)})_n$ weakly converges towards x in E. The embedding $E \subset G$ is continuous, therefore we deduce that $(x_{\varphi(n)})_n$ weakly converges to x in G. It follows that $y = x$ and the claim is proved.

□

Remark II.2.1. This result can be easily adapted to the case of the weak-$\star$ convergence.

We can now give a useful criterion of strong convergence for weakly convergent sequences in a Hilbert space.

Proposition II.2.11. *Let H be a Hilbert space and $(u_n)_n$ be a sequence of elements of H which weakly converges towards u in H. Let us assume that*

$$\limsup_{n\to\infty} \|u_n\|_H \leq \|u\|_H,$$

then the sequence $(u_n)_n$ strongly converges towards u in H.

Proof.

It is sufficient to write

$$\|u - u_n\|_H^2 = \|u\|_H^2 + \|u_n\|_H^2 - 2(u, u_n)_H.$$

Since the weak convergence gives $(u_n, u)_H \xrightarrow[n\to\infty]{} \|u\|_H^2$, we have

$$\limsup_{n\to\infty} \|u - u_n\|_H^2 = \|u\|_H^2 + \limsup_{n\to\infty} \|u_n\|_H^2 - 2\|u\|_H^2 \leq 0,$$

by using the assumption.

□

We later show (Proposition II.2.32) that this result is also valid in some Banach spaces (e.g. in the spaces $L^p(\Omega)$ with $1 < p < +\infty$).

Unfortunately, the concept of weak convergence, although easier to use, does not generally allow passing to the limit in nonlinear terms. As an example (see Section 2.3 for the main properties of Lebesgue spaces), let the sequence of functions $(u_n)_n$ be defined on $]0,1[$ by $u_n(x) = \sin(nx)$. Then $(u_n)_n$ weakly converges towards 0 in $L^2(]0,1[)$ (Riemann–Lebesgue lemma) and $\displaystyle\int_0^1 u_n^2\,dx$ converges towards $1/2$. Hence the sequence $(u_n^2)_n$ does not weakly converge towards 0 in $L^2(]0,1[)$. However, we note that the sequence $(u_n^2)_n$ does weakly converge in $L^2(]0,1[)$ but its limit is the constant function equal to $1/2$ and not 0.

Nevertheless, we prove in the following result that the product of a strongly converging sequence with a weakly converging one is a sequence which weakly converges towards the product of the limits.

Proposition II.2.12. *Let E, F, and G be three Banach spaces and let B be a continuous bilinear function of $E\times F$ in G. If $(u_n)_n$ is a sequence of elements of E which strongly converges towards u and $(v_n)_n$ is a sequence of elements of F which weakly converges towards v, then the sequence $(B(u_n, v_n))_n$ weakly converges towards $B(u, v)$ in G.*

Proof.

Let $\varphi \in G'$; we need to show that

$$\langle \varphi, B(u_n, v_n) \rangle_{G',G} \xrightarrow[n\to\infty]{} \langle \varphi, B(u,v) \rangle_{G',G}.$$

By using the bilinearity of B, we have

$$\begin{aligned} &|\langle \varphi, B(u_n, v_n) - B(u,v) \rangle_{G',G}| \\ \leq &\|\varphi\|_{G'} \|B(u-u_n, v_n)\|_G + |\langle \varphi, B(u, v_n - v) \rangle_{G',G}| \,. \end{aligned}$$

From Corollary II.2.8, the sequence $(v_n)_n$ is bounded. Hence, since the function B is continuous, the first term is estimated by

$$\|\varphi\|_{G'} \|B(u-u_n, v_n)\|_G \leq \|\varphi\|_{G'} \|B\| \|u-u_n\|_E \|v_n\|_F \leq C\|u-u_n\|_E \xrightarrow[n\to\infty]{} 0.$$

The function $x \in F \mapsto \langle \varphi, B(u,x) \rangle_{G',G}$ is a continuous linear functional on F because u is fixed in E and B is continuous. Hence from the definition of weak convergence, the second term also tends towards 0 when n tends towards infinity.

□

Remark II.2.2. If the space G is reflexive and if $(u_n)_n$ and $(v_n)_n$ converge only weakly towards u (or, respectively, v), then the sequence $(B(u_n, v_n))_n$ is bounded in G (because B is continuous and $(u_n)_n$ and $(v_n)_n$ are bounded). Hence, Theorem II.2.7 shows us that we can extract a subsequence which weakly converges in G towards a certain g.

The problem is that without the property of strong convergence, we cannot in general conclude that g is equal to the expected limit, which would be $B(u,v)$ as shown in the example given above.

As shown in later chapters, we need to establish strong convergence properties of the sequence studied in larger spaces in order to identify the limit g to be the product $B(u,v)$.

Therefore, in order to deal with nonlinearities, it is necessary to obtain strong convergence in one way or another. One way to do this is to use the compactness properties described in Section 3.

2.3 Lebesgue spaces

2.3.1 Definitions and main properties

Definition II.2.13 (Conjugate exponent). *For all $1 \leq p \leq +\infty$, we define* the conjugate exponent p' *of p by*

$$\frac{1}{p} + \frac{1}{p'} = 1,$$

with the obvious conventions for $p = 1$ and $p = +\infty$.

This notation is used systematically throughout this text. We also note that for all p, we have $(p')' = p$.

- For $1 \leq p < +\infty$, the space $L^p(\Omega)$ is the set of Lebesgue-measurable functions on any open set Ω with real values, for which the pth power of the absolute value is integrable for the Lebesgue measure. For each $f \in L^p(\Omega)$ we set $\|f\|_{L^p} = \left(\int_\Omega |f|^p\,dx\right)^{1/p}$.
- The space $L^\infty(\Omega)$ is the set of Lebesgue-measurable functions which are essentially bounded on Ω. For each $f \in L^\infty(\Omega)$, we set $\|f\|_{L^\infty} = \text{esssup}_\Omega |f|$.

In fact, the elements of these spaces have to be considered as the classes of functions which coincide except over null Lebesgue measure sets.

It can be shown (see Proposition II.2.21 and Remark II.2.3) that $\|.\|_{L^p}$ is a norm on $L^p(\Omega)$. Moreover, these spaces are Banach spaces.

- For $1 < p < +\infty$, the space $L^p(\Omega)$ is separable and reflexive. Moreover its dual is isomorphic with $L^{p'}(\Omega)$ where p' is the conjugate exponent of p.
- The space $L^1(\Omega)$ is separable but not reflexive, its dual being isomorphic with $L^\infty(\Omega)$.
- By contrast, the space $L^\infty(\Omega)$ is neither separable nor reflexive and its dual is strictly larger than $L^1(\Omega)$.

We conclude this introduction by recalling the following version of the change of variable theorem.

Definition II.2.14. *Let Ω, $\tilde{\Omega}$ be two open sets in $\mathbb{R}^d$. A map $T : \tilde{\Omega} \to \Omega$ is said to be a* Lipschitz diffeomorphism *if and only if*

- *T is a bijection.*
- *T and T^{-1} are Lipschitz-continuous.*

Notice that such a map is not in general a diffeomorphism in the usual sense because, in particular, it is not necessarily differentiable everywhere.

Proposition II.2.15. *Let Ω, $\tilde{\Omega}$ be two open sets in $\mathbb{R}^d$ and $T : \tilde{\Omega} \to \Omega$ a Lipschitz diffeomorphism.*

For any measurable function $u : \Omega \to \mathbb{R}$ and $1 \leq p \leq \infty$, we have

$$u \in L^p(\Omega) \Longleftrightarrow u \circ T \in L^p(\tilde{\Omega}).$$

Moreover, we have

$$C_1 \|u\|_{L^p(\Omega)} \leq \|u \circ T\|_{L^p(\tilde{\Omega})} \leq C_2 \|u\|_{L^p(\Omega)},$$

for some $C_1, C_2 > 0$ depending only on T.

2.3.2 Elementary inequalities

We give here rather general versions of Young's and Hölder's inequalities, without proof. We use these repeatedly in the following sections, without necessarily explicitly referencing them.

Proposition II.2.16 (Young's inequality). *Let $n \geq 2$, and $x_1, \ldots, x_n$ be non-negative real numbers. Also, let $p_1, \ldots, p_n$ be positive real numbers such that*

$$\frac{1}{p_1} + \cdots + \frac{1}{p_n} = 1.$$

We then have:

$$x_1 \cdots x_n \leq \frac{x_1^{p_1}}{p_1} + \cdots + \frac{x_n^{p_n}}{p_n}.$$

The proof of this inequality is a simple application of the concavity of the logarithm function. We can directly deduce an useful version of this inequality.

Corollary II.2.17. *Let $p_1, \ldots, p_n$ be real numbers satisfying the hypothesis of the preceding proposition. For all positive $\varepsilon_1, \ldots, \varepsilon_{n-1}$, there exists a $C(\varepsilon_1, \ldots, \varepsilon_{n-1}) > 0$, such that for all positive $x_1, \ldots, x_n$, we have*

$$x_1 \cdots x_n \leq \varepsilon_1 x_1^{p_1} + \cdots + \varepsilon_{n-1} x_{n-1}^{p_{n-1}} + C(\varepsilon_1, \ldots, \varepsilon_{n-1}) x_n^{p_n}.$$

In other words, in Young's inequality, all the coefficients can be fixed except for one. Of course, the coefficient $C(\varepsilon_1, \ldots, \varepsilon_{n-1})$ blows up when one of the ε_i tends towards 0.

From Young's inequality we can deduce Hölder's inequality which is stated in the following way.

Proposition II.2.18 (Hölder's inequality). *Let Ω be an open set of $\mathbb{R}^d$ and let $p_1, \ldots, p_n$ be positive real numbers (possibly infinite). Let $r \in [1, +\infty]$ such that*

$$\frac{1}{r} = \frac{1}{p_1} + \cdots + \frac{1}{p_n}.$$

For all functions $f_1, \ldots, f_n$, with $f_i \in L^{p_i}(\Omega)$, the product $f_1 \cdots f_n$ belongs to $L^r(\Omega)$ and we have

$$\|f_1 \cdots f_n\|_{L^r} \leq \|f_1\|_{L^{p_1}} \cdots \|f_n\|_{L^{p_n}}.$$

We also need the following generalisation of Fubini's theorem.

Proposition II.2.19. *Let $d \geq 2$ and $f_1, \ldots, f_d : \mathbb{R}^{d-1} \to \mathbb{R}$ be d functions belonging to $L^{d-1}(\mathbb{R}^{d-1})$. We define the following product*

$$f(x) = f_1(x_2, \ldots, x_d) f_2(x_1, x_3, \ldots, x_d) \cdots f_d(x_1, \ldots, x_{d-1}), \forall x \in \mathbb{R}^d,$$

where the term with f_i depends on all the variables except x_i.

Then, f belongs to $L^1(\mathbb{R}^d)$ and we have

$$\|f\|_{L^1(\mathbb{R}^d)} \leq \prod_{i=1}^{d} \|f_i\|_{L^{d-1}(\mathbb{R}^{d-1})}.$$

Proof.

In the case $d = 2$, we have $f(x_1, x_2) = f_1(x_1)f_2(x_2)$ and by assumption $f_1, f_2 \in L^1(\mathbb{R})$. Therefore, Fubini's theorem implies that $f \in L^1(\mathbb{R}^2)$ and that $\|f\|_{L^1(\mathbb{R}^2)} = \|f_1\|_{L^1(\mathbb{R})}\|f_2\|_{L^1(\mathbb{R})}$. This proves the result (in this particular case the claimed inequality is an equality).

Let us only prove the result for $d = 3$ because the general case follows by a simple induction using Hölder's inequality (see [27], for instance, for a complete proof). Let us integrate the definition of $|f|$ with respect to the variable x_3 and apply the Cauchy–Schwarz inequality

$$\begin{aligned}
&\int_{\mathbb{R}} |f|(x_1, x_2, x_3)\,dx_3\\
=&|f_3|(x_1, x_2)\left(\int_{\mathbb{R}} |f_1|(x_2, x_3)|f_2|(x_1, x_3)\,dx_3\right)\\
\leq&|f_3|(x_1, x_2)\underbrace{\left(\int_{\mathbb{R}} |f_1|^2(x_2, x_3)\,dx_3\right)^{1/2}}_{=(g_1(x_1))^{1/2}}\underbrace{\left(\int_{\mathbb{R}^3} |f_2|^2(x_1, x_3)\,dx_3\right)^{1/2}}_{=(g_2(x_2))^{1/2}}.
\end{aligned}$$

We apply once more the Cauchy–Schwarz inequality to get

$$\int_{R^3} |f|\,dx \leq \left(\int_{\mathbb{R}^2} |f_3|^2(x_1, x_2)\,dx_1\,dx_2\right)^{1/2}\left(\int_{\mathbb{R}^2} g_1(x_1)g_2(x_2)\,dx_1\,dx_2\right)^{1/2}.$$

The last term is estimated by using the induction assumption (i.e., Fubini's theorem) to get the claim

$$\|f\|_{L^1(\mathbb{R}^3)} \leq \|f_3\|_{L^2(\mathbb{R}^2)}\|g_1\|_{L^1(\mathbb{R})}^{1/2}\|g_2\|_{L^1(\mathbb{R})}^{1/2} = \|f_3\|_{L^2(\mathbb{R}^2)}\|f_1\|_{L^2(\mathbb{R}^2)}\|f_2\|_{L^2(\mathbb{R}^2)}.$$

□

We also need the following version of Jensen's inequality.

Proposition II.2.20 (Jensen's inequality). *Let Ω be an open set of $\mathbb{R}^d$ and $\eta \in L^1(\Omega)$ a nonnegative function. For any function f such that $|f|^p\eta \in L^1(\Omega)$, for some $1 \leq p < +\infty$, we have $f\eta \in L^1(\Omega)$ and*

$$\left|\int_{\Omega} f\eta\,dx\right|^p \leq \|\eta\|_{L^1}^{p-1}\int_{\Omega} |f|^p\eta\,dx.$$

Proof.

We write $f\eta = (f\eta^{1/p})\eta^{1/p'}$ and we use the Hölder inequality with exponents p and $p' = p/(p-1)$.

□

Let us now establish a general version of the classic Minkowski inequality that we need below.

Proposition II.2.21 (Minkowski's inequality). *Let (X_1, μ_1) and (X_2, μ_2) be two σ-finite measure spaces. Then for any nonnegative measurable function f defined on $X_1 \times X_2$ and any $r \geq 1$ we have*

$$\left(\int_{X_1}\left(\int_{X_2} f(x_1,x_2)\,d\mu_2\right)^r d\mu_1\right)^{1/r} \leq \int_{X_2}\left(\int_{X_1} f(x_1,x_2)^r d\mu_1\right)^{1/r} d\mu_2.$$

Remark II.2.3. If one takes $X_2 = \{0,1\}$ and μ_2 the counting measure, the above inequality can be written as

$$\|f+g\|_{L^r} \leq \|f\|_{L^r} + \|g\|_{L^r},$$

for any nonnegative f, g and any $r \geq 1$.

Proof.

For all $x_1 \in X_1$ we denote $J(x_1) = \displaystyle\int_{X_2} f(x_1,x_2)\,d\mu_2$. Then, by using the Hölder inequality and the Fubini theorem, we have

$$\begin{aligned}
\int_{X_1} J(x_1)^r\,d\mu_1 &= \int_{X_1} J(x_1)^{r-1}\left(\int_{X_2} f(x_1,x_2)\,d\mu_2\right) d\mu_1\\
&= \int_{X_1}\int_{X_2} J(x_1)^{r-1} f(x_1,x_2)\,d\mu_2\,d\mu_1\\
&= \int_{X_2}\int_{X_1} J(x_1)^{r-1} f(x_1,x_2)\,d\mu_1\,d\mu_2\\
&\leq \int_{X_2}\left(\int_{X_1} J(x_1)^r\,d\mu_1\right)^{(r-1)/r}\left(\int_{X_1} f(x_1,x_2)^r\,d\mu_1\right)^{1/r} d\mu_2\\
&= \left(\int_{X_1} J(x_1)^r\,d\mu_1\right)^{(r-1)/r}\int_{X_2}\left(\int_{X_1} f(x_1,x_2)^r\,d\mu_1\right)^{1/r} d\mu_2.
\end{aligned}$$

From where we deduce the claim.

□

Let us also mention the following reverse Minkowski inequality that we state in a simple framework sufficient for our purposes.

Proposition II.2.22. *Let $0 < q < 1$ and Ω an open set of $\mathbb{R}^d$. For any nonnegative measurable functions $f, g : \Omega \to \mathbb{R}$, we have*

$$\|f+g\|_{L^q} \geq \|f\|_{L^q} + \|g\|_{L^q}.$$

Proof.

For any $x \in \Omega$ we write

$$\frac{f(x)+g(x)}{\|f\|_{L^q}+\|g\|_{L^q}} = \frac{\|f\|_{L^q}}{\|f\|_{L^q}+\|g\|_{L^q}}\frac{f(x)}{\|f\|_{L^q}} + \frac{\|g\|_{L^q}}{\|f\|_{L^q}+\|g\|_{L^q}}\frac{g(x)}{\|g\|_{L^q}},$$

that is to say,

$$\frac{f(x)+g(x)}{\|f\|_{L^q}+\|g\|_{L^q}} = \theta\frac{f(x)}{\|f\|_{L^q}} + (1-\theta)\frac{g(x)}{\|g\|_{L^q}},$$

with $\theta \in [0,1]$. From the assumption on q we know that the map $s \mapsto s^q$ is concave on $\mathbb{R}^+$. Therefore we get

$$\left(\frac{f(x)+g(x)}{\|f\|_{L^q}+\|g\|_{L^q}}\right)^q \geq \theta\left(\frac{f(x)}{\|f\|_{L^q}}\right)^q + (1-\theta)\left(\frac{g(x)}{\|g\|_{L^q}}\right)^q.$$

By integrating this inequality on Ω, we observe that we obtain 1 in the right-hand side. The claim follows immediately.

□

2.3.3 Mollifying kernels. Density result

Mollifying is a central procedure in functional analysis. It in particular allows us to prove density results in suitable functional spaces related to mathematical fluid mechanics. It is also crucial in the renormalized solutions theory for the transport equation that we describe in detail in Chapter VI.

Definition II.2.23. *A map $\eta : \mathbb{R}^d \to \mathbb{R}$ is called a mollifying kernel if*

- $\eta \in \mathcal{C}_c^\infty(\mathbb{R}^d)$, *with* $\operatorname{Supp} \eta \subset B$, *the unit ball of* $\mathbb{R}^d$.
- $\eta \geq 0$ *and* $\int_{\mathbb{R}^d} \eta \, dx = \int_B \eta \, dx = 1$.
- $\eta(x)$ *only depends on* $|x|$.

Note first that the last condition is not necessary but it sometimes allows simplifications in some computations. It is also worth noticing that such a function actually exists.

For any $\varepsilon > 0$, we can now define

$$\eta_\varepsilon(x) = \frac{1}{\varepsilon^d}\eta\left(\frac{x}{\varepsilon}\right) \text{ and } (\nabla\eta)_\varepsilon(x) = \frac{1}{\varepsilon^d}(\nabla\eta)\left(\frac{x}{\varepsilon}\right),$$

in such a way that $\nabla\eta_\varepsilon = (1/\varepsilon)(\nabla\eta)_\varepsilon$.

Definition II.2.24. *For any $f \in L^p(\mathbb{R}^d)$, $1 \leq p \leq \infty$ and any $\varepsilon > 0$ we define the convolution*

$$
\begin{aligned}
(f \star \eta_\varepsilon)(x) &= \int_{\mathbb{R}^d} f(y)\eta_\varepsilon(x-y)\,dy \\
&= \int_{\mathbb{R}^d} f(x-y)\eta_\varepsilon(y)\,dy = \int_B f(x-\varepsilon z)\eta(z)\,dz.
\end{aligned}
\tag{II.5}
$$

Notice that, inasmuch as η is bounded and compactly supported, all the integrals in the definition above make sense.

Proposition II.2.25. *For any $\varepsilon > 0$, we have*

$$
\begin{aligned}
f \star \eta_\varepsilon &\in \mathcal{C}^\infty(\mathbb{R}^d), \\
\|f \star \eta_\varepsilon\|_{L^\infty} &\le \frac{C}{\varepsilon^{d/p}}\|f\|_{L^p}, \\
\|\nabla(f \star \eta_\varepsilon)\|_{L^\infty} &\le \frac{C}{\varepsilon^{1+d/p}}\|f\|_{L^p}, \\
\|f \star \eta_\varepsilon\|_{L^p} &\le C\|f\|_{L^p},
\end{aligned}
\tag{II.6}
$$

for some $C > 0$ depending only on η and p. Finally, if $p < +\infty$ we have

$$
f \star \eta_\varepsilon \xrightarrow[\varepsilon \to 0]{} f, \quad \textit{in } L^p(\mathbb{R}^d).
$$

Proof.

The regularity of $f \star \eta_\varepsilon$ follows from the regularity of the kernel η and usual results of differentiation under the integral sign. The L^∞ estimates simply follow from Hölder's inequality and the fact that $\|\eta_\varepsilon\|_{L^{p'}} = \|\eta\|_{L^{p'}}/\varepsilon^{d/p}$.

To prove the L^p estimate (II.6) for $p < +\infty$, we first use the Jensen inequality (Proposition II.2.20) to get, for any $x \in \mathbb{R}^d$,

$$
|(f \star \eta_\varepsilon)(x)|^p \le \int_B |f(x-\varepsilon z)|^p \eta(z)\,dz.
$$

By integrating with respect to x and using Fubini's theorem, we get

$$
\begin{aligned}
\|f \star \eta_\varepsilon\|_{L^p}^p &\le \int_{\mathbb{R}^d}\int_B |f(x-\varepsilon z)|^p \eta(z)\,dz\,dx \\
&= \int_B \eta(z)\left(\int_{\mathbb{R}^d} |f(x-\varepsilon z)|^p\,dx\right)dz = \|f\|_{L^p}^p.
\end{aligned}
$$

Let us now show the convergence property. Since $p < +\infty$, we can use the density of $\mathcal{C}_c^0(\mathbb{R}^d)$ in $L^p(\mathbb{R}^d)$ (this property comes from the regularity of the Lebesgue measure). Therefore, there exists a sequence $(f_n)_n$ of functions in $\mathcal{C}_c^0(\mathbb{R}^d)$ which converges towards f in $L^p(\mathbb{R}^d)$. Each function f_n is uniformly continuous and we denote by ω_n its modulus of continuity.

Using the properties of the kernel η, we observe that for each n we have

$$
|f_n \star \eta_\varepsilon(x) - f_n(x)| \le \int_B |f_n(x-\varepsilon z) - f_n(x)|\eta(z)\,dz \le \omega_n(\varepsilon),
$$

and therefore, since f_n is compactly supported we have

$$\|f_n \star \eta_\varepsilon - f_n\|_{L^p} \leq C_n \omega_n(\varepsilon).$$

Using the triangle inequality and (II.6) we get

$$\begin{aligned}\|f \star \eta_\varepsilon - f\|_{L^p} &\leq \|(f - f_n) \star \eta_\varepsilon\|_{L^p} + \|f_n \star \eta_\varepsilon - f_n\|_{L^p} + \|f_n - f\|_{L^p} \\ &\leq (1 + C)\|f - f_n\|_{L^p} + C_n \omega_n(\varepsilon).\end{aligned}$$

Taking the superior limit as $\varepsilon \to 0$, we obtain for any n that

$$\limsup_{\varepsilon \to 0} \|f \star \eta_\varepsilon - f\|_{L^p} \leq (1 + C)\|f - f_n\|_{L^p}.$$

Taking now the limit $n \to \infty$, we finally get $\limsup_{\varepsilon \to 0} \|f \star \eta_\varepsilon - f\|_{L^p} = 0$ and the claim is proved.

□

Theorem II.2.26. *For any open set Ω in $\mathbb{R}^d$, the set $\mathcal{D}(\Omega)$ is dense in $L^p(\Omega)$ for any $1 \leq p < +\infty$.*

Proof.

Let $f \in L^p(\Omega)$. For any $n \geq 1$ we define the open set $\Omega_n = \{x \in \Omega, d(x, \partial\Omega) > 1/n\}$. By the dominated convergence theorem we see that $f_n = f 1_{\overline{\Omega_n}}$ converges to f in $L^p(\Omega)$. We consider now the function $f_{n,\varepsilon} = f_n \star \eta_\varepsilon$. By the previous proposition, we know that $f_{n,\varepsilon} \in \mathcal{C}^\infty(\mathbb{R}^d)$; moreover, for any $\varepsilon < 1/n$, we observe that the support of $f_{n,\varepsilon}$ is contained in Ω, therefore $f_{n,\varepsilon} \in \mathcal{D}(\Omega)$. The result follows because we have $\lim_{\varepsilon \to \infty} (\lim_{n \to \infty} f_{n,\varepsilon}) = f$ in $L^p(\Omega)$.

□

2.3.4 Weak and weak-$\star$ convergences in Lebesgue spaces

From the recap at the beginning of this section, and in particular from the characterisation of the dual space of $L^p(\Omega)$, we can write the L^p-version of Theorem II.2.7.

Proposition II.2.27. *Let $(u_n)_n$ be a bounded sequence of $L^p(\Omega)$, $1 < p < +\infty$; then we can extract a weakly converging subsequence from the sequence $(u_n)_n$; that is*

$$\exists (u_{n_k})_k, \exists u \in L^p(\Omega), \lim_{k \to \infty} \int_\Omega u_{n_k} \varphi \, dx = \int_\Omega u \varphi \, dx, \forall \varphi \in L^{p'}(\Omega).$$

This result does not hold in $L^1(\Omega)$ because that space is not reflexive. Nevertheless, we have a similar result in $L^\infty(\Omega)$ provided that we consider the weak-$\star$ topology on this space, because it is the dual of the separable space $L^1(\Omega)$.

Proposition II.2.28. *Let $(u_n)_n$ be a bounded sequence of $L^\infty(\Omega)$; then, from the sequence $(u_n)_n$, we can extract a subsequence which is weakly-$\star$ convergent; that is*

$$\exists (u_{n_k})_k, \exists u \in L^\infty(\Omega), \lim_{k\to\infty} \int_\Omega u_{n_k}\varphi\,dx = \int_\Omega u\varphi\,dx, \forall \varphi \in L^1(\Omega).$$

With this characterisation of the weak-$\star$ convergence in $L^\infty(\Omega)$, we can extend the density result given in Theorem II.2.26.

Theorem II.2.29. *For any open set Ω of $\mathbb{R}^d$, the set $\mathcal{D}(\Omega)$ is dense in $L^\infty(\Omega)$ for the weak-$\star$ topology.*

Proof.

Let $f \in L^\infty(\Omega)$. We set $\psi_n = 1_{B(0,n)}$ so that $f\psi_n \in L^1(\Omega) \cap L^\infty(\Omega)$. By using Theorem II.2.26, we know that for each n there exists a function $f_n \in \mathcal{D}(\Omega)$ such that $\|f_n - f\psi_n\|_{L^1} \le 1/n$. Observe in the proof of this theorem that we have the additional property $\|f_n\|_{L^\infty} \le \|f\psi_n\|_{L^\infty} \le \|f\|_{L^\infty}$.

Let now $\varphi \in \mathcal{D}(\Omega)$. We have

$$\left|\int_\Omega f_n\varphi\,dx - \int_\Omega f\varphi\,dx\right| \le \underbrace{\int_\Omega |f_n - f\psi_n||\varphi|\,dx}_{\|f_n - f\psi_n\|_{L^1}\|\varphi\|_{L^\infty}} + \int_\Omega |\psi_n - 1||f||\varphi|\,dx.$$

The first term in the right-hand side tends to 0 by construction of $(f_n)_n$ and the second one also tends to 0 thanks to the Lebesgue dominated convergence theorem.

Finally, since $(f_n)_n$ is bounded in $L^\infty(\Omega)$ and $\mathcal{D}(\Omega)$ is dense in $L^1(\Omega)$, we deduce that

$$\left|\int_\Omega f_n g\,dx - \int_\Omega fg\,dx\right| \xrightarrow[n\to\infty]{} 0, \forall g \in L^1(\Omega),$$

which proves the theorem.

□

By applying Proposition II.2.12 within the framework of L^p-spaces, and by using Hölder's inequality, we obtain the following useful result.

Proposition II.2.30. *Let p, q, and r be three real numbers in $[1, +\infty[$ such that*

$$\frac{1}{r} = \frac{1}{p} + \frac{1}{q}.$$

If $(u_n)_n$ is a sequence of $L^p(\Omega)$ which strongly converges towards u in $L^p(\Omega)$ and if $(v_n)_n$ is a sequence of $L^q(\Omega)$ which weakly converges towards v in $L^q(\Omega)$, then the product sequence $(u_nv_n)_n$ weakly converges towards uv in $L^r(\Omega)$.

We now state the classic inequalities in L^p spaces which prove that these spaces are uniformly convex ([27], [69]) except for $p = 1$ and $p = +\infty$. We can view these inequalities as generalisations of the parallelogram law in Hilbert space.

Lemma II.2.31 (Clarkson's inequalities). *Let $1 < p < +\infty$, and let f, g be in $L^p(\Omega)$.*

- *If $p \geq 2$, we have*

$$\left\| \frac{f+g}{2} \right\|_{L^p}^p + \left\| \frac{f-g}{2} \right\|_{L^p}^p \leq \frac{1}{2}\|f\|_{L^p}^p + \frac{1}{2}\|g\|_{L^p}^p.$$

- *If $p < 2$, we have*

$$\left\| \frac{f+g}{2} \right\|_{L^p}^{p'} + \left\| \frac{f-g}{2} \right\|_{L^p}^{p'} \leq \left(\frac{1}{2}\|f\|_{L^p}^p + \frac{1}{2}\|g\|_{L^p}^p \right)^{1/(p-1)}.$$

We can now prove the strong convergence criterion for a weakly converging sequence in L^p spaces.

Proposition II.2.32. *Let $1 < p < +\infty$, and let $(u_n)_n$ be a sequence of functions of $L^p(\Omega)$ which weakly converges towards u in $L^p(\Omega)$. If we assume*

$$\limsup_{n \to \infty} \|u_n\|_{L^p} \leq \|u\|_{L^p},$$

then the sequence $(u_n)_n$ strongly converges towards u.

According to Corollary II.2.8, this hypothesis is equivalent to saying that the sequence of norms $(\|u_n\|_{L^p})_n$ converges towards $\|u\|_{L^p}$.

For L^p spaces, this result generalises Proposition II.2.11 which dealt with the Hilbertian case (i.e., $p = 2$).

Proof.

The Clarkson inequalities given by the previous lemma can be written in the general form

$$\left\| \frac{f+g}{2} \right\|_{L^p}^{\alpha p} + \left\| \frac{f-g}{2} \right\|_{L^p}^{\alpha p} \leq \left(\frac{1}{2}\|f\|_{L^p}^p + \frac{1}{2}\|g\|_{L^p}^p \right)^{\alpha},$$

where $\alpha = 1$ if $p \geq 2$ and $\alpha = 1/(p-1)$ if $p < 2$. Let us apply this inequality to $f = u_n$ and $g = u$. We obtain

$$\left\| \frac{u_n+u}{2} \right\|_{L^p}^{\alpha p} + \left\| \frac{u_n-u}{2} \right\|_{L^p}^{\alpha p} \leq \left(\frac{1}{2}\|u_n\|_{L^p}^p + \frac{1}{2}\|u\|_{L^p}^p \right)^{\alpha},$$

and we denote the left-hand side of this inequality as a_n. If we pass to the upper limit in this inequality, then by using the hypothesis, we find that:

$$\limsup_{n\to\infty} a_n \leq \|u\|_{L^p}^{p\alpha}. \tag{II.7}$$

However, we have:

$$\left\|\frac{u_n - u}{2}\right\|_{L^p}^{\alpha p} = a_n - \left\|\frac{u_n + u}{2}\right\|_{L^p}^{\alpha p},$$

which gives

$$\limsup_{n\to\infty} \left\|\frac{u_n - u}{2}\right\|_{L^p}^{\alpha p} \leq \limsup_{n\to\infty} a_n - \liminf_{n\to\infty} \left\|\frac{u_n + u}{2}\right\|_{L^p}^{\alpha p}. \tag{II.8}$$

Moreover, the sequence $(u_n + u)/2$ also weakly converges towards u, so that the Corollary II.2.8 shows us that

$$\|u\|_{L^p}^{p\alpha} \leq \liminf_{n\to\infty} \left\|\frac{u_n + u}{2}\right\|_{L^p}^{\alpha p}. \tag{II.9}$$

By combining (II.7), (II.8), and (II.9), we finally obtain:

$$\limsup_{n\to\infty} \left\|\frac{u_n - u}{2}\right\|_{L^p}^{\alpha p} \leq 0,$$

which concludes the proof.

□

2.3.5 Interpolation between L^p spaces

We now establish an interpolation inequality which is nothing but a convexity property.

Lemma II.2.33. *Let Ω be any open set of $\mathbb{R}^d$ and let $u \in L^p(\Omega) \cap L^q(\Omega)$ with $1 \leq p, q \leq +\infty$. Then for all r such that*

$$\frac{1}{r} = \frac{\theta}{p} + \frac{1-\theta}{q}, 0 \leq \theta \leq 1,$$

we have $u \in L^r(\Omega)$ and

$$\|u\|_{L^r} \leq \|u\|_{L^p}^{\theta} \|u\|_{L^q}^{1-\theta}.$$

Proof.

We note that

$$1 = \frac{\theta r}{p} + \frac{(1-\theta)r}{q},$$

and we can therefore apply the Hölder inequality in the following way:

$$\begin{aligned}\int_\Omega |u|^r dx &= \int_\Omega |u|^{r\theta}|u|^{r(1-\theta)}dx\\&\leq \left(\int_\Omega |u|^p dx\right)^{r\theta/p}\left(\int_\Omega |u|^q dx\right)^{r(1-\theta)/q}\\&\leq \|u\|_{L^p}^{r\theta}\|u\|_{L^q}^{r(1-\theta)}.\end{aligned}$$

□

The preceding inequality allows us to obtain convergence properties in the intermediate spaces from the convergences in suitable L^p spaces.

Corollary II.2.34. *Let Ω be any open set of $\mathbb{R}^d$. Let $p_1, p_2 \in [1,+\infty]$ and let $(u_n)_n$ be a sequence of functions which strongly converges towards u in $L^{p_1}(\Omega)$ and which weakly converges (weakly-$\star$ if $p_2 = +\infty$) in $L^{p_2}(\Omega)$. Then, for all p included strictly between p_1 and p_2, the sequence $(u_n)_n$ strongly converges towards u in $L^p(\Omega)$.*

Proof.

Since p is strictly included between p_1 and p_2, there exists a $\theta \in]0,1[$ such that

$$\frac{1}{p} = \frac{\theta}{p_1} + \frac{1-\theta}{p_2}.$$

From the interpolation inequality given by the preceding lemma, we have

$$\|u-u_n\|_{L^p} \leq \|u-u_n\|_{L^{p_1}}^{\theta}\|u-u_n\|_{L^{p_2}}^{1-\theta}.$$

However, the weak (or weak-$\star$) convergence of $(u_n)_n$ in $L^{p_2}(\Omega)$, shows that the sequence $(u-u_n)_n$ is bounded in this space, as well as that the strong convergence in $L^{p_1}(\Omega)$ ensures convergence towards 0 of the first term, because θ is not zero.

□

2.3.6 Local Lebesgue spaces

Definition II.2.35. *For all open sets Ω of $\mathbb{R}^d$ and for all $p \in [1,+\infty[$, we denote as $L^p_{loc}(\Omega)$ the set of measurable functions for which the p-th power of the absolute value is locally integrable, that is, its integral over all compact subsets included in Ω is finite. Similarly, we denote as $L^\infty_{loc}(\Omega)$ the set of measurable functions essentially bounded over all compact sets included in Ω.*

We can say that a sequence $(u_n)_n$ converges towards u in $L^p_{loc}(\Omega)$, if $(u_n)_n$ converges towards u in $L^p(\omega)$ for any bounded open set ω such that $\overline{\omega} \subset \Omega$.

It is clear that $L^p(\Omega) \subset L^p_{loc}(\Omega)$, the inverse inclusion being certainly false. A frequently useful property of sequences of functions in $L^p_{loc}(\Omega)$ is the following.

Proposition II.2.36. *Let Ω be a bounded open set of $\mathbb{R}^d$, $q > 1$ and let $(u_n)_n$ be a sequence of bounded functions in $L^q(\Omega)$. We assume that $(u_n)_n$ converges towards u in $L^p_{loc}(\Omega)$ with $1 \leq p < q$; then we have*

$$u \in L^q(\Omega),$$

and

$$u_n \xrightarrow[n\to\infty]{} u, \quad \text{in } L^p(\Omega).$$

Proof.

The sequence $(u_n)_n$ being bounded in $L^q(\Omega)$, we know from Propositions II.2.27 and II.2.28 that we can extract a subsequence $(u_{n_k})_k$ which weakly converges (weakly-$\star$ if $q = +\infty$) towards a function $v \in L^q(\Omega)$. In particular, we deduce that for any $\omega \subset \overline{\omega} \subset \Omega$, $(u_{n_k})_k$ converges weakly (or weakly-$\star$) towards v in $L^q(\omega) \subset L^p(\omega)$. The convergence in $L^p_{loc}(\Omega)$ implies strong convergence in $L^p(\omega)$, thus we deduce that $u = v \in L^q(\Omega)$.

For any $k \geq 1$, we set $\omega_k = \{x \in \Omega, d(x, \partial\Omega) > 1/k\}$. We have $\overline{\omega_k} \subset \Omega$ so that, by assumption,

$$\|u_n - u\|_{L^p(\omega_k)} \xrightarrow[n\to\infty]{} 0.$$

Moreover, by using the Hölder inequality we get

$$\|u_n - u\|_{L^p(\Omega\setminus\omega_k)} \leq \|u_n - u\|_{L^q(\Omega)} |\Omega\setminus\omega_k|^{(q-p)/q)} \leq 2C|\Omega\setminus\omega_k|^{(q-p)/q},$$

where C is a bound of the sequence $(u_n)_n$ and of the function u in $L^q(\Omega)$. We then set $\varepsilon > 0$ and choose k to be sufficiently large so that $2C|\Omega\setminus\omega_k|^{(q-p)/q} < \varepsilon$. We then choose n_0 sufficiently large that

$$\|u_n - u\|_{L^p(\Omega_k)} \leq \varepsilon, \forall n \geq n_0,$$

and, therefore

$$\|u_n - u\|_{L^p(\Omega)} \leq 2\varepsilon, \forall n \geq n_0.$$

$\square$

2.4 Partitions of unity

Let us start with a useful lemma when studying the local properties of functions.

Lemma II.2.37. *Let Ω be a nonempty open set of $\mathbb{R}^d$ and let ω be a bounded open set of $\mathbb{R}^d$ satisfying $\overline{\omega} \subset \Omega$. Then there exists a function $\varphi \in \mathcal{D}(\Omega)$, such that*

$$0 \leq \varphi \leq 1,$$
$$\varphi(x) = 1, \forall x \in \overline{\omega}.$$

Proof.

By hypothesis $\overline{\omega}$ is compact and disjoint from the closed set $\mathbb{R}^d \backslash \Omega$, hence

$$\delta = d(\overline{\omega}, \mathbb{R}^d \backslash \Omega) > 0.$$

We then introduce the open set $\mathcal{U} = \{x \in \Omega,\ d(x, \overline{\omega}) < \delta/2\}$. It is clear that

$$\overline{\omega} \subset \mathcal{U} \text{ and } \overline{\mathcal{U}} \subset \Omega.$$

The reader can easily convince her- or himself that the function

$$\varphi = 1_{\mathcal{U}} \star \eta_{\frac{\delta}{4}},$$

obtained by convolution with a mollifying kernel of the characteristic function of $\mathcal{U}$ satisfies the stated result.

□

We can now show the essential result of this section.

Lemma II.2.38 (Partition of unity). *Let A be a nonempty set of $\mathbb{R}^d$. We suppose given a covering of A by any family of open sets,*

$$A \subset \bigcup_{i \in I} \omega_i.$$

There exists a family $(\psi_i)_{i \in I}$ of nonnegative functions of $\mathcal{C}^\infty(\mathbb{R}^d)$, indexed on I such that

$$\operatorname{Supp} \psi_i \subset \omega_i, \forall i \in I,$$
$$\sum_{i \in I} \psi_i(x) = 1, \forall x \in A,$$

this sum being locally finite. Moreover, the ψ_i are identically zero except for a countable number of indices $i \in I$.

One such family of functions is called a *partition of unity* associated with the covering $(\omega_i)_{i \in I}$.

Proof.

- Let us consider the set S of points of A with rational coordinates. We then consider the family $(B_j)_{j \in J}$ of spheres centred on S, for which the radius is rational and which are contained in one of the $(\omega_i)_i$. This family is, of course, countable and we therefore index it with the integers $n \in \mathbb{N}$, and by the density of $\mathbb{Q}$ in $\mathbb{R}$ we clearly have

$$A \subset \bigcup_{n=0}^{+\infty} B_n.$$

- For all $n \geq 0$, we let V_n denote the ball with its centre at the same point as B_n and for which the radius is half that of B_n.
  According to Lemma II.2.37, there exists a positive regular function φ_n with compact support included in B_n and which is identically equal to 1 on V_n.
  We then define $\alpha_0 = \varphi_0$ and for all $n \geq 1$,
  $$\alpha_n = (1-\varphi_0)\cdots(1-\varphi_{n-1})\varphi_n.$$
  Is is clear that α_n is smooth and nonnegative. Furthermore, by definition, α_n has its support contained in B_n which is itself contained in one ω_i for some $i \in I$.
  Moreover, a straightforward computation implies that
  $$\sum_{n=0}^{N} \alpha_n(x) = 1 - (1-\varphi_0)\cdots(1-\varphi_N).$$
  This shows firstly, since $0 \leq \varphi_i \leq 1$, that for all N we have
  $$\sum_{n=0}^{N} \alpha_n(x) \leq 1.$$
  Furthermore, since $\varphi_i = 1$ on V_i, we see that
  $$\sum_{n=0}^{N} \alpha_n(x) = 1, \quad \forall x \in V_1 \cup \cdots \cup V_N.$$
  Inasmuch as the α_i are nonnegative, this implies that for all $n \geq N+1$, α_n is zero on $V_1 \cup \cdots V_N$, which indeed proves that the sum $\sum_{n\geq 0} \alpha_n$ is locally finite and that
  $$\sum_{n\in\mathbb{N}} \alpha_n(x) = 1, \forall x \in A. \tag{II.10}$$
- For any n, we denote as $i(n) \in I$ an index such that $\operatorname{Supp} \alpha_n \subset \omega_{i(n)}$. We then note that
  $$A \subset \bigcup_{n=0}^{+\infty} \omega_{i(n)}.$$
  Indeed, Equation (II.10), shows that any point of A belongs to the support of at least one function α_n and therefore lies in $\omega_{i(n)}$.
  We now set $\psi_i = 0$ for any $i \in I \backslash \{i(n), n \in \mathbb{N}\}$. It remains to define the functions $\psi_{i(n)}$. To do this, we define

$$\psi_{i(0)}(x) = \sum_{\substack{k\in\mathbb{N},\text{s.t.}\\ \text{Supp}\,(\alpha_k)\subset\omega_{i(0)}}} \alpha_k(x).$$

This sum is perfectly defined because the sum of the family $(\alpha_n)_n$ is locally finite in A. Furthermore, it is clear that $\beta_{i(0)}$ is nonnegative and that its support is contained in $\omega_{i(0)}$. We then define for $n \geq 1$,

$$\psi_{i(n)}(x) = \sum_{\substack{k\in\mathbb{N},\text{s.t. }\text{Supp}\,(\alpha_k)\subset\omega_{i(n)}\\ \forall p\leq n-1,\text{Supp}\,(\alpha_k)\not\subset\omega_{i(p)}}} \alpha_k(x).$$

It is then obvious to check that those $(\psi_i)_{i\in I}$ solve the problem.

□

2.5 A short introduction to distribution theory

Let us first describe the sequential topology of $\mathcal{D}(\Omega)$. A sequence $(\varphi_n)_n \subset \mathcal{D}(\Omega)$ is said to be convergent towards some $\varphi \in \mathcal{D}(\Omega)$ if there is a compact $K \subset \mathbb{R}^d$ which contains the support of φ and of all the functions φ_n and if for any multi-index $\alpha \in \mathbb{N}^d$, the sequence $(\partial^\alpha \varphi_n)_n$ uniformly converges towards $\partial^\alpha \varphi$.

Definition II.2.39 (Distributions). *A linear map $T : \mathcal{D}(\Omega) \to \mathbb{R}$ is called a distribution if it is continuous in the sense that $T(\varphi_n) \xrightarrow[n\to\infty]{} T(\varphi)$ for any sequence $(\varphi_n)_n$ converging towards φ in $\mathcal{D}(\Omega)$.*

The set of distributions is denoted by $\mathcal{D}'(\Omega)$.

Even though $\mathcal{D}(\Omega)$ is not a Banach space, by similarity with the usual duality theory, we also adopt the notation

$$\langle T, \varphi\rangle_{\mathcal{D}',\mathcal{D}} = T(\varphi).$$

Definition II.2.40 (Convergence of distributions). *A sequence of distributions $(T_n)_n \subset \mathcal{D}'(\Omega)$ is said to converge towards a distribution $T \in \mathcal{D}'(\Omega)$ if for any $\varphi \in \mathcal{D}(\Omega)$ we have*

$$\langle T_n, \varphi\rangle_{\mathcal{D}',\mathcal{D}} \xrightarrow[n\to\infty]{} \langle T, \varphi\rangle_{\mathcal{D}',\mathcal{D}}.$$

Notice that the limit of a sequence of distributions $(T_n)_n$, if it exists, is necessarily unique.

Definition II.2.41 (Derivatives of distributions). *For any distribution $T \in \mathcal{D}'(\Omega)$ and any multi-index $\alpha \in \mathbb{N}^d$, the derivative of T in the distribution sense is the distribution $\partial^\alpha T$ defined by*

$$\langle \partial^\alpha T, \varphi\rangle_{\mathcal{D}',\mathcal{D}} = (-1)^{|\alpha|}\langle T, \partial^\alpha \varphi\rangle_{\mathcal{D}',\mathcal{D}}, \forall \varphi \in \mathcal{D}(\Omega).$$

We can associate the distribution $T_f \in \mathcal{D}'(\Omega)$ defined by,

$$\langle T_f, \varphi\rangle_{\mathcal{D}',\mathcal{D}} = \int_\Omega f\varphi\, dx$$

with any $f \in L^1_{loc}(\Omega)$. The following property is fundamental.

Proposition II.2.42. *The map*

$$\mathcal{T} : f \in L^1_{loc}(\Omega) \mapsto T_f \in \mathcal{D}'(\Omega)$$

is injective and sequentially continuous.

Proof.

Let $f, g \in L^1_{loc}(\Omega)$ such that $T_f = T_g$. Let us show that $f = g$ almost everywhere.

Let ω be any bounded open subset of Ω. We set $h = \operatorname{sgn}(f - g) \in L^\infty(\omega)$. By using Theorem II.2.29 we can find a sequence $\varphi_n \in \mathcal{D}(\omega)$ such that $(\varphi_n)_n$ converges to h in $L^\infty(\omega)$ weak-$\star$. By extending φ_n by zero on Ω, we see that $\varphi_n \in \mathcal{D}(\Omega)$ and therefore by assumption we have $\langle T_f, \varphi_n\rangle_{\mathcal{D}',\mathcal{D}} = \langle T_g, \varphi_n\rangle_{\mathcal{D}',\mathcal{D}}$; that is,

$$0 = \int_\Omega (f - g)\varphi_n\, dx = \int_\omega (f - g)\varphi_n\, dx.$$

Since $f - g \in L^1(\omega)$ and $(\varphi_n)_n$ converges in $L^\infty(\omega)$ weak-$\star$, we can pass to the limit in this formula and finally obtain

$$0 = \int_\omega (f - g)\operatorname{sgn}(f - g)\, dx = \int_\omega |f - g|\, dx.$$

It follows that $f = g$ almost everywhere in ω. This is true for any such ω, thus we have $f = g$ in Ω.

Let $(f_n)_n \subset L^1_{loc}(\Omega)$ which converges towards some $f \in L^1_{loc}(\Omega)$. For any $\varphi \in \mathcal{D}(\Omega)$, the sequence $(\varphi f_n)_n$ converges to φf in $L^1(\Omega)$ because φ is compactly supported. This implies that $\langle T_{f_n}, \varphi\rangle_{\mathcal{D}',\mathcal{D}} \xrightarrow[n\to\infty]{} \langle T_f, \varphi\rangle_{\mathcal{D}',\mathcal{D}}$, and the claim is proved.

□

Thanks to the previous result we see that the map $\mathcal{T}$ let us identify $L^1_{loc}(\Omega)$ to a subspace of $\mathcal{D}'(\Omega)$. By abuse of notation we say that $L^1_{loc}(\Omega) \subset \mathcal{D}'(\Omega)$ and we systematically identify f and the distribution T_f. Reciprocally, if a distribution $T \in \mathcal{D}'(\Omega)$ is such that $T = T_f$ for some $f \in L^1_{loc}(\Omega)$ we say that $T \in L^1_{loc}(\Omega)$.

Note also that, as soon as f is a smooth enough function, a simple integration by parts shows that we have

$$\partial^\alpha(T_f) = T_{\partial^\alpha f}, \text{in } \mathcal{D}'(\Omega),$$

so that the derivative in the distribution sense coincides with the derivative in the usual sense.

It is also fundamental to recall that the convergence of functions in the distribution sense is weaker than all the weak and weak-$\star$ convergences that we defined above. The precise result, whose proof is straightforward, is the following.

Proposition II.2.43. • *Let $1 \leq p < +\infty$, and $(f_n)_n$ be a sequence in $L^p(\Omega)$ which converges weakly towards $f \in L^p(\Omega)$. Then we have*

$$f_n \xrightarrow[n\to\infty]{} f, \ \text{in } \mathcal{D}'(\Omega).$$

• *Let $(f_n)_n$ be a sequence in $L^\infty(\Omega)$ which converges weakly-$\star$ towards $f \in L^\infty(\Omega)$. Then we have*

$$f_n \xrightarrow[n\to\infty]{} f, \ \text{in } \mathcal{D}'(\Omega).$$

We conclude the presentation of the distribution theory with the following useful lemma. Despite its very simple statement, the proof is not so straightforward.

Lemma II.2.44. *Let Ω be a connected open set of $\mathbb{R}^d$ and let $T \in \mathcal{D}'(\Omega)$ be a distribution such that $\nabla T = 0$ (in other words $\partial T/\partial x_i = 0$ in $\mathcal{D}'(\Omega)$ for all i). Then, T is constant; that is, there exists some $\alpha \in \mathbb{R}$ such that*

$$T = \alpha.$$

Proof.

- Let us start with the case where Ω is the cube $]0,1[^d$. We fix a function $\theta \in \mathcal{D}(]0,1[)$ to be nonnegative with integral equal to 1. Now let $\varphi \in \mathcal{D}(\Omega)$. We then denote

$$m_i(\varphi)(x_{i+1},\dots,x_d) = \int_0^1 \cdots \int_0^1 \varphi(u_1,\dots,u_i,x_{i+1},\dots,x_d)\,du_1\dots du_i.$$

We set

$$\Phi_1(x_1,\dots,x_d) = \int_0^{x_1} \varphi(t,x_2,\dots,x_d)\,dt - m_1(\varphi)(x_2,\dots,x_d)\int_0^{x_1}\theta(t)\,dt.$$

It is clear that Φ_1 is regular and, by choice of θ, this function has compact support in Ω. By hypothesis we have

$$\begin{aligned} 0 = \left\langle \frac{\partial T}{\partial x_1}, \Phi_1 \right\rangle_{\mathcal{D}',\mathcal{D}} &= -\left\langle T, \frac{\partial \Phi_1}{\partial x_1} \right\rangle_{\mathcal{D}',\mathcal{D}} \\ &= -\langle T, \varphi - m_1(\varphi)(x_2,\dots,x_d)\theta(x_1)\rangle_{\mathcal{D}',\mathcal{D}}. \end{aligned}$$

We have therefore shown that for any $\varphi \in \mathcal{D}(\Omega)$, we have

$$\langle T, \varphi\rangle_{\mathcal{D}',\mathcal{D}} = \langle T, m_1(\varphi)\theta(x_1)\rangle_{\mathcal{D}',\mathcal{D}}.$$

We now set

$$\begin{aligned}\Phi_2(x_1,\dots,x_d) = \theta(x_1)\int_0^{x_2} m_1(\varphi)(t,x_3,\dots,x_d)\,dt\\ - \theta(x_1)m_2(\varphi)(x_3,\dots,x_d)\int_0^{x_2}\theta(t)\,dt.\end{aligned}$$

By using the fact that $\partial T/\partial x_2 = 0$ for this test function (which belongs indeed to $\mathcal{D}(\Omega)$), we find

$$\begin{aligned}\langle T, \varphi\rangle_{\mathcal{D}',\mathcal{D}} &= \langle T, \theta(x_1)m_1(\varphi)(x_2,\dots,x_d)\rangle_{\mathcal{D}',\mathcal{D}}\\ &= \langle T, \theta(x_1)\theta(x_2)m_2(\varphi)(x_3,\dots,x_d)\rangle_{\mathcal{D}',\mathcal{D}}.\end{aligned}$$

Hence, by induction we obtain that

$$\langle T, \varphi\rangle_{\mathcal{D}',\mathcal{D}} = \langle T, \theta(x_1)\cdots\theta(x_d)m_d(\varphi)\rangle_{\mathcal{D}',\mathcal{D}}.$$

However, $m_d(\varphi)$ is a constant which is simply the integral of φ on Ω. If we define

$$\alpha = \langle T, \theta(x_1)\cdots\theta(x_d)\rangle_{\mathcal{D}',\mathcal{D}},$$

then we obtain

$$\langle T, \varphi\rangle_{\mathcal{D}',\mathcal{D}} = \alpha m_d(\varphi) = \int_\Omega \alpha\varphi\,dx_1\cdots dx_n,$$

which proves the result in the case of the unique cube. It is clear that by translations and homothety this proves the result for all the cubes.

- The case of any connected open set:
  We start by covering Ω with a locally finite family $(\omega_i)_i$ of open cubes. For all i, the distribution T restricted to ω_i has zero gradient in $\mathcal{D}'(\omega_i)$ and is therefore constant on ω_i. In other words, there exists some α_i such that for all φ with support in ω_i we have

$$\langle T, \varphi\rangle_{\mathcal{D}',\mathcal{D}} = \int_{\omega_i} \alpha_i\varphi(x)\,dx = \alpha_i\int_\Omega \varphi(x)\,dx.$$

We now consider a locally finite $\mathcal{C}^\infty$ partition of unity $(\psi_i)_i$ (see Lemma II.2.38) associated with the covering of Ω under consideration. Let $\varphi \in \mathcal{D}(\Omega)$, then since the support of φ is compact, it is included in a finite union of the open sets of the family $(\omega_i)_i$. We therefore obtain the following equality

$$\varphi = \sum_i \varphi\psi_i,$$

the summation being, in fact, finite. Hence, we have

$$\langle T, \varphi \rangle_{\mathcal{D}',\mathcal{D}} = \sum_i \langle T, \varphi\psi_i \rangle_{\mathcal{D}',\mathcal{D}}.$$

However, the functions $\varphi\psi_i$ have support in the cube ω_i, such that we obtain

$$\langle T, \varphi \rangle_{\mathcal{D}',\mathcal{D}} = \sum_i \alpha_i \left(\int_\Omega \varphi(x)\psi_i(x)\, dx \right).$$

The summation on i is in reality finite, therefore we have

$$\langle T, \varphi \rangle_{\mathcal{D}',\mathcal{D}} = \int_\Omega \varphi(x) \left(\sum_i \alpha_i \psi_i(x) \right) dx.$$

The fact that this is valid for all φ shows that the distribution T coincides with the function of class $\mathcal{C}^\infty$ defined by

$$T(x) = \sum_i \alpha_i \psi_i(x).$$

However, by hypothesis, this function has a gradient (in the classic sense) which is 0 on Ω. Since Ω is connected, this shows that the function T is indeed constant on Ω.

□

2.6 Lipschitz continuous functions

This class of function is important in the sequel because we mainly study the equations of fluid mechanics in a domain whose boundary has a Lipschitz regularity (including, in particular, polygonal/polyhedral domains). That is why we need to state here some basic results concerning those functions.

We first give a very simple extension theorem in this class.

Proposition II.2.45 (McShane–Whitney extension). *Let $A \subset \mathbb{R}^d$ be any nonempty set and $f : A \to \mathbb{R}$ be a Lipschitz continuous function on A. There exists a Lipschitz continuous function $F : \mathbb{R}^d \to \mathbb{R}$ such that $F_{|A} = f$ and $\mathrm{Lip}(F) = \mathrm{Lip}(f)$.*

Proof.

If we set $L = \mathrm{Lip}(f)$, it is a simple exercise to check that

$$F(x) = \inf_{y \in A} \left(f(y) + L|x - y| \right),$$

satisfies the required property.

□

In the sequel of this book (in particular in Chapter III), we often use the fact that Lipschitz continuous functions are almost-differentiable functions. The precise result, whose proof is given in [68], for instance, is the following.

Theorem II.2.46 (Rademacher). *Any locally Lipschitz continuous function f defined on an open set of $\mathbb{R}^d$ is differentiable (in the classic sense) almost everywhere.*

We need to analyse carefully the action of mollifying operators on Lipschitz continuous maps. We suppose given a mollifying kernel η as in Definition II.2.23 and we recall that, for any $\varepsilon > 0$, $f \star \eta_\varepsilon$ is defined in (II.5).

Proposition II.2.47. *Assume that f is Lipschitz continuous on $\mathbb{R}^d$, then*

1. *For any $\varepsilon > 0$, we have $\mathrm{Lip}(f \star \eta_\varepsilon) \leq \mathrm{Lip}(f)$.*
2. *$f \star \eta_\varepsilon$ uniformly converges in $\mathbb{R}^d$ towards f as $\varepsilon \to 0$.*
3. *For any $x \in \mathbb{R}^d$ such that f is differentiable at x, we have*

$$\nabla(f \star \eta_\varepsilon)(x) \xrightarrow[\varepsilon \to 0]{} \nabla f(x).$$

Proof.

1. The regularity of $f \star \eta_\varepsilon$ comes from that of kernel η. The estimate of the Lipschitz seminorm is given by the following simple computation

$$\begin{aligned} |f \star \eta_\varepsilon(x) - f \star \eta_\varepsilon(y)| &\leq \int_B |f(x - \varepsilon z) - f(y - \varepsilon z)| \eta(z)\, dz \\ &\leq \mathrm{Lip}(f)|x - y|, \quad \forall x, y \in \mathbb{R}^d. \end{aligned}$$

2. Since $\int_{\mathbb{R}^d} \eta(z)\, dz = 1$, we have

$$\begin{aligned} |f \star \eta_\varepsilon(x) - f(x)| &= \left| \int_B (f(x - \varepsilon z) - f(x)) \eta(z)\, dz \right| \\ &\leq \int_B |f(x - \varepsilon z) - f(x)| \eta(z)\, dz \leq \varepsilon\, \mathrm{Lip}(f) \int_B |z| \eta(z)\, dz, \end{aligned}$$

   and the claim is proved.
3. Let x be a point such that f is differentiable at x. Then, there exists $\tau : \mathbb{R}^d \to \mathbb{R}$ such that $\lim_{h \to 0} \tau(h) = 0$ and

$$|f(x + h) - f(x) - \nabla f(x) \cdot h| \leq |h| \tau(h), \forall h \in \mathbb{R}^d.$$

   Let $i \in \{1, \ldots, d\}$. From (II.5), we see that

$$\partial_{x_i}(f \star \eta_\varepsilon)(x) = \frac{1}{\varepsilon} \int_B f(x - \varepsilon z) \partial_{z_i} \eta(z)\, dz.$$

Moreover, by integration by parts, we observe that

$$\int_B \partial_{z_i}\eta(z)\,dz = 0, \text{ and } \int_B z\partial_{z_i}\eta(z)\,dz = -e_i,$$

so that we can write

$$\begin{aligned}&\partial_{x_i}(f \star \eta_\varepsilon)(x) - \partial_{x_i} f(x)\\ =&\frac{1}{\varepsilon}\left(\int_B \Big[f(x-\varepsilon z) - f(x) + \nabla f(x).(\varepsilon z)\Big]\partial_{z_i}\eta(z)\,dz\right),\end{aligned}$$

and then we can conclude, by using the dominated convergence theorem, that

$$|\partial_{x_i}(f \star \eta_\varepsilon)(x) - \partial_{x_i} f(x)| \le \int_B |z|\tau(\varepsilon z)\partial_{z_i}\eta(z)\,dz \xrightarrow[\varepsilon\to 0]{} 0.$$

□

3 Basic compactness results

As we show below, highlighting of the compactness properties of certain sets, or of certain maps, is often a crucial step in proving the existence of solutions to certain nonlinear partial differential equations. In this section we summarise essential definitions and results which are used later.

3.1 Compact sets in function spaces

Ascoli's theorem is one of the fundamental tools of nonlinear analysis. It allows relatively compact sets in $C^0(E, F)$ to be simply characterised, where E is a compact space. This result is central because it underpins the majority of the compactness results used later in this text. A very classic proof of this theorem can be found, for example, in [99].

Theorem II.3.1 (Ascoli). *Let E be a compact metric space, and let F be a metric space. Let $\mathcal{C}^0(E, F)$ be the metric space formed from the continuous functions of E in F equipped with the uniform distance:*

$$d(f, g) = \sup_{x\in E} d(f(x), g(x)).$$

Let $\mathscr{K}$ be a subset of $\mathcal{C}^0(E, F)$. We assume that:

1. For any $x \in E$, the subset of F, $\mathscr{K}(x)$ defined by:

$$\mathscr{K}(x) = \{f(x), f \in \mathscr{K}\}$$

is relatively compact in F.

2. *The set $\mathscr{K}$ is equicontinuous; that is for all $x \in E$ and all $\varepsilon > 0$, there exists an $\eta > 0$ such that*

$$d(f(x), f(y)) < \varepsilon,\ \forall y \in E,\ \text{such that } d(x,y) < \eta,\ \forall f \in \mathscr{K}.$$

Then, $\mathscr{K}$ is relatively compact (i.e., has compact adherence) in $\mathcal{C}^0(E, F)$.

Unfortunately, it is very rare in the analysis of partial differential equations that we work in the set of continuous functions on a compact space. The following theorem, which follows from Ascoli's theorem, gives us a compactness criterion similar to that of Ascoli for the bounded subsets in $L^p(\Omega)$ spaces.

Theorem II.3.2 (Kolmogorov). *Let Ω be any open set of $\mathbb{R}^d$, and let $\mathcal{F}$ be a bounded subset of $L^p(\Omega)$, with $1 \leq p < +\infty$. We assume that*

1. *For all $\varepsilon > 0$, and for all bounded open sets ω such that $\overline{\omega} \subset \Omega$, there exists an $\alpha > 0$, with $\alpha < d(\omega, \mathbb{R}^d \backslash \Omega)$ such that*

$$\|\tau_h f - f\|_{L^p(\omega)} \leq \varepsilon,\ \forall f \in \mathcal{F}, \forall h \in \mathbb{R}^d, |h| \leq \alpha. \tag{II.11}$$

2. *For all $\varepsilon > 0$, there exists a bounded open set ω, such that $\overline{\omega} \subset \Omega$ and such that*

$$\|f\|_{L^p(\Omega \backslash \omega)} \leq \varepsilon, \forall f \in \mathcal{F}. \tag{II.12}$$

Then, $\mathcal{F}$ is relatively compact in $L^p(\Omega)$.

In this theorem $\tau_h f$ denotes the translated function defined by

$$\tau_h f(x) = f(x+h).$$

The first condition looks like the equicontinuity condition of Ascoli's theorem; the second tells us that the functions of $\mathcal{F}$ must be "uniformly small" in the L^p norm near the boundary of Ω and near infinity.

Proof.

- We set $\varepsilon > 0$ and choose an open set ω satisfying (II.12). Let $\alpha > 0$ with $\alpha < d(\omega, \mathbb{R}^d \setminus \Omega)$ satisfying (II.11). We observe that, for any $f \in \mathcal{F}$ and any $x \in \omega$ we have

$$|\bar{f} \star \eta_\alpha(x) - f(x)| \leq \int_B |\bar{f}(x - \alpha z) - f(x)| \eta(z)\, dz.$$

Using the Jensen inequality, integrating on ω and using Fubini's theorem leads to

$$\|\bar{f} \star \eta_\alpha - f\|_{L^p(\omega)} \leq \int_B \eta(z) \|\tau_{-\alpha z} f - f\|_{L^p(\omega)}\, dz.$$

Note that we have used here the fact that, by assumption on α, $x-\alpha z$ belongs to Ω as soon as $x\in\omega$ and $z\in B$ so that $\bar f(x-\alpha z)=f(x-\alpha z)$. For any $z\in B$ we have $|x-\alpha z|\leq\alpha$, therefore we deduce from (II.11) that

$$\|\bar f\star\eta_\alpha - f\|_{L^p(\omega)}\leq\varepsilon,\ \forall f\in\mathcal{F}. \tag{II.13}$$

- Let us define $\mathcal{F}_\alpha=\{\bar f\star\eta_\alpha,\ f\in\mathcal{F}\}$ and $F_\omega=\{f_{|\omega},\ \ f\in\mathcal{F}\}$. Each function in $\mathcal{F}_\alpha$ is continuous on the compact $\overline{\omega}$ and satisfies

$$\|\nabla(\bar f\star\eta_\alpha)\|_{L^\infty}\leq\frac{C}{\alpha^{1+d/p}}\|f\|_{L^p}\leq\frac{C'}{\alpha^{1+d/p}},$$

because $\mathcal{F}$ is a bounded set of $L^p(\Omega)$.
The number $\alpha>0$ being fixed, we have shown that $\mathcal{F}_\alpha$ satisfies the assumptions of the Ascoli theorem. It follows that $\mathcal{F}_\alpha$ is relatively compact in $\mathcal{C}^0(\overline{\omega})$ and thus in $L^p(\omega)$ by continuity of the embedding $\mathcal{C}^0(\overline{\omega})\subset L^p(\omega)$.
- As a consequence, there exist a finite number of balls in $L^p(\omega)$ with radius ε which cover $\mathcal{F}_\alpha$ and with (II.13) we deduce that there exist a finite number of balls in $L^p(\omega)$ with radius 2ε which cover $\mathcal{F}_{|\omega}$ in $L^p(\omega)$. We denote such a covering as $(B_{L^p(\omega)}(g_i,2\varepsilon))_{1\leq i\leq N}$.
- We finally prove that the balls $(B_{L^p(\Omega)}(\overline{g_i},3\varepsilon))_{1\leq i\leq N}$ actually cover $\mathcal{F}$. Indeed, for any $f\in\mathcal{F}$ there is a $1\leq i\leq N$ such that $f_{|\omega}\in B_{L^p(\omega)}(g_i,2\varepsilon)$ and thus

$$\|f-\overline{g_i}\|^p_{L^p(\Omega)}=\|f\|^p_{L^p(\Omega\setminus\omega)}+\|f-g_i\|^p_{L^p(\omega)}\leq\varepsilon^p+(2\varepsilon)^p\leq(3\varepsilon)^p,$$

by using (II.12).
The claim is proved because, for any $\varepsilon>0$, we have built a finite covering of $\mathcal{F}$ in $L^p(\Omega)$ made of balls of radius 3ε.

□

3.2 Compact maps

Definition II.3.3. *Let E and F be two Banach spaces. We say that a map S from E into F is compact if the image of any bounded subset of E by S is a relatively compact set of F; that is, it is a set having compact closure in F.*

Of course, any compact linear function is continuous because compactness implies that it is bounded in the neighborhood of 0. We generally use the compactness properties of maps in the following form.

Let $(u_n)_n$ be a bounded sequence of points in E then, if S is compact, there exists a subsequence $(u_{n_k})_k$ such that $(Su_{n_k})_k$ converges in F.

In particular, we have the following result.

Proposition II.3.4. *Let $(u_n)_n$ be a sequence of points in E which weakly converges towards u in E, and let $S : E \to F$ be a compact linear map, then the sequence $(Su_n)_n$ strongly converges towards Su in F.*

Proof.

We note first that the sequence $(Su_n)_n$ weakly converges towards Su. Indeed, for $f \in F'$,

$$\langle f, Su_n\rangle_{F',F} = \langle {}^tSf, u_n\rangle_{E',E} \xrightarrow[n\to\infty]{} \langle {}^tSf, u\rangle_{E',E} = \langle f, Su\rangle_{F',F}.$$

Moreover, since $(u_n)_n$ is weakly convergent, it is a bounded sequence (Corollary II.2.8). Hence $(u_n)_n$ belongs to a bounded subset B of E. Inasmuch as S is compact, $\overline{S(B)}$ is compact, and therefore the sequence $(Su_n)_n$ lies in a compact space of F.

However, in a compact metric space, a sequence converges if and only if it has a unique accumulation point. Therefore let $v = \lim_{k\to\infty} Su_{n_k}$ be an accumulation point of $(Su_n)_n$ in F. As we have seen above, the sequence $(Su_{n_k})_k$ weakly converges towards Su. From the uniqueness of the weak limit, this means $v = Su$. Hence, Su is the unique accumulation point of the sequence $(Su_n)_n$, which is therefore convergent.

□

In particular, this makes it possible to recover strong convergence from weak convergence, but in a larger space than the initial space. Indeed, if a space E is embedded into a space F with a compact embedding (we say that E is embedded in a compact way into F), then any sequence of elements of E which is weakly convergent in E, is strongly convergent in F.

On the other hand, the compactness of linear functions is a stable concept by composition and by passing to the adjoint. More precisely, we have the following results.

Lemma II.3.5. *Let E, F, G be three Banach spaces, let S be a continuous linear map from E to F, and let T be a continuous linear map from F to G. If S is compact* or *if T is compact, then $T \circ S$ is compact.*

Proof.

This is essentially a consequence of the fact that the image of a bounded (resp., compact) set by a continuous linear map is another bounded (resp., compact) set.

□

Lemma II.3.6. *Let E and F be two Banach spaces and let S be a compact linear map from E into F. Then the adjoint map ${}^tS$ from F' into E' is compact.*

Proof.

Let $(f_n)_n$ be a bounded sequence of F'. From the definition of the adjoint of S, we have for all $u \in B_E(0,1)$,

$$\langle {}^tSf_n, u\rangle_{E',E} = \langle f_n, Su\rangle_{F',F}.$$

Since S is compact, the set

$$\mathcal{K} = \overline{S\,(B(0,1)_E)},$$

is a compact subset of F. The sequence $(f_n)_n$, being bounded in F', is therefore also bounded in $\mathcal{C}^0(\mathcal{K},\mathbb{R})$. Moreover, for all points $v_1, v_2 \in \mathcal{K}$ we have

$$|\langle f_n, v_1 - v_2\rangle_{F',F}| \leq \|f_n\|_{F'}\|v_1 - v_2\|_F,$$

which, since the sequence $(f_n)_n$ is bounded in F', proves that $(f_n)_n$ is an equicontinuous family on $\mathcal{K}$.

Hence, from Ascoli's theorem (Theorem II.3.1), there exists an extracted sequence $(f_{n_k})_k$ which converges uniformly on $\mathcal{K}$ towards a continuous function f from $\mathcal{K}$ to $\mathbb{R}$.

Therefore, by transposition, for all $u \in B_E(0,1)$ we have

$$\langle {}^tSf_{n_k}, u\rangle_{E',E} = \langle f_{n_k}, Su\rangle_{F',F} \xrightarrow[k\to\infty]{} f(Su),$$

and, moreover, the convergence is uniform in u on $B_E(0,1)$. By homogeneity (i.e., because of the linearity of the functions ${}^tSf_{n_k}$) we deduce that for all $u \in E$, the sequence $(\langle {}^tSf_{n_k}, u\rangle_{E',E})_k$ converges and, furthermore, the convergence is uniform on all the bounded sets of E. The functions ${}^tSf_{n_k}$ are linear and continuous, therefore the limit obtained is necessarily linear and continuous. All this demonstrates that convergence does indeed occur in E' (for which the strong topology is simply the uniform convergence on bounded sets).

□

Let us apply this result in the case where there is a continuous embedding of one Banach space into another.

Proposition II.3.7. *Let E and F be two Banach spaces. We assume that E is continuously embedded into F and that the range of E is dense in F (we say incorrectly that E is dense in F); then the map*

$$T : f \in F' \mapsto T_f \in E',$$

defined by

$$\langle T_f, u\rangle_{E',E} = \langle f, u\rangle_{F',F},\ \ \forall u \in E,$$

is an embedding (said to be canonical *with respect to the considered embedding from E into F) from F' into E'.*

Moreover, if the embedding of E into F is compact, then T is a compact embedding. Finally, if E is reflexive then the range of T is dense in E'.

Proof.

Since the embedding of E into F is continuous, there exists a constant $C > 0$ such that for any $u \in E$ we have $\|u\|_F \leq C\|u\|_E$. Hence, for all $f \in F'$, the function T_f is indeed continuous and linear on E, that is, an element of E'.

Let us prove the injectivity of the function $f \mapsto T_f$. Let $f \in F'$ such that $T_f = 0$. We therefore have $\langle f, u\rangle_{F',F} = 0$ for all u in E, but, since E is dense in F, we can deduce that $f = 0$.

If the embedding from E into F is compact, then the compactness of T results directly from Lemma II.3.6.

Let us now assume that E is reflexive. We need to show that $T(F')$ is dense in E'. To do this, we use Proposition II.2.2. Any continuous linear functional on E' is of the form $f \mapsto \langle f, u\rangle_{E',E}$ for a certain u in E. Let us suppose that one such functional cancels on $T(F')$ and let us show that it cancels on all of E'. To say that this linear functional cancels on $T(F')$ means that

$$\langle f, u\rangle_{F',F} = 0, \ \forall f \in F'.$$

Proposition II.2.1 then shows that $\|u\|_F = 0$ and hence $u = 0$, which proves the result.

□

The Riesz theorem allows this result to be specified in the Hilbertian case.

Corollary II.3.8. *Let V and H be two Hilbert spaces such that V embeds densely into H. According to the Riesz theorem we can identify H and its dual via its scalar product. We then have a double dense embedding*

$$V \subset H \subset V',$$

the second embedding being defined by

$$f \in H \mapsto T_f \in V', \ \textit{with } \langle T_f, v\rangle_{V',V} = (f, v)_H, \ \forall v \in V.$$

If the embedding of V into H is compact, then the embedding of H into V' is also compact.

For obvious reasons, in the situation described by the corollary, the space H is called the *pivot space*. Furthermore, since T is injective, we systematically identify $f \in H$ with its image $T_f \in V'$ so that the duality (V', V) can be expressed, using the scalar product of H, by

$$\langle f, v\rangle_{V',V} = (f, v)_H, \ \forall f \in H, \ \forall v \in V.$$

3.3 The Schauder fixed-point theorem

For solving nonlinear partial differential equations, one can often use a rather classic fixed-point technique. In this book, we follow this strategy for studying the steady Navier–Stokes equations (Section 3 of Chapter V) and for studying the unsteady Navier–Stokes equations for a nonhomogeneous flow in Chapter VI.

The key of this technique lies in the following theorem for which the rather tricky proof can be found, for example, in [114]. It relies on the concept of *topological degree* which is beyond the scope and objectives of this book.

Theorem II.3.9 (Schauder fixed-point theorem). *Let E be a Banach space and let C be a convex compact set in E. If T is a continuous (nonlinear) function from C into C, then it has at least one fixed-point in C.*

We note that this theorem says nothing about the uniqueness of the fixed-point and that in general uniqueness does not hold (consider the identity function). Moreover, the fact that the function T maps the set C into itself is, of course, a crucial fact.

In the particular case where $E = \mathbb{R}$, we recover an elementary result which says that a continuous function from $\mathbb{R}$ into $\mathbb{R}$ which maps a compact interval $[a, b]$ onto itself contains a fixed-point in this interval.

This result also exists in a slightly different form which is given below.

Theorem II.3.10. *Let E be a Banach space and let C be a convex, closed and bounded region of E. If T is a compact and continuous (nonlinear) function from C into C, then it has at least one fixed-point in C.*

In the finite-dimensional framework, this theorem is known as the Brouwer theorem and is equivalent to the following result.

Proposition II.3.11. *Let P be a continuous function from $\mathbb{R}^N$ to $\mathbb{R}^N$, such that there exists a $\rho > 0$ satisfying*

$$\xi \cdot P(\xi) \geq 0, \forall \xi \in \mathbb{R}^N, |\xi| = \rho.$$

Then, there exists $\overline{\xi} \in \mathbb{R}^N$, $|\overline{\xi}| \leq \rho$ such that $P(\overline{\xi}) = 0$.

Proof.

Suppose, by contradiction, that for all $\xi \in \overline{B(0, \rho)}$, $P(\xi) \neq 0$; then the continuous map $Q : \xi \in \mathbb{R}^N \mapsto -(\rho/|P(\xi)|)P(\xi)$ maps the ball $\overline{B(0, \rho)}$ which is compact and convex into itself. Then by application of the Brouwer/Schauder fixed-point theorem, there exists $\xi^\star \in \overline{B(0, \rho)}$ such that

$$\xi^\star = Q(\xi^\star). \tag{II.14}$$

Necessarily, we have $|\xi^\star| = \rho$. By taking the scalar product of each term of (II.14) by $\xi^\star$, one obtains:

$$\rho^2 = -\frac{\rho}{|P(\xi^\star)|}(\xi^\star \cdot P(\xi^\star)).$$

Therefore $\xi^\star \cdot P(\xi^\star) < 0$, which is in contradiction with the hypothesis.

□

4 Functions of one real variable

In this section, we primarily review the links that exist between the concepts of weak derivatives and derivatives in the usual sense. We do this in a limited but sufficient way, for the case that concerns us, for functions of one real variable.

We conclude the section by reviewing Gronwall-type inequalities which are a useful tool for obtaining a priori estimates for solutions of evolution partial differential equations.

4.1 Differentiation and antiderivatives

Let $[a,b]$ be a compact interval of $\mathbb{R}$. We recall that $W^{1,1}(]a,b[)$ is the set of functions of $L^1(]a,b[)$ for which the derivative in the sense of distributions is a function of $L^1(]a,b[)$ (see Chapter III for a more complete study of Sobolev spaces). A fundamental question that we can ask for such a function is if it can be differentiated in the usual sense and if we can write the fundamental theorem of calculus

$$f(y) = f(x) + \int_x^y f'(t)\,dt,\ \forall x, y \in [a,b].$$

Here, we recall some results that concern this question. This material is useful in the sequel of the book in order to justify the validity of the time evolution of the kinetic energy for weak solutions of the Navier–Stokes equations (see in particular Section 1.4 of Chapter V).

Lemma II.4.1. *Let $g \in L^1(]a,b[)$ and $C \in \mathbb{R}$. We consider the function f defined by*

$$f(t) = C + \int_a^t g(s)\,ds.$$

Then, f is continuous on $[a,b]$. Moreover, $f \in W^{1,1}(]a,b[)$ and its derivative in the sense of distributions is g.

Proof.

For all $t_0 \in [a,b[$ and $h > 0$ but sufficiently small, we have

$$f(t_0+h)-f(t_0)=\int_{t_0}^{t_0+h} g(s)\,ds=\int_a^b g(s)1_{[t_0,t_0+h]}(s)\,ds \xrightarrow[h\to 0]{} 0,$$

from Lebesgue's dominated convergence theorem. A similar argument with $h<0$ shows the continuity of f.

We now need to verify that the derivative of f in the sense of distributions is the function g. Let $\varphi \in \mathcal{D}(]a,b[)$; we have

$$-\int_a^b f(t)\varphi'(t)\,dt=-C\int_a^b \varphi'(t)\,dt-\int_a^b\left(\int_a^t g(s)\varphi'(t)\,ds\right)dt.$$

The first term is zero because $\varphi(a)=\varphi(b)=0$ and we apply Fubini's theorem to the second term (the function $(t,s)\mapsto \varphi'(t)g(s)$ is integrable with respect to the two variables). It follows that

$$\begin{aligned}
-\int_a^b f(t)\varphi'(t)\,dt &= -\int_a^b\left(\int_a^b 1_{[a\le s\le t]}g(s)\varphi'(t)\,ds\right)\,dt\\
&= -\int_a^b\left(\int_a^b 1_{[a\le s\le t]}g(s)\varphi'(t)\,dt\right)\,ds\\
&= -\int_a^b\left(g(s)\int_s^b \varphi'(t)\,dt\right)ds=-\int_a^b\left(g(s)(\varphi(b)-\varphi(s))\right)\,ds\\
&= \int_a^b g(s)\varphi(s)\,ds.
\end{aligned}$$

This indeed proves that $g=f'$ in the sense of distributions and therefore $f\in W^{1,1}(]a,b[)$.

$\square$

Corollary II.4.2. *Any function f of $W^{1,1}(]a,b[)$ is equal almost everywhere to a continuous function $\tilde{f}$ on $[a,b]$ and we have for all $x,y\in[a,b]$,*

$$\tilde{f}(y)=\tilde{f}(x)+\int_x^y f'(s)\,ds;$$

in other words, we have for almost every $x,y\in[a,b]$,

$$f(y)=f(x)+\int_x^y f'(s)\,ds.$$

Proof.

Let $f\in W^{1,1}(]a,b[)$. We introduce

$$g(t)=\int_a^t f'(s)\,ds.$$

- According to Lemma II.4.1, the function g is continuous in $W^{1,1}(]a,b[)$ and its derivative in the sense of distributions is f'. Hence, $f-g$ is a function for which the distribution derivative is zero. We then know that there exists a real number C such that $f-g=C$ almost everywhere (see Lemma II.2.44 for a more general case of this result). If we define $\tilde{f}=C+g$, we have shown that f coincides with the continuous function $\tilde{f}$, almost everywhere.
- From the definition of $\tilde{f}$, it is clear that for all $x,y\in[a,b]$, we have

$$\tilde{f}(y)-\tilde{f}(x)=\left(C+\int_a^y f'(s)\,ds\right)-\left(C+\int_a^x f'(s)\,ds\right)=\int_x^y f'(s)\,ds.$$

□

For any point $t_0\in[a,b]$, we denote as $\mathcal{V}_\eta(t_0)$ the set of open neighborhoods ω of t_0 in $[a,b]$ whose Lebesgue measure $|\omega|$ is less than η.

Definition II.4.3 (Lebesgue points). *Let f be a function of $L^1(]a,b[)$ and $t_0\in]a,b[$. We say that t_0 is a Lebesgue point of f if*

$$\sup_{\omega\in\mathcal{V}_\eta(t_0)}\frac{1}{|\omega|}\int_\omega |f(t)-f(t_0)|\,dt \underset{\eta\to 0}{\longrightarrow} 0.$$

With this definition at hand, we have the following result.

Proposition II.4.4. *Let f be a function of $L^1(]a,b[)$ and $t_0\in]a,b[$. If t_0 is a Lebesgue point of f, then any antiderivative of f defined by*

$$F(t)=C+\int_a^t f(s)\,ds,$$

can be differentiated in the classic sense at t_0 and moreover we have

$$F'(t_0)=f(t_0).$$

Proof.

It is sufficient to write

$$\begin{aligned}\left|\frac{F(t_0+h)-F(t_0)}{h}-f(x_0)\right| &= \left|\frac{1}{h}\int_{t_0}^{t_0+h}(f(s)-f(t_0))\,ds\right| \\ &\leq \frac{1}{h}\left|\int_{t_0}^{t_0+h}|f(s)-f(t_0)|\,ds\right|;\end{aligned}$$

this last quantity tends towards 0 when h tends towards 0, by definition of a Lebesgue point.

□

The fundamental theorem in this section is a rather difficult result from measure theory, which we do not prove. However, a proof can be found, for example, in [69] or in [100].

Theorem II.4.5. *If $f \in L^1(]a,b[)$, then almost every point of $]a,b[$ is a Lebesgue point of f.*

The immediate consequence of the preceding two results is that the antiderivative F of any function f in $L^1(]a,b[)$ is differentiable almost everywhere and satisfies $F' = f$ and the fundamental theorem of calculus

$$F(y) = F(x) + \int_x^y F'(t)\,dt = F(x) + \int_x^y f(t)\,dt, \quad \forall x, y \in [a,b].$$

Finally, the following elementary result is useful.

Proposition II.4.6. *Let $f \in L^1(]a,b[)$. Any point of continuity of f is a Lebesgue point of f and, in particular, any antiderivative of f is differentiable at any point t_0 where f is continuous and its derivative is $f(t_0)$.*

Remark II.4.1. We can prove [69] that the functions of $W^{1,1}(]a,b[)$ are none other than the absolutely continuous functions on $]a,b[$.

We conclude this section by the following result and its corollary.

Lemma II.4.7 (Hardy's inequality). *For any $1 < p < +\infty$ and any nonnegative $f \in L^p(]0,+\infty[)$ we have*

$$\int_0^M \left(\frac{1}{x}\int_0^x f(s)\,ds\right)^p dx \leq \left(\frac{p}{p-1}\right)^p \int_0^M f^p(s)\,ds, \ \forall M \in [0,+\infty].$$

Note that a similar inequality does not hold for $p = 1$.

Proof.

We prove the inequality for $M = +\infty$. The general case follows by taking $f_M(s) = 1_{[0,M]}(s)f(s)$.

By density, it is enough to prove this result for functions $f \in \mathcal{C}_c^\infty(]0,+\infty[)$. For such an f, we set $F(x) = (1/x)\int_0^x f(s)\,ds$ and we note that $F = 0$ in the neighborhood of 0 and that $F(x) = C/x$ for some $C \in \mathbb{R}$ and x large enough. In particular, we have $F \in L^p(]0,+\infty[)$. We remark that $d(xF(x))/dx = f(x)$ so that we can integrate by parts as follows

$$\begin{aligned}
\int_0^{+\infty} F(x)^p\,dx &= \int_0^{+\infty} (xF(x))^p \frac{1}{x^p}\,dx \\
&= \frac{p}{p-1}\int_0^{+\infty} (xF(x))^{p-1} f(x).\frac{1}{x^{p-1}}\,dx \\
&= \frac{p}{p-1}\int_0^{+\infty} F(x)^{p-1} f(x)\,dx.
\end{aligned}$$

We conclude the proof by using the Hölder inequality

$$\int_0^{+\infty} F(x)^p \, dx \leq \frac{p}{p-1} \|F\|_{L^p}^{p-1} \|f\|_{L^p}.$$

□

We immediately deduce from this inequality the following result which is important in the sequel (see in particular the multidimensional version in Proposition III.2.40).

Corollary II.4.8. *Let $f \in W^{1,p}(]a,b[) \subset W^{1,1}(]a,b[)$ with $1 < p < +\infty$ and such that $f(a) = 0$. Then, the function $g : x \mapsto f(x)/(x-a)$ belongs to $L^p(]a,b[)$ and satisfies*

$$\|g\|_{L^p} \leq C\|f'\|_{L^p}.$$

4.2 Differential inequalities and Gronwall's lemma

The following lemmas, concerning ordinary differential equations and inequalities, are very useful ingredients for studying time-dependent partial differential equations, in particular for proving energy estimates.

Lemma II.4.9. *Let there be two real numbers $\alpha > 0$ and $\beta \geq 0$ and let y be a function in $\mathcal{C}^1([0,+\infty[,\mathbb{R})$ satisfying the differential inequality:*

$$y'(t) + \alpha y(t) \leq \beta, \ \forall t \geq 0.$$

Then we have

$$y(t) \leq y(0)e^{-\alpha t} + \frac{\beta}{\alpha}, \ \forall t \geq 0.$$

Proof.

We multiply the two sides of the differential inequality by $e^{\alpha t}$ and then integrate.

□

The following lemma, known as Gronwall's inequality (even though its proof in the present form is due to Bellman [15]), is central in proving a priori estimates on solutions of (partial) differential equations.

Lemma II.4.10. *Let us consider a function $y \in L^\infty(]0,T[)$, a nonnegative function $g \in L^1(]0,T[)$ and $y_0 \in \mathbb{R}$, such that*

$$y(t) \leq y_0 + \int_0^t g(s)y(s)\, ds, \ \textit{for almost all } t \in]0,T[,$$

we then have

$$y(t) \leq y_0 \exp\left(\int_0^t g(s)\, ds\right), \ \textit{for almost all } t \in]0,T[.$$

Proof.

We set

$$h(t) = y_0 + \int_0^t g(s)y(s)\,ds,$$

that is, the second term of the inequality from the hypothesis. Since y belongs to $L^\infty(]0,T[)$ and g to $L^1(]0,T[)$, the function h lies in $W^{1,1}(]0,T[)$ and is therefore differentiable almost everywhere and its derivative is gy (see Section 4.1). Furthermore, for almost all t, we have

$$h'(t) = g(t)y(t) \leq g(t)h(t),$$

from the hypothesis and because g is nonnegative. Then, if we set $z(t) = h(t)e^{-\int_0^t g}$, we immediately see that z belongs to $W^{1,1}(]0,T[)$ and that

$$z'(t) = \big(h'(t) - g(t)h(t)\big)\exp\left(-\int_0^t g(s)\,ds\right).$$

Therefore, for almost all t, we have $z'(t) \leq 0$. From Corollary II.4.2, this implies that the function z is nonincreasing and therefore we have

$$z(t) \leq z(0) = h(0) = y_0, \forall t \in [0,T],$$

which can be written as

$$h(t) \leq y_0 \exp\left(\int_0^t g(s)\,ds\right),\ \forall t \in [0,T].$$

This proves the claim since, by hypothesis, we have $y \leq h$ almost everywhere.

□

Lemma II.4.11 (Uniform Gronwall lemma [121]). *Let g_1 and g_2 be two nonnegative functions of $L^1_{loc}(\mathbb{R}^+)$ satisfying:*

$$\exists k_1,\ \int_t^{t+1} g_1(s)\,ds \leq k_1, \forall t \in \mathbb{R}^+,$$

$$\exists k_2,\ \int_t^{t+1} g_2(s)\,ds \leq k_2, \forall t \in \mathbb{R}^+.$$

Let y be a function of $\mathcal{C}^1([0,+\infty[,\mathbb{R}^+)$ satisfying

$$y'(t) \leq g_1(t) + g_2(t)y(t), \textit{ for almost all } t \geq 0, \tag{II.15}$$

$$y(0) \leq k_3 \textit{ and } \int_t^{t+1} y(s)\,ds \leq k_3,\ \forall t \geq 0. \tag{II.16}$$

Then y is bounded on $\mathbb{R}^+$ and we have the following upper bound,

$$y(t) \leq (k_1 + k_3)e^{k_2}, \forall t \geq 0.$$

Proof.

We integrate (II.15) between s and t (with $0 \leq s \leq t$) and we get

$$y(t) \leq y(s) + \int_s^t g_1(\tau)\,d\tau + \int_s^t g_2(\tau)y(\tau)\,d\tau.$$

From Lemma II.4.10, we deduce

$$y(t) \leq \left(y(s) + \int_s^t g_1(\tau)\,d\tau\right)\exp\left(\int_s^t g_2(\tau)\,d\tau\right).$$

- For $t \leq 1$, we take $s = 0$ and we directly obtain the result.
- For $t \geq 1$, we take $s \in [t-1, t]$, and we apply (II.16) to obtain

$$y(t) \leq (y(s) + k_1)e^{k_2}.$$

We integrate this last inequality with respect to s between $t-1$ and t (with t fixed), which gives

$$y(t) \leq (k_3 + k_1)e^{k_2}.$$

□

We conclude this section by giving a result of the same type which is useful in the study of some nonlinear equations. This result is similar to the usual comparison theorems between differential inequalities, except that the hypothesis is formulated in an integral form, which is weaker. This explains the necessity of some monotonicity assumption for the nonlinear term.

Lemma II.4.12 (Bihari's inequality [16]). *Let $f : [0, +\infty[\mapsto [0, +\infty[$ be a nondecreasing continuous function such that $f > 0$ on $]0, +\infty[$ and $\int_1^{+\infty} 1/f(x)\,dx < +\infty$. We denote the antiderivative of $-1/f$ which cancels at $+\infty$ as F.*

Let y be a continuous function which is nonnegative on $[0, +\infty[$ and let g be a nonnegative function in $L^1_{loc}([0, +\infty[)$. We assume that there exists a $y_0 > 0$ such that for all $t \geq 0$ we have the inequality

$$y(t) \leq y_0 + \int_0^t g(s)\,ds + \int_0^t f(y(s))\,ds.$$

Then, there exists a unique T^* which satisfies the equation

$$T^* = F\left(y_0 + \int_0^{T^*} g(s)\,ds\right), \tag{II.17}$$

and, for any $T < T^*$ we have

$$\sup_{t\leq T} y(t) \leq F^{-1}\left(F\left(y_0+\int_0^T g(s)\,ds\right)-T\right).$$

Proof.

The existence and uniqueness of T^* satisfying (II.17) arises from the fact that the function F is nonincreasing and tends towards 0 at $+\infty$, and that g is nonnegative. Let us set T such that $0<T<T^*$. For all $t\leq T$, since g is nonnegative, we have

$$y(t)\leq y_0+\int_0^T g(s)\,ds+\int_0^t f(y(s))\,ds. \tag{II.18}$$

Let us denote the right-hand side of this inequality as $z_T(t)$. Since f and y are continuous, the function z_T is of class $\mathcal{C}^1$ and we have $z_T(0)=y_0+\int_0^T g(s)\,ds$ and for all $t<T$

$$z_T'(t)=f(y(t))\leq f(z_T(t)),$$

because f is nondecreasing. We note that z_T is an nondecreasing function and since $y_0>0$, the function z_T does not cancel. Hence, we have

$$\frac{z_T'(t)}{f(z_T(t))}\leq 1, \forall t<T,$$

which, after integration between times 0 and T, gives

$$F(z_T(T))-F(z(0))\geq -T.$$

F is nonincreasing, thus it follows that

$$z_T(T)\leq F^{-1}(F(z_T(0))-T)=F^{-1}\left(F\left(y_0+\int_0^T g(s)\,ds\right)-T\right). \tag{II.19}$$

We note that this makes sense because the definition of T^* and the condition $T<T^*$ imply that $F(z_T(0))-T$ belongs to the range of F. From (II.18), and since z_T is nondecreasing, we have

$$y(t)\leq z_T(t)\leq z_T(T), \forall t<T.$$

Inequality (II.19) therefore provides the claim.

□

5 Spaces of Banach-valued functions

5.1 Definitions and main properties

Definition II.5.1. *Let X be a Banach space and let I be an interval of $\mathbb{R}$; we say that a function f from I in X is Lebesgue measurable, if*

- *The inverse image under f of all open sets of X is a Borel set of I.*
- *We can change f on a subset of zero Lebesgue measure of I, so that f takes its values into a separable subspace of X.*

In the case where X is separable, this definition is identical to the traditional definition of measurability. In the case where X is not separable, this definition ensures that one such function is indeed the limit almost everywhere of a sequence of simple functions with values in X, which makes it possible to define clearly the integral of f when it exists. This theory is known as the Bochner integral.

Proposition II.5.2. *Let X be a Banach space and let I be an interval of $\mathbb{R}$. For all $p \in [1, +\infty[$, we denote as $L^p(]0, T[, X)$, the set of Lebesgue measurable functions defined on I and with values in X, such that $t \mapsto \|f(t)\|_X^p$ is integrable on I. This is a Banach space for the norm*

$$\|f\|_{L^p(I,X)} = \left(\int_I \|f(t)\|_X^p \, dt \right)^{1/p}.$$

In the same way, we define, for $p = +\infty$, a Banach space $L^\infty(I, X)$ provided by the norm

$$\|f\|_{L^\infty(I,X)} = \operatorname{esssup}_{t \in I} \|f(t)\|_X.$$

Proposition II.5.3. *If $p < +\infty$, the set of continuous functions on I with values in X is dense in $L^p(I, X)$.*

For all $f \in L^p(I, X)$, we denote as $\tilde{f}$ the extension by 0 of f to the whole time interval $\mathbb{R}$; then, for all $h \in \mathbb{R}$, we denote as $\tau_h f$ the translated function of $\tilde{f}$ defined by

$$\tau_h f(\cdot) = \tilde{f}(\cdot + h). \tag{II.20}$$

The restriction of $\tau_h f$ to the interval I is of course in $L^p(I, X)$ and we have the following result (the proof being identical to the classic case where $X = \mathbb{R}$; see [69] for example).

Corollary II.5.4 (Continuity of the translation operator). *If $p < +\infty$, then for all $f \in L^p(I, X)$ we have*

$$\tau_h f \xrightarrow[h \to 0]{} f, \quad \textit{in } L^p(I, X).$$

These results are of course false if $p = +\infty$. In the case of $p < +\infty$, the proposition above shows that we can also define $L^p(I, X)$ as the completion of $\mathcal{C}^0(I, X)$ for the norm $\|\cdot\|_{L^p(I,X)}$.

For any function $f \in L^1(I, X)$ (and hence also if $f \in L^p(I, X)$ and if I is bounded), we can define the integral

$$\int_I f(t)\,dt \in X,$$

in a similar way to the Lebesgue integrals of real-valued functions, that is, by constructing the integral over simple measurable functions (i.e., taking a finite number of values) and by passing to the simple limit. We assume this result, as well as all the usual properties of the integral: Chasles' linearity theorem, and so on. Moreover, for all linear forms $\varphi \in X'$, we have

$$\int_I \langle \varphi, f(t)\rangle_{X',X}\,dt = \left\langle \varphi, \int_I f(t)\,dt \right\rangle_{X',X}.$$

The first examples for such spaces that are very useful in the sequel are given for $p, q \in [1, +\infty]$, by $L^p(I, L^q(\Omega))$.

The properties of L^p spaces, which we gave at the start of this section, naturally transpose into these spaces and we use these later without giving more details. In particular, if $p < +\infty$ and $q < +\infty$, then we have

$$(L^p(I, L^q(\Omega)))' \equiv L^{p'}(I, L^{q'}(\Omega)),$$

the identification of the two spaces being achieved via the natural inner product of the Hilbert space $L^2(I, L^2(\Omega)) \approx L^2(I \times \Omega)$. Hence, Propositions II.2.27, II.2.28, II.2.30, and II.2.32, can be immediately transposed to these spaces.

Among the particularly useful results to keep in mind, we give the following interpolation result as well as its corollary which gives the convergence properties in some intermediate spaces.

Theorem II.5.5. *Let I be an interval of $\mathbb{R}$, let Ω be an open set of $\mathbb{R}^d$, and let p_1,q_1,p_2,q_2 be four real numbers in $[1, +\infty]$. If $f \in L^{p_1}(I, L^{q_1}(\Omega)) \cap L^{p_2}(I, L^{q_2}(\Omega))$ then for all $\theta \in]0, 1[$, the function f belongs to $L^p(I, L^q(\Omega))$ for p and q defined by*

$$\frac{1}{p} = \frac{\theta}{p_1} + \frac{1-\theta}{p_2}, \text{ and } \frac{1}{q} = \frac{\theta}{q_1} + \frac{1-\theta}{q_2}, \tag{II.21}$$

and we have

$$\|f\|_{L^p(I,L^q(\Omega))} \leq \|f\|^{\theta}_{L^{p_1}(I,L^{q_1}(\Omega))} \|f\|^{1-\theta}_{L^{p_2}(I,L^{q_2}(\Omega))}.$$

Proof.

From Lemma II.2.33, for almost all $t \in I$ we have:

$$\|f(t)\|_{L^q}^p \leq \|f(t)\|_{L^{q_1}}^{p\theta} \|f(t)\|_{L^{q_2}}^{p(1-\theta)}.$$

If we assume that p_1 and p_2 are finite, then the Hölder inequality applied with the conjugate exponents $p_1/(p\theta)$ and $p_2/(p(1-\theta))$ shows that

$$\int_I \|f(t)\|_{L^q}^p \, dt \leq \left(\int_I \|f(t)\|_{L^{q_1}}^{p_1} \, dt \right)^{p\theta/p_1} \left(\int_I \|f(t)\|_{L^{q_2}}^{q_1} \, dt \right)^{p(1-\theta)/p_2},$$

which gives the desired result. The case where p_1 and/or p_2 are infinite is straightforward.

□

Corollary II.5.6. *We consider the same notation as in the previous theorem and we assume further that p_1 and q_1 are finite and that p_2 and q_2 are strictly larger than 1.*

If $(u_n)_n$ is a sequence of functions which strongly converges towards u in $L^{p_1}(I, L^{q_1}(\Omega))$ and weakly (or weakly-$\star$ if p_2 and/or q_2 are infinite) in $L^{p_2}(I, L^{q_2}(\Omega))$, then for all θ such that $0 < \theta \leq 1$ the sequence $(u_n)_n$ strongly converges towards u in $L^p(I, L^q(\Omega))$, where p and q are given by (II.21).

Proof.

From the preceding theorem we have for all n,

$$\|u - u_n\|_{L^p(I, L^q(\Omega))} \leq \|u - u_n\|_{L^{p_1}(I, L^{q_1}(\Omega))}^{\theta} \|u - u_n\|_{L^{p_2}(I, L^{q_2}(\Omega))}^{1-\theta}.$$

The weak convergence in $L^{p_2}(I, L^{q_2}(\Omega))$ shows that the sequence $(u - u_n)_n$ is bounded in this space and the strong convergence in $L^{p_1}(I, L^{q_1}(\Omega))$ allows us to reach our conclusion, given that $\theta > 0$.

□

All of these results are used systematically in this book, without necessarily referencing them.

5.2 Regularity in time

5.2.1 Weak time derivative

In the study of parabolic partial differential equations, one independent variable (usually time) plays a particular role with respect to the other variables (typically space variables). This is why we work in $L^p(]0, T[, X)$ spaces where X is the functional space in the space variables.

In this section, we therefore generalise the concept of weak derivatives for functions defined on an interval of $\mathbb{R}$ and with values in a Banach space. In a general way, it is possible to define and study distribution spaces with values in X, but this theory is not required here and we refer the reader to [113] for more details.

For reasons which become clear later, it is useful to construct a theory in which the weak derivative of the function being considered can exist in a space that is larger than the initial space.

Definition II.5.7. *Let I be an interval of $\mathbb{R}$, and $X \subset Y$ be two Banach spaces, $1 \leq p, q \leq +\infty$. We say that a function $u \in L^p(I, X)$ has a weak derivative in $L^q(I, Y)$ if there exists a function $g \in L^q(I, Y)$ such that*

$$\int_I \varphi'(t) u(t)\, dt = -\int_I \varphi(t) g(t)\, dt,\ \forall \varphi \in \mathcal{D}(I). \tag{II.22}$$

If such a function g exists, it is unique and we denote

$$\frac{du}{dt} = g(t).$$

We should note that in (II.22), the left-hand term is an element of X and the right-hand term is an element of Y. However, since $X \subset Y$ this equality makes sense.

Remark II.5.1. A priori this definition depends on the space $L^q(I, Y)$ in which we seek the weak derivative. We can show that if $Y \subset Z$ in a dense way, if Z' is separable and if g and h are the weak derivatives of u in $L^q(I, Y)$ and $L^r(I, Z)$, respectively, then $g = h$ almost everywhere. This, therefore, justifies the notation du/dt.

5.2.2 Weak continuity

Definition II.5.8. *Let Y be a Banach space; we say that a function $u : [0, T] \to Y$ is weakly continuous if for all $\psi \in Y'$, the function defined by $t \in [0, T] \mapsto \langle \psi, u(t) \rangle_{Y', Y} \in \mathbb{R}$ is continuous. We denote by $\mathcal{C}^0([0, T], Y_{\text{weak}})$, the set of functions defined on $[0, T]$ with values in Y which are weakly continuous.*

We will now show the following important result (see, e.g., [85]).

Lemma II.5.9. *Let X be a separable and reflexive Banach space, and let Y be a Banach space, such that $X \subset Y$ with continuous embedding. Then*

$$L^\infty(]0, T[, X) \cap \mathcal{C}^0([0, T], Y_{\text{weak}}) = \mathcal{C}^0([0, T], X_{\text{weak}}).$$

Proof.

The space X is embedded into Y in a continuous way, therefore the restrictions to X of elements of Y' are in X'.

- Let us show that $\mathcal{C}^0([0,T], X_{\text{weak}}) \subset L^\infty(]0,T[, X) \cap \mathcal{C}^0([0,T], Y_{\text{weak}})$.

  Let $u \in \mathcal{C}^0([0,T], X_{\text{weak}})$ and let $\psi \in Y'$. Since $\psi_{|X} \in X'$, the function $t \mapsto \langle \psi, u(t)\rangle$ is continuous by definition, which shows that $u \in \mathcal{C}^0([0,T], Y_{\text{weak}})$.
  Let us show that $u \in L^\infty(]0,T[, X)$. First, we note that u is measurable. Indeed, any sphere B of X which is closed for the strong topology is also closed for the weak topology of X, because it is convex (see [27]). Therefore, $u^{-1}(B)$ is a closed set and hence a Borel set of $]0,T[$, because u is continuous on $]0,T[$ with values in X for the weak topology. However, X being separable, any open set of X is a countable union of closed spheres. Indeed, if $(x_n)_n$ is a dense sequence in X, it is obvious that any open set U is the union of all closed spheres centred on a point of the sequence $(x_n)_n$, with a rational radius and contained in U. Hence, for any open set U, $u^{-1}(U)$ is a Borel set of $]0,T[$. This proves the measurability.
  Let us now introduce the family of elements of X'' indexed by $t \in [0,T]$, defined by
  $$\Phi_t : \psi \in X' \mapsto \langle \psi, u(t)\rangle.$$
  By hypothesis, for all $\psi \in X'$, the function $t \mapsto \Phi_t(\psi)$ is continuous on $[0,T]$ and therefore bounded. From the Banach–Steinhaus theorem (Theorem II.2.4), we know that the family of operators $(\Phi_t)_{t\in]0,T[}$ is bounded in the sense of the norm of X''. Alternatively, we can say that there exists $C > 0$ such that
  $$|\langle \psi, u(t)\rangle_{X',X}| = |\Phi_t(\psi)| \leq C\|\psi\|_{X'},\ \forall t \in]0,T[, \forall \psi \in X'.$$
  If we apply Proposition II.2.1, this gives
  $$\|u(t)\|_X = \sup_{\psi\in X', \psi\neq 0} \frac{|\langle \psi, u(t)\rangle_{X',X}|}{\|\psi\|_{X'}} \leq C,\ \forall t \in]0,T[.$$
  This demonstrates that $u \in L^\infty(]0,T[, X)$.
- Let us show that $L^\infty(]0,T[, X) \cap \mathcal{C}^0([0,T], Y_{\text{weak}}) \subset \mathcal{C}^0([0,T], X_{\text{weak}})$.

  Let $u \in L^\infty(]0,T[, X) \cap \mathcal{C}^0([0,T], Y_{\text{weak}})$. Let us first verify that for all $t \in [0,T]$, $u(t) \in X$. A priori, we know only that $u(t) \in Y$ for all t, and that $u(t) \in X$ for almost all t.
  First, let us extend u to all of $\mathbb{R}$ (e.g., by successive reflections performed by setting $u(t) = u(-t)$ for $t \in [-T, 0]$, etc.). It is then obvious that $u \in L^\infty(\mathbb{R}, X) \cap \mathcal{C}^0(\mathbb{R}, Y_{\text{weak}})$. Let $\eta : \mathbb{R} \to \mathbb{R}$ be a mollifying kernel (see Definition II.2.23). We set $u_n = u \star \eta_{1/n}$ which is defined for all t and takes its values in X.
  Let $t_0 \in \mathbb{R}$ be fixed. For all $n \geq 1$, we have
  $$\|u_n(t_0)\|_X = \|(u \star \eta_{1/n})(t_0)\|_X \leq \|u\|_{L^\infty(\mathbb{R},X)}.$$

The sequence $(u_n(t_0))_n$ is bounded in X, which is reflexive, therefore we can extract a subsequence $(u_{n_k}(t_0))_k$ which weakly converges in X towards a certain $\tilde{u}(t_0)$ (Theorem II.2.7).
However, for all $\psi \in Y'$, we have

$$\langle \psi, (u \star \eta_{1/n})(t_0) - u(t_0) \rangle_{Y',Y} = \big((\langle \psi, u \rangle_{Y',Y} \star \eta_{1/n})(t_0) - \langle \psi, u \rangle_{Y',Y}(t_0) \big) \xrightarrow[n\to\infty]{} 0,$$

because, by hypothesis, $t \mapsto \langle \psi, u(t) \rangle_{Y',Y}$ is a continuous function on $\mathbb{R}$ (the extension by reflection preserves this property). We have hence shown that $(u_n(t_0))_n$ weakly converges in Y towards $u(t_0)$. Through the uniqueness of the weak limit in Y, we obtain

$$u(t_0) = \tilde{u}(t_0) \in X,$$

which indeed proves that the function u takes its values in X for all t and that there exists $C > 0$ such that

$$\|u(t)\|_X \leq C, \; \forall t \in \mathbb{R}. \tag{II.23}$$

We can now define the function $t \mapsto \langle \psi, u(t) \rangle_{X',X}$ for all $\psi \in X'$. Let us show that it is continuous. Let $(t_n)_n$ be a sequence of real numbers which converges towards $t \in \mathbb{R}$. From (II.23) the sequence $(u(t_n))_n$ is bounded in X, and we can therefore extract a subsequence which weakly converges towards a certain x in X. Furthermore $(u(t_n))_n$ weakly converges towards $u(t)$ in Y, and through the uniqueness of the weak limit in Y, we obtain $x = u(t)$. This proves that the sequence $(u(t_n))_n$ is relatively weakly compact in X and has only one accumulation point. We therefore know that all of the sequence weakly converges towards its unique accumulation point $u(t)$.

□

Remark II.5.2. If X and Y are two separable Banach spaces such that Y is embedded in a dense way into X, then we have

$$L^\infty(]0,T[, X') \cap \mathcal{C}^0([0,T], Y'_{\text{weak}-\star}) \subset \mathcal{C}^0([0,T], X'_{\text{weak}-\star}).$$

This result is proven in an equivalent way to the preceding one. The converse embedding is of course true from that which has gone before, if we add the reflexivity hypothesis.

5.2.3 Strong continuity

Let X and Y be two Banach spaces such that X is embedded in a continuous and dense way into Y, and let $T > 0$ and p, q satisfy $1 \leq p, q \leq +\infty$. We denote:

$$E_{p,q} = \left\{ u \in L^p(]0,T[,\, X),\ \frac{du}{dt} \in L^q(]0,T[,\, Y) \right\}.$$

Lemma II.5.10. *The space $E_{p,q}$ endowed with the norm*

$$\|u\|_{E_{p,q}} = \|u\|_{L^p(]0,T[,\, X)} + \left\| \frac{du}{dt} \right\|_{L^q(]0,T[,\, Y)},$$

is a Banach space. Moreover, if p and q are finite then $\mathcal{C}^\infty([0,T], X)$ is dense in $E_{p,q}$.

Proof.

Let θ_1, θ_2 be two nonnegative functions of $\mathcal{C}^\infty([0,T], \mathbb{R})$, having sum 1 with

$$\mathrm{supp}(\theta_1) \subset \left[0, \frac{2}{3}T\right], \text{ and } \mathrm{supp}(\theta_2) \subset \left[\frac{1}{3}T, T\right].$$

Then let $u \in E$. To approximate u by regular functions, it is sufficient to separately approximate $\theta_1 u$ and $\theta_2 u$, because $u = \theta_1 u + \theta_2 u$.

The function $v = \theta_1 u$ is an element of

$$\left\{ f \in L^p(]0,+\infty[, X),\ \frac{df}{dt} \in L^q(]0,+\infty[, Y) \right\}.$$

Let us set $v_h(t) = v(t+h)$; then $v_{h,\varepsilon} = v_h \star \eta_\varepsilon$ with $\varepsilon < h$ where $\eta : \mathbb{R} \to \mathbb{R}$ is a mollifying kernel. The claim follows from Corollary II.5.4. A similar argument holds for the function $\theta_2 u$.

$\square$

Remark II.5.3. Let us assume that $p = +\infty$, $q < +\infty$, and that X is the dual of a Banach space E. Then we can easily see that the family of regular functions constructed in the preceding proof satisfies

$$v_{h,\epsilon} \xrightarrow[(h,\varepsilon)\to 0]{} v, \qquad \text{weakly-}\star \text{ in } L^\infty(]0,+\infty[, E'),$$

$$\frac{d}{dt} v_{h,\epsilon} \xrightarrow[(h,\varepsilon)\to 0]{} \frac{d}{dt} v, \quad \text{in } L^q(]0,T[, Y).$$

In other words, the density property of the regular functions still occurs by taking the weak-star topology on $L^\infty(]0,T[, E')$. We can deal, in the same way, with the case where p is finite, $q = +\infty$, and Y is the dual of F as well as the case where $p = q = +\infty$ and $X = E'$, $Y = F'$.

Proposition II.5.11. *Any element u of $E_{p,q}$ (defined almost everywhere) possesses a continuous representation on $[0,T]$ with values in Y, and the embedding of $E_{p,q}$ into $\mathcal{C}^0([0,T],Y)$ is continuous.*

Moreover, for all $t_1,t_2\in[0,T]$, we have

$$u(t_2)-u(t_1)=\int_{t_1}^{t_2}\frac{du}{dt}\,dt,$$

where it is understood that we have identified u and its continuous representation.

Proof.

This result is proven in an entirely similar way to those established in Section 4.

□

In the Hilbertian case we can improve the preceding result in the following way.

Theorem II.5.12 (Lions–Magenes [85]). *Let V and H be two Hilbert spaces such that V is embedded in a continuous and dense way into H. We then identify H with its dual such that we have $V\subset H\subset V'$, the duality bracket between V and V' being given by the scalar product of H. Let $1\leq p,q\leq+\infty$ and let u,v be two functions such that*

$$u\in E_{p,q'}=\left\{f\in L^p(]0,T[,V),\ \frac{df}{dt}\in L^{q'}(]0,T[,V')\right\},$$

$$v\in E_{q,p'}=\left\{f\in L^q(]0,T[,V),\ \frac{df}{dt}\in L^{p'}(]0,T[,V')\right\}.$$

Then the function $t\mapsto(u(t),v(t))_H$ has a continuous representation on $[0,T]$ and we have for all $t_1,t_2\in[0,T]$,

$$\begin{aligned}&(u(t_2),v(t_2))_H-(u(t_1),v(t_1))_H\\&=\int_{t_1}^{t_2}\left\langle\frac{du}{dt}(t),v(t)\right\rangle_{V',V}+\left\langle\frac{dv}{dt}(t),u(t)\right\rangle_{V',V}dt.\end{aligned}$$

Proof.

Here, we give the proof of this result when $1<p,q<+\infty$. The argument can be adapted to other cases by applying Remark II.5.3 and Proposition II.2.12.

Let us consider the following bilinear forms defined on $E_{p,q'}\times E_{q,p'}$ with values in $L^1(]0,T[)$ by

$$\begin{aligned}\Psi_1(f,g)&=\big(t\mapsto(f(t),g(t))_H\big),\\\Psi_2(f,g)&=\left(t\mapsto\left\langle\frac{df}{dt}(t),g(t)\right\rangle_{V',V}+\left\langle\frac{dg}{dt}(t),f(t)\right\rangle_{V',V}\right).\end{aligned}$$

These two maps are well-defined because the exponents p, q and p', q' are conjugate, respectively. Moreover, we have for all $(f, g) \in E_{p,q'} \times E_{q,p'}$:

$$\begin{aligned}\|\Psi_1(f,g)\|_{L^1(]0,T[)} &\leq \|f\|_{L^p(]0,T[,V)}\|g\|_{L^{p'}(]0,T[,V')} \\ &\leq T^{1/p'}\|f\|_{L^p(]0,T[,V)}\|g\|_{\mathcal{C}^0(]0,T[,V')} \\ &\leq C\|f\|_{E_{p,q'}}\|g\|_{E_{q,p'}},\end{aligned}$$

and

$$\begin{aligned}\|\Psi_2(f,g)\|_{L^1(]0,T[)} \leq &\left\|\frac{df}{dt}\right\|_{L^{q'}(]0,T[,V')}\|g\|_{L^q(]0,T[,V)} \\ &+\left\|\frac{dg}{dt}\right\|_{L^{p'}(]0,T[,V')}\|f\|_{L^p(]0,T[,V)} \\ \leq &C\|f\|_{E_{p,q'}}\|g\|_{E_{q,p'}}.\end{aligned}$$

This proves that Ψ_1 and Ψ_2 are continuous bilinear forms. Using Lemma II.5.10, let us consider $(u_n)_n$ and $(v_n)_n$, two sequences of $\mathcal{C}^\infty([0,T],V)$ which converge towards u and v, respectively, in $E_{p,q'}$ and $E_{q,p'}$. The functions u_n and v_n being regular, we can differentiate the scalar product $(u_n(t), v_n(t))_H$ in the classic sense which implies, in particular, that for all functions $\varphi \in \mathcal{D}(]0,T[)$, we have

$$\begin{aligned}&-\int_0^T \varphi'(t)(u_n(t), v_n(t))_H\,dt \\ &=\int_0^T \left(\left\langle\frac{du_n}{dt}(t), v_n(t)\right\rangle_{V',V} + \left\langle\frac{dv_n}{dt}(t), u_n(t)\right\rangle_{V',V}\right)\varphi(t)\,dt.\end{aligned}$$

In other words, we have

$$-\int_0^T \varphi'(t)\Psi_1(u_n,v_n)\,dt = \int_0^T \varphi(t)\Psi_2(u_n,v_n)\,dt.$$

By continuity of Ψ_1 and Ψ_2, we can pass to the limit in this expression and obtain

$$\begin{aligned}&-\int_0^T \varphi'(t)(u(t), v(t))_H\,dt \\ &=\int_0^T \left(\left\langle\frac{du}{dt}(t), v(t)\right\rangle_{V',V} + \left\langle\frac{dv}{dt}(t), u(t)\right\rangle_{V',V}\right)\varphi(t)\,dt.\end{aligned}$$

The function $\Psi_1(u,v) : t \mapsto (u(t), v(t))_H$ belongs to $L^1(]0,T[)$, therefore this last expression, valid for all regular functions with compact support φ, shows us that $\Psi_1(u,v)$ belongs to $W^{1,1}(]0,T[)$ and that its weak derivative is $\Psi_2(u,v)$.

Corollary II.4.2 then allows us to conclude the proof.

$\square$

Theorem II.5.13. *Let V and H be two Hilbert spaces satisfying the hypotheses of the preceding theorem; then the space*

$$E_{2,2} = \left\{ u \in L^2(]0,T[,\, V),\ \frac{du}{dt} \in L^2(]0,T[,\, V') \right\}$$

is continuously embedded into $\mathcal{C}^0([0,T],H)$.

Proof.

We apply the preceding theorem, with $p = q = 2$ and $u = v$, and we immediately obtain that the function $t \mapsto \frac{1}{2}\|u(t)\|_H^2$ is continuous on $[0,T]$ and that for all $t, s \in [0,T]$, we have

$$\begin{aligned}\frac{1}{2}\|u(t)\|_H^2 &= \frac{1}{2}\|u(s)\|_H^2 + \int_s^t \left\langle \frac{du}{dt}, u \right\rangle_{V',V} d\tau \\ &\leq \frac{1}{2}\|u(s)\|_H^2 + C\|u\|_{E_{2,2}}^2.\end{aligned}$$

By integrating this with respect to s, we find

$$\begin{aligned}\frac{1}{2}\|u(t)\|_H^2 &\leq \frac{1}{2}\|u\|^2_{L^2(]0,T[,H)} + C\|u\|^2_{E_{2,2}} \\ &\leq C\|u\|^2_{L^2(]0,T[,V)} + C\|u\|^2_{E_{2,2}} \leq C\|u\|^2_{E_{2,2}}.\end{aligned} \tag{II.24}$$

This proves that the function u lies in $L^\infty(]0,T[,H)$. Furthermore, Proposition II.5.11 shows that u is continuous with values in V'. We can then apply Lemma II.5.9, to obtain that u is weakly continuous with values in H (we should not forget that all Hilbert spaces are reflexive).

The strong continuity of u with values in H is now a consequence of the weak continuity in H, of the continuity of the function $t \mapsto \|u(t)\|_H^2$, and Proposition II.2.11. Moreover, the estimate (II.24) leads to

$$\|u\|_{\mathcal{C}^0([0,T],H)} \leq C\|u\|_{E_{2,2}}.$$

$\square$

This situation is a special case of a more general result (see [85]) which is the following.

Theorem II.5.14. *Let V and W be two Hilbert spaces; then*

$$E_{2,2} = \left\{ v \in L^2(]0,T[,\, V),\ \frac{dv}{dt} \in L^2(]0,T[,\, W) \right\}$$

is continuously embedded in $\mathcal{C}^0([0,T],\,[V,W]_{\frac{1}{2}})$ where $[V,W]_{\frac{1}{2}}$ is the interpolated space of order $\frac{1}{2}$ of V and W.

We refer the reader to [85] for a precise definition of interpolated space $[V,W]_{\frac{1}{2}}$. In Chapter IV, we present the proof of a special case of this statement (see Theorem IV.5.11).

5.3 Compactness theorems

Let us start by proving a now classic lemma (due to J.-L. Lions [84]), which is the basis of a large part of all the following compactness results.

Lemma II.5.15. *Let $B_0 \subset B_1$and $\subset B_2$ be three Banach spaces. We assume that the embedding of B_1 in B_2 is continuous and the embedding of B_0 in B_1 is compact. Then, for all $\varepsilon > 0$, there exists a constant $C(\varepsilon)$, such that for all $u \in B_0$, we have*

$$\|u\|_{B_1} \leq \varepsilon \|u\|_{B_0} + C(\varepsilon)\|u\|_{B_2}.$$

Proof.

Let us assume that the claim is false; then there exists $\varepsilon_0 > 0$ and a sequence $(u_n)_n \subset B_0$, such that

$$\|u_n\|_{B_1} \geq \varepsilon_0 \|u_n\|_{B_0} + n\|u_n\|_{B_2},\ \forall n \geq 1.$$

By homogeneity we can take $\|u_n\|_{B_1} = 1$ in the above inequality. Hence, the sequence $(u_n)_n$ is bounded in B_0 and satisfies $\|u_n\|_{B_2} \leq 1/n$. The embedding of B_0 into B_1 is compact, thus we can extract a subsequence $(u_{n_k})_k$ which converges in B_1 towards an element denoted u_∞. Of course $\|u_\infty\|_{B_1} = 1$, and furthermore $\|u_\infty\|_{B_2} = 0$. This is the contradiction.

□

We can now prove one of the fundamental results of compactness in the study of nonlinear evolution problems.

Theorem II.5.16 (Aubin–Lions–Simon). *Let $B_0 \subset B_1 \subset B_2$ be three Banach spaces. We assume that the embedding of B_1 in B_2 is continuous and that the embedding of B_0 in B_1 is compact. Let p, r such that $1 \leq p, r \leq +\infty$. For $T > 0$, we define*

$$E_{p,r} = \left\{ v \in L^p(]0,T[,\,B_0),\ \frac{dv}{dt} \in L^r(]0,T[,\,B_2) \right\}.$$

i) If $p < +\infty$, the embedding of $E_{p,r}$ in $L^p(]0,T[,\,B_1)$ is compact.
ii) If $p = +\infty$ and if $r > 1$, the embedding of $E_{p,r}$ in $\mathcal{C}^0([0,T],\,B_1)$ is compact.

The proof which follows comes from [109] and does not assume that the spaces considered are reflexive, in contrast to the proof in [14]. It thus applies to the spaces L^1 and L^∞, for example.

Proof.

We only demonstrate point i) of the theorem the second point being treated in an entirely similar fashion. Furthermore, it is clear that we do not lose any generality by assuming that $r = 1$.

To demonstrate the theorem, we establish that if a sequence $(u_n)_n$ satisfies:

$$\begin{aligned} (u_n)_n &\text{ is bounded in } \ L^p(]0,T[,\, B_0), \\ \left(\frac{du_n}{dt}\right)_n &\text{ is bounded in } \ L^1(]0,T[,\, B_2), \end{aligned}$$

then we can extract a Cauchy subsequence in $L^p(]0,T[,\, B_1)$.

• **Step 1:**

From Lemma II.5.15, it suffices to find a Cauchy sequence in $L^p(]0,T[,\, B_2)$. Indeed, if a sequence $(v_n)_n$ satisfies the Cauchy criterion in $L^p(]0,T[,\, B_2)$ and is bounded in $L^p(]0,T[,\, B_0)$, then for all $\varepsilon > 0$ we have

$$\begin{aligned} \|v_n - v_m\|_{L^p(]0,T[,\, B_1)} &\le \varepsilon \|v_n - v_m\|_{L^p(]0,T[,\, B_0)} + C(\varepsilon)\|v_n - v_m\|_{L^p(]0,T[,\, B_2)} \\ &\le 2K\varepsilon + C(\varepsilon)\|v_n - v_m\|_{L^p(]0,T[,\, B_2)}, \end{aligned}$$

where K is some bound of $(v_n)_n$ in $L^p(]0,T[,\, B_0)$. Hence, if $(v_n)_n$ is a Cauchy sequence in $L^p(]0,T[,\, B_2)$, it follows

$$\limsup_{n,m\to\infty} \|v_n - v_m\|_{L^p(]0,T[,\, B_1)} \le 2K\varepsilon,$$

which proves the claim, ε being arbitrary.

• **Step 2:**

Let $\theta \in \mathcal{C}^\infty([0,T],\mathbb{R}),\ \theta(T) = 0$, such that we have

$$u_n = \theta u_n + (1-\theta)u_n \equiv v_n + w_n.$$

We show that we are able to extract from $(v_n)_n$ a Cauchy sequence in $L^p(]0,T[, B_2)$. We would proceed in a similar way for the sequence $(w_n)_n$.

We extend v_n by continuity to $\mathbb{R}_+$ by setting $v_n(t) = 0,\ \forall t \ge T$, and for all $h > 0$ we break down v_n into

$$v_n(t) = \left(\frac{1}{h}\int_t^{t+h} v_n(s)ds\right) + \left(\frac{1}{h}\int_t^{t+h} (v_n(t) - v_n(s))ds\right) \equiv a_{n,h}(t) + b_{n,h}(t).$$

Let h be positive; then we show that the sequence $(a_{n,h}(t))_n$ is uniformly bounded and equicontinuous with values in a compact set of B_2 (we know that $a_{n,h}$ is continuous from Proposition II.5.11). For this, we have

$$\sup_{t\in\mathbb{R}^+} \|a_{n,h}(t)\|_{B_0} \leq \frac{1}{h} h^{1/p'} \|v_n\|_{L^p(]0,T[,\,B_0)},$$

which proves that $t \mapsto a_{n,h}(t)$ takes its values, independently from n, in a bounded set of B_0 (i.e., in a compact of B_2, because the embedding of B_0 in B_2 is compact). Moreover, we have

$$\frac{da_{n,h}}{dt} = \frac{1}{h}\left(v_n(t+h) - v_n(t)\right) = \frac{1}{h}\int_t^{t+h} \frac{dv_n}{dt}(\tau) d\tau,$$

from which, for all $t > 0$,

$$\begin{aligned}\left\|\frac{d}{dt} a_{n,h}(t)\right\|_{B_2} &\leq \frac{1}{h}\int_t^{t+h} \left\|\frac{dv_n}{dt}(\tau)\right\|_{B_2} d\tau,\\ &\leq \frac{1}{h}\left\|\frac{dv_n}{dt}(\tau)\right\|_{L^1(]0,T[,B_2)}.\end{aligned}$$

Hence, h being fixed, the sequence $(a_{n,h})_n$ is equicontinuous with values in a compact set of B_2. Ascoli's theorem (Theorem II.3.1) then shows that we can extract from the sequence $(a_{n,h})_n$ a subsequence which is convergent in $C^0([0,T],B_2)$ and thus in $L^p(]0,T[,B_2)$. Furthermore, we have

$$\begin{aligned}\|b_{n,h}(t)\|_{B_2} &\leq \frac{1}{h}\int_t^{t+h} \|v_n(t) - v_n(s)\|_{B_2}\, ds\\ &\leq \frac{1}{h}\int_t^{t+h} \left(\int_t^s \left\|\frac{dv_n}{dt}(\tau)\right\|_{B_2} d\tau\right) ds.\end{aligned}$$

Hence, by using Jensen's inequality (Proposition II.2.20) and Fubini's theorem we obtain

$$\begin{aligned}\int_0^T \|b_{n,h}(t)\|_{B_2}^p\, dt &\leq \int_0^T \frac{1}{h}\int_t^{t+h} \left(\int_t^s \left\|\frac{dv_n}{dt}(\tau)\right\|_{B_2} d\tau\right)^p ds\, dt\\ &\leq \left\|\frac{dv_n}{dt}\right\|_{L^1(]0,T[,B_2)}^{p-1} \int_0^T \frac{1}{h}\int_t^{t+h}\int_t^s \left\|\frac{dv_n}{dt}(\tau)\right\|_{B_2} d\tau\, ds\, dt\\ &\leq \left\|\frac{dv_n}{dt}\right\|_{L^1(]0,T[,B_2)}^{p-1} \int_0^T \frac{1}{h}\int_t^{t+h} \left\|\frac{dv_n}{dt}(\tau)\right\|_{B_2} (t+h-\tau)\, d\tau\, dt\\ &\leq h\left\|\frac{dv_n}{dt}\right\|_{L^1(]0,T[,B_2)}^{p},\end{aligned}$$

and hence

$$\|b_{n,h}\|_{L^p(]0,T[,B_2)} \leq C_T h^{1/p}. \tag{II.25}$$

- **Step 3:**

We now use a diagonal process to construct a convergent subsequence of $(v_n)_n$. For this, for $k \geq 1$, we set $h_k = 1/k$.

For $k = 1$ we have seen that we can extract a convergent subsequence $(a_{\varphi_1(n),h_1})_n$ from $(a_{n,h_1})_n$. For $k = 2$ we can again extract a subsequence of the sequence $(a_{\varphi_1(n),h_2})_n$, denoted $(a_{\varphi_1\circ\varphi_2(n),h_2})_n$, which converges. We hence proceed with successive extractions of subsequences such that for all $k \geq 1$, the sequence $(a_{\varphi_1\circ\cdots\circ\varphi_k(n),h_k})_n$ converges.

We now define $\psi(k) = \varphi_1 \circ \varphi_2 \circ \cdots \circ \varphi_k(k)$. Let us verify that the sequence $(v_{\psi(k)})_k$ is indeed a Cauchy sequence. Let $\varepsilon > 0$; according to (II.25) there exists $k_0 \geq 1$ such that for all n and for all $k \geq k_0$, we have

$$\|b_{n,h_k}\|_{L^p(]0,T[,B_2)} \leq \varepsilon.$$

Let us now write

$$v_{\psi(k)} = a_{\psi(k),h_{k_0}} + b_{\psi(k),h_{k_0}}.$$

From the diagonal extraction process employed, the sequence $(a_{\psi(k),h_{k_0}})_{k\geq k_0}$ is a sequence extracted from the sequence $(a_{\varphi_1\circ\cdots\circ\varphi_{k_0}(n),h_{k_0}})_n$ which, by definition, is a convergent sequence and thus it satisfies the Cauchy criterion. Hence, there exists $k_1 \geq k_0$ such that for all $k, k' \geq k_1$, we have

$$\|a_{\psi(k),h_{k_0}} - a_{\psi(k'),h_{k_0}}\|_{L^p(]0,T[,B_2)} \leq \varepsilon.$$

Thus, finally, we have for all $k, k' \geq k_1$,

$$\begin{aligned}\|v_{\psi(k)} - v_{\psi(k')}\|_{L^p(]0,T[,B_2)} &\leq \|a_{\psi(k),h_{k_0}} - a_{\psi(k'),h_{k_0}}\|_{L^p(]0,T[,B_2)} \\ &\quad + \|b_{\psi(k),h_{k_0}}\|_{L^p(]0,T[,B_2)} + \|b_{\psi(k'),h_{k_0}}\|_{L^p(]0,T[,B_2)} \\ &\leq 3\varepsilon.\end{aligned}$$

This, indeed, proves that the sequence $(v_{\psi(k)})_k$ is a Cauchy sequence in $L^p(]0,T[,B_2)$.

□

In certain cases, the preceding theorem does not apply and we need to use sharper results.

Let E be a Banach space. For $f \in L^1(]0,T[,E)$ we denote the translated function of f defined by (II.20) as $\tau_h f$. For $1 \leq q < +\infty$ and $0 < \sigma < 1$, we define the Nikolskii spaces $N_q^\sigma(]0,T[,\ E)$ by:

$$N_q^\sigma(]0,T[,\ E) = \left\{ f \in L^q(]0,T[,E),\ \sup_{0<h<T} \frac{\|\tau_h f - f\|_{L^q(]0,T-h[,E)}}{h^\sigma} < +\infty \right\}, \tag{II.26}$$

and for $f \in N_q^\sigma(]0,T[,E)$ we introduce the norm

$$\|f\|_{N_q^\sigma(]0,T[,E)} = \left(\|f\|^q_{L^q(]0,T[,E)} + \sup_{0<h<T} \left(\frac{1}{h^\sigma} \|\tau_h f - f\|_{L^q(]0,T-h[,E)} \right)^q \right)^{1/q}.$$

Intuitively, this leads to replacing a condition on the derivative with respect to time of f by a condition of the Hölder type, which is weaker but proves to be sufficient. We give, for example, the following theorem for which the reader will find a proof in [109].

Theorem II.5.17 (Simon). *Let B_0, B_1, B_2 be three Banach spaces with $B_0 \subset B_1 \subset B_2$. We assume that the embedding of B_1 into B_2 is continuous and that the embedding of B_0 into B_1 is compact.*

Then, for all $1 \leq q \leq +\infty$ and $0 < \sigma < 1$, the embedding

$$L^q(]0,T[,\ B_0) \cap N_q^\sigma(]0,T[,\ B_2) \hookrightarrow L^q(]0,T[,\ B_1),$$

is compact.

5.4 Banach-valued Fourier transform

To obtain compactness, we have seen that it is necessary to establish estimates of the derivatives with respect to time or of the translated functions with respect to time from the sequences of functions involved. As we show in the following, several methods are available to obtain these estimates. One of these consists in using the Fourier transform with respect to the time variable. We demonstrate, in this sense, the Proposition II.5.23 which gives a characterisation of the sequences of bounded functions in the Nikolskii spaces using the Fourier transform. We illustrate this technique in Section 1 of Chapter VII for the investigation of the Navier–Stokes equations with nonstandard boundary conditions.

Before this, we need to recall the definition of the essential properties of the Fourier transform of functions of the time variable with values in a Banach space.

Definition II.5.18. *Let X be a complex Banach space and let $f \in L^1(\mathbb{R}, X)$. We call the Fourier transform of f the function $\mathcal{F}(f) \in L^\infty(\mathbb{R}, X)$ defined by*

$$\mathcal{F}(f)(\tau) \equiv \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t) e^{-it\tau}\, dt.$$

In the case where X is a finite-dimensional space, we have the following classic and fundamental theorem, for which the reader will find a proof, for example, in [100].

Theorem II.5.19 (Hausdorff–Young). *We assume that $X = \mathbb{C}^n$.*

- *For all f in $L^1(\mathbb{R}, \mathbb{C}^n) \cap L^2(\mathbb{R}, \mathbb{C}^n)$ we have*

$$\mathcal{F}(f) \in L^2(\mathbb{R}, \mathbb{C}^n) \text{ and } \|\mathcal{F}(f)\|_{L^2} = \|f\|_{L^2},$$

and hence $\mathcal{F}$ extends in a unique way as an isometry of $L^2(\mathbb{R},\mathbb{C}^n)$ on itself.

- *For any $p \in [1,2]$, there exists a $C > 0$ such that for all f in $L^1(\mathbb{R},\mathbb{C}^n) \cap L^p(\mathbb{R},\mathbb{C}^n)$ we have*

$$\mathcal{F}(f) \in L^{p'}(\mathbb{R},\mathbb{C}^n) \text{ and } \|\mathcal{F}(f)\|_{L^{p'}} \leq C\|f\|_{L^p}.$$

In our context, we shall need to use the Fourier transform in the case where X is a Banach space. In this case, we pay attention to the fact that the preceding theorem is not in general true! Nevertheless, in the particular case where X is a space containing integrable functions, we can find a suitable framework in which the Hausdorff–Young inequality holds. We refer to [8, 64, 96] for more complete and more precise results on this subject.

Theorem II.5.20 (Hausdorff–Young for Lebesgue spaces). *Let (E,μ) be a compact, locally-separated topological space equipped with a regular measure μ on its Borel sets. Let $q \in [1,+\infty]$ and we set $X = L^q(E,\mu)$.*

For all $p \in [1,2]$ such that $p \leq q \leq p'$ there exists a $C > 0$ such that for all f in $L^1(\mathbb{R},L^q(E,\mu)) \cap L^p(\mathbb{R},L^q(E,\mu))$ we have

$$\mathcal{F}(f) \in L^{p'}(\mathbb{R},L^q(E,\mu)) \text{ and } \|\mathcal{F}(f)\|_{L^{p'}(\mathbb{R},L^q(E,\mu))} \leq C\|f\|_{L^p(\mathbb{R},L^q(E,\mu))}.$$

We note that the condition $p \leq q \leq p'$ is optimal (see [96] for a counter example).

Proof.

By density, it is sufficient to prove the result for smooth functions f. We then successively use the Minkowski inequality with $r = p'/q \geq 1$, followed by the Hausdorff inequality given by Theorem II.5.19 for the scalar function $t \mapsto f(t,x)$ with x fixed and finally use the Minkowski inequality again, with $r = q/p \geq 1$. This gives

$$\begin{aligned}
\|\mathcal{F}(f)\|^{p'}_{L^{p'}(\mathbb{R},L^q(E,\mu))} &= \int_{\mathbb{R}} \left(\int_E |\mathcal{F}(f)(\tau,x)|^q \,d\mu \right)^{p'/q} d\tau \\
&\leq \left(\int_E \left(\int_{\mathbb{R}} |\mathcal{F}(f)(\tau,x)|^{p'} \,d\tau \right)^{q/p'} d\mu \right)^{p'/q} \\
&\leq C \left(\int_E \left(\int_{\mathbb{R}} |f(t,x)|^p \,dt \right)^{q/p} d\mu \right)^{p'/q} \\
&\leq C \left(\int_{\mathbb{R}} \left(\int_E |f(t,x)|^q \,d\mu \right)^{p/q} dt \right)^{p'/p} \\
&= C\|f\|^{p'}_{L^p(\mathbb{R},L^q(E,\mu))},
\end{aligned}$$

which gives the claim.

□

Corollary II.5.21. *If $X = H$ is a Hilbert space then the Fourier transform $\mathcal{F}$ extends as an isometry on $L^2(\mathbb{R}, H)$. Moreover $\mathcal{F}$ is invertible on $L^2(\mathbb{R}, H)$ and its inverse is given by*

$$\mathcal{F}^{-1}(g)(t) \equiv \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g(\tau) e^{i\tau t}\, d\tau,\ \forall g \in L^1(\mathbb{R}, H) \cap L^2(\mathbb{R}, H).$$

Proof.

This result is deduced from the preceding theorem because any Hilbert space is isomorphic with an $L^2(E, \mu)$ space for well-chosen (E, μ).

We note that this property characterises Hilbert spaces: if X is a Banach space such that $\mathcal{F}$ maps $L^2(\mathbb{R}, X)$ into itself then X is isomorphic to a Hilbert space.

□

Finally, we use the following result whose proof is a straightforward integration by parts.

Proposition II.5.22. *Let X be a Banach space and $f \in L^1(\mathbb{R}, X)$ such that $\frac{df}{dt} \in L^1(\mathbb{R}, X)$ (we say that $f \in W^{1,1}(\mathbb{R}, X)$); then we have*

$$\mathcal{F}\left(\frac{df}{dt}\right)(\tau) = i\tau \mathcal{F}(f)(\tau).$$

The first difficulty that we encounter is the fact that the functions with which we deal with are only defined on a bounded interval of time $]0, T[$ and not on the whole real axis. As a consequence, we proceed in the following manner.

Let $\tilde{f}$ be the extension of a function f of $W^{1,1}(]0, T[, X)$ by zero outside $]0, T[$. This function is not in $W^{1,1}(\mathbb{R}, X)$ but nevertheless we can derive it in the sense of distributions and obtain

$$\frac{\partial \tilde{f}}{\partial t} = \widetilde{\frac{\partial f}{\partial t}} + f(0)\delta_0 - f(T)\delta_T.$$

Let us recall that $f(0)$ and $f(T)$ are perfectly defined because the functions of $W^{1,1}(]0, T[, X)$ are continuous on $[0, T]$ with values in X (see Corollary II.4.2).

Furthermore, we can generalise the Fourier transform to the tempered distributions (in particular Dirac mass; see, e.g., [101]) and we show that we have

$$i\tau \mathcal{F}\left(\tilde{f}\right) = \mathcal{F}\left(\widetilde{\frac{\partial f}{\partial t}}\right) + \frac{1}{\sqrt{2\pi}} f(0) - \frac{e^{-i\tau T}}{\sqrt{2\pi}} f(T). \tag{II.27}$$

In view of the functions which interest us for analysis of nonlinear partial differential equations, the essential result is the following.

Proposition II.5.23. *Let H be a Hilbert space and let $f \in L^2(]0,T[,H)$ be a function such that for a certain $0 < \sigma \leq 1$ we have*

$$\int_{\mathbb{R}} |\tau|^{2\sigma} \|\mathcal{F}(\tilde{f})(\tau)\|_H^2 d\tau \leq C^2. \tag{II.28}$$

Then f belongs to the Nikolskii space $N_2^\sigma(]0,T[,H)$ defined by (II.26) and we have

$$\|f\|_{N_2^\sigma(]0,T[,H)} \leq M_\sigma(1+C),$$

where M_σ depends only on σ and T.

Proof.

Let f be a function satisfying (II.28), and let $h \in]0,T[$. We set $g_h(t) = \tilde{f}(t+h) - \tilde{f}(t)$ such that

$$\mathcal{F}(g_h)(\tau) = (e^{i\tau h} - 1)\mathcal{F}(\tilde{f})(\tau).$$

We then write for $\tau \neq 0$

$$\mathcal{F}(g_h)(\tau) = \frac{e^{i\tau h} - 1}{\tau^\sigma h^\sigma} h^\sigma \tau^\sigma \mathcal{F}(\tilde{f})(\tau).$$

However, the function $x \mapsto (e^{ix} - 1)/x^\sigma$ is bounded on $\mathbb{R}$ as soon as $\sigma \leq 1$. If K_σ is the bound of this function, we have

$$\|\mathcal{F}(g_h)(\tau)\|_H \leq K_\sigma h^\sigma |\tau|^\sigma \|\mathcal{F}(\tilde{f})(\tau)\|_H,$$

such that

$$\int_{\mathbb{R}} \|\mathcal{F}(g_h)(\tau)\|_H^2 d\tau \leq K_\sigma^2 h^{2\sigma} \int_{\mathbb{R}} |\tau|^{2\sigma} \|\mathcal{F}(\tilde{f})(\tau)\|_H^2 d\tau \leq K_\sigma^2 C^2 h^{2\sigma}.$$

From Corollary II.5.21 the Fourier transform is an isometry of $L^2(\mathbb{R},H)$, therefore we obtain by the definition of g_h

$$\int_{\mathbb{R}} \|\tilde{f}(t+h) - \tilde{f}(t)\|_H^2 dt \leq K_\sigma^2 C^2 h^{2\sigma},$$

which implies that

$$\int_0^{T-h} \|f(t+h) - f(t)\|_H^2 dt \leq K_\sigma^2 C^2 h^{2\sigma},$$

and proves that f belongs in the Nikolskii space $N_2^\sigma(]0,T[,H)$ as well as the stated estimate.

□

Hence, to obtain compactness for a family of approximate solutions to an evolution problem, we can attempt to obtain uniform bounds on the Fourier transform with respect to time for these solutions of type (II.28). This ensures a bound in a Nikolskii space and hence the compactness property through Theorem II.5.17.

6 Some results in spectral analysis of unbounded operators

To generalise the well-known spectral theory in finite dimension to linear operators in infinite dimension, it is convenient to work with operators which map a Hilbert space H into itself (of course one could also work with Banach spaces; see [27, 101]). Indeed, to give meaning to the definition of an eigenvector

$$Au = \lambda u$$

it is clear that u and Au must coexist in the same space. Unfortunately, the common operators which appear in problems from physics are, in general, differential operators and these do not map the common Sobolev spaces $H^s(\Omega)$ onto themselves, because of the loss of derivatives (see Chapter III for the definition of Sobolev spaces).

In the sequel of this book, we particularly apply this theory to the Stokes operator (see Section 5.2 of Chapter IV).

6.1 Definitions

We need to look at these so-called *unbounded* operators A, as operators in H which are only defined on a subset of H, known as the *domain* of the operator and denoted by $D(A)$. For example, we can define the Laplace operator as an unbounded operator on $L^2(\Omega)$ with domain $H^2(\Omega) \cap H^1_0(\Omega)$.

Hence, to be given an unbounded operator is to be given:

- A Hilbert space, H.
- A linear subspace, $D(A) \subset H$.
- A linear mapping $A : D(A) \to H$.

In our context, we assume that all the unbounded operators have a dense domain in H and are closed, which means that the graph of A, defined by $G(A) = \{(u, Au),\, u \in D(A)\}$ is a closed subset of $H \times H$.

Remark II.6.1. The term "unbounded operator", comes from the fact that if A is closed with a dense domain $D(A)$, which is not equal to H, then A cannot be bounded; that is, there does not exist some $C > 0$ such that

$$\|Au\|_H \leq C\|u\|_H,\ \forall u \in D(A).$$

Indeed, let us suppose that this inequality is true for some $C > 0$. Any $u \in H$ is the limit of a sequence $(u_n)_n$ of elements of $D(A)$. This sequence being a Cauchy sequence, from the inequality above, $(Au_n)_n$ is also a Cauchy sequence in H and therefore is convergent. Hence $(u_n, Au_n)_n$ is a sequence of elements in the graph of A which converges, and, therefore, this graph being assumed to be closed, we obtain that $u \in D(A)$. We have therefore shown that $H = D(A)$, which is not the case.

We now define the fundamental concept of self-adjoint operator, which again generalises the usual concept in finite-dimensional spaces.

Definition II.6.1. *Let $A : D(A) \subset H \to H$, be an unbounded operator with a dense domain. We then introduce*

$$D(A^*) = \{u \in H,\ v \in D(A) \mapsto (Av, u)_H \text{ is continuous for the norm of } H\}.$$

For $u \in D(A^*)$, the mapping $v \mapsto (Av, u)_H$ can therefore be extended by a continuous linear functional on H which may be represented by an element denoted as $A^*u \in H$:

$$(Av, u)_H = (v, A^*u)_H, \forall u \in D(A^*), \forall v \in D(A).$$

The operator A^*, whose domain is $D(A^*)$, is called the adjoint operator of A.

This definition is consistent because the density of $D(A)$ into H ensures the uniqueness of the extension which is used in the definition. We can now define a fundamental class of operators.

Definition II.6.2. *An unbounded operator A is said to be self-adjoint if it satisfies*

$$D(A^*) = D(A) \text{ and } Au = A^*u, \forall u \in D(A).$$

Proposition II.6.3. *Let A be an unbounded self-adjoint operator. We assume that A is a bijection from $D(A)$ onto H and that A^{-1} is continuous from H into H. Then, the (bounded) operator A^{-1} is self-adjoint.*

Proof.

The domain of A^{-1} is H. We therefore first need to show that $D((A^{-1})^*)$ is also equal to H. Let $u \in H$; then, since A^{-1} is continuous, it is clear that $v \mapsto (A^{-1}v, u)_H$ is continuous on all of H and therefore by definition $u \in D((A^{-1})^*)$.

Let $u, v \in H$, then $A^{-1}u$ and $A^{-1}v$ are in $D(A)$, and since A is self-adjoint we have

$$\big(A(A^{-1}u), A^{-1}v\big)_H = \big(A^{-1}u, A(A^{-1}v)\big)_H;$$

in other words

$$(u, A^{-1}v)_H = (A^{-1}u, v)_H,$$

which shows that A^{-1} is indeed self-adjoint.

□

In general, unbounded operators are not continuous from $D(A)$ equipped with the norm of H into H. This means that the norm of H is not the correct norm for us to set on $D(A)$. The following proposition, whose proof is straightforward, is the consequence of the fact that the graph of A is closed.

Proposition II.6.4. *Let A be a closed unbounded operator in a Hilbert space H of domain $D(A)$. We provide $D(A)$ with the following scalar product,*

$$(u, v)_{D(A)} = (u, v)_H + (Au, Av)_H, \forall u, v \in D(A), \tag{II.29}$$

and the associated norm. Then, $D(A)$ is a Hilbert space, the embedding from $D(A)$ into H is continuous and A is continuous from $D(A)$ into H.

Remark II.6.2. It is clear that the norm introduced above is equivalent to the norm known as the "graph norm" defined by

$$\|u\|_{\text{graph}} = \|u\|_H + \|Au\|_H, \forall u \in D(A).$$

6.2 Elementary results of spectral theory

The fundamental application of the concepts above, to the subject of interest to us, resides in the following result.

Theorem II.6.5 (Compact self-adjoint operators). *Let H be a separable (infinite-dimensional) Hilbert space and let T be a (bounded, i.e., defined and continuous on all H) compact self-adjoint operator from H to H. Then H has an orthonormal basis formed from eigenvectors of T. Moreover, the set of its eigenvalues (which are real numbers) can be ordered in a sequence tending towards* 0.

We do not give the proof here and we refer for instance to [27].

When A is an unbounded operator which is a bijection from $D(A)$ onto H, then the open mapping theorem tells us that A is an isomorphism of $D(A)$ (equipped with the graph norm) onto H. In this case A^{-1} is a continuous operator from H to $D(A)$. The embedding from $D(A)$ into H being continuous, we can also consider A^{-1} as a continuous operator from H into H.

If, moreover, the embedding from $D(A)$ into H is compact, which is often the case in the applications, then A^{-1}, seen as an operator from H to H, is compact (see Lemma II.3.5). We can therefore apply the preceding theorem and obtain the following result.

Theorem II.6.6 (Operators with compact inverse). *Let H be a separable, infinite-dimensional Hilbert space. Let $A : D(A) \subset H \to H$ be an unbounded* *operator. We assume that A is self-adjoint, bijective from $D(A)$ onto H and that the canonical embedding from $D(A)$ into H is compact.*

Then there exists an orthonormal basis $(w_k)_{k\geq 1}$ of H formed by eigenvectors of A, that is, such that for all $k \geq 1$,

$$w_k \in D(A), \text{ and } Aw_k = \lambda_k w_k,$$

where the eigenvalues $(\lambda_k)_{k\geq 1}$ of A are real numbers that we can order in such a way that $(|\lambda_k|)_k$ is increasing and tends towards $+\infty$ when k tends towards infinity.

Finally, the eigenvectors $(w_k)_{k\geq 1}$ form a complete orthogonal family of $D(A)$.

Proof.

As we have remarked above, under the hypotheses of the theorem, the operator A^{-1} can be viewed as a bounded, compact self-adjoint operator. From Theorem II.6.5, there exists an orthonormal basis $(w_k)_{k\geq 1}$ of H formed from eigenvectors of A^{-1} for the eigenvalues μ_k with, moreover, $\mu_k \to 0$.

Let us now note that, since A^{-1} is injective (Beware! It is not surjective on H), 0 cannot be an eigenvalue of A^{-1}. Hence, for all $k \geq 0$, $\mu_k \neq 0$. Moreover, since $A^{-1}w_k = \mu_k w_k$, we can clearly see that w_k belongs to the image of A^{-1}, that is to in $D(A)$.

If we now set $\lambda_k = 1/\mu_k$, we immediately obtain

$$w_k \in D(A) \text{ and } Aw_k = \lambda_k w_k.$$

The fact that $|\lambda_k| \to +\infty$ is a clear consequence of the fact that $\mu_k \to 0$. This demonstrates the first part of the theorem.

For $k, l \geq 0$, we have

$$(w_k, w_l)_{D(A)} = (w_k, w_l)_H + (Aw_k, Aw_l)_H = (1 + \lambda_k\lambda_l)(w_k, w_l)_H,$$

which shows, $(w_k)_{k\geq 1}$ being a Hilbertian basis of H, that when $k \neq l$, $(w_k, w_l)_{D(A)} = 0$ and that $\|w_k\|_{D(A)} = \sqrt{1 + \lambda_k^2}$. The family $(w_k)_k$ is therefore an orthogonal family of $D(A)$.

To establish that this is also a complete family, it is necessary to show that the only vector of $D(A)$ orthogonal to all the w_k is the null vector. Therefore, let $u \in D(A)$ such that $(u, w_k)_{D(A)} = 0$ for all k. By using the self-adjoint characteristic of A, this gives

$$\begin{aligned}
0 = (u, w_k)_{D(A)} &= (u, w_k)_H + (Au, Aw_k)_H \\
&= (u, w_k)_H + \lambda_k (Au, w_k)_H \\
&= (u, w_k)_H + \lambda_k (u, Aw_k)_H \\
&= (1 + \lambda_k^2)(u, w_k)_H,
\end{aligned}$$

which demonstrates that $(u, w_k)_H = 0$ for all k. Since $(w_k)_k$ is an orthonormal basis of H, we have indeed shown that $u = 0$.

$\square$

We know that all $u \in H$ can be expressed uniquely in the form

$$u = \sum_{k \geq 1} u_k w_k,$$

with the convergence being taken in the sense of H and moreover, $u_k = (u, w_k)_H$. Using this expression we can recognise which of the elements of H are in $D(A)$.

Proposition II.6.7. *Let us take an operator A which satisfies the hypotheses of the preceding theorem. We then have*

$$D(A) = \left\{ u \in H, \text{ such that } \sum_{k \geq 1} \lambda_k^2 (u, w_k)_H^2 < +\infty \right\}.$$

Proof.

- Let $u \in D(A)$; $(w_k)_k$ is a complete orthogonal family of $D(A)$, thus we can see that
$$\left(\frac{w_k}{\|w_k\|_{D(A)}} \right)_k$$
is an orthonormal basis of $D(A)$. Hence, we know that
$$\sum_{k \geq 1} \frac{(u, w_k)_{D(A)}^2}{\|w_k\|_{D(A)}^2} < +\infty.$$
However, for all k, we have
$$\begin{aligned}(u, w_k)_{D(A)} &= (u, w_k)_H + (Au, Aw_k)_H = (u, w_k)_H + (u, A^2 w_k)_H \\ &= (1 + \lambda_k^2)(u, w_k)_H,\end{aligned}$$
and in particular
$$\|w_k\|_{D(A)}^2 = (1 + \lambda_k^2)\|w_k\|_H^2 = (1 + \lambda_k^2).$$
Hence, we have obtained
$$\sum_{k \geq 1} (1 + \lambda_k^2)(u, w_k)_H^2 < +\infty,$$
which proves the desired assertion.
- Now, let $u \in H$, we assume that

$$\sum_{k\geq 1} \lambda_k^2 (u, w_k)_H^2 < +\infty.$$

The hypothesis implies that

$$\sum_{k\geq 1} \|(u, w_k)_H w_k\|_{D(A)}^2 < +\infty,$$

and since the vectors $\big((u, w_k)_H w_k\big)_k$ are pairwise orthogonal in $D(A)$, this shows that the series

$$\sum_{k\geq 1} (u, w_k)_H w_k$$

converges towards a certain $\tilde{u} \in D(A)$, in the sense of the norm of $D(A)$. However, since the embedding from $D(A)$ to H is continuous, the convergence also takes place in H and hence we obtain

$$(\tilde{u}, w_k)_H = (u, w_k)_H,$$

which proves that $u - \tilde{u}$ is orthogonal to all w_k. Inasmuch as $(w_k)_k$ is complete in H, this shows that $u = \tilde{u}$ and therefore that $u \in D(A)$.

□

We have shown in passing that for $u \in D(A)$, we have

$$\|u\|_H^2 = \sum_{k\geq 1} (u, w_k)_H^2 \text{ and } \|u\|_{D(A)}^2 = \sum_{k\geq 1} (1 + \lambda_k^2)(u, w_k)_H^2,$$

and moreover, since the absolute values of the eigenvalues $(|\lambda_k|)_k$ are bounded below by a positive real number, the norm in $D(A)$ is equivalent to the norm defined by

$$u \in D(A) \mapsto \left(\sum_{k\geq 1} \lambda_k^2 (u, w_k)_H^2\right)^{1/2}.$$

We now wish to define the powers of the operator A. We could, for example, define the operator A^2 in the following natural way

$$D(A^2) = \{u \in D(A), \text{ such that } Au \in D(A)\}, \text{ and } A^2 u = A(Au), \forall u \in D(A^2).$$

However, we choose another definition which allows us to define the fractional powers of an operator. To this end, we have to assume that the operator is nonnegative (i.e., such that $(Au, u)_H \geq 0$ for all $u \in D(A)$). This is equivalent to assuming that the eigenvalues of A are nonnegative. From now on, we make this assumption and for all nonnegative real numbers s, we introduce

$$D(A^s) = \left\{ u \in H, \quad \text{such that } \sum_{k\geq 1} \lambda_k^{2s} (u, w_k)_H^2 < +\infty \right\},$$

and for all $u \in D(A^s)$, we set

$$A^s u = \sum_{k\geq 1} \lambda_k^s (u, w_k)_H w_k \in H.$$

Finally, we equip $D(A^s)$ with the natural scalar product defined by

$$(u, v)_{D(A^s)} = \sum_{k\geq 1} (1 + \lambda_k^{2s})(u, w_k)_H (v, w_k)_H.$$

We can easily verify the following properties.

Proposition II.6.8. *1. The operator A^1 is simply the operator A. Moreover, $D(A^0) = H$ and A^0 is the identity operator. The norms on $D(A^1)$ and $D(A^0)$ are equivalent to the usual norms on these spaces.*

2. For all $s > 0$, $D(A^s)$ is a Hilbert space and A^s is a nonnegative self-adjoint operator which is an isomorphism from $D(A^s)$ onto H. Moreover, $(w_k)_k$ is a complete family in $D(A^s)$.

3. For all $0 \leq s < s'$, we have $D(A^{s'}) \subset D(A^s)$, the inclusion being strict and the embedding being compact.

Proof.

The first two points are trivial, as is the strict inclusion $D(A^{s'}) \subset D(A^s)$. We only prove the compactness of the embedding of $D(A^{s'})$ in $D(A^s)$.

Let $(u^k)_k$ be a bounded sequence in $D(A^{s'})$, then there exists a $C > 0$ such that

$$\sum_n \lambda_n^{2s'} |u_n^k|^2 \leq C, \, \forall k \in \mathbb{N},$$

where we have denoted the coordinates of u^k in the basis $(w_n)_n$ as $u_n^k = (u^k, w_n)_H$.

We show that for all $\varepsilon > 0$, we can cover the sequence $(u^k)_k$ by a finite number of spheres with radius ε in $D(A^s)$. For any $\varepsilon > 0$, the sequence $(\lambda_n)_n$ tends towards $+\infty$, thus there exists a $n_0 \geq 0$ such that

$$\lambda_n \geq \left(\frac{2C}{\varepsilon^2} \right)^{1/(2(s'-s))}, \, \forall n \geq n_0.$$

Hence, for all k, we have

$$\sum_{n\geq n_0} \lambda_n^{2s} |u_n^k|^2 = \sum_{n\geq n_0} \lambda_n^{2(s-s')} \lambda_n^{2s'} |u_n^k|^2 \leq \frac{\varepsilon^2}{2C} \sum_{n\geq n_0} \lambda_n^{2s'} |u_n^k|^2 \leq \frac{\varepsilon^2}{2}. \quad \text{(II.30)}$$

Since n_0 is fixed, we see that for all $n < n_0$, we have

$$|u_n^k|^2 \leq \frac{C}{\lambda_n^{2s'}}, \quad \forall k.$$

Thus, the sequences of real numbers $(u_n^k)_k$ for $0 \leq n \leq n_0 - 1$ are bounded. The closed spheres of $\mathbb{R}^{n_0}$ are compact sets, thus there exists a finite family of elements of $\mathbb{R}^{n_0}$, denoted $(v^i)_{i \in I}$, with $v^i = (v_n^i)_{0 \leq n \leq n_0 - 1}$ such that

$$\forall k \in \mathbb{N}, \exists i \in I, \text{ such that } |u_n^k - v_n^i|^2 \leq \frac{\varepsilon^2}{2 \sum_{n=0}^{n_0-1} \lambda_n^{2s}}, \forall n \leq n_0 - 1. \tag{II.31}$$

For all $i \in I$, we consider the element $\tilde{v}^i$ of $D(A^s)$ defined by

$$\tilde{v}^i = \sum_{n=0}^{n_0-1} v_n^i w_n.$$

Let us now show that the sequence considered, $(u^k)_k$, is covered by the spheres of $D(A^s)$ with centres at $\tilde{v}^i$ and having radius ε. Indeed, if $k \in \mathbb{N}$ is fixed, we consider the index $i \in I$ given by (II.31), such that we have

$$\|u^k - \tilde{v^i}\|_{D(A^s)}^2 = \sum_{n \leq n_0 - 1} \lambda_n^{2s} |u_n^k - v_n^i|^2 + \sum_{n \geq n_0} \lambda_n^{2s} |u_n^k|^{2s} \leq \frac{\varepsilon^2}{2} + \frac{\varepsilon^2}{2} = \varepsilon^2,$$

the first term being bounded above by (II.31) and the second by (II.30).

□

We now wish to define similar concepts for $s < 0$. Unfortunately, in this case the preceding definitions do not apply (because the spaces $D(A^s)$ would all be equal to H and none would be complete). The spaces $D(A^s)$ with $s < 0$ must be larger than H. Hence, for all $u \in H$, we define

$$\|u\|_{D(A^s)}^2 = \sum_{k \geq 1} \lambda_k^{2s} (u, w_k)_H^2.$$

This is a norm on H and if we define $D(A^s)$ as the completion of H for this norm, then the operator A^s is naturally defined. Indeed, $D(A^s)$ is a Hilbert space for the scalar product obtained by completion, and the preceding properties of positive real powers of A adapt without a problem. We then accept the following result.

Proposition II.6.9. *For all $s \geq 0$, if we identify H with its dual, we have*

$$D(A^s)' \approx D(A^{-s}),$$

and the duality can be written using the scalar product of H in the following way,

$$\langle u, v\rangle_{D(A^{-s}),D(A^s)} = \sum_{k\geq 1}(u, w_k)_H(v, w_k)_H, \; \forall u \in H, \forall v \in D(A^s).$$

6.3 Applications to the semigroup theory

Let $(A, D(A))$ be an unbounded operator in H which is self-adjoint, bijective from $D(A)$ onto H, nonnegative, and such that the canonical embedding of $D(A)$ into H is compact.

Our goal is to show, with the elements given above, how to solve the infinite-dimensional linear differential equation

$$\begin{cases} \dfrac{du}{dt} + Au = 0, \\ u(0) = u_0. \end{cases} \tag{II.32}$$

Let $(w_k)_{k\geq 1}$ be the spectral basis associated with A and $(\lambda_k)_{k\geq 1}$ the (positive) eigenvalues of A. We assume that the sequence $(\lambda_k)_{k\geq 1}$ is sorted in a nondecreasing way.

Definition and Proposition II.6.10. *Let $u_0 \in H$ so that we write $u_0 = \sum_{k\geq 1} u_{0,k}w_k$. For any $t \geq 0$ we can define*

$$e^{-tA}u_0 = \sum_{k\geq 1} u_{0,k}e^{-t\lambda_k}w_k \in H.$$

Moreover, for any $s \geq 0$ and $t > 0$, we have $e^{-tA}u_0 \in D(A^s)$.

Remark II.6.3. We obviously have the property

$$e^{-(t+s)A} = e^{-tA}e^{-sA} = e^{-sA}e^{-tA}, \; \forall s, t \geq 0.$$

That's the reason why the family of continuous operator in H defined by $(e^{-tA})_{t\geq 0}$ is called *the semigroup associated with $-A$.*

Proof.

Since $\lambda_k \geq 0$ and $t \geq 0$, it is clear that this sum is well-defined in H. Moreover, for a fixed $t > 0$ and any $s > 0$, we have

$$\lambda_k^s e^{-t\lambda_k} = (t\lambda_k)^s e^{-t\lambda_k}t^{-s} \leq C_s t^{-s},$$

where $C_s = \sup_{[0,+\infty[} y^s e^{-y} < +\infty$.

It follows that

$$\sum_{k\geq 1} \left(\lambda_k^s u_{0,k}e^{-t\lambda_k}\right)^2 \leq C_s^2 t^{-2s} \sum_{k\geq 1} |u_{0,k}|^2 < +\infty,$$

and thus that $e^{-tA}u_0 \in D(A^s)$ for any $t > 0$ and any $s \geq 0$.

$\square$

Theorem II.6.11. *For any $u_0 \in H$, there exists a unique $u \in \mathcal{C}^0([0,+\infty[,H) \cap \mathcal{C}^1(]0,+\infty[,D(A))$ which solves (II.32). It is defined by the formula*

$$u(t) = e^{-tA}u_0, \ \forall t \geq 0. \tag{II.33}$$

Proof.

- Let us first show the uniqueness property. The problem is linear, therefore it is enough to show that any solution u for the initial data $u_0 = 0$ is necessarily equal to 0.
  Let $0 < \varepsilon < T$ be given. We apply Theorem II.5.12 (and Corollary II.3.8) to obtain

$$\begin{aligned}\|u(T)\|_H^2 - \|u(\varepsilon)\|_H^2 &= 2\int_\varepsilon^T \left(\frac{du}{dt}(t), u(t)\right)_H \, dt \\ &= -2\int_\varepsilon^T (Au(t), u(t))_H \, dt \leq 0,\end{aligned}$$

  inasmuch as A is a nonnegative operator. It follows that

$$\|u(T)\|_H \leq \|u(\varepsilon)\|_H,$$

  but since u is continuous with values in H and satifies $u(0) = 0$, we can let ε go to 0 in the inequality above and obtain that $\|u(T)\|_H = 0$, which gives $u(T) = 0$. This being true for any $T > 0$, the claim is proved.
- We easily check that the function $t \in [0,+\infty[\mapsto u(t)$ defined by (II.33) satisfies the claimed regularity property (notice that u is not necessarily differentiable at $t = 0$ with values in H) and $u(0) = u_0$.
  For any $t > 0$, we can differentiate the series to obtain

$$\begin{aligned}\frac{du}{dt}(t) &= \sum_{k\geq 1}(-\lambda_k)u_{0,k}e^{-t\lambda_k}w_k \\ &= -\sum_{k\geq 1} u_{0,k}e^{-t\lambda_k}Aw_k = -A\left(\sum_{k\geq 1} u_{0,k}e^{-t\lambda_k}w_k\right) = -Au(t),\end{aligned}$$

  the last equality being true because the series converges in $D(A)$ and A is continuous from $D(A)$ in H.

$\square$

When we add a source term f to the problem (II.32), the semigroup associated with $-A$ still allows us to solve the problem. More precisely, one can prove, for instance, the following result (see [48] or [95]) which is an infinite-dimensional version of a very standard result for ordinary differential equations.

Theorem II.6.12. *Let* $u_0 \in H$ *and* $f \in \mathcal{C}^1([0,+\infty[,H)$*. There exists a unique solution* $u \in \mathcal{C}^0([0,+\infty[,H) \cap \mathcal{C}^1(]0,+\infty[,H) \cap \mathcal{C}^0(]0,+\infty[,D(A))$ *to*

$$\begin{cases} \dfrac{du}{dt} + Au = f, \\ u(0) = u_0. \end{cases} \tag{II.34}$$

This solution is given by the Duhamel formula

$$u(t) = e^{-tA}u_0 + \int_0^t e^{-(t-s)A} f(s)\, ds, \; \forall t \geq 0. \tag{II.35}$$

Proof.

Using the change of variable $s \to t - s$ in the integral in (II.35) we get

$$u(t) = e^{-tA}u_0 + \int_0^t e^{-sA} f(t-s)\, ds.$$

Inasmuch as f is assumed to be of class $\mathcal{C}^1$ with values in H, we can justify the derivation in all the terms and then conclude by integration by parts.

□

Remark II.6.4. The above result, in particular Formula (II.35), still holds for less regular source terms but this needs to weaken the notion of solution we consider.

Chapter III
Sobolev spaces

The first section of this chapter is dedicated to the basic definitions and properties of domains in $\mathbb{R}^d$. We particularly focus our attention on the case of Lipschitz domains for which we can easily define an integration theory on $\partial\Omega$, the outward unit normal on $\partial\Omega$, and finally prove the Stokes formula which is the keystone of the study of partial differential equations on domains.

In the second section we define and study Sobolev spaces on such domains. We are first interested in proving the density of smooth functions in those spaces and in studying the associated trace theory. We then introduce useful tools in the analysis of PDEs such as Sobolev embedding theorems or Poincaré and Hardy inequalities.

Section 3 concerns the study of suitable parametrisations of neighborhoods of the boundary of sufficiently smooth domains. The idea is to be able to separate tangential and normal coordinates near the boundary in a somewhat intrinsic way. To achieve this objective, we need to carefully study the properties and the regularity of the *distance to the boundary* function $\delta = d(., \partial\Omega)$. In fact, it is also necessary to define and study a suitable regularised distance function. We are then able to build a normal/tangential coordinate system near the boundary and to write the expressions of usual differential operators in those variables. These elements are widely used in the proof of the elliptic regularity properties of Stokes problems in Chapter IV. This section can certainly be skipped for a first reading because the material it contains is not strictly necessary for understanding many results in this book.

The last section of the chapter is dedicated to the study of the very classical Laplace problem with Dirichlet or Neumann boundary conditions. Existence, uniqueness, and regularity are proved in each case. These results are used at various places in the sequel of the book. Moreover, it is also an opportunity for us to illustrate the concepts introduced in Section 3 for a quite simple problem before applying similar ideas to the Stokes and Navier–Stokes equations.

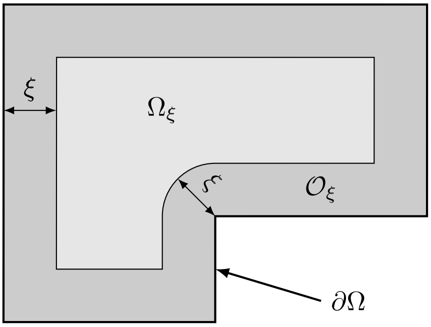
Following the general philosophy of this book, we decided to provide complete proofs for almost all results given in this chapter. As a consequence, this chapter should be self-contained and useful for any reader interested in

F. Boyer and P. Fabrie, *Mathematical Tools for the Study of the Incompressible Navier-Stokes Equations and Related Models*, Applied Mathematical Sciences 183,
DOI 10.1007/978-1-4614-5975-0_3,

the study of partial differential equations in bounded domains, not only in the framework of fluid mechanics.

1 Domains

For any open set Ω, we recall that $\delta(x)$, defined in (II.1) denotes the signed distance from any point $x \in \mathbb{R}^d$ to the boundary $\partial\Omega$. For any $\xi \geq 0$ we use the following notation,



$$\mathcal{O}_\xi = \{x \in \Omega,\ \delta(x) < \xi\},$$

$$\Omega_\xi = \{x \in \Omega,\ \delta(x) > \xi\}.$$

1.1 General definitions

In this section k designates a nonnegative integer and α designates a real number in $[0, 1]$. Moreover we assume that $k + \alpha \geq 1$, which implies that functions of class $\mathcal{C}^{k,\alpha}$ are at least locally Lipschitz continuous.

We say that an open set $\Omega \subset \mathbb{R}^d$ is a domain of class $\mathcal{C}^{k,\alpha}$ if its boundary $\partial\Omega$ is locally the graph of a function of class $\mathcal{C}^{k,\alpha}$ such that Ω is situated on one single side of this graph. Let us emphasise the fact that this last condition is crucial (particularly in allowing us to define the concept of an outward normal to the domain and hence an orientation on $\partial\Omega$ as we show below).

Let us now give more precise definitions.

Definition III.1.1 ($\mathcal{C}^{k,\alpha}$ half-spaces). *For any map $a : \mathbb{R}^{d-1} \to \mathbb{R}$, of class $\mathcal{C}^{k,\alpha}$ and compactly supported we define the following open set*

$$\mathcal{H}_a = \{(\bar{x}, x_d) \in \mathbb{R}^d, \quad x_d > a(\bar{x})\},$$

which is called a $\mathcal{C}^{k,\alpha}$ half-space *in $\mathbb{R}^d$.*

The set $\mathbb{R}^d_+ = \mathbb{R}^{d-1} \times \mathbb{R}^+_* = \mathcal{H}_0$ is called the flat half-space*.*

Definition III.1.2. *An open set Ω of $\mathbb{R}^d$ is said to be a domain of class $\mathcal{C}^{k,\alpha}$ if for any point $\sigma \in \partial\Omega$, there exists an open neighborhood U_σ of σ in $\mathbb{R}^d$, a rotation $R_\sigma \in O_d^+(\mathbb{R})$, and a $\mathcal{C}^{k,\alpha}$ half-space $\mathcal{H}_{a_\sigma}$ such that*

$$\Omega \cap U_\sigma = (R_\sigma \mathcal{H}_{a_\sigma}) \cap U_\sigma,$$

$$\partial\Omega \cap U_\sigma = (R_\sigma \partial\mathcal{H}_{a_\sigma}) \cap U_\sigma,$$

$$\Omega^c \cap U_\sigma = \big(R_\sigma \mathcal{H}_{a_\sigma}^c\big) \cap U_\sigma.$$

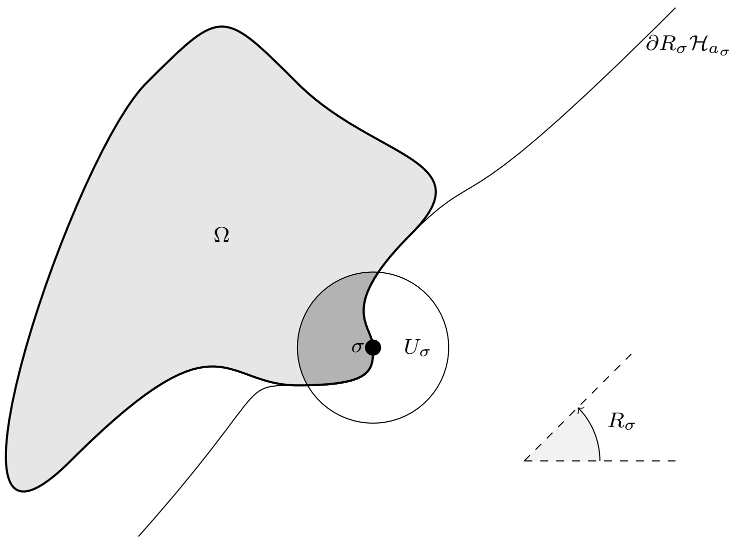


Fig. III.1 A $\mathcal{C}^{k,\alpha}$ domain is locally a $\mathcal{C}^{k,\alpha}$ half-space

In the case where $k = 0$ and $\alpha = 1$, we simply say that the domain Ω is *Lipschitz*. In the case $\alpha = 0$ we simply say that the domain Ω is of class $\mathcal{C}^k$.

1.2 Lipschitz domains

The more general class of domains that we consider here is the one of Lipschitz domains. We refer for instance to the monographs [67, 89] for specific information about more complex domains.

We also want to mention, without proof, that the class of Lipschitz domains contains, for example, all the convex open sets.

1.2.1 Lipschitz half-spaces

In this section we prove that any Lipschitz half-space $\mathcal{H}_a$ as defined in Definition III.1.1 can be mapped onto the flat half-space $\mathbb{R}^d_+ = \mathbb{R}^{d-1} \times \mathbb{R}^+_*$ thanks to a suitable Lipschitz diffeomorphism. Therefore, many properties of functional spaces on $\mathcal{H}_a$ are deduced from the corresponding property on $\mathbb{R}^d_+$.

We suppose given a mollifying kernel $\eta : \mathbb{R}^{d-1} \to \mathbb{R}$ as defined in Definition II.2.23. We introduce the following positive numbers associated with the kernel η,

$$M_\eta = \int_B |z|\eta(z)\,dz \text{ and } M_{\nabla\eta,k} = \int_B |z|^k |\nabla\eta(z)|\,dz, \forall k \in \{0,1,2\},$$

where B is the unit ball of $\mathbb{R}^{d-1}$. Let now $\gamma > 0$ be given such that $\gamma \operatorname{Lip}(a) M_\eta \leq 1/2$. For any $x = (\underbrace{x_1, \ldots, x_{d-1}}_{=\bar{x}}, x_d)$ in $\mathbb{R}^d$ we define

$$a_\gamma(\bar{x}, x_d) = \left(a \star \eta_{\gamma|x_d|}\right)(\bar{x}) = \int_B a(\bar{x} - \gamma|x_d|z)\eta(z)\,dz. \tag{III.1}$$

Proposition III.1.3. *The following properties hold.*

1. a_γ *is* $\mathcal{C}^\infty$ *on* $\{(\bar{x}, x_d) \in \mathbb{R}^d, \quad x_d \neq 0\}$ *and we have* $a_\gamma(\bar{x}, 0) = a(\bar{x})$*, for all* $\bar{x} \in \mathbb{R}^{d-1}$*.*
2. a_γ *is Lipschitz continuous on* $\mathbb{R}^d$ *and we have*

$$\begin{aligned} |\partial_{x_d} a_\gamma| &\leq \gamma \operatorname{Lip}(a) M_\eta, \\ |\partial_{x_i} a_\gamma| &\leq \operatorname{Lip}(a), \forall i \in \{1, \ldots, d-1\}. \\ |\partial_{x_i}\partial_{x_j} a_\gamma| &\leq C_\gamma \frac{1}{|x_d|}, \forall i,j \in \{1, \ldots, d\}. \end{aligned}$$

3. *For any* $\bar{x} \in \mathbb{R}^{d-1}$ *such that* a *is differentiable at* $\bar{x}$*, we have*

$$\partial_{x_i} a_\gamma(\bar{x}, x_d) \xrightarrow[x_d \to 0]{} \partial_{x_i} a(\bar{x}), \forall i \in \{1, \ldots, d-1\}.$$

4. *For any* $\bar{x} \in \mathbb{R}^{d-1}$*, the map*

$$x_d \in \mathbb{R} \mapsto x_d + a_\gamma(\bar{x}, x_d) \in \mathbb{R},$$

   is increasing and maps $\mathbb{R}$ *onto* $\mathbb{R}$*.*
5. *The map*

$$T_a : (\bar{x}, x_d) \in \mathbb{R}^d \mapsto (\bar{x}, x_d + a_\gamma(\bar{x}, x_d)) \in \mathbb{R}^d,$$

   is a Lipschitz diffeomorphism from $\mathbb{R}^d$ *onto* $\mathbb{R}^d$*. Its Jacobian determinant* J_{T_a} *satisfies the estimates*

$$\frac{1}{2} \leq J_{T_a}(x) \leq \frac{3}{2} \text{ and } |\nabla J_{T_a}(x)| \leq \frac{C_\gamma}{|x_d|}, \forall x \in \mathbb{R}^d,$$

and the Jacobian determinant $J_{T_a^{-1}}$ of T_a^{-1} satisfies

$$\frac{2}{3} \leq J_{T_a^{-1}}(y) \leq 2 \text{ and } |\nabla J_{T_a^{-1}}(y)| \leq \frac{C_\gamma}{|(T_a^{-1}(y))_d|}, \forall y \in \mathbb{R}^d,$$

Moreover, we have

$$T_a(\mathbb{R}^d_+) = \mathcal{H}_a,\ T_a(\mathbb{R}^d_-) = \overline{\mathcal{H}_a}^c,\ T_a(\partial\mathbb{R}^d_+) = \partial\mathcal{H}_a.$$

Proof.

1. This comes from Proposition II.2.47.
2. Let $(\bar{x}, x_d) \in \mathbb{R}^d$ and $(\bar{y}, y_d) \in \mathbb{R}^d$; we have

$$\begin{aligned}|a_\gamma(\bar{x}, x_d) - a_\gamma(\bar{y}, y_d)| &\leq \int_B |a(\bar{x} - \gamma|x_d|z) - a(\bar{y} - \gamma|y_d|z)|\eta(z)\, dz \\ &\leq \mathrm{Lip}(a)|\bar{x} - \bar{y}| + \mathrm{Lip}(a)\gamma M_\eta |x_d - y_d|,\end{aligned}$$

which gives the estimates for the first derivatives of a_γ. We now compute

$$\begin{aligned}\partial_{x_i} a_\gamma &= \frac{1}{\gamma|x_d|}\left(\int_B a(\bar{x} - \gamma|x_d|z)\partial_{z_i}\eta(z)\, dz\right) \\ &= \frac{1}{\gamma|x_d|}\left(\int_B \Big[a(\bar{x} - \gamma|x_d|z) - a(\bar{x})\Big]\partial_{z_i}\eta(z)\, dz\right), \forall i \in \{1, \dots, d-1\},\end{aligned}$$

because $\int_B \partial_{z_i}\eta(z)\, dz = 0$. Similarly, we have

$$\begin{aligned}\partial_{x_d} a_\gamma =& \frac{-1}{|x_d|}\left(\int_B a(\bar{x} - \gamma|x_d|z) z \cdot \nabla\eta(z)\, dz\right) - \frac{d-1}{|x_d|} a_\gamma(\bar{x}, x_d) \\ =& \frac{-1}{|x_d|}\left(\int_B \big[a(\bar{x} - \gamma|x_d|z) - a(\bar{x})\big] z \cdot \nabla\eta(z)\, dz\right) \\ &- \frac{d-1}{|x_d|}(a_\gamma(\bar{x}, x_d) - a(\bar{x})),\end{aligned}$$

because by integration by parts we have $\int_B z \cdot \nabla\eta(z)\, dz = -(d-1)$. By using the Lipschitz regularity of a and a_γ, it follows that

$$\begin{aligned}|\partial_{x_d}\partial_{x_i} a_\gamma| &\leq 2\frac{\mathrm{Lip}(a)}{|x_d|} M_{\nabla\eta,1}, \forall i \in \{1, \dots, d-1\}, \\ |\partial_{x_j}\partial_{x_i} a_\gamma| &\leq \frac{\mathrm{Lip}(a)}{\gamma|x_d|} M_{\nabla\eta,0}, \forall i, j \in \{1, \dots, d-1\},\end{aligned}$$

and that

$$|\partial^2_{x_d} a_\gamma| \leq \frac{2\gamma \operatorname{Lip}(a)}{|x_d|} M_{\nabla\eta,2} + \frac{2\gamma |d-1| \operatorname{Lip}(a)}{|x_d|} M_\eta.$$

3. This point comes directly from Proposition II.2.47.
4. For any $\bar{x} \in \mathbb{R}^{d-1}$ and $y_1 > y_2$ given we have

$$\begin{aligned}\big[y_1 + a_\gamma(\bar{x}, y_1)\big] - \big[y_2 + a_\gamma(\bar{x}, y_2)\big] &\geq (y_1 - y_2) - \|\partial_{x_d} a_\gamma\|_\infty |y_1 - y_2| \\ &\geq (1 - \gamma \operatorname{Lip}(a) M_\eta)\, |y_1 - y_2|,\end{aligned}$$

and the clweaim is proved.
5. We easily compute $J_{T_a}(x) = (1 + \partial_{x_d} a_\gamma(x))$. The properties of T_a, J_{T_a}, $J_{T_a^{-1}}$ immediately follows from all the above estimates.

□

We later show (see Theorem III.2.13) that the map T_a allows us to build isomorphisms between the Sobolev spaces on $\mathcal{H}_a$ and the Sobolev spaces on the flat half-space $\mathbb{R}^d_+$. At that point, by using Proposition II.2.15, we can just state that $L^p(\mathcal{H}_a)$ and $L^p(\mathbb{R}^d_+)$ are isomorphic.

1.2.2 The cone property and its consequences

Most of the properties of Sobolev spaces that we establish below rely on the following property which is referred to as the *cone property* (see [1, 67, 33]).

Theorem III.1.4. *Any Lipschitz domain $\Omega \subset \mathbb{R}^d$ with compact boundary satisfies the cone property: There exists a finite open covering $(U_i)_{1\leq i\leq N}$ of $\partial\Omega$, and there exists $\theta \in]0, \pi/2[$ and $\delta_0 > 0$, such that*

- $\mathcal{O}_\delta = \{x \in \Omega, \delta(x) < \delta_0\} \subset \bigcup_{i=1}^N U_i$.
- *For each $1 \leq i \leq N$, there is a point $\sigma_i \in \partial\Omega$ such that $U_i \subset U_{\sigma_i}$ (where U_{σ_i} is introduced in Definition III.1.2).*
- *For any $1 \leq i \leq N$, there exists a unit vector $\tilde{\nu}_i$ such that*

$$x + C_i \subset \Omega, \forall x \in \Omega \cap U_i, \tag{III.2}$$

$$x - C_i \subset \Omega^c, \forall x \in \Omega^c \cap U_i,$$

where C_i is the truncated cone

$$C_i = \{z \in \mathbb{R}^d, \quad \sin(\theta)|z| \leq z \cdot \tilde{\nu}_i \leq \delta_0\}.$$

Moreover, we have

$$x - t\tilde{\nu}_i \in U_i, \forall x \in \Omega \cap U_i, \forall t \in [0, \delta_0/2]. \tag{III.3}$$

Proof.

By using Definitions III.1.1, III.1.2, and the fact that $\partial\Omega$ is compact, we see that it is enough to assume that $\sigma = 0$, for instance and to consider a Lipschitz half-space domain $\mathcal{H}_a$ such that $\sigma = 0 \in \partial\mathcal{H}_a$ because all the required properties are invariant by rotation.

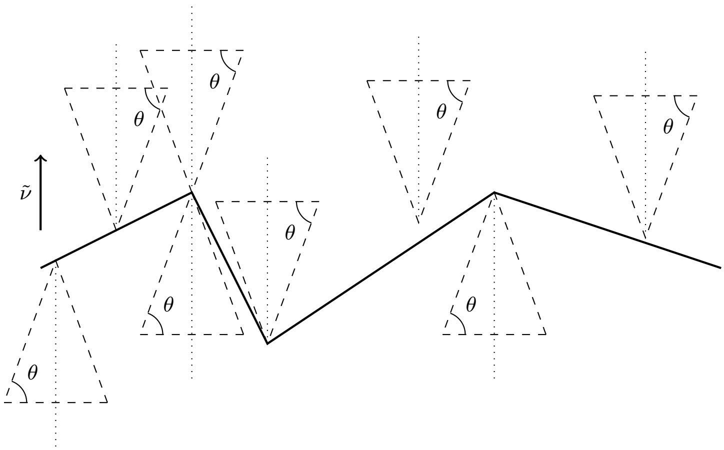


Fig. III.2 The half-space $\mathcal{H}_a$

For such a Lipschitz half-space we prove the result by choosing the unit vector $\tilde{\nu} = {}^t(0, \ldots, 0, 1)$, the angle $\theta = \arctan(\mathrm{Lip}(a))$, and the open neighborhood of $\sigma = 0$ defined by

$$U = \bigcup_{\substack{\bar{x} \in \mathbb{R}^{d-1} \\ |\bar{x}| < \alpha}} \{\bar{x}\} \times \left]a(\bar{x}) - \alpha, a(\bar{x}) + \alpha\right[, \tag{III.4}$$

for an $\alpha > 0$ chosen small enough in such a way that $R_\sigma U \subset U_\sigma$.

Let us introduce the infinite cone

$$C = \{z = (\bar{z}, z_d) \in \mathbb{R}^d, \sin(\theta)|z| \leq z_d\}.$$

Note that we have

$$z = (\bar{z}, z_d) \in C \Rightarrow |z| \geq \frac{1}{\cos\theta}|\bar{z}|.$$

Let now $x \in \mathcal{H}_a$ and $y \in x + C$. Since $y - x \in C$, we have

$$\begin{aligned} y_d &= x_d + (y-x)_d \geq x_d + \sin(\theta)|y-x| \\ &> a(\bar{x}) + \sin(\theta)|y-x| \\ &= a(\bar{y}) + \big(a(\bar{x}) - a(\bar{y})\big) + \sin(\theta)|y-x| \\ &\geq a(\bar{y}) - \mathrm{Lip}(a)|\bar{x}-\bar{y}| + \tan(\theta)|\bar{x}-\bar{y}| = a(\bar{y}), \end{aligned}$$

and thus $y \in \mathcal{H}_a$. A similar computation shows that $x - C \subset (\mathcal{H}_a)^c$ for any $x \in (\mathcal{H}_a)^c$.

Let us finally check the last property. By (III.4), any $x \in \Omega \cap U$ can be written $x = (\bar{x}, a(\bar{x}) + z)$ with $|\bar{x}| < \alpha$ and $0 < z < \alpha$. By construction of $\tilde{\nu}$ we deduce that

$$x - t\tilde{\nu} = (\bar{x}, a(\bar{x}) + z - t),$$

so that for $t \geq 0$ we have $x - t\tilde{\nu} \in U$ if and only if $z - t > -\alpha$. This is true for any $0 < z < \alpha$ and any $0 < t < \delta_0/2$ if and only if we have $\delta_0 < 2\alpha$.

Therefore, (III.3) holds if we choose δ_0 small enough.

□

We now set $U_0 = \Omega_{\delta_0/2} = \{x \in \Omega, \delta(x) > \delta_0/2\}$ and $\nu_0 = 0 \in \mathbb{R}^d$. As a consequence, we just have built a finite open covering of $\overline{\Omega}$.

$$\overline{\Omega} \subset \bigcup_{i=0}^{N} U_i.$$

Proposition III.1.5. *Consider the same notation as in Theorem III.1.4; there exists $M > 0$, and $\varepsilon_0 > 0$ such that if one defines $\nu_i = M\tilde{\nu}_i$ for any $i \in \{1, \ldots, N\}$, the following properties hold for any $0 < \varepsilon < \varepsilon_0$.*

- *For any $i \in \{0, \ldots, N\}$ and any $y \in U_i \cap \Omega$ we have*

$$B(y + \varepsilon\nu_i, \varepsilon) \subset \overline{B(y + \varepsilon\nu_i, 2\varepsilon)} \subset \Omega. \tag{III.5}$$

- *For any $i \in \{1, \ldots, N\}$ and any $y \in U_i$ such that $d(y, \partial\Omega) \leq \varepsilon/M$, we have*

$$B(y - \varepsilon\nu_i, \varepsilon) \subset \overline{\Omega}^c. \tag{III.6}$$

Notice that (III.6) is obviously false for $i = 0$. We also want to point out the fact that we do not require that $y \in \Omega$ in (III.6).

Proof.

Let us consider $M > 2$ such that $\sin(\theta)^2 = (M^2 - 4)/M^2$ and define $\varepsilon_0 = \delta_0/(M + 2)$.

- For $i = 0$, since $\nu_0 = 0$ we only have to observe that for any $y \in U_0$ we have
$$\overline{B(y, 2\varepsilon)} \subset \Omega,$$
because, by definition of $U_0 = \Omega_{\delta_0/2}$, we have $d(y, \partial\Omega) > \delta_0/2 \geq 2\varepsilon_0 \geq 2\varepsilon$.
- Let $i \in \{1, \ldots, N\}$ and $y \in U_i$.
  - If we assume that $y \in \Omega$, by (III.2) we have $y + C_i \subset \Omega$. Therefore, we only need to show that $\overline{B(\varepsilon\nu_i, 2\varepsilon)} \subset C_i$ (see Figure III.3).
    Let $z \in \overline{B(\varepsilon\nu_i, 2\varepsilon)} = \overline{B(\varepsilon M\tilde{\nu}_i, 2\varepsilon)}$. We first have
$$0 < (M - 2)\varepsilon \leq z \cdot \tilde{\nu}_i \leq (M + 2)\varepsilon \leq (M + 2)\varepsilon_0 \leq \delta_0.$$

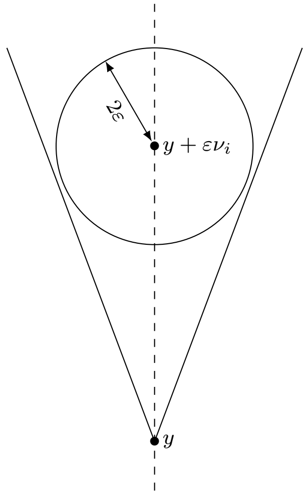


Fig. III.3 The ball $B(y + \varepsilon\nu_i, 2\varepsilon)$ is included in the cone C_i

Second, we perform the following computation by using the Young inequality (Corollary II.2.17)

$$\begin{aligned}|z|^2 &= |z - \varepsilon M\tilde{\nu_i} + \varepsilon M\tilde{\nu_i}|^2 \\ &\leq (4 - M^2)\varepsilon^2 + 2\varepsilon M z \cdot \tilde{\nu_i} \leq \frac{M^2}{M^2 - 4}|z \cdot \tilde{\nu_i}|^2,\end{aligned}$$

which, by definition of M and the fact that $z \cdot \tilde{\nu_i} \geq 0$, exactly gives $\sin(\theta)|z| \leq (z \cdot \tilde{\nu_i})$ and the claim is proved.

– Let $y \in U_i$ such that $d(y, \partial\Omega) \leq \varepsilon/M$. We claim that there exists $0 \leq \alpha \leq d(y, \partial\Omega)/2$ such that $y - \alpha\nu_i \in U_i \cap \Omega^c$.
  · If $y \in \Omega^c$, this is obvious by chosing $\alpha = 0$.
  · If $y \in \Omega$, since $d(y, \partial\Omega) \leq \varepsilon$, we deduce that

$$\frac{Md(y, \partial\Omega)}{2} \leq \frac{M\varepsilon_0}{2} < \frac{\delta_0}{2},$$

by definition of ε_0. Hence, we can use (III.3) to assert that the point y^* defined by

$$y^* = y - \frac{d(y, \partial\Omega)}{2} M\tilde{\nu_i}$$

belongs to U_i. Let us now prove that $y^* \in \Omega^c$. By contradiction, we assume that $y^* \in \Omega$. By using (III.5) (with ε replaced by $d(y, \partial\Omega)/2$ which is less than ε_0) we find

$$\overline{B\left(\underbrace{y^* + \frac{d(y,\partial\Omega)}{2} M\tilde{\nu}_i}_{=y}, d(y,\partial\Omega)\right)} \subset \Omega,$$

which implies that $d(y,\partial\Omega) > d(y,\partial\Omega)$. This is obviously a contradiction.

We know that $y - \alpha\nu_i \in U_i \cap \Omega^c$, therefore we can use the same argument as in the proof of (III.5) to obtain that

$$B(y - \alpha\nu_i - \varepsilon\nu_i, 2\varepsilon) \subset \Omega^c,$$

and we conclude by noting that

$$B(y - \varepsilon\nu_i, \varepsilon) \subset B(y - \varepsilon\nu_i - \alpha\nu_i, \varepsilon + \alpha M) \subset B(y - \varepsilon\nu_i - \alpha\nu_i, 2\varepsilon),$$

because $\alpha M \leq M d(y,\partial\Omega)/2 \leq \varepsilon/2$.

□

A first consequence of the cone property is the following result which shows that every bounded Lipschitz domain in $\mathbb{R}^d$ is locally star-shaped.

Definition III.1.6. *A set $A \subset \mathbb{R}^d$ is said to be star-shaped if there is a point $x_0 \in A$ such that for any $x \in A$, the segment $[x_0, x]$ is included in A. We say that A is star-shaped with respect to x_0.*

Lemma III.1.7. *Let Ω be a bounded Lipschitz domain of $\mathbb{R}^d$. There exists a finite family $(\omega_i)_{i\in I}$ of star-shaped open sets of $\mathbb{R}^d$ such that*

$$\Omega = \bigcup_{i\in I} \omega_i.$$

If Ω is a nonbounded Lipschitz domain, we can cover it with a locally finite family of star-shaped open sets.

Proof.

It is enough to prove that for any point $a \in \overline{\Omega}$, there exists an open set O of $\overline{\Omega}$ containing a such that $O \cap \Omega$ is star-shaped. Indeed, since $\overline{\Omega}$ is compact, it is possible to find a countable, locally finite, covering of $\overline{\Omega}$ with such open sets. We also observe that the result is obvious for any point $a \in \Omega$, by considering a ball centred at a and included in Ω.

Let us then consider a point $a \in \partial\Omega$ and a $1 \leq i \leq N$ such that $a \in U_i$, where the covering $(U_i)_i$ is given by Theorem III.1.4. Let $\varepsilon > 0$ be given and let us define

$$O = \bigcup_{\substack{x\in\partial\Omega\cap U_i \\ |x-a|<\varepsilon}} \left(x + \left(\overset{\circ}{C}_i \cup \{0\}\right)\right),$$

where $\overset{\circ}{C}_i$ is the open truncated cone

$$\overset{\circ}{C}_i = \{z \in \mathbb{R}^d, \sin(\theta)|z| < z \cdot \tilde{\nu}_i < \delta_0\}.$$

One can easily check that O is an open set of $\overline{\Omega}$. We prove that for a suitable $\varepsilon > 0$, O is star-shaped with respect to the point $y = a + (\delta_0/2)\tilde{\nu}_i$. Each set $x + \big(\overset{\circ}{C}_i \cup \{0\}\big)$ is convex, thus it is sufficient to show that y belongs to each of these sets.

Let $x \in \partial\Omega \cap U_i$ such that $|x - a| < \varepsilon$ and let us show that $y \in x + \overset{\circ}{C}_i$, that is to say that $y - x = a - x + (\delta_0/2)\tilde{\nu}_i \in \overset{\circ}{C}_i$.

- By construction, we have

  $$(y - x) \cdot \tilde{\nu}_i = (a - x) \cdot \tilde{\nu}_i + \frac{\delta_0}{2},$$

  so that

  $$\frac{\delta_0}{2} - \varepsilon \leq (y - x) \cdot \tilde{\nu}_i \leq \frac{\delta_0}{2} + \varepsilon.$$

  Provided that $\varepsilon < \delta_0/2$, we thus have $(y - x) \cdot \tilde{\nu}_i < \delta_0$.
- Moreover, if we choose $\varepsilon > 0$ such that

  $$\sin(\theta)\left(\varepsilon + \frac{\delta_0}{2}\right) < \frac{\delta_0}{2} - \varepsilon,$$

  which is possible because $\sin(\theta) < 1$, we conclude that

  $$\sin(\theta)|y - x| < \sin(\theta)\left(\varepsilon + \frac{\delta_0}{2}\right) < \frac{\delta_0}{2} - \varepsilon \leq (y - x) \cdot \tilde{\nu}_i,$$

  and the claim is proved.

□

1.2.3 Integration on the boundary of a Lipschitz domain

Let $\mathcal{H}_a$ be a Lipschitz half-space as defined in Definition III.1.1 and φ a compactly supported continuous function on $\mathbb{R}^d$. The integral of φ on $\partial\mathcal{H}_a$ is defined as

$$\int_{\partial\mathcal{H}_a} \varphi \, d\sigma = \int_{\mathbb{R}^{d-1}} \varphi(\bar{x}, a(\bar{x}))\, J_a(\bar{x}) d\bar{x}, \tag{III.7}$$

where

$$J_a(\bar{x}) = \sqrt{1 + \sum_{i=1}^{d-1} \left(\frac{\partial a}{\partial x_i}(\bar{x})\right)^2}.$$

Notice that, by Theorem II.2.46, this quantity is well-defined for almost every $\bar{x}$ and satisfies

$$1 \leq J_a(\bar{x}) \leq \sqrt{1 + (d-1)\,\mathrm{Lip}(a)^2};$$

as a consequence, the integral (III.7) is well-defined.

For almost any $\bar{x}$ we define the following unit vector

$$\nu_{\mathcal{H}_a}(\bar{x}, a(\bar{x})) = \frac{1}{J_a(\bar{x})} \begin{pmatrix} \frac{\partial a}{\partial x_1}(\bar{x}) \\ \vdots \\ \frac{\partial a}{\partial x_{d-1}}(\bar{x}) \\ -1 \end{pmatrix}, \tag{III.8}$$

which is called the *unit outward normal* of $\mathcal{H}_a$ at the point $(\bar{x}, a(\bar{x})) \in \partial\mathcal{H}_a$.

Finally, it is easy to check that the integral (III.7) and the unit outward normal (III.8) are invariant by rotation of the orthonormal reference frame in which they are computed.

Let now Ω be a Lipschitz domain in $\mathbb{R}^d$ such that $\partial\Omega$ is bounded. We consider the finite open covering $(U_i)_{1\leq i\leq N}$ of $\partial\Omega$ introduced in Theorem III.1.4. By construction, for any i there exists a rotation R_i such that $\Omega \cap U_i$ is equal to $(R_i\mathcal{H}_{a_i}) \cap U_i$ for a suitable Lipschitz continuous map a_i. Let us then consider a partition of unity $(\psi_i)_i$ associated with this open covering of $\partial\Omega$. Then, for any continuous function φ on $\mathbb{R}^d$, we define

$$\int_{\partial\Omega} \varphi\, d\sigma = \sum_{i=1}^{N} \int_{\partial\mathcal{H}_{a_i}} \big(\varphi \circ R_i\big)\big(\psi_i \circ R_i\big)\, d\sigma.$$

Moreover, for any $1 \leq i \leq N$ and any $\sigma \in \partial\Omega \cap U_i$, we can define the unit outward normal

$$\nu_\Omega(\sigma) = R_i \nu_{\mathcal{H}_{a_i}}(R_i^{-1}\sigma).$$

It can be easily checked that those definitions depend neither on the covering $(U_i)_i$, nor on the choice of the half-spaces $\mathcal{H}_{a_i}$, nor on the partition of unity that we used.

Finally, the map $\varphi \in \mathcal{C}^0(\partial\Omega) \mapsto \int_{\partial\Omega} \varphi\, d\sigma$ defines a Radon measure $d\sigma$ on $\partial\Omega$ which lets us define the Lebesgue spaces $L^p(\partial\Omega)$, $1 \leq p \leq +\infty$ and gives a sense to the integral $\int_{\partial\Omega} \varphi\, d\sigma$ for any $d\sigma$-integrable function. Those spaces enjoy similar properties as the ones of the spaces $L^p(\Omega)$ stated in Section 2.3 of Chapter II.

Notice for instance that, with this definition, we have $\nu_\Omega \in (L^\infty(\partial\Omega))^d$. When no confusion is possible, we simply denote the outward unit normal ν_Ω as ν.

1.2.4 Stokes formula

In this section we prove the following fundamental theorem in the study of partial differential equations.

Theorem III.1.8 (Stokes formula. Divergence theorem). *Let Ω be a Lipschitz domain in $\mathbb{R}^d$ with compact boundary. For any vector field $\Phi \in (\mathcal{C}^1_c(\mathbb{R}^d))^d$ we have*

$$\int_\Omega \operatorname{div} \Phi \, dx = \int_{\partial\Omega} (\Phi \cdot \nu) \, d\sigma. \tag{III.9}$$

Let us first mention that using the formulas in Appendix A, we can state many equivalent forms of this formula. For instance, we shall often use that for any $\Phi \in (\mathcal{C}^1_c(\mathbb{R}^d))^d$ and $\psi \in \mathcal{C}^1_c(\mathbb{R}^d)$ we have

$$\int_\Omega \psi \operatorname{div} \Phi \, dx + \int_\Omega \Phi \cdot \nabla\psi \, dx = \int_{\partial\Omega} \psi(\Phi \cdot \nu) \, d\sigma. \tag{III.10}$$

This is obtained by simply applying (III.9) to the vector field $\psi\Phi$.

Proof.

We first assume that the result holds for any Lipschitz half-space $\mathcal{H}_a$ and then we prove it for a Lipschitz domain Ω with compact boundary. We use the fact that formula (III.9) is linear with respect to Φ.

- If Φ is compactly supported in Ω, then the boundary integral in (III.9) obviously vanishes and the volume integral reads

$$\int_\Omega \operatorname{div} \Phi \, dx = \int_{\mathbb{R}^d} \operatorname{div} \overline{\Phi} \, dx = \sum_{i=1}^d \int_{\mathbb{R}^d} \frac{\partial}{\partial x_i} \overline{\Phi_i} \, dx = 0,$$

  by integration by parts in $\mathbb{R}^d$.
- Let now $(U_i)_{0 \leq i \leq N}$ be the open covering of $\overline{\Omega}$ introduced in Section 1.2.2 and $(\psi_i)_{0 \leq i \leq N}$ an associated partition of unity. We can check that $\psi_0\Phi$ is compactly supported in $U_0 \subset \Omega$ so that we just have seen that it satisfies the Stokes formula. It remains to prove that (III.9) holds for any $\psi_i\Phi$, for $1 \leq i \leq N$. Since ψ_i is supported in U_i we have, by a change of variables

$$\begin{aligned}\int_\Omega \operatorname{div}(\psi_i\Phi) \, dx &= \int_{R_i\mathcal{H}_{a_i}} \operatorname{div}(\psi_i\Phi) \, dx = \int_{\mathcal{H}_{a_i}} \operatorname{div}(\psi_i\Phi) \circ R_i \, dy \\ &= \int_{\mathcal{H}_{a_i}} \operatorname{div} \Big((\psi_i \circ R_i) \, (R_i^{-1}\Phi \circ R_i) \Big) \, dy.\end{aligned}$$

  We have used here that for any vector field v we have $\operatorname{div}(v) \circ R = \operatorname{div}(R^{-1}.v \circ R)$. We assumed that the Stokes formula holds for Lipschitz half-spaces, and using that R_i^{-1} is an isometry of $\mathbb{R}^d$, we therefore deduce that

$$\begin{aligned}\int_\Omega \operatorname{div}(\psi_i\Phi)\,dx &= \int_{\partial\mathcal{H}_{a_i}} \psi_i\circ R_i\big(R_i^{-1}\Phi\circ R_i\big)\cdot\nu_{\mathcal{H}_{a_i}}\,d\sigma\\ &= \int_{\partial\mathcal{H}_{a_i}} \psi_i\circ R_i\left[\big(R_i^{-1}\Phi\circ R_i\big)\cdot\big(R_i^{-1}\nu_\Omega\circ R_i\big)\right]\,d\sigma\\ &= \int_{\partial\mathcal{H}_{a_i}} \psi_i\circ R_i\left[\big(\Phi\circ R_i\big)\cdot\nu_\Omega\circ R_i\right]\,d\sigma = \int_{\partial\Omega}\psi_i\Phi\cdot\nu_\Omega\,d\sigma,\end{aligned}$$

and the claim is proved.

- It remains to show the formula in the case of Lipschitz half-spaces. We first prove the result for a smooth half-space $\mathcal{H}_a$ with $a\in\mathcal{C}^\infty$. In this case, by the Fubini theorem, we have

$$\begin{aligned}\int_{\mathcal{H}_a}\operatorname{div}\Phi\,dx &= \int_{\mathbb{R}^{d-1}}\int_{a(\bar{x})}^{+\infty}\left(\sum_{i=1}^{d-1}\frac{\partial}{\partial x_i}\Phi_i+\frac{\partial}{\partial x_d}\Phi_d\right)dx_d\,d\bar{x}\\ &= \int_{\mathbb{R}^{d-1}}\sum_{i=1}^{d-1}\frac{\partial}{\partial x_i}\left(\int_{a(\bar{x})}^{+\infty}\Phi_i\,dx_d\right)d\bar{x}\\ &\quad+\int_{\mathbb{R}^{d-1}}\sum_{i=1}^{d-1}\frac{\partial a}{\partial x_i}(\bar{x})\Phi_i(\bar{x},a(\bar{x}))\,d\bar{x}-\int_{\mathbb{R}^{d-1}}\Phi_d(\bar{x},a(\bar{x}))\,d\bar{x}.\end{aligned}$$

Since Φ is compactly supported, an integration by parts shows that the first integral in the right-hand side vanishes. Using (III.8) and (III.7), it remains

$$\int_{\mathcal{H}_a}\operatorname{div}\Phi\,dx = \int_{\mathbb{R}^{d-1}}\Phi(\bar{x},a(\bar{x}))\cdot\nu_{\mathcal{H}_a}(\bar{x},a(\bar{x}))J_a(\bar{x})\,d\bar{x} = \int_{\partial\mathcal{H}_a}\Phi\cdot\nu_{\mathcal{H}_a}\,d\sigma,$$

and the claim is proved.

- We finally deal with the general case of a Lipschitz half-space domain $\mathcal{H}_a$. We consider the map a_γ defined in (III.1) and studied in Proposition III.1.3. For any $\varepsilon>0$, we define the smooth half-space

$$\mathcal{H}^\varepsilon_{a_\gamma} = \{x=(\bar{x},x_d)\in\mathbb{R}^d,\ x_d>a_\gamma(\bar{x},\varepsilon)\},$$

which is an approximation of $\mathcal{H}_a$.
We apply the Stokes formula in $\mathcal{H}^\varepsilon_{a_\gamma}$

$$\int_{\mathcal{H}^\varepsilon_{a_\gamma}}\operatorname{div}\Phi\,dx = \int_{\partial\mathcal{H}^\varepsilon_{a_\gamma}}\Phi\cdot\nu_{\mathcal{H}^\varepsilon_{a_\gamma}}\,d\sigma.$$

When ε goes to 0, we have $a_\gamma(\bar{x},\varepsilon)\to a(\bar{x})$ for any $\bar{x}$. Thus the characteristic function of $\mathcal{H}^\varepsilon_{a_\gamma}$ converges almost everywhere to the characteristic function of $\mathcal{H}_a$. By using the dominated convergence theorem, we deduce that the integral in the left-hand side converges to $\int_{\mathcal{H}_a}\operatorname{div}\Phi\,dx$. It remains

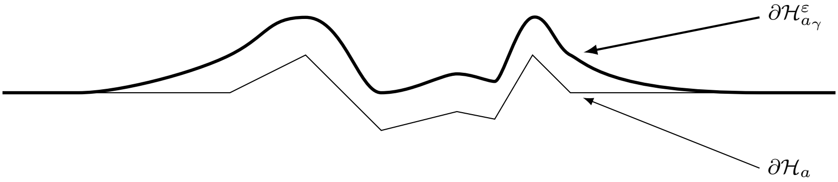


Fig. III.4 Lipschitz half-space $\mathcal{H}_a$ and its smooth approximation $\mathcal{H}^{\varepsilon}_{a_\gamma}$

to pass to the limit in the right-hand side, that is, in the formula

$$\int_{\mathbb{R}^{d-1}} \Phi(\bar{x}, a_\gamma(\bar{x},\varepsilon)) \cdot {}^t\left(\partial_{x_1} a_\gamma(\bar{x},\varepsilon), \dots, \partial_{x_{d-1}} a_\gamma(\bar{x},\varepsilon), -1\right)\, d\bar{x}.$$

This can also be justified by the Lebesgue dominated convergence theorem, as follows.

- Since Φ is continuous and compactly supported, and $\mathrm{Lip}(a_\gamma) \le \mathrm{Lip}(a)$, we can easily dominate the function which is integrated uniformly with respect to ε by some integrable function.
- From Proposition III.1.3, we know that, for any $\bar{x}$ we have $a_\gamma(\bar{x},\varepsilon) \to a(\bar{x})$ when $\varepsilon \to 0$ and that, for almost every $\bar{x}$ (using Theorem II.2.46), we have $\partial_{x_i} a_\gamma(\bar{x},\varepsilon) \to \partial_{x_i} a(\bar{x})$.

We finally find that

$$\begin{aligned}\int_{\mathcal{H}_a} \operatorname{div}\Phi\, dx &= \int_{\mathbb{R}^{d-1}} \Phi(\bar{x}, a(\bar{x})) \cdot {}^t\left(\partial_{x_1} a(\bar{x}), ..., \partial_{x_{d-1}} a(\bar{x}), -1\right)\, d\bar{x}\\ &= \int_{\mathbb{R}^{d-1}} \Phi(\bar{x}, a(\bar{x})) \cdot \nu_{\mathcal{H}_a}(\bar{x}, a(\bar{x})) J_a(\bar{x})\, d\bar{x} = \int_{\partial\mathcal{H}_a} \Phi \cdot \nu_{\mathcal{H}_a}\, d\sigma,\end{aligned}$$

and the theorem is proved.

□

2 Sobolev spaces on Lipschitz domains

We now give the main definitions and results related to Sobolev spaces (see, e.g., [1, 27, 92, 117, 33]).

2.1 Definitions

Definition III.2.1. *Let Ω be an open set of $\mathbb{R}^d$, let m be a nonnegative integer, and let $1 \leq p \leq +\infty$. We define the Sobolev spaces by*

$$W^{m,p}(\Omega) = \{f \in L^p(\Omega), \quad \partial^\alpha f \in L^p(\Omega), \quad \forall |\alpha| \leq m\},$$

where $\partial^\alpha f$ are the derivatives of f in the sense of distributions as described in Section 2.5 of Chapter II.

The Sobolev space $W^{m,p}(\Omega)$ is a Banach space for the natural norm

$$\begin{cases} \|f\|_{W^{m,p}} = \left(\sum_{|\alpha| \leq m} \|\partial^\alpha f\|_{L^p}^p \right)^{1/p}, & \text{for } p < +\infty, \\ \|f\|_{W^{m,\infty}} = \sup_{|\alpha| \leq m} \|\partial^\alpha f\|_{L^\infty}, & \text{for } p = +\infty, \end{cases}$$

or for any other equivalent norm constructed from the norms in $L^p(\Omega)$ of the weak derivatives of f. For $p = 2$, the space $W^{m,2}(\Omega)$ is denoted as $H^m(\Omega)$ and its norm is a Hilbertian norm.

For all $p \in [1, +\infty[$, we denote as $W_0^{m,p}(\Omega)$ (or $H_0^m(\Omega)$), the closure of $\mathcal{D}(\Omega)$ in $W^{m,p}(\Omega)$ (or in $H^m(\Omega)$, resp.). Of course, we have for all $p < +\infty$, $W^{0,p}(\Omega) = W_0^{0,p}(\Omega) = L^p(\Omega)$.

Remark III.2.1. In the whole space $\Omega = \mathbb{R}^d$, we can define the spaces $H^s(\mathbb{R}^d)$ (for all $s \geq 0$, not being necessarily an integer) using the Fourier transform $(f \mapsto \hat{f})$ by

$$H^s(\mathbb{R}^d) = \left\{ f \in \mathcal{S}'(\mathbb{R}^d), \quad \xi \mapsto (1 + |\xi|^2)^{s/2} \hat{f}(\xi) \in L^2(\mathbb{R}^d) \right\}, \tag{III.11}$$

where $\mathcal{S}'(\mathbb{R}^d)$ denotes the set of tempered distributions on $\mathbb{R}^d$.

We may also define and characterise local Sobolev spaces as follows.

Definition III.2.2. *Let Ω be an open set of $\mathbb{R}^d$, f be a measurable function on Ω, m be a nonnegative integer, and $1 \leq p \leq +\infty$. We say that $f \in W_{loc}^{m,p}(\Omega)$ if for any bounded open set ω such that $\overline{\omega} \subset \Omega$, we have*

$$f_{|\omega} \in W^{m,p}(\omega).$$

Proposition III.2.3. *With the same notation as in the previous definition, we have*

$$f \in W_{loc}^{m,p}(\Omega) \Longleftrightarrow \overline{\varphi f} \in W^{m,p}(\mathbb{R}^d), \, \forall \varphi \in \mathcal{D}(\Omega).$$

As usual the notation $\overline{\varphi f}$ is used for the extension by 0 of $f\varphi$ outside Ω. The proof of this result is straightforward by using Lemma II.2.37.

To begin with, we start to investigate the relationship between the functions in $W^{1,\infty}(\Omega)$ and the Lipschitz-continuous functions.

Proposition III.2.4. *Let $\Omega \subset \mathbb{R}^d$ be any open set and $f : \Omega \to \mathbb{R}$ be a bounded Lipschitz continuous function. We denote by ∇f the gradient of f, which is defined almost everywhere thanks to the Rademacher theorem (Theorem II.2.46).*

Then, we have $f \in W^{1,\infty}(\Omega)$ and its gradient in the sense of distributions is ∇f.

Moreover, if there is a continuous function $G : \Omega \to \mathbb{R}^d$ such that $G = \nabla f$ almost everywhere, then f is of class $\mathcal{C}^1$ in Ω and G is the gradient of f in the classic sense.

Proof.

We consider a Lipschitz continuous extension $F : \mathbb{R}^d \to \mathbb{R}$ of f given by Proposition II.2.45. Let $\varphi \in (\mathcal{D}(\Omega))^d$ be a test function. We have

$$\int_\Omega F \star \eta_\varepsilon (\operatorname{div} \Phi)\, dx = -\int_\Omega \nabla(F \star \eta_\varepsilon) \cdot \Phi \, dx,$$

where the convolution is defined in (II.5). By using Proposition II.2.47, and the dominated convergence theorem, we can pass to the limit in this formula to obtain

$$\int_\Omega F(\operatorname{div} \Phi)\, dx = -\int_\Omega (\nabla F) \cdot \Phi \, dx.$$

Since $F = f$ on Ω and Φ is supported in Ω, we have

$$\int_\Omega f(\operatorname{div} \Phi)\, dx = -\int_\Omega (\nabla f) \cdot \Phi \, dx,$$

which proves that ∇f is the gradient of f in the distribution sense. Hence, we have $\nabla f \in (L^\infty(\Omega))^d$ and thus $f \in W^{1,\infty}(\Omega)$.

For any $\varepsilon > 0$ we have $\nabla(F \star \eta_\varepsilon) = (\nabla F) \star \eta_\varepsilon$. Since G is continuous and $\nabla F = G$ almost everywhere in Ω, we easily see that $\nabla(F \star \eta_\varepsilon) = G \star \eta_\varepsilon$ converges locally uniformly towards G in Ω. This implies that f, which is the uniform limit of $F \star \eta_\varepsilon$, is differentiable everywhere on Ω and that its gradient is G, so that $f \in \mathcal{C}^1(\Omega)$.

□

The converse result (namely that $W^{1,\infty}(\Omega) \subset \operatorname{Lip}(\Omega)$) is not true in general. For instance if we consider $\Omega = \mathbb{R}^d_+ \cup \mathbb{R}^d_-$ and $f = \operatorname{sgn}(x_d)$, we can check that $f \in W^{1,\infty}(\Omega)$ but f is not Lipschitz continuous on Ω. This comes from the fact that Ω lies on both sides of its boundary. However, we show below, in Proposition III.2.9, that this converse inclusion is true provided that Ω is a Lipschitz domain with a compact boundary, for instance.

2.2 Mollifying operators and Friedrichs commutator estimates

From now on, we are given a Lipschitz domain Ω in $\mathbb{R}^d$ with compact boundary.

We introduce suitable mollifying operators that enable a complete study of functional spaces on Lipschitz domains in a general framework. This idea comes back to the works of Friedrichs (see, for instance, [61]). Note that using these operators is also crucial in other places in this book (see, in particular, Theorem IV.4.7 and Section 1 of Chapter VI). The presentation we give here is particularly inspired by [17].

We first need to introduce a truncation function. More precisely, let $\beta : \mathbb{R} \to \mathbb{R}$ be a nonnegative decreasing $\mathcal{C}^\infty$ function such that $\beta = 1$ on $\mathbb{R}^-$ and $\beta = 0$ on $]1, +\infty[$. For any $0 < \varepsilon < 1$, we define $\beta_\varepsilon : \mathbb{R}^d \to \mathbb{R}$ as follows, $\beta_\varepsilon(x) = \beta(|x| + \ln \varepsilon)$, so that the following properties hold,

$$\beta_\varepsilon = 1, \text{ on } B(0, |\ln \varepsilon|),$$

$$\operatorname{Supp}(\beta_\varepsilon) \subset B(0, 1 + |\ln \varepsilon|), \tag{III.12}$$

$$\|\beta_\varepsilon\|_{L^p(\mathbb{R}^d)} \leq C_{p,d}(1 + |\ln \varepsilon|)^{d/p}, \forall p \in [1, +\infty[, \tag{III.13}$$

$$\|\partial^\alpha \beta_\varepsilon\|_{L^\infty(\mathbb{R}^d)} \leq \|\partial^\alpha \beta\|_{L^\infty(\mathbb{R}^d)}, \forall \varepsilon > 0, \forall \alpha \in \mathbb{N}^d.$$

Finally, by the dominated convergence theorem we see that

$$\|\beta_\varepsilon f - f\|_{L^p(\Omega)} \xrightarrow[\varepsilon \to 0]{} 0, \quad \forall f \in L^p(\Omega), \tag{III.14}$$

and

$$\|(\partial^\alpha \beta_\varepsilon) f\|_{L^p(\Omega)} \xrightarrow[\varepsilon \to 0]{} 0, \quad \forall f \in L^p(\Omega), \forall \alpha \in \mathbb{N}^d, |\alpha| \geq 1. \tag{III.15}$$

Let $\mathcal{U} = (U_i)_{0 \leq i \leq N}$ be the open covering of $\overline{\Omega}$ that we constructed in Section 1.2.2. We consider an associated partition of unity denoted by $(\psi_i)_{0 \leq i \leq N}$ (see Lemma II.2.38).

Lemma III.2.5. *For any $y \in \overline{\Omega}$, we have*

$$\sum_{i=0}^N \psi_i(y) \int_\Omega \eta_\varepsilon(y - x + \varepsilon \nu_i)\, dx = 1, \tag{III.16}$$

$$\sum_{i=0}^N \psi_i(y) \int_{\mathbb{R}^d} \eta_\varepsilon(y - x - \varepsilon \nu_i)\, dx = 1, \tag{III.17}$$

Proof.

Let $y \in \Omega$ and $i \in \{0, \dots, N\}$. If $\psi_i(y) > 0$, then $y \in U_i$ and by (III.5), we deduce that $B(y + \varepsilon \nu_i, \varepsilon) \subset \Omega$; then

$$\int_\Omega \eta_\varepsilon(y-x+\varepsilon\nu_i)\,dx = \int_{B(y+\varepsilon\nu_i,\varepsilon)} \eta_\varepsilon(y-x+\varepsilon\nu_i)\,dx = \int_{\mathbb{R}^d} \eta(z)\,dz = 1.$$

The property (III.16) follows because $\sum_{i=0}^N \psi_i = 1$. The proof of (III.17) is even simpler because all the integrals are taken on the whole space $\mathbb{R}^d$ and thus, by a change of variable, are all equal to 1.

□

Definition III.2.6. *For any distribution $f \in \mathcal{D}'(\Omega)$ and any $\varepsilon < \varepsilon_0$, we define a function $\mathcal{S}_\varepsilon f$ by*

$$\mathcal{S}_\varepsilon f(y) = \sum_{i=0}^N \beta_\varepsilon(y)\psi_i(y)\Big\langle f, \eta_\varepsilon(y+\varepsilon\nu_i - \cdot)\Big\rangle_{\mathcal{D}',\mathcal{D}}, \forall y \in \overline{\Omega}. \tag{III.18}$$

Remark III.2.2. This definition makes sense inasmuch as for any $y \in \overline{\Omega}$ such that $\psi_i(y) \neq 0$ we have $y \in U_i$ so that by (III.5) we have $B(y+\varepsilon\nu_i,\varepsilon) \subset \Omega_{\varepsilon/2}$ which implies that

$$x \in \mathbb{R}^d \mapsto \eta_\varepsilon(y-x+\varepsilon\nu_i),$$

is compactly supported in $\Omega_{\varepsilon/2}$.

Notice that this argument shows in fact that $\mathcal{S}_\varepsilon f$ is well-defined on an open neighborhood of $\overline{\Omega}$.

Definition III.2.7. *For any function $f \in L^1(\Omega) + L^\infty(\Omega)$, and any $\varepsilon < \varepsilon_0$, we define the following function*

$$\widetilde{\mathcal{S}}_\varepsilon f(y) = \sum_{i=0}^N \beta_\varepsilon(y)\psi_i(y) \int_\Omega f(x)\eta_\varepsilon(y-x-\varepsilon\nu_i)\,dx,\ \forall y \in \mathbb{R}^d.$$

Remark III.2.3. The difference between the definition of $\mathcal{S}_\varepsilon$ and $\widetilde{\mathcal{S}}_\varepsilon$ is the sign in front of the normal ν_i in the term $\eta_\varepsilon(y-x-\varepsilon\nu_i)$. As a consequence, when y is near $\partial\Omega$, the support of $x \mapsto \eta_\varepsilon(y-x-\varepsilon\nu_i)$ is located *outside* Ω. It follows that

- Contrary to $\mathcal{S}_\varepsilon f$, the function $\widetilde{\mathcal{S}}_\varepsilon f$ is compactly supported in Ω (see Proposition III.2.8).
- The definition of $\widetilde{\mathcal{S}}_\varepsilon f$ cannot be generalised to distributions f.

These definitions are extended to vector-valued functions in a straightforward way. It is useful to notice the following equivalent definition of $\widetilde{\mathcal{S}}_\varepsilon$,

$$\widetilde{\mathcal{S}}_\varepsilon f(y) = \sum_{i=0}^N \beta_\varepsilon(y)\psi_i(y) \int_{\mathbb{R}^d} \overline{f}(x)\eta_\varepsilon(y-x-\varepsilon\nu_i)\,dx,\ \forall y \in \mathbb{R}^d. \tag{III.19}$$

Moreover, if $f \in L^1_{loc}(\Omega)$, we can simply write

$$\mathcal{S}_\varepsilon f(y) = \sum_{i=0}^{N} \beta_\varepsilon(y)\psi_i(y) \int_\Omega f(x)\eta_\varepsilon(y - x + \varepsilon\nu_i)\, dx,\ \forall y \in \overline{\Omega}. \tag{III.20}$$

The first properties of these mollifying operators are summed up in the following result.

Proposition III.2.8. *1. We have*

$$\mathcal{S}_\varepsilon f \in \mathcal{C}_c^\infty(\overline{\Omega}),\ \forall f \in \mathcal{D}'(\Omega),$$
$$\widetilde{\mathcal{S}}_\varepsilon f \in \mathcal{C}_c^\infty(\Omega),\ \forall f \in L^1(\Omega) + L^\infty(\Omega).$$

Moreover, for any $p \in [1, +\infty]$, and any $f \in L^p(\Omega)$, we have

$$\|\mathcal{S}_\varepsilon f\|_{L^p(\Omega)} \leq \|f\|_{L^p(\Omega)} \text{ and } \|\widetilde{\mathcal{S}}_\varepsilon f\|_{L^p(\Omega)} \leq \|f\|_{L^p(\Omega)}, \tag{III.21}$$

$$\|\nabla(\mathcal{S}_\varepsilon f)\|_{L^p(\Omega)} \leq \frac{C}{\varepsilon}\|f\|_{L^p(\Omega)} \text{ and } \|\nabla(\widetilde{\mathcal{S}}_\varepsilon f)\|_{L^p(\Omega)} \leq \frac{C}{\varepsilon}\|f\|_{L^p(\Omega)}.$$

2. Let $p, q \in [1, +\infty]$ with $(p, q) \neq (+\infty, +\infty)$, and $q \geq p'$. Let $r \in [1, +\infty[$ such that $1/r = 1/p + 1/q$. For any $\alpha \in L^q(\Omega)$ and $f \in L^p(\Omega)$ we have

$$\|\mathcal{S}_\varepsilon(\alpha f) - \alpha(\mathcal{S}_\varepsilon f)\|_{L^r} \xrightarrow[\varepsilon \to 0]{} 0,$$

$$\|\widetilde{\mathcal{S}}_\varepsilon(\alpha f) - \alpha(\widetilde{\mathcal{S}}_\varepsilon f)\|_{L^r} \xrightarrow[\varepsilon \to 0]{} 0,$$

so that in particular, for any $p \in [1, +\infty[$, $\mathcal{S}_\varepsilon f$ and $\widetilde{\mathcal{S}}_\varepsilon f$ both converge towards f in $L^p(\Omega)$ when ε goes to 0.

It can also be shown that if $f \in L^\infty(\Omega)$ then $\mathcal{S}_\varepsilon f$ and $\widetilde{\mathcal{S}}_\varepsilon f$ both weakly-$\star$ converge to f in $L^\infty(\Omega)$.

Proof.

1. The truncation function β and the kernel η are smooth, thus it is clear that $\mathcal{S}_\varepsilon f$ and $\widetilde{\mathcal{S}}_\varepsilon f$ both belong to $\mathcal{C}^\infty(\overline{\Omega})$. Moreover, by (III.12), their supports are bounded.
   For $f \in L^p(\Omega)$, we observe that, because of (III.16)–(III.17) and (III.19)–(III.20), the values of $|\mathcal{S}_\varepsilon f|$ and $|\widetilde{\mathcal{S}}_\varepsilon f|$ are bounded by convex combinations of values of $|f|$, as soon as $f \in L^p(\Omega)$. As a consequence, the Jensen inequality (Proposition II.2.20) gives that for any $1 \leq p < +\infty$ and any $f \in L^p(\Omega)$,

   $$|\mathcal{S}_\varepsilon f(y)|^p \leq \mathcal{S}_\varepsilon(|f|^p)(y), \text{ and } |\widetilde{\mathcal{S}}_\varepsilon f(y)|^p \leq \widetilde{\mathcal{S}}_\varepsilon(|f|^p)(y),\ \forall y \in \Omega$$

   so that estimates (III.21) hold.
   Using (III.20), we can compute

$$\nabla \mathcal{S}_\varepsilon f(y) = \sum_{i=0}^{N} \nabla(\beta_\varepsilon \psi_i)(y) \int_\Omega f(x)\eta_\varepsilon(y - x + \varepsilon\nu_i)\,dx$$
$$+ \frac{1}{\varepsilon}\sum_{i=0}^{N} \beta_\varepsilon(y)\psi_i(y) \int_\Omega f(x)(\nabla\eta)_\varepsilon(y - x + \varepsilon\nu_i)\,dx. \quad \text{(III.22)}$$

Thus, the claimed estimates follow by using once more the Jensen inequality and the fact that for $y \in U_i$ we have

$$\int_\Omega |(\nabla\eta)_\varepsilon|(y - x + \varepsilon\nu_i)\,dx = \int_{\mathbb{R}^d} |\nabla\eta(x)|\,dx,$$

which does not depend on ε. A similar computation leads to the corresponding estimate for $\widetilde{\mathcal{S}}_\varepsilon f$.

It remains to show that $\widetilde{\mathcal{S}}_\varepsilon f$ is compactly supported in Ω. We already know that its support is contained in $B(0, 1 + |\ln\varepsilon|)$, therefore it is enough to prove now that its support is contained in $\Omega_{\varepsilon/M} \cup U_0$, where we recall that $U_0 = \Omega_{\delta_0/2}$. For $y \in \overline{\Omega} \setminus (\Omega_{\varepsilon/M} \cup U_0)$, we know that $\psi_0(y) = 0$ and moreover, for any $i \in \{1, \dots, n\}$ such that $\psi_i(y) \neq 0$, we have $y \in U_i$ and $d(y, \partial\Omega) \le \varepsilon/M$ by assumption. Thus, the support of $x \mapsto \eta_\varepsilon(y - x - \varepsilon\nu_i)$ which is included in the ball $B(y - \varepsilon\nu_i, \varepsilon)$ is itself included in Ω^c, thanks to (III.6).

As a consequence, all the terms of the sum in the definition of $\widetilde{\mathcal{S}}_\varepsilon f(y)$ vanish for such a point y.

2. • We first focus on the case $\alpha = 1$, $q = +\infty$ and $p < +\infty$.
By the previous estimates, it is enough to show the convergence property for any f in a dense subset of $L^q(\Omega)$. For instance we consider a function $f \in \mathcal{C}_c^0(\Omega)$ and, by (III.16), we find that for any $y \in \overline{\Omega}$, we have

$$\mathcal{S}_\varepsilon f(y) - \beta_\varepsilon(y) f(y) = \sum_{i=0}^{N} \beta_\varepsilon(y)\psi_i(y) \int_\Omega (f(x) - f(y))\eta_\varepsilon(y - x + \varepsilon\nu_i)\,dx,$$

from which we deduce that

$$|\mathcal{S}_\varepsilon f(y) - \beta_\varepsilon f(y)| \le \mathrm{Lip}(f)(1 + M)\varepsilon\beta_\varepsilon(y),$$

and thus by (III.13) and (III.14) we have

$$\|\mathcal{S}_\varepsilon f - f\|_{L^p(\Omega)} \le C\varepsilon\|\beta_\varepsilon\|_{L^p(\mathbb{R}^d)} + \|\beta_\varepsilon f - f\|_{L^p(\Omega)} \xrightarrow[\varepsilon\to 0]{} 0,$$

so that the claim is proved for the operator $\mathcal{S}_\varepsilon$. A similar computation gives the result for the operator $\widetilde{\mathcal{S}}_\varepsilon$, by using the expression (III.19), the formula (III.17), and the fact that, since f is compactly supported in Ω, its extension $\bar{f}$ is Lipschitz continuous on the whole space $\mathbb{R}^d$.

- It remains to treat the general case $\alpha \in L^q(\Omega)$ and $f \in L^p(\Omega)$. We only give the proof for $\mathcal{S}_\varepsilon$ because the one for $\widetilde{\mathcal{S}}_\varepsilon$ is the same.
  - If $p < +\infty$, then we have

$$\begin{aligned}\|\mathcal{S}_\varepsilon(\alpha f) - \alpha\mathcal{S}_\varepsilon(f)\|_{L^r} &\leq \|\mathcal{S}_\varepsilon(\alpha f) - \alpha f\|_{L^r} + \|\alpha f - \alpha\mathcal{S}_\varepsilon(f)\|_{L^r} \\ &\leq \|\mathcal{S}_\varepsilon(\alpha f) - \alpha f\|_{L^r} + \|\alpha\|_{L^q}\|f - \mathcal{S}_\varepsilon(f)\|_{L^p},\end{aligned}$$

    and we can conclude by applying the convergence we just obtained to the functions $\alpha f \in L^r(\Omega)$ and $f \in L^q(\Omega)$.
  - If $p = +\infty$ (so that $r = q < +\infty$), we need to use a slightly different argument. We take $\varepsilon' > 0$, $\varepsilon'' > 0$ and we write

$$\begin{aligned}\|\mathcal{S}_\varepsilon(\alpha f) - \alpha\mathcal{S}_\varepsilon(f)\|_{L^q} \leq &\|\mathcal{S}_\varepsilon(\alpha f) - \alpha f\|_{L^q} + \|(\alpha - \mathcal{S}_{\varepsilon'}\alpha)(f - \mathcal{S}_\varepsilon f)\|_{L^q} \\ &+ \|\mathcal{S}_{\varepsilon'}\alpha(f - \mathcal{S}_\varepsilon f)\|_{L^q} \\ \leq &\|\mathcal{S}_\varepsilon(\alpha f) - \alpha f\|_{L^q} + C\|f\|_{L^\infty}\|\alpha - \mathcal{S}_{\varepsilon'}\alpha\|_{L^q} \\ &+ \|\mathcal{S}_{\varepsilon'}\alpha\|_{L^\infty}\|f - \mathcal{S}_\varepsilon f\|_{L^q} \\ \leq &\|\mathcal{S}_\varepsilon(\alpha f) - \alpha f\|_{L^q} + C\|f\|_{L^\infty}\|\alpha - \mathcal{S}_{\varepsilon'}\alpha\|_{L^q} \\ &+ \|\mathcal{S}_{\varepsilon'}\alpha\|_{L^\infty}\|\beta_{\varepsilon''}f - \mathcal{S}_\varepsilon(\beta_{\varepsilon''}f)\|_{L^q} \\ &+ 2\|\mathcal{S}_{\varepsilon'}\alpha\|_{L^\infty}\|f - \beta_{\varepsilon''}f\|_{L^q}.\end{aligned}$$

    Using the previous convergence results on $\mathcal{S}_\varepsilon$ in $L^q(\Omega)$ with finite q and since $\beta_{\varepsilon''}f \in L^q(\Omega)$, we get for any $\varepsilon', \varepsilon'' > 0$ the inequality

$$\begin{aligned}\limsup_{\varepsilon\to 0}\|\mathcal{S}_\varepsilon(\alpha f) - \alpha\mathcal{S}_\varepsilon(f)\|_{L^q} \leq C\|f\|_{L^\infty}&\|\alpha - \mathcal{S}_{\varepsilon'}\alpha\|_{L^p} \\ &+ 2\|\mathcal{S}_{\varepsilon'}\alpha\|_{L^\infty}\|f - \beta_{\varepsilon''}f\|_{L^q}.\end{aligned}$$

    We first take the limit when ε'' goes to 0, using (III.14), and then the limit ε' goes to 0. We obtain that

$$\limsup_{\varepsilon\to 0}\|\mathcal{S}_\varepsilon(\alpha f) - \alpha\mathcal{S}_\varepsilon(f)\|_{L^q} = 0,$$

    and the proof is complete.

□

We can now prove the following result, which is the reciprocal property of the one given in Proposition III.2.4. It implies that, for a smooth enough domain, the set of bounded Lipschitz continuous functions can be identified to the Sobolev space $W^{1,\infty}$.

Proposition III.2.9. *Let Ω be a Lipschitz domain in $\mathbb{R}^d$ with compact boundary. Any $f \in W^{1,\infty}(\Omega)$ is equal almost everywhere to a Lipschitz continuous function in Ω, still referred to as f, and we have*

$$\mathrm{Lip}(f) \leq C\|f\|_{W^{1,\infty}(\Omega)},$$

for some $C > 0$ depending only on Ω.

Proof.

We have seen that $\mathcal{S}_\varepsilon f \in \mathcal{C}_c^\infty(\mathbb{R}^d)$ and by integrating by parts the last term in formula (III.22) we get that

$$\|\nabla \mathcal{S}_\varepsilon f\|_{L^\infty} \le \sum_{i=0}^{N} (\|\nabla\beta\|_{L^\infty}\|\psi_i\|_{L^\infty} + \|\nabla\psi_i\|_{L^\infty})\|f\|_{L^\infty} + \sum_{i=0}^{N} \|\psi_i\|_{L^\infty}\|\nabla f\|_{L^\infty}.$$

Therefore, $\mathcal{S}_\varepsilon f$ is Lipschitz continuous on $\mathbb{R}^d$ and using the mean-value theorem we get

$$\mathrm{Lip}(\mathcal{S}_\varepsilon f) \le C\|f\|_{W^{1,\infty}}. \tag{III.23}$$

By the Ascoli theorem (see Theorem II.3.1), we deduce that $(\mathcal{S}_\varepsilon f)_\varepsilon$ has a subsequence which converges locally uniformly towards some continuous function F. By the results of Proposition III.2.8, this limit is certainly equal to f almost everywhere in Ω. We conclude that f is continuous in Ω, and passing to the limit in the following property (obtained from (III.23))

$$|\mathcal{S}_\varepsilon f(x) - \mathcal{S}_\varepsilon f(y)| \le C\|f\|_{W^{1,\infty}}|x-y|, \ \forall x, y \in \Omega,$$

we conclude that f is Lipschitz continuous in Ω and satisfies the required Lipschitz estimate.

□

We can now investigate the properties of the commutators between the mollifying operators $\mathcal{S}_\varepsilon$, $\widetilde{\mathcal{S}}_\varepsilon$ and first-order differential operators. The main result in this direction (which is often called Friedrichs' lemma) is the following. We give it in a quite general form which is useful to us in Chapter VI.

Theorem III.2.10. *Let Ω be a Lipschitz domain of $\mathbb{R}^d$ with compact boundary. Let $k \in \mathbb{N}^*$, $p, q \in [1, +\infty]$, with $q \ge p'$ and $(p, q) \ne (+\infty, +\infty)$ and let $r \in [1, +\infty[$ be defined by $1/r = 1/p + 1/q$.*

For $j \in \{1, \dots, d\}$, let $\alpha_j \in (W^{1,q}(\Omega))^k$ and let $D : (L^p(\Omega))^k \to \mathcal{D}'(\Omega)$ be the differential operator defined by

$$Du = \sum_{j=1}^{d} \frac{\partial}{\partial x_j}(\alpha_j \cdot u), \ \forall u \in (L^p(\Omega))^k. \tag{III.24}$$

For any $u \in (L^p(\Omega))^k$, we have

$$\|\mathcal{S}_\varepsilon Du - D\mathcal{S}_\varepsilon u\|_{L^r} \le C\|u\|_{L^p}\|\alpha\|_{W^{1,q}}, \tag{III.25}$$

for some $C > 0$ independent of u, α and ε, and moreover

$$\|\mathcal{S}_\varepsilon Du - D\mathcal{S}_\varepsilon u\|_{L^r} \xrightarrow[\varepsilon\to 0]{} 0.$$

In particular, if $Du \in L^r(\Omega)$ then we have $D\mathcal{S}_\varepsilon u \xrightarrow[\varepsilon\to 0]{} Du$ in $L^r(\Omega)$.

Remark III.2.4. A similar result holds with $L^p(\Omega)$, $L^q(\Omega)$, and $L^r(\Omega)$ replaced with $W^{m,p}(\Omega)$, $W^{m,q}(\Omega)$, and $W^{m,r}(\Omega)$ that we do not state for simplicity.

Before proving the theorem above, let us deduce the following density result.

Theorem III.2.11. *Let Ω be a Lipschitz domain in $\mathbb{R}^d$ with compact boundary and $1 \leq p < +\infty$, $m \in \mathbb{N}^*$. The set $\mathcal{C}_c^\infty(\overline{\Omega})$ is dense in $W^{m,p}(\Omega)$.*

Proof.

We only give the proof for $m = 1$ because the one for $m > 1$ uses Theorem III.2.10 in the case that we have not presented (see Remark III.2.4).

For $u \in W^{1,p}(\Omega)$, and $l \in \{1, \dots, d\}$, we can apply Theorem III.2.10 with $k = 1$ and $\alpha_j(x) = \delta_{jl}$ so that $D = \partial/\partial x_l$. We obtain

$$u \in L^p(\Omega) \text{ and } \frac{\partial u}{\partial x_l} \in L^p(\Omega) \implies \frac{\partial \mathcal{S}_\varepsilon u}{\partial x_l} \xrightarrow[\varepsilon\to 0]{} \frac{\partial u}{\partial x_l} \text{ in } L^p(\Omega),$$

and thus $\mathcal{S}_\varepsilon u \in \mathcal{C}_c^\infty(\overline{\Omega})$ converges to u in $W^{1,p}(\Omega)$ as $\varepsilon \to 0$.

$\square$

Remark III.2.5. The proof of Theorem III.2.11 gives in fact a little more precise result that can be useful in some cases. Indeed, using the properties of the mollifying operator $\mathcal{S}_\varepsilon$, one can check that if $u \in W^{1,p}(\Omega)$ is nonnegative, then it can be approximated by nonnegative functions in $\mathcal{C}_c^\infty(\overline{\Omega})$. Similarly, if $u \in W^{1,p}(\Omega) \cap L^\infty(\Omega)$, then it can be approximated in $W^{1,p}(\Omega)$ by a sequence of functions in $\mathcal{C}_c^\infty(\overline{\Omega})$ which is, additionally, bounded in $L^\infty(\Omega)$.

Proof (of Theorem III.2.10).

For any $y \in \Omega$, we build a test function $\varphi_{y,\varepsilon} \in \mathcal{D}(\Omega)$ defined by

$$\varphi_{y,\varepsilon}(x) = \Phi(y) \sum_{i=0}^{N} \beta_\varepsilon(y)\psi_i(y)\eta_\varepsilon(y - x + \varepsilon\nu_i), \; \forall x \in \Omega.$$

The fact that $\varphi_{y,\varepsilon}$ is compactly supported in Ω comes from (III.5). Indeed, if $x \in \overline{\Omega}$ is such that $\varphi_{y,\varepsilon}(x) \neq 0$, there exists $i \in \{0, \dots, N\}$ such that $\psi_i(y) \neq 0$ and $\eta_\varepsilon(y - x + \varepsilon\nu_i) \neq 0$. This implies that $y \in U_i$ and $x \in B(y + \varepsilon\nu_i, \varepsilon)$ so that, by (III.5),

$$B(x, \varepsilon) \subset B(y - x + \varepsilon\nu_i, 2\varepsilon) \subset \Omega.$$

We deduce that $x \in \Omega$ and $\delta(x) \geq \varepsilon$. We have proved that the support of $\varphi_{y,\varepsilon}$ is contained in Ω_ε.

By definition of $\mathcal{S}_\varepsilon$, we have

$$\mathcal{S}_\varepsilon(Du)(y) = \langle Du, \varphi_{y,\varepsilon} \rangle_{\mathcal{D}',\mathcal{D}}.$$

It follows that

$$\mathcal{S}_\varepsilon(Du)(y) = -\sum_{j=1}^{d}\sum_{i=0}^{N} \beta_\varepsilon(y)\psi_i(y)\int_\Omega \frac{\partial}{\partial x_j}\big(\eta_\varepsilon(y-x+\varepsilon\nu_i)\big)\alpha_j(x)\cdot u(x)\,dx.$$

Moreover, by definition, we have

$$\begin{aligned}
D(\mathcal{S}_\varepsilon u)(y) &= \sum_{j=1}^{d} \frac{\partial}{\partial y_j}\big(\alpha_j(y)\cdot \mathcal{S}_\varepsilon u(y)\big)\\
&= \sum_{j=1}^{d}\sum_{i=0}^{N}\int_\Omega u(x)\cdot \frac{\partial}{\partial y_j}\big(\alpha_j(y)\beta_\varepsilon(y)\psi_i(y)\eta_\varepsilon(y-x+\varepsilon\nu_i)\big)\,dx.
\end{aligned}$$

Therefore, if we define $R_\varepsilon = D(\mathcal{S}_\varepsilon u) - \mathcal{S}_\varepsilon(D)u$, we have obtained

$$\begin{aligned}
R_\varepsilon(y) =& \sum_{j=1}^{d}\sum_{i=0}^{N}\int_\Omega u(x)\cdot\Bigg(\alpha_j(x)\beta_\varepsilon(y)\psi_i(y)\frac{\partial}{\partial x_j}\big(\eta_\varepsilon(y-x+\varepsilon\nu_i)\big)\\
&+\frac{\partial}{\partial y_j}\big(\alpha_j(y)\beta_\varepsilon(y)\psi_i(y)\eta_\varepsilon(y-x+\varepsilon\nu_i)\big)\Bigg)\,dx.
\end{aligned}$$

We now prove the estimate (III.25).

We first observe that, since $\alpha_j \in (W^{1,q}(\Omega))^k$, we have $R_\varepsilon \in L^q(\Omega) \subset L^r(\Omega)$ because $\operatorname{Supp} R_\varepsilon \subset \operatorname{Supp}\beta_\varepsilon$ is bounded. By using that

$$\frac{\partial}{\partial x_j}\big(\eta_\varepsilon(y-x+\varepsilon\nu_i)\big) = -\frac{\partial}{\partial y_j}\big(\eta_\varepsilon(y-x+\varepsilon\nu_i)\big),$$

we can write $R_\varepsilon = R_{\varepsilon,0} + \sum_{j=1}^d R_{\varepsilon,j}$ with

$$\begin{aligned}
R_{\varepsilon,0}(y) &= \sum_{j=1}^{d}\sum_{i=0}^{N}\int_\Omega u(x)\cdot\frac{\partial(\alpha_j\beta_\varepsilon\psi_i)}{\partial y_j}(y)\eta_\varepsilon(y-x+\varepsilon\nu_i)\,dx,\\
R_{\varepsilon,j}(y) &= \sum_{i=0}^{N}\int_\Omega \beta_\varepsilon(y)\psi_i(y)u(x)\cdot(\alpha_j(y)-\alpha_j(x))\frac{\partial}{\partial y_j}\big(\eta_\varepsilon(y-x+\varepsilon\nu_i)\big)\,dx.
\end{aligned}$$

Let us study those terms separately.

- The term $R_{\varepsilon,0}$ is written

$$R_{\varepsilon,0}(y) = \sum_{j=1}^{d}\sum_{i=0}^{N}\frac{\partial}{\partial y_j}(\alpha_j(y)\beta_\varepsilon(y)\psi_i(y))\cdot\left(\int_\Omega u(x)\eta_\varepsilon(y-x+\varepsilon\nu_i)\,dx\right).$$

  By using Proposition II.2.25 (with the translated kernel $\eta(.+\nu_i)$) we deduce that

$$R_{\varepsilon,0} \xrightarrow[\varepsilon\to 0]{} \sum_{j=1}^{d}\sum_{i=0}^{N} \frac{\partial}{\partial y_j}(\alpha_j\psi_i)\cdot u = \sum_{j=1}^{d}\left(\frac{\partial}{\partial y_j}\alpha_j\right)\cdot u, \text{ in } L^r(\Omega), \tag{III.26}$$

where we have used that $\sum_{i=0}^{N} \partial\psi_i/\partial y_j = 0$, for any $j \in \{1,\dots,d\}$, inasmuch as $(\psi_i)_i$ is a partition of unity.
Note that we can immediately prove the bound

$$\|R_{\varepsilon,0}\|_{L^r} \le C\|u\|_{L^p}\sum_{j=1}^{d}\|\alpha_j\|_{W^{1,q}}. \tag{III.27}$$

- Since ψ_i is supported in U_i and using (III.5), we can write

$$\begin{aligned}
&R_{\varepsilon,j}(y)\\
=&\sum_{i=0}^{N}\beta_\varepsilon(y)\psi_i(y)\int_{B(y+\varepsilon\nu_i,\varepsilon)} u(x)\cdot(\alpha_j(y)-\alpha_j(x))\frac{\partial}{\partial y_j}\big(\eta_\varepsilon(y-x+\varepsilon\nu_i)\big)\,dx\\
=&\sum_{i=0}^{N}\beta_\varepsilon(y)\psi_i(y)\int_{B} u(y+\varepsilon\nu_i-\varepsilon z)\cdot\frac{\alpha_j(y)-\alpha_j(y+\varepsilon\nu_i-\varepsilon z)}{\varepsilon}\frac{\partial\eta}{\partial z_j}(z)\,dz\\
=&\sum_{i=0}^{N}\beta_\varepsilon(y)\psi_i(y)\\
&\times\int_B\int_0^1 u(y+\varepsilon\nu_i-\varepsilon z)\cdot\Big[\nabla\alpha_j(y+t\varepsilon\nu_i-t\varepsilon z).(z-\nu_i)\Big]\frac{\partial\eta}{\partial z_j}(z)\,dz\,dt.
\end{aligned}$$

It follows that

$$\begin{aligned}
&\int_\Omega |R_{\varepsilon,j}(y)|^r\,dy\\
\le& C\sum_{i=0}^{N}\int_{U_i}\int_B\int_0^1 |u|^r(y+\varepsilon\nu_i-\varepsilon z)|\nabla\alpha_j|^r(y+t\varepsilon\nu_i-t\varepsilon z)\,dz\,dt\,dy\\
\le& C\sum_{i=0}^{N}\left(\int_{U_i}\int_B |u|^p(y+\varepsilon\nu_i-\varepsilon z)\,dy\,dz\right)^{r/p}\\
&\times\left(\int_{U_i}\int_B\int_0^1 |\nabla\alpha_j|^q(y+t\varepsilon\nu_i-t\varepsilon z)\,dz\,dt\,dy\right)^{p/q}.
\end{aligned}$$

Notice that, for any $t\in[0,1]$ and $z\in B$, the map $y\mapsto y+t\varepsilon\nu_i-t\varepsilon z$ maps U_i onto some open subset of Ω. Thus, by change of variables, we conclude that

$$\|R_{\varepsilon,j}\|_{L^r} \le C\|u\|_{L^p}\|\nabla\alpha_j\|_{L^q}. \tag{III.28}$$

Furthermore, by integration by parts, we observe that

$$\int_B (z-\nu_i)\frac{\partial \eta}{\partial z_j}\,dz = -e_j,$$

where e_j is the jth vector of the canonical basis of $\mathbb{R}^d$. Hence, when u and α_j are smooth enough we immediately see, by using the dominated convergence theorem, that

$$\begin{aligned} R_{\varepsilon,j}(y) \xrightarrow[\varepsilon\to 0]{} & \sum_{i=0}^N \psi_i(y)u(y)\cdot\left[\nabla\alpha_j(y).\left(\int_B (z-\nu_i)\frac{\partial\eta}{\partial z_j}\,dz\right)\right] \\ &= -\sum_{i=0}^N \psi_i(y)u(y)\cdot\frac{\partial\alpha_j}{\partial y_j}(y) = -u(y)\cdot\frac{\partial\alpha_j}{\partial y_j}(y), \end{aligned}$$

uniformly with respect to y, and thus in particular we have

$$R_{\varepsilon,j} \xrightarrow[\varepsilon\to 0]{} -u\cdot\frac{\partial\alpha_j}{\partial y_j},\text{ in } L^r(\Omega). \tag{III.29}$$

- Let us denote by $\mathcal{R}_\varepsilon$ the bilinear map which, to any $u \in L^p(\Omega)$, $\alpha = (\alpha_i)_i \in (W^{1,q}(\mathbb{R}))^k$ associates the term $R_\varepsilon \in L^r(\Omega)$ defined above. Gathering (III.27)–(III.28) and (III.26)–(III.29), we finally proved that

$$\|\mathcal{R}_\varepsilon(u,\alpha)\|_{L^r} \le C\|u\|_{L^p}\|\alpha\|_{W^{1,q}}, \tag{III.30}$$

$$\|\mathcal{R}_\varepsilon(u,\alpha)\|_{L^r} \xrightarrow[\varepsilon\to 0]{} 0, \text{ as soon as } u\in\mathcal{C}^\infty(\overline{\Omega}), \alpha\in\mathcal{C}^\infty(\overline{\Omega})^{kd}.$$

We are now able to conclude the proof of this theorem.

- In the case $p < +\infty$ and $q < +\infty$, we know that $\mathcal{C}^\infty(\overline{\Omega})$ is dense in $L^p(\Omega)$ and in $W^{1,q}(\Omega)$. It is then obvious to deduce that $\|\mathcal{R}_\varepsilon(u,\alpha)\|_{L^r} \to 0$ for any $(u,\alpha) \in L^p(\Omega)\times(W^{1,q}(\Omega))^{kd}$.
- In the case $p = +\infty$ and $q < +\infty$ (and thus $r = q$), $\mathcal{C}^\infty(\overline{\Omega})$ is not dense in $L^\infty(\Omega)$ but we can argue as follows. For any $n \ge 1$, there exists an $\alpha^n \in (\mathcal{C}^\infty(\overline{\Omega}))^k$ such that $\|\alpha-\alpha^n\|_{W^{1,q}} \le 1/n$. We have $\alpha^n \in (H^1(\Omega))^k$, $\forall n \ge 1$, and $u \in L^\infty(\Omega) \subset L^2(\Omega)$, thus we can find a sequence $(u^n)_n \subset \mathcal{C}^\infty(\overline{\Omega})$ such that

$$\|u^n - u\|_{L^2} \le \frac{1}{n\|\alpha^n\|_{H^1}}.$$

Using (III.30) with the chosen exponents (p,q) but also with the exponents $(2,2)$, we obtain that

$$\begin{aligned}\|\mathcal{R}_\varepsilon(u,\alpha)\|_{L^q} \leq& \|\mathcal{R}_\varepsilon(u,\alpha-\alpha^n)\|_{L^q} + \underbrace{\|\mathcal{R}_\varepsilon(u-u^n,\alpha^n)\|_{L^q}}_{\leq C\|\mathcal{R}_\varepsilon(u-u^n,\alpha^n)\|_{L^\infty}}\\ &+ \|\mathcal{R}_\varepsilon(u^n,\alpha^n)\|_{L^q}\\ \leq& C\|u\|_{L^\infty}\|\alpha-\alpha^n\|_{W^{1,q}} + C' \underbrace{\|u-u^n\|_{L^2}}_{\leq (n\|\alpha^n\|_{H^1})^{-1}} \|\alpha^n\|_{H^1}\\ &+ \|\mathcal{R}_\varepsilon(u^n,\alpha^n)\|_{L^q}\\ \leq& \frac{C\|u\|_{L^\infty}}{n} + \frac{C'}{n} + \|\mathcal{R}_\varepsilon(u^n,\alpha^n)\|_{L^q}.\end{aligned}$$

For a fixed n, the last term of this inequality tends to 0 when $\varepsilon \to 0$ because u^n and α^n are smooth. We deduce that

$$0 \leq \limsup_{\varepsilon\to 0} \|\mathcal{R}_\varepsilon(u,\alpha)\|_{L^q} \leq \frac{C\|u\|_{L^\infty}}{n} + \frac{C'}{n}.$$

We can now let $n \to +\infty$ to deduce that $\limsup_{\varepsilon\to 0} \|\mathcal{R}_\varepsilon(u,\alpha)\|_{L^q} = 0$, which proves the claim.

– The case $p < +\infty$ and $q = +\infty$ can be treated in the same way.

□

Remark III.2.6. Notice that the density argument used at the end of the above proof is only needed when the coefficients α_j are not constant. This is coherent with the fact that, in order to prove the density Theorem III.2.11, we only used Theorem III.2.10 in the constant coefficient case.

We now state and prove a result similar to Theorem III.2.10 for the operator $\widetilde{\mathcal{S}}_\varepsilon$. For this result to be true, we need a stronger assumption on the function u, namely that $\bar{u}$, the extension of u by 0, satisfies $D\bar{u} \in L^r(\mathbb{R}^d)$. As we show later, this assumption is equivalent to asking that some well-adapted trace of u vanishes on $\partial\Omega$ (see Theorem III.2.45).

Theorem III.2.12. *Under the same assumptions as in Theorem III.2.10, for any $u \in (L^p(\Omega))^k$ such that $D\bar{u} \in L^r(\mathbb{R}^d)$, we have*

$$\|\widetilde{\mathcal{S}}_\varepsilon Du - D\widetilde{\mathcal{S}}_\varepsilon u\|_{L^r(\Omega)} \xrightarrow[\varepsilon\to 0]{} 0.$$

In particular we have $D\widetilde{\mathcal{S}}_\varepsilon u \xrightarrow[\varepsilon\to 0]{} Du$ in $L^r(\Omega)$.

Proof.

The computation of $D\bar{u}$ needs a priori to extend the coefficients α_j to the whole space $\mathbb{R}^d$, but in fact the result does not depend on the particular extension we choose. More precisely, the statement that $D\bar{u} = f \in L^r(\mathbb{R}^d)$ means that we have

$$-\int_\Omega \alpha_j(x)\cdot u(x)\frac{\partial\varphi}{\partial x_j}(x)\,dx = \int_\Omega f(x)\varphi(x)\,dx,\ \forall\varphi\in\mathcal{C}_c^\infty(\overline{\Omega}). \tag{III.31}$$

The important point is that this formula holds even if φ is not compactly supported in Ω. As a consequence, we can follow the same line of proof as for Theorem III.2.10 by taking

$$\tilde{\varphi}_{y,\varepsilon}(x) = \Phi(y) \sum_{i=0}^{N} \beta_\varepsilon(y)\psi_i(y)\eta_\varepsilon(y - x - \varepsilon\nu_i),$$

as a test function in (III.31). Note that, since the sign in front of $\varepsilon\nu_i$ has been changed, this function $\tilde{\varphi}_{y,\varepsilon}$ is not compactly supported in Ω.

It remains to integrate the result with respect to y in Ω. We finally get a formula for $\tilde{R}_\varepsilon = D(\widetilde{\mathcal{S}}_\varepsilon u) - \widetilde{\mathcal{S}}_\varepsilon(D\bar{u})$ which is similar to the one obtained in the proof of Theorem III.2.10 and we can conclude in the very same way as in this proof.

□

2.3 Change of variables

Theorem III.2.13. *Let Ω and $\widetilde{\Omega}$ be two Lipschitz domains (not necessarily bounded) in $\mathbb{R}^d$ and $m \in \mathbb{N}^*$. Assume that we are given an homeomorphism $T : \tilde{\Omega} \to \Omega$ such that $T \in (\mathcal{C}^{m-1,1}(\widetilde{\Omega}))^d$ and $T^{-1} \in (\mathcal{C}^{m-1,1}(\Omega))^d$ with bounded derivatives up to the order m.*

For any measurable function $u : \Omega \to \mathbb{R}$ and any $1 \leq p \leq +\infty$, we have

$$u \in W^{m,p}(\Omega) \Longleftrightarrow u \circ T \in W^{m,p}(\widetilde{\Omega}),$$

and there exist $C_1, C_2 > 0$ which do not depend on u such that

$$C_1 \|u\|_{W^{m,p}(\Omega)} \leq \|u \circ T\|_{W^{m,p}(\widetilde{\Omega})} \leq C_2 \|u\|_{W^{m,p}(\Omega)}.$$

Proof.

We only prove the case $m = 1$, because the general case can be deduced by successive differentiations of $u \circ T$. Moreover, since T^{-1} satisfies the same assumptions as T, it is enough to prove that for $u \in W^{1,p}(\Omega)$ we have $u \circ T \in W^{1,p}(\widetilde{\Omega})$ with a suitable estimate of the norm.

For $u \in \mathcal{C}_c^\infty(\overline{\Omega})$, we observe that $u \circ T$ is Lipschitz continuous and that for any $x \in \widetilde{\Omega}$ where T is differentiable (i.e., for almost every x; see Theorem II.2.46), $u \circ T$ is also differentiable and we have

$$\nabla(u \circ T) = {}^t(\nabla T).\big((\nabla u) \circ T\big), \text{ almost everywhere.}$$

We deduce that

$$|\nabla(u \circ T)| \leq \mathrm{Lip}(T)|\nabla u \circ T|,$$

and we can conclude that $\nabla(u \circ T)$ belongs to $L^p(\widetilde{\Omega})$ by using Proposition II.2.15.

Since $\mathcal{C}_c^\infty(\overline{\Omega})$ is dense in $W^{1,p}(\Omega)$, the claim is proved.

□

2.4 Extension operator

The goal of this section is to build, for a given Lipschitz domain Ω with compact boundary a linear extension operator, referred to as E_Ω, such that for any given $1 < p < +\infty$, E_Ω maps $W^{1,p}(\Omega)$ into $W^{1,p}(\mathbb{R}^d)$ and satisfies for any such p the properties

$$E_\Omega \text{ is continuous from } W^{1,p}(\Omega) \text{ into } W^{1,p}(\mathbb{R}^d), \tag{III.32}$$

$$(E_\Omega u)_{|\Omega} = u, \ \forall u \in W^{1,p}(\Omega), \tag{III.33}$$

$$(E_\Omega u)_{|\Omega^c} = 0, \ \forall u \in W_0^{1,p}(\Omega). \tag{III.34}$$

Notice that the operator E_Ω does not depend on p.

Remark III.2.7. Here also the case $p = 1$ is specific. It can be shown that in that case, such an extension operator E_Ω does not exist. Nevertheless, it is possible to build an extension operator $\widetilde{E}_\Omega$ from $W^{1,1}(\Omega)$ into $W^{1,1}(\mathbb{R}^d)$ satisfying (III.32) and (III.33) but which does not satisfy (III.34); see, for instance, [33]. Particularities due to the case $p = 1$ are also observed in Remark III.2.10.

We first need to build such an extension operator on the flat half-space $\mathbb{R}^d_+$. Let $\eta : \mathbb{R}^{d-1} \to \mathbb{R}$ be a mollifying kernel supported in the unit ball B of $\mathbb{R}^{d-1}$. We define the extension operator on $\mathbb{R}^d_+$, for any $u \in \mathcal{C}^{0,1}(\overline{\mathbb{R}^d_+})$, as follows

$$\left(E_{\mathbb{R}^d_+} u\right)(x) = \begin{cases} u(x), & \text{for } x \in \mathbb{R}^d_+, \\ \varphi(x_d)\left(\displaystyle\int_B u(\bar{x} - x_d z, 0)\eta(z)\,dz\right), & \text{for } x = (\bar{x}, x_d) \in \mathbb{R}^d_-, \end{cases}$$

where $\varphi : \mathbb{R} \to \mathbb{R}$ is smooth, compactly supported, and satisfies $\varphi(0) = 1$.

Proposition III.2.14. *For any $1 < p < +\infty$, the operator $E_{\mathbb{R}^d_+}$ satisfies properties* (III.32)–(III.34).

Proof.

Property (III.33) is clear from the definition of $E_{\mathbb{R}^d_+}$. It is also clear that $E_{\mathbb{R}^d_+} u$ vanishes on $\mathbb{R}^d_-$ for any $u \in \mathcal{D}(\mathbb{R}^d_+)$. Assume that we can prove the inequality

$$\|E_{\mathbb{R}^d_+}u\|_{W^{1,p}(\mathbb{R}^d)} \leq C\|u\|_{W^{1,p}(\mathbb{R}^d_+)}, \ \forall u \in \mathcal{C}^\infty_c(\overline{\mathbb{R}^d_+});$$

then, by density, the operator E_Ω can be uniquely extended to the whole Sobolev space $W^{1,p}(\mathbb{R}^d_+)$ and all the required properties are proved. Let u be any element in $\mathcal{C}^\infty_c(\overline{\mathbb{R}^d_+})$.

- It is straightforward to check that, for such a u, $E_{\mathbb{R}^d_+}u$ is Lipschitz continuous on $\mathbb{R}^d$ and compactly supported.
- We first estimate the $L^p(\mathbb{R}^d_-)$-norm by using the Fubini theorem and the fact that $\int_B \eta(z)\,dz = 1$

$$\begin{aligned}&\int_{\mathbb{R}^d_-} \left|\left(E_{\mathbb{R}^d_+}u\right)(x)\right|^p\,dx\\ &= \int_{-\infty}^0 |\varphi(x_d)|^p \int_{\mathbb{R}^{d-1}}\int_B |u(\bar{x}-x_dz,0)|^p\eta(z)\,d\bar{x}\,dz\\ &=\|\varphi\|^p_{L^p(\mathbb{R})}\|u(.,0)\|^p_{L^p(\partial\mathbb{R}^d_+)} \leq C\|u\|^p_{W^{1,p}(\mathbb{R}^d_+)}\end{aligned}$$

- We now estimate the $L^p(\mathbb{R}^d_-)$-norm of $\partial_{x_i}(E_{\mathbb{R}^d_+}u)$ for $i = 1,\dots,d-1$. To begin with, we write

$$\partial_{x_i}\left(E_{\mathbb{R}^d_+}u\right)(\bar{x},x_d) = \frac{\varphi(x_d)}{x_d}\int_B u(\bar{x}-x_dz,0)\partial_{z_i}\eta(z)\,dz,$$

  and since $\int_B \partial_{z_i}\eta(z)\,dz = 0$, we deduce that

$$\begin{aligned}\partial_{x_i}\left(E_{\mathbb{R}^d_+}u\right)(\bar{x},x_d) =&\frac{\varphi(x_d)}{x_d}\int_B \Big(u(\bar{x}-x_dz,0)-u(\bar{x}-x_dz,-x_d)\\ &+u(\bar{x}-x_dz,-x_d)-u(\bar{x},-x_d)\Big)\partial_{z_i}\eta(z)\,dz,\end{aligned}$$

  and then

$$\begin{aligned}\partial_{x_i}\left(E_{\mathbb{R}^d_+}u\right)(\bar{x},x_d) =& \frac{\varphi(x_d)}{x_d}\int_B\int_0^{|x_d|}\partial_{x_d}u(\bar{x}-x_dz,\xi)\partial_{z_i}\eta(z)\,d\xi\,dz\\ &+\varphi(x_d)\int_B\int_0^1 \partial_{x_i}u(\bar{x}-tx_dz,-x_d)\partial_{z_i}\eta(z)\,dt\,dz.\end{aligned}$$

  Let us refer to the two terms in the right-hand side as $v_1(x)$ and $v_2(x)$, respectively.

  - We perform the change of variables $z \mapsto x_dy$ in the definition of v_1 to obtain

$$v_1(x) = \varphi(x_d)\int_{\mathbb{R}^{d-1}}\left(\frac{1}{x_d}\int_0^{|x_d|}\partial_{x_d}u(\bar{x}-y,\xi)\,d\xi\right)\frac{\partial_{z_i}\eta(y/x_d)}{x_d^{d-1}}\,d\xi\,dy;$$

then we use the Jensen inequality (Proposition II.2.20) to get

$$|v_1(x)|^p \leq \|\partial_{z_i}\eta\|_{L^1}^{p-1} \int_{\mathbb{R}^{d-1}} \left| \frac{1}{x_d} \int_0^{|x_d|} \partial_{x_d} u(\bar{x}-y,\xi)\, d\xi \right|^p \frac{\partial_{z_i}\eta(y/x_d)}{x_d^{d-1}}\, d\xi\, dy.$$

We integrate with respect to $\bar{x} \in \mathbb{R}^{d-1}$ to get

$$\int_{\mathbb{R}^{d-1}} |v_1(\bar{x},x_d)|^p\, d\bar{x} \leq C \int_{\mathbb{R}^{d-1}} \left| \frac{1}{x_d} \int_0^{|x_d|} \partial_{x_d} u(\bar{x},\xi)\, d\xi \right|^p d\bar{x}.$$

Finally, we integrate with respect to $x_d \in]-\infty, 0]$ and we use the Hardy inequality (Lemma II.4.7) so that we finally have

$$\|v_1\|_{L^p(\mathbb{R}^d_-)}^p \leq C \|\partial_{x_d} u\|_{L^p(\mathbb{R}^d_+)}^p.$$

– For estimating v_2 we first use the Jensen inequality to get

$$|v_2(x)|^p \leq C \int_B \int_0^1 |\partial_{x_i} u(\bar{x}-tx_d z, -x_d)|^p |\partial_{z_i}\eta(z)|\, dt\, dz,$$

and we first integrate with respect to $\bar{x}$ to find

$$\int_{\mathbb{R}^{d-1}} |v_2(\bar{x},x_d)|^p\, d\bar{x} \leq C' \int_{\mathbb{R}^{d-1}} |\partial_{x_i} u(\bar{x},-x_d)|^p\, d\bar{x};$$

then we integrate with respect to x_d to conclude that

$$\|v_2\|_{L^p(\mathbb{R}^d_-)}^p \leq C' \|\partial_{x_i} u\|_{L^p(\mathbb{R}^d_+)}^p.$$

- It remains to study the derivative of $E_{\mathbb{R}^d_+} u$ with respect to x_d. A straightforward computation shows that

$$\begin{aligned}\partial_{x_d}\left(E_{\mathbb{R}^d_+} u\right)(\bar{x},x_d) =& \varphi'(x_d) \int_B u(\bar{x}-x_d z, 0)\eta(z)\, dz \\ &- \frac{\varphi(x_d)}{x_d} \int_B u(\bar{x}-x_d z,0)\, \mathrm{div}_z \big(\eta(z) z\big)\, dz,\end{aligned}$$

so that the estimate of $\|\partial_{x_d} E_{\mathbb{R}^d_+} u\|_{L^p(\mathbb{R}^d_-)}$ can be done in a similar way as before.

$\square$

We may now build an extension operator for any Lipschitz half-space $\mathcal{H}_a$ as defined in Definition III.1.1. Let T_a be the Lipschitz diffeomorphism that we introduced in Proposition III.1.3.

Proposition III.2.15. *The operator $E_{\mathcal{H}_a}$ defined by*

$$E_{\mathcal{H}_a} u = \left(E_{\mathbb{R}^d_+}(u \circ T_a) \right) \circ T_a^{-1}, \, \forall u \in W^{1,p}(\mathcal{H}_a),$$

satisfies the properties (III.32)–(III.34).

Proof.

This is a straightforward consequence of Proposition III.1.3, Theorem III.2.13, and Proposition III.2.14.

□

Finally, for any Lipschitz domain with compact boundary Ω, we can use once more a suitable open covering $(U_i)_{1\leq i\leq N}$ of $\partial\Omega$ and an associated partition of unity $(\psi_i)_{1\leq i\leq N}$.

Theorem III.2.16. *The operator E_Ω defined by*

$$E_\Omega u = \sum_{i=1}^{d} E_{\mathcal{H}_{a_i}}((u\psi_i) \circ R_i^{-1}) \circ R_i, \, \forall u \in W^{1,p}(\Omega),$$

satisfies the properties (III.32)–(III.34).

Proof.

The main point is to observe that for any $i = 1, \ldots, N$, the function $u\psi_i$ is supported in $\Omega \cap U_i = R_i\mathcal{H}_{a_i} \cap U_i$, so that the extension of $(u\psi_i) \circ R_i^{-1}$ by zero on $\mathcal{H}_{a_i}$ is a function in $W^{1,p}(\mathcal{H}_{a_i})$ whose norm is controlled by $\|u\|_{W^{1,p}(\Omega)}$.

□

2.5 Trace and trace lifting operators

In this section, we study the trace theory for functions belonging to a Sobolev space $W^{1,p}(\Omega)$.

2.5.1 Trace operator and trace spaces

Once more, we begin by studying the case of a flat half-space.

Proposition III.2.17. *Let $1 \leq p < +\infty$. There exists $C > 0$ such that, for any $\varphi \in \mathcal{C}^{0,1}(\mathbb{R}^d)$ which is compactly supported, we have*

$$\|\varphi_{|\partial\mathbb{R}^d_+}\|_{L^p(\partial\mathbb{R}^d_+)} \leq C \|\varphi\|^{1-1/p}_{L^p(\mathbb{R}^d_+)} \|\varphi\|^{1/p}_{W^{1,p}(\mathbb{R}^d_+)}.$$

Proof.

Let us first prove the result for $\varphi \in \mathcal{C}^\infty_c(\mathbb{R}^d)$. For any $x = (\bar{x}, x_d)$, we compute

$$|\varphi(\bar{x}, 0)|^p = |\varphi(\bar{x}, x_d)|^p - p \int_0^{x_d} |\varphi(\bar{x}, s)|^{p-2} \varphi(\bar{x}, s) \frac{\partial \varphi}{\partial x_d}(\bar{x}, s) \, ds.$$

Since φ is compactly supported, we can take the limit $x_d \to +\infty$; then we integrate with respect to $\bar{x} \in \mathbb{R}^{d-1}$ to find that

$$\int_{\mathbb{R}^{d-1}} |\varphi(\bar{x},0)|^p \, d\bar{x} \le p \int_{\mathbb{R}^{d-1}} \int_0^{+\infty} |\varphi(\bar{x},s)|^{p-1} \left| \frac{\partial \varphi}{\partial x_d}(\bar{x},s) \right| \, ds \, d\bar{x}.$$

By using the Hölder inequality we find that

$$\int_{\mathbb{R}^{d-1}} |\varphi(\bar{x},0)|^p \, d\bar{x} \le p \|\varphi\|_{L^p(\mathbb{R}^d_+)}^{p-1} \left\| \frac{\partial \varphi}{\partial x_d} \right\|_{L^p(\mathbb{R}^d_+)},$$

which implies the required inequality.

In the case where φ is only Lipschitz continuous and compactly supported, we can apply the above inequality to the function $\varphi \star \eta_\varepsilon$ defined in (II.5). We observe that the functions $\varphi \star \eta_\varepsilon$ are compactly supported, uniformly with respect to ε. By using Proposition II.2.47, we deduce that $\varphi \star \eta_\varepsilon$ converges to φ in $W^{1,p}(\mathbb{R}^d_+)$ and that the traces of $\varphi \star \eta_\varepsilon$ on $\partial\mathbb{R}^d_+$ also converge towards the trace of φ in $L^p(\partial\mathbb{R}^d_+)$. The claim is proved.

□

We can now extend the previous result to any Lipschitz half-space.

Proposition III.2.18. *Let $1 \le p < +\infty$ and $\mathcal{H}_a$ be a Lipschitz half-space of $\mathbb{R}^d$. There exists $C_a > 0$ such that for any $\varphi \in \mathrm{Lip}(\mathbb{R}^d)$ compactly supported, we have*

$$\|\varphi_{|\partial\mathcal{H}_a}\|_{L^p(\partial\mathcal{H}_a)} \le C_a \|\varphi\|_{L^p(\mathcal{H}_a)}^{1-1/p} \|\varphi\|_{W^{1,p}(\mathcal{H}_a)}^{1/p}.$$

Proof.

We consider the Lipschitz diffeomorphism $T_a : \mathbb{R}^d_+ \to \mathcal{H}_a$ that we studied in Proposition III.1.3. By construction, $\varphi \circ T_a$ is Lipschitz continuous and compactly supported, so that we can apply Proposition III.2.17 to obtain

$$\|\varphi \circ T_{a|\partial\mathbb{R}^d_+}\|_{L^p(\partial\mathbb{R}^d_+)} \le C \|\varphi \circ T_a\|_{L^p}^{1-1/p} \|\varphi \circ T_a\|_{W^{1,p}(\mathbb{R}^d_+)}^{1/p}.$$

By Proposition II.2.15 and Theorem III.2.13, we can bound the right-hand side by corresponding norms of φ in $L^p(\mathcal{H}_a)$ and in $W^{1,p}(\mathcal{H}_a)$. It remains to observe that

$$\begin{aligned}
\|\varphi \circ T_{a|\partial\mathbb{R}^d_+}\|_{L^p(\partial\mathbb{R}^d_+)}^p &= \int_{\mathbb{R}^{d-1}} |\varphi(T_a(\bar{x},0))|^p \, d\bar{x} = \int_{\mathbb{R}^{d-1}} |\varphi(\bar{x},a(\bar{x}))|^p \, d\bar{x} \\
&= \int_{\mathbb{R}^{d-1}} |\varphi(\bar{x},a(\bar{x}))|^p \, J_a(\bar{x}) \frac{1}{J_a(\bar{x})} d\bar{x} \\
&\ge \frac{1}{\sqrt{1+(d-1)\,\mathrm{Lip}(a)^2}} \int_{\partial\mathcal{H}_a} |\varphi|^p \, d\sigma.
\end{aligned}$$

□

Finally, we can prove the main result of this section.

Theorem III.2.19. *Let Ω be a Lipschitz domain of $\mathbb{R}^d$ with compact boundary and $1 \leq p < +\infty$. There exists $C > 0$, such that for any $\varphi \in \mathcal{D}(\mathbb{R}^d)$ we have*

$$\|\varphi_{|\partial\Omega}\|_{L^p(\partial\Omega)} \leq C\|\varphi\|_{L^p(\Omega)}^{1-1/p}\|\varphi\|_{W^{1,p}(\Omega)}^{1/p}.$$

As a consequence, there exists a unique linear continuous operator

$$\gamma_0 : W^{1,p}(\Omega) \to L^p(\partial\Omega),$$

such that $\gamma_0(\varphi) = \varphi_{|\partial\Omega}$ for any $\varphi \in \mathcal{C}_c^\infty(\overline{\Omega})$. This operator is called the trace operator *on $W^{1,p}(\Omega)$.*

Remark III.2.8. We show in Theorem III.2.46 that the kernel of the operator γ_0 is nothing but the space $W_0^{1,p}(\Omega)$. That is the reason why we can say that $W_0^{1,p}(\Omega)$ is exactly the set of functions in $W^{1,p}(\Omega)$ which vanish on the boundary $\partial\Omega$.

Proof.

We consider the open covering $(U_i)_{1\leq i\leq N}$ of $\partial\Omega$ that we introduced in Theorem III.1.4 and $(\psi_i)_{1\leq i\leq N}$ an associated partition of unity. Using that $\sum_{i=1}^N \psi_i = 1$ on $\partial\Omega$ we have

$$\|\varphi_{|\partial\Omega}\|_{L^p(\partial\Omega)} \leq \sum_{i=1}^N \|(\psi_i\varphi)_{|\partial\Omega}\|_{L^p(\partial\Omega)}.$$

Since each $\psi_i\varphi$ is supported in U_i and $\partial\Omega \cap U_i = R_i\partial\mathcal{H}_{a_i} \cap U_i$, $\Omega \cap U_i = R_i\mathcal{H}_{a_i} \cap U_i$, we can apply Proposition III.2.18 to find that

$$\begin{aligned}\|(\psi_i\varphi)_{|\partial\Omega}\|_{L^p(\partial\Omega)} =& \|(\psi_i\varphi)_{|\partial R_i\mathcal{H}_{a_i}}\|_{L^p(\partial R_i\mathcal{H}_{a_i})} \\ \leq& C_i\|\varphi\psi_i\|_{L^p(R_i\mathcal{H}_{a_i})}^{1-1/p}\|\varphi\psi_i\|_{W^{1,p}(R_i\mathcal{H}_{a_i})}^{1/p} \\ \leq& C_i\|\varphi\psi_i\|_{L^p(\Omega)}^{1-1/p}\|\varphi\psi_i\|_{W^{1,p}(\Omega)}^{1/p}.\end{aligned}$$

Each ψ_i belongs to $\mathcal{D}(\mathbb{R}^d)$, thus we have $\|\varphi\psi_i\|_{W^{1,p}(\Omega)} \leq C_i\|\varphi\|_{W^{1,p}(\Omega)}$ which gives the inequality.

The existence and uniqueness of the trace operator now follows from the density of $\mathcal{C}_c^\infty(\overline{\Omega})$ in $W^{1,p}(\Omega)$; see Theorem III.2.11.

□

We later show (Remark III.2.18) that the trace operator γ_0 is not surjective onto $L^p(\partial\Omega)$ for $1 < p < +\infty$. This justifies the following definition.

Definition III.2.20. *Let Ω be a Lipschitz domain with compact boundary and $1 < p < +\infty$. The range of the trace operator is referred to as*

$$W^{1-1/p,p}(\partial\Omega) = \gamma_0\left(W^{1,p}(\Omega)\right).$$

This is a Banach space for the following norm,

$$\|v\|_{W^{1-1/p,p}(\partial\Omega)} = \inf_{\substack{u\in W^{1,p}(\Omega)\\ \gamma_0(u)=v}} \|u\|_{W^{1,p}(\Omega)}.$$

Remark III.2.9. • As suggested by the notation, the space $W^{1-1/p,p}(\partial\Omega)$ only depends on the manifold $\partial\Omega$ and not on the domain Ω itself. This is due to the existence of the extension operator E_Ω. Indeed, it implies in particular that the trace space associated with $W^{1,p}(\Omega)$ and with $W^{1,p}(\overline{\Omega}^c)$ are the same, and that the associated norms are equivalent. The result follows because Ω and $\overline{\Omega}^c$ share the same boundary.
In fact, trace spaces can be defined in an intrinsic way as fractional Sobolev spaces but this is not needed here (see, for instance, [33]).

- In the case $p = 1$, the trace operator γ_0 is surjective, that is $\gamma_0(W^{1,1}(\Omega)) = L^1(\partial\Omega)$ but this is quite difficult to show (see Remark III.2.10). As an extension of the previous notation we set $W^{0,1}(\partial\Omega) = L^1(\partial\Omega)$.

Theorem III.2.21. *With the same notation as in Definition III.2.20 and $1 \leq p < +\infty$, we have the following properties.*

- *The trace operator γ_0 is continuous from $W^{1,p}(\Omega)$ onto $W^{1-1/p,p}(\partial\Omega)$.*
- *The space $\mathcal{C}^{0,1}(\partial\Omega)$ is included and dense in $W^{1-1/p,p}(\partial\Omega)$.*
- *The space $W^{1-1/p,p}(\partial\Omega)$ is dense in $L^p(\partial\Omega)$.*

Proof.

- The first property is obvious by definition of the norm on $W^{1-1/p,p}(\partial\Omega)$.
- Note first that, thanks to Proposition II.2.45, any Lipschitz-continuous function v on $\partial\Omega$ is the restriction on $\partial\Omega$ of some compactly supported Lipschitz continuous function u on $\mathbb{R}^d$. In particular such a u belongs to $W^{1,p}(\Omega)$ and therefore $v \in \mathcal{C}^{0,1}(\partial\Omega) \subset W^{1-1/p,p}(\Omega)$.
Let now $v \in W^{1-1/p,p}(\partial\Omega)$, with $v = \gamma_0(u)$, $u \in W^{1,p}(\Omega)$. The trace of $\mathcal{S}_\varepsilon u$ on $\partial\Omega$ is, at least, Lipschitz-continuous and converges to v in $W^{1-1/p,p}(\partial\Omega)$, because the trace operator is continuous and $\mathcal{S}_\varepsilon u$ converges to u in $W^{1,p}(\Omega)$.
- The last property follows from the density of $\mathcal{C}^{0,1}(\partial\Omega)$ into $L^p(\partial\Omega)$.

□

As a consequence of the above properties of trace operators and of the density of smooth functions in Sobolev spaces, we can extend the Stokes formula (III.10) to any $\varphi \in W^{1,p}(\Omega)$ and $\Psi \in (W^{1,p'}(\Omega))^d$ as follows

$$\int_\Omega \varphi \operatorname{div} \Psi \, dx + \int_\Omega \Phi \cdot \nabla\psi \, dx = \int_{\partial\Omega} \gamma_0(\psi)(\gamma_0(\Phi)\cdot\nu)\, d\sigma. \tag{III.35}$$

2.5.2 Trace lifting operator

Let Ω be a Lipschitz domain with compact boundary. We set $\widetilde{\Omega} = \overline{\Omega}^c$ which is also a Lipschitz domain with the same compact boundary as $\Omega : \partial\widetilde{\Omega} = \partial\Omega$.

We define a continuous operator $\tilde{R}_0$ from $W^{1,p}(\Omega)$ into itself as

$$\tilde{R}_0 u = E_{\widetilde{\Omega}}\left((E_\Omega u)_{|\widetilde{\Omega}}\right)_{|\Omega}, \ \forall u \in W^{1,p}(\Omega).$$

Since E_Ω and $E_{\widetilde{\Omega}}$ are continuous (as well as the restriction operators), the linear operator $\tilde{R}_0$ is also continuous. Moreover, by (III.34) we see that $\tilde{R}_0 u = 0$ for any $u \in W_0^{1,p}(\Omega)$.

We later show (Theorem III.2.46) that $W_0^{1,p}(\Omega) = \operatorname{Ker}\gamma_0$. As a consequence, if $u_1, u_2 \in W^{1,p}(\Omega)$ are such that $\gamma_0(u_1) = \gamma_0(u_2)$ then we have $u_1 - u_2 \in \operatorname{Ker}\gamma_0 \in W_0^{1,p}(\Omega)$ and then we have $\tilde{R}_0 u_1 = \tilde{R}_0 u_2$.

As a consequence, $\tilde{R}_0 u$ only depends on the trace $\gamma_0(u) \in W^{1-1/p,p}(\partial\Omega)$ and we can define a new operator $R_0 : W^{1-1/p,p}(\partial\Omega) \to W^{1,p}(\Omega)$ by $R_0 v = \tilde{R}_0 u$ for any $u \in W^{1,p}(\Omega)$ such that $\gamma_0(u) = v$.

The properties of R_0 are now summed up in the following theorem.

Theorem III.2.22 (Lifting operator). *Let Ω be a Lipschitz domain with compact boundary and $1 < p < +\infty$. There exists a linear continuous operator R_0 from $W^{1-1/p,p}(\partial\Omega)$ into $W^{1,p}(\Omega)$ which is a right-inverse of the trace operator, that is, which satisfies*

$$\gamma_0 \circ R_0 = \operatorname{Id}_{W^{1-1/p,p}(\partial\Omega)}.$$

Proof.

The only point that remains to be proved is the continuity of the operator R_0. For any $v = \gamma_0(u) \in W^{1-1/p,p}(\partial\Omega)$ with $u \in W^{1,p}(\Omega)$ we have

$$\|R_0 v\|_{W^{1,p}(\Omega)} = \|\tilde{R}_0 u\|_{W^{1,p}(\Omega)} \leq \|\tilde{R}_0\| \|u\|_{W^{1,p}(\Omega)}.$$

Inasmuch as this estimate is true for any $u \in W^{1,p}(\Omega)$ satisfying $\gamma_0 u = v$ we can take the infimum to obtain

$$\|R_0 v\|_{W^{1,p}(\Omega)} \leq C\|v\|_{W^{1-1/p,p}(\partial\Omega)}.$$

□

Remark III.2.10. Such a linear continuous lifting operator does not exist in the case $p = 1$. This is closely related to Remark III.2.7.

Nevertheless, it is possible to show that there exists a continuous *nonlinear* lifting operator from $L^1(\partial\Omega)$ into $W^{1,1}(\Omega)$; see, for instance, [33]. In particular the trace operator $\gamma_0 : W^{1,1}(\Omega) \to L^1(\partial\Omega)$ is onto.

2.5.3 Higher-order traces

By extension of the previous definitions we define, for any $k \geq 1$ and any $1 < p < +\infty$, the space

$$W^{k-1/p,p}(\partial\Omega) = \gamma_0(W^{k,p}(\Omega)),$$

equipped with the norm

$$\|v\|_{W^{k-1/p,p}} = \inf_{\substack{u\in W^{k,p}(\Omega)\\ \gamma_0(u)=v}} \|u\|_{W^{k,p}(\Omega)}.$$

This is a Banach space (and even a Hilbert space when $p = 2$).

We can show that, as soon as Ω is a $\mathcal{C}^{k-1,1}$ domain, the lifting operator R_0 that we built in the previous section is continuous from $W^{k-1/p,p}(\partial\Omega)$ onto $W^{k,p}(\Omega)$.

To this end, it is enough to prove that the extension operator E_Ω satisfies the additional property: for any $u \in W^{k,p}(\Omega)$, we have $(E_\Omega u)_{\overline{\Omega}^c}$ belongs to $W^{k,p}(\overline{\Omega}^c)$ and there is a $C > 0$ independent of u, so that we have

$$\|(E_\Omega u)_{|\overline{\Omega}^c}\|_{W^{k,p}(\overline{\Omega}^c)} \leq C\|u\|_{W^{k,p}(\Omega)}. \tag{III.36}$$

By using the regularity assumptions on Ω, we see that it is enough to prove (III.36) in the case of the flat half-space. This can be done in the very same way as we did for the case $k = 1$. We leave this calculation to the reader.

Remark III.2.11. It is not true that $E_\Omega u \in W^{k,p}(\mathbb{R}^d)$. However, it is possible to build extension operators satisfying this additional property but not (III.34). We do not present this construction which is not useful here (see [1], for instance).

Assume now that Ω is of class $\mathcal{C}^{1,1}$. We observe that, for any $u \in W^{2,p}(\Omega)$, we have $\nabla u \in (W^{1,p}(\Omega))^d$ so that we can define the normal derivative

$$\gamma_\nu(u) = \frac{\partial u}{\partial \nu} = \gamma_0(\nabla u)\cdot \nu \in L^p(\partial\Omega).$$

In fact, from the smoothness hypothesis on the domain, it can be shown (see Section 3.2) that ν can be extended to a Lipschitz continuous function on $\overline{\Omega}$. Therefore, the product $(\nabla u)\cdot\nu$ belongs to $W^{1,p}(\Omega)$ and thus, the normal derivative $\gamma_\nu(u)$ belongs to $W^{1-1/p,p}(\partial\Omega)$.

We have shown that γ_ν is a continuous linear operator from $W^{2,p}(\Omega)$ into $W^{1-1/p,p}(\partial\Omega)$. In fact, it can be shown that this operator has a continuous right inverse.

Theorem III.2.23. *Let Ω be a domain of $\mathbb{R}^d$ of class $\mathcal{C}^{1,1}$ with compact boundary. There exists a continuous linear lifting operator R_ν from $W^{1-1/p,p}(\partial\Omega)$ into $W^{2,p}(\Omega)$ such that*

$$\gamma_\nu \circ R_\nu = \mathrm{Id}_{W^{1-1/p,p}(\partial\Omega)}, \text{ and } \gamma_0\circ R_\nu = 0.$$

The proof of this result is postponed to Section 3.3.

Remark III.2.12. Let us define the following operator

$$
\begin{aligned}
R : (g_0, g_1) \in W^{2-1/p,p}(\partial\Omega) \times W^{1-1/p,p}(\partial\Omega) \\
\mapsto R(g_0, g_1) = R_0 g_0 + R_\nu \left(g_1 - \gamma_\nu(R_0 g_0)\right) \in W^{2,p}(\Omega).
\end{aligned}
$$

By the properties of R_0 and R_ν, and the definition of $W^{2-1/p,p}(\partial\Omega)$, we see that this operator is linear and continuous.

Using that $\gamma_0 R_\nu = 0$, $\gamma_0 R_0 = \mathrm{Id}$, and $\gamma_\nu R_\nu = \mathrm{Id}$, we get

$$
\begin{aligned}
&\gamma_0(R(g_0, g_1)) = \gamma_0 R_0 g_0 = g_0, \\
&\gamma_\nu(R(g_0, g_1)) = \gamma_\nu(R_0 g_0) + \gamma_\nu R_\nu(g_1 - \gamma_\nu(R_0 g_0)) = g_1.
\end{aligned}
$$

We have thus obtained an operator which is a simultaneous lifting operator for the usual trace and for the trace of the normal derivative in $W^{2,p}(\Omega)$.

2.6 Duality theory for Sobolev spaces

2.6.1 Definitions and characterisation

For all $p \in [1, +\infty]$ and $m \geq 1$, we define the following distribution space,

$$
W^{-m,p'}(\Omega) = \left\{ u \in \mathcal{D}'(\Omega), u = \sum_{|\alpha| \leq m} \partial^\alpha f_\alpha, \text{ with } f_\alpha \in L^{p'}(\Omega) \right\}, \tag{III.37}
$$

which is equipped with the norm

$$
\begin{cases}
\|u\|_{W^{-m,p'}} = \inf \left(\displaystyle\sum_{|\alpha| \leq m} \|f_\alpha\|_{L^{p'}}^{p'} \right)^{1/p'}, & \text{for } p' < +\infty \\
\|u\|_{W^{-m,\infty}} = \inf \left(\displaystyle\sup_{|\alpha| \leq m} \|f_\alpha\|_{L^\infty} \right), & \text{for } p' = +\infty
\end{cases} \tag{III.38}
$$

where the infimum is taken on all the families $(f_\alpha)_\alpha$ such that $u = \sum_{|\alpha| \leq m} \partial^\alpha f_\alpha$.

These spaces play an important role in the duality theory of $W_0^{m,p}(\Omega)$ spaces. Let us make this point precise.

Theorem III.2.24. *1. For all $m \geq 1$ and $p \in [1, +\infty]$, the space $W^{-m,p'}(\Omega)$ is a Banach space.*

2. For all $m \geq 1$ and $p \in]1, +\infty]$, the dual of $W^{-m,p'}(\Omega)$ is isomorphic with $W_0^{m,p}(\Omega)$.

3. For all $m \geq 1$ and $p \in [1, +\infty[$, the dual of $W_0^{m,p}(\Omega)$ is isomorphic with $W^{-m,p'}(\Omega)$.

Proof.

1. We denote as N_m the number of multi-indices $\alpha \in \mathbb{N}^d$ such that $|\alpha| \le m$. Then, let Φ be the map

$$\Phi : \left(L^{p'}(\Omega)\right)^{N_m} \longrightarrow W^{-m,p'}(\Omega)$$
$$(f_\alpha)_\alpha \mapsto \sum_{|\alpha|\le m} \partial^\alpha f_\alpha.$$

By definition, this map is linear, continuous, and surjective. Hence, Φ naturally infers a linear, continuous, and bijective function $\overline{\Phi}$ between the quotient space $X = (L^{p'}(\Omega))^{N_m} / \operatorname{Ker}\Phi$ and $W^{-m,p'}(\Omega)$. Since $\operatorname{Ker}\Phi$ is closed, X, provided with the quotient norm, is a Banach space. Furthermore, from the definitions of the quotient norm and the norm (III.38) chosen on $W^{-m,p'}(\Omega)$, we see that $\overline{\Phi}$ is an isometry between these two spaces which proves that $W^{-m,p'}(\Omega)$ is a Banach space.

2. We assume $p > 1$, in other words $p' < +\infty$. By duality, the adjoint function ${}^t\overline{\Phi}$ produces an isomorphism between $(W^{-m,p'}(\Omega))'$ and X'. However, the dual of X is naturally identified with the subspace of continuous linear functionals on $(L^{p'}(\Omega))^{N_m}$ which cancel on $\operatorname{Ker}\Phi$. Since p' is finite, the dual of $L^{p'}(\Omega)$ is canonically isomorphic with $L^p(\Omega)$ such that the dual of X is made up of elements of $(L^p(\Omega))^{N_m}$ which cancel on the kernel of Φ. One such element can be written as $u = (u_\alpha)_{|\alpha|\le m} \in (L^p(\Omega))^{N_m}$ with

$$\int_\Omega \sum_{|\alpha|\le m} u_\alpha f_\alpha \, dx = 0,$$

for all families $(f_\alpha)_\alpha \in (L^{p'}(\Omega))^{N_m}$ such that $\sum_\alpha \partial^\alpha f_\alpha = 0$. If we apply this to the families of the form $(-\partial^\alpha\varphi, 0, \dots, 0, \varphi, 0, \dots)$ for any $\varphi \in \mathcal{D}(\Omega)$, we obtain the result that the family $(u_\alpha)_\alpha$ has the form

$$u_\alpha = (-1)^{|\alpha|}\partial^\alpha v.$$

In particular, the function v belongs to $W^{m,p}(\Omega)$. However, we can also take the φ in $\mathcal{D}(\overline{\Omega})$, which shows that v belongs to $W_0^{m,p}(\Omega)$. In summary, the function

$$S : v \in W_0^{m,p}(\Omega) \mapsto (v, \dots, (-1)^{|\alpha|}\partial^\alpha v, \dots) \in X',$$

produces an isomorphism between $W_0^{m,p}(\Omega)$ and X' which is itself isomorphic with the dual of $W^{-m,p'}(\Omega)$.

3. The proof of this point is similar to that of the previous example.

□

In this proof, we have used the identification of the dual of $L^p(\Omega)$ with $L^{p'}(\Omega)$ through the bilinear form $(f, g) \in L^p(\Omega) \times L^{p'}(\Omega) \mapsto \int_\Omega fg \, dx$. It is useful to understand that, when it occurs, the duality $(W^{-m,p'}, W_0^{m,p})$ is written using this bilinear form as follows,

$$\left\langle \sum_{|\alpha|\le m} \partial^\alpha f_\alpha, \varphi \right\rangle_{W^{-m,p'},W_0^{m,p}} = \int_\Omega \sum_{|\alpha|\le m} (-1)^{|\alpha|} f_\alpha \partial^\alpha \varphi \, dx, \ \forall \varphi \in W_0^{m,p}(\Omega). \tag{III.39}$$

Of course, the term on the right-hand side of this equality only depends on the sum $\sum_{|\alpha|\le m} \partial^\alpha f_\alpha$.

Remark III.2.13. For any $u \in W^{-m,p'}(\Omega)$, we have

$$\|u\|_{W^{-m,p'}} = \sup_{\varphi \in W_0^{m,p}(\Omega)} \frac{\langle u, \varphi\rangle_{W^{-m,p'},W_0^{m,p}}}{\|\varphi\|_{W_0^{m,p}}}, \tag{III.40}$$

even for $p = +\infty$, thanks to Proposition II.2.1.

In the case $p = 2$, it can be shown that the norm of $H^{-m}(\Omega)$ defined by (III.38) and characterised by (III.40) is an Hilbert norm.

Definition III.2.25. *For any Lipschitz domain Ω in $\mathbb{R}^d$ with compact boundary and any $1 \le p < +\infty$, we define $W^{-1/p',p'}(\partial\Omega)$ to be the dual space of $W^{1-1/p,p}(\partial\Omega)$.*

Note that in the case $p = 1$ we have seen (Remark III.2.9) that $W^{0,1}(\partial\Omega) = L^1(\partial\Omega)$, so that we adopt the convention that $W^{0,\infty}(\partial\Omega) = L^\infty(\partial\Omega)$.

By using Theorem III.2.21, $W^{1-1/p,p}(\partial\Omega)$ is dense in $L^p(\partial\Omega)$. Therefore, by using Proposition II.3.7 we can identify $L^{p'}(\partial\Omega)$ as a dense subspace of $W^{-1/p',p'}(\partial\Omega)$. Indeed, for any $v \in L^{p'}(\partial\Omega)$, the mapping

$$T_v : w \in W^{1-1/p,p}(\partial\Omega) \mapsto \int_{\partial\Omega} vw \, d\sigma,$$

is linear continuous so that $T_v \in W^{-1/p',p'}(\partial\Omega)$ and moreover

$$\|T_v\|_{W^{-1/p',p'}(\partial\Omega)} \le \|w\|_{L^p(\partial\Omega)} \|v\|_{L^{p'}(\partial\Omega)} \le C\|w\|_{W^{1-1/p,p}(\partial\Omega)} \|v\|_{L^{p'}(\partial\Omega)}.$$

Therefore, the map $v \in L^{p'}(\partial\Omega) \mapsto T_v \in W^{-1/p',p'}(\partial\Omega)$ is a canonical continuous embedding of $L^{p'}(\partial\Omega)$ into $W^{-1/p',p'}(\partial\Omega)$.

In other words, we often use the following duality formula on the boundary in the sequel,

$$\langle v, w\rangle_{W^{-1/p',p'}(\partial\Omega),W^{1-1/p,p}(\partial\Omega)} = \int_{\partial\Omega} vw \, d\sigma,$$

for any $v \in L^{p'}(\partial\Omega)$ and $w \in W^{1-1/p,p}(\partial\Omega) \subset L^p(\partial\Omega)$.

2.6.2 Mollifying operators and Friedrichs commutation estimates

We study here the properties of the mollifying operator $\mathcal{S}_\varepsilon$ defined in Section 2.2 when applied to distributions belonging to dual Sobolev spaces. These results are useful in the study of the functional spaces of fluid mechanics that we give in Chapter IV, in particular in the proof of the Nečas inequality (see Remark IV.1.2). We suppose we are given a Lipschitz domain Ω with compact boundary.

Proposition III.2.26. *For any $f \in W^{-1,p}(\Omega)$, with $p \in [1, +\infty[$, we have*

$$\|\mathcal{S}_\varepsilon f - f\|_{W^{-1,p}} \xrightarrow[\varepsilon \to 0]{} 0.$$

Proof.

From Definition III.37, any distribution in $W^{-1,p}(\Omega)$ can be written as $f = g_0 + \sum_{j=1}^{d} \partial g_j / \partial x_j$, with $g_0, \ldots, g_d \in L^p(\Omega)$.

By using Proposition III.2.8, we know that

$$\mathcal{S}_\varepsilon g_0 \xrightarrow[\varepsilon \to 0]{} g_0, \text{ in } L^p(\Omega),$$

which implies the convergence in $W^{-1,p}(\Omega)$.

Since $\mathcal{S}_\varepsilon$ is linear, it remains to show that $\mathcal{S}_\varepsilon(\partial g_j / \partial x_j)$ converges towards $\partial g_j / \partial x_j$ in $W^{-1,p}(\Omega)$ for any j. To this end, we write

$$\begin{aligned}
\left\| \mathcal{S}_\varepsilon \frac{\partial g_j}{\partial x_j} - \frac{\partial g_j}{\partial x_j} \right\|_{W^{-1,p}} \leq & \left\| \mathcal{S}_\varepsilon \frac{\partial g_j}{\partial x_j} - \frac{\partial}{\partial x_j}(\mathcal{S}_\varepsilon g_j) \right\|_{W^{-1,p}} \\
& + \left\| \frac{\partial}{\partial x_j}(\mathcal{S}_\varepsilon g_j - g_j) \right\|_{W^{-1,p}} \\
\leq & C \left\| \mathcal{S}_\varepsilon \frac{\partial g_j}{\partial x_j} - \frac{\partial}{\partial x_j}(\mathcal{S}_\varepsilon g_j) \right\|_{L^p} + \|\mathcal{S}_\varepsilon g_j - g_j\|_{L^p}.
\end{aligned}$$

Theorem III.2.10 proves that the first term converges to zero and Proposition III.2.8 shows that the second term also tends to zero. The proof is complete. □

Remark III.2.14. This result proves in particular that, for any Lipschitz domain Ω with compact boundary, the set $\mathcal{C}_c^\infty(\overline{\Omega})$ is dense in $W^{-1,p}(\Omega)$, $1 \leq p < +\infty$.

We can now prove a commutator estimate similar to the one given in Theorem III.2.10 in the framework of the dual space $W^{-1,p}(\Omega)$. For reasons of simplicity and because it is sufficient for our purposes in this book (i.e., for the proof of the Nečas inequality in Chapter IV; see Remark IV.1.2), we only consider here the case of constant coefficient differential operators.

Theorem III.2.27. *Let $k \in \mathbb{N}^*$. For any $j \in \{1, \dots, d\}$, let $\alpha_j \in \mathbb{R}^k$ be a constant vector and let $D : (W^{-1,p}(\Omega))^k \to \mathcal{D}'(\Omega)$ be the differential operator defined by*

$$Df = \sum_{j=1}^{d} \frac{\partial}{\partial x_j}(\alpha_j \cdot f), \forall f \in (W^{-1,p}(\Omega))^k.$$

For any $f \in (W^{-1,p}(\Omega))^k$, we have

$$\|\mathcal{S}_\varepsilon Df - D\mathcal{S}_\varepsilon f\|_{W^{-1,p}} \xrightarrow[\varepsilon \to 0]{} 0.$$

In particular, if $Df \in W^{-1,p}(\Omega)$ we have $D\mathcal{S}_\varepsilon f \xrightarrow[\varepsilon\to 0]{} Df$ in $W^{-1,p}(\Omega)$.

Proof.

Let $\Phi \in \mathcal{D}(\Omega)$ be a test function. When f is smooth, we can use the computations done during the proof of Theorem III.2.10, and find that

$$\begin{aligned}
&\int_\Omega (\mathcal{S}_\varepsilon Df - D\mathcal{S}_\varepsilon f)(y)\Phi(y)\, dx \\
&= \int_\Omega R_{\varepsilon,0}(y) \cdot \Phi(y)\, dy \\
&= \int_B \eta(z) \sum_{i=0}^{n} \int_\Omega f(x) \sum_{j=1}^{d} \alpha_j \cdot \left(\Phi \frac{\partial(\beta_\varepsilon \psi_i)}{\partial x_j}\right)(x - \varepsilon\nu_i + \varepsilon z)\, dx\, dz,
\end{aligned}$$

the last equality being obtained by performing the change of variable $z = (y - x + \varepsilon\nu_i)/\varepsilon$. Note that there is no term $R_{\varepsilon,j}$ in this formula because we assume here that the coefficients α_j are constant.

By density of $\mathcal{C}_c^\infty(\overline{\Omega})$ in $W^{-1,p}(\Omega)$, the above formula finally gives, for any $f \in W^{-1,p}(\Omega)$,

$$\begin{aligned}
&\langle \mathcal{S}_\varepsilon Df - D\mathcal{S}_\varepsilon f, \Phi\rangle_{W^{-1,p},W_0^{1,p'}} \\
&= \int_B \eta(z) \sum_{i=0}^{n} \left\langle f, \sum_{j=1}^{d} \alpha_j \cdot \left(\Phi \frac{\partial(\beta_\varepsilon\psi_i)}{\partial x_j}\right)(\cdot - \varepsilon\nu_i + \varepsilon z)\right\rangle_{W^{-1,p},W_0^{1,p'}} dz.
\end{aligned}$$

We introduce $\mathcal{R}_\varepsilon : (W^{-1,p}(\Omega))^k \to W^{-1,p}(\Omega)$ the map which associates to any $f \in (W^{-1,p}(\Omega))^k$ the right-hand side of this formula.

We first observe that

$$\begin{aligned}
\langle \mathcal{R}_\varepsilon f, \Phi\rangle_{W^{-1,p},W_0^{1,p'}} &\leq C\|f\|_{W^{-1,p}} \sum_{i=1}^{N}\sum_{j=1}^{d} \left\| \Phi \frac{\partial(\beta_\varepsilon\psi_i)}{\partial x_j}\right\|_{W_0^{1,p'}} \\
&\leq C\|f\|_{W^{-1,p}} \|\Phi\|_{W_0^{1,p'}},
\end{aligned}$$

so that, by (III.40), we get

$$\|\mathcal{R}_\varepsilon f\|_{W^{-1,p}} \leq C\|f\|_{W^{-1,p}}.$$

Moreover, if $f \in (\mathcal{C}_c^\infty(\overline{\Omega}))^k$, we know by Theorem III.2.10 that $\mathcal{S}_\varepsilon Df - D\mathcal{S}_\varepsilon f$ converges to 0 in $L^p(\Omega)$ and then

$$\|\mathcal{R}_\varepsilon f\|_{W^{-1,p}} \xrightarrow[\varepsilon\to 0]{} 0, \, \forall f \in (\mathcal{C}^\infty(\overline{\Omega}))^k.$$

We can now conclude that $\mathcal{R}_\varepsilon f$ converges to 0 for any $f \in (W^{-1,p}(\Omega))^k$ by using the density of $(\mathcal{C}_c^\infty(\overline{\Omega}))^k$ in $(W^{-1,p}(\Omega))^k$ that we just proved before (see Remark III.2.14).

In the case where $Df \in W^{-1,p}(\Omega)$, we know from Proposition III.2.26 that $\mathcal{S}_\varepsilon Df$ converges towards Df in $W^{-1,p}(\Omega)$ so that we can conclude that $D\mathcal{S}_\varepsilon f$ also converges towards Df in $W^{-1,p}(\Omega)$.

□

2.7 Translation estimates

Definition III.2.28. *Let $h \in \mathbb{R}^d$. The translation operator along h, referred to as τ_h, is defined for any measurable function f on $\mathbb{R}^d$ by*

$$\tau_h(f) = \big(x \mapsto f(x+h)\big).$$

We also define the difference operator δ_h for any function f by

$$\delta_h(f) = \tau_h(f) - f.$$

We recall the notation $\Omega_\xi = \{x \in \Omega, \ \delta(x) > \xi\}$ that we introduced at the beginning of Section 1.

Theorem III.2.29. *Let Ω be any open set in $\mathbb{R}^d$ and $1 \leq p \leq +\infty$. For any $h \in \mathbb{R}^d$ and any $f \in W^{1,p}(\Omega)$ we have*

$$\|\tau_h f - f\|_{L^p(\Omega_{2|h|})} \leq |h| \|\nabla f\|_{L^p(\Omega)}.$$

Proof.

Let $h \in \mathbb{R}^d$ with $|h|$ small enough such that $\Omega_{|h|}$ is nonempty. The case $p = +\infty$ comes directly from Proposition III.2.9. We assume now that p is finite. By using the same strategy as in the proof of Lemma II.2.37, we can find a function $\varphi \in \mathcal{C}^\infty(\mathbb{R}^d)$ such that $0 \leq \varphi \leq 1$, $\varphi_{|\Omega_{|h|}} = 1$, and $\varphi_{|\Omega^c} = 0$. We set $g = \bar{f}\varphi$ and we observe that $g \in W^{1,p}(\mathbb{R}^d)$. Hence, there exists a sequence $(g_\varepsilon)_\varepsilon \subset \mathcal{D}(\mathbb{R}^d)$ which converges towards g in $W^{1,p}(\mathbb{R}^d)$. For any $x \in \mathbb{R}^d$ we have

$$g_\varepsilon(x+h) - g_\varepsilon(x) = \int_0^1 \nabla g_\varepsilon(x+th) \cdot h \, dt,$$

and then

$$|g_\varepsilon(x+h)-g_\varepsilon(x)|^p \leq |h|^p \int_0^1 |\nabla g_\varepsilon(x+th)|^p\,dt,$$

so that by integration on $\Omega_{2|h|}$, we find

$$\int_{\Omega_{2|h|}} |g_\varepsilon(x+h)-g_\varepsilon(x)|^p\,dx \leq |h|^p \int_0^1 \left(\int_{\Omega_{2|h|}} |\nabla g_\varepsilon(x+th)|^p\,dx\right) dt.$$

For any $t\in[0,1]$, and any $x\in\Omega_{2|h|}$ we have $x+th\in\Omega_{|h|}$ so that we finally get

$$\|\tau_h g_\varepsilon - g_\varepsilon\|^p_{L^p(\Omega_{2|h|})} \leq |h|^p \|\nabla g_\varepsilon\|^p_{L^p(\Omega_{|h|})}.$$

We can now pass to the limit when $\varepsilon\to 0$ and find that

$$\|\tau_h g - g\|^p_{L^p(\Omega_{2|h|})} \leq |h|^p \|\nabla g\|^p_{L^p(\Omega_{|h|})}.$$

Since $g=f$ on $\Omega_{|h|}$ we have obtained

$$\|\tau_h f - f\|_{L^p(\Omega_{2|h|})} \leq |h|\|\nabla f\|_{L^p(\Omega_{|h|})} \leq |h|\|\nabla f\|_{L^p(\Omega)}.$$

□

In fact, one can characterise Sobolev spaces by means of translation operators. We first consider only the case of the whole space $\Omega=\mathbb{R}^d$. We address the problem of such characterisations for Sobolev spaces on bounded domains in Section 3.5.

Lemma III.2.30. *Let $1\leq p\leq+\infty$, $f\in L^p(\mathbb{R}^d)$, $g\in L^{p'}(\mathbb{R}^d)$, and $h\in\mathbb{R}^d$. We have*

$$\int_{\mathbb{R}^d} \tau_h(f) g\,dx = \int_{\mathbb{R}^d} f\tau_{-h}(g)\,dx,$$

and

$$\int_{\mathbb{R}^d} \delta_h(f) g\,dx = \int_{\mathbb{R}^d} f\delta_{-h}(g)\,dx.$$

Proof.

Straightforward by using an affine change of variable.

□

Lemma III.2.31. *Let $1\leq p\leq+\infty+\infty$, $k\geq 0$, and $f\in W^{k,p}(\mathbb{R}^d)$. For any $h\in\mathbb{R}^d$, we have*

$$\|\delta_h f\|_{W^{k-1,p}} \leq |h|\,\|\nabla f\|_{W^{k-1,p}} \leq |h|\,\|f\|_{W^{k,p}}.$$

Proof.

- In the case $k\geq 1$, the result is nothing but a consequence of Theorem III.2.29 because the boundary of $\mathbb{R}^d$ is empty.

- Let us now deal with the slightly more specific case where $k = 0$. This concerns an inequality with a negative Sobolev norm, thus it is convenient to argue by duality. Hence, let $g \in W^{1,p'}(\mathbb{R}^d)$ and, according to the preceding lemma, we have
$$\int_{\mathbb{R}^d} \delta_h(f) g\, dx = \int_{\mathbb{R}^d} f \delta_{-h} g\, dx.$$
Hence
$$\left| \int_{\mathbb{R}^d} \delta_h(f) g\, dx \right| \leq \|f\|_{L^p} \|\delta_{-h} g\|_{L^{p'}} \leq |h| \|f\|_{L^p} \|g\|_{W^{1,p'}},$$
from the case $k = 1$ applied to the function g. This inequality, valid for any function g in $W^{1,p'}(\mathbb{R}^d)$, proves the result.

□

Let $(e_1, \ldots, e_d)$ be the canonical basis of $\mathbb{R}^d$, k be a nonnegative integer, and $1 \leq p \leq \infty$. For $f \in W^{k,p}(\mathbb{R}^d)$, we define
$$|||f|||_{k+1,p} = \|f\|_{W^k} + \sum_{i=1}^{d} \left(\sup_{0<h<1} \frac{1}{h} \|\delta_{he_i} f\|_{W^{k,p}} \right).$$

We can characterise the functions of $W^{k+1,p}(\mathbb{R}^d)$ using these norms.

Proposition III.2.32. *We have the following equality:*
$$W^{k+1,p}(\mathbb{R}^d) = \left\{ f \in W^{k,p}(\mathbb{R}^d),\ |||f|||_{k+1,p} < +\infty \right\},$$
and we have
$$\|\nabla f\|_{W^{k,p}} \leq |||f|||_{k+1,p},\ \forall f \in W^{k+1,p}(\mathbb{R}^d).$$

Proof.

- If $f \in W^{k+1,p}(\mathbb{R}^d)$, Lemma III.2.31 asserts that $|||f|||_{k+1,p}$ is finite and also that
$$|||f|||_{k+1,p} \leq \|f\|_{W^{k,p}} + \sum_{i=1}^{d} \left\| \frac{\partial f}{\partial x_i} \right\|_{W^{k,p}}.$$

- Now, let $f \in W^{k,p}(\mathbb{R}^d)$ such that $|||f|||_{k+1,p} < +\infty$.
Since $|||f|||_{k+1,p} < +\infty$ we know that for all $i \in \{1, \ldots, d\}$ the family $\big((\delta_{he_i} f)/h\big)_{0<h<1}$ is bounded in $W^{k,p}(\mathbb{R}^d)$; we can therefore find a sequence $(h_n)_n$ of reals which tends towards 0, such that $0 < h_n < 1$ and such that for all i, the sequence $\big((\delta_{h_n e_i} f)/h_n\big)_n$ weakly (or weakly-$\star$) converges in $W^{k,p}(\mathbb{R}^d)$ towards a certain function g_i (Theorem II.2.7). Furthermore, we have
$$\frac{\delta_{he_i} f}{h} \xrightarrow[h \to 0]{} \frac{\partial f}{\partial x_i}, \text{ in } \mathcal{D}'(\mathbb{R}^d).$$
Indeed, from Lemma III.2.30 and the dominated convergence theorem, we have for any function $\varphi \in \mathcal{D}(\mathbb{R}^d)$,

$$\int_{\mathbb{R}^d} \frac{\delta_{he_i} f}{h} \varphi \, dx = \int_{\mathbb{R}^d} f \frac{\delta_{-he_i} \varphi}{h} \, dx \xrightarrow[h \to 0]{} - \int_{\mathbb{R}^d} f \frac{\partial \varphi}{\partial x_i} \, dx = \left\langle \frac{\partial f}{\partial x_i}, \varphi \right\rangle_{\mathcal{D}', \mathcal{D}}.$$

We know from Proposition II.2.43 that the weak (or weak-$\star$) convergence in $W^{k,p}(\mathbb{R}^d)$ implies convergence in the sense of distributions. Thus, by uniqueness of the limit in $\mathcal{D}'(\mathbb{R}^d)$, we deduce that $\partial f / \partial x_i = g_i \in W^{k,p}(\mathbb{R}^d)$, which proves that f belongs to $W^{k+1,p}(\mathbb{R}^d)$. However, we have for all n,

$$\|f\|_{W^{k,p}} + \sum_{i=1}^{d} \left\| \frac{\delta_{h_n e_i} f}{h_n} \right\|_{W^{k,p}} \leq |||f|||_{k+1,p}.$$

By passing to the lower limit with respect to n in this inequality and by using Corollary II.2.8, we obtain

$$\|f\|_{W^{k,p}} + \sum_{i=1}^{d} \left\| \frac{\partial f}{\partial x_i} \right\|_{W^{k,p}} \leq |||f|||_{k+1,p},$$

and the claim is proved.

$\square$

2.8 Sobolev embeddings

We now demonstrate some results concerning Sobolev embeddings.

Definition III.2.33. *For all $1 \leq p \leq +\infty$, we define p^*,* the critical exponent associated with p *by*

$$\begin{cases} \dfrac{1}{p^*} = \dfrac{1}{p} - \dfrac{1}{d}, & \text{for } p < d, \\ \text{any } p^* \text{ such that } 1 \leq p^* < +\infty, & \text{for } p = d, \\ p^* = +\infty, & \text{for } p > d, \end{cases}$$

where d is the space dimension.

Theorem III.2.34 (Sobolev embeddings). *Let Ω be a bounded Lipschitz domain in $\mathbb{R}^d$.*

1. For $1 \leq p < +\infty$ and $1 \leq q \leq p^*$, then

$$W^{1,p}(\Omega) \subset L^q(\Omega), \tag{III.41}$$

with compact embedding if $1 \leq q < p^*$.
Furthermore, for $d < p \leq +\infty$ and $0 \leq \alpha \leq 1 - d/p$, we have

$$W^{1,p}(\Omega) \subset \mathcal{C}^{0,\alpha}(\overline{\Omega}), \tag{III.42}$$

with compact embedding for $0 \leq \alpha < 1 - d/p$.

2. • *For all* $q \in [1, d'[$, *we have*

$$L^1(\Omega) \subset W^{-1,q}(\Omega).$$

• *For all* $p \in]1, d[$, *we have*

$$L^p(\Omega) \subset W^{-1,p^*}(\Omega).$$

• *For all* $p \geq d$, *we have*

$$L^p(\Omega) \subset W^{-1,\infty}(\Omega).$$

Proof.

1. • We only prove (III.41) in the case $p < d$. We refer to [27] for the proof of (III.42).
By using the extension operator E_Ω studied in Section 2.4, we observe that it is enough to prove the embedding of $W^{1,p}(\mathbb{R}^d)$ into $L^{p^*}(\mathbb{R}^d)$. We first consider the case $p = 1$, $p^* = 1^* = d/(d-1)$. By density of $\mathcal{D}(\mathbb{R}^d)$ in $W^{1,1}(\mathbb{R}^d)$ it is enough to prove the inequality

$$\|u\|_{L^{d/(d-1)}} \leq C\|\nabla u\|_{L^1}, \forall u \in \mathcal{D}(\mathbb{R}^d), \tag{III.43}$$

for some $C > 0$ independent of u. For such a function u, which is compactly supported we have, for any $1 \leq i \leq d$,

$$|u(x)| \leq \int_{\mathbb{R}} \left| \frac{\partial u}{\partial x_i} \right| dx_i, \forall x \in \mathbb{R}^d.$$

We denote the right-hand side member of this inequality, which does not depend on the variable x_i, as $\tilde{f}_i(x) = f_i(x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_d)$. Considering the product of all such inequalities for i varying from 1 to d we obtain

$$|u(x)|^d \leq \prod_{i=1}^d \tilde{f}_i(x),$$

and thus

$$|u(x)|^{d/(d-1)} \leq \prod_{i=1}^d (\tilde{f}_i(x))^{1/(d-1)}.$$

Each f_i belongs to $L^1(\mathbb{R}^{d-1})$, thus we see that $f_i^{1/(d-1)}$ belongs to $L^{d-1}(\mathbb{R}^{d-1})$. We may therefore apply Proposition II.2.19 to conclude that $|u|^{d/(d-1)} \in L^1(\mathbb{R}^d)$, which gives $u \in L^{d/(d-1)}(\mathbb{R}^d)$, and that

$$\|u\|_{L^{d/(d-1)}(\mathbb{R}^d)} \leq \prod_{i=1}^{d-1} \|f_i\|_{L^1(\mathbb{R}^{d-1})}^{1/d}.$$

Coming back to the definition of f_i we conclude that $\|f_i\|_{L^1(\mathbb{R}^{d-1})} \leq \|\nabla u\|_{L^1(\mathbb{R}^d)}$ for any i and the inequality (III.43) is proved.
Let us now prove the result for the general case $1 < p < d$. For any $u \in \mathcal{D}(\mathbb{R}^d)$ we set $v = |u|^{\alpha-1}u$ for some $\alpha > 1$ to be determined later. We observe that $v \in \mathcal{C}_c^1(\mathbb{R}^d) \subset W^{1,1}(\mathbb{R}^d)$ and that

$$\nabla v = \alpha |u|^{\alpha-1} \nabla u.$$

Applying (III.43) to v and using the Hölder inequality leads to

$$\|u\|_{L^{\alpha d/(d-1)}}^{\alpha} \leq \alpha \| |u|^{\alpha-1} \nabla u\|_{L^1} \leq \alpha \|\nabla u\|_{L^p} \|u\|_{L^{p'(\alpha-1)}}^{\alpha-1}. \tag{III.44}$$

Let us now observe that we can choose α in such a way that the two norms of u in this inequality are the same. To this end, it is needed that

$$\frac{\alpha d}{d-1} = \frac{(\alpha-1)p}{p-1};$$

that is

$$\alpha = \frac{p(d-1)}{d-p}.$$

This number is greater than 1 since $1 < p < d$ and, with this choice, we compute

$$\frac{\alpha d}{d-1} = \frac{pd}{d-p} = p^*,$$

and finally (III.44) gives the required inequality.

- Let us now prove the compactness of the embeddings in the case where $1 \leq q < r^*$. This amounts to showing that the bounded sets of $W^{1,p}(\Omega)$ are relatively compact in $L^q(\Omega)$. Hence, let $\mathcal{F}$ be the unit sphere of $W^{1,p}(\Omega)$. We use the Kolmogorov theorem (Theorem II.3.2) to prove that $\mathcal{F}$ is relatively compact in $L^q(\Omega)$.
  – Since $q < p^*$, there exists a real number s such that $q < s < r^*$. From the Sobolev embeddings above, we know that $W^{1,p}(\Omega)$ is embedded in a continuous way into $L^s(\Omega)$ and that a constant $C > 0$ exists, such that

  $$\|f\|_{L^s} \leq C\|f\|_{W^{1,s}}, \forall f \in W^{1,p}(\Omega).$$

  Hence, for all $f \in \mathcal{F}$, we have

  $$\|f\|_{L^s} \leq C'.$$

  As a consequence, for all bounded open sets ω, such that $\overline{\omega} \subset \Omega$, via the Hölder inequality, we have

$$\|f\|_{L^q(\Omega\setminus\omega)} \leq |\Omega\setminus\omega|^{(s-q)/(sq)}\|f\|_{L^s(\Omega\setminus\omega)} \leq C'|\Omega\setminus\omega|^{(s-q)/(sq)}, \, \forall f \in \mathcal{F}.$$

This shows that by choosing ω sufficiently close to Ω, the second assumption of the Kolmogorov theorem is satisfied. Furthermore, we see here why this proof requires the strict condition $q < p^*$. Indeed, we need to be able to say that $\mathcal{F}$ is bounded in an L^s space with s a little larger than q. We show in Remark III.2.16, which follows the proof of this theorem, that the Sobolev embedding in the case $q = p^*$ is never compact even in a bounded domain.

– To satisfy the first condition of the Kolmogorov theorem, we fix a bounded open set ω, such that $\overline{\omega} \subset \Omega$. For sufficiently small $\alpha > 0$ we have $\omega \subset \Omega_\alpha$, so that for any $h \in \mathbb{R}^d$ such that $|h| < \alpha/2$, we have

$$\|\tau_h f - f\|_{L^p(\omega)} \leq \|\tau_h f - f\|_{L^p(\Omega_\alpha)} \leq \|\tau_h f - f\|_{L^p(\Omega_{2|h|})} \leq |h|\|\nabla f\|_{L^p(\Omega)},$$

thanks to Theorem III.2.29. Hence, for all $f \in \mathcal{F}$, we have

$$\|\tau_h f - f\|_{L^p(\omega)} \leq C|h|.$$

We now establish a result of the same type in $L^q(\omega)$, by using the interpolation inequality given by Lemma II.2.33. Hence

$$\|\tau_h f - f\|_{L^q(\omega)} \leq \|\tau_h f - f\|^{\alpha}_{L^p(\omega)}\|\tau_h f - f\|^{(1-\alpha)}_{L^s(\omega)},$$

for a certain $\alpha \in]0, 1[$ (here, we also use the fact that $q < p^*$). However, $\mathcal{F}$ is bounded in $W^{1,p}(\Omega)$ and, therefore, in $L^s(\Omega)$. In particular there exists a $D > 0$ such that

$$\|\tau_h f - f\|_{L^s(\omega)} \leq 2\|f\|_{L^s(\Omega)} \leq D, \, \forall f \in \mathcal{F}.$$

Hence, we finally obtain

$$\|\tau_h f - f\|_{L^q(\omega)} \leq |h|^{\alpha} D^{1-\alpha}, \, \forall f \in \mathcal{F},$$

which certainly shows that $\mathcal{F}$ satisfies the first hypothesis of the Kolmogorov theorem. In conclusion, we have shown that $\mathcal{F}$ is relatively compact in $L^q(\Omega)$.

2. This involves arguing by duality from the previous Sobolev embedding.

- The case where $q = 1$ is trivial from the definition of $W^{-1,1}(\Omega)$.
  Now, let $q \in]1, d'[$; then $q' \in]d, +\infty[$ and therefore $(q')^* = +\infty$. In other words, we have the continuous embedding

$$W_0^{1,q'}(\Omega) \subset W^{1,q'}(\Omega) \subset L^\infty(\Omega). \tag{III.45}$$

We note that $W_0^{1,q'}(\Omega)$ is not dense in $L^\infty(\Omega)$ and that the dual of $L^\infty(\Omega)$ is not $L^1(\Omega)$, so we cannot directly invoke a duality argument. Nevertheless, for any $f \in L^1(\Omega)$, we can define a linear functional T_f on $W_0^{1,q'}(\Omega)$ by

$$T_f(g) = \int_\Omega f g \, dx,$$

and from (III.45) we have

$$|T_f(g)| \leq C\|f\|_{L^1}\|g\|_{W_0^{1,q'}},$$

and therefore $T_f \in W^{-1,q}(\Omega)$ and $\|T_f\|_{W^{-1,q}} \leq C\|f\|_{L^1}$ from Theorem III.2.24. It remains to observe that $f \mapsto T_f$ is injective. Therefore, let $f \in L^1(\Omega)$ such that $T_f = 0$. This implies that for all $\varphi \in \mathcal{D}(\Omega)$ we have

$$\int_\Omega f\varphi \, dx = 0.$$

However, since $\mathcal{D}(\Omega)$ is $\star$-weakly dense in $L^\infty(\Omega)$ (Theorem II.2.29), we deduce that

$$\int_\Omega f g \, dx = 0,$$

for all $g \in L^\infty$. By taking $g = \mathrm{sgn}(f)$, it gives $f = 0$ which proves the injectivity of the map T.

- Let $p \in]1, d[$, we set $q = p^* \in]d/(d-1), +\infty[$, such that $q' \in]1, d[$. From the first point, we have the dense embedding

$$W_0^{1,q'}(\Omega) \subset L^{(q')^*}(\Omega).$$

We note that since $q' < d$, $(q')^*$ is a finite real number which is perfectly defined (we are not in a critical case). By using Theorem III.2.24, it follows that

$$L^{((q')^*)'}(\Omega) \subset W^{-1,q}(\Omega).$$

It just remains to identify $((q')^*)'$, for which we use the fact that $q' < d$ and we write

$$\frac{1}{((q')^*)'} = 1 - \frac{1}{(q')^*} = 1 - \left(\frac{1}{q'} - \frac{1}{d}\right) = \frac{1}{q} + \frac{1}{d},$$

and since $q = p^*$ with $p < d$, it follows that

$$((q')^*)' = p,$$

which is the desired result.

- If $d \leq p \leq +\infty$ and $p > 1$, we have $1 \leq p' \leq d/(d-1) = 1^*$. Therefore, the Sobolev embedding theorem gives us the dense embedding

$$W_0^{1,1}(\Omega) \subset L^{p'}(\Omega),$$

which, using again Theorem III.2.24, gives

$$L^p(\Omega) \subset W^{-1,\infty}(\Omega).$$

The case where $p = d = 1$ is somewhat special because the "critical case" of Sobolev embedding is valid, as we show in Section 4: we have

$$W^{1,1}(I) \subset \mathcal{C}^0(\bar{I}) \subset L^\infty(I),$$

for any bounded interval of $\mathbb{R}$. In consequence, we can make the same argument as above and obtain

$$L^1(I) \subset W^{-1,\infty}(I).$$

□

Remark III.2.15. The first part of the preceding result remains valid in a nonbounded domain provided it is restricted to exponents q satisfying $p \leq q \leq p^*$. Furthermore, we lose the compactness of the embeddings. Let us show, for example, that the embedding of $H^1(\mathbb{R}^d)$ in $L^2(\mathbb{R}^d)$ is not compact. Indeed, if $f \in H^1(\mathbb{R}^d)$ is not zero, we consider the sequence of translated functions $f_n(x) = f(x - n\,e)$ where e is a nonnull vector of $\mathbb{R}^d$. It is obvious that the sequence $(f_n)_n$ is bounded in $H^1(\mathbb{R}^d)$ and that $\|f_n\|_{L^2} = \|f\|_{L^2} = \text{const.} \neq 0$. Furthermore, we can easily show that $(f_n)_n$ weakly converges towards 0 in $L^2(\mathbb{R}^d)$. This proves that the embedding of $H^1(\mathbb{R}^d)$ into $L^2(\mathbb{R}^d)$ cannot be compact.

The second and third parts of the theorem are valid in unbounded domains, without modification of the statement.

Remark III.2.16. In the case where $p < d$, and therefore $p^* < +\infty$, the Sobolev embedding of $W^{1,p}(\Omega)$ into $L^{p^*}(\Omega)$ is never compact. As an example, assume that Ω is the unit ball of $\mathbb{R}^d$. Let us take a nonnull function $f \in \mathcal{D}(\Omega)$ that we extend by 0 on the whole space $\mathbb{R}^d$ and, for all $n \in \mathbb{N}$, we set

$$f_n(x) = n^{d/p^*} f\,(nx)\,.$$

We can easily show by change of variable that

$$\begin{aligned} \|f_n\|_{L^q} &= \|f\|_{L^q}.n^{d(1/p^*-1/q)},\ \forall q < +\infty, \\ \|\nabla f_n\|_{L^p} &= \|\nabla f\|_{L^p} = \text{cte}. \end{aligned}$$

Hence, the sequence $(f_n)_n$ is bounded in $W^{1,p}(\Omega)$ and converges towards zero in $L^q(\Omega)$ for all $q < p^*$.

If the embedding of $W^{1,p}(\Omega)$ into $L^{p^*}(\Omega)$ were compact, there would exist a subsequence $(f_{n_k})_k$ which would converge in $L^{p^*}(\Omega)$ towards a certain function g. However, Ω being bounded, we know that this convergence also occurs in $L^1(\Omega)$. This proves that necessarily $g = 0$ because $\|f_n\|_{L^1} = \|f\|_{L^1} n^{d(1/p^*-1)} \xrightarrow[n\to\infty]{} 0$. However, we have shown above that

$$\|f_{n_k}\|_{L^{p^*}} = \|f\|_{L^{p^*}} \neq 0, \, \forall k,$$

which contradicts the fact that $g = 0$.

In one particular case, we now make the inequalities provided by certain Sobolev embeddings more precise. Beyond the result itself, it is very instructive to have in mind the technique of the proof which can be used in many circumstances in order to obtain many Sobolev inequalities (such as anisotropic Sobolev inequalities, for instance).

Proposition III.2.35. *Let Ω be a Lipschitz domain of $\mathbb{R}^d$ with compact boundary. Let $p \in [1, +\infty]$ and $q \in [p, p^*]$. There is a $C > 0$ such that*

$$\|u\|_{L^q} \leq C\|u\|_{L^p}^{1+d/q-d/p} \|u\|_{W^{1,p}}^{d/p-d/q}, \, \forall u \in W^{1,p}(\Omega).$$

Proof.

Thanks to the assumptions on Ω, we can use the extension operator E_Ω from $W^{1,p}(\Omega)$ into $W^{1,p}(\mathbb{R}^d)$ which allows us to bring back the proof to the case of $\mathbb{R}^d$ (see Section 2.4).

Thanks to the assumptions on q, the Sobolev embedding theorem shows that there exists a $C > 0$ such that for any $v \in W^{1,p}(\mathbb{R}^d)$ we have

$$\|v\|_{L^q(\mathbb{R}^d)} \leq C(\|v\|_{L^p(\mathbb{R}^d)} + \|\nabla v\|_{L^p(\mathbb{R}^d)}).$$

The idea is now to proceed by homogeneity. For any $u \in W^{1,p}(\mathbb{R}^d)$ and any $\alpha > 0$ the function $v_\alpha(x) = u(\alpha x)$ also belongs to $W^{1,p}(\mathbb{R}^d)$ and we can therefore apply the inequality above to get

$$\|v_\alpha\|_{L^q} \leq C(\|v_\alpha\|_{L^p} + \|\nabla v_\alpha\|_{L^p}). \tag{III.46}$$

We now need to calculate the various norms presented by a change of variables. This gives:

$$\|v_\alpha\|_{L^r}^r = \int_{\mathbb{R}^d} |u(\alpha x)|^r \, dx = \frac{1}{\alpha^d} \int_{\mathbb{R}^d} |u(y)|^r \, dy = \frac{1}{\alpha^d} \|u\|_{L^r}^r, \, \forall r < +\infty$$

and

$$\|\nabla v_\alpha\|_{L^p}^p = \frac{1}{\alpha^{d-p}} \|\nabla u\|_{L^p}^p.$$

This is substituted into inequality (III.46) and we find, for all $\alpha > 0$

$$\frac{1}{\alpha^{\frac{d}{q}}} \|u\|_{L^q} \leq C \left(\frac{1}{\alpha^{d/p}} \|u\|_{L^p} + \frac{1}{\alpha^{d/p-1}} \|\nabla u\|_{L^p} \right);$$

that is,

$$\|u\|_{L^q} \leq C\left(\alpha^{d(1/q-1/p)}\|u\|_{L^p} + \alpha^{d(1/q-1/p)+1}\|\nabla u\|_{L^p}\right).$$

Inasmuch as we wish to obtain an inequality for which the right-hand side is a product of various terms, it is reasonable to seek to equalise the two terms of the sum present in the inequality above. Hence, we choose

$$\alpha = \frac{\|u\|_{L^p}}{\|\nabla u\|_{L^p}},$$

which is also equivalent to minimising the right-hand side of the inequality with respect to α. This gives

$$\begin{aligned}\|u\|_{L^q} &\leq 2C\alpha^{d/q}\frac{\|\nabla u\|_{L^p}^{d/p}}{\|u\|_{L^p}^{(d/p)-1}} = 2C\|u\|_{L^p}^{(d/q)-(d/p)+1}\|\nabla u\|_{L^p}^{(d/p)-(d/q)}\\ &\leq C'\|u\|_{L^p}^{(d/q)-(d/p)+1}\|u\|_{W^{1,p}}^{(d/p)-(d/q)}.\end{aligned}$$

□

Remark III.2.17. We might believe that we could have obtained a little better result from the inequality

$$\|u\|_{L^q} \leq C'\|u\|_{L^p}^{(d/q)-(d/p)+1}\|\nabla u\|_{L^p}^{(d/p)-(d/q)}.$$

However, we need to remember that this inequality does not remain in this form when we make use of the extension operator to study the case of a generic Lipschitz bounded domain Ω. We immediately see that in this case the inequality above is false for constant functions. By contrast, this result will be true for the functions of $W_0^{1,p}(\Omega)$ or for functions with a zero mean-value, thanks to the Poincaré inequalities (see Propositions III.2.38 and III.2.39).

By using the Sobolev embeddings we proved above, it is possible to deduce new properties of the trace space $W^{1-1/p,p}(\partial\Omega)$.

Theorem III.2.36. *Let Ω be a Lipschitz domain in $\mathbb{R}^d$ with compact boundary and $1 < p < d$. Then, for any*

$$r \in \left[p, \frac{p(d-1)}{d-p}\right],$$

we have

$$W^{1-1/p,p}(\partial\Omega) \subset L^r(\partial\Omega),$$

and moreover there exists $C > 0$ such that for any $u \in W^{1,p}(\Omega)$ we have

$$\|\gamma_0(u)\|_{L^r(\partial\Omega)} \leq C\|u\|_{L^p(\Omega)}^{1-(d/p)+((d-1)/r)}\|u\|_{W^{1,p}(\Omega)}^{(d/p)-((d-1)/r)}. \tag{III.47}$$

In the case where $p = d$, the previous result holds true for any $r \in [p, +\infty[$.

Proof.

By using the same arguments as in Section 2.5, it is enough to prove the result in the flat half-space $\mathbb{R}^d_+$. For any $\varphi \in \mathcal{C}^\infty_c(\mathbb{R}^d)$ we have

$$|\varphi(\bar{x}, 0)|^r = |\varphi(\bar{x}, x_d)|^r - r \int_0^{x_d} |\varphi(\bar{x}, s)|^{r-2} \varphi(\bar{x}, s) \frac{\partial \varphi}{\partial x_d}(\bar{x}, s)\, ds,$$

so that by using the Hölder inequality with exponents p and $p' = p/(p-1)$ we get

$$\|\varphi(., 0)\|^r_{L^r(\mathbb{R}^{d-1})} \leq r \|\varphi\|_{L^{(p(r-1))/(p-1)}(\mathbb{R}^d_+)} \left\| \frac{\partial \varphi}{\partial x_d} \right\|_{L^p(\mathbb{R}^d_+)}.$$

Observe now that

$$p \leq \frac{p(r-1)}{p-1} \leq \frac{dp}{d-p} = p^*,$$

so that we have

$$\frac{p-1}{p(r-1)} = \frac{\theta}{p} + \frac{1-\theta}{p^*}, \text{ with } \theta = 1 - \frac{d}{p} + \frac{d-1}{r}.$$

Using Lemma II.2.33, we deduce that

$$\|\varphi(., 0)\|^r_{L^r(\mathbb{R}^{d-1})} \leq r \|\varphi\|^\theta_{L^p(\mathbb{R}^d_+)} \|\varphi\|^{1-\theta}_{L^{p^*}(\mathbb{R}^d_+)} \left\| \frac{\partial \varphi}{\partial x_d} \right\|_{L^p(\mathbb{R}^d_+)},$$

and the claim is proved by using the Sobolev embedding $W^{1,p}(\mathbb{R}^d_+) \subset L^{p^*}(\mathbb{R}^d_+)$.

$\square$

Remark III.2.18.

- This theorem shows, in particular, that $W^{1-1/p,p}(\partial\Omega)$ is continuously embedded into $L^r(\partial\Omega)$ for any suitable r. In particular, it proves that $W^{1-1/p,p}(\partial\Omega)$ is a strict subspace of $L^p(\partial\Omega)$, for $p > 1$.
- Let us prove that it also implies that the embedding of $W^{1-1/p,p}(\partial\Omega)$ into $L^r(\partial\Omega)$ is compact as soon as $r < (p(d-1))/(d-p)$.

  Let us assume first that Ω is bounded. Let $(v_n)_n$ be a bounded sequence in $W^{1-1/p,p}(\partial\Omega)$. By continuity of the lifting operator then the sequence $u_n = R_0(v_n)$ is bounded in $W^{1,p}(\Omega)$. The embedding of $W^{1,p}(\Omega)$ in $L^p(\Omega)$ is compact, therefore we can find a subsequence $(u_{n_k})_k$ which converges towards some u weakly in $W^{1,p}(\Omega)$ and strongly in $L^p(\Omega)$. We define $v = \gamma_0(u) \in W^{1-1/p,p}(\partial\Omega)$. From (III.47), we deduce that, for some $\theta \in]0, 1[$, we have

$$\begin{aligned}\|v_{n_k} - v\|_{L^r(\partial\Omega)} &\leq C\|u_{n_k} - u\|^{\theta}_{L^p(\Omega)}\|u_{n_k} - u\|^{1-\theta}_{W^{1,p}(\Omega)}\\ &\leq C'\|u_{n_k} - u\|^{\theta}_{L^p(\Omega)} \xrightarrow[k\to\infty]{} 0,\end{aligned}$$

which proves the claim.

In the case where Ω is unbounded (with compact boundary), we can take an $R > 0$ such that $\partial\Omega \subset B(0, R)$ and then define $\widetilde{\Omega} = \Omega \cap B(0, R)$. We easily check that $\widetilde{\Omega}$ is a bounded Lipschitz domain and that $W^{1-1/p,p}(\partial\Omega)$ is a subspace of $W^{1-1/p,p}(\partial\widetilde{\Omega})$ so that the claim is proved by applying the result on the bounded domain $\widetilde{\Omega}$.

As a consequence of the Sobolev embeddings we can now study the regularity of the product of two functions (or distributions) in Sobolev spaces (possibly with negative exponents). For convenience reasons, we only give the following result in the case of dimension $d = 3$. A similar result obviously holds in any dimension.

In particular, this result is often used in conjunction with Proposition II.2.12 to pass to the weak limit in a product of convergent sequences of functions (one convergence being strong and the other one being weak) in the correct Sobolev spaces.

We recall that the notation p^* is introduced in Definition III.2.33.

Theorem III.2.37 (Product theorem). *Let Ω be a bounded Lipschitz domain of $\mathbb{R}^3$.*

1. *Let $1 \leq p \leq q \leq +\infty$ such that $1/q^* + 1/p \leq 1$. We define $t \in [1, +\infty]$ by $1/t = 1/s^* + 1/p$, then the product mapping*

$$(u, v) \in W^{1,p}(\Omega) \times W^{1,q}(\Omega) \mapsto uv \in W^{1,t}(\Omega),$$

is well-defined and continuous.

2. *For $1 \leq p, q \leq +\infty$, such that $1/p + 1/q \leq 1$, we define t_1 by $1/t_1 = 1/p^* + 1/q^*$, then the product mapping*

$$(u, v) \in W^{1,p}(\Omega) \times W^{-1,q}(\Omega) \mapsto uv \in W^{-1,\sigma}(\Omega),$$

is well-defined and continuous for any $\sigma < t_1$ in the case where $1/p+1/q = 1$ and $p \leq 3$, and for any $\sigma \leq t_1$ in all the other cases.

Proof.

1. Since q^* is always greater than q, it is obvious that we have

$$\|uv\|_{L^t} \leq C\|u\|_{L^p}\|v\|_{L^q} \leq C\|u\|_{W^{1,p}}\|v\|_{W^{1,q}},$$

for all $u \in W^{1,p}(\Omega)$ and $v \in W^{1,q}(\Omega)$.

Now let $u \in \mathcal{C}^\infty(\overline{\Omega})$ and let $v \in W^{1,s}(\Omega)$, by integration by parts; we then have for all $\varphi \in \mathcal{D}(\Omega)$,

$$\begin{aligned}\left\langle \frac{\partial(uv)}{\partial x_i}, \varphi \right\rangle_{\mathcal{D}',\mathcal{D}} &= -\int_\Omega uv \frac{\partial \varphi}{\partial x_i}\, dx \\ &= -\int_\Omega v \frac{\partial(u\varphi)}{\partial x_i}\, dx + \int_\Omega v \frac{\partial u}{\partial x_i}\varphi\, dx, \\ &= \int_\Omega \frac{\partial v}{\partial x_i} u\varphi\, dx + \int_\Omega v \frac{\partial u}{\partial x_i}\varphi\, dx,\end{aligned}$$

which shows that we have

$$\frac{\partial(uv)}{\partial x_i} = u\frac{\partial v}{\partial x_i} + v\frac{\partial u}{\partial x_i}, \tag{III.48}$$

in the sense of distributions. Furthermore, by using Hölder's inequality, we obtain

$$\left\| v\frac{\partial u}{\partial x_i}\right\|_{L^t} \leq C\|v\|_{L^{q^*}}\|u\|_{W^{1,p}},$$

and

$$\left\| u\frac{\partial v}{\partial x_i}\right\|_{L^{t_2}} \leq C\|u\|_{L^{p^*}}\|v\|_{W^{1,q}},$$

where t_2 is defined by

$$\frac{1}{t_2} = \frac{1}{p^*} + \frac{1}{q}.$$

Since $p \leq q$, we always have $t \leq t_2$ such that, by using the Sobolev embeddings given by Theorem III.2.34, we have indeed shown that

$$\|uv\|_{W^{1,t}} \leq C\|u\|_{W^{1,p}}\|v\|_{W^{1,q}},$$

for all $v \in W^{1,q}(\Omega)$ and u smooth enough. In the case where p is finite, we can make the conclusion from the density of $\mathcal{C}^\infty(\overline{\Omega})$ in $W^{1,p}(\Omega)$. In the case where p is infinite (which implies $q = +\infty$), the formula (III.48) is justified in $L^t(\Omega)$ for all finite t because of the above arguments. However, since all the terms on the right-hand side are bounded, we can conclude that the estimate above is also true for $p = q = t = +\infty$.

2. We recall that a distribution v belongs to $W^{-1,q}(\Omega)$ if and only if

$$v = v_0 + \sum_{j=1}^{3} \frac{\partial v_j}{\partial x_j},$$

where $v_j \in L^q(\Omega)$, $j = 1, \ldots, 3$ (see Definition (III.37)). Hence, the product uv is defined in the following way

$$\begin{aligned}uv &= uv_0 + \sum_{j=1}^{3} \frac{\partial(uv_j)}{\partial x_j} - \sum_{j=1}^{3} v_j \frac{\partial u}{\partial x_j}, \\ &= f_0 + f_1 + f_2.\end{aligned}$$

We note that this definition does not depend on the choice of the v_ks which define the distribution v. Let us now look at each term. We have

$$f_0 \in L^{t_1}(\Omega),$$
$$f_1 \in W^{-1,t_1}(\Omega),$$
$$f_2 \in L^{t_2}(\Omega), \text{ with } \frac{1}{t_2} = \frac{1}{p} + \frac{1}{q}.$$

- Let us first assume that $t_2 = 1$ (i.e., $q = p'$); then the preceding theorem tells us that
$$L^{t_2}(\Omega) \subset W^{-1,q}(\Omega),$$
for all $1 < q < \frac{3}{2}$. Let us now calculate t_1.
  - If $p > 3$, then $p^* = +\infty$ and therefore
$$\frac{1}{t_1} = \frac{1}{q} = 1 - \frac{1}{p} \in \left]\frac{2}{3}, 1\right[,$$
or even
$$t_1 \in \left]1, \frac{3}{2}\right[.$$
In this case, we therefore have
$$L^{t_2}(\Omega) \subset W^{-1,t_1}(\Omega),$$
and by bringing together the results obtained on f_0, f_1, and f_2, we indeed have
$$uv \in W^{-1,t_1}(\Omega).$$
  - If $p = 3$, $q = \frac{3}{2}$, and p^* is finite and as large as we wish, then
$$\frac{1}{t_1} = \frac{1}{p^*} + \frac{2}{3}$$
where t_1 is any value strictly less than $\frac{3}{2}$ and uv is indeed in $W^{-1,\sigma}(\Omega)$ for all $\sigma < t_1$.
  - If $p < 3$, then $t_1 = \frac{3}{2}$ and in this case $L^{t_2}(\Omega) = L^1(\Omega)$ is only embedded in the $W^{-1,\sigma}(\Omega)$ for all $\sigma < \frac{3}{2}$.
- If now $t_2 \in]1, 3[$, then
$$L^{t_2}(\Omega) \subset W^{-1,t_2^*}(\Omega),$$
and therefore
$$uv \in W^{-1,t_2^*}(\Omega) + W^{-1,t_1}(\Omega) = W^{-1,\min(t_2^*,t_1)}(\Omega).$$
However, $t_2 < 3$, hence we have

$$\frac{1}{t_2^*} = \frac{1}{t_2} - \frac{1}{3} = \frac{1}{p} - \frac{1}{3} + \frac{1}{q} \leq \frac{1}{p^*} + \frac{1}{q} = \frac{1}{t_1},$$

hence $t_2^* \geq t_1$ which proves the result.

- Finally, if $t_2 \geq 3$ then

$$L^{t_2}(\Omega) \subset W^{-1,\infty}(\Omega) \subset W^{-1,t_1}(\Omega),$$

which allows the conclusion to be reached again.

□

2.9 Poincaré and Hardy inequalities

In the first part of this section we present two Poincaré-type inequalities. The idea is that, in a bounded domain, if we possess a minimum of a priori information on a function u (essentially saying that u cannot be a nonzero constant), then the $W^{1,p}$-norm of u is equivalent to the L^p-norm of the gradient of u. The first result is valid when we control the trace of u on a nontrivial part of the boundary and the second result when we control the mean value of u on Ω. Note that, with the same strategy of proof (i.e., by using a compactness argument) we prove similar results in slightly different contexts in Chapter IV (see Propositions IV.1.7 and IV.7.7).

The second part of this section is dedicated to the proof of the Hardy inequality, which says that if $u \in W_0^{1,p}(\Omega)$ (this means that $u = 0$ on the boundary in some sense; see Remark III.2.8), then the function u/δ belongs to $L^p(\Omega)$, where δ is the *distance to the boundary* function.

Proposition III.2.38 (Generalised Poincaré inequality). *Let Ω be a bounded, connected, Lipschitz domain of $\mathbb{R}^d$. Let Γ_1 be a part of the boundary $\partial\Omega$ with a nonzero surface measure. For $1 \leq p < +\infty$, we define*

$$W_{0,\Gamma_1}^{1,p}(\Omega) = \left\{u \in W^{1,p}(\Omega),\ \gamma_0(u)_{|\Gamma_1} = 0\right\}.$$

There exists a $C > 0$ such that, for all functions $u \in W_{0,\Gamma_1}^{1,p}(\Omega)$ we have:

$$\|u\|_{L^p} \leq C\|\nabla u\|_{L^p};$$

that is, in other words,

$$\|u\|_{W^{1,p}} \leq C'\|\nabla u\|_{L^p}.$$

Proof.

From Section 1.2.3, the fact that Γ_1 has a nonzero surface measure means that

$$\int_{\partial\Omega} 1_{\Gamma_1}\, d\sigma > 0. \tag{III.49}$$

Let us assume that the first inequality is not satisfied; there then exists a sequence $(u_n)_n$ of $W^{1,p}_{0,\Gamma_1}(\Omega)$ such that:

$$\|u_n\|_{L^p} \geq n\|\nabla u_n\|_{L^p(\Omega)}.$$

By homogeneity, we can assume that $\|u_n\|_{L^p}$ has value 1, which then implies that $\|\nabla u_n\|_{L^p} \leq 1/n$. Hence, the sequence $(u_n)_n$ is bounded in $W^{1,p}(\Omega)$ and there exists a subsequence $(u_{n_k})_k$ which weakly converges towards u in $W^{1,p}(\Omega)$. The embedding of $W^{1,p}(\Omega)$ in $L^p(\Omega)$ is compact, therefore we know that this sequence strongly converges in $L^p(\Omega)$ (Proposition II.3.4). Hence, at the limit, u satisfies

$$\|u\|_{L^p} = 1. \tag{III.50}$$

Since $(\nabla u_n)_n$ converges towards ∇u in the sense of distributions and towards 0 in $L^p(\Omega)$, we therefore have $\nabla u = 0$ in the sense of distributions. Since the open set Ω is connected, we can deduce from Lemma II.2.44 that u is a constant referred to as $\bar{u} \in \mathbb{R}$. Moreover we have

$$\bar{u}\int_{\partial\Omega} 1_{\Gamma_1}\, d\sigma = \int_{\partial\Omega} 1_{\Gamma_1}\gamma_0(u)\, d\sigma = 0,$$

and thus $\bar{u} = 0$ thanks to (III.49). Therefore, we proved that $u = 0$. The contradiction with (III.50) then follows directly.

□

This result is particularly useful in the case $\Gamma_1 = \partial\Omega$. It implies that $u \mapsto \|\nabla u\|_{L^p(\Omega)}$ is a norm on $W^{1,p}_0(\Omega)$ which is equivalent to the $W^{1,p}(\Omega)$-norm.

Proposition III.2.39 (Poincaré–Wirtinger inequality). *Let Ω be a connected, bounded Lipschitz domain and $1 \leq p < +\infty$. There exists $C > 0$ such that, for all functions u of $W^{1,p}(\Omega)$, we have*

$$\left\| u - \frac{1}{|\Omega|}\int_\Omega u(x)\, dx \right\|_{W^{1,p}} \leq C\|\nabla u\|_{L^p},$$

or equivalently

$$\|u\|_{L^p} \leq C\left(\|\nabla u\|_{L^p} + \frac{1}{|\Omega|}\left|\int_\Omega u(x)\, dx\right|\right).$$

Proof.

The proof is entirely similar to that of the preceding proposition. If the inequality above is false, then there exists a sequence $(u_n)_n$ of functions of $W^{1,p}(\Omega)$ such that:

$$\|u_n\|_{L^p} \geq n \left(\|\nabla u_n\|_{L^p} + \frac{1}{|\Omega|} \left| \int_\Omega u_n(x)\, dx \right| \right).$$

By homogeneity, we can assume that $\|u_n\|_{L^p} = 1$. Hence, the sequence $(u_n)_n$ is a bounded sequence of $W^{1,p}(\Omega)$ from which we can extract a subsequence $(u_{n_k})_k$ which is weakly converging in $W^{1,p}(\Omega)$ and strongly converging in $L^p(\Omega)$ (by compactness of the Sobolev embedding) towards a certain function u of $W^{1,p}(\Omega)$.

However, since $\|\nabla u_n\|_{L^p} \leq 1/n$, the limit u satisfies $\nabla u = 0$ in the sense of distributions, which is equivalent to saying via Lemma II.2.44, that u is constant because Ω is connected. This constant is necessarily zero because at the limit we have

$$\frac{1}{|\Omega|} \left| \int_\Omega u(x)\, dx \right| = 0.$$

This contradicts the fact that $\|u_n\|_{L^p} = 1$ for all n.

□

Proposition III.2.40 (Hardy inequality in $W_0^{1,p}(\Omega)$). *Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain in $\mathbb{R}^d$ with compact boundary and $1 < p < +\infty$. There exists a $C > 0$, such that for any function u belonging to $W_0^{1,p}(\Omega)$, one has*

$$\left\| \frac{u}{\delta} \right\|_{L^p} \leq C \|\nabla u\|_{L^p},$$

where we recall that $\delta(x) = d(x, \partial\Omega)$ is the distance from any point $x \in \Omega$ to the boundary.

Let us observe once more that this result does not hold when $p = 1$.

Proof.

- Let us first prove the result for the flat half-space $\Omega = \mathbb{R}^d_+$. It is enough to prove the inequality for functions $u \in \mathcal{D}(\mathbb{R}^d_+)$. The general claim follows by the usual density argument.
  Since u vanishes on $\partial\Omega = \mathbb{R}^{d-1} \times \{0\}$, for any $x = (\bar{x}, x_d) \in \mathbb{R}^d_+$, we have

  $$\frac{u(\bar{x}, x_d)}{x_d} = \frac{u(\bar{x}, x_d) - u(\bar{x}, 0)}{x_d} = \frac{1}{x_d} \int_0^{x_d} \frac{\partial u}{\partial s}(\bar{x}, s)\, ds.$$

  By using Lemma II.4.7, we deduce

  $$\int_0^{+\infty} \left| \frac{u(\bar{x}, x_d)}{x_d} \right|^p dx_d \leq C \int_0^{+\infty} \left| \frac{\partial u}{\partial s}(\bar{x}, s) \right|^p ds.$$

  We integrate this formula with respect to $\bar{x} \in \mathbb{R}^{d-1}$ to get

  $$\int_{\mathbb{R}^d_+} \left| \frac{u(x)}{x_d} \right|^p dx \leq C \int_{\mathbb{R}^d_+} |\nabla u|^p\, dx,$$

and the claim is proved because $x_d = d(x, \mathbb{R}^{d-1} \times \{0\})$, for any $x \in \mathbb{R}^d_+$.

- We deduce now the result for any Lipschitz half-space $\Omega = \mathcal{H}_a$ by using the Lipschitz diffeomorphism T_a that we introduced in Proposition III.1.3, and Theorem III.2.13.

  For $u \in W^{1,p}_0(\mathcal{H}_a)$ we apply the Hardy inequality in the flat half-space to $v = u \circ T_a \in W^{1,p}_0(\mathbb{R}^d_+)$. It follows that

$$\left\| \frac{u \circ T_a}{x_d} \right\|_{L^p(\mathbb{R}^d_+)} \leq C \|\nabla(u \circ T_a)\|_{L^p(\mathbb{R}^d_+)}.$$

  Since $\nabla(u \circ T_a) = {}^t(\nabla T_a).(\nabla u) \circ T_a$ and ∇T_a is bounded, a simple change of variables gives that the right-hand side of this inequality is bounded by $\|\nabla u\|_{L^p(\mathcal{H}_a)}$.

  We may also perform a change of variables in the left-hand side and we get

$$\left\| \frac{u}{(T_a^{-1}(.))_d} \right\|_{L^p(\mathcal{H}_a)} \leq C \|\nabla u\|_{L^p(\mathcal{H}_a)}.$$

  In order to prove the result we just have to show that there is some $\alpha > 0$ such that $|(T_a^{-1}(x))_d| \leq \alpha \delta(x)$. In other words we have to show that

$$|y_d| \leq \alpha \delta(T_a(y)), \; \forall y \in \mathbb{R}^d_+.$$

  Let $\bar{x} \in \mathbb{R}^{d-1}$ we have

$$\begin{aligned} |T_a(y) - (\bar{x}, a(\bar{x}))| &= |(\bar{y} - \bar{x}, y_d + a_\gamma(\bar{y}, y_d) - a(\bar{x}))| \\ &\geq |\bar{y} - \bar{x}| + |y_d + a_\gamma(\bar{y}, y_d) - a(\bar{x})|. \end{aligned}$$

  Let $\xi > 0$ be determined later.

  – For $\bar{x}$ such that $|\bar{y} - \bar{x}| \geq \xi y_d$, we obviously deduce

$$|T(y) - (\bar{x}, a(\bar{x}))| \geq \xi y_d.$$

  – For $\bar{x}$ such that $|\bar{y} - \bar{x}| \leq \xi y_d$, we get

$$\begin{aligned} |T(y) - (\bar{x}, a(\bar{x}))| &\geq y_d - |a_\gamma(\bar{y}, y_d) - a(\bar{y})| - |a(\bar{y}) - a(\bar{x})| \\ &\geq y_d - \gamma \operatorname{Lip}(a) M_\eta y_d - \operatorname{Lip}(a)|\bar{y} - \bar{x}| \\ &\geq (1 - \gamma \operatorname{Lip}(a) M_\eta - \operatorname{Lip}(a)\xi)\, y_d. \end{aligned}$$

  If we choose ξ such that

$$\gamma \operatorname{Lip}(a) M_\eta + \operatorname{Lip}(a)\xi < 1,$$

  we see that we obtained

$$|T(y) - (\bar{x}, a(\bar{x}))| \geq \alpha y_d, \; \forall \bar{x} \in \mathbb{R}^{d-1},$$

with
$$\alpha = \min\left(\xi, 1 - \gamma\,\mathrm{Lip}(a)M_\eta - \mathrm{Lip}(a)\xi\right).$$

This gives exactly
$$\delta(T(y)) \geq \alpha y_d,$$

and the claim is proved.

- We consider a finite open covering $(U_i)_{1\leq i\leq N}$ of $\partial\Omega$ as given by Theorem III.1.4. We also further assume that for each $1 \leq i \leq N$ we have

$$\delta(x) = d(x, \partial\Omega) = d(x, R_{\sigma_i}\partial\mathcal{H}_{a_i}),\ \forall x \in U_i \cap \Omega. \tag{III.51}$$

This property can always be achieved by considering a finer open covering, if necessary. We finally consider an open set $U_0 \subset \overline{U_0} \subset \Omega$ such that $(U_i)_{0\leq i\leq N}$ is an open covering of $\overline{\Omega}$, and we choose an associated partition of unity $(\psi_i)_{0\leq i\leq N}$.
For any $u \in \mathcal{D}(\Omega)$, we write $u = u\psi_0 + \sum_{i=1}^N u\psi_i$.

– Since $u\psi_0$ is compactly supported in U_0, and $\overline{U_0} \subset \Omega$, we have

$$\inf_{U_0} \delta > 0,$$

so that
$$\left\|\frac{u\psi_0}{\delta}\right\|_{L^p(\Omega)} = \left\|\frac{u\psi_0}{\delta}\right\|_{L^p(U_0)} \leq \frac{1}{\inf_{U_0}\delta}\|u\|_{L^p(\Omega)}.$$

– Since $u\psi_i$ is compactly supported in $U_i \cap \Omega = (R_{\sigma_i}\mathcal{H}_{a_i}) \cap U_i$, we can apply to $u\psi_i$ the Hardy inequality we have proved above

$$\left\|\frac{u\psi_i}{\delta_i}\right\|_{L^p(R_{\sigma_i}\mathcal{H}_{a_i})} \leq C_i\|\nabla(u\psi_i)\|_{L^p(R_{\sigma_i}\mathcal{H}_{a_i})},$$

where $\delta_i(x) = d(x, R_{\sigma_i}\partial\mathcal{H}_{a_i})$. By using Property (III.51), we deduce that

$$\left\|\frac{u\psi_i}{\delta}\right\|_{L^p(\Omega)} = \left\|\frac{u\psi_i}{\delta_i}\right\|_{L^p(R_{\sigma_i}\mathcal{H}_{a_i})} \leq C_i'(\|u\|_{L^p(\Omega)} + \|\nabla u\|_{L^p(\Omega)}).$$

Summing all the inequalities above we finally obtain

$$\left\|\frac{u}{\delta}\right\|_{L^p(\Omega)} \leq C(\|u\|_{L^p(\Omega)} + \|\nabla u\|_{L^p(\Omega)}),$$

and the conclusion follows by using the Poincaré inequality of Proposition III.2.38 and the density of $\mathcal{D}(\Omega)$ in $W_0^{1,p}(\Omega)$.

We sometimes need the following corollary of the Hardy inequality.

Corollary III.2.41. *With the same assumptions as in the previous proposition, assume that f is a function such that $f \in L^\infty(\Omega)$ and that $\delta\nabla f \in L^\infty(\Omega)$; then for any $u \in W_0^{1,p}(\Omega)$ we have $fu \in W_0^{1,p}(\Omega)$ and*

$$\|fu\|_{W_0^{1,p}} \leq C\|u\|_{W_0^{1,p}},$$

where C does not depend on u.

Proof.

As usual, by a density argument, it is enough to prove the result for any $u \in \mathcal{D}(\Omega)$.

Since f is bounded, it is clear that $\|fu\|_{L^p} \leq \|f\|_{L^\infty}\|u\|_{L^p}$. Moreover, we have $\nabla(fu) = f(\nabla u) + (\nabla f)u = f(\nabla u) + \delta(\nabla f)(u/\delta)$. Since f and $\delta\nabla f$ are bounded and using the Hardy inequality, we find that

$$\|\nabla(uf)\|_{L^p} \leq C\left(\|f\|_{L^\infty} + \|\delta\nabla f\|_{L^\infty}\|\nabla u\|_{L^p}\right),$$

and the claim is proved.

□

2.10 Domains of first-order differential operators

Let D be a first-order differential operator as defined in (III.24) whose coefficients α_j belong to $W^{1,q}(\Omega)$, $1 \leq q \leq \infty$. For any $1 \leq p \leq +\infty$ such that $p' \geq q$, we observe that $D\big((W^{1,p}(\Omega))^k\big) \subset L^r(\Omega)$ where $r \in [1, +\infty]$ is defined by $1/p + 1/q = 1/r$.

It is then natural to consider D as an unbounded operator from $(L^p(\Omega))^k$ into $L^r(\Omega)$ with a domain defined by

$$W_D^{p,r}(\Omega) = \{u \in (L^p(\Omega))^k,\ Du \in L^r(\Omega)\},$$

and which is equipped with the graph norm

$$\|u\|_{W_D^{p,r}} = \|u\|_{L^p} + \|Du\|_{L^r},\ \forall u \in W_D^{p,r}(\Omega).$$

Theorem III.2.42. *With the same assumption as in Theorem III.2.10, the space $W_D^{p,r}(\Omega)$ is a Banach space which contains $(W^{1,p}(\Omega))^k$.*

Moreover, if $p < +\infty$, the space $(\mathcal{C}_c^\infty(\overline{\Omega}))^k$ is dense in $W_D^{p,r}(\Omega)$.

Proof.

Any Cauchy sequence $(u_n)_n$ in $W_D^{p,r}(\Omega)$ is such that $(u_n)_n$ and $(Du_n)_n$ are Cauchy sequences in $(L^p(\Omega))^k$ and $L^r(\Omega)$, respectively. Therefore, $(u_n)_n$ converges towards some u in $(L^p(\Omega))^k$ and $(Du_n)_n$ converges towards some g in $L^r(\Omega)$.

We observe that $(Du_n)_n$ converges to Du in the sense of distributions. This implies that $Du = g \in L^r(\Omega)$ and finally that $(u_n)_n$ converges towards u in $W^{p,r}_D(\Omega)$.

Proposition III.2.8 and Theorem III.2.10 exactly imply that the function $\mathcal{S}_\varepsilon u \in (\mathcal{C}^\infty_c(\overline{\Omega}))^k$ converges towards u in $W^{p,r}_D(\Omega)$ as ε goes to 0.

□

For any $x \in \Omega$, we denote by $\alpha(x)$ the $k \times d$ matrix whose columns are the $\alpha_j(x)$, $j = 1, \ldots, d$.

Theorem III.2.43. *Let us assume that $1 < p \leq +\infty$ and $q \geq p'$ with $(p,q) \neq (+\infty,+\infty)$. We define $\overline{t} \in [1,+\infty[$ by*

$$\frac{1}{\overline{t}} = \min\left(\frac{1}{p'}, \frac{1}{r'} + \frac{1}{d}\right).$$

For any $t > \overline{t}$, there is a unique linear continuous operator from $W^{p,r}_D(\Omega)$ into $W^{-1/t',t'}(\partial\Omega)$, referred to as $\gamma_{\alpha.\nu}$, such that

$$\gamma_{\alpha.\nu}(u) = \gamma_0(u) \cdot (\gamma_0(\alpha).\nu), \quad \forall u \in (W^{1,p}(\Omega))^k. \tag{III.52}$$

Moreover, we have the following generalised Stokes formula

$$\int_\Omega (Du)\varphi\, dx + \int_\Omega \sum_{j=1}^d (u \cdot \alpha_j)\frac{\partial \varphi}{\partial x_j}\, dx = \langle \gamma_{\alpha.\nu} u, \gamma_0 \varphi\rangle_{W^{-1/t',t'},W^{1-1/t,t}}, \tag{III.53}$$

for any $u \in W^{p,r}_D(\Omega)$ and $\varphi \in W^{1,t}(\Omega)$.

Moreover the same result holds with $t = \overline{t}$ except in the two cases $q = d$ or $q = p' > d$.

Proof.

For any $u \in (\mathcal{C}^\infty_c(\overline{\Omega}))^k$ and any $\varphi \in W^{1,\infty}(\overline{\Omega}) = \mathcal{C}^{0,1}(\overline{\Omega})$, the Stokes formula (III.35) gives

$$\int_\Omega (Du)\varphi\, dx + \int_\Omega \sum_{j=1}^d (u \cdot \alpha_j)\frac{\partial \varphi}{\partial x_j}\, dx = \int_{\partial\Omega} \varphi u \cdot (\gamma_0(\alpha).\nu)\, d\sigma. \tag{III.54}$$

For any $u \in W^{p,r}_D(\Omega)$ we define

$$T_\varepsilon u = (\mathcal{S}_\varepsilon u) \cdot (\gamma_0(\alpha).\nu) \in L^q(\partial\Omega).$$

It is clear that, for $u \in W^{1,p}(\Omega)$, we have $T_\varepsilon u \xrightarrow[\varepsilon\to 0]{} \gamma_0(u) \cdot (\gamma_0(\alpha).\nu)$ in $L^1(\partial\Omega)$, for instance.

Let $\theta \in \mathcal{D}(\mathbb{R}^d)$ be a smooth function such that $\theta = 1$ on $\partial\Omega$. Let $u \in W^{p,r}_D(\Omega)$ and $\varphi \in \mathcal{C}^\infty_c(\overline{\Omega})$; we deduce from (III.54) that

$$\begin{aligned}\left|\int_{\partial\Omega}(T_{\varepsilon_1}u-T_{\varepsilon_2}u)\varphi\,d\sigma\right| = &\left|\int_{\partial\Omega}(T_{\varepsilon_1}u-T_{\varepsilon_2}u)\theta\varphi\,d\sigma\right|\\ \leq &\|D\mathcal{S}_{\varepsilon_1}u-D\mathcal{S}_{\varepsilon_2}u\|_{L^r}\|\theta\varphi\|_{L^{r'}}\\ &+\sum_{j=1}^d\|(\mathcal{S}_{\varepsilon_1}u-\mathcal{S}_{\varepsilon_2}u)\cdot\alpha_j(|\theta|+|\nabla\theta|)\|_{L^{t'}}\|\varphi\|_{W^{1,t}}.\end{aligned} \tag{III.55}$$

Notice that, by definition of $\overline{t}$ we have

$$\frac{1}{t}-\frac{1}{d}<\frac{1}{\overline{t}}-\frac{1}{d}\leq\frac{1}{r'},$$

and thus, since $\theta\varphi$ is supported in a fixed compact of $\mathbb{R}^d$, we have

$$\|\theta\varphi\|_{L^{r'}}\leq C\|\theta\varphi\|_{W^{1,t}}\leq C'\|\varphi\|_{W^{1,t}}.$$

By density of $\mathcal{C}_c^\infty(\overline{\Omega})$ into $W^{1,t}(\Omega)$ (because t is finite), the estimate (III.55), proves that

$$\begin{aligned}\|T_{\varepsilon_1}u-T_{\varepsilon_2}u\|_{W^{-1/t',t'}(\partial\Omega)}\leq &C\|D\mathcal{S}_{\varepsilon_1}u-D\mathcal{S}_{\varepsilon_2}u\|_{L^r}\\ &+C\sum_{j=1}^d\|(\mathcal{S}_{\varepsilon_1}u-\mathcal{S}_{\varepsilon_2}u)\cdot\alpha_j(|\theta|+|\nabla\theta|)\|_{L^{t'}}.\end{aligned}$$

By definition of $\overline{t}$, we have $0<1/p'-1/t<1$, therefore we can introduce $s\in]1,+\infty[$ such that $1/s=1/p'-1/t$. Using once more the assumption $t>\overline{t}$ and the definition of $\overline{t}$ we see that

$$\frac{1}{q}-\frac{1}{d}<\frac{1}{p'}-\frac{1}{t}=\frac{1}{s},$$

and thus, since $|\theta|+|\nabla\theta|$ is supported in a fixed compact set, we deduce that $\alpha_j(|\theta|+|\nabla\theta|)\in L^s(\Omega)$ for any j with

$$\|\alpha_j(|\theta|+|\nabla\theta|)\|_{L^s}\leq C\|\alpha_j\|_{W^{1,q}}.$$

By using Proposition III.2.8 (because s is finite), we know that $(\mathcal{S}_\varepsilon u)\cdot\alpha_j(|\theta|+|\nabla\theta|)$ is convergent in $L^{t'}$ because we have $1/t'=1/s+1/p$ by definition of s.

We finally deduce from all the above estimates that $(T_\varepsilon u)_\varepsilon$ is a Cauchy sequence in $W^{-1/t',t'}(\partial\Omega)$. We define $\gamma_{\alpha.\nu}(u)$ to be its limit. It is clear that $\gamma_{\alpha.\nu}$ is linear and the same estimates as above show that it is continuous from $W_D^{p,r}(\Omega)$ into $W^{-1/t',t'}(\partial\Omega)$.

In the case $t=\overline{t}$, it may happen that some Sobolev embeddings that we used above (namely $W^{1,t}\subset L^{r'}$ and $W^{1,q}\subset L^s$) are in fact critical L^∞-embeddings which do not hold.

- The fact that the Sobolev embedding $W^{1,\bar{t}} \subset L^{r'}$ is critical means that $r' = +\infty$ and $\bar{t} = d$. This means that $q = p'$ and $p' \geq d$.
- The fact that the Sobolev embedding $W^{1,q} \subset L^s$ is critical means that $q = d$ and $s = +\infty$ (those two equalities being in fact equivalent).

□

Definition III.2.44. *We define $W^{p,r}_{D,0}(\Omega)$ to be the closure of $(\mathcal{D}(\Omega))^k$ in $W^{p,r}_D(\Omega)$.*

Theorem III.2.45. *Let $u \in W^{p,r}_D(\Omega)$. We have the following equivalences*

$$u \in W^{p,r}_{D,0}(\Omega) \iff u \in \operatorname{Ker} \gamma_{\alpha.\nu} \iff \bar{u} \in W^{p,r}_D(\mathbb{R}^d).$$

Notice that in this statement we have extended the definition of the operator D to the whole space $\mathbb{R}^d$. This is possible by applying the extension operator introduced in Section 2.4 to the coefficients α_j.

Proof.

- By construction, it is clear that $\gamma_{\alpha.\nu}(u) = u \cdot (\alpha.\nu) = 0$ on $\partial\Omega$ for any $u \in (\mathcal{D}(\Omega))^k$. The operator $\gamma_{\alpha.\nu}$ is continuous, therefore we deduce that it vanishes identically on $W^{p,r}_{D,0}(\Omega)$, which is the closure of $(\mathcal{D}(\Omega))^k$. This proves that $W^{p,r}_{D,0}(\Omega) \subset \operatorname{Ker} \gamma_{\alpha.\nu}$.
- Let us consider $u \in \operatorname{Ker} \gamma_{\alpha.\nu}$. By the Stokes formula (III.53) we have, for any $\varphi \in W^{1,r'}(\Omega)$,

$$\int_\Omega (Du)\varphi \, dx + \int_\Omega \sum_{j=1}^d (u \cdot \alpha_j) \frac{\partial \varphi}{\partial x_j} \, dx = 0,$$

  and thus, by extending u and Du by zero to the whole space $\mathbb{R}^d$, we get

$$\int_{\mathbb{R}^d} \overline{Du} \, \varphi \, dx + \int_{\mathbb{R}^d} \sum_{j=1}^d (\bar{u} \cdot \alpha_j) \frac{\partial \varphi}{\partial x_j} \, dx = 0,$$

  for any $\varphi \in \mathcal{D}(\mathbb{R}^d)$. This formula exactly means that $D\bar{u}$, in the sense of distributions on $\mathbb{R}^d$, is equal to $\overline{Du}$ which belongs to $L^r(\mathbb{R}^d)$. We thus have proved that $\bar{u} \in W^{p,r}_D(\mathbb{R}^d)$.
- Let us finally assume that u is such that $\bar{u} \in W^{p,r}_D(\mathbb{R}^d)$. We can thus use Theorem III.2.12 to affirm that the family $(\widetilde{\mathcal{S}}_\varepsilon u)_\varepsilon \subset (\mathcal{D}(\Omega))^k$ converges towards u in $W^{p,r}_D(\Omega)$, so that u belongs to $W^{p,r}_{D,0}(\Omega)$.

□

By applying the above results to the elementary differential operators ∂_{x_i} (in which case we have $q = +\infty$ and $r = p$) we get the following.

Theorem III.2.46. *Let Ω be a Lipschitz domain with compact boundary and $1 \leq p < +\infty$. We have the equalities*

$$W^{1,p}(\Omega) = \bigcap_{i=1}^{d} W^{p,p}_{\partial_{x_i}}(\Omega), \tag{III.56}$$

$$W^{1,p}_0(\Omega) = \bigcap_{i=1}^{d} W^{p,p}_{\partial_{x_i},0}(\Omega). \tag{III.57}$$

Moreover, for any $u \in W^{1,p}(\Omega)$, we have

$$\gamma_0(u)\,\nu_i = \gamma_{\nu_i}(u), \ \forall i \in \{1,\dots,d\}, \tag{III.58}$$

and

$$\gamma_0(u) = \sum_{i=1}^{d} \gamma_{\nu_i}(u)\nu_i. \tag{III.59}$$

Finally, for any $u \in W^{1,p}(\Omega)$, we have

$$u \in W^{1,p}_0(\Omega) \iff u \in \operatorname{Ker}\gamma_0 \iff \bar{u} \in W^{1,p}(\mathbb{R}^d).$$

Proof.

- The first equality is nothing but the definition of the Sobolev space $W^{1,p}(\Omega)$.
- For any $u \in W^{1,p}_0(\Omega)$, there is a sequence $(u_n)_n \subset \mathcal{D}(\Omega)$ which converges towards u in $W^{1,p}(\Omega)$. In particular we have $u_n \xrightarrow[n\to\infty]{} u$ in $L^p(\Omega)$ and $\partial_{x_i}u_n \xrightarrow[n\to\infty]{} \partial_{x_i}u$ in $L^p(\Omega)$, which implies that $u_n \xrightarrow[n\to\infty]{} u$ in $W^{p,p}_{\partial_{x_i}}(\Omega)$ for any $i \in \{1,\dots,d\}$. This proves that $u \in W^{p,p}_{\partial_{x_i},0}(\Omega)$ for any i.
  Conversely, if $u \in W^{p,p}_{\partial_{x_i},0}(\Omega)$, Theorem III.2.45 shows that $\bar{u} \in W^{p,p}_{\partial_{x_i},0}(\Omega)$ so that, by Theorem III.2.12, the family $(\widetilde{\mathcal{S}}_\varepsilon u)_\varepsilon \subset \mathcal{D}(\Omega)$ converges to u in $W^{p,p}_{\partial_{x_i}}(\Omega)$. This being true for any $i \in \{1,\dots,d\}$, we conclude that $(\widetilde{\mathcal{S}}_\varepsilon u)_\varepsilon$ converges to u in $W^{1,p}(\Omega)$. This proves that $u \in W^{1,p}_0(\Omega)$.
  Notice that it is absolutely fundamental here that *the same* mollifying operator $\widetilde{\mathcal{S}}_\varepsilon$ can be used in all the spaces $W^{p,r}_{D,0}(\Omega)$.
- We apply (III.52) with $D = \partial_{x_i}$, that is $\alpha_j = \delta_{i,j}$. We exactly obtain (III.58). Formula (III.59) comes from the fact that $|\nu| = 1$ almost everywhere, which means that $\sum_{i=1}^d \nu_i^2 = 1$ almost everywhere on $\partial\Omega$.
  It follows from (III.58) and (III.59) that

$$\operatorname{Ker}\gamma_0 = \bigcap_{i=1}^{d} \operatorname{Ker}\gamma_{\nu_i} = \bigcap_{i=1}^{d} W^{p,p}_{\partial_{x_i},0}(\Omega) = W^{1,p}_0(\Omega).$$

  The characterisation of $W^{1,p}_0(\Omega)$ which uses the extension $\bar{u}$ of u comes from Theorem III.2.45, (III.56) and (III.57).

$\square$

3 Calculus near the boundary of domains

3.1 Local charts description of the boundary

From now on, we consider a given domain Ω of $\mathbb{R}^d$ of class $\mathcal{C}^{k+1,1}$, $k \geq 0$, with compact boundary. An orientation in $\mathbb{R}^d$ is chosen. For reasons of clarity, in the sequel of this chapter and contrary to the convention in other chapters of this book, we denote by $\langle \cdot, \cdot \rangle$ the usual Euclidean scalar product of $\mathbb{R}^d$ (and of $\mathbb{R}^{d-1}$). To simplify the notation a little, we set $\Gamma = \partial\Omega$.

From the implicit function theorem and using Definition III.1.2, we can state the following.

Definition and Proposition III.3.1. *For any point $\sigma \in \Gamma$, there exists a convex open set U_σ of $\mathbb{R}^{d-1}$ containing 0, an open set V_σ of $\mathbb{R}^d$ containing σ, and a function $\varphi_\sigma : U_\sigma \to V_\sigma$ of class $\mathcal{C}^{k+1,1}$ such that $\varphi_\sigma(U_\sigma) = V_\sigma \cap \Gamma$, $\varphi_\sigma(0) = \sigma$, and*

$$|u_1 - u_2| \leq C|\varphi_\sigma(u_1) - \varphi_\sigma(u_2)|, \, \forall u_1, u_2 \in U_\sigma, \tag{III.60}$$

for some $C > 0$ independent of σ.

The pair $(U_\sigma, \varphi_\sigma)$ is referred to as a local chart *of the manifold Γ around the point σ.*

Furthermore, we can choose the maps associated with the various points of the manifold Γ in a compatible way; that is, that if σ_1 and σ_2 are in Γ and such that $\Gamma_{\sigma_1\sigma_2} \stackrel{def}{=} V_{\sigma_1} \cap V_{\sigma_2} \cap \Gamma \neq \emptyset$ then the change of map function $\Phi_1^2 \stackrel{def}{=} \varphi_{\sigma_1}^{-1} \circ \varphi_{\sigma_2}$ defined on $\varphi_{\sigma_2}^{-1}(\Gamma_{\sigma_1\sigma_2})$ and with values in $\varphi_{\sigma_1}^{-1}(\Gamma_{\sigma_1\sigma_2})$ is a $\mathcal{C}^{k+1,1}$ diffeomorphism.

Finally, since Γ is compact, we can describe the whole manifold Γ by a finite number of charts.

We therefore possess local coordinates of the manifold Γ in the neighborhood of any point σ. In particular, the family of vectors of $\mathbb{R}^d$ defined by $(\partial\varphi_\sigma/\partial u_i)_{1\leq i\leq d-1}$ is linearly independent and generates an hyperplane of $\mathbb{R}^d$ called the vector tangent hyperplane Γ at the point σ and denoted as $T_\sigma\Gamma$. We easily see that this hyperplane does not depend on the local map chosen around σ.

For any chart (U, φ), we set

$$N_\varphi(u) = \frac{\partial\varphi}{\partial u_1}(u) \wedge \cdots \wedge \frac{\partial\varphi}{\partial u_{d-1}}(u),$$

so that the outward unit normal ν can be expressed in the considered chart as follows

$$\nu(\varphi(u)) = \varepsilon_\varphi \frac{N_\varphi(u)}{|N_\varphi(u)|},$$

where $\varepsilon_\varphi \in \{-1, 1\}$ is a sign which depends on the orientation. It follows in particular, since $\varphi \in \mathcal{C}^{k+1,1}$, that $\nu \circ \varphi \in \mathcal{C}^{k,1}(U)$. At each point $\sigma = \varphi(u)$ (or at almost each point in the case $k = 0$), we can define the endomorphism $d\nu(\sigma) : T_\sigma\Gamma \to T_\sigma\Gamma$ by the formula

$$d\nu(\sigma).\frac{\partial \varphi}{\partial u_i}(u) = \frac{\partial}{\partial u_i}(\nu \circ \varphi)(u).$$

The fact that $d\nu(\sigma)$ maps $T_\sigma\Gamma$ into itself comes from the fact that $|\nu| = 1$. Indeed, we have

$$0 = \frac{\partial}{\partial u_i}\left(|\nu \circ \varphi|^2\right) = 2\left\langle d\nu(\sigma).\frac{\partial \varphi}{\partial u_i}(u), \nu \circ \varphi(u)\right\rangle,$$

so that $d\nu(\sigma).(\partial\varphi/\partial u_i)(u)$ is orthogonal to $N_\varphi(u)$ which proves that it belongs to the tangent space $T_\sigma\Gamma$.

Definition III.3.2. *We call the first fundamental quadratic form of the manifold Γ in the chart (U, φ), the quadratic form in $\mathbb{R}^{d-1}$ associated with the matrix*

$$\mathfrak{g}(u) = (\mathfrak{g}_{ij}(u))_{1\leq i,j\leq d-1} = \left(\left\langle \frac{\partial \varphi}{\partial u_i}(u), \frac{\partial \varphi}{\partial u_j}(u)\right\rangle\right)_{1\leq i,j\leq d-1},$$

for any $u \in U$.

This is the Gram matrix associated with the basis of the tangent plane canonically associated with the chart being considered.

Notice that $\mathfrak{g}$ is of class $\mathcal{C}^{k,1}$ in U.

The first fundamental quadratic form allows us to compute the norm of a tangent vector to Γ in terms of its coordinates in the basis of $T_\sigma\Gamma$ issued from the chart. More precisely, we have the following proposition.

Proposition III.3.3. *Let $\sigma \in \Gamma$, (U, φ) be a chart of the neighborhood of σ and $u \in U$ such that $\sigma = \varphi(u)$.*

Let $F = \sum_{i=1}^{d-1} f_i(\partial\varphi/\partial u_i)(u)$ be a vector of $T_\sigma\Gamma$. We denote as $f = (f_i)_i$ the vector of the coordinates of F in the basis of $T_\sigma\Gamma$ associated with the map (U, φ). We then have

$$|F|^2 = \langle \mathfrak{g}(u).f, f\rangle = {}^t f.\mathfrak{g}(u).f,$$

and

$$f = \mathfrak{g}(u)^{-1}.\left(\left\langle F, \frac{\partial \varphi}{\partial u_i}(u)\right\rangle\right)_{1\leq i\leq d-1}.$$

Corollary III.3.4. *The first fundamental form is positive definite. Therefore, the matrix $\mathfrak{g}(u)$ is invertible and its determinant is strictly positive for any $u \in U$.*

In all that follows we denote the coefficients of the inverse matrix $\mathfrak{g}(u)^{-1}$ by
$$\mathfrak{g}(u)^{-1} = (\mathfrak{g}^{ij}(u))_{1\leq i,j\leq d-1}.$$

The definition of the first fundamental form depends explicitly on the chart in which it is considered. Nevertheless, one can show that if (U_1, φ_1), (U_2, φ_2) are two charts around the same point of Γ, the two associated first fundamental forms $\mathfrak{g}_1$ and $\mathfrak{g}_2$ are related through the formula
$$\mathfrak{g}_2 = {}^t(\nabla \Phi_1^2) \cdot \mathfrak{g}_1 \cdot (\nabla \Phi_1^2),$$
where $\Phi_1^2 = \varphi_1^{-1} \circ \varphi_2$ is the change of variable between the two charts.

3.2 Distance to the boundary. Projection on the boundary

We recall that Ω is a domain of $\mathbb{R}^d$ of class $\mathcal{C}^{k+1,1}$, $k \geq 0$, with compact boundary.

In this section we investigate further the regularity properties of the signed distance function δ defined in (II.1).

We consider the locally Lipschitz continuous map
$$\psi_0 : (\sigma, s) = \Gamma \times \mathbb{R} \mapsto \sigma - s\nu(\sigma) \in \mathbb{R}^d.$$
We are going to show that, through this map ψ_0, one can use the coordinates (σ, s), that we call tangent/normal coordinates, to describe points in a suitable neighborhood of Γ. We recall that we have defined
$$\mathcal{O}_\gamma = \{x \in \Omega, \quad \delta(x) < \gamma\}.$$

Theorem III.3.5. *There exists a $\gamma > 0$ and a $C > 0$ such that*
$$\begin{aligned} |\sigma_1 - \sigma_2| + |s_1 - s_2| \leq & C|\psi_0(\sigma_1, s_1) - \psi_0(\sigma_2, s_2)|, \\ & \forall \sigma_1, \sigma_2 \in \Gamma, \forall s_1, s_2 \in]-\gamma, \gamma[, \end{aligned} \tag{III.61}$$
and moreover
$$\psi_0(\Gamma \times]0, \gamma[) = \mathcal{O}_\gamma,$$
$$\psi_0(\Gamma \times]-\gamma, 0[) \subset \overline{\Omega}^c.$$

This result shows that ψ_0 is, at least, a Lipschitz diffeomorphism from $\Gamma \times]0, \gamma[$ onto $\mathcal{O}_\gamma$.

Proof.

We recall that Γ can be covered by a finite number of charts (U_i, φ_i), $i = 1, \ldots, N$, of class $\mathcal{C}^{k+1,1}$.

- We first observe that there exists a $\gamma > 0$ such that for any $\sigma_1, \sigma_2 \in \Gamma$ such that $|\sigma_1 - \sigma_2| \leq \gamma$, there exists a chart (U_i, φ_i) such that σ_1, and σ_2 both belong to $\varphi_i(U_i)$. Then we can write $\sigma_1 = \varphi_i(u_1), \sigma_2 = \varphi_i(u_2)$, and since $\varphi_i \in \mathcal{C}^{1,1}(U_i)$ and U_i is convex we can write

$$\langle \sigma_2 - \sigma_1, \nu(\sigma_1)\rangle = \int_0^1 \big\langle (\nabla\varphi_i)(u_1 + t(u_2 - u_2)).(u_1 - u_2), \nu(\varphi(u_1))\big\rangle \, dt.$$

Since $(\nabla\varphi_i)(u_1+t(u_2-u_2)).(u_1-u_2)$ is orthogonal to $\nu(\varphi_i(u_1+t(u_2-u_2)))$ we can write

$$\begin{aligned}&\langle \sigma_2 - \sigma_1, \nu(\sigma_1)\rangle \\ &= \int_0^1 \big\langle (\nabla\varphi_i)(u_1+t(u_2-u_2)).(u_1-u_2), \nu(\varphi(u_1+t(u_2-u_1))-\nu(\varphi(u_1))\big\rangle \, dt,\end{aligned}$$

and then

$$|\langle \sigma_2 - \sigma_1, \nu(\sigma_1)\rangle| \leq \|\nabla\varphi_i\|_\infty \operatorname{Lip}(\nu \circ \varphi_i)|u_1 - u_2|^2,$$

and finally, using (III.60), we find that

$$|\langle \sigma_2 - \sigma_1, \nu(\sigma_1)\rangle| \leq C|\sigma_1 - \sigma_2|^2. \tag{III.62}$$

Since Γ is compact, this inequality remains true even if $|\sigma_1 - \sigma_2| \geq \gamma$ with a possibly different C.
- Let us prove (III.61) by contradiction. Assume that, for any $n \geq 1$, there exists $\sigma_1^n, \sigma_2^n \in \Gamma$, $s_1^n, s_2^n \in \mathbb{R}$ such that $s_1^n \to 0$, $s_2^n \to 0$ and

$$\frac{1}{n}\left(|\sigma_1^n - \sigma_2^n| + |s_1^n - s_2^n|\right) > |\psi_0(\sigma_1^n, s_1^n) - \psi_0(\sigma_2^n, s_2^n)|. \tag{III.63}$$

Γ is compact, therefore we can assume that $(\sigma_1^n)_n$ and $(\sigma_2^n)_n$ are convergent towards $\sigma_1 \in \Gamma$, $\sigma_2 \in \Gamma$, respectively. Passing to the limit in (III.63) shows that $\varphi(\sigma_1, 0) = \varphi(\sigma_2, 0)$ so that $\sigma_1 = \sigma_2$. We set $\sigma = \sigma_1 = \sigma_2$.
- Computing $\langle \psi_0(\sigma_1^n, s_1^n) - \psi_0(\sigma_2^n, s_2^n), \nu(\sigma_2^n)\rangle$ and using the fact that $|\nu| = 1$, we find that

$$\begin{aligned}s_2^n - s_1^n =& s_1^n\langle \nu(\sigma_1^n) - \nu(\sigma_2^n), \nu(\sigma_2^n)\rangle \\ &+ \langle \sigma_2^n - \sigma_1^n, \nu(\sigma_2^n)\rangle + \langle \psi_0(\sigma_1^n, s_1^n) - \psi_0(\sigma_2^n, s_2^n), \nu(\sigma_2^n)\rangle.\end{aligned}$$

Moreover, since $|\nu(\sigma_1^n)| = |\nu(\sigma_2^n)| = 1$, we have

$$\langle \nu(\sigma_1^n) - \nu(\sigma_2^n), \nu(\sigma_2^n)\rangle = -\frac{1}{2}\langle \nu(\sigma_1^n) - \nu(\sigma_2^n), \nu(\sigma_1^n) - \nu(\sigma_2^n)\rangle,$$

and therefore by using (III.62) and (III.63), we deduce that

$$\begin{aligned}|s_1^n - s_2^n| &\leq C'|\sigma_1^n - \sigma_2^n|^2 + |\psi_0(\sigma_1^n, s_1^n) - \psi_0(\sigma_2^n, s_2^n)| \\ &\leq C'|\sigma_1^n - \sigma_2^n|^2 + \frac{1}{n}|s_1^n - s_2^n| + \frac{1}{n}|\sigma_1^n - \sigma_2^n|,\end{aligned}$$

and then, for n large enough,

$$|s_1^n - s_2^n| \leq C'|\sigma_1^n - \sigma_2^n|^2 + \frac{1}{n}|\sigma_1^n - \sigma_2^n|. \tag{III.64}$$

- Computing now $\psi_0(\sigma_1^n, s_1^n) - \psi_0(\sigma_2^n, s_2^n)$ and using (III.63) we get

$$\begin{aligned}|\sigma_2^n - \sigma_1^n| &= |\psi_0(\sigma_2^n, s_2^n) - \psi_0(\sigma_1^n, s_1^n)| + |s_2^n\nu(\sigma_2^n) - s_1^n\nu(\sigma_1^n)|, \\ &\leq \frac{1}{n}|s_1^n - s_2^n| + \frac{1}{n}|\sigma_1^n - \sigma_2^n| + |s_1^n - s_2^n| + |s_1^n|\,\mathrm{Lip}(\nu)|\sigma_1^n - \sigma_2^n|.\end{aligned}$$

  Since $s_1^n \to 0$ and using (III.64), we get

$$|\sigma_2^n - \sigma_1^n| \leq C''|\sigma_2^n - \sigma_1^n|^2,$$

  This inequality implies that $\sigma_1^n = \sigma_2^n$ for n large enough, because $\sigma_1^n - \sigma_2^n \to 0$, and therefore $s_1^n = s_2^n$ for n large enough (thanks to (III.64)).
  This is a contradiction with the strict inequality (III.63) so that (III.61) is proved for a $\gamma > 0$ small enough.
- It is clear that for $\gamma > 0$ small enough we have

$$\psi_0(\Gamma\times]0, \gamma[) \subset \Omega, \quad \psi_0(\Gamma\times] - \gamma, 0[) \subset \overline{\Omega}^c.$$

  – If $x = \psi_0(\sigma, s)$ with $\sigma \in \Gamma$ and $s \in]0, \gamma[$, we have $|x - \sigma| = s$ so that $\delta(x) \leq s < \gamma$ and thus $x \in \mathcal{O}_\gamma$.
  – Conversely, let $x \in \mathcal{O}_\gamma$ and let $\sigma \in \Gamma$ such that $|x - \sigma| = \delta(x) = \inf_{\sigma' \in \Gamma} |x - \sigma'|$. Notice that it is possible a priori to have many such σ. Consider a chart (U, φ) such that $\sigma = \varphi(u_0) \in \varphi(U)$. By definition of σ, the map $u \in U \mapsto |\varphi(u) - x|^2$ has a minimum at $u = u_0$ and is of class $\mathcal{C}^1$. It follows that its derivative has to vanish at $u = u_0$ which means that $\sigma - x = \varphi(u_0) - x$ is orthogonal to the tangent hyperplane $T_\sigma\Gamma$ and therefore is proportional to $\nu(\sigma)$. It follows that $x = \sigma - s\nu(\sigma)$ for some s. It is easy to check that $|s| = \delta(x) < \gamma$, therefore we have $x = \psi_0(\sigma, s) \in \psi_0(\Gamma\times] - \gamma, \gamma[)$.

□

It follows from the last part of this proof that, for $x \in \mathcal{O}_\gamma$, there is a unique point $\sigma \in \Gamma$ such that $|x - \sigma| = \delta(x)$: this point is called the projection of x onto Γ and is referred to as $P_0(x)$. Moreover, we have the inversion formula

$$x = \psi_0(P_0(x), \delta(x)) = P_0(x) - \delta(x)\nu(P_0(x)), \ \forall x \in \mathcal{O}_\gamma. \tag{III.65}$$

In addition (III.61) shows that the projection P_0 and the distance function $d(., \Gamma)$ are, at least, Lipschitz continuous in $\mathcal{O}_\gamma$.

Let $x \in \mathcal{O}_\gamma$ be a point where δ is differentiable and $h \in \mathbb{R}$ small enough. We then have

$$(x - h\nu(P_0(x))) = P_0(x) - (\delta(x) + h)\nu(P_0(x)) = \psi_0(P_0(x), \delta(x) + h),$$

and then $\delta(x - h\nu(P_0(x))) = \delta(x) + h$. Letting $h \to 0$ we deduce that

$$\langle \nabla\delta(x), \nu(P_0(x))\rangle = -1,$$

and since we know that $|\nabla\delta| \leq 1$, we find that, for such a point x we have

$$\nabla\delta(x) = -\nu(P_0(x)). \tag{III.66}$$

Since $\nu \circ P_0$ is continuous, by using Proposition III.2.4, we deduce that δ is $\mathcal{C}^1$ and its gradient is $-\nu \circ P_0$ so that we have finally proved that δ is $\mathcal{C}^{1,1}$.

More generally, if $k \geq 1$ then we can show that δ is $\mathcal{C}^{k+1,1}$ and P is $\mathcal{C}^{k,1}$ in $\mathcal{O}_\gamma$. For simplicity, let us only consider the case $k = 1$. Taking the derivative of (III.65) with respect to x_i we get

$$\frac{\partial P_0}{\partial x_i}(x) = e_i + \frac{\partial \delta}{\partial x_i}\nu(P_0(x)) + \delta(x) d\nu(P_0(x)).\frac{\partial P_0}{\partial x_i}(x),$$

which gives

$$\frac{\partial P_0}{\partial x_i}(x) = \left(\mathrm{Id} - \delta(x) d\nu(P_0(x))\right)^{-1}\left(e_i + \frac{\partial \delta}{\partial x_i}\nu(P_0(x))\right), \tag{III.67}$$

where we recall that $d\nu(P_0(x))$ is an endomorphism of the tangent space $T_{P_0(x)}\Gamma$. Since Γ is of class $\mathcal{C}^{2,1}$, $\sigma \to s\nu(\sigma)$ is a Lipschitz continuous map. It follows from (III.67) that $\partial P_0/\partial x_i$ is itself Lipschitz continuous. Therefore, we have $P_0 \in \mathcal{C}^{1,1}$ and $\nu \circ P_0 \in \mathcal{C}^{1,1}$. By (III.66), we also get that $\delta \in \mathcal{C}^{2,1}$.

3.3 Regularised distance

The regularity of δ away from $\partial\Omega$ that we proved in the previous section is not sufficient for our purposes. In particular it is the case when we consider the problem of proving elliptic regularity properties of the Stokes operator (or even for the Laplace operator) with optimal generic regularity assumptions on the domain Ω. However, in most of the results that we prove in the next chapters, one can use the original distance δ instead of the regularised distance as soon as Ω is more regular than necessary (one level of regularity higher is in general sufficient). In that case, some computations can be slightly simplified.

With this objective in mind, we need now to build a regularised distance function as follows. This construction is taken from [82]. We choose a mollifying kernel as defined in Definition II.2.23. For any $x \in \mathbb{R}^d$ and any $\tau \in \mathbb{R}$ we set

$$G(x,\tau) = \int_B \delta\left(x + \frac{\tau}{2} z\right) \eta(z)\, dz.$$

Since η is smooth, we see that G is of class $\mathcal{C}^\infty$ on $\mathbb{R}^d \times \mathbb{R}^*$. Moreover, using that $\mathrm{Lip}(\delta) = 1$, we have the estimate

$$\begin{aligned} |G(x,\tau_1) - G(x,\tau_2)| &\leq \int_B \left|\delta\left(x + \frac{\tau_1}{2} z\right) - \delta\left(x + \frac{\tau_2}{2} z\right)\right| \eta(z)\, dz \\ &\leq \frac{1}{2}|\tau_1 - \tau_2| \int_B |z|\eta(z)\, dz \leq \frac{1}{2}|\tau_1 - \tau_2|, \end{aligned}$$

and thus, by using the Banach fixed-point theorem, we obtain that for any $x \in \mathbb{R}^d$, there exists a unique $\rho(x) \in \mathbb{R}$ such that

$$\rho(x) = G(x, \rho(x)). \tag{III.68}$$

Definition III.3.6. *The function ρ defined by (III.68) is said to be a regularised distance function for the domain Ω.*

We now investigate the main properties of ρ, showing in particular why it can plays the role of a distance function.

Proposition III.3.7. *1. For any $x \in \mathbb{R}^d$, we have*

$$\rho(x) = 0 \iff \delta(x) = 0 \iff x \in \partial\Omega.$$

Moreover, there exists $C_1, C_2 > 0$ such that

$$C_1 \leq \frac{\delta(x)}{\rho(x)} \leq C_2,\ \forall x \in \mathbb{R}^d \setminus \partial\Omega. \tag{III.69}$$

2. The function ρ is of class $\mathcal{C}^{k+1,1}$ in $\mathbb{R}^d$ and of class $\mathcal{C}^\infty$ in $\mathbb{R}^d \setminus \partial\Omega$. Moreover for any multi-index $\alpha \in \mathbb{N}^d$ such that $|\alpha| = k+3$, we have the estimate

$$|\partial^\alpha \rho(x)| \leq \frac{C}{\rho(x)},\ \forall x \in \mathbb{R}^d \setminus \partial\Omega.$$

3. On the boundary $\partial\Omega$, we have

$$\nabla\rho(x) = \nabla\delta(x) = -\nu(x),\ \forall x \in \partial\Omega.$$

Moreover, there is an open neighborhood U of $\partial\Omega$ in $\mathbb{R}^d$, such that

$$\inf_U |\nabla\rho| > 0.$$

Remark III.3.1. It is not surprising that, in general, the derivatives of order $k+3$ of ρ are blowing up on $\partial\Omega$ inasmuch as, if it were not the case, the boundary $\partial\Omega$ would be a submanifold of class $\mathcal{C}^{k+2,1}$.

Remark III.3.2. Even though, in some cases, the regularity of the original distance function δ near the boundary $\partial\Omega$ is sufficient for our needs, it may happen that δ is not smooth inside the whole domain Ω. This leads to complication of some computations, and therefore it is preferable, even in this case, to use the regularised function ρ instead. This is for instance the case in the proof of Theorem III.2.23 below.

Proof.

1. Since $\int_B \eta\, dz = 1$ we have

$$\rho(x) - \delta(x) = \int_B \left(\delta\left(x + \frac{\rho(x)}{2} z\right) - \delta(x)\right) \eta(z)\, dz.$$

so that

$$|\rho(x) - \delta(x)| \leq \frac{1}{2}|\rho(x)| \int_B |z|\eta(z)\, dz \leq \frac{1}{2}|\rho(x)|.$$

It follows that $\rho(x) = 0$ if and only if $\delta(x) = 0$ and that ρ and δ have the same signs away from $\partial\Omega$. We also get that, for any $x \notin \partial\Omega$,

$$\frac{1}{2} \leq \frac{\delta(x)}{\rho(x)} \leq \frac{3}{2}.$$

2. By the implicit function theorem and the regularity of G, we see that ρ is $\mathcal{C}^\infty$ on $\mathbb{R}^d \setminus \{\rho = 0\} = \mathbb{R}^d \setminus \{\delta = 0\} = \mathbb{R}^d \setminus \partial\Omega$.
Moreover for any $x, y \in \mathbb{R}^d$, we have

$$\begin{aligned}
|\rho(x) - \rho(y)| &= |G(x, \rho(x)) - G(y, \rho(y))| \\
&\leq |G(x, \rho(x)) - G(x, \rho(y))| + |G(x, \rho(y)) - G(y, \rho(y))| \\
&\leq \frac{1}{2}|\rho(x) - \rho(y)| + |x - y|,
\end{aligned}$$

so that

$$|\rho(x) - \rho(y)| \leq 2|x - y|,$$

and thus ρ is Lipschitz continuous on $\mathbb{R}^d$ with $\mathrm{Lip}(\rho) \leq 2$.
For simplicity we only consider here the case $k = 0$. By differentiation of (III.68), we find that

$$\begin{aligned}
\frac{\partial\rho}{\partial x_i}(x) &= \int_B \frac{\partial\delta}{\partial x_i}\left(x + \frac{\rho(x)}{2} z\right) \eta(z)\, dz \\
&\quad + \frac{1}{2}\frac{\partial\rho}{\partial x_i}(x) \int_B \left\langle z, \nabla\delta\left(x + \frac{\rho(x)}{2} z\right)\right\rangle \eta(z)\, dz.
\end{aligned}$$

It follows that

$$\frac{\partial\rho}{\partial x_i}(x)=\frac{\int_B \frac{\partial\delta}{\partial x_i}\left(x+\frac{\rho(x)}{2}z\right)\eta(z)\,dz}{1-\frac{1}{2}\int_B\left\langle z,\nabla\delta\left(x+\frac{\rho(x)}{2}z\right)\right\rangle\eta(z)\,dz}=\frac{T_1(x)}{1-\frac{1}{2}T_2(x)}.$$

Notice that, since $|\nabla\delta|\le 1$, the denominator $1-\frac{1}{2}T_2(x)$ in the above formula does not vanish.
By the usual computations, we find that for $l=1,2$ we have $\mathrm{Lip}(T_l)\le C\,\mathrm{Lip}(\nabla\delta)(1+\mathrm{Lip}(\rho))$, therefore $\nabla\rho$ is Lipschitz continuous on $\mathbb{R}^d$; that is, $\rho\in\mathcal{C}^{1,1}$.
Let now $x\in\Omega$; we can compute the derivatives of T_1 as follows

$$\begin{aligned}\frac{\partial T_1}{\partial x_j}=&-\frac{2}{\rho(x)}\int_B\frac{\partial\delta}{\partial x_i}\left(x+\frac{\rho(x)}{2}z\right)\frac{\partial\eta}{\partial z_j}(z)\,dz\\&-\frac{1}{\rho(x)}\frac{\partial\rho}{\partial x_j}(x)\int_B\frac{\partial\delta}{\partial x_i}\left(x+\frac{\rho(x)}{2}z\right)\mathrm{div}_z(\eta(z)z)\,dz.\end{aligned}$$

We want to study the second derivatives of T_1 (inasmuch as these terms are involved in the computation of third derivatives of ρ). Since $\partial_{x_j}\rho$ is smooth, it is enough to study the regularity of terms of the form

$$F(x)=\frac{1}{\rho(x)}\int_B\frac{\partial\delta}{\partial x_i}\left(x+\frac{\rho(x)}{2}z\right)\psi(z)\,dz,$$

where ψ is smooth, compactly supported in B and satisfies $\int_B\psi\,dz=0$. We have

$$\begin{aligned}\frac{\partial F}{\partial x_l}(x)=&-\frac{\partial\rho}{\partial x_l}(x)\frac{1}{\rho(x)^2}\int_B\frac{\partial\delta}{\partial x_i}\left(x+\frac{\rho(x)}{2}z\right)\psi(z)\,dz\\&+\frac{1}{\rho(x)}\int_B\frac{\partial^2\delta}{\partial x_i\partial x_l}\left(x+\frac{\rho(x)}{2}z\right)\psi(z)\,dz\\&+\frac{1}{2}\frac{1}{\rho(x)}\frac{\partial\rho}{\partial x_l}\int_B\left\langle z,\underbrace{\left(\nabla\frac{\partial\delta}{\partial x_i}\right)\left(x+\frac{\rho(x)}{2}z\right)}_{=\frac{2}{\rho(x)}\nabla_z\left(\frac{\partial\delta}{\partial x_i}\left(x+\frac{\rho(x)}{2}z\right)\right)}\right\rangle\psi(z)\,dz.\end{aligned}$$

Integrating the last term by parts, and using that

$$\int_B\psi(z)\,dz=\int_B\mathrm{div}(\psi(z)z)\,dz=0,$$

we have

$$\begin{aligned}\frac{\partial F}{\partial x_l}(x) = &- \frac{\partial \rho}{\partial x_l}(x)\frac{1}{\rho(x)^2}\int_B \left(\frac{\partial \delta}{\partial x_i}\left(x+\frac{\rho(x)}{2}z\right) - \frac{\partial \delta}{\partial x_i}(x)\right)\psi(z)\,dz\\ &+ \frac{1}{\rho(x)}\int_B \frac{\partial^2\delta}{\partial x_i \partial x_l}\left(x+\frac{\rho(x)}{2}z\right)\psi(z)\,dz\\ &- \frac{1}{\rho(x)^2}\frac{\partial \rho}{\partial x_l}\int_B \left(\frac{\partial \delta}{\partial x_i}\left(x+\frac{\rho(x)}{2}z\right) - \frac{\partial \delta}{\partial x_i}(x)\right)\mathrm{div}_z(\psi(z)z)\,dz,\end{aligned}$$

and then

$$\left|\frac{\partial F}{\partial x_l}(x)\right| \le C\frac{(1+\mathrm{Lip}(\rho))\,\mathrm{Lip}(\nabla\delta)}{\rho(x)}.$$

This proves that the second derivatives of T_1 are bounded by C/ρ. The same computations can be made for the term T_2 which gives the estimate for third derivatives of ρ.

3. For $x \in \partial\Omega$ we have $\rho(x) = 0$ and then we find that

$$T_1(x) = \frac{\partial \delta}{\partial x_i}(x)\int_B \eta(z)\,dz = \frac{\partial \delta}{\partial x_i}(x),$$
$$T_2(x) = \left\langle \nabla\delta(x), \int_B z\eta(z)\,dz \right\rangle = 0,$$

because η is symmetric. Finally, it follows that

$$\frac{\partial \rho}{\partial x_i}(x) = \frac{\partial \delta}{\partial x_i}(x).$$

As a consequence we have $|\nabla\rho| = 1$ on $\partial\Omega$ and since $\nabla\rho$ is continuous on $\mathbb{R}^d$, the existence of such an open set U is straightforward.

□

We are now in position to prove the lifting theorem for normal derivatives that we presented in Section 2.5.3.

Proof (of Theorem III.2.23).

For any $g \in W^{1-1/p,p}(\partial\Omega)$, we define

$$R_\nu g(x) = -\rho(x)\int_B (R_0 g)(x+\alpha\rho(x)z)\eta(z)\,dz.$$

In this definition $R_0 g \in W^{1,p}(\Omega)$ is the lifting of g introduced in Theorem III.2.22 and $\alpha > 0$ is such that

$$\alpha \le C_1,\ \alpha\,\mathrm{Lip}(\rho) < 1,$$

where $C_1 > 0$ is the constant appearing in (III.69). Note that, for any $x \in \Omega$ and any $z \in B$, we have $x + \alpha\rho(x)z \in \Omega$, so that $R_\nu g$ is well-defined.

Since ρ is $\mathcal{C}^\infty$ in Ω and η is smooth, we see that $R_\nu g \in \mathcal{C}^\infty(\Omega)$. Let us show that $R_\nu g \in W^{2,p}(\Omega)$. Let $1 \le i \le d$, we have

$$\begin{aligned}\frac{\partial R_\nu g}{\partial x_i}(x) =& - \frac{\partial \rho}{\partial x_i}(x) \int_B (R_0 g)(x+\alpha\rho(x)z)\eta(z)\,dz \\ & - \rho(x)\int_B \left\langle \nabla(R_0 g)(x+\alpha\rho(x)z), e_i + \alpha\frac{\partial\rho}{\partial x_i}(x)z\right\rangle \eta(z)\,dz \\ =& - \frac{\partial \rho}{\partial x_i}(x)\int_B (R_0 g)(x+\alpha\rho(x)z)\underbrace{(\eta(z) - \mathrm{div}_z(\eta(z)z))}_{=\psi(z)}\,dz \\ & - \frac{1}{\alpha^d\rho^{d-1}(x)}\int_{\mathbb{R}^d} \frac{\partial(R_0 g)}{\partial x_i}(y)\eta\left(\frac{y-x}{\alpha\rho(x)}\right)dy.\end{aligned} \tag{III.70}$$

By the same kind of computation we get

$$\begin{aligned}\frac{\partial^2(R_\nu g)}{\partial x_i \partial x_j}(x) =& - \frac{\partial^2\rho}{\partial x_i\partial x_j}(x)\int_B (R_0 g)(x+\alpha\rho(x)z)\psi(z)\,dz \\ & - \frac{\partial\rho}{\partial x_i}(x)\int_B \frac{\partial(R_0 g)}{\partial x_j}(x+\alpha\rho(x)z)\psi(z)\,dz \\ & + \frac{1}{\rho(x)}\frac{\partial\rho}{\partial x_i}(x)\frac{\partial\rho}{\partial x_j}(x)\int_B (R_0 g)(x+\alpha\rho(x)z)\,\mathrm{div}_z(\psi(z)z)\,dz \\ & + (d-1)\frac{\partial\rho}{\partial x_j}(x)\int_B \frac{\partial(R_0 g)}{\partial x_i}(x+\alpha\rho(x)z)\eta(z)\,dz \\ & + \frac{1}{\alpha}\int_B \frac{\partial(R_0 g)}{\partial x_i}(x+\alpha\rho(x)z)\frac{\partial\eta}{\partial z_j}(z)\,dz \\ & + \frac{\partial\rho}{\partial x_j}(x)\int_B \frac{\partial(R_0 g)}{\partial x_i}(x+\alpha\rho(x)z)\left\langle\nabla\eta(z), z\right\rangle\,dz\end{aligned}$$

We can integrate by parts the integral in the third term to obtain

$$\begin{aligned}&\frac{1}{\rho(x)}\frac{\partial\rho}{\partial x_i}(x)\frac{\partial\rho}{\partial x_j}(x)\int_B (R_0 g)(x+\alpha\rho(x)z)\,\mathrm{div}_z(\psi(z)z)\,dz \\ &= -\alpha\frac{\partial\rho}{\partial x_i}(x)\frac{\partial\rho}{\partial x_j}(x)\int_B \left\langle\nabla(R_0 g)(x+\alpha\rho(x)z), \psi(z)z\right\rangle\,dz.\end{aligned}$$

All the derivatives of ρ up to second-order are bounded, therefore it remains to study terms of the form

$$F(x) = \int_B f(x+\alpha\rho(x)z)\widetilde{\psi}(z)\,dz,\ x\in\Omega,$$

where $f \in L^p(\Omega)$ and $\widetilde{\psi}$ is smooth and compactly supported in the unit ball B. By the Jensen inequality we get

$$|F(x)|^p \le \|\widetilde{\psi}\|_{L^1}^p \int_B |f(x+\alpha\rho(x)z)|^p|\widetilde{\psi}(z)|\,dz,$$

so that we have

$$\|F\|_{L^p(\Omega)}^p \leq C_{\psi,p} \int_B \left(\int_\Omega |f(x+\alpha\rho(x)z)|^p \, dx \right) dz.$$

For any given $z \in B$, by the choice of α, the map $x \mapsto x + \alpha\rho(x)z$ is a diffeomorphism from Ω onto some subset of Ω whose Jacobian determinant is bounded from below by $1 - \alpha \operatorname{Lip}(\rho) > 0$. It follows that

$$\|F\|_{L^p(\Omega)}^p \leq C_{\psi,p,\rho,\alpha} \|f\|_{L^p(\Omega)}^p.$$

Gathering all the previous results we have finally proved that $R_\nu g \in W^{2,p}(\Omega)$ and that

$$\|R_\nu g\|_{W^{2,p}(\Omega)} \leq C\|R_0 g\|_{W^{1,p}(\Omega)} \leq C'\|g\|_{W^{1-1/p,p}(\partial\Omega)}.$$

Since $\rho = 0$ and $\nabla\rho = -\nu$ on the boundary $\partial\Omega$, we immediately see from (III.70) that for $x \in \partial\Omega$ we have

$$\frac{\partial R_\nu g}{\partial x_i}(x) = \nu_i(x)(R_0 g)(x) = \nu_i(x) g(x),$$

so that

$$\gamma_\nu(R_\nu g) = \langle \nabla(R_\nu g), \nu\rangle = \left(\sum_{i=1}^d \nu_i^2\right) g = g,$$

and the claim is proved.

□

3.4 Parametrisation of a neighborhood of $\partial\Omega$

Using the regularised distance ρ that we have constructed above, we set $\mathcal{O}_{s_0}^\rho = \{x \in \mathbb{R}^d, \ \ 0 < \rho(x) < s_0\}$. For $s_0 > 0$ small enough we have $\overline{\mathcal{O}_{s_0}^\rho} \subset U$ so that $\nabla\rho$ does not vanish in $\mathcal{O}_{s_0}^\rho$.

For $0 \leq s \leq s_0$ we let $\Gamma_s = \{x \in \mathbb{R}^d, \ \rho(x) = s\}$. We have $\Gamma_0 = \Gamma = \partial\Omega$ and $\Gamma_s \subset \Omega$ for any $0 < s \leq s_0$. Moreover Γ_0 is a $\mathcal{C}^{k+1,1}$ compact manifold (by assumption on Ω) and Γ_s, $s > 0$ is a $\mathcal{C}^\infty$ compact manifold because ρ is smooth in a neighborhood of Γ_s and its gradient does not vanish on Γ_s.

We now set

$$\nu(x) = -\frac{\nabla\rho(x)}{|\nabla\rho(x)|}, \ \forall x \in \overline{\mathcal{O}_{s_0}^\rho}.$$

This notation is consistent with the fact that $\nabla\rho$ is equal to the outward unit normal denoted by ν on $\partial\Omega$. For any vector field $w : \mathcal{O}_{s_0}^\rho \to \mathbb{R}^d$, we define its normal and tangential parts as follows

$$
\begin{aligned}
w_N(x) &= \langle w(x), \nu(x)\rangle \in \mathbb{R}, && \forall x \in \mathcal{O}^\rho_{s_0},\\
w_T(x) &= w(x) - w_N(x)\nu(x) \in \mathbb{R}^d, && \forall x \in \mathcal{O}^\rho_{s_0}.
\end{aligned}
$$

By construction, at each point x the vector $w_T(x)$ is tangent to the manifold $\Gamma_{\rho(x)}$. Actually, $w_T(x)$ is the orthogonal projection of $w(x)$ onto the tangent space $T\Gamma_{\rho(x)}$.

- In the case where $w = \nabla v$ is a gradient, we rather use the notation

$$\nabla_T v = (\nabla v)_T,\ \nabla_N v = (\nabla v)_N.$$

- In the case where $v : \mathcal{O}^\rho_{s_0} \to \mathbb{R}^d$ is a vector field and thus ∇v is a tensor field, the formulas above can be generalised naturally as follows

$$
\begin{aligned}
\nabla_N v &= (\nabla v).\nu,\\
\nabla_T v &= \nabla v - (\nabla_N v) \otimes \nu.
\end{aligned}
$$

By using the properties of ρ established above, we see that ν is $\mathcal{C}^\infty$ away from $\partial\Omega$ and of class $\mathcal{C}^{k,1}$ on $\overline{\mathcal{O}^\rho_{s_0}}$. Moreover the second derivatives of ν behave as does $1/\rho$ in a neighborhood of $\partial\Omega$ (see Proposition III.3.7).

Associated with ν we set

$$\tilde{\nu} = \frac{\nu}{|\nabla\rho|},$$

and the corresponding flow ζ defined on $\mathbb{R}\times\mathbb{R}^d$ by

$$
\begin{cases}
\dfrac{d}{ds}\zeta(s,x) = \tilde{\nu}(\zeta(s,x)), & \forall (s,x) \in \mathbb{R}\times\mathbb{R}^d,\\
\zeta(0,x) = x, & \forall x \in \mathbb{R}^d.
\end{cases}
\tag{III.71}
$$

This flow is well-defined thanks to the Cauchy–Lipschitz theorem. Moreover we observe that

$$\frac{d}{ds}\rho(\zeta(s,x)) = \left\langle \nabla\rho(\zeta(s,x)), \frac{d}{ds}\zeta(s,x)\right\rangle = -1,$$

which leads to the formula

$$\rho(\zeta(s,x)) = \rho(x) - s,\ \forall s \in \mathbb{R}, \forall x \in \mathbb{R}^d. \tag{III.72}$$

Proposition III.3.8. *The flow $(s,x) \mapsto \zeta(s,x)$ is of class $\mathcal{C}^{k,1}$ on $\mathbb{R}\times\mathbb{R}^d$, and of class $\mathcal{C}^\infty$ in the set*

$$E = \{(s,x) \in \mathbb{R}\times\mathcal{O}^\rho_{s_0},\quad s < \rho(x)\},$$

and satisfies for any $\alpha \in \mathbb{N}^d$, $|\alpha| = k+2$, and any $(s,x) \in E$, the estimates

$$|\partial^\alpha \zeta(s,x)| \le C|\ln(\rho(x) - s)|, \tag{III.73}$$

$$\left|\frac{\partial}{\partial s}\partial^\alpha \zeta(s,x)\right| \leq \frac{C}{\rho(x)-s}. \tag{III.74}$$

Proof.

Since $\tilde{\nu}$ is of class $\mathcal{C}^{k,1}$, the first part of the result is simply the usual regularity result for solutions of ordinary differential equations.

For simplicity, let us only establish (III.73) and (III.74) the second estimate in the case $k=0$. Using (III.72), we first note that for any $(s,x)\in E$, we have $\zeta(t,x)\in \mathcal{O}^\rho_{s_0}$ for any $t\in[0,s]$. The vector field $\tilde{\nu}$ is of class $\mathcal{C}^\infty$ in $\mathcal{O}^\rho_{s_0}$, so that the differential equation satisfied by second derivatives of ζ is given by

$$\frac{d}{ds}\frac{\partial^2\zeta}{\partial x_i\partial x_j} = \nabla\tilde{\nu}(\zeta).\frac{\partial^2\zeta}{\partial x_i\partial x_j} + D^2\tilde{\nu}(\zeta)\cdot\left(\frac{\partial\zeta}{\partial x_i},\frac{\partial\zeta}{\partial x_j}\right), \tag{III.75}$$

so that, using (III.72), the properties of $\tilde{\nu}$ and the fact that $\nabla\zeta$ is bounded, we get

$$\left|\frac{d}{ds}\frac{\partial^2\zeta}{\partial x_i\partial x_j}\right| \leq C\left|\frac{\partial^2\zeta}{\partial x_i\partial x_j}\right| + \frac{C}{\rho(x)-s}.$$

By integration between 0 and s, and the Gronwall lemma, we deduce (III.73). The estimate (III.74) now comes from (III.75).

□

Using the flow ζ we can now define a suitable parametrisation of $\overline{\mathcal{O}^\rho_{s_0}}$.

Proposition III.3.9. *The map*

$$\psi : (\sigma,s)\in\Gamma_{s_0}\times[0,s_0] \to \zeta(s_0-s,\sigma)\in\overline{\mathcal{O}^\rho_{s_0}},$$

is a bijection and its inverse is given by

$$x\in\overline{\mathcal{O}^\rho_{s_0}} \to (P(x),\rho(x))\in\Gamma_{s_0}\times[0,s_0],$$

where P is the projection onto Γ_{s_0} defined by

$$P(x)=\zeta(\rho(x)-s_0,x),\ \forall x\in\overline{\mathcal{O}^\rho_{s_0}}.$$

Furthermore, ψ is of class $\mathcal{C}^\infty$ on $\Gamma_{s_0}\times]0,s_0]$ and of class $\mathcal{C}^{k,1}$ on $\Gamma_{s_0}\times[0,s_0]$. Similarly, P is $\mathcal{C}^\infty$ in $\mathcal{O}^\rho_{s_0}$.

Proof.

The regularity properties directly follow from the ones of the flow ζ (see Proposition III.3.8). It remains to show that ψ is a bijection and to characterise its inverse.

- Assume that $\psi(\sigma_1,s_1)=\psi(\sigma_2,s_2)$. Let us denote this point by x. Using (III.72) we have

$$\rho(x)=\rho(\zeta(s_0-s_1,\sigma_1)) = \underbrace{\rho(\sigma_1)}_{=s_0}-(s_0-s_1)=s_1,$$

because $\sigma_1 \in \Gamma_{s_0}$. Similarly we have $\rho(x) = s_2$ and thus $s_1 = s_2$. We now have $\zeta(s_0 - s_1, \sigma_1) = \zeta(s_0 - s_1, \sigma_2)$ and thus $\sigma_1 = \sigma_2$ because ζ is the flow associated with an ordinary differential equation.

- Let $x \in \overline{\mathcal{O}^\rho_{s_0}}$. By assumption we have $0 \leq \rho(x) \leq s_0$. We define $P(x) = \zeta(\rho(x) - s_0, x)$ so that, by (III.72) we get that $\rho(P(x)) = s_0$, (i.e., $P(x) \in \Gamma_{s_0}$) and it is clear that $x = \psi(P(x), \rho(x))$.

□

Remark III.3.3. With this parametrisation at hand, we can see that

$$(\nabla_N v) \circ \psi = \langle (\nabla v) \circ \psi, \nu \circ \psi \rangle = -\frac{1}{|\tilde{\nu}|} \frac{\partial}{\partial s} (v \circ \psi). \tag{III.76}$$

Proposition III.3.10. *For any $s \in [0, s_0]$, there exists a function $\mathcal{J}_s : \sigma \in \Gamma_{s_0} \to \mathbb{R}$ such that*

$$\int_{\Gamma_s} v(\sigma)\, d\sigma = \int_{\Gamma_{s_0}} v \circ \psi(\sigma, s) \mathcal{J}_s(\sigma)\, d\sigma,$$

for any integrable function $v : \Gamma_s \to \mathbb{R}$.

Proof.

Let (U, φ) be a chart of Γ_{s_0}. We observe that $(U, \psi(\varphi(.), s))$ is a chart on the manifold Γ_s for any $s \in [0, s_0]$. Assume that v is supported in $\psi(\varphi(U), s)$. For any $u \in U$, we define

$$\mathcal{J}_s(\varphi(u)) = \sqrt{\frac{\det \mathfrak{g}_s(u)}{\det \mathfrak{g}_{s_0}(u)}},$$

where for any s, $\mathfrak{g}_s$ is the matrix of the first fundamental form of Γ_s in the chart $(U, \psi(\varphi(.), s))$. We can easily check that this quantity only depends on the point $\varphi(u)$ and not on the particular chart (φ, U) used in this computation. By definition of the integral we have

$$\begin{aligned}
\int_{\Gamma_s} v(\sigma)\, d\sigma &= \int_U v \circ \psi(\varphi(u), s) \sqrt{\det \mathfrak{g}_s(u)}\, du \\
&= \int_U v \circ \psi(\varphi(u), s) \sqrt{\frac{\det \mathfrak{g}_s(u)}{\det \mathfrak{g}_{s_0}(u)}} \sqrt{\det \mathfrak{g}_0(u)}\, du \\
&= \int_{\Gamma_{s_0}} v \circ \psi(\sigma, s) \mathcal{J}_s(\sigma)\, d\sigma.
\end{aligned}$$

□

Definition III.3.11. *We define the Jacobian $\mathcal{J} : \overline{\mathcal{O}^\rho_{s_0}} \to \mathbb{R}$ as*

$$\mathcal{J}(x) = \mathcal{J}_{\rho(x)}(P(x)),\ \forall x \in \overline{\mathcal{O}^\rho_{s_0}}.$$

In particular we have $\mathcal{J}_s(\sigma) = \mathcal{J} \circ \psi(\sigma, s)$ for any $s \in [0, s_0]$ and $\sigma \in \Gamma_{s_0}$.

Proposition III.3.12. *For any integrable function v, we have the following change of variable formula*

$$\begin{aligned}\int_{\mathcal{O}^\rho_{s_0}} v(x)\,dx &= \int_0^{s_0} \left(\int_{\Gamma_s} |\tilde{\nu}| v\, d\sigma \right) ds \\ &= \int_0^{s_0} \left(\int_{\Gamma_{s_0}} \left(|\tilde{\nu}| \mathcal{J} v \right) \circ \psi(\sigma, s)\, d\sigma \right) ds.\end{aligned}$$

Moreover $\mathcal{J}$ is of class $\mathcal{C}^\infty$ in $\mathcal{O}^\rho_{s_0}$ and satisfies, for any $\alpha \in \mathbb{N}^d$

$$\begin{aligned}\partial^\alpha \mathcal{J} &\in L^\infty(\mathcal{O}^\rho_{s_0}), \text{for } |\alpha| \leq k,\\ \frac{1}{|\ln \rho|}\partial^\alpha \mathcal{J} &\in L^\infty(\mathcal{O}^\rho_{s_0}), \text{for } |\alpha| = k+1,\\ \rho \nabla_N \partial^\alpha \mathcal{J} &\in L^\infty(\mathcal{O}^\rho_{s_0}), \text{for } |\alpha| = k+1.\end{aligned}$$

Remark III.3.4. In some situations we do not need additional regularity properties of the distance function, and one can use the original distance function δ in those computations. In particular, for a small enough $s_1 > 0$ we can obtain the following formula

$$\int_{\mathcal{O}_{s_1}} v(x)\,dx = \int_0^{s_1} \left(\int_{\partial\Omega} (\tilde{\mathcal{J}} v)(\sigma - s\nu(\sigma))\, d\sigma \right) ds,$$

where $\tilde{\mathcal{J}}$ is a Jacobian term which satisfies

$$\partial^\alpha \tilde{\mathcal{J}} \in L^\infty(\mathcal{O}_{s_1}), \text{ for } |\alpha| \leq k+1,$$

but not the other regularity properties of $\mathcal{J}$.

Proof.

It is enough to show the first equality because the second one comes from Proposition III.3.10.

Let $U \subset \mathbb{R}^{d-1}$ and $\varphi : U \to \mathbb{R}^d$ such that (U, φ) is a chart of the manifold Γ_{s_0}. Assume that v is supported in $\psi(\varphi(U) \times [0, s_0])$ and let us write

$$\int_{\mathcal{O}^\rho_{s_0}} v(x)\,dx = \int_0^{s_0} \int_U v(\psi(\varphi(u), s)) |\operatorname{Jac} \psi(\varphi(u), s)|\, du\, ds.$$

The Jacobian matrix J of $(u, s) \in \mathbb{R}^{d-1} \times \mathbb{R} \mapsto \psi(\varphi(u), s)$ is given by

$$J = \left(\frac{\partial \psi(\varphi(.), s)}{\partial u_1}, \dots, \frac{\partial \psi(\varphi(.), s)}{\partial u_{d-1}}, \frac{\partial \psi(\varphi(.), s)}{\partial s} \right).$$

We notice that, by definition of the flow we have

$$\frac{\partial \psi(\varphi(.), s)}{\partial s} = -\tilde{\nu}(\psi(\varphi(.), s)),$$

and since $\tilde{\nu}$ is orthogonal to the tangent space of Γ_s at each point it follows

$${}^tJ.J = \begin{pmatrix} \begin{pmatrix} \mathfrak{g}_s(u) \end{pmatrix} & 0 \\ 0 & |\tilde{\nu}(\psi(\varphi(u), s))|^2 \end{pmatrix},$$

where $\mathfrak{g}_s(u)$ is the matrix of the first quadratic form of Γ_s in the chart $(U, \psi(\varphi(.), s))$. All this gives that

$$\operatorname{Jac} \psi(\varphi(u), s) = \det J = |\tilde{\nu}| \sqrt{\det \mathfrak{g}_s(u)}.$$

We deduce that

$$\int_{\mathcal{O}^\rho_{s_0}} v(x)\, dx = \int_0^{s_0} \left(\int_U |\tilde{\nu}(\psi(\varphi(u), s))| v(\psi(\varphi(u), s)) \sqrt{\det \mathfrak{g}_s(u)}\, du \right) ds,$$

and the result follows by definition of the integral on the manifold Γ_s.

If the support of v is not included in $\psi(\varphi(U) \times [0, s_0])$, we can prove the result by considering a covering of Γ_{s_0} by such charts (U, φ) and by using a partition of unity.

The regularity properties of $\mathcal{J}$ immediately come from the ones of the flow ζ; see Proposition III.3.8.

□

As a first application of the previous results, we can prove the following generalised Hardy inequality which holds for functions which do not necessarily vanish on the boundary.

Proposition III.3.13. *Let Ω be a domain of $\mathbb{R}^d$ of class $\mathcal{C}^{1,1}$ and with compact boundary and $1 < p < +\infty$. There exists a $C > 0$ such that for any $u \in W^{1,p}(\Omega)$ we have*

$$\left\| \frac{u - (\gamma_0 u) \circ \psi(P(.), 0)}{\rho} \right\|_{L^p(\mathcal{O}^\rho_{s_0})} \leq C \|u\|_{W^{1,p}}.$$

Remark III.3.5. Note that for any $x \in \mathcal{O}^\rho_{s_0}$, $\psi(P(x), 0)$ is nothing but the point on the boundary $\partial\Omega$ which is obtained by following the flow ζ starting from x.

In the case where Ω is smooth enough, we can replace the regularised distance ρ by the usual distance δ. In that case $\psi(P(.), 0)$ is exactly the usual orthogonal projection P_0 on $\partial\Omega$, and the same Hardy inequality holds.

Proof.

By density of $\mathcal{C}^\infty_c(\overline{\Omega})$ in $W^{1,p}(\Omega)$, it is enough to prove the inequality for any smooth u. For such a u, we have

$$\begin{aligned}u(x)-(\gamma_0 u)\circ\psi(P(x),0) &= u(x)-u(\zeta(\rho(x),x))\\ &= u(\zeta(0,x))-u(\zeta(\rho(x),x)) = \int_0^{\rho(x)}\frac{d}{ds}u(\zeta(s,x))\,ds\\ &= \int_0^{\rho(x)}(\nabla u)(\zeta(s,x))\tilde{\nu}(s,x)\,ds.\end{aligned}$$

It follows that

$$\left|\frac{u(x)-(\gamma_0 u)\circ\psi(P(x),0)}{\rho(x)}\right| \leq \frac{1}{\rho(x)}\int_0^{\rho(x)}|\nabla u|(\zeta(s,x))\,ds.$$

By Proposition III.3.12, we get

$$\begin{aligned}&\int_{\mathcal{O}^\rho_{s_0}}\left|\frac{u-(\gamma_0 u)\circ\psi(P(.),0)}{\rho}\right|^p dx\\ \leq& C\int_0^{s_0}\int_{\Gamma_{s_0}}\left(\frac{1}{s}\int_0^s|\nabla u|(\zeta(\tau,\psi(\sigma,s)))\,d\tau\right)^p d\sigma\,ds.\end{aligned}$$

By definition of ψ, we have $\zeta(\tau,\psi(\sigma,s))=\zeta(s_0+\tau-s,\sigma)$. We can perform the change of variable $\tau\to s-\tau$ and then use the Hardy inequality for functions defined on $\mathbb{R}$ (Lemma II.4.7) to obtain

$$\begin{aligned}\int_{\mathcal{O}^\rho_{s_0}}\left|\frac{u-(\gamma_0 u)\circ\psi(P(.),0)}{\rho}\right|^p dx &\leq C\int_{\Gamma_{s_0}}\int_0^{s_0}|\nabla u|^p(\psi(\sigma,s))\,ds\,d\sigma\\ &\leq C\int_{\mathcal{O}^\rho_{s_0}}|\nabla u|^p\,dx,\end{aligned}$$

where we used the properties of $\tilde{\nu}$ and of $\mathcal{J}$.

□

3.5 Tangential Sobolev spaces

3.5.1 Definitions

Let us first characterise the Sobolev regularity in $\mathcal{O}^\rho_{s_0}$ by means of tangent and normal parts of the gradient of functions and of lower-order Sobolev spaces.

Theorem III.3.14. *Let Ω be a $\mathcal{C}^{k+1,1}$ domain of $\mathbb{R}^d$, $k\geq 0$, with compact boundary.*

Let $0\leq l\leq k+1$, $1\leq p<+\infty$, and $f\in W^{l,p}(\Omega)\cap W^{l+1,p}_{loc}(\Omega)$. We have

$$f\in W^{l+1,p}(\Omega)\iff \nabla_T f\in (W^{l,p}(\mathcal{O}^\rho_{s_0}))^d \text{ and } \nabla_N f\in W^{l,p}(\mathcal{O}^\rho_{s_0}).$$

Proof.

- Assume that $\nabla f \in W^{l,p}(\Omega)$. By definition we have $\nabla_N f = \langle \nabla f, \nu\rangle$ in $\mathcal{O}^\rho_{s_0}$. Let α, β be multi-indices such that $|\alpha| + |\beta| = l$ and let us show that $\langle \partial^\alpha \nabla f, \partial^\beta \nu\rangle$ belongs to $L^p(\mathcal{O}^\rho_{s_0})$.
  Since $|\beta| \leq k+1$, we know that $\partial^\beta \nu$ is bounded and moreover $\partial^\alpha f \in L^p(\Omega)$ by assumption on f; the result follows. The tangential part has the same regularity because $\nabla_T f = \nabla f - (\nabla_N f)\nu$.
- Assume now that $\nabla_T f$ and $\nabla_N f$ belong to $W^{l,p}(\mathcal{O}^\rho_{s_0})$; the same argument as above shows the result.

$\square$

We can now introduce the tangential Sobolev spaces which are useful in the study of regularity properties for solutions of elliptic problems, in particular for the Stokes problem.

Definition III.3.15. *Let Ω be a $\mathcal{C}^{k+1,1}$ domain, $k \geq 0$, with compact boundary.*
For $0 \leq l \leq k+1$, $1 \leq p < +\infty$, and $f \in W^{l,p}(\Omega) \cap W^{l+1,p}_{loc}(\Omega)$. We say that $f \in W^{l+1,p}_{tang}(\Omega)$ if and only if

$$\nabla_T f \in (W^{l,p}(\mathcal{O}^\rho_{s_0}))^d \text{ and } \rho \nabla_N f \in W^{l,p}(\mathcal{O}^\rho_{s_0}).$$

Observe that, in this definition, we require that $\nabla_N f$ blows up at most like $1/\rho$ near the boundary of Ω, ρ being the (regularised) distance to the boundary.

Proposition III.3.16. *Using the assumptions and notations of the previous definition, we have that*

$$f \in W^{l+1,p}_{tang}(\Omega) \Longrightarrow \partial^\alpha f \in W^{1,p}_{tang}(\Omega), \forall \alpha \in \mathbb{N}^d, s.t.\ |\alpha| = l.$$

Proof.

Let $\alpha \in \mathbb{N}^d$ such that $|\alpha| = l$, the Leibniz formula leads to

$$\partial^\alpha(\nabla_T f) = \underbrace{\nabla(\partial^\alpha f) - \langle \nabla(\partial^\alpha f), \nu\rangle \nu}_{=\nabla_T \partial^\alpha f} + \sum_{\substack{\alpha',\alpha'',\alpha''' \in \mathbb{N}^d \\ |\alpha'|+|\alpha''|+|\alpha'''| \leq |\alpha| \\ |\alpha'| < |\alpha|}} c_{\alpha',\alpha'',\alpha'''} \langle \nabla(\partial^{\alpha'} f), \partial^{\alpha''}\nu\rangle \partial^{\alpha'''}\nu,$$

for suitable coefficients $c_{\alpha',\alpha'',\alpha'''}$. By assumption, the left-hand side term belongs to L^p and all the terms in the last sum also belong to L^p because the derivatives of f involved are at most of order l and all the derivatives of ν are at most of order $l \leq k+1$ and thus are also bounded. It follows that $\nabla_T(\partial^\alpha f) \in L^p(\Omega)$.

The same argument shows that $\rho \nabla_N(\partial^\alpha f) \in L^p(\Omega)$.

$\square$

3.5.2 Tangent translation operators

We introduce translation operators which act tangentially to the boundary of Ω. To this end, we consider the class of vector fields $\theta : \Omega \to \mathbb{R}^d$ such that

$$\begin{aligned}\theta \in \mathcal{C}^{k,1}(\overline{\Omega}) \cap \mathcal{C}^{k+1,1}(\Omega), \text{ with } \langle \theta, \nu\rangle = 0, \text{ on } \partial\Omega,\\ \text{and } |\partial^\alpha \theta(x)| \leq \frac{C_\theta}{\rho(x)}, \forall \alpha \in \mathbb{N}^d, |\alpha| = k+2, \ \forall x \in \Omega.\end{aligned} \tag{III.77}$$

For all $x \in \overline{\Omega}$, we define $\tau^\theta(h,x)$ as the solution of the ordinary Cauchy problem

$$\begin{cases} \dfrac{d}{dh}\tau^\theta(h,x) = \theta(\tau^\theta(h,x)),\\ \tau^\theta(0,x) = x.\end{cases} \tag{III.78}$$

In other words, $\tau^\theta(h,\cdot)$ defines a translation of step h along the field θ. We can check that $x \mapsto \tau(h,x)$ is also of class $\mathcal{C}^{k,1}$ uniformly with respect to $h \in [-1,1]$. Moreover, for fixed h, $\tau^\theta(h,\cdot)$ is a bijection which preserves Ω and $\partial\Omega$. Finally, for any function f and any $h \in \mathbb{R}$, we introduce the notation

$$\begin{aligned}\tau_h^\theta f &= f \circ \tau^\theta(h,\cdot),\\ \delta_h^\theta f &= \tau_h^\theta f - f.\end{aligned}$$

For all $0 \leq l \leq k+1$, and any function $f \in W^{l,p}(\Omega)$, and for all vector fields θ satisfying (III.77), we define

$$|||f|||_{\theta,l+1,p} = \sup_{0<h<1} \frac{1}{h}\|\delta_h^\theta f\|_{W^{l,p}} = \sup_{0<h<1}\frac{1}{h}\|\tau_h^\theta f - f\|_{W^{l,p}}.$$

We begin with a simple lemma concerning properties of the normal part of θ.

Lemma III.3.17. *Let θ be a vector field satisfying* (III.77)*. Then we have*

$$\langle\theta,\nu\rangle \in \mathcal{C}^{k,1}(\overline{\Omega}), \tag{III.79}$$

$$\frac{\langle\theta,\nu\rangle}{\rho} \in \begin{cases} L^\infty(\Omega) & \textit{for } k=0,\\ \mathcal{C}^{k-1,1}(\overline{\Omega}) & \textit{for } k \geq 1.\end{cases} \tag{III.80}$$

Proof.
The property (III.79) immediately follows from the regularity of ν and θ.
For any $x \in \Omega$, using the fact that $\langle\theta,\nu\rangle$ vanishes on $\partial\Omega$ and the Definition (III.71) of the flow ζ, we write

$$\begin{aligned}\frac{\langle\theta,\nu\rangle(x)}{\rho(x)} &= \frac{\langle\theta,\nu\rangle(x)-\langle\theta,\nu\rangle(\zeta(\rho(x),x))}{\rho(x)}\\ &= \frac{1}{\rho(x)}\int_0^1 \frac{\partial}{\partial u}\Big(\langle\theta,\nu\rangle(\zeta(u\rho(x),x))\Big)\,du\\ &= \int_0^1 \langle\nabla(\langle\theta,\nu\rangle),\tilde{\nu}\rangle\,(\zeta(u\rho(x),x))\,du.\end{aligned}$$

Therefore, we obtain (III.80) from (III.79).

□

We can now prove the following properties of the flow τ^θ.

Proposition III.3.18. *For any field θ satisfying (III.77), there exists a constant $C(\theta)>0$ such that for all $h\in\mathbb{R}$, satisfying $|h|\le 1$, we have:*

$$\|\tau^\theta(h,\cdot)-\mathrm{Id}_\Omega\|_{W^{k+1,\infty}}\le C(\theta)|h|.$$

In particular, we have

$$\|\nabla\tau^\theta(h,\cdot)-\mathrm{Id}\|_{W^{k,\infty}}\le C(\theta)|h|, \tag{III.81}$$

$$\|\mathrm{Jac}(\tau^\theta(h,\cdot))-1\|_{W^{k,\infty}}\le C(\theta)|h|, \tag{III.82}$$

$$\|\rho\nabla\,\mathrm{Jac}(\tau^\theta(h,\cdot))\|_{W^{k,\infty}}\le C(\theta)|h|, \tag{III.83}$$

where $\mathrm{Jac}(\tau^\theta)$ designates the determinant of the Jacobian matrix $\nabla\tau^\theta$.

Proof.

The first inequality results directly from the theory of ordinary differential equations, by noting that $\mathrm{Id}_\Omega=\tau(0,\cdot)$. Estimates (III.81) and (III.82) result directly from this.

We need an estimate on $\rho(\tau^\theta(h,x))$, that is, of the distance to the boundary of $\tau^\theta(h,x)$. To this end, we compute

$$\begin{aligned}\frac{d}{dh}\rho(\tau^\theta(h,x)) &= \left\langle\nabla\rho(\tau^\theta(h,x)),\frac{d}{dh}(\tau^\theta(h,x))\right\rangle\\ &= -|\nabla\rho(\tau^\theta(h,x))|\langle\nu,\theta\rangle(\tau^\theta(h,x)).\end{aligned} \tag{III.84}$$

Recall now that $\nabla\rho$ is bounded, that ν and θ are, at least, Lipschitz continuous, and that $\langle\nu,\theta\rangle$ vanishes on the boundary. We deduce that, for any $y\in\partial\Omega$, we have

$$|\langle\nu,\theta\rangle(\tau^\theta(h,x))|=\big|\langle\nu,\theta\rangle(\tau^\theta(h,x))-\langle\nu,\theta\rangle(y)\big|\le\mathrm{Lip}(\langle\nu,\theta\rangle)|\tau^\theta(h,x)-y|,$$

so that taking the infimum over y, we get

$$|\langle\nu,\theta\rangle(\tau^\theta(h,x))|\le\mathrm{Lip}(\langle\nu,\theta\rangle)\,\delta(\tau^\theta(h,x)),$$

and then, using (III.69), we finally get

$$|\langle \nu, \theta \rangle (\tau^\theta(h,x))| \leq C\rho(\tau^\theta(h,x)).$$

Coming back to (III.84) we finally get

$$\frac{d}{dh}\rho(\tau^\theta(h,x)) \geq -C'\rho(\tau^\theta(h,x)),$$

and by integration, we deduce that

$$\rho(\tau^\theta(h,x)) \geq \rho(x)e^{-C'h} \geq C''\rho(x), \ \forall 0 < h < 1, \tag{III.85}$$

and a similar estimate is valid for $-1 < h < 0$.

We can now prove the estimate (III.83). For simplicity, we only consider the case $k = 0$. We recall that the Jacobian determinant solves the following problem,

$$\frac{d}{dh}\operatorname{Jac}(\tau^\theta(h,\cdot)) = (\operatorname{div}\theta)(\tau^\theta(h,\cdot))\operatorname{Jac}(\tau^\theta(h,\cdot)), \ \operatorname{Jac}(\tau^\theta(0,x)) = 1.$$

By applying the differential operator ∂^α to this equation (recall that $|\alpha| = 1$ in that case), we find that

$$\begin{aligned}\frac{d}{dh}\partial^\alpha \operatorname{Jac}(\tau^\theta(h,\cdot)) =& (\operatorname{div}\theta)(\tau^\theta(h,\cdot))\partial^\alpha \operatorname{Jac}(\tau^\theta(h,\cdot)) \\ &+ \nabla(\operatorname{div}\theta)(\tau^\theta(h,\cdot)).\partial^\alpha\tau^\theta(h,\cdot)\operatorname{Jac}(\tau^\theta(h,\cdot)).\end{aligned}$$

The assumptions on θ show that $\operatorname{div}\theta$ is bounded and that $\nabla(\operatorname{div}\theta)$ behaves as does $1/\rho$. Moreover, since $|\alpha| = 1$, (III.81) gives that $\partial^\alpha\tau^\theta$ is also bounded. We thus have the inequality

$$\left|\frac{d}{dh}\partial^\alpha \operatorname{Jac}(\tau^\theta(h,\cdot))\right| \leq C|\partial^\alpha \operatorname{Jac}(\tau^\theta(h,\cdot))| + \frac{C_\theta}{\rho(\tau^\theta(h,\cdot))}.$$

By the Gronwall lemma, and the fact that $\partial^\alpha \operatorname{Jac}(\tau^\theta(h,\cdot))$ vanishes at $h = 0$, we deduce that

$$\left|\partial^\alpha \operatorname{Jac}(\tau^\theta(h,\cdot))\right| \leq \left|\int_0^h \left(\frac{C_\theta}{\rho(\tau^\theta(s,\cdot))}\,ds\right)\right| e^{C|h|}.$$

By using (III.85) and the fact that $|h| < 1$, we conclude that the claimed inequality holds; that is,

$$\left|\partial^\alpha \operatorname{Jac}(\tau^\theta(h,\cdot))\right| \leq C|h|\frac{C_\theta}{\rho(x)}.$$

□

Let us now investigate the behavior of translation operators $f \mapsto \tau_h^\theta f$ in Sobolev spaces. To this end, we first introduce the following notation for the commutator of translation operators and differential operators

$$\left[D, \tau_h^\theta\right] f = D(\tau_h^\theta f) - \tau_h^\theta(Df) = D(\delta_h^\theta f) - \delta_h^\theta(Df),$$

where the last equality is obtained by adding and subtracting Df to the definition of the commutator. For any two functions f, g we define

$$\{f, g\}_h = (\delta_h^\theta f)g - f(\delta_{-h}^\theta g). \tag{III.86}$$

We make use of the following straightforward algebraic formulas, for any functions f, g,

$$\begin{aligned}
&\{f, g\}_h = (\tau_h^\theta f)g - f(\tau_{-h}^\theta g), \\
&\delta_h^\theta(fg) = (\tau_h^\theta f)(\delta_h^\theta g) + (\delta_h^\theta f)g = f(\delta_h^\theta g) + (\delta_h^\theta f)(\tau_h^\theta g).
\end{aligned}$$

Proposition III.3.19. *Let l be an integer such that $-1 \leq l \leq k+1$.*

1. For any function $f \in W^{l,p}(\Omega)$, we have

$$\sup_{0<h<1} \|\tau_h^\theta f\|_{W^{l,p}} \leq C(\theta)\|f\|_{W^{l,p}}, \tag{III.87}$$

and

$$\tau_h^\theta f \xrightarrow[h\to 0]{} f, \quad \text{in } W^{l,p}(\Omega). \tag{III.88}$$

2. For any $f \in L^p(\Omega)$ and $g \in L^{p'}(\Omega)$ we have $\{f, g\}_h \in L^1(\Omega)$ and

$$\left|\int_\Omega \{f, g\}_h \, dx\right| \leq C(\theta)|h|\|f\|_{L^p}\|g\|_{L^{p'}}. \tag{III.89}$$

Similarly, for any $f \in L^p(\partial\Omega)$,$g \in L^{p'}(\partial\Omega)$, we have

$$\left|\int_{\partial\Omega} \{f, g\}_h \, d\sigma\right| \leq C(\theta)|h|\|f\|_{L^p(\partial\Omega)}\|g\|_{L^{p'}(\partial\Omega)}. \tag{III.90}$$

3. For any function $f \in W^{l+1,p}(\Omega)$, we have

$$\sup_{0<h<1}\left(\frac{1}{h}\left\|\left[\nabla, \tau_h^\theta\right] f\right\|_{W^{l,p}}\right) \leq C(\theta)\|f\|_{W^{l+1,p}}. \tag{III.91}$$

4. For any function $f \in W^{l+1,p}(\Omega)$, we have

$$|||f|||_{\theta,l+1,p} \leq C(\theta)\|f\|_{W^{l+1,p}}. \tag{III.92}$$

Proof.

1. For $l \geq 0$, the proof of (III.87) is obvious, we simply differentiate the map $f \circ \tau^\theta(h, \cdot)$ a sufficient number of times and then make a change of variables $y = \tau^\theta(h, x)$ for fixed h. We make use of the fact that the Jacobian of the change of variables is continuous with respect to (h, x) and therefore never vanishes. Hence, it is bounded below by some positive constant on $[-1, 1] \times \overline{\Omega}$. We also make use of the fact that the mapping τ^θ is at least continuous with respect to the two variables (h, x) on the compact set $[-1, 1] \times \overline{\Omega}$.
   For $l = -1$, the statement has to be precised, because we need to define $\tau_h^\theta f = f \circ \tau^\theta(h, \cdot)$ when f is only in $W^{-1,p}$ and is therefore not a function. Of course, this is achieved by duality in the case where f is regular (and the conclusion follows by density of smooth functions in $W^{-1,p}(\Omega)$). To this end, for all $g \in W_0^{1,p'}(\Omega)$, we write
$$\begin{aligned}\langle f \circ \tau^\theta(h, \cdot), g\rangle_{W^{-1,p}, W_0^{1,p'}} &= \int_\Omega f(y) g \circ \tau^\theta(-h, y) \operatorname{Jac} \tau^\theta(-h, y)\, dy \\ &= \langle f, g \circ \tau^\theta(-h, \cdot) \operatorname{Jac} \tau^\theta(-h, \cdot)\rangle_{W^{-1,p}, W_0^{1,p'}}, \\ &\leq C\|f\|_{W^{-1,p}} \|g \circ \tau^\theta(-h, \cdot)\|_{W_0^{1,p'}}\end{aligned}$$
   where we used Corollary III.2.41 and the properties of $\operatorname{Jac} \tau^\theta$ that we proved above, in particular the estimate (III.83). The proof of (III.87) in $W^{-1,p}$ therefore follows directly by applying the result for $l = 1$ to the test function g.
   The convergence property (III.88) is obvious for smooth functions (say in $\mathcal{C}_c^\infty(\overline{\Omega})$) by using the properties of the flow proved in Proposition III.3.18. Finally, the uniform estimate (III.87), shows that the general result can be proved by using the density of $\mathcal{C}_c^\infty(\overline{\Omega})$ in $W^{l,p}(\Omega)$.
2. We perform a change of variable $x = \tau^\theta(-h, y)$ (which is equivalent to $y = \tau^\theta(h, x)$) in the following integral,
$$\begin{aligned}\int_\Omega f \tau_h^\theta g\, dx &= \int_\Omega f(x) g \circ \tau^\theta(h, x)\, dx \\ &= \int_\Omega f \circ \tau^\theta(-h, y) g(y) \operatorname{Jac}(\tau^\theta(-h, y))\, dy \\ &= \int_\Omega \tau_{-h}^\theta f g\, dy + \int_\Omega f \circ \tau^\theta(-h, y) g(y) \big(\operatorname{Jac}(\tau^\theta(-h, y)) - 1\big)\, dy,\end{aligned}$$
   and the claim follows by using (III.82), the Hölder inequality, and (III.87). The property (III.90) is proved in the same way by observing that the flow τ^θ maps the boundary into itself, because θ is assumed to be tangent to the boundary.
3. Let us now prove (III.91) for $l = 0$. Let $f \in W^{1,p}(\Omega)$. By the chain rule we have
$$\nabla\big(f \circ \tau^\theta(h, \cdot)\big) = {}^t\big(\nabla \tau^\theta(h, \cdot)\big).(\nabla f) \circ \tau^\theta(h, \cdot).$$

Hence

$$\nabla\big(f\circ\tau^\theta(h,\cdot)\big)-(\nabla f)\circ\tau^\theta(h,\cdot)=\Big({}^t\big(\nabla\tau^\theta(h,\cdot)\big)-\mathrm{Id}\Big)\cdot(\nabla f)\circ\tau^\theta(h,\cdot).$$

We therefore have

$$\begin{aligned}&\|\nabla\big(f\circ\tau^\theta(h,\cdot)\big)-(\nabla f)\circ\tau^\theta(h,\cdot)\|_{L^p}\\ \leq&\|\nabla\tau^\theta(h,\cdot)-\mathrm{Id}\|_{L^\infty}\|(\nabla f)\circ\tau^\theta(h,\cdot)\|_{L^p}\\ \leq&C(\theta)|h|\|\nabla f\|_{L^p},\end{aligned}$$

from (III.82) and (III.87).
The proof in the case where $l\geq 1$ is similar and we leave it as an exercise for the reader.
In the case $l=-1$, we again proceed by duality. Hence, if g belongs to $(W_0^{1,p'}(\Omega))^d$ then, by definition, we have

$$\begin{aligned}&\big\langle\nabla\big(f\circ\tau^\theta(h,\cdot)\big)-(\nabla f)\circ\tau^\theta(h,\cdot),g\big\rangle_{W^{-1,p},W_0^{1,p'}}\\ =&-\int_\Omega f\circ\tau^\theta(h,x)\,\mathrm{div}\,g(x)\,dx-\langle\nabla f,g\circ\tau^\theta(-h,\cdot)\,\mathrm{Jac}\,\tau^\theta(-h,\cdot)\rangle_{W^{-1,p'},W_0^{1,p}}\\ =&-\int_\Omega f(y)(\mathrm{div}\,g)\circ\tau^\theta(-h,y)\ \mathrm{Jac}(\tau^\theta(-h,y))\,dy\\ &+\int_\Omega f(y)\,\mathrm{div}\big(\,\mathrm{Jac}\,\tau^\theta(-h,y)g(\tau^\theta(-h,y))\big)\,dy.\end{aligned}$$

Hence, we have

$$\begin{aligned}&\big\langle\nabla\big(f\circ\tau^\theta(h,\cdot)\big)-(\nabla f)\circ\tau^\theta(h,\cdot),g\big\rangle_{W^{-1,p},W_0^{1,p'}}\\ \leq&\|f\|_{L^p}\big\|\,\mathrm{Jac}(\tau^\theta(-h,\cdot))(\mathrm{div}\,g)\circ\tau^\theta(-h,\cdot)\\ &-\mathrm{div}\,\big(\,\mathrm{Jac}\,\tau^\theta(-h,\cdot)g\circ\tau^\theta(-h,\cdot)\big)\big\|_{L^{p'}}.\end{aligned}$$

We now need to set an upper bound for this last quantity as a function of the $W^{1,p'}$ norm of g. To do this, we write

$$\begin{aligned}&\|\,\mathrm{Jac}(\tau^\theta(-h,\cdot))(\mathrm{div}\,g)\circ\tau^\theta(-h,\cdot)-\mathrm{div}\big(\,\mathrm{Jac}\,\tau^\theta(-h,\cdot)g\circ\tau^\theta(-h,\cdot)\big)\|_{L^{p'}}\\ \leq&\|\,\mathrm{Jac}(\tau^\theta(-h,\cdot))(\mathrm{div}\,g)\circ\tau^\theta(-h,\cdot)-\mathrm{Jac}\,\tau^\theta(-h,\cdot)\,\mathrm{div}\big(g\circ\tau^\theta(-h,\cdot)\big)\|_{L^{p'}}\\ &+\|\big(\nabla\,\mathrm{Jac}\,\tau^\theta(-h,\cdot)\big)\cdot g\circ\tau^\theta(-h,\cdot)\|_{L^{p'}}\\ \leq&\|\,\mathrm{Jac}(\tau^\theta(-h,\cdot))\|_{L^\infty}\|(\mathrm{div}\,g)\circ\tau^\theta(-h,\cdot)-\mathrm{div}\big(g\circ\tau^\theta(-h,\cdot)\big)\|_{L^{p'}}\\ &+\|\rho\circ\tau^\theta(-h,\cdot)\nabla\,\mathrm{Jac}\,\tau^\theta(-h,\cdot)\|_{L^\infty}\underbrace{\left\|\frac{g\circ\tau^\theta(-h,\cdot)}{\rho\circ\tau^\theta(-h,\cdot)}\right\|_{L^{p'}}}_{\leq C\|\frac{g}{\rho}\|_{L^{p'}}}\\ \leq&C(\theta)|h|\|g\|_{W^{1,p'}}+C(\theta)|h|\|g\|_{W^{1,p'}},\end{aligned}$$

by using (III.83), (III.87), and (III.91) for the function g with $l = 1$ and the Hardy inequality (Proposition III.2.40).

4. Let us now first show (III.92) for $p = 0$. To achieve this, we consider a sufficiently regular function f and the result follows by a density argument. Let us use the Taylor formula for all $x \in \Omega$,

$$f(\tau^\theta(h,x)) - f(x) = h \int_0^1 \left\langle \nabla f(\tau^\theta(hu,x)), \theta(\tau^\theta(hu,x)) \right\rangle du. \tag{III.93}$$

Hence, since θ is bounded, we have

$$|f(\tau^\theta(h,x)) - f(x)|^p \le C_1(\theta) h^p \int_0^1 |\nabla f(\tau^\theta(hu,x))|^p \, du.$$

By integrating for $x \in \Omega$ and then using the Fubini theorem and a change of variables, we get

$$\begin{aligned} \|\delta_h^\theta f\|_{L^p}^p &\le C_1(\theta) h^p \int_0^1 \int_\Omega |\nabla f(\tau^\theta(hu,x))|^p \, dx du \\ &= C_1(\theta) h^p \int_0^1 \int_\Omega |\nabla f(y)|^p \ \operatorname{Jac} \tau^\theta(-hu, y) \, dy \, du \\ &\le C(\theta) h^p \int_\Omega |\nabla f(y)|^p \, dy, \end{aligned}$$

which proves the claim.

We now show the result for $l \ge 1$ by induction. Hence, we have

$$\begin{aligned} \|\delta_h^\theta f\|_{W^{l,p}} &\le \|\delta_h^\theta f\|_{W^{l-1,p}} + \|\nabla(\delta_h^\theta f)\|_{W^{l-1,p}} \\ &\le \|\delta_h^\theta f\|_{W^{l-1,p}} + \|\delta_h^\theta \nabla f\|_{W^{l-1,p}} + \| \left[\nabla, \tau_h^\theta\right] f\|_{W^{l-1,p}}. \end{aligned}$$

We then apply the induction hypothesis to f and to ∇f as well as (III.91) to obtain

$$\|\delta_h^\theta f\|_{W^{l,p}} \le C(\theta)|h|(\|f\|_{W^{l,p}} + \|\nabla f\|_{W^{l,p}}) \le C(\theta)|h|\|f\|_{W^{l+1,p}}.$$

Finally, the result for $l = -1$ is obtained in the now usual manner of an argument from duality, the details of which we leave to the reader.

□

3.5.3 Characterisation through translation operators

We can now give a characterisation of tangential Sobolev spaces in terms of translation operations (see [108]).

Theorem III.3.20. *Let Ω be a domain of class $\mathcal{C}^{k+1,1}$, $k \geq 0$, with compact boundary and $1 < p < +\infty$, then for all $0 \leq l \leq k+1$, we have the following characterisation of the space $W^{l+1,p}_{tang}(\Omega)$:*

$$W^{l+1,p}_{tang}(\Omega) = \left\{ f \in W^{l,p}(\Omega),\ |||f|||_{\theta,l+1,p} < +\infty,\ \textit{for all } \theta \textit{ satisfying } (\text{III.77}) \right\}.$$

Proof.

- Let $f \in W^{l,p}(\Omega)$ and assume that $|||f|||_{\theta,l+1,p} < +\infty$ for any θ satisfying (III.77).
  - We first show that $f \in W^{l+1,p}_{loc}(\Omega)$ by using Proposition III.2.3. Let $\varphi \in \mathcal{D}(\Omega)$; we want to show that $\overline{\varphi f} \in W^{l+1,p}(\mathbb{R}^d)$. Let ω be an open set such that $\operatorname{Supp}\varphi \subset \omega \subset \overline{\omega} \subset \Omega$, and $\psi \in \mathcal{D}(\Omega)$ such that $\psi = 1$ on ω and $\|\psi\|_{L^\infty} \leq 1$.
    We define $h_0 = d(\operatorname{Supp}\varphi, \omega^c) > 0$ and we consider the vector field $\theta = \psi e_i$, for some $1 \leq i \leq d$. It is clear that θ (which vanishes near the boundary of Ω) satisfies (III.77). By assumption, we deduce that $|||f|||_{\theta,l+1,p} < +\infty$.
    By construction of θ, we can easily see that
    $$\tau^\theta(h,x) = x + he_i, \forall x \in \operatorname{Supp}(\varphi), \forall |h| < h_0. \tag{III.94}$$
    Let now $x \in \mathbb{R}^d$ and $0 < h < h_0$; we show that
    $$(\overline{f\varphi})(\tau^\theta(h,x)) = (\overline{f\varphi})(x + he_i). \tag{III.95}$$
    - In the case where $\tau^\theta(h,x) \notin \operatorname{Supp}(\varphi)$ and $x + he_i \notin \operatorname{Supp}(\varphi)$, Formula (III.95) certainly holds inasmuch as both terms are zero.
    - Assume that $x + he_i \in \operatorname{Supp}(\varphi)$. By using (III.94), we have
      $$\tau^\theta(-h, x + he_i) = x + he_i - he_i = x,$$
      which means that $\tau^\theta(h,x) = x + he_i$. This proves (III.95).
    - In the same way, if $\tau^\theta(h,x) \in \operatorname{Supp}(\varphi)$, then we can use (III.94) to find
      $$x = \tau^\theta(-h, \tau^\theta(h,x)) = \tau^\theta(h,x) - he_i,$$
      that is, $\tau^\theta(h,x) = x + he_i$, and the conclusion follows.

    Using (III.95), we deduce that
    $$\delta_{he_i}(\overline{f\varphi}) = \overline{\delta^\theta_h(f\varphi)} = \overline{(\delta^\theta_h f)(\tau^\theta_h \varphi) + f(\delta^\theta_h \varphi)}.$$
    It follows that

$$\sup_{0<h<h_0} \frac{1}{h} \left\|\delta_{he_i}(\overline{f\varphi})\right\|_{W^{l,p}(\mathbb{R}^d)} \leq \|\|f\|\|_{\theta,l+1,p}\|\varphi\|_{W^{l,\infty}(\Omega)} + C_\theta \|f\|_{W^{l,p}(\Omega)}\|\varphi\|_{W^{l+1,\infty}(\Omega)} < +\infty.$$

Note that a similar estimate holds for $h_0 \leq h < 1$, by using that $1/h \leq 1/h_0$.
This being true for any $i \in \{1, \ldots, d\}$, Theorem III.2.32 gives that $\overline{f\varphi} \in W^{l+1,p}(\mathbb{R}^d)$ and the claim is proved.

– Let us show now that $f \in W^{l+1,p}_{tang}(\Omega)$.
  · Let θ satisfy (III.77). We set

$$G = \langle \nabla f, \theta \rangle = \mathrm{div}(f\theta) - f\,\mathrm{div}(\theta) \in W^{l-1,p}(\Omega) \cap W^{l,p}_{loc}(\Omega),$$

and from (III.93) we write

$$\left\| \frac{\delta_h^\theta f}{h} - G \right\|_{W^{l-1,p}} \leq \int_0^1 \|G \circ \tau^\theta(sh, .) - G\|_{W^{l-1,p}}\, ds \leq \sup_{0\leq s\leq h} \|G \circ \tau^\theta(s, .) - G\|_{W^{l-1,p}} \xrightarrow[h\to 0]{} 0,$$

this last convergence being given in (III.88). We deduce that G is the limit in $W^{l-1,p}(\Omega)$ of $(\delta_h^\theta f)/h$.
By assumption, this family is known to be bounded in $W^{l,p}(\Omega)$ and thus, up to a subsequence, it has a weak limit in the space $W^{l,p}(\Omega)$. By uniqueness of the limit, we deduce that $\langle \nabla f, \theta \rangle = G$ necessarily belongs to $W^{l,p}(\Omega)$.
  · For any given $1 \leq i \leq d$ we choose $\theta = e_i - \langle e_i, \nu \rangle \nu$. By construction, and using the properties of the regularised distance, we see that θ exactly satisfies (III.77). For this particular choice of θ we observe that

$$\langle \nabla f, \theta \rangle = \langle \nabla f, e_i \rangle - \langle e_i, \nu \rangle \langle \nabla f, \nu \rangle = \langle \nabla_T f, e_i \rangle.$$

We have just proved that this quantity belongs to $W^{l,p}(\Omega)$. This is true for any $1 \leq i \leq d$, therefore we conclude that $\nabla_T f \in (W^{l,p}(\Omega))^d$.
  · We choose now $\theta = \rho\nu$. Using the properties of ρ and ν (in particular the fact that ρ vanishes on the boundary), we immediately obtain that θ satisfies the assumptions (III.77).
We conclude that $\langle \nabla f, \theta \rangle \in W^{l,p}(\Omega)$. This exactly means that $\rho(\nabla_N f) = \rho \langle \nabla f, \nu \rangle \in W^{l,p}(\Omega)$.

• Assume that $f \in W^{l+1,p}_{tang}(\Omega)$ and let θ be a vector field satisfying (III.77).
  – Let us first show that $\langle \nabla f, \theta \rangle$ belongs to $W^{l,p}(\Omega)$. To this end, we write

$$\langle \nabla f, \theta \rangle = \langle \nabla_T f, \theta \rangle + (\rho \nabla_N f) \frac{\langle \theta, \nu \rangle}{\rho}.$$

By assumption, $\nabla_T f \in W^{l,p}(\Omega)$ and all the derivatives of θ up to order $k+1$ are bounded, therefore the first term actually belongs to $W^{l,p}(\Omega)$. It remains to study the second term. In the case $l = 0$, using (III.80), it is clear that this term belongs to $L^p(\Omega)$.
In the case $l \geq 1$, we compute

$$\begin{aligned}&\nabla\left((\rho\nabla_N f)\frac{\langle\theta,\nu\rangle}{\rho}\right)\\ =&\nabla(\rho\nabla_N f)\frac{\langle\theta,\nu\rangle}{\rho} + (\nabla_N f)\nabla(\langle\theta,\nu\rangle) + (\nabla_N f)\frac{\langle\theta,\nu\rangle}{\rho}\nabla\rho.\end{aligned}$$

Since $f \in W^{l,p}(\Omega)$, we know that $\nabla_N f \in W^{l-1,p}(\Omega)$ and we also have, by assumption, that $\nabla(\rho\nabla_N f) \in W^{l-1,p}(\Omega))^d$. Finally, using (III.79) and (III.80), we conclude that

$$\nabla\left((\rho\nabla_N f)\frac{\theta\cdot\nu}{\rho}\right) \in (W^{l-1,p}(\Omega))^d.$$

This concludes the proof of the fact that $\langle\nabla f, \theta\rangle$ belongs to $W^{l,p}(\Omega)$ and we obtain the estimate on $|||f|||_{\theta,l+1,p}$ by using (III.93) and (III.87).

□

3.6 Differential operators in tangential/normal coordinates

Let $w : \Gamma_s \to \mathbb{R}^d$ be a smooth vector field. We define the divergence operator on the manifold Γ_s by the formula

$$\int_{\Gamma_s} (\operatorname{div}_{\Gamma_s} w)v\, d\sigma = -\int_{\Gamma_s} \langle w, \nabla_T v\rangle\, d\sigma,$$

which is required to hold for any smooth test function $v : \mathbb{R}^d \to \mathbb{R}$. If (U, φ_s) is any chart on Γ_s and v is supported in $\varphi_s(U)$ this formula reads

$$\begin{aligned}&\int_U v(\varphi_s(u))(\operatorname{div}_{\Gamma_s} w)(\varphi_s(u))\sqrt{\det \mathfrak{g}_s(u)}\, du\\ &= -\int_U \langle w(\varphi_s(u)), (\nabla_T v)(\varphi_s(u))\rangle\sqrt{\det \mathfrak{g}_s(u)}\, du.\end{aligned}$$

Writing $w(\varphi_s(u)) = \sum_{i=1}^{d-1} w_i(u)(\partial\varphi_s/\partial u_i)(u)$ the above integral is written

$$
\begin{aligned}
&-\int_U \sum_{i=1}^{d-1} w_i(u)\left\langle \frac{\partial \varphi_s}{\partial u_i}(u), (\nabla_T v)(\varphi_s(u))\right\rangle \sqrt{\det \mathfrak{g}_s(u)}\, du\\
&= -\int_U \sum_{i=1}^{d-1} w_i(u) \frac{\partial v\circ \varphi_s}{\partial u_i}(u) \sqrt{\det \mathfrak{g}_s(u)}\, du\\
&= \int_U \sum_{i=1}^{d-1} v\circ\varphi_s(u) \frac{\partial}{\partial u_i}\left(w_i(u)\sqrt{\det \mathfrak{g}_s(u)}\right) du\\
&= \int_U v\circ\varphi_s(u)\left(\frac{1}{\sqrt{\det \mathfrak{g}_s(u)}} \sum_{i=1}^{d-1} \frac{\partial}{\partial u_i}\left(w_i(u)\sqrt{\det \mathfrak{g}_s(u)}\right)\right) \sqrt{\det \mathfrak{g}_s(u)}\, du.
\end{aligned}
$$

By using Proposition III.3.3 we get

$$\left(\left\langle w(\varphi_s(u)), \frac{\partial \varphi_s}{\partial u_j}\right\rangle\right)_j = \mathfrak{g}_s(u).(w_i(u))_i,$$

so that

$$(w_i(u))_i = \mathfrak{g}_s^{-1}(u).\left(\left\langle w(\varphi_s(u)), \frac{\partial \varphi_s}{\partial u_j}\right\rangle\right)_j .$$

Putting all together we get

$$
\begin{aligned}
&(\operatorname{div}_{\Gamma_s} w)(\varphi_s(u))\\
=&\frac{1}{\sqrt{\det \mathfrak{g}_s(u)}} \sum_{i=1}^{d-1}\sum_{j=1}^{d-1} \frac{\partial}{\partial u_i}\left(\mathfrak{g}_s^{ij}(u)\left\langle w(\varphi_s(u)), \frac{\partial \varphi_s}{\partial u_j}\right\rangle \sqrt{\det \mathfrak{g}_s(u)}\right).
\end{aligned}
\tag{III.96}
$$

Definition III.3.21. *The tangential divergence operator is defined as*

$$(\operatorname{div}_T w)(x) = (\operatorname{div}_{\Gamma_{\rho(x)}} w)(x),\ \forall x \in \mathcal{O}_{s_0}^\rho,$$

for any smooth enough vector field $w : \mathcal{O}_{s_0}^\rho \to \mathbb{R}^d$.

The above expression can be written in a more compact form which, in particular, clearly shows that $\operatorname{div}_T w$ only contains tangential derivatives of w.

Proposition III.3.22. *There exists a vector field $\mathcal{G} : \mathcal{O}_{s_0}^\rho \to \mathbb{R}^d$ such that for any $w : \mathcal{O}_{s_0}^\rho \to \mathbb{R}^d$ we have*

$$\operatorname{div}_T w = \operatorname{Tr}(\nabla_T w) + \langle w, \mathcal{G}\rangle, \text{ in } \mathcal{O}_{s_0}^\rho, \tag{III.97}$$

and also

$$\operatorname{div}_T w = \operatorname{Tr}(\nabla_T w_T) + \langle w_T, \mathcal{G}\rangle, \text{ in } \mathcal{O}_{s_0}^\rho \tag{III.98}$$

Moreover, $\mathcal{G}$ is smooth in $\mathcal{O}_{s_0}^\rho$, and for any $\alpha \in \mathbb{N}^d$ we have

$$\partial^\alpha \mathcal{G} \in (L^\infty(\mathcal{O}_{s_0}^\rho))^d, \textit{ for } |\alpha| \leq k-1,$$
$$\frac{1}{|\ln \rho|}\partial^\alpha \mathcal{G} \in (L^\infty(\mathcal{O}_{s_0}^\rho))^d, \textit{ for } |\alpha| = k,$$
$$\rho \nabla_N \partial^\alpha \mathcal{G} \in (L^\infty(\mathcal{O}_{s_0}^\rho))^d, \textit{ for } |\alpha| = k.$$

Proof.

Let $s \in [0, s_0]$ fixed and let us consider the case where $w_{|\Gamma_s}$ is supported in the image of one chart (U, φ_s), the general case being deduced by using, as usual, a suitable partition of the unity. In that case, we write (III.96) under the form

$$(\mathrm{div}_{\Gamma_s} w)(\varphi_s(u)) = \sum_{i=1}^{d-1}\sum_{j=1}^{d-1}\left(\mathfrak{g}_s^{ij}(u)\left\langle \frac{\partial w(\varphi_s(u))}{\partial u_i}, \frac{\partial \varphi_s}{\partial u_j}\right\rangle\right)$$
$$+\left\langle w(\varphi_s(u)), \frac{1}{\sqrt{\det \mathfrak{g}_s(u)}}\sum_{i=1}^{d-1}\sum_{j=1}^{d-1}\frac{\partial}{\partial u_i}\left(\mathfrak{g}_s^{ij}(u)\frac{\partial \varphi_s}{\partial u_j}\sqrt{\det \mathfrak{g}_s(u)}\right)\right\rangle. \tag{III.99}$$

The second term is of the form $\langle w, \mathcal{G}\rangle$ for a suitable definition of $\mathcal{G}$. It remains to prove that the first term above is actually the trace of the tangential part of the gradient of w. Consider the basis of $\mathbb{R}^d$ defined by

$$\frac{\partial \varphi_s}{\partial u_1}, \dots, \frac{\partial \varphi_s}{\partial u_{d-1}}, \nu,$$

observe that

$$\nabla_T w(\varphi_s(u)).\nu = 0,$$

and let us write

$$\nabla_T w(\varphi_s(u)).\frac{\partial \varphi_s}{\partial u_i} = \alpha_i \nu + \sum_{k=1}^{d-1}\alpha_{ik}\frac{\partial \varphi_s}{\partial u_k}.$$

It follows that the trace of $\nabla_T w(\varphi_s(u))$ is nothing but the sum $\sum_{j=1}^{d-1}\alpha_{ii}$. Observe now that, with this notation, the first term in (III.99) is written

$$\sum_{i=1}^{d-1}\sum_{j=1}^{d-1}\sum_{k=1}^{d-1}\alpha_{ik}\left(\mathfrak{g}_s^{ij}(u)\left\langle \frac{\partial \varphi_s}{\partial u_k}, \frac{\partial \varphi_s}{\partial u_j}\right\rangle\right)$$
$$= \sum_{i=1}^{d-1}\sum_{k=1}^{d-1}\alpha_{ik}\underbrace{\sum_{j=1}^{d-1}\left(\mathfrak{g}_s^{ij}(u)\mathfrak{g}_{s,jk}(u)\right)}_{=\delta_{ik}} = \sum_{i=1}^{d-1}\alpha_{ii},$$

and (III.97) is proved. Formula (III.98) comes from the observation that only the tangential part of w enters the definition of div_T. Therefore, we have $\operatorname{div}_T w = \operatorname{div}_T w_T$ and we can use (III.97) with w_T in place of w.

It remains to observe that $\mathcal{G}$ contains second derivatives of the map ψ to deduce its regularity properties from the ones of ψ.

□

We have defined above the tangential divergence of a vector field which only involves derivatives of the vector field in directions parallel to the boundary. We now want to express the full divergence operator by separating normal and tangential components and derivatives.

Proposition III.3.23. *Let $w : \mathcal{O}^\rho_{s_0} \to \mathbb{R}^d$ be a smooth vector field. The following formulas hold.*

$$\operatorname{div} w = \operatorname{div} w_T + \nabla_N w_N + \frac{\nabla_N \mathcal{J}}{\mathcal{J}} w_N,$$
$$\operatorname{div} w = \frac{1}{|\tilde{\nu}|} \operatorname{Tr}(\nabla_T(|\tilde{\nu}| w)) + \nabla_N w_N + \langle w, \mathcal{G}\rangle + \frac{\nabla_N \mathcal{J}}{\mathcal{J}} w_N,$$
$$\operatorname{div} w = \operatorname{Tr}(\nabla_T w) + \nabla_N w_N + \left\langle w, \tilde{\mathcal{G}} \right\rangle,$$

where

$$\tilde{\mathcal{G}} = \mathcal{G} + \frac{\nabla_T |\tilde{\nu}|}{|\tilde{\nu}|} + \frac{\nabla_N \mathcal{J}}{\mathcal{J}} \nu$$

is a vector field which enjoys the same regularity properties as $\mathcal{G}$.

Proof.

For any test function $v \in \mathcal{D}(\mathcal{O}^\rho_{s_0})$ we write

$$\int_{\mathcal{O}^\rho_{s_0}} \langle w(x), \nabla v(x)\rangle\, dx = \int_{\mathcal{O}^\rho_{s_0}} \langle w_T(x), \nabla_T v(x)\rangle\, dx + \int_{\mathcal{O}^\rho_{s_0}} w_N(x) \nabla_N v(x)\, dx.$$

Let us study separately the two terms T_1 and T_2 in the right-hand side.

- We use Proposition III.3.12 to express T_1 as follows

$$\begin{aligned} T_1 &= \int_0^{s_0} \left(\int_{\Gamma_s} |\tilde{\nu}| \langle w_T, \nabla_T v\rangle \, d\sigma \right) ds \\ &= -\int_0^{s_0} \left(\int_{\Gamma_s} \operatorname{div}_{\Gamma_s}(|\tilde{\nu}| w_T) v \, d\sigma \right) ds \\ &= -\int_{\mathcal{O}^\rho_{s_0}} \frac{1}{|\tilde{\nu}|} \operatorname{div}_T(|\tilde{\nu}| w_T) v \, dx. \end{aligned}$$

- Similarly we use Propositions III.3.10 and III.3.12 to write

$$
\begin{aligned}
T_2 &= \int_0^{s_0} \left(\int_{\Gamma_s} \underbrace{w_N |\tilde{\nu}| \langle \nabla v, \nu \rangle}_{= w_N \langle \nabla v, \tilde{\nu} \rangle} \, d\sigma \right) ds \\
&= - \int_0^{s_0} \int_{\Gamma_{s_0}} w_N \circ \psi(\sigma, s) \frac{\partial v \circ \psi(\sigma, s)}{\partial s} \mathcal{J}_s(\sigma) \, d\sigma \, ds \\
&= \int_0^{s_0} \int_{\Gamma_{s_0}} \frac{\partial}{\partial s} \left(w_N \circ \psi(\sigma, s) \mathcal{J} \circ \psi(\sigma, s) \right) v \circ \psi(\sigma, s) \, d\sigma \, ds \\
&= - \int_0^{s_0} \int_{\Gamma_{s_0}} |\tilde{\nu} \circ \psi(\sigma, s)| \nabla_N (w_N \mathcal{J}) \circ \psi v \circ \psi(\sigma, s) \, d\sigma \, ds \\
&= - \int_{\mathcal{O}_{s_0}^{\rho}} \frac{1}{\mathcal{J}} \nabla_N (w_N \mathcal{J}) v \, dx.
\end{aligned}
$$

Gathering the two terms T_1 and T_2, we finally find that

$$
\operatorname{div}(w) = \frac{1}{|\tilde{\nu}|} \operatorname{div}_T (|\tilde{\nu}| w_T) + \frac{1}{\mathcal{J}} \nabla_N (w_N \mathcal{J});
$$

that is,

$$
\operatorname{div}(w) = \frac{1}{|\tilde{\nu}|} \operatorname{div}_T (|\tilde{\nu}| w_T) + \nabla_N w_N + \frac{\nabla_N \mathcal{J}}{\mathcal{J}} w_N.
$$

Moreover, since $(w_T)_N = 0$ and $(w_T)_T = w_T$, the above formula gives

$$
\operatorname{div}(w_T) = \frac{1}{|\tilde{\nu}|} \operatorname{div}_T (|\tilde{\nu}| w_T),
$$

so that we finally deduce that

$$
\operatorname{div}(w) = \operatorname{div}(w_T) + \nabla_N w_N + \frac{\nabla_N \mathcal{J}}{\mathcal{J}} w_N.
$$

The other formulas come directly from (III.97).

□

We have already seen that by definition the gradient operator is written

$$
\nabla v = \nabla_T v + (\nabla_N v) \nu.
$$

Therefore, we can derive the same kind of formulas for the Laplace operator because, by definition, we have $\Delta v = \operatorname{div}(\nabla v)$.

Proposition III.3.24. *For any $v : \mathcal{O}_{s_0}^{\rho} \to \mathbb{R}$ sufficiently smooth, the following formulas hold*

$$
\Delta v = \operatorname{div}(\nabla_T v) + \nabla_N (\nabla_N v) + \frac{\nabla_N \mathcal{J}}{\mathcal{J}} \nabla_N v,
$$

and

$$\Delta v = \mathrm{Tr}(\nabla_T \nabla v) + \nabla_N \nabla_N v + \left\langle \nabla v, \tilde{\mathcal{G}} \right\rangle .$$

4 The Laplace problem

We prove in this section the main results concerning the Laplace problem. We consider first the Dirichlet boundary conditions, then the Neumann boundary conditions.

The existence and uniqueness come from the Lax–Milgram theorem and the trace lifting results. The regularity properties can be deduced from the general theory for elliptic systems of Agmon, Douglis, and Nirenberg; see [2, 3]. Nevertheless, it appears to us to be more instructive to provide a complete self-contained proof here. It is based on the Nirenberg translations method, for which we have introduced a suitable framework in the previous sections.

The same approach is used for proving regularity results for Stokes problems in the next chapter.

4.1 Dirichlet boundary conditions

Theorem III.4.1. *Let Ω be a connected, bounded, Lipschitz domain of $\mathbb{R}^d$. Let $f \in H^{-1}(\Omega)$ and let $u_b \in H^{1/2}(\partial\Omega)$. There exists a unique solution $u \in H^1(\Omega)$ of*

$$\begin{cases} -\Delta u = f, & \text{in } \Omega, \\ \quad u = u_b, & \text{on } \partial\Omega. \end{cases} \tag{III.100}$$

It satisfies

$$\|u\|_{H^1} \leq C(\|f\|_{H^{-1}} + \|u_b\|_{H^{1/2}}),$$

for some $C > 0$ depending only on Ω.

Proof.

- We first use the lifting operator given by Theorem III.2.22 and observe that $u \in H^1(\Omega)$ is a solution of (III.100) if and only if $\tilde{u} = u - R_0 u_b \in H^1_0(\Omega)$ is a solution to the homogeneous problem

$$\begin{cases} -\Delta \tilde{u} = f + \Delta(R_0 u_b), & \text{in } \Omega, \\ \quad \tilde{u} = 0, & \text{on } \partial\Omega, \end{cases}$$

  with $\tilde{f} = f + \Delta(R_0 u_b) \in H^{-1}(\Omega)$. Since

$$\|\Delta(R_0 u_b)\|_{H^{-1}} \leq \|R_0 u_b\|_{H^1} \leq C\|u_b\|_{H^{1/2}},$$

we have that $\|\tilde{f}\|_{H^{-1}} \leq \|f\|_{H^{-1}} + \|u_b\|_{H^{1/2}}$. Moreover, if $\tilde{u}$ exists, we have

$$\|u\|_{H^1} \leq \|\tilde{u}\|_{H^1} + \|R_0 u_b\|_{H^1} \leq \|\tilde{u}\|_{H^1} + C\|u_b\|_{H^{1/2}}.$$

As a consequence of this analysis we see that it is enough to consider the case of the homogeneous boundary data $u_b = 0$.

- In the case $u_b = 0$, we look for the solution u in the Hilbert space $H_0^1(\Omega)$. Let us define the bilinear form

$$a(u,v) = \int_\Omega \nabla u \cdot \nabla v \, dx, \; \forall u, v \in H_0^1(\Omega).$$

By the Cauchy–Schwarz inequality it is clear that a is continuous. Moreover, by using the Poincaré inequality (Theorem III.2.38), we know that a is coercive in $H_0^1(\Omega)$. We can therefore use the Lax–Milgram theorem (Theorem II.2.5) and deduce that there is a unique solution $u \in H_0^1(\Omega)$ to the variational formulation

$$\int_\Omega \nabla u \cdot \nabla v \, dx = \langle f, v \rangle_{H^{-1}, H_0^1}, \; \forall v \in H_0^1(\Omega).$$

Since $\mathcal{D}(\Omega) \subset H_0^1(\Omega)$, this formula implies that u solves the equation $-\Delta u = f$ in the distribution sense, the boundary condition being satisfied by definition of the space $H_0^1(\Omega)$ (and the characterisation given in Theorem III.2.46). The estimate of $\|u\|_{H^1}$ follows from (II.3).

□

Theorem III.4.2. *We use the same notation as in Theorem III.4.1. Assume moreover that, for some $k \geq 0$, Ω is of class $\mathcal{C}^{k+1,1}$, $f \in H^k(\Omega)$ and $u_b \in H^{k+3/2}(\partial\Omega)$ then the solution u of (III.100) belongs to $H^{k+2}(\Omega)$ and we have*

$$\|u\|_{H^{k+2}} \leq C \left(\|f\|_{H^k} + \|u_b\|_{H^{k+3/2}} \right),$$

for some $C > 0$ depending only on Ω.

Proof.

We only consider here the case $k = 0$ because the proof of the general case follows the same lines.

We assume now that Ω is of class $\mathcal{C}^{1,1}$, $f \in L^2(\Omega)$, and $u_b \in H^{3/2}(\partial\Omega)$. The assumptions on Ω and u_b show that $R_0 u_b \in H^2(\Omega)$ and that $\|R_0 u_b\|_{H^2} \leq C\|u_b\|_{H^{3/2}}$, see Section 2.5.3. Therefore, by using the same argument as in the previous proof, it is enough to consider the homogeneous case $u_b = 0$.

- Let us first show that $u \in H^2_{loc}(\Omega)$. In order to use Proposition III.2.3, we consider a function $\varphi \in \mathcal{D}(\Omega)$ and we want to show that $\overline{u\varphi} \in H^2(\mathbb{R}^d)$. Let us compute $-\Delta(\overline{u\varphi})$ in $\mathcal{D}'(\mathbb{R}^d)$. To this end, we take a test function $\psi \in \mathcal{D}(\mathbb{R}^d)$ and using that

$$\varphi\Delta\psi = \Delta(\varphi\psi) - \psi\Delta\varphi - 2\nabla\varphi\cdot\nabla\psi,$$

and that $\varphi\psi \in \mathcal{D}(\Omega)$, we can write

$$\begin{aligned}
\langle -\Delta(\overline{u\varphi}), \psi\rangle_{\mathcal{D}'(\mathbb{R}^d),\mathcal{D}(\mathbb{R}^d)} &= \langle \overline{u\varphi}, -\Delta\psi\rangle_{\mathcal{D}'(\mathbb{R}^d),\mathcal{D}(\mathbb{R}^d)} \\
&= \int_{\mathbb{R}^d} \overline{u\varphi}(-\Delta\psi)\,dx = \int_\Omega u\varphi(-\Delta\psi)\,dx \\
&= -\int_\Omega u\Delta(\varphi\psi)\,dx + 2\int_\Omega u(\nabla\varphi\cdot\nabla\psi)\,dx + \int_\Omega u(\Delta\varphi)\psi\,dx \\
&= \langle u, -\Delta(\varphi\psi)\rangle_{\mathcal{D}'(\Omega),\mathcal{D}(\Omega)} - 2\int_\Omega (\nabla u\cdot\nabla\varphi)\psi\,dx - \int_\Omega u(\Delta\varphi)\psi\,dx \\
&= \langle -\Delta u, \varphi\psi\rangle_{\mathcal{D}'(\Omega),\mathcal{D}(\Omega)} - 2\int_{R^d} \overline{(\nabla u\cdot\nabla\varphi)}\psi\,dx - \int_{\mathbb{R}^d} \overline{u(\Delta\varphi)}\psi\,dx \\
&= \int_{\mathbb{R}^d}\Big(\overline{f}\varphi - 2\overline{(\nabla u\cdot\nabla\varphi)} - \overline{u(\Delta\varphi)}\Big)\,\psi\,dx.
\end{aligned}$$

This proves that, in $\mathbb{R}^d$, the function $v = \overline{u\varphi}$, which belongs to $H^1(\mathbb{R}^d)$, solves the equation

$$-\Delta v = \underbrace{\overline{f}\varphi - 2\overline{(\nabla u\cdot\nabla\varphi)} - \overline{u(\Delta\varphi)}}_{=F},$$

where F belongs to $L^2(\mathbb{R}^d)$ and satisfies $\|F\|_{L^2(\mathbb{R}^d)} \leq C_\varphi(\|f\|_{L^2(\Omega)} + \|u\|_{H^1(\Omega)})$.
Let $h \in \mathbb{R}^d$. Using the notation introduced in Section 2.7, we see that

$$-\Delta(\delta_h v) = \delta_h F.$$

Note that $\delta_h v$ is compactly supported, so that we can take $\delta_h v$ as a test function in this equation:

$$\begin{aligned}
\|\nabla\delta_h v\|_{L^2}^2 &= \langle -\Delta(\delta_h v), \delta_h v\rangle_{H^{-1}(\mathbb{R}^d),H^1(\mathbb{R}^d)} \\
&= \langle \delta_h F, \delta_h v\rangle_{H^{-1},H^1} \leq \|\delta_h F\|_{H^{-1}}\|\delta_h v\|_{H^1}.
\end{aligned}$$

Using Lemma III.2.31, we get

$$\|\nabla\delta_h v\|_{L^2}^2 \leq |h|\|F\|_{L^2}(|h|\|v\|_{H^1} + \|\delta_h v\|_{H^1}),$$

and then, by the Young inequality, we get

$$\|\nabla\delta_h v\|_{L^2}^2 \leq C|h|^2(\|F\|_{L^2}^2 + \|v\|_{L^2}^2).$$

This being true for any h, Proposition III.2.32 gives that $\nabla v \in (H^1(\mathbb{R}^d))^d$, that is $v \in H^2(\mathbb{R}^d)$, as well as the claimed estimate. The proof is complete.

- We now prove that $u \in H^2_{tang}(\Omega)$. We recall that we consider only the case $u_b = 0$ so that $u \in H^1_0(\Omega)$ satisfies

$$\int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx, \ \forall v \in H^1_0(\Omega). \tag{III.101}$$

Let θ be a vector field satisfying (III.77). Since θ is tangent, the flow associated with θ preserves the boundary of Ω. As a consequence, the function $v = \delta^\theta_{-h}\delta^\theta_h u$ also belongs to $H^1_0(\Omega)$ and can be used as a test function in (III.101). Using the notation introduced in Section 3.5.2, we obtain

$$\int_\Omega \nabla(\delta^\theta_h u) \cdot \nabla(\delta^\theta_h u) \, dx = \int_\Omega (\delta^\theta_h f)(\delta^\theta_h u) \, dx - \int_\Omega \{f, \delta^\theta_h u\}_h \, dx \\ + \int_\Omega \{\nabla u, \nabla \delta^\theta_h u\}_h \, dx + \int_\Omega \left(\left[\nabla, \tau^\theta_h\right] u \right) \cdot \nabla(\delta^\theta_h u) \, dx.$$

Using Proposition III.3.19, we get

$$\|\nabla \delta^\theta_h u\|^2_{L^2} \leq \underbrace{\|\delta^\theta_h f\|_{H^{-1}}}_{\leq C|h|\|f\|_{L^2}} \|\delta^\theta_h u\|_{H^1_0} + C|h|\|f\|_{L^2}\|\delta^\theta_h u\|_{L^2} \\ + C|h|\|\nabla u\|_{L^2}\|\nabla \delta^\theta_h u\|_{L^2}.$$

By using the Poincaré and Young inequalities, we deduce

$$\|\delta^\theta_h u\|^2_{H^1_0} \leq C|h|^2(\|f\|^2_{L^2} + \|\nabla u\|^2_{L^2}).$$

It follows that $|||u|||_{\theta,2,2} < +\infty$ for any θ. Using Theorem III.3.20, we get $u \in H^2_{tang}(\Omega)$.

- The last step is now to show the complete H^2 regularity of the solution from the local and tangential regularity properties that we established above.

To this end, we consider the equation in the neighborhood of the boundary $\mathcal{O}^\rho_{s_0}$ defined in Section 3.4, and the expressions of the Laplace operator given in Proposition III.3.24. We obtain

$$\mathrm{Tr}(\nabla_T \nabla u) + \nabla_N \nabla_N u + (\nabla u) \cdot \tilde{\mathcal{G}} = f, \text{ in } \mathcal{O}^\rho_{s_0}. \tag{III.102}$$

From the definition of tangential Sobolev spaces (Definition III.3.15) and the fact that $u \in H^2_{tang}(\Omega)$, we know that $\mathrm{Tr}(\nabla_T \nabla u) \in L^2(\mathcal{O}^\rho_{s_0})$ and we also have, by assumption, that $f \in L^2(\mathcal{O}^\rho_{s_0})$. By the regularity properties of the vector field $\tilde{\mathcal{G}}$ which are the same as $\mathcal{G}$ (see Proposition III.3.22), we know that

$$\frac{1}{|\ln \rho|}\tilde{\mathcal{G}} \in (L^\infty(\mathcal{O}^\rho_{s_0}))^d. \tag{III.103}$$

Since $\nabla u \in L^2(\mathcal{O}^\rho_{s_0})$, it follows from the integrability properties of the logarithm that

$$(\nabla u) \cdot \tilde{\mathcal{G}} \in L^q(\mathcal{O}^\rho_{s_0}),\ \forall q < 2. \tag{III.104}$$

As a consequence, we deduce from (III.102) that

$$\nabla_N \nabla_N u \in L^q(\mathcal{O}^\rho_{s_0}),\ \forall q < 2.$$

It follows that

$$\nabla(\nabla_N u) = \nabla_T(\nabla_N u) + (\nabla_N \nabla_N u)\nu = \nabla_T((\nabla u)\cdot \nu) + (\nabla_N \nabla_N u)\nu.$$

Since $\nabla u \in H^1_{tang}(\Omega)$ and ν is Lipschitz-continuous on $\overline{\Omega}$, the first term in the right-hand side belongs to $(L^2(\Omega))^d$, and we have shown above that the second term belongs to $L^q(\Omega)^d$ for any $q < 2$. Therefore, we have proved that

$$\nabla_N u \in W^{1,q}(\Omega),\ \forall q < 2.$$

By definition of the tangential Sobolev spaces, we already know that $\nabla_T u \in (H^1(\Omega))^d \subset (W^{1,q}(\Omega))^d$ for $q < 2$. Theorem III.3.14 thus proves that $u \in W^{2,q}(\Omega)$ for any $q < 2$.
We finally use the Sobolev embeddings (Theorem III.2.34) to deduce that there is a $r > 2$ such that $\nabla u \in (L^r(\Omega))^d$. Using once more (III.103), we deduce that (III.104) still holds with $q = 2$.
All the steps of the proof above now lead to the expected regularity, that is, $u \in H^2(\Omega)$.

□

4.2 Neumann boundary conditions

Theorem III.4.3. *Let Ω be a connected, bounded, Lipschitz domain of $\mathbb{R}^d$. Let $f \in L^2(\Omega)$ and $f_b \in H^{-1/2}(\partial\Omega)$ satisfying the compatibility condition*

$$\int_\Omega f(x)\,dx + \langle f_b, 1\rangle_{H^{-1/2},H^{1/2}} = 0. \tag{III.105}$$

There exists a unique solution $u \in H^1(\Omega) \cap L^2_0(\Omega)$ of

$$\begin{cases} -\Delta u = f, & \text{in } \Omega,\\ \dfrac{\partial u}{\partial \nu} = f_b, & \text{on } \partial\Omega, \end{cases}$$

and it satisfies, for some $C > 0$ depending only on Ω,

$$\|u\|_{H^1} \leq C(\|f\|_{L^2} + \|f_b\|_{H^{-1/2}}). \tag{III.106}$$

Moreover, if for $k \geq 0$, Ω is of class $\mathcal{C}^{k+1,1}$, $f \in H^k(\Omega)$ and $f_b \in H^{k+1/2}(\partial\Omega)$ then $u \in H^{k+2}(\Omega)$ and we have

$$\|u\|_{H^{k+2}} \leq C\left(\|f\|_{H^k} + \|f_b\|_{H^{k+1/2}}\right).$$

Remark III.4.1. Any $u \in H^1(\Omega)$ which is solution of $-\Delta u = f$ in the distribution sense, satisfies that $\nabla u \in H_{\text{div}}(\Omega)$, where $H_{\text{div}}(\Omega)$ is the space that we study in Section 3.2 of Chapter IV. In particular, we are able to give a weak sense to the normal derivative of u in the dual space $H^{-1/2}(\partial\Omega)$ and to prove the generalised Stokes formula (IV.10). This formula implies, in particular that

$$\int_\Omega \operatorname{div}(\nabla u)\, dx = \left\langle \frac{\partial u}{\partial \nu}, 1 \right\rangle_{H^{-1/2},H^{1/2}}.$$

Therefore, it is clear that the compatibility condition (III.105), is necessary for ensuring the solvability of the problem. Moreover, the hypothesis that u has a zero mean value is, of course, required to ensure uniqueness of the solution.

Proof.

We define the Hilbert space

$$H^1_m(\Omega) = \left\{ v \in H^1(\Omega),\ \int_\Omega v\, dx = 0 \right\},$$

the bilinear form

$$a : (u,v) \in H^1_m(\Omega) \times H^1_m(\Omega) \mapsto \int_\Omega \nabla u \cdot \nabla v\, dx,$$

and the linear form

$$L : v \in H^1_m(\Omega) \mapsto \int_\Omega f v\, dx + \langle f_b, \gamma_0(v)\rangle_{H^{-1/2},H^{1/2}}.$$

It is clear that a and L are continuous. Moreover, since Ω is connected, the Poincaré–Wirtinger inequality (see Proposition III.2.39) implies that a is coercive on $H^1_m(\Omega)$.

Therefore, the Lax–Milgram theorem applies and gives that there exists a unique solution $u \in H^1_m(\Omega)$ to the problem

$$a(u,v) = L(v),\ \forall v \in H^1_m(\Omega).$$

The compatibility condition gives that $L(1) = 0$, and moreover we have $a(u,1) = 0$. Thus, since any $v \in H^1(\Omega)$ can be written as a sum of a constant function and of an element of $H^1_m(\Omega)$, we deduce that u actually solves the problem

$$a(u,v) = L(v),\ \forall v \in H^1(\Omega).$$

Taking test functions $v \in \mathcal{D}(\Omega)$ gives that $-\Delta u = f$ in the distribution sense. Then, using the generalised Stokes formula (IV.10), we can deduce that the boundary condition is satisfied in the weak sense.

Let us prove the regularity properties. Using Theorem III.2.23, we can consider a lifting of the boundary data $R_\nu(f_b)$ which belong to $H^2(\Omega)$. Therefore, it is enough to consider the case where $f_b = 0$.

- We first observe that the local regularity properties that we obtained during the proof of Theorem III.4.2 only use the fact that $u \in H^1(\Omega)$ and that the equation $-\Delta u = f$ is satisfied in the distribution sense. It does not depend on the kind of boundary condition which is considered. Therefore, the solution of the Laplace–Neumann problem belongs to $H^2_{loc}(\Omega)$.
- Second, we observe that the demonstration of the tangential regularity property given in the proof of Theorem III.4.2 only relies on the variational formulation of the problem, on the fact that one can take $v = \delta^\theta_{-h}\delta^\theta_h u$ as a test function and on the Poincaré inequality. All these computations still hold, except the Poincaré inequality because $\delta^\theta_h u$ does not vanish on the boundary. Nevertheless, since $u \in H^1_m(\Omega)$, we can use (III.89) to obtain

$$\left|\int_\Omega \delta^\theta_h u\,dx\right| = \underbrace{\left|\int_\Omega u\,dx\right|}_{=0} + \left|\int_\Omega \{u,1\}_h\,dx\right| \le C|h|\|u\|_{L^2}.$$

  Therefore we can apply the Poincaré–Wirtinger inequality (Proposition III.2.39) to get

$$\|\delta^\theta_h u\|_{H^1} \le C\|\nabla\delta^\theta_h u\|_{L^2} + C|h|\|u\|_{L^2}.$$

  The rest of the proof can be applied to get that $u \in H^2_{tang}(\Omega)$.
  Finally, we can obtain the complete regularity of u in the same way as in the case of Dirichlet boundary conditions because this does not make use of the kind of boundary conditions satisfied by the solution.

$\square$

Chapter IV
Steady Stokes equations

The first section of this chapter is dedicated to the proof of the Nečas inequality which says that, in the space $L^2(\Omega)$, the L^2-norm is equivalent to the sum of the H^{-1}-norm of the function and of its gradient. Even if this seems to be a very natural property, the proof (given here in any Lipschitz domain with compact boundary) is far from being straightforward.

Using the Nečas inequality, we are able in Section 2 to give a fundamental characterization of a gradient field, which is often called de Rham's theorem: this result is an expression of the fact that a vector field of $(H^{-1}(\Omega))^d$ which is orthogonal (in the dual sense) to any divergence-free element of $(H_0^1(\Omega))^d$, is necessarily a gradient field. This theorem is indispensable for establishing the existence of the pressure in the Stokes and Navier–Stokes systems.

In Section 3, we investigate various properties of the divergence operator and of some related spaces. In particular we prove the existence of a continuous right-inverse to the divergence operator with values in $(H_0^1(\Omega))^d$. The conclusion of this section is the study of the so-called Leray (or Hodge–Helmholtz) decomposition for the vector fields in $(L^2(\Omega))^d$ into a divergence-free part and a gradient part.

Section 4 is dedicated to the study of the curl operator and related spaces, in the case of space dimension $d = 3$. The main result of the section is the study of suitable assumptions that ensure that a divergence-free vector field can be written as the curl of another vector field in a Sobolev framework. These results are useful in the study of the steady Navier–Stokes equations with nonhomogeneous Dirichlet boundary data.

We then turn to the study of the Stokes problem (i.e., the steady linear model for the flow of incompressible viscous fluids) with Dirichlet boundary conditions. In Section 5 we formulate and prove the well-posedness theorem for this system and investigate the properties of the (unbounded) Stokes operator that we can build from this system. This makes it possible to establish the existence of a basis of eigenfunctions for the Stokes problem that we use to construct the Galerkin approximations for the Navier–Stokes equations in

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the next chapter. We also use these spectral properties to build solutions of the unsteady linear Stokes equations.

Section 6 is devoted to the complete proof of elliptic regularity properties for the Stokes problem and related systems. We make use here of the Nirenberg translation method which is developed using the formalism introduced in the previous chapter.

The last three sections 7, 8, and 9 are dedicated to the study of the Stokes equations with other kinds of boundary and transmission conditions. We first consider Neumann-like boundary conditions and then the physically relevant stress boundary conditions. Then we study the interface Stokes problem for which we assumed that the viscosity is piecewise smooth and that the source term models stress jumps in the domain. Finally, we investigate the case where a boundary condition on the vorticity is considered. This last framework has the particularity that the computation of the pressure can be decoupled from the one of the velocity, contrary to other cases.

In each of these three sections, we prove existence, uniqueness, and regularity results.

An important part of the material of this chapter is inspired by classic references on the subject such as [92, 91, 54, 118, 119, 65, 122, 120].

Convention:

In the usual way, when we establish energy estimates, a certain number of constants appear in the inequalities. However, so as not to make the notation too cumbersome, these constants are always denoted as C even though their value may possibly change from one line to another. The dependence of these constants on various parameters of the problem is only referred to when necessary.

1 Nečas inequality

For any Lipschitz domain Ω in $\mathbb{R}^d$, we define the space

$$\chi(\Omega) = \left\{ p \in H^{-1}(\Omega), \nabla p \in \big(H^{-1}(\Omega)\big)^d \right\},$$

endowed with the norm

$$\|p\|_{\chi(\Omega)} = \|p\|_{H^{-1}} + \|\nabla p\|_{H^{-1}}.$$

Our goal is to prove that this space is algebraically and topologically equal to the space $L^2(\Omega)$. This result is initially due to Nečas (see [91],[92]). It is a possible starting point to the study of the functional spaces that we need for the analysis of the Stokes and Navier–Stokes equations, as presented in particular in [118, 119].

Theorem IV.1.1. *Let Ω be a Lipschitz domain of $\mathbb{R}^d$ with compact boundary; then we have $\chi(\Omega) = L^2(\Omega)$ and, moreover, there is a $C > 0$ such that*

$$\|p\|_{L^2(\Omega)} \leq C\|p\|_{\chi(\Omega)}, \forall p \in L^2(\Omega). \tag{IV.1}$$

Remark IV.1.1. For any $1 < q < +\infty$, we also have

$$\|p\|_{L^q(\Omega)} \leq C(\|p\|_{W^{-1,q}(\Omega)} + \|\nabla p\|_{W^{-1,q}(\Omega)}), \forall p \in L^q(\Omega).$$

The proof of Theorem IV.1.1 which is given below can be easily adapted to this framework. The main new difficulty is to prove the inequality for $\Omega = \mathbb{R}^d$, as in Proposition IV.1.2. Indeed, in the case $q \neq 2$, this requires us to characterise the Fourier multipliers that continuously map $L^q(\mathbb{R}^d)$ into itself, which is far from the main scope of this book (see [91]). However, all the other steps of the proof can be adapted in a straightforward way.

Remark IV.1.2. To prove this theorem, it is enough to prove (IV.1) for any $p \in \mathcal{C}_c^\infty(\overline{\Omega})$. Indeed, assume that this inequality holds for any such functions; then we know from Proposition III.2.26 and Theorem III.2.27 that for any $p \in \chi(\Omega)$, the family $(\mathcal{S}_\varepsilon p) \subset \mathcal{C}_c^\infty(\overline{\Omega})$ converges towards p in $\chi(\Omega)$. Therefore, by assumption we have that

$$\|\mathcal{S}_{\varepsilon_1} p - \mathcal{S}_{\varepsilon_2} p\|_{L^2(\Omega)} \leq C\|\mathcal{S}_{\varepsilon_1} p - \mathcal{S}_{\varepsilon_2} p\|_{\chi(\Omega)},$$

from which we deduce that $(\mathcal{S}_\varepsilon p)_\varepsilon$ is a Cauchy sequence in $L^2(\Omega)$. We already know that $\mathcal{S}_\varepsilon p \xrightarrow[\varepsilon \to 0]{} p$ in $H^{-1}(\Omega)$, therefore we deduce that p necessarily belongs to $L^2(\Omega)$ and that (IV.1) holds by passing to the limit in the inequality

$$\|\mathcal{S}_\varepsilon p\|_{L^2(\Omega)} \leq C\|\mathcal{S}_\varepsilon p\|_{\chi(\Omega)}.$$

Following this remark, our goal is now to prove (IV.1) for any sufficiently smooth enough function. We start by studying the case where $\Omega = \mathbb{R}^d$ and follow with the case of the flat half-space $\Omega = \mathbb{R}^d_+$. We then deduce the result for any Lipschitz half-space $\mathcal{H}_a$ and finally for any Lipschitz domain with compact boundary.

1.1 Proof of the inequality

1.1.1 The case of the whole space

This case becomes entirely elementary by using the characterisation of Sobolev spaces through the Fourier transform given by (III.11).

Proposition IV.1.2. *Theorem IV.1.1 holds for $\Omega = \mathbb{R}^d$.*

Proof.

We follow Remark IV.1.2 and we consider a $p \in \mathcal{D}(\mathbb{R}^d)$. Denoting as $\hat{p}$ the Fourier transform of p, we have from (III.11) that

$$\|p\|_{H^{-1}}^2 = C \int_{\mathbb{R}^d} (1+|\xi|^2)^{-1} |\hat{p}(\xi)|^2 \, d\xi,$$
$$\|\nabla p\|_{H^{-1}}^2 = C \int_{\mathbb{R}^d} (1+|\xi|^2)^{-1} |\xi|^2 |\hat{p}(\xi)|^2 \, d\xi.$$

Thus, using that

$$|\hat{p}(\xi)|^2 = (1+|\xi|^2)^{-1}(1+|\xi|^2)|\hat{p}(\xi)|^2 = (1+|\xi|^2)^{-1}|\hat{p}(\xi)|^2 + (1+|\xi|^2)^{-1}|\xi|^2|\hat{p}(\xi)|^2,$$

we deduce, by the Hausdorff–Young theorem, that

$$\|p\|_{L^2}^2 = C\|\hat{p}\|_{L^2}^2 \leq C\|p\|_{H^{-1}}^2 + C\|\nabla p\|_{H^{-1}}^2 \leq C\|p\|_{\chi(\Omega)}^2.$$

The claim is proved.

□

1.1.2 The flat half-space case

Lemma IV.1.3. *The operator*

$$\pi : p \in L^2(\mathbb{R}^d_+) \mapsto \bar{p} \in L^2(\mathbb{R}^d),$$

which associates with any $p \in L^2(\mathbb{R}^d_+)$ its extension by 0 on the whole space is continuous.

Moreover, it maps $H^1_0(\mathbb{R}^d_+)$ continuously into $H^1(\mathbb{R}^d)$.

Proof.

The continuity of π in L^2 is straightforward. We know from Theorem III.2.46, that π actually maps $H^1_0(\mathbb{R}^d_+)$ into $H^1(\mathbb{R}^d)$. Moreover, we observe that $\nabla\bar{p} = \overline{\nabla p}$ (because $p \in H^1_0(\Omega)$) so that we have

$$\|\bar{p}\|_{L^2(\mathbb{R}^d)} = \|p\|_{L^2(\mathbb{R}^d_+)},$$
$$\|\nabla\bar{p}\|_{L^2(\mathbb{R}^d)} = \|\overline{\nabla p}\|_{L^2(\mathbb{R}^d)} = \|\nabla p\|_{L^2(\mathbb{R}^d_+)},$$

and therefore we get

$$\|\pi p\|_{H^1(\mathbb{R}^d)} = \|p\|_{H^1_0(\mathbb{R}^d_+)}.$$

□

We consider now the adjoint operator ${}^t\pi : L^2(\mathbb{R}^d) \to L^2(\mathbb{R}^d_+)$ defined by

$$({}^t\pi p, q)_{L^2(\mathbb{R}^d_+)} = (p, \pi q)_{L^2(\mathbb{R}^d)}, \forall p \in L^2(\mathbb{R}^d), \forall q \in L^2(\mathbb{R}^d_+).$$

For any $p \in L^2(\mathbb{R}^d)$ and any $q \in H^1_0(\mathbb{R}^d)$, using the previous lemma and (III.39), we deduce that

$$({}^t\pi p, q)_{L^2(\mathbb{R}^d_+)} \leq \|p\|_{H^{-1}(\mathbb{R}^d)} \|\pi q\|_{H^1(\mathbb{R}^d)} \leq C\|p\|_{H^{-1}(\mathbb{R}^d)} \|q\|_{H^1_0(\mathbb{R}^d)},$$

which proves that

$$\|{}^t\pi p\|_{H^{-1}(\mathbb{R}^d_+)} \leq C\|p\|_{H^{-1}(\mathbb{R}^d)}, \forall p \in L^2(\mathbb{R}^d).$$

Since $L^2(\mathbb{R}^d)$ is dense in $H^{-1}(\mathbb{R}^d)$, we deduce that ${}^t\pi$ has a unique continuous extension defined from $H^{-1}(\mathbb{R}^d)$ into $H^{-1}(\mathbb{R}^d_+)$. We still denote this extension by ${}^t\pi$.

We easily observe that for all $p \in L^2(\mathbb{R}^d)$ we have ${}^t\pi p = p_{|\mathbb{R}^d_+}$ so that ${}^t\pi$ can also be viewed as a "restriction to $\mathbb{R}^d_+$" operator for the elements of $H^{-1}(\mathbb{R}^d)$.

Lemma IV.1.4. *There exists a linear continuous operator P from $\chi(\mathbb{R}^d_+)$ into $\chi(\mathbb{R}^d)$ which satisfies*

$${}^t\pi \circ P = Id_{\chi(\mathbb{R}^d_+)}.$$

Proof.

We define this extension operator by duality.

- Step one:
  Let us begin by introducing a projection operator from $H^1(\mathbb{R}^d)$ onto $H^1_0(\mathbb{R}^d_+)$.
  We first define this operator on smooth functions $p \in \mathcal{D}(\mathbb{R}^d)$, for all $x = (x_i)_{1\leq i\leq d} \in \mathbb{R}^d_+$ by

  $$Qp(x) = p(x) + \sum_{j=1}^{2} \alpha_j p(x_1, \dots, x_{d-1}, -jx_d).$$

  It is clear that the function Qp is smooth, and that if we impose the condition

  $$1 + \sum_{j=1}^{2} \alpha_j = 0, \tag{IV.2}$$

  then its trace on $\partial\mathbb{R}^d_+ = \mathbb{R}^{d-1} \times \{0\}$ is zero and, consequently, $Qp \in H^1_0(\mathbb{R}^d_+)$. Moreover, we can easily verify that for all $p \in \mathcal{D}(\mathbb{R}^d)$,we have

  $$\|Qp\|_{H^1_0(\mathbb{R}^d_+)} \leq C\|p\|_{H^1(\mathbb{R}^d)},$$

  where $C > 0$ only depends on the $(\alpha_i)_i$. Hence, by the density of $\mathcal{D}(\mathbb{R}^d)$ in $H^1(\mathbb{R}^d)$, this operator can be extended in a unique way as a continuous operator from $H^1(\mathbb{R}^d)$ into $H^1_0(\mathbb{R}^d_+)$.
  Finally we can show (first on smooth functions and then by density) that

$$Q \circ \pi = Id_{H_0^1(\mathbb{R}_+^d)}.$$

- Step two:
  By taking the adjoint of the above identity, we get

$$^t\pi \circ {}^tQ = Id_{H^{-1}(\mathbb{R}_+^d)}.$$

  Hence, by restricting this equality in the subspace $\chi(\mathbb{R}_+^d)$ of $H^{-1}(\mathbb{R}_+^d)$, we find

$$^t\pi \circ P = Id_{\chi(\mathbb{R}_+^d)},$$

  where we have defined $P = {}^tQ_{|\chi(\mathbb{R}_+^d)}$.
  The operator $P : \chi(\mathbb{R}_+^d) \mapsto H^{-1}(\mathbb{R}^d)$, thus defined, is a suitable candidate to be the sought-after extension operator. To complete the proof, we need to prove that P takes its values in $\chi(\mathbb{R}^d)$ and that it is continuous for the respective topologies of $\chi(\mathbb{R}_+^d)$ and $\chi(\mathbb{R}^d)$.
- Step three:
  Let now v be any element in $\chi(\mathbb{R}_+^d)$. By definition, the operator ${}^tQ$ is linear and continuous from $H^{-1}(\mathbb{R}_+^d)$ into $H^{-1}(\mathbb{R}^d)$ and hence we have

$$\|Pv\|_{H^{-1}(\mathbb{R}^d)} = \|{}^tQv\|_{H^{-1}(\mathbb{R}^d)} \leq C\|v\|_{H^{-1}(\mathbb{R}_+^d)}.$$

  We now need to show that the gradient of Pv belongs to the space $(H^{-1}(\mathbb{R}^d))^d$. For all $\varphi \in \mathcal{D}(\mathbb{R}^d)$ and for all $i \in \{1, \dots, d\}$,we have

$$\begin{aligned}\langle \partial_{x_i} Pv, \varphi\rangle_{H^{-1},H_0^1} &= \langle \partial_{x_i} {}^tQv, \varphi\rangle_{H^{-1},H_0^1} = -\langle {}^tQv, \partial_{x_i}\varphi\rangle_{H^{-1},H_0^1}\\ &= -\langle v, Q\partial_{x_i}\varphi\rangle_{H^{-1},H_0^1}.\end{aligned}$$

  To demonstrate the result, it is necessary to make use of the assumption that $\nabla v \in (H^{-1}(\mathbb{R}_+^d))^d$ and thus, we need to express the operator $Q\partial_{x_i}$ as a divergence operator applied to some functions of $H_0^1(\mathbb{R}_+^d)$. To do this, we must consider two different cases. For all $x = (x_i)_i \in \mathbb{R}_+^d$, we denote $\bar{x} = (x_1, \dots, x_{d-1})$.

  – If $i < d$:
    We have from the definition of Q:

$$\begin{aligned}Q\partial_{x_i}\varphi(x) &= \partial_{x_i}\varphi(x) + \sum_{j=1}^{2} \alpha_j (\partial_{x_i}\varphi)(\bar{x}, -jx_d)\\ &= \partial_{x_i} Q\varphi(x).\end{aligned}$$

    We then have

$$\langle \partial_{x_i} Pv, \varphi\rangle_{H^{-1},H_0^1} = -\langle v, \partial_{x_i} Q\varphi\rangle_{H^{-1},H_0^1} = \langle \partial_{x_i} v, Q\varphi\rangle_{H^{-1},H_0^1},$$

which shows that

$$\begin{aligned}|\langle \partial_{x_i} Pv, \varphi\rangle_{H^{-1},H_0^1}| &\leq \|\partial_{x_i} v\|_{H^{-1}(\mathbb{R}^d_+)} \|Q\varphi\|_{H_0^1(\mathbb{R}^d_+)} \\ &\leq C\|\nabla v\|_{H^{-1}(\mathbb{R}^d_+)} \|\varphi\|_{H^1(\mathbb{R}^d)}.\end{aligned}$$

This indeed proves that $\partial_{x_i} Pv \in H^{-1}(\mathbb{R}^d)$ and that we have the estimate $\|\partial_{x_i} Pv\|_{H^{-1}(\mathbb{R}^d)} \leq C\|\nabla v\|_{H^{-1}(\mathbb{R}^d_+)}$.

– If $i = d$:
In this case, we have

$$\begin{aligned}Q\partial_{x_d}\varphi(x) &= \partial_{x_d}\varphi(x) + \sum_{j=1}^{2} \alpha_j (\partial_{x_d}\varphi)(\bar{x}, -jx_d) \\ &= \partial_{x_d}\varphi(x) - \sum_{j=1}^{2} \frac{1}{j}\alpha_j \partial_{x_d}\big(\varphi(\bar{x}, -jx_d)\big) \\ &= \partial_{x_d} R_d\varphi(x),\end{aligned}$$

where the operator R_d is defined by

$$R_d\varphi(x) = \varphi(x) - \sum_{j=1}^{2} \frac{1}{j}\alpha_j \varphi(\bar{x}, -jx_d).$$

We can verify that R_d maps $H^1(\mathbb{R}^d)$ into $H^1(\mathbb{R}^d_+)$ in a continuous way. It just remains to ensure that $R_d\varphi$ is indeed in $H_0^1(\mathbb{R}^d_+)$. We immediately see that this is true as soon as the condition

$$1 - \sum_{j=1}^{2} \frac{\alpha_j}{j} = 0, \tag{IV.3}$$

is satisfied. By assuming this condition is satisfied, we can conclude by writing that

$$\begin{aligned}|\langle \partial_{x_d} Pv, \varphi\rangle_{H^{-1},H_0^1}| &= |\langle \partial_{x_d} v, R_d\varphi\rangle_{H^{-1},H_0^1}| \\ &\leq \|\partial_{x_d} v\|_{H^{-1}(\mathbb{R}^d_+)} \|R_d\varphi\|_{H_0^1(\mathbb{R}^d_+)} \\ &\leq C\|\partial_{x_d} v\|_{H^{-1}(\mathbb{R}^d_+)} \|\varphi\|_{H^1(\mathbb{R}^d)},\end{aligned}$$

which simultaneously shows that $\partial_{x_d} Pv \in H^{-1}(\mathbb{R}^d)$ and demonstrates the estimate $\|\partial_{x_d} Pv\|_{H^{-1}(\mathbb{R}^d)} \leq C\|\nabla v\|_{H^{-1}(\mathbb{R}^d_+)}$.

It remains, of course, to observe that we can simultaneously fulfill the two conditions (IV.2) and (IV.3) by taking $(\alpha_1, \alpha_2) = (3, -4)$.

□

Proposition IV.1.5. *Theorem IV.1.1 holds for $\Omega = \mathbb{R}^d_+$.*

Proof.

Let p be an element of $\chi(\mathbb{R}^d_+)$; then, by definition of the extension operator $Pp \in \chi(\mathbb{R}^d)$. By Proposition IV.1.2, we deduce that $Pp \in L^2(\mathbb{R}^d)$ and that

$$\|Pp\|_{L^2(\mathbb{R}^d)} \leq C\|Pp\|_{\chi(\mathbb{R}^d)} \leq C\|p\|_{\chi(\mathbb{R}^d_+)}.$$

By using Lemma IV.1.4, we get $p = {}^t\pi Pp$, and since $Pp \in L^2(\mathbb{R}^d)$ we have seen that ${}^t\pi(Pp) = (Pp)_{|\mathbb{R}^d_+}$.

It follows that $p \in L^2(\mathbb{R}^d_+)$ and that $\|p\|_{L^2(\mathbb{R}^d_+)} \leq C\|p\|_{\chi(\mathbb{R}^d_+)}$, which proves the result.

□

1.1.3 The Lipschitz half-space case

Proposition IV.1.6. *Theorem IV.1.1 holds for any Lipschitz half-space $\mathcal{H}_a$ as defined in Definition III.1.1.*

Proof.

We use the Lipschitz diffeomorphism $T_a : \mathbb{R}^d_+ \to \mathcal{H}_a$ introduced in Section 1.2.1 of Chapter III.

Let $\varphi \in H^1_0(\mathbb{R}^d_+)$. By using Corollary III.2.41 and the properties of T_a given in Proposition III.1.3 we deduce that

$$\varphi \circ T_a^{-1} J_{T_a^{-1}} \in H^1_0(\mathcal{H}_a), \text{ and } \left\|\varphi \circ T_a^{-1} J_{T_a^{-1}}\right\|_{H^1_0(\mathcal{H}_a)} \leq C\|\varphi\|_{H^1_0(\mathbb{R}^d_+)}.$$

For any $p \in \mathcal{C}^\infty_c(\overline{\mathcal{H}_a})$, we can write

$$\begin{aligned}
\langle p \circ T_a, \varphi\rangle_{H^{-1}(\mathbb{R}^d_+), H^1_0(\mathbb{R}^d_+)} &= \int_{\mathbb{R}^d_+} (p \circ T_a)\varphi \, dx \\
&= \int_{\mathcal{H}_a} p(\varphi \circ T_a^{-1}) J_{T_a^{-1}} \, dy \\
&= \left\langle p, (\varphi \circ T_a^{-1}) J_{T_a^{-1}} \right\rangle_{H^{-1}(\mathcal{H}_a), H^1_0(\mathcal{H}_a)},
\end{aligned}$$

and then it follows

$$\langle p \circ T_a, \varphi\rangle_{H^{-1}(\mathbb{R}^d_+), H^1_0(\mathbb{R}^d_+)} \leq C\|p\|_{H^{-1}(\mathcal{H}_a)}\|\varphi\|_{H^1_0(\mathbb{R}^d_+)},$$

so that we have proved

$$\|p \circ T_a\|_{H^{-1}(\mathbb{R}^d_+)} \leq C\|p\|_{H^{-1}(\mathcal{H}_a)}. \tag{IV.4}$$

We use now the chain rule to get

$$\nabla(p \circ T_a) = {}^t(\nabla T_a)(\nabla p) \circ T_a,$$

so that for any $\psi \in (H_0^1(\mathbb{R}_+^d))^d$ we have

$$\langle \nabla(p \circ T_a), \psi \rangle_{H^{-1}(\mathbb{R}_+^d), H_0^1(\mathbb{R}_+^d)} = \int_{\mathbb{R}_+^d} ((\nabla p) \circ T_a) \cdot ((\nabla T_a).\psi)\, dx.$$

Since ∇T_a is bounded and has first derivatives which behave as does $1/\rho$ we can apply once more Corollary III.2.41 to deduce that $((\nabla T_a).\psi)$ belongs to $(H_0^1(\mathbb{R}_+^d))$ with an estimate $\|(\nabla T_a).\psi\|_{H_0^1} \leq C\|\psi\|_{H_0^1}$. It follows that

$$\begin{aligned}&\langle \nabla(p \circ T_a), \psi \rangle_{H^{-1}(\mathbb{R}_+^d), H_0^1(\mathbb{R}_+^d)} = \langle ((\nabla p) \circ T_a), (\nabla T_a).\psi \rangle_{H^{-1}(\mathbb{R}_+^d), H_0^1(\mathbb{R}_+^d)} \\ &\leq \|(\nabla p) \circ T_a\|_{H^{-1}(\mathbb{R}_+^d)} \|(\nabla T_a).\psi\|_{H_0^1(\mathbb{R}_+^d)} \leq C\|(\nabla p) \circ T_a\|_{H^{-1}(\mathbb{R}_+^d)} \|\psi\|_{H_0^1(\mathbb{R}_+^d)}.\end{aligned}$$

Using (IV.4) with p replaced by ∇p, we finally deduce

$$\|\nabla(p \circ T_a)\|_{H^{-1}(\mathbb{R}_+^d)} \leq C\|\nabla p\|_{H^{-1}(\mathcal{H}_a)}. \tag{IV.5}$$

Finally, by (IV.4) and (IV.5) we get that

$$\|p \circ T_a\|_{\chi(\mathbb{R}_+^d)} \leq C\|p\|_{\chi(\mathcal{H}_a)}.$$

Using Proposition IV.1.5, we deduce that for any such smooth p, we have

$$\|p \circ T_a\|_{L^2(\mathbb{R}_+^d)} \leq C\|p\|_{\chi(\mathcal{H}_a)},$$

and finally, with Proposition II.2.15, we get that

$$\|p\|_{L^2(\mathcal{H}_a)} \leq C\|p \circ T_a\|_{L^2(\mathbb{R}_+^d)} \leq C'\|p\|_{\chi(\mathcal{H}_a)},$$

which concludes the proof, following Remark IV.1.2.

□

1.1.4 The general case

We can now perform the proof of the main theorem of this section.

Proof (of Theorem IV.1.1).

We consider a finite open covering $(U_i)_{1\leq i\leq N}$ of $\partial\Omega$ as given by Theorem III.1.4, and another open set $U_0 \subset \overline{U_0} \subset \Omega$ such that $(U_i)_{0\leq i\leq N}$ is a finite open covering of $\overline{\Omega}$. Let $(\psi_i)_{0\leq i\leq N}$ be an associated partition of unity, and $p \in \mathcal{C}_c(\overline{\Omega})$.

- The function $p\psi_0$ is compactly supported in Ω. Therefore, its extension by 0, $\overline{p\psi_0}$ satisfies $\nabla(\overline{p\psi_0}) = \overline{\nabla(p\psi_0)}$ and we can apply Proposition IV.1.2 to get

$$\begin{aligned}\|p\psi_0\|_{L^2(\Omega)} = \|\overline{p\psi_0}\|_{L^2(\mathbb{R}^d)} &\leq C_0(\|\overline{p\psi_0}\|_{H^{-1}(\mathbb{R}^d)} + \|\overline{\nabla(p\psi_0)}\|_{H^{-1}(\mathbb{R}^d)}) \\ &\leq C_0(\|p\|_{H^{-1}(\Omega)} + \|\nabla p\|_{H^{-1}(\Omega)}).\end{aligned}$$

- For any $1 \leq i \leq N$, the function $p\psi_i$ is supported in a (rotated) Lipschitz half-space $R_{\sigma_i}\mathcal{H}_{a_i}$. By Proposition IV.1.6, we can deduce that

$$\begin{aligned}\|p\psi_i\|_{L^2(\Omega)} &= \|p\psi_i\|_{L^2(R_{\sigma_i}\mathcal{H}_{a_i})} \\ &\leq C_i(\|p\psi_i\|_{H^{-1}(R_{\sigma_i}\mathcal{H}_{a_i})} + \|\nabla(p\psi_i)\|_{H^{-1}(R_{\sigma_i}\mathcal{H}_{a_i})}) \qquad \text{(IV.6)} \\ &= C_i(\|p\psi_i\|_{H^{-1}(\Omega)} + \|\nabla(p\psi_i)\|_{H^{-1}(\Omega)}).\end{aligned}$$

We observe now that for any $\varphi \in H_0^1(\Omega)$, we have

$$\begin{aligned}\langle p\psi_i, \varphi\rangle_{H^{-1}(\Omega),H_0^1(\Omega)} &= \int_\Omega p\psi_i\varphi\,dx = \langle p, \psi_i\varphi\rangle_{H^{-1}(\Omega),H_0^1(\Omega)} \\ &\leq \|p\|_{H^{-1}(\Omega)}\|\psi_i\varphi\|_{H_0^1(\Omega)} \leq C_{\psi_i}\|p\|_{H^{-1}(\Omega)}\|\varphi\|_{H_0^1(\Omega)},\end{aligned}$$

so that

$$\|p\psi_i\|_{H^{-1}(\Omega)} \leq C_{\psi_i}\|p\|_{H^{-1}(\Omega)}.$$

Similarly, we can prove that

$$\begin{aligned}\|\nabla(p\psi_i)\|_{H^{-1}(\Omega)} &\leq \|(\nabla p)\psi_i\|_{H^{-1}(\Omega)} + \|(\nabla\psi_i)p\|_{H^{-1}(\Omega)} \\ &\leq C'_{\psi_i}(\|p\|_{H^{-1}(\Omega)} + \|\nabla p\|_{H^{-1}(\Omega)}).\end{aligned}$$

Hence, (IV.6) gives

$$\|p\psi_i\|_{L^2(\Omega)} \leq C'_i(\|p\|_{H^{-1}(\Omega)} + \|\nabla p\|_{H^{-1}(\Omega)}) = C'_i\|p\|_{X(\Omega)}.$$

Gathering all the above inequalities, we have finally proved that there exists a $C > 0$ such that

$$\|p\|_{L^2(\Omega)} \leq C\|p\|_{X(\Omega)},\ \forall p \in \mathcal{C}_c^\infty(\overline{\Omega}).$$

By using Remark IV.1.2, the proof is complete.

□

1.2 Related Poincaré inequalities

The Nečas inequality now allows us to demonstrate a new Poincaré-type inequality for the functions of $L^2(\Omega)$.

Proposition IV.1.7. *Let Ω be a connected, bounded, Lipschitz domain of $\mathbb{R}^d$. There exists a $C > 0$ such that for all $p \in L^2(\Omega)$, we have*

$$\|p\|_{H^{-1}} \le C\left(\frac{1}{|\Omega|}\left|\int_\Omega p\,dx\right| + \|\nabla p\|_{H^{-1}}\right).$$

Proof.

This is the same principle as for Propositions III.2.38 and III.2.39. We argue by contradiction, by initially assuming that there exist functions $(p_n)_n$ in $L^2(\Omega)$ such that

$$\|p_n\|_{H^{-1}} \ge n\left(\frac{1}{|\Omega|}\left|\int_\Omega p_n\,dx\right| + \|\nabla p_n\|_{H^{-1}}\right).$$

By homogeneity, we assume that $\|p_n\|_{H^{-1}} = 1$. Then, since we have the inequality $\|\nabla p_n\|_{H^{-1}} \le 1/n$, we can deduce that $(p_n)_n$ is bounded in $L^2(\Omega)$ according to Theorem IV.1.1.

The embedding of $H_0^1(\Omega)$ into $L^2(\Omega)$ is compact, thus we know that the adjoint embedding of $L^2(\Omega)$ into $H^{-1}(\Omega)$ is also compact (Lemma II.3.6). We can therefore extract from $(p_n)_n$ a sequence $(p_{n_k})_k$ which weakly converges in $L^2(\Omega)$ and strongly in $H^{-1}(\Omega)$ towards a function p of $L^2(\Omega)$.

Since $\|\nabla p_n\|_{H^{-1}} \le 1/n$, we obtain $\nabla p = 0$ in the distribution sense and therefore, the open set Ω being connected, the limit p is necessarily a constant $\alpha \in \mathbb{R}$ (Lemma II.2.44). However, we also have

$$\frac{1}{|\Omega|}\left|\int_\Omega p_{n_k}\,dx\right| \le \frac{\|p_{n_k}\|_{H^{-1}}}{n_k} = \frac{1}{n_k},$$

and therefore at the limit (we can pass to the limit in the integral by weak convergence in $L^2(\Omega)$), we obtain

$$|\alpha| \le 0.$$

This shows that p is zero, which is in contradiction with the fact that

$$\|p\|_{H^{-1}} = \lim_{k\to\infty} \|p_{n_k}\|_{H^{-1}} = 1,$$

the convergence in this space being strong.

□

Of course, we cannot obtain a similar result to Proposition III.2.38, because the functions of $L^2(\Omega)$ do not have traces on $\partial\Omega$.

We now define a quotient space which is useful in our study because as we show later in incompressible models, the pressure is only defined apart from a constant, as soon as the domain Ω is connected.

Definition IV.1.8. *Let Ω be a connected, bounded open set of $\mathbb{R}^d$. We define the quotient space*

$$L_0^2(\Omega) = L^2(\Omega)/\mathbb{R},$$

made up of classes of functions of $L^2(\Omega)$ which differ by a constant. We equip this space with the quotient norm

$$\|u\|_{L_0^2} = \inf_{\alpha \in \mathbb{R}} \|u + \alpha\|_{L^2}.$$

Furthermore, for any function u of $L^2(\Omega)$, we introduce the mean of u on Ω defined by

$$m(u) = \frac{1}{|\Omega|} \int_\Omega u(x)\, dx.$$

In the usual way, we identify a class of $L_0^2(\Omega)$ with any of its representatives. It is easy to verify that, provided with the quotient norm above, the space $L_0^2(\Omega)$ is a Banach space. This is in fact a Hilbert space for the scalar product

$$(u, v)_{L_0^2} = \int_\Omega (u - m(u))(v - m(v))\, dx.$$

Indeed, the norm coming from this scalar product is equivalent to the quotient norm thanks to the following result.

Lemma IV.1.9. *For any bounded, connected, and Lipschitz domain Ω of $\mathbb{R}^d$, there exists a $C > 0$ such that for all $p \in L_0^2(\Omega)$, we have*

$$\|p\|_{L_0^2} \leq \|p - m(p)\|_{L^2} \leq 2\|p\|_{L_0^2}, \tag{IV.7}$$

$$\frac{1}{C}\|p\|_{L_0^2} \leq \|\nabla p\|_{H^{-1}} \leq C\|p\|_{L_0^2}. \tag{IV.8}$$

Proof.

The first inequality of (IV.7) is a direct consequence of the definition of the quotient norm. The second inequality of (IV.7) is obtained by writing for all $\alpha \in \mathbb{R}$

$$\|p - m(p)\|_{L^2} = \|p + \alpha - m(p + \alpha)\|_{L^2} \leq \|p + \alpha\|_{L^2} + |\Omega|^{1/2}|m(p + \alpha)|.$$

However, for any function v of $L^2(\Omega)$, the Hölder inequality gives us

$$|m(v)| \leq \frac{1}{|\Omega|} \int_\Omega |v|\, dx \leq |\Omega|^{-1/2}\|v\|_{L^2},$$

from which we get the desired result by taking the infimum in α of the inequality obtained.

The inequalities (IV.8) are obtained by using (IV.7), Theorem IV.1.1, and the Poincaré inequality of Proposition IV.1.7.

□

This lemma proves that the space $L_0^2(\Omega)$ is isomorphic with the closed subspace of $L^2(\Omega)$ made up of functions with a zero mean. Often, we choose the unique representative with zero mean as the representative of the $L_0^2(\Omega)$ classes.

2 Characterisation of gradient fields. De Rham's theorem

The aim of this section is to provide a useful characterisation of gradient fields among the set of all vector fields.

Definition IV.2.1. *Let E be a Banach space, whose dual space is E'. For all A subset of E, we define the* orthogonal *of A to be the linear subspace of E' defined by*

$$A^\perp = \{\varphi \in E',\ \forall x \in A,\ \langle \varphi, x\rangle_{E',E} = 0\}.$$

In the case where E is a Hilbert space, by using the Riesz representation theorem, it is easily seen that $A^\perp$ is also the orthogonal of A with respect to the scalar product of E.

Remark IV.2.1.

- The orthogonal of A is a closed linear subspace of E'. When E is reflexive, $A^{\perp\perp}$ can be identified with a linear subspace of E. This is then the smallest closed linear subspace of E containing A; that is, $A^{\perp\perp} = \overline{\operatorname{Span} A}$. Therefore, for any vector subspace A of E we have

$$A \text{ is closed in } E \Longleftrightarrow A^{\perp\perp} = A.$$

- If $A \subset B$ then $B^\perp \subset A^\perp$.

Theorem IV.2.2. *Let Ω be a Lipschitz domain of $\mathbb{R}^d$.*

- *If Ω is bounded, the image of the gradient operator*

$$\nabla\big(L^2(\Omega)\big) = \big\{f \in (H^{-1}(\Omega))^d,\ \exists p \in L^2(\Omega), f = \nabla p\big\},$$

is closed in $(H^{-1}(\Omega))^d$.

- *If Ω is not bounded, the set*

$$\nabla\big(L^2_{loc}(\Omega)\big) \cap (H^{-1}(\Omega))^d = \big\{f \in (H^{-1}(\Omega))^d,\ \exists p \in L^2_{loc}(\Omega), f = \nabla p\big\},$$

is closed in $(H^{-1}(\Omega))^d$.

Note that in the unbounded case, the gradient (in the distribution sense) of a function of $L^2_{loc}(\Omega)$ is not, in general, an element of $(H^{-1}(\Omega))^d$.

Proof (of Theorem IV.2.2).

We can only consider the case of a connected domain. Indeed, if Ω is not connected the result follows by applying the theorem on each of its connected components.

- Let us first assume that Ω is bounded. Let $(\nabla p_n)_n$ with $p_n \in L^2(\Omega)$ be a sequence which converges in $(H^{-1}(\Omega))^d$. Since Ω is bounded we can assume that $m(p_n) = 0$ without changing ∇p_n.

We can now apply the Nečas inequality (Theorem IV.1.1) and Poincaré's inequality (Proposition IV.1.7) to obtain

$$\|p_n - p_{n+k}\|_{L^2} \leq C\|\nabla(p_n - p_{n+k})\|_{H^{-1}}.$$

Since $(\nabla p_n)_n$ is a Cauchy sequence in $(H^{-1}(\Omega))^d$, we deduce that $(p_n)_n$ is a Cauchy sequence in $L^2(\Omega)$. It follows that there exists a $p \in L^2(\Omega)$ such that $p_n \to p$ as $n \to \infty$. Therefore, $\nabla p_n \xrightarrow[n\to\infty]{} \nabla p$ in $(H^{-1}(\Omega))^d$ and the claim is proved.

- In the unbounded case, we deduce the result from the one in the bounded case. To do this we consider an increasing sequence of regular, bounded, and connected open sets, $(\Omega_k)_k$ having union Ω.

  Now, let $(f_n)_n$ be a sequence of elements of $\left(H^{-1}(\Omega)\right)^d$ which is expressed as $f_n = \nabla p_n$ with $p_n \in L^2_{loc}(\Omega)$ and which converges towards an element f of $\left(H^{-1}(\Omega)\right)^d$.

  For all $k \geq 1$, the restriction of f to Ω_k is well-defined (by extending the functions of $H^1_0(\Omega_k)$ by 0 on all Ω). Furthermore, we can easily see that the sequence $\left(f_{n|\Omega_k}\right)_n$ converges towards $f_{|\Omega_k}$ in $\left(H^{-1}(\Omega_k)\right)^d$. Hence, from the result obtained in the bounded case, we know that $f_{|\Omega_k}$ is the gradient of a function $q_k \in L^2(\Omega_k)$.

  Since the Ω_k are connected, the function q_k is uniquely defined apart from a constant. We choose this function such that q_k has a zero mean on Ω_1 (which is contained in all the Ω_k).

  It is therefore evident that if $k_1 < k_2$, then the function $q_{k_1} - q_{k_2}$ is zero on Ω_{k_1}. Indeed, this function having a zero gradient in the sense of distributions (because $f_{|\Omega_{k_1}} = \nabla q_1$ and $f_{|\Omega_{k_2}} = \nabla q_2$), is therefore constant according to Lemma II.2.44. However, since q_{k_1} and q_{k_2} both have zero mean on Ω_1, this constant is necessarily zero.

  Hence, the functions $(q_k)_k$ are constructed in a coherent way on the $(\Omega_k)_k$, which gives us a function q defined on all Ω and which is indeed in $L^2_{loc}(\Omega)$. Moreover, it is clear that the gradient of q is also equal to f in $\left(H^{-1}(\Omega)\right)^d$.

□

Theorem IV.2.3. *Let Ω be a connected, Lipschitz domain of $\mathbb{R}^d$ with compact boundary. Let f be in $\left(H^{-1}(\Omega)\right)^d$ such that*

$$\langle f, \varphi\rangle_{H^{-1},H^1_0} = 0, \quad \forall \varphi \in \left(H^1_0(\Omega)\right)^d, \textit{ such that } \operatorname{div}\varphi = 0.$$

Then there exists an element p in $L^2_{loc}(\Omega)$, unique apart from a constant, such that $f = \nabla p$.

Furthermore, if Ω is bounded, then p belongs to $L^2(\Omega)$.

Proof.

Let Y be the subspace of $\left(H^{-1}(\Omega)\right)^d$ defined by

$$
\begin{aligned}
Y &= \left\{\nabla p,\ p \in L^2(\Omega)\right\}, && \text{if } \Omega \text{ is bounded,}\\
Y &= \left\{f \in \left(H^{-1}(\Omega)\right)^d,\ \exists p \in L^2_{loc}(\Omega),\ f = \nabla p\right\}, && \text{if } \Omega \text{ is unbounded.}
\end{aligned}
$$

Now, let Z be the subspace of $\left(H_0^1(\Omega)\right)^d$ defined by

$$Z = \left\{\varphi \in \left(H_0^1(\Omega)\right)^d,\ \operatorname{div}\varphi = 0\right\}.$$

From Definition IV.2.1, the theorem that we wish to prove affirms that if $f \in Z^\perp$, then $f \in Y$. However, Theorem IV.2.2 says that Y is a closed subspace of $\left(H^{-1}(\Omega)\right)^d$ which is reflexive, and hence we have $Y^{\perp\perp} = Y$ from Remark IV.2.1. Therefore, it is sufficient to establish the inclusion $Y^\perp \subset Z$ (the other inclusion is obviously true).

Let $u \in Y^\perp \subset \left(H_0^1(\Omega)\right)^d$. By definition this implies that

$$\langle \nabla p, u\rangle_{H^{-1},H_0^1} = 0,\ \forall p \in L^2(\Omega).$$

However, for any $p \in L^2(\Omega)$ we have by definition

$$\langle \nabla p, u\rangle_{H^{-1},H_0^1} = -\int_\Omega p \operatorname{div} u \, dx.$$

Hence, we have proved that any $u \in Y^\perp$, satisfies

$$\int_\Omega p(\operatorname{div} u)\, dx = 0,\ \forall p \in L^2(\Omega).$$

Choosing $p = \operatorname{div} u$ in this formula exactly gives that $\operatorname{div} u = 0$ and thus $u \in Z$.

□

We can now prove a stronger version of Theorem IV.2.3 which gives the same result but with weaker assumptions. Indeed, we can now assume that f cancels against the divergence-free test functions of $(\mathcal{D}(\Omega))^d$ but no longer against all divergence-free functions of $\left(H_0^1(\Omega)\right)^d$.

Theorem IV.2.4 (de Rham). *Let Ω be a connected, bounded, Lipschitz domain of $\mathbb{R}^d$. Let f be an element in $\left(H^{-1}(\Omega)\right)^d$, such that for any function $\varphi \in (\mathcal{D}(\Omega))^d$ satisfying $\operatorname{div}\varphi = 0$, we have $\langle f, \varphi\rangle_{H^{-1},H_0^1} = 0$. Then, there exists a unique function p belonging to $L_0^2(\Omega)$ such that $f = \nabla p$.*

Proof.

- Step one:
  Let $(\Omega_n)_n$ be an increasing sequence of regular and connected open sets included in Ω (i.e., $\overline{\Omega_n} \subset \Omega_{n+1}$) such that $\cup_n \Omega_n = \Omega$.

For all $u_n \in \left(H^1_0(\Omega_n)\right)^d$ such that $\operatorname{div} u_n = 0$, and for any $\varepsilon > 0$, we set $u_{n,\varepsilon} = \overline{u_n} \star \eta_\varepsilon$. We observe that $u_{n,\varepsilon}$ is compactly supported in Ω for ε small enough, because u_n is supported in Ω_n. Therefore we have $u_{n,\varepsilon} \in (\mathcal{D}(\Omega))^d$ and $\operatorname{div} u_{n,\varepsilon} = 0$. By hypothesis, we have

$$\langle f, u_n \rangle_{H^{-1},H^1_0} = \lim_{\varepsilon \to 0} \langle f, u_{n,\varepsilon} \rangle_{H^{-1},H^1_0} = 0.$$

Hence, from Theorem IV.2.3, $f_{|\Omega_n}$ as an element of $(H^{-1}(\Omega_n))^d$, is the gradient of some function p_n of $L^2(\Omega_n)$. Furthermore, for all n, the restriction of p_{n+1} to Ω_n has the same gradient as p_n, hence $p_{n+1} - p_n$ is a constant function in Ω_n, and we can choose p_{n+1} so that this constant is zero.

All things considered, the sequence of functions $(p_n)_n$ defines a function $p \in L^2_{loc}(\Omega)$ such that

$$f = \nabla p.$$

- Step two:

  It remains to show that indeed $p \in L^2(\Omega)$.

  To do this, we use Lemma III.1.7. Hence, we can write

  $$\Omega = \bigcup_{i=1}^{k} \omega_i,$$

  where the ω_i are a finite number of star-shaped open sets. It is sufficient to show that $p \in L^2(\omega_i)$ for all i.

  We now place ourselves in the star-shaped open set ω_i and, to simplify the notation, we assume (up to a translation) that it is star-shaped with respect to 0. For all $u_i \in (H^1_0(\omega_i))^d$, such that $\operatorname{div}(u_i) = 0$ (which we extend by 0 to all $\mathbb{R}^d$), and for all $\theta \in [0,1[$, we set $u_{i,\theta}(x) = u_i(x/\theta)$ which is compactly supported in ω_i and therefore in Ω, and divergence-free. We now proceed again through regularisation by introducing $u_{i,\theta,\varepsilon} = u_{i,\theta} \star \eta_\varepsilon$ which is divergence-free and compactly supported in Ω for $\varepsilon > 0$ small enough, so that the assumption gives

  $$\langle f, u_{i,\theta} \rangle_{H^{-1},H^1_0} = \lim_{\varepsilon \to 0} \langle f, u_{i,\theta,\varepsilon} \rangle_{H^{-1},H^1_0} = 0,$$

  and then we can easily show that $u_{i,\theta}$ converges towards u_i (when θ tends towards 1) in $(H^1_0(\Omega))^d$ which, finally, gives

  $$\langle f, u_i \rangle_{H^{-1},H^1_0} = \lim_{\theta \to 1} \langle f, u_{i,\theta} \rangle_{H^{-1},H^1_0} = 0.$$

  This being true for all u_i in $\left(H^1_0(\omega_i)\right)^d$, Theorem IV.2.3 applies and we therefore know that $f = \nabla p_i$ with $p_i \in L^2(\omega_i)$. As in step one, it is clear that $p - p_i$ is constant and that we can choose this constant to be zero. Hence, we have proved that

$$p_{|\omega_i} \in L^2(\omega_i),\ \forall i \in \{1, \ldots, k\},$$

and thus that $p \in L^2(\Omega)$. We can of course now replace p by $p - m(p)$ to ensure that $p \in L^2_0(\Omega)$.

□

We have, in fact, a more general result due to de Rham [49], one consequence of which is the following result (see also [111, 112, 124]).

Theorem IV.2.5. *Let Ω be any open set of $\mathbb{R}^d$ and let $f \in \big(\mathcal{D}'(\Omega)\big)^d$ such that for all $\varphi \in \big(\mathcal{D}(\Omega)\big)^d$ with zero divergence, $\langle f, \varphi\rangle_{\mathcal{D}',\mathcal{D}} = 0$; then there exists $\pi \in \mathcal{D}'(\Omega)$ such that $f = \nabla\pi$.*

3 The divergence operator and related spaces

3.1 Right-inverse for the divergence

The results above imply that the divergence operator defined from $(H^1_0(\Omega))^d$ into $L^2_0(\Omega)$ has a continuous right-inverse.

Theorem IV.3.1. *Let Ω be a connected, bounded, Lipschitz domain of $\mathbb{R}^d$. There exists a continuous linear operator Π from $L^2_0(\Omega)$ into $(H^1_0(\Omega))^d$, such that for all $q \in L^2_0(\Omega)$, the function $u = \Pi(q)$ satisfies*

$$\operatorname{div} u = q.$$

Remark IV.3.1. Notice that there exists a somewhat more explicit proof of this result using the theory of singular integrals. This is a construction due to Bogovskii (see [18, 19, 63, 93]).

We also give below a simpler proof in the case of a smooth 2D domain.

Remark IV.3.2. Following Remark IV.1.1, it can be shown that the same result holds if we replace $L^2_0(\Omega)$ by $L^p_0(\Omega)$ and $(H^1_0(\Omega))^d$ by $(W^{1,p}_0(\Omega))^d$ for $1 < p < +\infty$. Nevertheless, it does not hold for $p = 1$ or $p = +\infty$. We refer, for instance, to [21] for a detailed discussion on those questions, in particular for some alternative proof of the above theorem.

Proof.

We have previously seen (Theorem IV.2.2) that the map ∇ from $L^2(\Omega)$ into $(H^{-1}(\Omega))^d$ has a closed range and thus is a Banach space. Let N be the kernel of this map; then we know that the map ∇ can be considered as a continuous linear bijection of $L^2(\Omega)/N$ on $\nabla(L^2(\Omega))$. From the open mapping theorem (Theorem II.2.3) we know, therefore, that ∇ produces an isomorphism of Banach spaces between $L^2(\Omega)/N$ and $\nabla(L^2(\Omega))$.

Hence, the adjoint of this map produces an isomorphism from $(\nabla(L^2(\Omega)))'$ onto $(L^2(\Omega)/N)'$. Let us identify the objects involved:

- $(\nabla(L^2(\Omega)))'$ is the set of the continuous linear functionals on $\nabla(L^2(\Omega))$; it is naturally identified with the quotient of $\big((H^{-1}(\Omega))^d\big)'$ by all the linear functionals which cancel on $\nabla(L^2(\Omega))$, that is on Z using the notation of the proof of Theorem IV.2.3. Furthermore, $(H_0^1(\Omega))^d$ is reflexive, thus $((H^{-1}(\Omega))^d)'$ is naturally identified with $(H_0^1(\Omega))^d$. In summary, we have

$$(\nabla(L^2(\Omega)))' = (H_0^1(\Omega))^d/Z.$$

- The elements of $(L^2(\Omega)/N)'$ are the linear functionals on $L^2(\Omega)$ which cancel on N which is the set of constant functions because Ω is connected. A continuous linear form on $L^2(\Omega)$ is represented by a function $q \in L^2(\Omega)$. It cancels over the constants if and only if we have

$$\int_\Omega q(x)\,dx = 0.$$

  Hence, we have

$$(L^2(\Omega)/N)' = \left\{q \in L^2(\Omega), \int_\Omega q(x)\,dx = 0\right\} = L_0^2(\Omega).$$

- Finally, for all $p \in L^2(\Omega)$ and $u \in (H_0^1(\Omega))^d$, using integration by parts we obtain

$$\langle \nabla p, u\rangle_{H^{-1},H_0^1} = -(p, \operatorname{div} u)_{L^2(\Omega)},$$

  which shows that the adjoint of the operator ∇ is the operator $(-\operatorname{div})$.

In summary, we have shown that the operator $(-\operatorname{div})$ produces an isomorphism between $(H_0^1(\Omega))^d/Z$ and $L_0^2(\Omega)$. We denote the reciprocal isomorphism as Π_0. Since Z is a closed linear subspace of $(H_0^1(\Omega))^d$, then we know that the orthogonal projection operator on $Z^\perp$, denoted by $\mathcal{P}_{Z^\perp}$, induces an isomorphism from $(H_0^1(\Omega))^d/Z$ onto $Z^\perp$. Hence, the operator $\Pi = \mathcal{P}_{Z^\perp} \circ \Pi_0$ from $L_0^2(\Omega)$ into $(H_0^1(\Omega))^d$ satisfies the required properties.

□

The previous result implies the following well-known inf-sup inequality (also known the as Ladyzhenskaya–Babuška–Brezzi (LBB) inequality)

$$\inf_{\substack{p \in L_0^2(\Omega)\\ p\neq 0}} \left(\sup_{\substack{v\in (H_0^1(\Omega))^d\\ v\neq 0}} \frac{\displaystyle\int_\Omega p(\operatorname{div} v)\,dx}{\|p\|_{L^2}\|v\|_{H^1}} \right) > 0.$$

In fact this inequality is equivalent to the existence of a continuous right-inverse for the divergence operator from $L_0^2(\Omega)$ into $(H_0^1(\Omega))^d$.

Related inequalities are particularly important in the numerical analysis of the Stokes problem (see for instance [65, 55]).

Let us now give an alternative proof of Theorem IV.3.1 in a particular case, see for instance [26].

Proof (in the case where Ω is a domain of $\mathbb{R}^2$ of class $\mathcal{C}^{1,1}$).

We choose the usual counterclockwise orientation on $\mathbb{R}^2$ and for any two vectors $a, b \in \mathbb{R}^2$ we define $a \wedge b = a_1 b_2 - a_2 b_1 \in \mathbb{R}$. Let $q \in L^2_0(\Omega)$, and consider the following Laplace–Neumann problem

$$\begin{cases} -\Delta\psi = q, & \text{in } \Omega, \\ -\dfrac{\partial \psi}{\partial \nu} = 0, & \text{on } \partial\Omega. \end{cases} \tag{IV.9}$$

Since Ω is of class $\mathcal{C}^{1,1}$ and $\int_\Omega q\,dx = 0$, Theorem III.4.3 implies that there exists a solution $\psi \in H^2(\Omega)$ of (IV.9) and that $\|\psi\|_{H^2} \leq C\|q\|_{L^2}$. We set $v = -\nabla\psi$ so that $v \in (H^1(\Omega))^2$, $\operatorname{div} v = q$ and $\|v\|_{H^1} \leq C\|q\|_{L^2}$. Unfortunately, v does not vanish on the boundary because we only have $v \cdot \nu = -\partial\psi/\partial\nu = 0$.

We set $g = v \wedge \nu$ on $\partial\Omega$, so that $g \in H^{1/2}(\partial\Omega)$. We use now the lifting operator R that we built in Remark III.2.12. We set $\varphi = R(0, g) \in H^2(\Omega)$, so that φ satisfies

$$\begin{aligned} &\|\varphi\|_{H^2} \leq C\|g\|_{H^{1/2}(\partial\Omega)} \leq C'\|v\|_{H^1} \leq C''\|q\|_{L^2}, \\ &\gamma_0(\varphi) = 0, \\ &\gamma_\nu(\varphi) = \frac{\partial\varphi}{\partial\nu} = g. \end{aligned}$$

We introduce now

$$w = (\nabla\varphi)^\perp = \left(-\frac{\partial\varphi}{\partial x_2}, \frac{\partial\varphi}{\partial x_1}\right) \in (H^1(\Omega))^2.$$

- Since φ is constant on $\partial\Omega$, we have $\nabla_T\varphi = 0$ on $\partial\Omega$. Therefore we have

  $$\nabla\varphi = (\nu \cdot \nabla\varphi)\nu, \text{ on } \partial\Omega,$$

  so that $w = (\nabla\varphi)^\perp$ is orthogonal to ν on the boundary; that is, $w \cdot \nu = 0$ on $\partial\Omega$.
- By construction we have

  $$w \wedge \nu = (\nabla\varphi)^\perp \wedge \nu = -(\nabla\varphi) \cdot \nu = -g,$$

  and $\operatorname{div} w = 0$.

Gathering the properties of v and w, we see that $u = v + w$ satisfies

$$\begin{aligned}
\operatorname{div} u &= \operatorname{div}(v+w) = q,\\
\|u\|_{H^1} &\leq \|v\|_{H^1} + \|w\|_{H^1} \leq C\|q\|_{L^2},\\
u\cdot\nu &= \underbrace{v\cdot\nu}_{=0} + \underbrace{w\cdot\nu}_{=0} = 0,\\
u\wedge\nu &= \underbrace{v\wedge\nu}_{=g} + \underbrace{w\wedge\nu}_{=-g} = 0.
\end{aligned}$$

It follows that $u \in (H^1_0(\Omega))^2$ and the claim is proved because the definition of u linearly depends on q.

□

3.2 The space $H_{\mathrm{div}}(\Omega)$

Definition IV.3.2. *Let Ω be a Lipschitz domain of $\mathbb{R}^d$ with compact boundary. We introduce the space*

$$H_{\mathrm{div}}(\Omega) = \left\{u \in (L^2(\Omega))^d,\ \operatorname{div} u \in L^2(\Omega)\right\},$$

equipped with the graph norm $u \mapsto \left(\|u\|^2_{L^2} + \|\operatorname{div} u\|^2_{L^2}\right)^{1/2}$.

Moreover, we define $H_{\mathrm{div},0}(\Omega)$ to be the closure of $(\mathcal{D}(\Omega))^d$ in $H_{\mathrm{div}}(\Omega)$.

With the notation introduced in Section 2.10, we observe that $H_{\mathrm{div}}(\Omega)$ is nothing but the domain in L^2 of the divergence operator. More precisely, we have

$$H_{\mathrm{div}}(\Omega) = W^{2,2}_{\mathrm{div}}(\Omega).$$

We can then reformulate with this notation the various results obtained in Section 2.10:

- The space $(\mathcal{C}^\infty_c(\overline{\Omega}))^d$ is dense in $H_{\mathrm{div}}(\Omega)$ (see Theorem III.2.42).
- There exists a continuous trace operator γ_ν from $H_{\mathrm{div}}(\Omega)$ into $H^{-1/2}(\partial\Omega)$ such that $\gamma_\nu(u) = u\cdot\nu$ for any $u \in (\mathcal{C}^\infty_c(\overline{\Omega}))^d$. Moreover the following Stokes formula holds,

$$\begin{aligned}
&\int_\Omega u\cdot\nabla w\, dx + \int_\Omega w \operatorname{div} u\, dx\\
=&\langle\gamma_\nu(u), \gamma_0(w)\rangle_{H^{-1/2},H^{1/2}},\ \forall u \in H_{\mathrm{div}}(\Omega), \forall w \in H^1(\Omega).
\end{aligned} \tag{IV.10}$$

  This comes from Theorem III.2.43.
- The space $H_{\mathrm{div},0}(\Omega)$ is the kernel of the trace operator γ_ν (see Theorem III.2.45).

To conclude we state an extension of the above results to tensor fields. We identify $(H_{\mathrm{div}}(\Omega))^d$ with the set of tensors $\sigma \in (L^2(\Omega))^{d\times d}$ whose divergence belongs to $(L^2(\Omega))^d$. By reasoning row by row, we can define a continuous linear operator still denoted by $\gamma_\nu : (H_{\text{div}}(\Omega))^d \to (H^{-1/2}(\partial\Omega))^d$ such that $\gamma_\nu(\sigma) = \sigma.\nu$ for any smooth tensor field σ. We immediately obtain the following result.

Lemma IV.3.3. *Let Ω be a bounded Lipschitz domain of $\mathbb{R}^d$. For any $v \in (H^1(\Omega))^d$ and any $\sigma \in (H_{\text{div}}(\Omega))^d$, we have the following Stokes formula*

$$\int_\Omega \sigma : \nabla v \, dx + \int_\Omega v \cdot \operatorname{div} \sigma \, dx = \langle \sigma.\nu, v \rangle_{H^{-1/2}, H^{1/2}}.$$

3.3 Divergence-free vector fields. Leray decomposition

Let us consider the set $\mathscr{V}$ of smooth divergence-free vector fields which are compactly supported in Ω,

$$\mathscr{V} = \left\{ \varphi \in (\mathcal{D}(\Omega))^d, \operatorname{div} \varphi = 0 \right\}.$$

We can then define the following fundamental spaces: V the closure of $\mathscr{V}$ in $(H^1_0(\Omega))^d$, and H the closure of $\mathscr{V}$ in $(L^2(\Omega))^d$. These two spaces are, of course, Hilbert spaces equipped with the scalar products respectively induced by those of $(H^1_0(\Omega))^d$ and of $(L^2(\Omega))^d$.

We can now characterise these two spaces in a more precise way.

Lemma IV.3.4. *Let Ω be a bounded Lipschitz domain of $\mathbb{R}^d$. We have*

$$V = \left\{ v \in (H^1_0(\Omega))^d, \quad \operatorname{div} v = 0 \right\}.$$

Proof.

We denote the space $\left\{ v \in (H^1_0(\Omega))^d, \quad \operatorname{div} v = 0 \right\}$ as $\widetilde{V}$. We need to prove the equality $V = \widetilde{V}$.

It is clear that $\mathscr{V} \subset \widetilde{V}$ and that $\widetilde{V}$ is closed in $(H^1_0(\Omega))^d$, hence the inclusion $\subset$ is evident. To demonstrate the other inclusion, we proceed by duality showing that any continuous linear functional on $\widetilde{V}$ which cancels on $\mathscr{V}$ also cancels on $\widetilde{V}$ (which is sufficient according to Proposition II.2.2).

Hence, if $f \in \left(H^{-1}(\Omega)\right)^d \cap \mathscr{V}^\perp$, then from Theorem IV.2.4, f is the gradient of a function p of $L^2(\Omega)$. We then have for any $v \in \widetilde{V}$,

$$\langle f, v \rangle_{H^{-1}, H^1_0} = \langle \nabla p, v \rangle_{H^{-1}, H^1_0} = -(p, \operatorname{div} v)_{L^2} = 0.$$

We have hence proved that f cancels on $\tilde{V}$, which concludes the proof.

□

Theorem IV.3.5. *Let Ω be a connected, bounded, Lipschitz domain of $\mathbb{R}^d$. Then we have*

$$H = \left\{u \in (L^2(\Omega))^d,\ \operatorname{div} u = 0,\ \gamma_\nu(u) = 0\right\},$$

and the orthogonal of H in $(L^2(\Omega))^d$ satisfies

$$H^\perp = \left\{u \in (L^2(\Omega))^d,\ \exists p \in H^1(\Omega),\ u = \nabla p\right\}. \tag{IV.11}$$

Proof.

- We first show the second equality which characterises the orthogonal of H in $(L^2(\Omega))^d$. For this, we denote

  $$\mathscr{H}^\star = \left\{u \in (L^2(\Omega))^d,\ \exists p \in H^1(\Omega),\ u = \nabla p\right\}.$$

  Let $u \in \mathscr{H}^\star$; then for all v of $\mathscr{V}$ we have

  $$(u, v)_{L^2} = \int_\Omega u \cdot v\, dx = \int_\Omega \nabla p \cdot v\, dx = -\int_\Omega p \operatorname{div} v\, dx = 0.$$

  This shows, by the density of $\mathscr{V}$ in H for the topology of $(L^2(\Omega))^d$, that $u \in H^\perp$, and that as a consequence $\mathscr{H}^\star \subset H^\perp$. Reciprocally, if $u \in H^\perp$ then, in particular, for all v of $\mathscr{V}$, we have

  $$\langle u, v\rangle_{H^{-1}, H_0^1} = (u, v)_{L^2} = 0,$$

  and by consequence, from Theorem IV.2.4, there exists $p \in L_0^2(\Omega)$ such that $u = \nabla p$. Since $u \in (L^2(\Omega))^d$, we indeed have $p \in H^1(\Omega)$. Hence, $u \in \mathscr{H}^\star$.
- Let us now prove the first equality. We denote

  $$\mathscr{H} = \left\{u \in (L^2(\Omega))^d,\ \operatorname{div} u = 0,\ \gamma_\nu(u) = 0\right\}.$$

  Let us demonstrate the inclusion $H \subset \mathscr{H}$. If $u \in H$, there exists a sequence $(u_n)_n$ of elements of $\mathscr{V}$, which converges towards u for the topology of $(L^2(\Omega))^d$. The functions u_n are divergence-free, thus it is the same for its limit u (in $\mathcal{D}'(\Omega)$). Consequently $u \in H_{\mathrm{div}}(\Omega)$ and

  $$\lim_{n\to\infty} \|u - u_n\|_{H_{\mathrm{div}}(\Omega)} = 0.$$

  From the continuity of the normal trace operator γ_ν, we can deduce that

  $$\lim_{n\to\infty} \|\gamma_\nu(u) - \gamma_\nu(u_n)\|_{H^{-1/2}(\partial\Omega)} = 0,$$

  and hence $\gamma_\nu(u) = 0$. This exactly states that u belongs to $\mathscr{H}$.
  We now know that H is included in $\mathscr{H}$, and therefore we denote the orthogonal complement of H into $\mathscr{H}$ as $\mathscr{H}^{\star\star}$. Let $u \in \mathscr{H}^{\star\star} \subset H^\perp$. By the characterisation (IV.11) we know that there exists a p in $H^1(\Omega)$ such that $u = \nabla p$, and since $u \in \mathscr{H}$, we see that p satisfies the problem

$$\begin{cases} \Delta p = \operatorname{div} u = 0, & \text{in } \Omega, \\ \dfrac{\partial p}{\partial \nu} = \gamma_\nu(u) = 0, & \text{on } \partial\Omega. \end{cases}$$

Then, using (IV.10) we can deduce that

$$\int_\Omega |\nabla p|^2 \, dx = 0,$$

and thus $u = \nabla p = 0$. Hence $\mathscr{H}^{\star\star} = \{0\}$. Therefore $\mathscr{H} = H$.

□

Hence, the space $(L^2(\Omega))^d$ decomposes orthogonally into two subspaces: the space H of vector fields with zero divergence and for which the normal trace $\gamma_\nu(u)$ is zero, and the space $H^\perp$ of gradient fields of functions in $H^1(\Omega)$.

Definition IV.3.6. *The decomposition*

$$(L^2(\Omega))^d = H \overset{\perp}{\oplus} H^\perp,$$

is called the Leray decomposition *of the space $(L^2(\Omega))^d$. The orthogonal projection from $(L^2(\Omega))^d$ onto H is known as the Leray projection and denoted as $\mathcal{P}$.*

When we work in all the space $\Omega = \mathbb{R}^d$, we can give an explicit expression for the Leray projection in Fourier variables. Unfortunately, in a bounded open set such an analysis is much more complicated. In particular, because of the boundary conditions, it is easy to see that, contrary to the case for the whole space or the periodic case, the Leray projection $\mathcal{P}$ does not commute with the usual differential operators. Furthermore, the operator $\mathcal{P}$ is nonlocal; the value of $\mathcal{P}u$ depends on the values of the function u on all Ω and not only in the neighborhood of the point at which we are looking.

In the case where the domain is smooth enough, we can prove that the Leray projection $\mathcal{P}$ maps $H^1(\Omega)^d$ into itself. More precisely, we have the following result.

Proposition IV.3.7. *Assume that Ω is a bounded domain of class $\mathcal{C}^{1,1}$. Then, for any $u \in (H^1(\Omega))^d$ we have $\mathcal{P}u \in (H^1(\Omega))^d$ and*

$$\|\mathcal{P}u\|_{H^1} \leq C\|u\|_{H^1},$$

where C depends only on Ω.

Proof.

By definition, we write $u = \mathcal{P}u + \nabla p$ where $p \in H^1(\Omega)$ and $\operatorname{div}(\mathcal{P}u) = 0, \gamma_\nu(\mathcal{P}u) = 0$. By taking the divergence of this equation, we get

$$\Delta p = \operatorname{div} u \in L^2(\Omega),$$

because we assumed that $u \in (H^1(\Omega))^d$. Moreover, we have

$$\gamma_\nu(\nabla p) = \gamma_\nu(u) + \gamma_\nu(\mathcal{P}u) = \gamma_\nu(u) = \gamma_0(u) \cdot \nu \in H^{1/2}(\partial\Omega).$$

Therefore, p is a solution to the Laplace equation with a source term in $L^2(\Omega)$ and Neumann boundary data which lie in $H^{1/2}(\partial\Omega)$. The elliptic regularity property for the Laplace–Neumann problem (see Theorem III.4.3) implies that $p \in H^2(\Omega)$ and that

$$\|p\|_{H^2} \leq C(\|\operatorname{div} u\|_{L^2} + \|\gamma_0(u) \cdot \nu\|_{H^{1/2}}) \leq C'\|u\|_{H^1}.$$

Finally, we get $\mathcal{P}u = u - \nabla p \in (H^1(\Omega))^d$ and $\|\mathcal{P}u\|_{H^1} \leq C\|u\|_{H^1}$.

□

4 The curl operator and related spaces

All the material in this section is given in the case of the space dimension 3 because this is the natural setting for the definition of the curl operator. Nevertheless, most of the results can be translated to the 2D case, by adapting the definition of this operator; in fact it is then needed to define two different curl-like operators in that case, one from $\mathbb{R}^2$ into $\mathbb{R}$, another from $\mathbb{R}$ into $\mathbb{R}^2$.

We have chosen to focus only on the main results that are needed in the sequel of this book. Many more results can be found in the references [54, 65, 47, 6, 60]. The main property we are interested in is the one given in Theorem IV.4.13 which says that, under suitable assumptions, a divergence-free vector field in $(H^1(\Omega))^3$ can be written as the curl of a vector potential field in $(H^2(\Omega))^3$. This result is particularly useful for studying the steady Navier–Stokes equations with nonhomogeneous boundary conditions, in Section 3 of Chapter V.

We denote by $S^2 = \{x \in \mathbb{R}^3, |x| = 1\}$ the unit sphere of $\mathbb{R}^3$. We recall that this sphere is simply connected (whereas the unit sphere of $\mathbb{R}^2$ is not). This means that any continuous closed curve on S^2 can be continuously deformed into a single point.

4.1 Poincaré's theorems

In this section we prove some results which are simple versions of the famous Poincaré theorem saying that any closed differential form in a star-shaped domain is an exact differential form.

Lemma IV.4.1. *Let $\mathcal{U} \subset \mathbb{R}^3$ be an open set which is star-shaped with respect to the origin and $k \geq 1$ an integer.*

1. *For any* $\Phi \in (\mathcal{C}^k(\mathcal{U}))^3$*, such that* $\operatorname{curl}\Phi = 0$ *in* $\mathcal{U}$*, the formula*

$$\psi(x) = \int_0^1 \Phi(tx)\cdot x\,dt, \tag{IV.12}$$

   defines a function $\psi \in \mathcal{C}^{k+1}(\mathcal{U})$ *such that* $\Phi = \nabla\psi$*.*
2. *For any* $\Phi \in (\mathcal{C}^k(\mathcal{U}))^3$*, such that* $\operatorname{div}\Phi = 0$ *in* $\mathcal{U}$*, the formula*

$$\Psi(x) = \int_0^1 t\Phi(tx)\wedge x\,dt,$$

   defines a function $\Psi \in (\mathcal{C}^k(\mathcal{U}))^3$ *such that* $\Phi = \operatorname{curl}\Psi$*.*

One can easily modify the above formulas by translation in the case where $\mathcal{U}$ is star-shaped with respect to a point different from the origin.

Proof.

1. Equation (IV.12) clearly defines a function in $\mathcal{C}^k(\mathcal{U})$. Let us compute its gradient

$$\begin{aligned}\frac{\partial\psi}{\partial x_i} &= \int_0^1 \frac{\partial\Phi}{\partial x_i}(tx)\cdot xt\,dt + \int_0^1 \Phi(tx)\cdot e_i\,dt\\ &= \sum_{j=1}^3 \int_0^1 \frac{\partial\Phi_j}{\partial x_i}(tx)x_j t\,dt + \int_0^1 \Phi_i(tx)\,dt.\end{aligned}$$

   Since $\operatorname{curl}\Phi = 0$, we have $\partial\Phi_j/\partial x_i = \partial\Phi_i/\partial x_j$ and thus

$$\begin{aligned}\frac{\partial\psi}{\partial x_i} &= \sum_{j=1}^3 \int_0^1 \frac{\partial\Phi_i}{\partial x_j}(tx)x_j t\,dt + \int_0^1 \Phi_i(tx)\,dt\\ &= \int_0^1 \frac{d}{dt}\left(t\Phi_i(tx)\right)\,dt = \Phi_i(x).\end{aligned}$$

   Therefore $\nabla\psi = \Phi$ and $\psi \in \mathcal{C}^{k+1}(\mathcal{U})$.
2. Let $i,j,k \in \{1,2,3\}$ such that $e_i \wedge e_j = e_k$, we have to show that

$$\Phi_k = \frac{\partial\Psi_j}{\partial x_i} - \frac{\partial\Psi_i}{\partial x_j}.$$

   To this end, we write

$$\Psi_i(x) = \int_0^1 tx_k\Phi_j(tx) - tx_j\Phi_k(tx)\,dt,$$

   and then

$$\frac{\partial \Psi_i}{\partial x_j}(x) = \int_0^1 t^2 x_k \frac{\partial \Phi_j}{\partial x_j}(tx) - t^2 x_j \frac{\partial \Phi_k}{\partial x_j}(tx) - t\Phi_k(tx)\, dt.$$

Similarly we get

$$\frac{\partial \Psi_j}{\partial x_i}(x) = \int_0^1 -t^2 x_k \frac{\partial \Phi_i}{\partial x_i}(tx) + t^2 x_i \frac{\partial \Phi_k}{\partial x_i}(tx) + t\Phi_k(tx)\, dt,$$

and it follows that

$$\begin{aligned}\frac{\partial \Psi_j}{\partial x_i}(x) - \frac{\partial \Psi_i}{\partial x_j}(x) &= \int_0^1 -t^2 x_k \left(\frac{\partial \Phi_i}{\partial x_i}(tx) + \frac{\partial \Phi_j}{\partial x_j}(tx)\right) dt \\ &\quad + \int_0^1 t^2 x_j \frac{\partial \Phi_k}{\partial x_j}(tx) + t^2 x_i \frac{\partial \Phi_k}{\partial x_i}(tx) + 2t\Phi_k(tx)\, dt.\end{aligned}$$

Since $\operatorname{div} \Phi = 0$, we have

$$\frac{\partial \Phi_i}{\partial x_i} + \frac{\partial \Phi_j}{\partial x_j} = -\frac{\partial \Phi_k}{\partial x_k},$$

and then

$$\begin{aligned}\frac{\partial \Psi_j}{\partial x_i}(x) - \frac{\partial \Psi_i}{\partial x_j}(x) &= \int_0^1 t^2 x \cdot \nabla \Phi_k(tx) + 2t\Phi_k(tx)\, dt \\ &= \int_0^1 \frac{d}{dt}\left(t^2 \Phi_k(tx)\right) dt = \Phi_k(x).\end{aligned}$$

Finally we have shown that $\operatorname{curl} \Psi = \Phi$.

$\square$

Lemma IV.4.2. *Let Ω be a simply connected open set in $\mathbb{R}^3$ and $k \geq 1$ an integer. For any $\Phi \in (\mathcal{C}^k(\Omega))^3$ satisfying $\operatorname{curl} \Phi = 0$ in Ω, there exists $\psi \in (\mathcal{C}^{k+1}(\Omega))^3$ such that $\Phi = \nabla \psi$.*

Proof.

Let $x_0 \in \Omega$. We define U to be the largest connected open set containing x_0 and included in Ω for which such a function ψ exists. By Lemma IV.4.1, we already know that U is not empty because we can find such a ψ in any ball included in Ω and centred at x_0. We denote by $\psi \in \mathcal{C}^{k+1}(U)$ a map such that $\Phi = \nabla \psi$ in U.

We now show by contradiction that $U = \Omega$.

- Assume that $V = \Omega \setminus U$ is not empty; then we have $\partial U \cap V \neq \emptyset$. Indeed, if $\partial U \cap V$ is empty, we have $V = \Omega \setminus \overline{U}$ and thus V is open. Therefore we have $\Omega = U \cup V$ and $U \cap V = \emptyset$, but this is impossible because Ω is connected.

Let $x \in \partial U \cap V$, and $B \subset \Omega$ be a ball centred at x. Since $x \in \partial U$, we have that $U \cap B \neq \emptyset$ and then we can find a point $\tilde{x} \in U \cap B$.

- By Lemma IV.4.1, there exists $\widetilde{\psi} \in \mathcal{C}^{k+1}(B)$ such that $\Phi = \nabla \widetilde{\psi}$ in B and, we can add a constant to $\widetilde{\psi}$ to ensure that $\widetilde{\psi}(\tilde{x}) = \psi(\tilde{x})$.
- Let us now show that $\psi = \widetilde{\psi}$ on $U \cap B$. Let y be any point in $U \cap B$. Since U is a connected open set, there exists a $\mathcal{C}^2$ path $\gamma_0 : [0,1] \to U$ such that $\gamma_0(0) = \tilde{x}$ and $\gamma_0(1) = y$. In a similar way there exists a $\mathcal{C}^2$ path $\gamma_1 : [0,1] \to B$ such that $\gamma_1(0) = \tilde{x}$ and $\gamma_1(1) = y$.
  The domain Ω being open and simply connected, there exists a $\mathcal{C}^2$ map $\gamma : [0,1] \times [0,1] \to \Omega$ such that $\gamma(0,.) = \gamma_0$, $\gamma(1,.) = \gamma_1$, $\gamma(.,0) = \tilde{x}$ and $\gamma(.,1) = y$. Let us now define the following quantity
$$I(s) = \int_0^1 \Phi(\gamma(s,t))\partial_t \gamma(s,t)\,dt,\ \forall s \in [0,1].$$
  - For $s = 0$, we have $\gamma(0,.) = \gamma_0$ which is a path in U and thus we have
$$\begin{aligned} I(0) &= \int_0^1 \Phi(\gamma_0(t))\gamma_0'(t)\,dt = \int_0^1 (\nabla\psi)(\gamma_0(t))\gamma_0'(t)\,dt \\ &= \psi(\gamma_0(1)) - \psi(\gamma_0(0)) = \psi(y) - \psi(\tilde{x}). \end{aligned}$$
  - For $s = 1$, we have $\gamma(1,.) = \gamma_1$ which is a path in B and thus we have
$$\begin{aligned} I(1) &= \int_0^1 \Phi(\gamma_1(t))\gamma_1'(t)\,dt = \int_0^1 (\nabla\widetilde{\psi})(\gamma_1(t))\gamma_1'(t)\,dt \\ &= \widetilde{\psi}(\gamma_0(1)) - \widetilde{\psi}(\gamma_0(0)) = \widetilde{\psi}(y) - \widetilde{\psi}(\tilde{x}). \end{aligned}$$
  - Let us compute the derivative $I'(s)$:
$$\begin{aligned} I'(s) = &\int_0^1 \big((\nabla\Phi)(\gamma(s,t)).(\partial_s\gamma(s,t))\big).\partial_t\gamma(s,t)\,dt \\ &+ \int_0^1 \Phi(\gamma(s,t)).\partial_s\partial_t\gamma(s,t)\,dt. \end{aligned}$$

  The fact that $\operatorname{curl}\Phi = 0$ implies that $\nabla\Phi$ is a symmetric matrix and then we find
$$\begin{aligned} I'(s) = &\int_0^1 \big((\nabla\Phi)(\gamma(s,t)).(\partial_t\gamma(s,t))\big).\partial_s\gamma(s,t)\,dt \\ &+ \int_0^1 \Phi(\gamma(s,t)).\partial_s\partial_t\gamma(s,t)\,dt \\ = &\int_0^1 \frac{d}{dt}\left(\Phi(\gamma(s,t)).\partial_s\gamma(s,t)\right)\,dt \\ = &\Phi(\gamma(s,1)).\partial_s\gamma(s,1) - \Phi(\gamma(s,0)).\partial_s\gamma(s,0) = 0, \end{aligned}$$

because $s \mapsto \gamma(s,1)$ and $s \mapsto \gamma(s,0)$ are constant.

– As a consequence of the previous computation we have $I(0) = I(1)$; that is,
$$\widetilde{\psi}(y) - \widetilde{\psi}(\tilde{x}) = \psi(y) - \psi(\tilde{x}).$$
By construction we have $\widetilde{\psi}(\tilde{x}) = \psi(\tilde{x})$ and then we deduce that $\psi(y) = \widetilde{\psi}(y)$.

- We have shown that $\psi = \widetilde{\psi}$ on $U \cap B$ and thus the map $\overline{\psi}$ defined on $U \cup B$ by
$$\overline{\psi} = \begin{cases} \psi, & \text{on } U, \\ \widetilde{\psi}, & \text{on } B, \end{cases}$$
is well-defined, smooth, and satisfies $\nabla\overline{\psi} = \Phi$ on $U \cup B$ which is open and connected. This is a contradiction with the maximality assumption on U.

□

Lemma IV.4.3. *Let $\mathcal{U} \subset \mathbb{R}^3$ be an open set in $\mathbb{R}^3$ which is star-shaped with respect to the origin and $k \geq 1$ an integer. For any $\Phi \in (\mathcal{C}_c^k(\mathcal{U}))^3$ such that* $\operatorname{div}\Phi = 0$, *there exists $\Psi \in (\mathcal{C}_c^k(\mathcal{U}))^3$ such that $\Phi = \operatorname{curl}\Psi$.*

Proof.

Since Φ is compactly supported, we can assume without loss of generality that $\mathcal{U}$ is bounded; then we know by Lemma IV.4.1 that there exists a $\widetilde{\Psi} \in (\mathcal{C}^k(\mathcal{U}))^3$ such that $\Phi = \operatorname{curl}\widetilde{\Psi}$ in $\mathcal{U}$, but $\widetilde{\Psi}$ is not necessarily compactly supported.

Let $\mathcal{U}_0 \subset \overline{\mathcal{U}_0} \subset \mathcal{U}$ be an open set star-shaped with respect to 0 which contains the support of Φ. By construction, we have $\operatorname{curl}\widetilde{\Psi} = 0$ in $\mathcal{U} \setminus \overline{\mathcal{U}_0}$.

Note that there exist two continuous maps $\xi_0, \xi : S^2 \mapsto]0,+\infty[$ with $\xi_0 < \xi$ and such that we have
$$\mathcal{U}_0 = \left\{ x \in \mathbb{R}^3,\ |x| < \xi_0\left(\frac{x}{|x|}\right) \right\}, \text{ and } \mathcal{U} = \left\{ x \in \mathbb{R}^3,\ |x| < \xi\left(\frac{x}{|x|}\right) \right\}.$$
In particular, the set $\widetilde{\mathcal{U}_0} = \mathcal{U} \setminus \overline{\mathcal{U}_0}$ satisfies
$$\widetilde{\mathcal{U}_0} = \left\{ x \in \mathbb{R}^3,\ \ \xi_0\left(\frac{x}{|x|}\right) < |x| < \xi\left(\frac{x}{|x|}\right) \right\},$$
and is then homeomorphic to $S^2 \times]0,1[$. As a consequence, $\widetilde{\mathcal{U}_0}$ is simply connected. By Lemma IV.4.2, there exists $\widetilde{\psi} \in \mathcal{C}^{k+1}(\widetilde{\mathcal{U}_0})$ such that $\widetilde{\Psi} = \nabla\widetilde{\psi}$ in $\widetilde{\mathcal{U}_0}$. Let $\mathcal{U}_1$ be any open set star-shaped with respect to 0 and such that $\overline{\mathcal{U}_0} \subset \mathcal{U}_1 \subset \overline{\mathcal{U}_1} \subset \mathcal{U}$. We can find a function $\psi \in \mathcal{C}^{k+1}(\mathcal{U})$ such that $\psi = \widetilde{\psi}$ in $\mathcal{U} \setminus \mathcal{U}_1$ and we define now $\Psi = \widetilde{\Psi} - \nabla\psi$.

By construction, we have $\operatorname{curl}\Psi = \Phi$ in $\mathcal{U}$ and $\Psi = 0$ in $\mathcal{U} \setminus \overline{\mathcal{U}_1}$, so that the claim is proved.

□

4.2 The space $H_{\mathrm{curl}}(\Omega)$

We can now study functional spaces of interest in which the curl operator plays a key role, just as we did in Section 3.2 with the space $H_{\mathrm{div}}(\Omega)$.

4.2.1 Definitions and first properties

Definition IV.4.4. *Let Ω be a Lipschitz domain of $\mathbb{R}^3$ with compact boundary. We define the space*

$$H_{\mathrm{curl}}(\Omega) = \left\{u \in (L^2(\Omega))^3, \quad \operatorname{curl} u \in (L^2(\Omega))^3\right\},$$

embedded with the norm

$$\|u\|_{H_{\mathrm{curl}}} = \left(\|u\|_{L^2}^2 + \|\operatorname{curl} u\|_{L^2}^2\right)^{1/2}.$$

It is straightforward to see that $H_{\mathrm{curl}}(\Omega)$ is a Hilbert space.

Moreover, we define $H_{\mathrm{curl},0}(\Omega)$ to be the closure of $(\mathcal{D}(\Omega))^3$ in $H_{\mathrm{curl}}(\Omega)$.

With the notation introduced in Section 2.10 of Chapter III, we observe that $H_{\mathrm{curl}}(\Omega)$ is nothing but the domain in L^2 of the curl operator. More precisely, we have

$$H_{\mathrm{curl}}(\Omega) = W^{2,2}_{\mathrm{curl}}(\Omega).$$

We can then reformulate with this notation the various results obtained in Section 2.10:

- The space $(\mathcal{C}_c^\infty(\overline{\Omega}))^3$ is dense in $H_{\mathrm{curl}}(\Omega)$ (see Theorem III.2.42).
- There exists a continuous trace operator $\gamma_{\wedge\nu}$ defined from $H_{\mathrm{curl}}(\Omega)$ into $(H^{-1/2}(\partial\Omega))^3$ such that $\gamma_{\wedge\nu}(u) = u \wedge \nu$ for any $u \in (\mathcal{C}_c^\infty(\overline{\Omega}))^d$. Moreover the following Stokes formula holds

$$\int_\Omega u \cdot (\operatorname{curl}\psi)\,dx = \int_\Omega (\operatorname{curl} u)\cdot\psi\,dx + \langle\gamma_{\wedge\nu}(u), \gamma_0(\psi)\rangle_{H^{-1/2},H^{1/2}},\ \forall\psi \in (H^1(\Omega))^3. \tag{IV.13}$$

  This comes from Theorem III.2.43. Note also that the trace $\gamma_{\wedge\nu}(u)$ is a tangent vector field in the following weak sense,

$$\langle\gamma_{\wedge\nu}(u), \gamma_0(\psi)\rangle_{H^{-1/2},H^{1/2}} = 0,\ \forall\psi \in (H^1(\Omega))^3, \text{ such that } \gamma_0(\psi)\wedge\nu = 0.$$

- The space $H_{\mathrm{curl},0}(\Omega)$ is the kernel of the trace operator $\gamma_{\wedge\nu}$ (see Theorem III.2.45).

We notice that for any function $p \in H^1(\Omega)$, we have $\operatorname{curl}(\nabla p) = 0$ and thus $\nabla p \in H_{\mathrm{curl}}(\Omega)$. Basically, we can imagine that $\nabla p \wedge \nu$ is nothing but the tangential part of the gradient of p, so that the condition $\nabla p \wedge \nu = 0$ should imply that p is locally constant on the boundary of Ω. Nevertheless, since the trace is understood in a weak sense this is not so obvious. The following proposition gives a detailed proof of this property when the domain is smooth enough.

Proposition IV.4.5. *Let Ω be a bounded domain in $\mathbb{R}^3$ of class $\mathcal{C}^{1,1}$. Let $p \in H^1(\Omega)$ such that $\gamma_{\wedge\nu}(\nabla p) = 0$; then $\gamma_0(p)$ is a constant on each connected component of $\partial\Omega$.*

Proof.

By (IV.13), we deduce that

$$\int_\Omega \nabla p \cdot \operatorname{curl} \psi \, dx = 0, \, \forall \psi \in (H^1(\Omega))^3. \tag{IV.14}$$

We show from this formula that $\gamma_0(p)$ is locally constant on $\partial\Omega$, and the claim is thus proved.

Let us consider any point $x_0 \in \partial\Omega$ and let Σ, Σ^* be two connected bounded open neighborhoods of x_0 in $\partial\Omega$ such that $\overline{\Sigma} \subset \Sigma^*$ (the closure being understood in the sense of the topological space $\partial\Omega$). Let then g be any function in $\mathcal{C}_c^\infty(\Sigma)$ such that $\int_{\partial\Omega} g \, d\sigma = \int_\Sigma g \, d\sigma = 0$. We extend g by 0 on $\partial\Omega \setminus \Sigma$. The claim is proved if we manage to show that for any such g, we have $\int_\Sigma \gamma_0(p) g \, d\sigma = 0$. Indeed, we deduce that $\gamma_0(p)$ is constant on Σ which is an open neighborhood of an arbitrary point of $\partial\Omega$.

We use the notation and the results of Section 3 of Chapter III. In particular, we consider the coordinate system

$$\psi : (\sigma, s) \in \Gamma_{s_0} \times [0, s_0] \mapsto \overline{\mathcal{O}^\rho_{s_0}},$$

which can be extended to $\Gamma_{s_0} \times [-s_0, s_0]$. We introduce

$$\Sigma_{s_0} = \psi(., s_0)^{-1}(\Sigma) \text{ and } \Sigma^*_{s_0} = \psi(., s_0)^{-1}(\Sigma^*),$$

and, for any ξ_1, ξ_2,

$$\begin{aligned} \mathcal{U}_{\xi_1,\xi_2} &= \psi(\Sigma_{s_0} \times]\xi_1, \xi_2[), \\ \mathcal{U}^*_{\xi_1,\xi_2} &= \psi(\Sigma^*_{s_0} \times]\xi_1, \xi_2[), \\ \Sigma_\xi &= \psi(\Sigma_{s_0} \times \{\xi\}). \end{aligned}$$

- Let us define in $\mathcal{O}^\rho_{s_0}$, the vector field

$$F(x) = \frac{\mathcal{J}(P(x))}{\mathcal{J}(x)} g(P(x)) \nu(x), \, \forall x \in \mathcal{O}^\rho_{s_0}.$$

By construction, the tangential part F_T of F is zero. Thus, Proposition III.3.23 gives

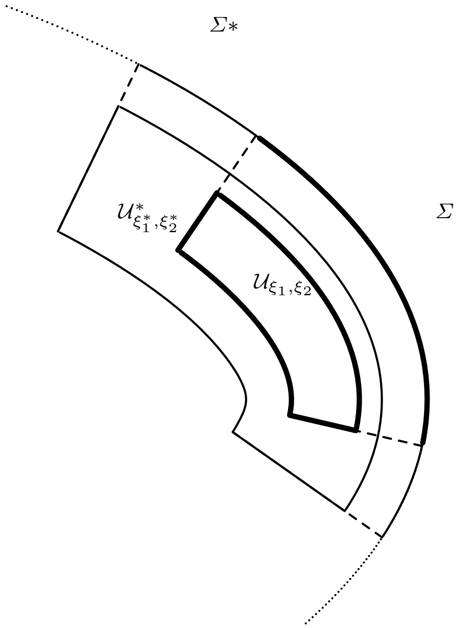

Fig. IV.1 Notations for the proof of Proposition IV.4.5

$$\operatorname{div} F = \nabla_N F_N + \frac{\nabla_N \mathcal{J}}{\mathcal{J}} F_N = \frac{1}{\mathcal{J}} \nabla_N (\mathcal{J} F_N).$$

By definition of F, we have

$$(\mathcal{J} F_N)(x) = \mathcal{J}(P(x)) g(P(x));$$

that is, the product $\mathcal{J} F_N$ only depends on the projection of x on the boundary. It follows from (III.76) that $\nabla_N(\mathcal{J} F_N) = 0$ and therefore that $\operatorname{div} F = 0$.

By construction, F is smooth in Ω, divergence-free, and satisfies $F \cdot \nu = g$ on the boundary $\partial\Omega$. Moreover, since g is supported in Σ_0, we see that F is supported in $\overline{\mathcal{U}_{0,s_0}}$

- Let us consider a truncation function $\beta \in \mathcal{C}_c^\infty(]-2s_0/3, 2s_0/3[)$ such that $\beta = 1$ on $]-s_0/3, s_0/3[$. We define $F^*(x) = \beta(\rho(x))F(x)$. We see that F^* still satisfies the boundary condition $F^* \cdot \nu = g$, is compactly supported in $\overline{\mathcal{U}_{0,2s_0/3}}$, and satisfies the divergence-free condition on $\mathcal{U}^*_{0,s_0/3}$. Unfortunately, F^* is no longer divergence-free on the whole set $\mathcal{U}^*_{0,s_0}$.
- We set $f^* = \operatorname{div}(F^*)$, which is compactly supported in $\mathcal{U}_{s_0/3,2s_0/3}$ and satisfies

$$\int_{\mathcal{U}_{\sigma_0/3,2s_0/3}} f^* \, dx = \int_{\Sigma_{s_0/3}} F^* \cdot \nu \, d\sigma = \int_{\Sigma} F^* \cdot \nu \, d\sigma = \int_{\Sigma} g \, d\sigma = 0,$$

because, by construction, $\operatorname{div} F^*$ vanishes in $\mathcal{U}_{0,s_0/3}$.
Hence, by Theorem IV.3.1, we know that there is a $\Phi^* \in (H_0^1(\mathcal{U}_{s_0/3,2s_0/3}))^3$ such that $\operatorname{div}\Phi^* = f^*$. We extend Φ^* by 0 outside its domain of definition and we set now $\overline{F} = F^* - \Phi^*$ which is divergence-free and compactly supported in $\overline{\mathcal{U}_{0,2s_0/3}}$ and moreover satisfies $\overline{F} = g\nu$ on Σ.

- By a similar construction in $\overline{\Omega}^c$, we may finally suppose that $\overline{F}$ is divergence-free and compactly supported in $\mathcal{U}_{-2s_0/3,2s_0/3}$. We introduce a mollifying kernel $\eta \in \mathcal{C}_c^\infty(B(0,1))$ and $\eta_\varepsilon = \varepsilon^{-3}\eta(./\varepsilon)$ then we define
$$\overline{F}_\varepsilon = \overline{F} \star \eta_\varepsilon,$$
so that $\overline{F}_\varepsilon$ is smooth, divergence-free, and compactly supported in $\mathcal{U}^*_{-s_0,s_0}$, as soon as ε is sufficiently small.
We may now assume that Σ and s_0 are sufficiently small so that $\mathcal{U}^*_{-s_0,s_0}$ is star-shaped since $\partial\Omega$ is smooth enough. Therefore, by using Lemma IV.4.3 we deduce that there exists $\Psi_\varepsilon \in (\mathcal{C}_c^\infty(\mathcal{U}^*_{-s_0,s_0}))^3$ such that $\overline{F_\varepsilon} = \operatorname{curl}\Psi_\varepsilon$.
- We extend Ψ_ε by 0 on the whole space $\mathbb{R}^3$ and we use it as a test function in (IV.14) to get
$$\begin{aligned}0 &= \int_\Omega \nabla p \cdot \operatorname{curl}\Psi_\varepsilon\, dx = \int_\Omega \nabla p \cdot \overline{F}_\varepsilon\, dx \\ &= -\int_\Omega p \underbrace{\operatorname{div}(\overline{F}_\varepsilon)}_{=0}\, dx + \int_{\Sigma^*} \gamma_0(p)\overline{F}_\varepsilon \cdot \nu\, d\sigma.\end{aligned}$$
Since $\overline{F}$ is smooth in the neighborhood of Σ^*, we can pass to the limit as $\varepsilon \to 0$ in this formula and find that
$$0 = \int_\Sigma \gamma_0(p)\overline{F}\cdot\nu\, d\sigma = \int_\Sigma \gamma_0(p) g\, d\sigma.$$
This property holds for any g whose mean value on Σ is zero, therefore we deduce as expected that $\gamma_0(p)$ is a constant on Σ.

□

Let us define now the following space
$$\begin{aligned}H_{\mathrm{div,curl},\nu}(\Omega) = \Big\{ & u \in (L^2(\Omega))^3, \operatorname{div} u \in L^2(\Omega), \\ & \operatorname{curl} u \in (L^2(\Omega))^3, \gamma_\nu(u) \in H^{1/2}(\partial\Omega)\Big\},\end{aligned}$$
equipped with the norm
$$\|u\|_{H_{\mathrm{div,curl},\nu}} = \left(\|u\|_{L^2}^2 + \|\operatorname{div} u\|_{L^2}^2 + \|\operatorname{curl} u\|_{L^2}^2 + \|\gamma_\nu(u)\|_{H^{1/2}}^2\right)^{1/2}.$$
We also define the following closed subspace of $H_{\mathrm{div,curl},\nu}(\Omega)$,

$$H_{\mathrm{div},\mathrm{curl},\nu,0}(\Omega) = \{u \in H_{\mathrm{div},\mathrm{curl},\nu}(\Omega), \gamma_\nu(u) = 0\}\,. \tag{IV.15}$$

Lemma IV.4.6. *Let Ω be a simply connected Lipschitz domain in $\mathbb{R}^3$. If $u \in (L^2(\Omega))^3$ is such that $\operatorname{div} u = 0$, $\operatorname{curl} u = 0$, and $\gamma_\nu(u) = 0$, then $u = 0$.*

Proof.

By assumption we have that $u \in H_{\mathrm{div}}(\Omega)$, so that the trace $\gamma_\nu(u)$ has to be understood in the sense of $H^{-1/2}(\partial\Omega)$.

Let B be any ball in Ω. Let $\Phi \in (\mathcal{C}_c^\infty(B))^3$ such that $\operatorname{div}\Phi = 0$. By Lemma IV.4.3, there exists $\Psi \in (\mathcal{C}_c^\infty(B))^3$ such that $\operatorname{curl}\Psi = \Phi$. Hence, we have

$$\langle u, \Phi\rangle_{H^{-1}(B),H_0^1(B)} = \int_B u \cdot \Phi\, dx = \int_B u \cdot (\operatorname{curl}\Psi)\, dx.$$

Using Stokes formula (IV.13), we deduce that

$$\langle u, \Phi\rangle_{H^{-1}(B),H_0^1(B)} = \int_B (\operatorname{curl} u)\cdot \Psi\, dx + \langle \gamma_{\wedge\nu}(u), \Psi\rangle_{H^{-1/2},H^{1/2}}.$$

Since $\operatorname{curl} u = 0$ and Ψ is compactly supported in B, we have $\langle u, \Phi\rangle_{H^{-1},H_0^1} = 0$ for any such Φ.

By Theorem IV.2.4, we know that there exists a $p_B \in L_0^2(B)$ such that $u = \nabla p_B$ in B. We have in fact $p_B \in H^1(B)$, because $u \in (L^2(B))^3$.

Moreover, we have $\Delta p_B = \operatorname{div}(\nabla p_B) = \operatorname{div} u = 0$, which means that p_B is harmonic in B and thus $p_B \in \mathcal{C}^\infty(B)$ (see, for instance, [56]). Finally, $u = \nabla p_B$ is also in $\mathcal{C}^\infty(B)$. Since this is true for any ball B in Ω, we have finally shown that $u \in (\mathcal{C}^\infty(\Omega))^3$.

We can now apply Lemma IV.4.2, and find that there exists a $p \in \mathcal{C}^\infty(\Omega)$ such that $u = \nabla p$ in Ω. Moreover, p is harmonic.

Note that $\nabla p \in (L^2(\Omega))^3$ and $p \in L^2_{loc}(\Omega)$. Thus, for any $\Phi \in (\mathcal{C}_c^\infty(\Omega))^3$, such that $\operatorname{div}\Phi = 0$, we have

$$\langle u, \Phi\rangle_{H^{-1},H_0^1} = \int_\Omega u \cdot \Phi\, dx = \int_\Omega \nabla p \cdot \Phi\, dx = -\langle p, \operatorname{div}\Phi\rangle_{\mathcal{D}',\mathcal{D}} = 0.$$

By Theorem IV.2.4, we deduce that there is a $q \in L_0^2(\Omega)$ such that $u = \nabla q$. Since Ω is connected, $p - q$ is constant and thus $p \in L^2(\Omega)$. Finally, we have $u = \nabla p$ with $p \in H^1(\Omega)$.

We conclude by applying the Stokes formula (IV.10)

$$\int_\Omega |\nabla p|^2\, dx = \int_\Omega u \cdot \nabla p\, dx = -\int_\Omega (\operatorname{div} u) p\, dx + \langle \gamma_\nu(u), \gamma_0(p)\rangle_{H^{-1/2},H^{1/2}},$$

and the two terms in the right-hand side are zero because $\operatorname{div} u = 0$ and $\gamma_\nu(u) = 0$. We deduce that $u = \nabla p = 0$. The claim is thus proved.

$\square$

4.2.2 Density result and applications

We are going to show that the spaces $H_{\text{div,curl},\nu}(\Omega)$ and $H_{\text{div,curl},\nu,0}(\Omega)$ are algebraically and topologically equal to $(H^1(\Omega))^3$ and $(H^1(\Omega))^3 \cap \operatorname{Ker}\gamma_\nu$, respectively.

To this end, we first need the following density result.

Theorem IV.4.7. *Let Ω be a bounded connected domain of class $\mathcal{C}^{1,1}$. The set*

$$\mathcal{H} = \left\{u \in (\mathcal{C}^{0,1}(\overline{\Omega}))^3 \cap (\mathcal{C}^\infty(\Omega))^3, \quad u \cdot \nu = 0 \textit{ in a neighborhood of } \partial\Omega\right\},$$

is dense in $H_{\text{div,curl},\nu,0}(\Omega)$ and in $(H^1(\Omega))^3 \cap \operatorname{Ker}\gamma_\nu$.

Proof.

1. We first prove the density of $\mathcal{H}$ in $(H^1(\Omega))^3 \cap \operatorname{Ker}\gamma_\nu$. Let $v \in (H^1(\Omega))^3$ such that $\gamma_\nu(v) = 0$ on $\partial\Omega$. Let $\mathcal{O}^\rho_{s_0}$ be a neighborhood of $\partial\Omega$ on which we can use tangential and normal coordinates (see Section 3 of Chapter III) and $\varphi \in \mathcal{C}^\infty_c(\Omega)$ such that $\operatorname{Supp}(1-\varphi) \subset \mathcal{O}^\rho_{s_0}$.
   We write $v = \varphi v + (1-\varphi)v$. By construction φv belongs to $(H^1_0(\Omega))^3$ and then can be approximated in the H^1-norm by a sequence of functions in $(\mathcal{D}(\Omega))^3 \subset \mathcal{H}$. It remains to deal with $u = (1-\varphi)v$. Since u is supported in $\mathcal{O}^\rho_{s_0}$ we can define the tangential part of u as $u_T = u - (u\cdot\nu)\nu$ which possesses the same regularity as u (because ν is Lipschitz continuous) and satisfies $u_T \cdot \nu = 0$ in $\mathcal{O}^\rho_{s_0}$. Since $\varphi = 0$ on $\partial\Omega$, we have
   $$u\cdot\nu = v\cdot\nu = 0, \text{ on } \partial\Omega.$$
   Therefore, $u\cdot\nu \in H^1_0(\mathcal{O}^\rho_{s_0})$ and $u_T \in (H^1(\mathcal{O}^\rho_{s_0}))^3$ and then, for any $\varepsilon > 0$ we can find $u_{T,\varepsilon} \in (\mathcal{C}^\infty(\overline{\mathcal{O}^\rho_{s_0}}))^3$ and $f_\varepsilon \in \mathcal{D}(\mathcal{O}^\rho_{s_0})$ such that
   $$\|f_\varepsilon - (u\cdot\nu)\|_{H^1(\mathcal{O}^\rho_{s_0})} + \|u_{T,\varepsilon} - u_T\|_{H^1(\mathcal{O}^\rho_{s_0})} \xrightarrow[\varepsilon\to 0]{} 0.$$
   We now set
   $$u_\varepsilon = u_{T,\varepsilon} - (u_{T,\varepsilon}\cdot\nu)\nu + f_\varepsilon \nu,$$
   and we observe that $u_\varepsilon \cdot \nu = f_\varepsilon$ in $\mathcal{O}^\rho_{s_0}$ so that $u_\varepsilon\cdot\nu$ is identically zero in a neighborhood of the boundary. Since ν is Lipschitz-continuous on $\overline{\Omega}$ and of class $\mathcal{C}^\infty$ in $\mathcal{O}^\rho_{s_0}$, we see that u_ε satisfies the required regularity properties. Moreover, since by construction we have $u_T\cdot\nu = 0$, we can write
   $$\begin{aligned}\|u - u_\varepsilon\|_{H^1} &= \|u_{T,\varepsilon} - u_T - ((u_{T,\varepsilon} - u_T)\cdot\nu)\nu + (\psi_\varepsilon - (u\cdot\nu))\nu\|_{H^1}\\ &\leq C\|u_{T,\varepsilon} - u_T\|_{H^1} + C\|\psi_\varepsilon - (u\cdot\nu)\|_{H^1} \xrightarrow[\varepsilon\to 0]{} 0,\end{aligned}$$
   which proves the first part of the result.
2. We deal now with the second density result. Let $v \in H_{\text{div,curl},\nu,0}(\Omega)$ and let us define, for any $\varepsilon > 0$, $\tilde{v}_\varepsilon = \mathcal{S}_\varepsilon v \in (\mathcal{C}^\infty(\overline{\Omega}))^3$, where $(\mathcal{S}_\varepsilon)_\varepsilon$ is the family of mollifying operators as defined in Section 2.2 of Chapter III.

By Theorem III.2.10, we have the following convergences

$$\begin{cases} \tilde{v}_\varepsilon \xrightarrow[\varepsilon\to 0]{} v, & \text{in } L^2(\Omega), \\ \operatorname{div} \tilde{v}_\varepsilon \xrightarrow[\varepsilon\to 0]{} \operatorname{div} v, & \text{in } L^2(\Omega), \\ \operatorname{curl} \tilde{v}_\varepsilon \xrightarrow[\varepsilon\to 0]{} \operatorname{curl} v, & \text{in } (L^2(\Omega))^3. \end{cases}$$

Notice that $\tilde{v}_\varepsilon \cdot \nu$ does not necessarily vanish on $\partial\Omega$. Let us now consider the following Neumann problem

$$\begin{cases} -\Delta f_\varepsilon = \dfrac{1}{|\Omega|} \displaystyle\int_{\partial\Omega} \tilde{v}_\varepsilon \cdot \nu \, d\sigma = \dfrac{1}{|\Omega|} \langle \tilde{v}_\varepsilon \cdot \nu, 1\rangle_{H^{-1/2}, H^{1/2}}, & \text{in } \Omega, \\ -\dfrac{\partial f_\varepsilon}{\partial \nu} = \tilde{v}_\varepsilon \cdot \nu, & \text{on } \partial\Omega, \\ \displaystyle\int_\Omega f_\varepsilon \, dx = 0. \end{cases}$$

- First of all, since $\tilde{v}_\varepsilon$ is smooth and ν is a Lipschitz vector field, we see that $\tilde{v}_\varepsilon \cdot \nu$ is Lipschitz continuous so that it belongs to $H^{1/2}(\partial\Omega)$. The source term in this Neumann problem is a constant function, thus the compatibility is satisfied; we deduce from Theorem III.4.3 that $f_\varepsilon \in H^2(\Omega)$, because Ω is supposed to be of class $\mathcal{C}^{1,1}$.
- Second, thanks to the continuity of the trace operator γ_ν, and to the fact that $\gamma_\nu(v) = 0$, we know that

  $$\|\tilde{v}_\varepsilon \cdot \nu\|_{H^{-1/2}} = \|\gamma_\nu(\tilde{v}_\varepsilon - v)\|_{H^{-1/2}} \le C(\|v - \tilde{v}_\varepsilon\|_{L^2} + \|\operatorname{div} v - \operatorname{div} \tilde{v}_\varepsilon\|_{L^2}),$$

  and thus we deduce that $\tilde{v}_\varepsilon \cdot \nu$ tends to 0 in $H^{-1/2}(\partial\Omega)$. The energy estimate (III.106) for the Laplace–Neumann problem gives

  $$\|\nabla f_\varepsilon\|_{L^2} \le C \|\tilde{v}_\varepsilon \cdot \nu\|_{H^{-1/2}} \xrightarrow[\varepsilon\to 0]{} 0.$$

  Moreover, by construction, we have

  $$\|\Delta f_\varepsilon\|_{L^2} = C_\Omega \left| \int_{\partial\Omega} \tilde{v}_\varepsilon \cdot \nu \, d\sigma \right| \xrightarrow[\varepsilon\to 0]{} 0.$$

- We can now define $v_\varepsilon = \tilde{v}_\varepsilon + \nabla f_\varepsilon$, which belongs to $(H^1(\Omega))^3$ and which satisfies

  $$\|v_\varepsilon - v\|_{L^2} \le \|\tilde{v}_\varepsilon - v\|_{L^2} + \|\nabla f_\varepsilon\|_{L^2} \xrightarrow[\varepsilon\to 0]{} 0,$$

  $$\|\operatorname{div} v_\varepsilon - \operatorname{div} v\|_{L^2} \le \|\operatorname{div} \tilde{v}_\varepsilon - \operatorname{div} v\|_{L^2} + \|\Delta f_\varepsilon\|_{L^2} \xrightarrow[\varepsilon\to 0]{} 0,$$

  and, since $\operatorname{curl} \nabla f_\varepsilon = 0$,

$$\|\operatorname{curl} v_\varepsilon - \operatorname{curl} v\|_{L^2} = \|\operatorname{curl} \tilde{v}_\varepsilon - \operatorname{curl} v\|_{L^2} \xrightarrow[\varepsilon\to 0]{} 0.$$

In conclusion, we just proved that $v_\varepsilon \in (H^1(\Omega))^3 \cap \operatorname{Ker}\gamma_\nu$ and satisfies

$$\|v - v_\varepsilon\|_{H_{\mathrm{div,curl},\nu}} \xrightarrow[\varepsilon\to 0]{} 0. \tag{IV.16}$$

- We now use the density of $\mathcal{H}$ into $(H^1(\Omega))^3 \cap \operatorname{Ker}\gamma_\nu$ that we proved above. We can then find $w_\varepsilon \in \mathcal{H}$ such that $\|v_\varepsilon - w_\varepsilon\|_{H^1} \xrightarrow[\varepsilon\to 0]{} 0$. But, since $v_\varepsilon, w_\varepsilon \in (H^1(\Omega))^3$, we obviously have

$$\|v_\varepsilon - w_\varepsilon\|_{H_{\mathrm{div,curl},\nu}} \leq C\|v_\varepsilon - w_\varepsilon\|_{H^1} \xrightarrow[\varepsilon\to 0]{} 0. \tag{IV.17}$$

  Using (IV.16) and (IV.17) we deduce that $w_\varepsilon \in \mathcal{H}$ converges to v in $H_{\mathrm{div,curl},\nu}(\Omega)$ which proves the claim.

□

Theorem IV.4.8. *Let Ω be a bounded domain of class $\mathcal{C}^{1,1}$ in $\mathbb{R}^3$. The following equalities hold*

$$\begin{cases} H_{\mathrm{div,curl},\nu,0}(\Omega) = (H^1(\Omega))^3 \cap \operatorname{Ker}\gamma_\nu, & \text{(IV.18)} \\ H_{\mathrm{div,curl},\nu}(\Omega) = (H^1(\Omega))^3, & \text{(IV.19)} \end{cases}$$

and there exists $C > 0$ depending only on Ω such that

$$\|u\|_{H^1} \leq C\Big(\|u\|_{L^2} + \|\operatorname{div} u\|_{L^2} + \|\operatorname{curl} u\|_{L^2} + \|\gamma_\nu(u)\|_{H^{1/2}}\Big), \tag{IV.20}$$

for any $u \in (H^1(\Omega))^3 = H_{\mathrm{div,curl},\nu}(\Omega)$. That is, the H^1 norm and the $H_{\mathrm{div,curl},\nu}$ norm are equivalent in the space $(H^1(\Omega))^3$.

Moreover, if Ω is simply connected, then (IV.20) can be replaced with

$$\|u\|_{H^1} \leq C\Big(\|\operatorname{div} u\|_{L^2} + \|\operatorname{curl} u\|_{L^2} + \|\gamma_\nu(u)\|_{H^{1/2}}\Big), \tag{IV.21}$$

for any $u \in (H^1(\Omega))^3 = H_{\mathrm{div,curl},\nu}(\Omega)$.

Proof.

- We first assume that (IV.20) holds for any vector field $u \in (H^1(\Omega))^3$; we then show that the equality (IV.18) holds.
  Consider the identity mapping I from $(H^1(\Omega))^3 \cap \operatorname{Ker}\gamma_\nu$ into $H_{\mathrm{div,curl},\nu,0}(\Omega)$.
  - By definition of the $H_{\mathrm{div,curl},\nu}$-norm, we obviously see that I is continuous.
  - The image of I is closed in $H_{\mathrm{div,curl},\nu,0}$. Indeed, if $(u_n)_n$ is a sequence in $(H^1(\Omega))^3 \cap \operatorname{Ker}\gamma_\nu$ which converges in $H_{\mathrm{div,curl},\nu,0}(\Omega)$, then by (IV.20)

we see that $(u_n)_n$ is a Cauchy sequence in $(H^1(\Omega))^3$; thus it converges towards some $u \in (H^1(\Omega))^3 \cap \operatorname{Ker}\gamma_\nu$ and it is clear that this function u is also the limit of $(u_n)_n$ in $H_{\mathrm{div},\mathrm{curl},\nu,0}(\Omega)$.

– By Theorem IV.4.7, we know that the space $\mathcal{H}$, which is included in the image of I, is dense in $H_{\mathrm{div},\mathrm{curl},\nu,0}(\Omega)$, therefore the image of I is dense in $H_{\mathrm{div},\mathrm{curl},\nu,0}(\Omega)$.

Finally, I has a closed and dense range in $H_{\mathrm{div},\mathrm{curl},\nu,0}(\Omega)$. This means that its range is nothing but the whole space $H_{\mathrm{div},\mathrm{curl},\nu,0}(\Omega)$ which proves (IV.18).

Then, we deduce that (IV.19) also holds. Indeed, for any u in $H_{\mathrm{div},\mathrm{curl},\nu}(\Omega)$, since $\gamma_\nu(u) \in H^{1/2}(\partial\Omega)$, we can use the lifting operator R_0 to set

$$v = u - R_0(\gamma_\nu(u)\nu).$$

since $R_0(\gamma_\nu(u)\nu) \in (H^1(\Omega))^3$, we see by construction that v belongs to $H_{\mathrm{div},\mathrm{curl},\nu,0}(\Omega)$. By (IV.18) we deduce that $v \in (H^1(\Omega))^3$ and then $u = v + R_0(\gamma_\nu(u)\nu)$ also belongs to $(H^1(\Omega))^3$.

- It remains to prove (IV.20) for any $u \in (H^1(\Omega))^3$. First of all, by the same reasoning as above, using the lifting operator R_0, we see that it is enough to prove the inequality for $u \in (H^1(\Omega))^3 \cap \operatorname{Ker}\gamma_\nu$.

  By Theorem IV.4.7, we can find a sequence $\varphi_n \in \mathcal{H}$ which converges to u in $(H^1(\Omega))^3$. Therefore, if we manage to prove (IV.20) for all the functions $\varphi \in \mathcal{H}$, the result will follow by density.

  Let $\varphi \in \mathcal{H}$. By (A.9) we have $-\Delta\varphi\cdot\varphi = \operatorname{curl}\operatorname{curl}\varphi\cdot\varphi - \nabla(\operatorname{div}\varphi)\cdot\varphi$. We integrate this equality on Ω and then use the Stokes formula for each term. It follows

$$\begin{aligned}\int_\Omega |\nabla\varphi|^2\,dx &= \int_\Omega |\operatorname{curl}\varphi|^2 + |\operatorname{div}\varphi|^2\,dx\\ &+ \int_{\partial\Omega} \big((\varphi\wedge\nu)\cdot(\operatorname{curl}\varphi) + \big[(\nu\cdot\nabla)\varphi\big]\cdot\varphi - (\operatorname{div}\varphi)(\varphi\cdot\nu)\big)\,d\sigma.\end{aligned} \tag{IV.22}$$

By using formulas (A.3), (A.4), (A.7), and (A.8) we can write

$$\begin{aligned}&(\varphi\wedge\nu)\cdot(\operatorname{curl}\varphi) + \big[(\nu\cdot\nabla)\varphi\big]\cdot\varphi\\ &=\operatorname{div}(\varphi\wedge(\varphi\wedge\nu)) + \varphi\cdot\big(\operatorname{curl}(\varphi\wedge\nu)\big) + (\nu\cdot\nabla)\frac{|\varphi|^2}{2}\\ &=\operatorname{div}((\varphi\cdot\nu)\varphi - |\varphi|^2\nu) + (\nu\cdot\nabla)\frac{|\varphi|^2}{2}\\ &\quad+\varphi\cdot\big((\operatorname{div}\nu)\varphi - (\operatorname{div}\varphi)\nu + (\nu\cdot\nabla)\varphi - (\varphi\cdot\nabla)\nu\big)\\ &=(\varphi\cdot\nabla)(\varphi\cdot\nu) - ((\varphi\cdot\nabla)\nu)\cdot\varphi\\ &=((\varphi\cdot\nabla)\varphi)\cdot\nu.\end{aligned}$$

Hence, the boundary terms can be written

$$\begin{aligned}&(\varphi \wedge \nu) \cdot (\operatorname{curl} \varphi) + \big[(\nu \cdot \nabla)\varphi\big] \cdot \varphi - (\operatorname{div} \varphi)(\varphi \cdot \nu)\\ =&((\varphi \cdot \nabla)\varphi) \cdot \nu - (\operatorname{div} \varphi)(\varphi \cdot \nu)\\ =&-((\varphi \cdot \nabla)\nu) \cdot \varphi + (\varphi \cdot \nabla)(\varphi \cdot \nu) - (\operatorname{div} \varphi)(\varphi \cdot \nu).\end{aligned}$$

Since $\varphi \in \mathcal{H}$, we know that $\varphi \cdot \nu = 0$ in a neighborhood of the boundary. As a consequence the last two terms in the formula above vanish and from (IV.22) we finally deduce

$$\int_\Omega |\nabla \varphi|^2 \, dx = \int_\Omega |\operatorname{curl} \varphi|^2 + |\operatorname{div} \varphi|^2 \, dx - \int_{\partial\Omega} ((\varphi \cdot \nabla)\nu) \cdot \varphi \, d\sigma. \quad \text{(IV.23)}$$

The last term of this equality, referred to as I, does not contain any derivative of φ, thus it follows that

$$|I| \leq \operatorname{Lip}(\nu) \|\varphi\|^2_{L^2(\partial\Omega)}.$$

By the trace inequality given in Theorem III.2.19, we have

$$\|\varphi\|_{L^2(\partial\Omega)} \leq C\|\varphi\|_{L^2} + C\|\varphi\|^{1/2}_{L^2} \|\nabla \varphi\|^{1/2}_{L^2},$$

and thus by using Young's inequality (see Corollary II.2.17) we find that

$$|I| \leq C'\|\varphi\|^2_{L^2} + \frac{1}{2}\|\nabla \varphi\|^2_{L^2}.$$

We use this estimate in (IV.23) and we obtain the claim.

- We assume now that Ω is simply connected and we want to prove (IV.21) by contradiction, just as in the proof of the Poincaré inequalities (see Propositions III.2.38 and III.2.39).
  We assume that (IV.21) does not hold, which implies the existence of a sequence $(u_n)_n \in (H^1(\Omega))^3$ such that

$$\|u_n\|_{H^1} = 1, \, \forall n \geq 0, \quad \text{(IV.24)}$$

$$\|\operatorname{div} u_n\|_{L^2} + \|\operatorname{curl} u_n\|_{L^2} + \|\gamma_\nu(u_n)\|_{H^{1/2}} \xrightarrow[n\to\infty]{} 0. \quad \text{(IV.25)}$$

Since Ω is bounded, the embedding of $H^1(\Omega)$ into $L^2(\Omega)$ is compact and then there exists a subsequence (still denoted as $(u_n)_n$) which strongly converges in $(L^2(\Omega))^3$ towards some $u \in (H^1(\Omega))^3$. By (IV.25) we see that, at the limit we have $\operatorname{div} u = 0$, $\operatorname{curl} u = 0$, and $\gamma_\nu(u) = 0$. Using the assumption that Ω is simply connected, Lemma IV.4.6 proves that $u = 0$. Therefore we necessarily have $\|u_n\|_{L^2} \xrightarrow[n\to\infty]{} 0$ in addition to (IV.25), and then by (IV.20) we deduce that $\|u_n\|_{H^1} \xrightarrow[n\to\infty]{} 0$, which is contradiction with (IV.24).

□

The same kind of result holds for higher regularity Sobolev spaces. For the sake of simplicity we only state the corresponding result for the H^2-regularity.

Theorem IV.4.9. *Let Ω be a bounded domain in $\mathbb{R}^3$ of class $\mathcal{C}^{2,1}$. For any $u \in (H^1(\Omega))^3$ such that*

$$\operatorname{div} u \in H^1(\Omega),\ \operatorname{curl} u \in (H^1(\Omega))^3,\ \gamma_\nu(u) \in H^{3/2}(\partial\Omega),$$

then $u \in (H^2(\Omega))^3$ and we have

$$\|u\|_{H^2} \leq C\left(\|u\|_{L^2} + \|\operatorname{div} u\|_{H^1} + \|\operatorname{curl} u\|_{H^1} + \|\gamma_\nu(u)\|_{H^{3/2}}\right),$$

for some $C > 0$ depending only on Ω.

The proof of this theorem is given in Section 6.4 because we need to use some tools that are introduced later during the study of regularity properties for the Stokes problem.

4.3 Kernel and image of the curl *operator*

Lemma IV.4.10. *Let Ω be a simply connected domain in $\mathbb{R}^3$ of class $\mathcal{C}^{1,1}$. For any $u \in (L^2(\Omega))^3$ such that $\operatorname{curl} u = 0$, there exists a $p \in H^1(\Omega)$ such that $u = \nabla p$. In other words we have the equality*

$$\operatorname{Ker}_{L^2}(\operatorname{curl}) = \nabla(H^1(\Omega)).$$

Proof.

We write the Leray decomposition (Definition IV.3.6) of such a u as

$$u = u_0 + \nabla p,$$

where $p \in H^1(\Omega)$ and $\operatorname{div} u_0 = 0$, $\gamma_\nu(u_0) = 0$. Taking the curl of this formula gives $\operatorname{curl} u_0 = \operatorname{curl} u = 0$. Therefore, by Lemma IV.4.6, we get $u_0 = 0$. Hence, $u = \nabla p$.

□

Remark IV.4.1. We emphasise the fact that, for elements in $\operatorname{Ker}_{L^2}(\operatorname{curl}) = \nabla(H^1(\Omega))$, the inequality (IV.21) thus reads

$$\|\nabla p\|_{H^1} \leq C\left(\|\Delta p\|_{L^2} + \left\|\frac{\partial p}{\partial \nu}\right\|_{H^{1/2}}\right),$$

for any $p \in H^1(\Omega)$ such that $\Delta p \in L^2(\Omega)$ and $\partial p/\partial\nu \in H^{1/2}(\partial\Omega)$. We observe that this result is nothing but the elliptic regularity theorem for the Laplace equation with Neumann boundary conditions (see Theorem III.4.3).

We can now characterise the image of the curl operator.

Theorem IV.4.11. *Let Ω be a simply connected bounded domain in $\mathbb{R}^3$ of class $\mathcal{C}^{1,1}$. We denote by $\Gamma_1, \ldots, \Gamma_m$ the connected components of the boundary $\partial\Omega$.*

1. *We have*
$$\operatorname{curl}\left((H^1(\Omega))^3\right) = \operatorname{curl}\left((H^1(\Omega))^3 \cap H\right). \tag{IV.26}$$
Moreover this space is closed in $(L^2(\Omega))^3$.
2. *We have in $(L^2(\Omega))^3$*
$$\begin{aligned} &\left[\operatorname{curl}\left((H^1(\Omega))^3\right)\right]^\perp \\ &= \left\{u \in (L^2(\Omega))^3, \operatorname{curl} u = 0,\ \gamma_{\wedge\nu} u = 0\right\} \\ &= \left\{u \in (L^2(\Omega))^3, u = \nabla p, p \in H^1(\Omega), \gamma_0(p) \textit{ is constant on each } \Gamma_i\right\}. \end{aligned} \tag{IV.27}$$
3. *We finally have*
$$\operatorname{curl}\left((H^1(\Omega))^3\right) = \left\{u \in (L^2(\Omega))^3, \operatorname{div} u = 0, \textit{ satisfying } \ (\text{IV.28})\right\},$$
where the following compatibility conditions are introduced
$$\langle \gamma_\nu(u), 1\rangle_{H^{-1/2}(\Gamma_i), H^{1/2}(\Gamma_i)} = 0, \forall i = 1, \ldots, m. \tag{IV.28}$$

Proof.

1. Let $u = \operatorname{curl} \widetilde{\Psi}$ with $\widetilde{\Psi} \in (H^1(\Omega))^3$. We introduce $\Psi = \mathcal{P}\widetilde{\Psi}$ the Leray projection of $\widetilde{\Psi}$ on H, so that we have $\widetilde{\Psi} = \Psi + \nabla q$, with $q \in H^1(\Omega)$ and then $\operatorname{curl}\Psi = \operatorname{curl}\widetilde{\Psi} = u$. Moreover, the Leray projection maps $(H^1(\Omega))^d$ into itself (see Proposition IV.3.7), therefore we know that $\Psi \in (H^1(\Omega))^3$. This proves the equality between the two spaces. Note that, by Lemma IV.4.6, Ψ is the unique element in $(H^1(\Omega))^3 \cap H$ such that $\operatorname{curl}\Psi = u$. By the inequality (IV.20), we deduce that
$$\|\Psi\|_{H^1} \leq C\|\operatorname{curl}\Psi\|_{L^2} = C\|u\|_{L^2}. \tag{IV.29}$$
It follows that $\operatorname{curl}\left((H^1(\Omega))^3\right)$ is closed. Indeed, if $(u_n)_n$ is a sequence of $\operatorname{curl}\left((H^1(\Omega))^3\right)$ which converges towards a $u \in (L^2(\Omega))^3$, we associate with each u_n, the unique $\Psi_n \in (H^1(\Omega))^3 \cap H$ such that $\operatorname{curl}\Psi_n = u_n$. By (IV.29), we see that Ψ_n is a Cauchy sequence in $(H^1(\Omega))^3$ and then converges towards some $\Psi \in (H^1(\Omega))^3$. It is clear that the limit u satisfies $u = \operatorname{curl}\Psi$ and the result is proved.
2. Let $u \in (L^2(\Omega))^3$ which is orthogonal to $\operatorname{curl}\left((H^1(\Omega))^3\right)$. In particular, for any $\Phi \in (\mathcal{D}(\Omega))^3$, we have
$$\langle \operatorname{curl} u, \Phi\rangle_{\mathcal{D}',\mathcal{D}} = \langle u, \operatorname{curl}\Phi\rangle_{\mathcal{D}',\mathcal{D}} = 0,$$

and thus $\operatorname{curl} u = 0$. We may now apply (IV.13) for any $v \in (H^1(\Omega))^3$ as follows,

$$0 = \int_\Omega u \cdot \operatorname{curl} v \, dx = \underbrace{\int_\Omega v \cdot \operatorname{curl} u \, dx}_{=0} + \langle \gamma_{\wedge\nu}(u), \gamma_0(v) \rangle_{H^{-1/2}, H^{1/2}},$$

which implies that $\gamma_{\wedge\nu}(u) = 0$ and the first part of the claim is proved. From Lemma IV.4.10, we know that $u = \nabla p$ for some $p \in (H^1(\Omega))^3$. The fact that $\gamma_{\wedge\nu}(\nabla p) = \gamma_{\wedge\nu}(u) = 0$ and Proposition IV.4.5 gives the result.

3. • For any $u \in \operatorname{curl}\big((H^1(\Omega))^3\big)$, we obviously have $\operatorname{div} u = 0$. For any $i \in \{1, \ldots, m\}$, we consider a function $p_i \in H^1(\Omega)$ such that $p_i = 1$ on Γ_i and $p_i = 0$ on Γ_j for $j \neq i$. Such a function is known to exist by the trace lifting theorem (Theorem III.2.22). Using (IV.27), the fact that $\operatorname{div} u = 0$ and the Stokes formula (IV.10) we get

$$\begin{aligned} 0 = \int_\Omega u \cdot \nabla p_i \, dx &= \langle \gamma_\nu(u), \gamma_0(p_i) \rangle_{H^{-1/2}(\partial\Omega), H^{1/2}(\partial\Omega)} \\ &= \langle \gamma_\nu(u), 1 \rangle_{H^{-1/2}(\Gamma_i), H^{1/2}(\Gamma_i)}, \end{aligned}$$

which proves that the conditions (IV.28) hold.

   • Conversely, assume that $u \in (L^2(\Omega))^3$ is such that $\operatorname{div} u = 0$ and that (IV.28) hold. We want to show that u belongs to the image of the curl operator. This image is closed in $(L^2(\Omega))^2$, therefore it is enough to show that u is orthogonal to any element in $\big(\operatorname{curl}\big((H^1(\Omega))^3\big)\big)^\perp$; see Remark IV.2.1. Therefore, we use (IV.27) and consider some $w = \nabla p$ with $p \in H^1(\Omega)$ and $\gamma_0(p)$ which is constant on each Γ_i. Using once more the Stokes formula (IV.10) we obtain

$$\begin{aligned} \int_\Omega u \cdot w \, dx &= \int_\Omega u \cdot \nabla p \, dx \\ &= -\int_\Omega (\operatorname{div} u) p \, dx + \sum_{i=1}^m p_i \langle \gamma_\nu(u), 1 \rangle_{H^{-1/2}(\Gamma_i), H^{1/2}(\Gamma_i)} = 0. \end{aligned}$$

This proves the claim.

□

4.4 *The* div/curl *problem*

Using all the previous results, we can now investigate the div/curl problem. We first study the case of normal boundary conditions, and then the case where we do not impose any boundary conditions. Notice that other kind of boundary conditions can be considered for this problem (this is useful, in particular, in electromagnetism) but we do not consider them here.

Theorem IV.4.12. *We use the same notation as in Theorem IV.4.11. We suppose given*

- *A vector field* $u \in (L^2(\Omega))^3$ *such that* $\operatorname{div} u = 0$ *and satisfying* (IV.28).
- *A scalar field* $v \in L^2(\Omega)$ *and a boundary data* $\Psi_\nu \in H^{1/2}(\partial\Omega)$ *satisfying the compatibility condition*

$$\int_\Omega v\,dx = \int_{\partial\Omega} \Psi_\nu\,d\sigma.$$

Then there exists a unique solution $\Psi \in (H^1(\Omega))^3$ *to the problem*

$$\begin{cases} \operatorname{div}\Psi = v, & \text{in } \Omega,\\ \operatorname{curl}\Psi = u, & \text{in } \Omega,\\ \Psi\cdot\nu = \Psi_\nu, & \text{on } \partial\Omega; \end{cases} \tag{IV.30}$$

moreover it satisfies

$$\|\Psi\|_{H^1} \le C(\|u\|_{L^2} + \|v\|_{L^2} + \|\Psi_\nu\|_{H^{1/2}}),$$

where $C > 0$ *depends only on* Ω.

Finally, if Ω *is of class* $\mathcal{C}^{2,1}$, $u \in (H^1(\Omega))^3$, $v \in H^1(\Omega)$, *and* $\Psi_\nu \in H^{3/2}(\partial\Omega)$, *then the solution* Ψ *of* (IV.30) *belongs to* $(H^2(\Omega))^3$ *and satisfies*

$$\|\Psi\|_{H^2} \le C(\|u\|_{H^1} + \|v\|_{H^1} + \|\Psi_\nu\|_{H^{3/2}}),$$

Proof.

We know from Theorem IV.4.11 and from the assumptions on u, that u belongs to the image of the curl operator. Moreover, by (IV.26), we know that $u \in \operatorname{curl}\big((H^1(\Omega))^3 \cap H\big)$. This exactly means that there exists a $\Psi_1 \in (H^1(\Omega))^3$ such that

$$\operatorname{div}\Psi_1 = 0, \operatorname{curl}\Psi_1 = u, \Psi_1\cdot\nu = 0, \text{ on } \partial\Omega.$$

We consider now the following Laplace–Neumann problem

$$\begin{cases} -\Delta\varphi = v, & \text{in } \Omega\\ -\dfrac{\partial\varphi}{\partial\nu} = \Psi_\nu, & \text{on } \partial\Omega. \end{cases}$$

Since Ω is of class $\mathcal{C}^{1,1}$, and thanks to the compatibility assumption between v and Ψ_ν, Theorem III.4.3 shows that there is a solution $\varphi \in H^2(\Omega)$ of this problem. We set $\Psi_2 = -\nabla\varphi$ and we observe that

$$\operatorname{div}\Psi_2 = v, \operatorname{curl}\Psi_2 = 0, \Psi_2\cdot\nu = \Psi_\nu, \text{ on } \partial\Omega.$$

As a consequence the sum $\Psi = \Psi_1 + \Psi_2 \in (H^1(\Omega))^3$ is a solution to Problem (IV.30).

The uniqueness of this solution, as well as the estimate of $\|\Psi\|_{H^1}$ immediately follows from (IV.21). Finally, the $H^2(\Omega)$ estimate in the case of smooth data follows from Theorem IV.4.9.

□

In the case where Ω is simply a Lipschitz bounded domain of $\mathbb{R}^d$ and if we do not impose any boundary condition on the solution of the div/curl problem, we can deduce the following result (see [63]), which will be useful in the study of the steady Navier–Stokes equations.

Theorem IV.4.13. *Let Ω be a simply connected bounded Lipschitz domain in $\mathbb{R}^3$. The connected components of its boundary are denoted by $\Gamma_1, \ldots, \Gamma_m$. Let $k \in \{0, 1\}$. We suppose given*

- *A vector field $u \in (H^k(\Omega))^3$ such that $\operatorname{div} u = 0$ and satisfying (IV.28).*
- *A scalar field $v \in H^k(\Omega)$.*

Then there exists at least one solution $\Psi \in (H^{k+1}(\Omega))^3$ to the problem

$$\begin{cases} \operatorname{div} \Psi = v, \ \text{in } \Omega, \\ \operatorname{curl} \Psi = u, \ \text{in } \Omega, \end{cases}$$

which satisfies

$$\|\Psi\|_{H^{k+1}} \leq C(\|u\|_{H^k} + \|v\|_{H^k}),$$

where $C > 0$ does not depend on u and v.

Remark IV.4.2. The solution Ψ is certainly not unique because we can add the gradient of any harmonic function to Ψ to obtain another solution.

Proof.

- Let $B = B(0, R)$ be a large ball which contains $\overline{\Omega}$. We consider the following Laplace–Dirichlet problem in B

$$\begin{cases} -\Delta\varphi = \tilde{v}, \text{ in } B, \\ \quad \varphi = 0, \text{ on } \partial B, \end{cases}$$

  where $\tilde{v} \in H^k(B)$ is an extension of v in B. Note that for $k = 0$ we can simply take $\tilde{v} = \bar{v}$ the extension by 0 of v, whereas for $k = 1$, we need to use the extension operator from $H^1(\Omega)$ into $H^1(\mathbb{R}^3)$.
  Since B is smooth and $\tilde{v} \in H^k(B)$, we know from Theorem III.4.2 that this problem has a unique solution $\varphi \in H^{k+2}(B)$, with the estimate $\|\varphi\|_{H^{k+2}} \leq C\|\tilde{v}\|_{H^k(B)} \leq C'\|v\|_{H^k(\Omega)}$.
  We now define Ψ_1 to be the restriction to Ω of $-\nabla\varphi$. It is clear that Ψ_1 satisfies in Ω the equations

$$\operatorname{div}\Psi_1 = v, \operatorname{curl}\Psi_1 = 0,$$

and that $\|\Psi_1\|_{H^{k+1}(\Omega)} \leq \|\varphi\|_{H^{k+2}(B)} \leq C'\|v\|_{H^k(\Omega)}$.
As a consequence, we are led now to prove the result in the case where $v = 0$.

- We set $\mathcal{U} = B \setminus \overline{\Omega}$. Notice that $\mathcal{U}$ can be multiply connected. We denote by $\mathcal{U}_1^*, \ldots, \mathcal{U}_k^*$ its connected components.
Let $1 \leq j \leq k$, we define $g_j^* \in H^{-1/2}(\partial\mathcal{U}_j^*)$ as follows

$$g_j^* = \begin{cases} 0, & \text{on } \partial\mathcal{U}_j^* \cap \partial B \\ \gamma_\nu(u), & \text{on } \partial\mathcal{U}_j^* \cap \partial\Omega. \end{cases}$$

The boundary of $\mathcal{U}_j^*$ is necessarily the union of some of the sets ∂B, Γ_1, ... Γ_m, thus we see that the conditions (IV.28) imply that

$$\langle g_j^*, 1\rangle_{H^{-1/2}(\partial\mathcal{U}_j^*), H^{1/2}(\partial\mathcal{U}_j^*)} = 0.$$

As a consequence, denoting by ν_j^* the outward normal to $\mathcal{U}_j^*$ (which is the opposite to the normal $\nu = \nu_\Omega$), Theorem III.4.3 shows that the Laplace–Neumann problem

$$\begin{cases} -\Delta\varphi_j = 0, & \text{in } \mathcal{U}_j^* \\ -\dfrac{\partial\varphi_j}{\partial\nu^*} = g_j^*, & \text{on } \partial\mathcal{U}_j^*, \end{cases}$$

has a solution $\varphi_j \in H^1(\mathcal{U}_j^*)$ which satisfies.

$$\|\nabla\varphi_j\|_{L^2(\mathcal{U}_j^*)} \leq C\|g_j^*\|_{H^{-1/2}(\partial\mathcal{U}_j^*)} \leq C'\|u\|_{H_{\mathrm{div}}(\Omega)} = C'\|u\|_{L^2(\Omega)},$$

the last equality being true because $\operatorname{div} u = 0$ in Ω.
As a consequence the function u^* defined by

$$u^* = \begin{cases} u, & \text{in } \Omega, \\ \nabla\varphi_j, & \text{in } \mathcal{U}_j^* \text{ for any } 1 \leq j \leq k, \end{cases}$$

belongs to $(L^2(B))^3$ and satisfies

$$\begin{aligned} \|u^*\|_{L^2(B)} &\leq C\|u\|_{L^2(\Omega)}, \\ \operatorname{div} u^* &= 0, && \text{in } B, \\ \gamma_\nu(u^*) &= 0, && \text{on } \partial B. \end{aligned}$$

The ball B is a smooth domain, therefore we can apply Theorem IV.4.11 in B and thus find a vector field $\Psi \in (H^1(B))^3$ such that

$$u^* = \operatorname{curl}\Psi \text{ and } \operatorname{div}\Psi = 0, \text{ in } B,$$

$$\|\Psi\|_{H^1(B)} \leq C\|u^*\|_{L^2(B)} \leq C'\|u\|_{L^2(\Omega)}.$$

It follows that the restriction of Ψ to Ω satisfies the required properties.

- In the case where $u \in (H^1(\Omega))^3$, following the same idea as before, we need to construct a suitable extension u^* of u in the ball B. To this end, on each connected component $\mathcal{U}_j^*$ of $\mathcal{U}$ we use the trace lifting operator to build a $u_j^* \in (H^1(\mathcal{U}_j^*))^3$ such that

$$\gamma_0(u_j^*) = \gamma_0(u) \text{ on } \partial\mathcal{U}_j^*$$
$$\|u_j^*\|_{H^1(\mathcal{U}_j^*)} \leq C\|\gamma_0(u)\|_{H^{1/2}} \leq C'\|u\|_{H^1(\Omega)}.$$

The Stokes formula and the conditions (IV.28) show that

$$\int_{\mathcal{U}_j^*} \operatorname{div}(u_j^*)\, dx = 0.$$

Therefore, using Theorem IV.3.1, we know that there exists a $u_j \in (H_0^1(\mathcal{U}_j^*))^3$ such that $\operatorname{div} u_j = \operatorname{div} u_j^*$ and

$$\|u_j\|_{H^1} \leq C\|\operatorname{div} u_j^*\|_{L_0^2} \leq C'\|u_j^*\|_{H^1}.$$

Gathering all these results, we see that the function u^* defined by

$$u^* = \begin{cases} u, & \text{in } \Omega, \\ u_j^* - u_j, & \text{in } \mathcal{U}_j^* \text{ for any } 1 \leq j \leq k, \end{cases}$$

satisfies

$$u^* \in (H^1(B))^3,\ \operatorname{div} u^* = 0 \text{ in } B,\ \|u^*\|_{H^1(B)} \leq C\|u\|_{H^1(\Omega)}.$$

Since B is smooth, we can apply Theorem IV.4.11 in B and thus find a vector field $\Psi \in (H^2(B))^3$ such that

$$u^* = \operatorname{curl}\Psi,\ \operatorname{div}\Psi = 0, \text{ in } B,$$

$$\|\Psi\|_{H^2(B)} \leq C\|u^*\|_{H^1(B)} \leq C'\|u\|_{H^1(\Omega)}.$$

Here also the claim is proved by considering the restriction of Ψ to Ω.

□

5 The Stokes problem

5.1 Well-posedness of the Stokes problem

We recall here the spaces introduced and studied in Section 3.3:

$$\mathscr{V} = \left\{\varphi \in (\mathcal{D}(\Omega))^d,\, \operatorname{div}\varphi = 0\right\},$$
$$V = \left\{v \in \left(H_0^1(\Omega)\right)^d,\, \operatorname{div} v = 0\right\},$$
$$H = \left\{v \in \left(L^2(\Omega)\right)^d,\, \operatorname{div} v = 0,\, \gamma_\nu(v) = 0\right\}.$$

We also recall that the vector fields of $(L^2(\Omega))^d$ with divergence in $L^2(\Omega)$ (i.e., belonging to $H_{\text{div}}(\Omega)$) have a normal trace that is well-defined in $H^{-1/2}(\partial\Omega)$ (see Section 3.2).

The dual of the space V, once the identification of H with its dual is made, is denoted as V'. As we saw in Chapter II, we have $V \subset H \subset V'$ with compact and dense embeddings (Lemma II.3.6 and Corollary II.3.8).

For all f in $(H^{-1}(\Omega))^d$, we refer to the following system of equations as the homogeneous *Stokes problem*, with Dirichlet boundary conditions,

$$\begin{cases} -\Delta v + \nabla p = f, & \text{in } \Omega, \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ v = 0, & \text{on } \partial\Omega. \end{cases} \tag{IV.31}$$

As we saw in Chapter I, these equations describe a laminar steady flow of a homogeneous incompressible fluid.

We can prove an existence and uniqueness theorem for weak solutions of the Stokes problem with homogeneous Dirichlet boundary conditions as follows.

Theorem IV.5.1. *Let Ω be a connected, bounded, Lipschitz domain of $\mathbb{R}^d$, and let f be in $\left(H^{-1}(\Omega)\right)^d$; then there exists a unique pair $(v,p) \in V \times L_0^2(\Omega)$, which is a solution of* (IV.31)*.*

Proof.

We consider the Hilbert space V defined at the beginning of this section and we define the bilinear form on V by

$$a(v,w) = \int_\Omega \nabla v : \nabla w \, dx.$$

This bilinear form is continuous and coercive on V (because from the Poincaré inequality $v \mapsto a(v,v)$ is a norm on $(H_0^1(\Omega))^d$). Now let the linear form L be defined by $L(w) = \langle f, w\rangle_{H^{-1},H_0^1}$. This linear form is continuous on V. From the Lax–Milgram theorem (Theorem II.2.5), we deduce that there exists a unique element $v \in V$ which is a solution of

$$a(v,w) = L(w),\, \forall w \in V. \tag{IV.32}$$

Hence, by the Stokes formula (see Theorem III.1.8 and its consequences) and from the definition of the duality bracket between $H^{-1}(\Omega)$ and $H_0^1(\Omega)$ via the scalar product of $L^2(\Omega)$ given by (III.39), it follows

$$\langle -\Delta v - f, w\rangle_{H^{-1},H_0^1} = 0,\ \forall w \in V.$$

Therefore, from Theorem IV.2.3 there exists a $p \in L^2(\Omega)$ such that

$$-\Delta v + \nabla p = f.$$

If we choose p with a zero mean value, we ensure the uniqueness of the pressure because Ω is connected.

□

Remark IV.5.1. It is easily observed that the velocity field v obtained above through the Lax–Milgram theorem is the unique minimiser on V of the energy functional

$$J(w) = \frac{1}{2}\int_\Omega |\nabla w|^2\, dx - \int_\Omega f \cdot w\, dx,\ \forall w \in V.$$

This is a general property of the solution of variational problems in the symmetric case.

The term $-\Delta v - f \in (H^{-1}(\Omega))^d$ is nothing but the gradient of J considered as a functional defined on $(H_0^1(\Omega))^d$. Since v minimises J on the subspace V of divergence-free vector fields, the Lagrange multiplier theorem shows that, at the optimum point v, the gradient of J has to be proportional to the gradient of the constraint in some sense. More precisely, the constraint here is linear and defined by the operator $\mathrm{div} : (H_0^1(\Omega))^d \to L_0^2(\Omega)$ and thus there must exist a $\Pi \in (L_0^2(\Omega))'$ such that

$$(\nabla J)(v) = \Pi \circ \mathrm{div} \in (H^{-1}(\Omega))^d.$$

Since $L_0^2(\Omega)'$ can be identified with $L_0^2(\Omega)$ itself, there exists a unique $p \in L_0^2(\Omega)$ such that

$$\langle \Pi, q\rangle_{(L_0^2)', L_0^2} = \int_\Omega pq\, dx,\ \forall q \in L_0^2(\Omega),$$

so that

$$\langle \Pi \circ \mathrm{div}, w\rangle_{H^{-1},H_0^1} = \int_\Omega p(\mathrm{div}\, w)\, dx = -\langle \nabla p, w\rangle_{H^{-1},H_0^1},\ \forall w \in (H_0^1(\Omega))^d;$$

that is $\Pi \circ \mathrm{div} = -\nabla p$. It follows from these computations that, for the Stokes problem, the pressure can be understood as the Lagrange multiplier related to the divergence-free constraint. This was already mentionned in Section 6 of Chapter I.

We can now study a slightly generalised Stokes problem. Its consists in assuming, instead of the divergence-free condition, that $\mathrm{div}\, v$ is a given function g which represents mass sources in the domain, and that the velocity v is equal to some given boundary data on $\partial\Omega$.

Theorem IV.5.2. *Let Ω be a connected, bounded, Lipschitz domain of $\mathbb{R}^d$. Let $f \in (H^{-1}(\Omega))^d$, $v_b \in (H^{1/2}(\partial\Omega))^d$, and $g \in L^2(\Omega)$ satisfying the compatibility condition*

$$\int_{\partial\Omega} v_b \cdot \nu \, d\sigma = \int_{\Omega} g \, dx, \tag{IV.33}$$

then there exists a unique pair (v, p) in $(H^1(\Omega))^d \times L^2_0(\Omega)$ which is a solution of

$$\begin{cases} -\Delta v + \nabla p = f, & \textit{in } \Omega, \\ \operatorname{div} v = g, & \textit{in } \Omega, \\ v = v_b, & \textit{on } \partial\Omega. \end{cases} \tag{IV.34}$$

Moreover, there exists a $C > 0$ depending only on Ω such that

$$\|v\|_{H^1} + \|p\|_{L^2_0} \leq C \left(\|f\|_{H^{-1}} + \|g\|_{L^2} + \|v_b\|_{H^{1/2}} \right).$$

Remark IV.5.2. By the divergence theorem (Theorem III.1.8), we know that if a solution to (IV.34) exists, then the compatibility condition IV.33 necessarily holds.

Proof.

The idea of the proof is to build a lifting of the trace and of the mass source term in order to reduce the problem to a classic Stokes problem (i.e., with homogeneous boundary data and divergence-free condition) with a modified source term belonging to $(H^{-1}(\Omega))^d$.

Thanks to Theorem III.2.22, $G = R_0(v_b) \in (H^1(\Omega))^d$ is such that $\gamma_0(G) = v_b$. This lifting G does not satisfy a priori the equation $\operatorname{div} G = g$. Nevertheless, thanks to the data compatibility condition and the divergence theorem, we have

$$\int_{\Omega} \operatorname{div} G \, dx = \int_{\partial\Omega} G \cdot \nu \, d\sigma = \int_{\partial\Omega} v_b \cdot \nu \, d\sigma = \int_{\Omega} g \, dx.$$

Hence, the function $g - \operatorname{div} G$ belongs to $L^2(\Omega)$ and has a zero mean value, and thus from Theorem IV.3.1, there exists a function H in $(H^1_0(\Omega))^d$ satisfying $\operatorname{div} H = g - \operatorname{div} G$. Hence, the function $G + H$ simultaneously satisfies the conditions

$$\operatorname{div}(G + H) = g, \text{ in } \Omega, \text{ and } \ (G + H) = v_b \text{ on } \partial\Omega.$$

We then seek v in the form $v = G + H + w$ where w must now satisfy the system

$$\begin{cases} -\Delta w + \nabla p = f + \Delta(G + H), & \text{in } \Omega, \\ \operatorname{div} w = 0, & \text{in } \Omega, \\ w = 0, & \text{on } \partial\Omega. \end{cases}$$

The source term $f + \Delta(G+H)$ belongs to $(H^{-1}(\Omega))^d$ so that Theorem IV.5.1 allows us to conclude the existence of a unique pair $(w, p) \in (H^1_0(\Omega))^d \times L^2_0(\Omega)$

which is a solution to the problem above. The pair $(v = G + H + w, p)$ is then the unique solution of system (IV.34). Furthermore, we have

$$\|w\|_{H^1} + \|p\|_{L^2_0} \leq C\|f + \Delta(G + H)\|_{H^{-1}} \\ \leq C\left(\|f\|_{H^{-1}} + \|G\|_{H^1} + \|H\|_{H^1_0}\right).$$

However, according to Theorem IV.3.1, the function H satisfies

$$\|H\|_{H^1_0} \leq C\|g - \operatorname{div} G\|_{L^2} \leq C(\|g\|_{L^2} + \|G\|_{H^1}).$$

Furthermore, by continuity of the trace lifting operator (Theorem III.2.22), we have

$$\|G\|_{H^1} \leq C\|v_b\|_{H^{1/2}}.$$

Finally, one obtains

$$\|w\|_{H^1} \leq C\left(\|f\|_{H^{-1}} + \|g\|_{L^2} + \|v_b\|_{H^{1/2}}\right).$$

And hence, since $v = w + G + H$, we have proved

$$\|v\|_{H^1} \leq C\left(\|f\|_{H^{-1}} + \|g\|_{L^2} + \|v_b\|_{H^{1/2}}\right).$$

To obtain the estimate of the pressure, we write the equation in the form

$$\nabla p = f + \Delta v,$$

from which we deduce

$$\|\nabla p\|_{H^{-1}} \leq \|f\|_{H^{-1}} + \|v\|_{H^1} \\ \leq C\left(\|f\|_{H^{-1}} + \|g\|_{L^2} + \|v_b\|_{H^{1/2}}\right).$$

Hence, according to the Poincaré inequality given in Proposition IV.1.7, and since p has a zero mean value, we indeed have

$$\|p\|_{L^2} \leq C\left(\|f\|_{H^{-1}} + \|g\|_{L^2} + \|v_b\|_{H^{1/2}}\right).$$

□

5.2 Stokes operator

In fact, in the previous section, we can apply the Lax–Milgram theorem for $f \in V'$ (which is larger than $(H^{-1}(\Omega))^d$), but then we can no longer interpret the solution as a solution to the Stokes problem because we cannot show the existence of the pressure. This is essentially due to the fact that the space V' is not a distribution space.

Nevertheless, the bilinear form a defined in the proof above being continuous on $V \times V$, we can define an operator A from V to V' by

$$\langle Av, u\rangle_{V',V} = \int_\Omega \nabla u : \nabla v \, dx, \ \forall u, v \in V. \tag{IV.35}$$

The operator A is called the *Stokes operator*. From the Lax–Milgram theorem this operator is an isomorphism from V onto V'.

We can now consider A as an unbounded operator in H with domain

$$D(A) = \{u \in V, \ Au \in H\}.$$

We check now that this operator $(A, D(A))$ enters the framework described in Section 6 of Chapter II.

Lemma IV.5.3. *The operator $(A, D(A))$ has a closed graph in $H \times H$.*

Proof.

Let $(u_n)_n$ be a sequence of elements of $D(A)$ which converges in H towards an element $u \in H$ and such that $(Au_n)_n$ converges in H towards a certain f. Since $f \in H$, there exists a unique $v \in D(A)$ such that $Av = f$. We then need to show that $v = u$.

The convergence in H implies that in V', therefore we deduce that $(Au_n)_n$ converges towards Av in V'. However, A is an isomorphism from V onto V' and hence $(u_n)_n$ converges towards v in V. The convergence in V implies the convergence in H, thus we have shown that $u = v$.

□

As we saw in the general theory of unbounded operators this property implies that $D(A)$, equipped with the scalar product defined in (II.29), is a Hilbert space and that the operator A is an isomorphism from $D(A)$ onto H. Furthermore for all $u \in D(A)$ we have

$$\|u\|_V \leq C\|Au\|_{V'} \leq \widetilde{C}\|Au\|_H \leq \widetilde{C}\|u\|_{D(A)},$$

which shows that the canonical embedding from $D(A)$ to V is continuous. Finally, since the embedding from V to H is compact, by using Lemma II.3.5 we see that the embedding from $D(A)$ to H is compact.

Last, we have the following fundamental result.

Lemma IV.5.4. *The Stokes operator $(A, D(A))$ is self-adjoint in H.*

Proof.

- First, by definition, for all $u, v \in D(A)$, we have

$$(Au, v)_H = \langle Au, v\rangle_{V',V} = \int_\Omega \nabla u : \nabla v \, dx,$$

and therefore it is clear that

$$(Au, v)_H = (u, Av)_H, \, \forall u, v \in D(A). \tag{IV.36}$$

This implies, in particular, that $D(A) \subset D(A^*)$ and that for all $u \in D(A)$, $A^*u = Au$.

- We now need to show that $D(A^*) = D(A)$. Let $u \in D(A^*)$. By definition, there exists $f = A^*u \in H$ such that for all $v \in D(A)$ we have

$$(Av, u)_H = (v, f)_H.$$

However, A is bijective from $D(A)$ onto H, and thus there exists $\tilde{u} \in D(A)$ such that $f = A\tilde{u}$. Let us show that $u = \tilde{u}$. To do this, let $w \in H$; there then exists $\tilde{w} \in D(A)$ such that $A\tilde{w} = w$. We then have

$$(w, u - \tilde{u})_H = (A\tilde{w}, u)_H - (A\tilde{w}, \tilde{u})_H. \tag{IV.37}$$

From the definition of the adjoint we have

$$(A\tilde{w}, u)_H = (\tilde{w}, A^*u)_H,$$

and from (IV.36), since $\tilde{u} \in D(A)$, we have

$$(A\tilde{w}, \tilde{u})_H = (\tilde{w}, A\tilde{u})_H.$$

By definition of $\tilde{u}$, we have $A\tilde{u} = A^*u$ and hence, (IV.37) becomes

$$(w, u - \tilde{u})_H = 0,$$

and this for all $w \in H$, which shows that $u = \tilde{u}$ and, in particular, that $u \in D(A)$.

□

In consequence of all the preceding results, the Stokes operator enters the general framework given in Section 6 of Chapter II. We can therefore construct a spectral decomposition of the Stokes operator. More precisely, we have the following result.

Theorem IV.5.5. *Let Ω be a bounded, connected, Lipschitz domain of $\mathbb{R}^d$. There exists an increasing sequence of positive real numbers $(\lambda_k)_{k\geq 1}$, which tends towards $+\infty$, and a sequence of functions $(w_k)_{k\geq 1}$ which is orthonormal in H, orthogonal in V and in $D(A)$, forming a complete family in H, in V, and in $D(A)$, and a sequence of functions $(p_k)_{k\geq 1}$ in $L^2_0(\Omega)$ satisfying*

$$\begin{cases} -\Delta w_k + \nabla p_k = \lambda_k w_k, & \text{in } \Omega, \\ \operatorname{div} w_k = 0, & \text{in } \Omega, \\ w_k = 0, & \text{on } \partial\Omega. \end{cases} \tag{IV.38}$$

Proof.

As we have just seen, the Stokes operator enters the framework of Theorem II.6.6. In consequence, there exists an orthonormal basis $(w_k)_k$ of H formed with eigenvectors of A for the eigenvalues $(\lambda_k)_k$, which is also a complete orthogonal family of $D(A)$.

It is clear, by construction of the Stokes operator, and via the preceding results of this chapter (Theorem IV.2.3) that there then exist pressures $p_k \in L^2_0(\Omega)$ such that the system (IV.38) is satisfied.

Moreover, we have

$$\lambda_k \|w_k\|_H^2 = (Aw_k, w_k)_H = \int_\Omega |\nabla w_k|^2 \, dx > 0,$$

which proves that the eigenvalues λ_k are positive and that they can therefore be ordered in a divergent monotonically increasing sequence.

It remains to prove that $(w_k)_k$ is a complete orthogonal family of V. First, since $D(A) \subset V$, we indeed have $w_k \in V$ for all k. Moreover, we have (from the identification of H with its dual we remember that we are in the situation where $V \subset H \subset V'$)

$$\begin{aligned}(w_k, w_l)_V = \int_\Omega \nabla w_k : \nabla w_l \, dx &= \langle Aw_k, w_l\rangle_{V',V} = (Aw_k, w_l)_H \\ &= \lambda_k (w_k, w_l)_H = \lambda_k \delta_{kl}.\end{aligned}$$

This proves that the family $(w_k)_k$ is orthogonal in V. Finally, $D(A)$ is dense in V, and $(w_k)_k$ is complete in $D(A)$, thus it is clear that this is a complete family in V.

□

Definition IV.5.6. *One such system of functions is called the special basis associated with the Stokes operator A of domain $D(A)$.*

We often use this family of functions to construct approximate solutions of unsteady problems in the next chapters.

Given the special basis of H made up of eigenfunctions of the Stokes operator, we can now give a suitable formula of the Leray projection $\mathcal{P}$ (Definition IV.3.6).

Lemma IV.5.7. *Let Ω be a bounded, connected, Lipschitz domain of $\mathbb{R}^d$. Let $(w_k)_{k\geq 1}$ be a basis of H made up of eigenfunctions of the Stokes operator. Then, for any function $u \in (L^2(\Omega))^d$, we have*

$$\mathcal{P}u = \sum_{k=1}^{+\infty} (u, w_k)_{L^2} \; w_k.$$

Proof.

Let us define

$$v = \sum_{k=1}^{+\infty} (u, w_k)_{L^2}\; w_k.$$

The sequence $(w_k)_k$ forms an orthogonal family of H (and hence an orthogonal family in $(L^2(\Omega))^d$), thus it is clear that $v \in H$. Furthermore, for all $n \geq 1$ we have

$$(v, w_n)_{L^2} = \sum_{k=1}^{+\infty} (u, w_k)_{L^2} (w_k, w_n)_{L^2} = (u, w_n)_{L^2}.$$

Hence, $u - v$ is orthogonal to all the w_n and therefore $u - v$ is orthogonal to all of H. This indeed proves that v is the orthogonal projection of u on H.

□

The Stokes operator satisfies elliptic regularity properties which are similar to those of the Laplace operator, for instance. More precisely, we can state the following result.

Theorem IV.5.8 (Regularity of the Stokes problem). *Let Ω be a bounded connected domain of class $\mathcal{C}^{k+1,1}$ in $\mathbb{R}^d$, with $k \geq 0$. Then, for any $f \in \left(H^k(\Omega)\right)^d$, $g \in H^{k+1}(\Omega)$, and $v_b \in (H^{k+3/2}(\partial\Omega))^d$ satisfying (IV.33), the unique solution of (IV.34) in $V \times L^2_0(\Omega)$ satisfies*

$$(v, p) \in \left(H^{k+2}(\Omega)\right)^d \times H^{k+1}(\Omega)$$

and we have

$$\|v\|_{H^{k+2}} + \|p\|_{H^{k+1}} \leq C \left(\|f\|_{H^k} + \|g\|_{H^{k+1}} + \|v_b\|_{H^{k+3/2}}\right),$$

where $C > 0$ depends only on Ω.

This result is admitted for the moment. The proof of this result is the subject of Section 6. We note that, for $k = 0$, this result is even true in convex polygonal domains, for example (see, e.g., [73]).

Remark IV.5.3. It is important to note that, as we might expect, the pressure p has one degree of regularity less than the velocity field v.

The preceding regularity theory enables a precise description of the domain of the Stokes operator in the following way.

Proposition IV.5.9 (Domain of the Stokes operator). *Let Ω be a bounded domain of $\mathbb{R}^d$ of class $\mathcal{C}^{1,1}$, then we have:*

$$D(A) = V \cap (H^2(\Omega))^d.$$

Moreover

$$Av = \mathcal{P}(-\Delta v),\ \forall v \in D(A),$$

where $\mathcal{P}$ is the Leray projection and there exists a $C > 0$ such that

$$\frac{1}{C}\|v\|_{H^2} \leq \|Av\|_H \leq C\|v\|_{H^2},\ \forall v \in D(A).$$

In particular, if Ω is of class $\mathcal{C}^{1,1}$ the eigenfunctions $(w_k)_k$ of the Stokes operator belong to $(H^2(\Omega))^d$ and the associated pressures $(p_k)_k$ belong to $H^1(\Omega)$.

Proof.

- Let us show that $V \cap (H^2(\Omega))^d \subset D(A)$. Let $v \in V \cap (H^2(\Omega))^d$, then a priori Av is an element of V'.
  For $\varphi \in \mathscr{V} \subset H$ and since v is smooth, we can integrate by parts and use the Leray projection $\mathcal{P}$ to obtain

$$\begin{aligned}\langle Av, \varphi\rangle_{V',V} &= \int_\Omega \nabla v : \nabla \varphi \, dx = \int_\Omega (-\Delta v) \cdot \varphi \, dx \\ &= (-\Delta v, \varphi)_{L^2} = (\mathcal{P}(-\Delta v), \varphi)_H = \langle \mathcal{P}(-\Delta v), \varphi\rangle_{V',V}.\end{aligned}$$

  By definition $\mathscr{V}$ is dense in V, thus this shows that $Av = \mathcal{P}(-\Delta v) \in H$ and hence that $v \in D(A)$. Moreover, since $\mathcal{P}$ is an orthogonal projection in $(L^2(\Omega))^d$ we have

$$\|Av\|_H \leq \| - \Delta v\|_{L^2} \leq C\|v\|_{H^2}.$$

- It therefore remains to show that $D(A) \subset V \cap (H^2(\Omega))^d$. Let $v \in D(A)$, that is such that $f = Av \in H$. We can now write for all $\varphi \in V$

$$\begin{aligned}\langle f, \varphi\rangle_{H^{-1},H^1_0} &= (f, \varphi)_{L^2} = (f, \varphi)_H = \langle Av, \varphi\rangle_{V',V} \\ &= \int_\Omega \nabla v : \nabla \varphi \, dx = \langle -\Delta v, \varphi\rangle_{H^{-1},H^1_0}.\end{aligned}$$

  Hence $-\Delta v - f$ is an element of $(H^{-1}(\Omega))^d$ such that for all $\varphi \in V$, we have $\langle -\Delta v - f, \varphi\rangle_{H^{-1},H^1_0} = 0$. From de Rham's theorem (Theorem IV.2.4), there exists $p \in L^2(\Omega)$ such that

$$-\Delta v + \nabla p = f.$$

  Since $f \in H \subset (L^2(\Omega))^d$, the regularity Theorem IV.5.8 implies that $v \in (H^2(\Omega))^d$ and that

$$\|v\|_{H^2} \leq C\|f\|_{L^2} = C\|Av\|_H,$$

  which ends the proof.

□

We now return to the definition of the fractional powers of the operator A described in Section 6 of Chapter II and we set for all $s \in \mathbb{R}$,

$$V_s = D(A^{s/2}).$$

We denote the scalar product and the norm on V_s as $(\cdot,\cdot)_s$ and $\|\cdot\|_s$, respectively. We then have the following result.

Proposition IV.5.10. *If Ω is of class $\mathcal{C}^{1,1}$ and if $1 \le s \le 2$, then*

$$V_s = V \cap (H^s(\Omega))^d.$$

If Ω is of class $\mathcal{C}^{k,1}$, $k \ge 2$ and if $2 < s \le k+1$, we have only

$$V_s \subset V \cap (H^s(\Omega))^d.$$

Proof.

The case $s = 2$ is given by the preceding proposition. We will only study the case $s = 1$, the rest of the proposition being admitted (see, for example, [44]).

We now wish to show that $V_1 = D(A^{1/2})$ is equal to V. To do this, we consider the basis of eigenfunctions of the Stokes operator $(w_k)_k$ which, as we have seen, all belong to V. Furthermore, we have

$$\begin{aligned}(w_k, w_l)_{D(A^{1/2})} &= \delta_{kl}(1+\lambda_k)(w_k, w_k)_H = \delta_{kl}(\|w_k\|_H^2 + (Aw_k, w_k)_H)\\ &= \delta_{kl}\left(\|w_k\|_H^2 + \int_\Omega |\nabla w_k|^2\,dx\right) = \delta_{kl}(\|w_k\|_H^2 + \|w_k\|_V^2)\\ &= (w_k, w_l)_H + (w_k, w_l)_V.\end{aligned} \tag{IV.39}$$

Any $u \in D(A^{1/2})$ is written as a limit in $D(A^{1/2})$ of finite sums of the type $S_N = \sum_{k=1}^N \alpha_k w_k$, this sequence $(S_N)_N$ is therefore a Cauchy sequence in $D(A^{1/2})$. From (IV.39), we get

$$\|S_N - S_{N+p}\|_V \le \|S_N - S_{N+p}\|_{D(A^{1/2})},$$

so that $(S_N)_N$ is a Cauchy sequence in V and thus converges in V. Its limit is necessarily equal to u because the embeddings of $D(A^{1/2})$ and of V in H are continuous and the limit of $(S_N)_N$ in H is unique. As a consequence, $D(A^{1/2}) \subset V$ and (IV.39) gives

$$(u, v)_{D(A^{1/2})} = (u, v)_H + (u, v)_V, \quad \forall u, v \in D(A^{1/2}).$$

Hence, the norms of V and of $D(A^{1/2})$ are equivalent in this last space and since $D(A^{1/2})$ is complete, this is a closed set of V.

Finally, we know that $D(A^{1/2})$ is dense in V, because the $(w_k)_k$, which form a complete family in V, all belong to $D(A^{1/2})$. This proves that $D(A^{1/2}) = V$.

□

Using these definitions, we can prove a continuity result by interpolation, which is nothing but the expression of Theorem II.5.14 in the particular case of interest to us.

Theorem IV.5.11. *Let (α, β) be two real numbers such that $\beta \leq \alpha$. Let*

$$E = \left\{ u \in L^2(]0,T[, V_\alpha),\ \frac{du}{dt} \in L^2(]0,T[, V_\beta) \right\}.$$

Then E is continuously embedded in $\mathcal{C}^0([0,T], V_{(\alpha+\beta)/2})$.

Proof.

- First, we take a function u written as

$$u(t) = \sum_{k=1}^{N} \alpha_k(t) w_k,$$

where $\alpha_k \in \mathcal{C}^\infty([0,T])$. We can write

$$\alpha_k(t) = \theta_1(t)\alpha_k(t) + (1-\theta_1(t))\alpha_k(t) = \beta_k(t) + \gamma_k(t),$$

with θ_1 being regular with support included in $[0, T/2]$ and, for example, $\theta_1 = 1$ identically on $[0, T/3]$. We set

$$v(t) = \sum_{k=1}^{N} \beta_k(t) w_k,$$

and we establish the result for v (the argument being made in an identical way for $u - v$). By noting that $\beta_k(t) = (v(t), w_k)_H$, we have the equality

$$\frac{1}{2}\frac{d}{dt}(v,v)_{(\alpha+\beta)/2} = \left(\frac{dv}{dt}, v\right)_{(\alpha+\beta)/2} = \sum_{k=1}^{N} \left((1+\lambda_k^{(\alpha+\beta)/2}) \beta_k'(t)\beta_k(t) \right).$$

Hence, since $v(+\infty) = 0$, we have for all $t \in [0,T]$,

$$\begin{aligned}
\frac{1}{2}||v(t)||^2_{(\alpha+\beta)/2} &= -\int_t^T \sum_{k=1}^{N} \left((1+\lambda_k^{(\alpha+\beta)/2}) \beta_k'(\tau)\beta_k(\tau) \right) d\tau \\
&\leq \frac{1}{2}\int_0^T \left(\sum_{k=1}^{N} (1+\lambda_k^{\alpha})\big(\beta_k(\tau)\big)^2 + \sum_{k=1}^{N} (1+\lambda_k^{\beta})\big(\beta_k'(\tau)\big)^2 \right) d\tau \\
&\leq \frac{1}{2}\int_0^T \left(||v(\tau)||^2_\alpha + \left\| \frac{dv}{dt}(\tau) \right\|^2_\beta \right) d\tau \leq \frac{1}{2}||v||^2_E.
\end{aligned}$$

The function u (and therefore v) being clearly continuous with values in $V_{(a+\beta)/2}$, this shows that

$$\|u\|_{\mathcal{C}^0([0,T],V_{(a+\beta)/2})} \leq C\|u\|_E.$$

- By density of $\mathcal{C}^\infty([0,T])$ in $H^1(]0,T[)$, we can extend the previous result to the case where all the functions α_k belong to $H^1(]0,T[)$. However, any function u of E decomposes in a unique way in the form

$$u(t) = \sum_{k=1}^{+\infty} \alpha_k(t)\omega_k,$$

where the α_k belong to $H^1(]0,T[)$ and the convergence of the series takes place in E. We can therefore extend the result to all functions of E.

□

Let us finish this section with a new sight to the Poincaré inequality.

Proposition IV.5.12. *Let Ω be a bounded, Lipschitz domain of $\mathbb{R}^d$. We have the following Poincaré inequalities*

$$\|v\|_H^2 \leq \frac{1}{\lambda_1}\|\nabla v\|_{L^2}^2,\ \forall v \in V, \tag{IV.40}$$

$$\|\nabla v\|_{L^2}^2 \leq \frac{1}{\lambda_1}\|Av\|_{L^2}^2,\ \forall v \in D(A),$$

where λ_1 is the smallest eigenvalue of the Stokes operator.

Moreover, the constant $1/\lambda_1$ is optimal in these two inequalities.

Proof.

We already proved the Poincaré inequality in a more general way (Proposition III.2.38). Inequality (IV.40) gives the value of the optimum constant in the case of divergence-free vector fields. We denote the special basis of the eigenfunctions of the Stokes operator as $(w_k)_{k\geq 1}$.

- Let us start by proving (IV.40).
  Let $v = \sum_{k\geq 1} v_k w_k \in V$, such that $Av = \sum_{k\geq 1} \lambda_k v_k w_k$. We then have

$$\int_\Omega |\nabla v|^2\, dx = \langle Av, v\rangle_{V',V} = \sum_{k\geq 1} \lambda_k |v_k|^2,$$

and since λ_1 is the smallest of the eigenvalues of A, we have

$$\int_\Omega |\nabla v|^2\, dx \geq \lambda_1 \sum_{k\geq 1} |v_k|^2 = \lambda_1 \|v\|_H^2.$$

It is clear that this constant is optimal, because if we take $v = w_1$ in (IV.40) it becomes an equality.

- Now, let $v \in D(A)$, which we again write in the special basis $v = \sum_{k\geq 1} v_k w_k$. We have

$$\|Av\|_{L^2}^2 = \sum_{k\geq 1} \lambda_k^2 v_k^2 \geq \lambda_1 \sum_{k\geq 1} \lambda_k v_k^2 = \lambda_1 \langle Av, v\rangle_{V',V} = \lambda_1 \|\nabla v\|_{L^2}^2.$$

The equality case is again obtained by taking $v = w_1$.

□

5.3 The unsteady Stokes problem

In this section, we use the elements introduced before concerning the Stokes operator in order to solve the following unsteady incompressible Stokes problem

$$\begin{cases} \dfrac{\partial v}{\partial t} - \Delta v + \nabla p = f, & \text{in } \Omega, \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ v = 0, & \text{on } \partial\Omega, \\ v(0) = v_0. \end{cases} \tag{IV.41}$$

Notice that the Reynolds number is chosen equal to 1. This is allowed without loss of generality because the problem is linear.

Theorem IV.5.13. *Let Ω be a connected, bounded, Lipschitz domain of $\mathbb{R}^d$. For any $v_0 \in H$, and $f \in L^2_{loc}([0,+\infty[,(H^{-1}(\Omega))^d)$, there exists a unique pair (v,p) defined on $\mathbb{R}^+$ which solves (IV.41) and such that for any $T > 0$ we have*

$$(v,p) \in \big(\mathcal{C}^0([0,T],H) \cap L^2(]0,T[,V)\big) \times W^{-1,\infty}(]0,T[,L^2_0(\Omega)),$$

and

$$\frac{dv}{dt} \in L^2(]0,T[,V').$$

Moreover, it satisfies the following energy equality for any $t \geq 0$.

$$\frac{1}{2}\|v(t)\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\int_0^t \|\nabla v(\tau)\|_{L^2}^2 d\tau = \frac{1}{2}\|v_0\|_{L^2}^2 + \int_0^t \langle f(\tau), v(\tau)\rangle_{H^{-1},H^1_0} d\tau.$$

Proof.

In order to illustrate the application of semigroup theory, here we only give the proof in the case of more regular source terms. The proof in the general case can be performed in a similar way as we do for the Navier–Stokes problem in Chapter V, that is, by using a suitable Galerkin approximation of the problem.

Let us assume that $f \in \mathcal{C}^1([0,+\infty[,(L^2(\Omega))^d)$. We can use Theorem II.6.12 which shows that there exists a unique solution $v \in \mathcal{C}^0([0,+\infty[,H) \cap \mathcal{C}^1(]0,+\infty[,H) \cap \mathcal{C}^0(]0,+\infty[,D(A))$ of the linear problem

$$\frac{dv}{dt} + Av = \mathcal{P}f,$$

with $v(0) = v_0$. We used here the Leray projection $\mathcal{P}$, so that $\mathcal{P}f \in \mathcal{C}^1([0, +\infty[, H)$.

Let $\varphi \in V$ be any time-independent test function. For any $t > 0$ we thus have

$$0 = \left(\frac{dv}{dt}(t) + Av(t) - \mathcal{P}f(t), \varphi\right)_H = \left(\frac{dv}{dt}(t) - \mathcal{P}f(t), \varphi\right)_{L^2} + \langle Av(t), \varphi\rangle_{V',V}.$$

By definition of the Leray projection and since $\varphi \in V \subset H$, we have

$$(\mathcal{P}f(t), \varphi)_{L^2} = (f(t), \varphi)_{L^2},$$

and by definition of the Stokes operator and integration by parts.

$$\langle Av(t), \varphi\rangle_{V',V} = \int_\Omega \nabla v(t) : \nabla\varphi\, dx = \langle -\Delta v(t), \varphi\rangle_{H^{-1},H_0^1}.$$

It follows

$$0 = \left\langle \frac{dv}{dt}(t) - \Delta v(t) - f(t), \varphi \right\rangle_{H^{-1},H_0^1}.$$

This being true for any $\varphi \in V$, we can use Theorem IV.2.3 to obtain that there exists a unique $p(t) \in L_0^2(\Omega)$ such that

$$\frac{dv}{dt}(t) - \Delta v(t) - f(t) = -\nabla p(t), \text{ in } H^{-1}(\Omega). \tag{IV.42}$$

From the regularity properties of v and f, it follows that the pressure gradient satisfies $\nabla p \in \mathcal{C}^0(]0, +\infty[, (H^{-1}(\Omega))^d)$.

Using the Nečas inequality (Theorem IV.1.1) and the Poincaré inequality (Proposition IV.1.7), we deduce that p belongs to $\mathcal{C}^0(]0, +\infty[, L_0^2(\Omega))$.

We conclude the proof by observing that (IV.42) is true for any $t > 0$, so that (v, p) solves the unsteady Stokes problem in the distribution sense.

□

Remark IV.5.4. We observe that the above result is independent on the space dimension d which is considered. We show in Chapter V that the situation is very different for the Navier–Stokes equations due to the presence of nonlinear terms in the system.

5.4 Penalty approximation of the Stokes problem

In this section, we present an alternative method to prove the well-posedness of the Stokes problem, that is to prove Theorem IV.5.1. In addition to its theoretical interest (it does not directly use de Rham's theorem, but rather the Nečas inequality), the method has a practical interest in particular as far as numerical approximation of the solution of the Stokes problem is concerned.

For any $\varepsilon > 0$ we consider the following penalty approximation of the system

$$\begin{cases} -\Delta v_\varepsilon + \nabla p_\varepsilon = f, & \text{in } \Omega, \\ \operatorname{div} v_\varepsilon + \varepsilon p_\varepsilon = 0, & \text{in } \Omega, \\ v_\varepsilon = 0, & \text{on } \partial\Omega. \end{cases} \tag{IV.43}$$

The main advantage of this new problem is that the pressure p_ε can be easily eliminated from the system (see (IV.45)). As a consequence, it is very easy to show that this system is well-posed.

Lemma IV.5.14. *Let Ω be a bounded, connected, Lipschitz domain in $\mathbb{R}^d$ and $f \in (H^{-1}(\Omega))^d$. For any $\varepsilon > 0$, there exists a unique pair $(v_\varepsilon, p_\varepsilon) \in (H^1_0(\Omega))^d \times L^2_0(\Omega)$ satisfying (IV.43).*

Proof.

We observe that the second equation of (IV.43) can be immediately solved by

$$p_\varepsilon = -\frac{1}{\varepsilon} \operatorname{div} v_\varepsilon, \tag{IV.44}$$

so that the first equation of (IV.43) becomes

$$-\Delta v_\varepsilon - \frac{1}{\varepsilon}\nabla(\operatorname{div} v_\varepsilon) = f. \tag{IV.45}$$

It simply remains to solve this new problem by using the Lax–Milgram theorem. Indeed, the continuous bilinear form $a_\varepsilon : (H^1_0(\Omega))^d \times (H^1_0(\Omega))^d \to \mathbb{R}$ defined by

$$a_\varepsilon(v, w) = \int_\Omega \nabla v : \nabla w \, dx + \frac{1}{\varepsilon}\int_\Omega (\operatorname{div} v)(\operatorname{div} w)\, dx, \; \forall v, w \in (H^1_0(\Omega))^d,$$

is clearly coercive on $(H^1_0(\Omega))^d$ and the claim is proved.

□

Proposition IV.5.15. *With the same assumption as in Lemma IV.5.14, there exists a $C > 0$ such that*

$$\|v_\varepsilon\|_{H^1} + \|p_\varepsilon\|_{H^1} \leq C\|f\|_{H^{-1}}, \; \forall \varepsilon > 0.$$

Proof.

The application of the Lax–Milgram theorem above leads to the estimate

$$\|\nabla v_\varepsilon\|^2_{L^2} + \frac{1}{\varepsilon}\|\operatorname{div} v_\varepsilon\|^2_{L^2} \leq \|f\|_{H^{-1}}\|v_\varepsilon\|_{H^1_0}.$$

By using the Poincaré inequality, and (IV.44), we deduce

$$\|\nabla v_\varepsilon\|_{L^2}^2 + 2\varepsilon\|p_\varepsilon\|_{L^2}^2 \le \|f\|_{H^{-1}}^2.$$

This estimate is not satisfactory for the pressure term. To obtain the bound on p_ε that we claimed, we use the Nečas inequality (Theorem IV.1.1) and more precisely the related Poincaré inequality (IV.8). This inequality gives

$$\|p_\varepsilon\|_{L^2} \le C\|\nabla p_\varepsilon\|_{H^{-1}} = C\|\Delta v_\varepsilon + f\|_{H^{-1}} \le C\|\nabla v_\varepsilon\|_{L^2} + C\|f\|_{H^{-1}}.$$

Combining this with the estimate above, we obtain the claim.

□

We can deduce from this result the existence of a solution $(v,p) \in (H_0^1(\Omega))^d \times L_0^2(\Omega)$ of the Stokes problem, that is, proving Theorem IV.5.1. Indeed, from the bounds obtained in Proposition IV.5.15, we can find a subsequence $(v_{\varepsilon_n})_n$ (resp., $(p_{\varepsilon_n})_n$) which weakly converges towards some v in $(H_0^1(\Omega))^d$ (resp., towards some p in $L_0^2(\Omega)$). These weak convergences let us pass to the limit in the approximate problem (IV.43) and prove that the limits (v,p) actually solve the Stokes problem. This solution is unique, therefore we can deduce the convergence of the whole families $(v_\varepsilon)_\varepsilon$ and $(p_\varepsilon)_\varepsilon$ (see Proposition II.2.9). Moreover, we have $\|\operatorname{div} v_\varepsilon\|_{L^2} \le C\varepsilon$.

We conclude this section by proving a first-order error estimate for this approximation method.

Proposition IV.5.16. *With the same assumption as in Lemma IV.5.14, there exists a $C > 0$ such that*

$$\|v - v_\varepsilon\|_{H^1} + \|p - p_\varepsilon\|_{L^2} \le C\varepsilon.$$

Proof.

We set $w_\varepsilon = v - v_\varepsilon$, $q_\varepsilon = p - p_\varepsilon$ and we write the equations satisfied by $(w_\varepsilon, q_\varepsilon)$

$$\begin{cases} -\Delta w_\varepsilon + \nabla q_\varepsilon = 0, & \text{in } \Omega, \\ \operatorname{div} w_\varepsilon - \varepsilon p_\varepsilon = 0, & \text{in } \Omega, \\ w_\varepsilon = 0, & \text{on } \partial\Omega. \end{cases} \tag{IV.46}$$

Using once more the inequality (IV.8), we obtain

$$\|q_\varepsilon\|_{L^2} \le C\|\nabla q_\varepsilon\|_{H^{-1}} \le C\|\Delta w_\varepsilon\|_{H^{-1}} \le C\|\nabla w_\varepsilon\|_{L^2}. \tag{IV.47}$$

Then, we use w_ε as a test-function in the first equation of (IV.46) to obtain

$$\|\nabla w_\varepsilon\|_{L^2}^2 = \int_\Omega (\operatorname{div} w_\varepsilon) q_\varepsilon \, dx \le \|\operatorname{div} w_\varepsilon\|_{L^2}\|q_\varepsilon\|_{L^2} \le \varepsilon\|p_\varepsilon\|_{L^2}\|q_\varepsilon\|_{L^2}.$$

From the estimate given in Proposition IV.5.15, from (IV.47) and from the Young inequality we get

$$\|\nabla w_\varepsilon\|_{L^2}^2 \le C\varepsilon^2.$$

Coming back to (IV.47) we get

$$\|q_\varepsilon\|_{L^2}^2 \leq C\varepsilon^2,$$

and the claim is proved.

□

6 Regularity of the Stokes problem

6.1 First degree of regularity

The goal of this section is to prove Theorem IV.5.8 for $k = 0$. We demonstrate the general version of Theorem IV.5.8 in Section 6.2, using an induction argument.

6.1.1 Local regularity

The first step in proving Theorem IV.5.8 is to establish a local regularity result. Notice that this result does not necessitate any assumption on the domain Ω. Moreover, it is independent on the boundary condition satisfied by the solution.

Theorem IV.6.1. *Let Ω be any open set of $\mathbb{R}^d$, $f \in (L^2_{loc}(\Omega))^d$, and $g \in H^1_{loc}(\Omega)$. For any (v,p) in $(H^1_{loc}(\Omega))^d \times L^2_{loc}(\Omega)$ which is a solution of*

$$\begin{cases} -\Delta v + \nabla p = f, & \textit{in } \Omega,\\ \operatorname{div} v = g, & \textit{in } \Omega, \end{cases}$$

then $(v,p) \in (H^2_{loc}(\Omega))^d \times H^1_{loc}(\Omega)$. More precisely, for any bounded open sets ω, ω_2 such that $\overline{\omega} \subset \omega_2 \subset \overline{\omega_2} \subset \Omega$, there exists a $C(\omega,\omega_2) > 0$, such that

$$\begin{aligned} \|v\|_{H^2(\omega)} + \|p\|_{H^1(\omega)} \leq & C(\omega,\omega_2)\big(\|f\|_{L^2(\omega_2)} + \|g\|_{H^1(\omega_2)} \\ & + \|v\|_{H^1(\omega_2)} + \|p\|_{L^2(\omega_2)}\big). \end{aligned}$$

Proof.

From Proposition III.2.3, we know that is sufficient to prove that, for any $\varphi \in \mathcal{D}(\Omega)$, we have $\overline{\varphi v} \in H^2(\mathbb{R}^d)$ and $\overline{\varphi p} \in H^1(\mathbb{R}^d)$ with a suitable estimate on $\|\overline{\varphi v}\|_{H^2(\mathbb{R}^d)} + \|\overline{\varphi p}\|_{H^1(\mathbb{R}^d)}$.

To this end, we set $w = \overline{\varphi v}$ and $q = \overline{\varphi p}$. Since φ is smooth, following the same computations as in the proof of Theorem III.4.2, we immediately find the equation satisfied by w and q in the sense of distributions of $\mathbb{R}^d$

$$\begin{cases} -\Delta w + \nabla q = \varphi\overline{f} - (\Delta\varphi)v - 2\overline{\nabla v}.\nabla\varphi + \overline{p}\nabla\varphi,\\ \operatorname{div} w = \varphi\overline{g} + \overline{v}\cdot\nabla\varphi. \end{cases}$$

If we denote the second terms of these equations as $\tilde{f}$ and $\tilde{g}$, it is clear that $\tilde{f}$ lies in $(L^2(\mathbb{R}^d))^d$, that $\tilde{g}$ belongs to $H^1(\mathbb{R}^d)$, and that

$$\begin{aligned}\|\tilde{f}\|_{L^2(\mathbb{R}^d)} \leq &\|\varphi\|_{L^\infty}\|f\|_{L^2(\omega_2)} + \|\Delta\varphi\|_{L^\infty}\|v\|_{L^2(\omega_2)} \\ &\quad + 2\|\nabla\varphi\|_{L^\infty}\|v\|_{H^1(\omega_2)} + \|\nabla\varphi\|_{L^\infty}\|p\|_{L^2(\omega_2)}, \\ \leq &C_1(\varphi)(\|f\|_{L^2(\omega_2)} + \|g\|_{H^1(\omega_2)} + \|v\|_{H^1(\omega_2)} + \|p\|_{L^2(\omega_2)}).\end{aligned} \tag{IV.48}$$

Similarly, we obtain

$$\|\tilde{g}\|_{H^1(\mathbb{R}^d)} \leq C_2(\varphi)(\|f\|_{L^2(\omega_2)} + \|g\|_{H^1(\omega_2)} + \|v\|_{H^1(\omega_2)} + \|p\|_{L^2(\omega_2)}). \tag{IV.49}$$

Moreover, we already have the estimate given by Theorem IV.5.2, which gives us here

$$\|w\|_{H^1(\mathbb{R}^d)} \leq (\|\varphi\|_{L^\infty} + \|\nabla\varphi\|_{L^\infty})\|v\|_{H^1(\omega_2)} \leq C(\varphi)\|v\|_{H^1(\omega_2)}, \tag{IV.50}$$

and

$$\|q\|_{L^2(\mathbb{R}^d)} \leq \|\varphi\|_{L^\infty}\|p\|_{L^2}. \tag{IV.51}$$

We are therefore led to studying the following Stokes system in the whole space $\mathbb{R}^d$

$$\begin{cases} -\Delta w + \nabla q = \tilde{f}, \\ \quad \operatorname{div} w = \tilde{g}. \end{cases} \tag{IV.52}$$

We note that all of the functions appearing in this system are zero outside the compact set $\overline{\Omega}$, which justifies all the operations which follow.

We use the characterisation of the Sobolev spaces $H^{k+1}(\mathbb{R}^d)$ by means of translation operators which is given in Proposition III.2.32. We use the notations introduced in Section 2.7 of Chapter III. We need to estimate $\delta_h w$ and $\delta_h q$. From (IV.52) we obtain

$$\begin{cases} -\Delta\delta_h w + \nabla\delta_h q = \delta_h\tilde{f}, \\ \quad \operatorname{div}\delta_h w = \delta_h\tilde{g}. \end{cases}$$

We use the Nečas inequality (Theorem IV.1.1) which gives that

$$\|\delta_h q\|_{L^2} \leq C(\|\delta_h q\|_{H^{-1}} + \|\nabla\delta_h q\|_{H^{-1}}).$$

The term $\|\delta_h q\|_{H^{-1}}$ is bounded by $|h|\|q\|_{L^2}$ according to Lemma III.2.31. The second term is expressed using the first equation of the system

$$\|\nabla\delta_h q\|_{H^{-1}} = \|\delta_h\tilde{f} + \Delta\delta_h w\|_{H^{-1}} \leq C|h|\|\tilde{f}\|_{L^2} + \|\nabla\delta_h w\|_{L^2},$$

so that we finally get

$$\|\delta_h q\|_{L^2} \leq C|h|(\|q\|_{L^2} + \|\tilde{f}\|_{L^2}) + C\|\nabla\delta_h w\|_{L^2}. \tag{IV.53}$$

We can now use $\delta_h w$ (which is compactly supported) as a test function in the first equation to obtain, with the help of the second equation,

$$\begin{aligned}\int_{\mathbb{R}^d} |\nabla \delta_h w|^2\, dx &= \langle \delta_h \tilde{f}, \delta_h w\rangle_{H^{-1},H^1} + \int_\Omega \delta_h q \delta_h \tilde{g}\, dx \\ &\le \|\delta_h \tilde{f}\|_{H^{-1}} \|\delta_h w\|_{H^1} + \|\delta_h q\|_{L^2} \|\delta_h \tilde{g}\|_{L^2} \\ &\le C|h|\|\tilde{f}\|_{L^2}(\|\nabla \delta_h w\|_{L^2} + |h|\|w\|_{H^1}) + C|h|\|\tilde{g}\|_{H^1}\|\delta_h q\|_{L^2}.\end{aligned}$$

Using (IV.53), it follows

$$\begin{aligned}\|\nabla \delta_h w\|_{L^2}^2 \le & C|h|\|\tilde{f}\|_{L^2}(\|\nabla \delta_h w\|_{L^2} + |h|\|w\|_{H^1}) \\ & + C|h|^2\|\tilde{g}\|_{H^1}(\|q\|_{L^2} + \|\tilde{f}\|_{L^2}) + C|h|\|\tilde{g}\|_{H^1}\|\nabla \delta_h w\|_{L^2}.\end{aligned}$$

Through Young's inequality, this gives

$$\|\nabla \delta_h w\|_{L^2}^2 \le C|h|^2 \left(\|\tilde{f}\|_{L^2}^2 + \|w\|_{H^1}^2 + \|\tilde{g}\|_{H^1}^2 + \|q\|_{L^2}^2\right).$$

We can now use the estimates (IV.48) through (IV.51) which finally give us, for any $h \in \mathbb{R}^d$,

$$\|\nabla \delta_h w\|_{L^2}^2 \le C(\varphi)|h|^2 \left(\|f\|_{L^2(\omega_2)}^2 + \|g\|_{H^1(\omega_2)}^2 + \|v\|_{H^1(\omega_2)}^2 + \|p\|_{L^2(\omega_2)}^2\right).$$

This allows us to conclude that $w \in (H^2(\mathbb{R}^d))^d$ from Proposition III.2.32 and that we have the required estimate on $\|w\|_{H^2(\mathbb{R}^d)}$. Furthermore, we can immediately see in the first equation of (IV.52) that it implies that ∇q lies in $(L^2(\Omega))^d$ and that we have

$$\begin{aligned}&\|v\|_{H^2(\omega_1)} + \|p\|_{H^1(\omega_1)} \le \|w\|_{H^2} + \|q\|_{H^1} \\ \le & C(\varphi)\big(\|f\|_{L^2(\omega_2)} + \|g\|_{H^1(\omega_2)} + \|v\|_{H^1(\omega_2)} + \|p\|_{L^2(\omega_2)}\big).\end{aligned}$$

□

Remark IV.6.1. Assume that $v \in (H^1(\Omega))^d$, $p \in L^2(\Omega)$, $f \in (L^2(\Omega))^d$, and $g \in H^1(\Omega)$. Let us consider the family of open sets of $\mathbb{R}^d$ defined by

$$\omega_k = \left\{x \in \Omega,\ d(x, \partial\Omega) > \frac{1}{k}\right\}.$$

For all sufficiently large k, we have

$$\omega_k \subset \overline{\omega_k} \subset \omega_{k+1} \subset \cdots.$$

The estimate given by the previous theorem can be specified in the following way,

$$\|v\|^2_{H^2(\omega_k)} + \|p\|^2_{H^1(\omega_k)} \leq C_k \left(\|f\|^2_{L^2(\Omega)} + \|g\|^2_{H^1(\Omega)} + \|v\|^2_{H^1(\Omega)} + \|p\|^2_{L^2(\Omega)} \right).$$

We note that the constant C_k which acts in this inequality can be easily estimated by

$$C_k \sim \|\nabla\varphi_k\|^2_{L^\infty} + \|\Delta\varphi_k\|^2_{L^\infty},$$

where φ_k is the truncature function which is supported in ω_{k+1} and which is equal to 1 on ω_k. A simple calculation shows that C_k behaves essentially as does k^8 when k tends towards infinity. This shows that the above technique does not allow us to pass to the limit, when k tends towards infinity, to obtain regularity of the solution on all Ω.

This is in particular because the regularity up to the boundary of the solution of an elliptic problem, as with the Stokes problem, strongly depends on the kind and the regularity of the boundary condition satisfied by the solution and also on the regularity of the domain Ω.

6.1.2 Regularity up to the boundary

We proved in the previous section the local regularity result for the Stokes problem. In order to complete the proof of the regularity result, it is now necessary to obtain regularity of the solution up to the boundary. As for the Laplace problem (see Section 4 of Chapter III), this is done in two main steps: we first obtain tangential regularity by using translations along tangent vector fields, then we deduce the complete regularity directly from the equations.

We first illustrate this strategy in the case; of the flat half-space $\mathbb{R}^d_+$ because the computations are much simpler; then will consider the general case by using the tools introduced in Section 3 of Chapter III.

6.1.2.1 The case of the half-space $\mathbb{R}^d_+$

In this paragraph, we illustrate the general strategy of proof in the case of the simple geometry of the half-space $\mathbb{R}^d_+$. The general case is considered in the next paragraph.

Let us consider here the following problem

$$\begin{cases} -\Delta v + v + \nabla p = f, & \text{in } \mathbb{R}^d_+, \\ \operatorname{div} v = 0, & \text{in } \mathbb{R}^d_+, \\ v = 0, & \text{on } \partial\mathbb{R}^d_+, \end{cases} \tag{IV.54}$$

where $f \in (L^2(\mathbb{R}^d_+))^d$. Because of the unbounded character of the domain $\mathbb{R}^d_+$ that we are considering in this example, and in order to simplify the description of the method, we have added a zero-order term v to the operator considered. This ensures the coercivity of the operator on $(H^1(\mathbb{R}^d_+))^d$ and hence the existence and uniqueness of solutions. This system is therefore a model problem that we use to illustrate the method that we use below on a sufficiently smooth bounded domain Ω.

Due to the coercivity of the operator, we know that there exists a unique pair $(v,p) \in (H^1_0(\mathbb{R}^d_+))^d \times L^2_{loc}(\mathbb{R}^d_+)$, the pressure being defined apart from a constant, providing a solution of the preceding problem and satisfying

$$\|v\|_{H^1_0} + \|\nabla p\|_{H^{-1}} \leq C\|f\|_{H^{-1}}. \tag{IV.55}$$

Remark IV.6.2. We pay attention to the fact that it is not possible to assert (see Proposition IV.2.4) that $p \in L^2(\mathbb{R}^d_+)$ or even to $H^{-1}(\mathbb{R}^d_+)$. In particular, we cannot consider a pressure here with a zero mean, unlike in the case of a bounded domain.

We want to use the characterisation of tangential Sobolev spaces through translation operators that we introduced in Section 3.5 of Chapter III.

In the case $\Omega = \mathbb{R}^d_+$ considered here, it is easily seen that we only need to consider constant tangential vector fields $\theta(x) = e_i$, for $i = 1, \ldots, d-1$. The corresponding flows $h \mapsto \tau^i(h,x)$, are thus defined with the differential equation

$$\begin{cases} \dfrac{d}{dh}\tau^i(h,x) = e_i, \\ \tau^i(0,x) = x. \end{cases}$$

This equation can of course be immediately solved to give $\tau^i(h,x) = x + he_i$.

The vectors e_i, $i = 1, \ldots, d-1$ are tangent to the boundary of $\mathbb{R}^d_+$, thus these flows satisfy the condition: for all $x \in \mathbb{R}^d_+$, and for all $h \in \mathbb{R}$, $\tau^i(h,x)$ also belongs to $\mathbb{R}^d_+$.

The geometry being very simple, the commutation estimates of Proposition III.3.19 are useless. Indeed, we easily check that in that case the translation operators and the differential operators exactly commute

$$\frac{\partial(v \circ \tau^i(h,x))}{\partial x_j} = \left(\frac{\partial v}{\partial x_j}\right) \circ \tau^i(h,x),\ \forall j \in \{1, \ldots, d\}, \forall i \in \{1, \ldots, d-1\}.$$

As we did in Chapter III, we then introduce the difference operators associated with these translation operators $(\delta^i_h v)(x) = v(\tau^i(h,x)) - v(x)$. Then, we can develop the same approach as we used for the Laplace problem in Section 4 of Chapter III. Of course, the presence of the pressure term and of the divergence-free constraint implies additional difficulties that we describe below. In particular, it should be noticed that tangential and normal components of the velocity field are not studied in a similar way.

Proposition IV.6.2. *Assuming that $f \in (L^2(\mathbb{R}^d_+))^d$, the solution (v,p) of System (IV.54), satisfies $v \in (H^2(\mathbb{R}^d_+))^d$ and $\nabla p \in L^2(\mathbb{R}^d_+)$. Moreover, there exists a $C > 0$ such that*

$$\|v\|_{H^2} + \|\nabla p\|_{L^2} \leq C\|f\|_{L^2}.$$

Proof.

- Step 1 : tangential regularity.
Let $\varphi \in (H_0^1(\mathbb{R}_+^d))^d$ such that $\operatorname{div} \varphi = 0$. The function $\delta_{-h}^i \varphi$ is also divergence-free (thanks to the commutation property described above) and may therefore be chosen as a test function in our problem. By using Lemma III.2.30, we obtain a variational problem satisfied by $\delta_h^i v$

$$\int_{\mathbb{R}_+^d} \nabla(\delta_h^i v) : \nabla\varphi \, dx + \int_{\mathbb{R}_+^d} (\delta_h^i v) \cdot \varphi \, dx = \int_{\mathbb{R}_+^d} (\delta_h^i f) \cdot \varphi \, dx,$$

for all $\varphi \in (H_0^1(\mathbb{R}_+^d))^d$, such that $\operatorname{div} \varphi = 0$. Taking $\varphi = \delta_h^i v$ in this equation gives

$$\|\delta_h^i v\|_{H^1}^2 = \langle \delta_h^i f, \delta_h^i v \rangle_{H^{-1}, H_0^1}.$$

Using the assumption that $f \in L^2(\mathbb{R}_+^d)$ and Lemma III.2.31, we obtain the inequality

$$\|\delta_h^i v\|_{H^1} \leq |h| \|f\|_{L^2},$$

which implies that v is tangentially regular on all $\mathbb{R}_+^d$. More precisely, by proceeding as in the proof of Proposition III.2.32, we deduce that $\partial v / \partial x_i$ belongs to $(H^1(\mathbb{R}_+^d))^d$ for all $i \in \{1, \ldots, d-1\}$. In other words, we have shown that $\partial^2 v / \partial x_i \partial x_j$ belongs to $L^2(\mathbb{R}_+^d)$ for all i, j except for $i = j = d$. We can now deduce the tangential regularity of the pressure. Indeed, for any $i \leq d-1$ we obtain from (IV.54) that

$$\begin{aligned} \frac{\partial}{\partial x_i}(\nabla p) &= \frac{\partial f}{\partial x_i} - \frac{\partial v}{\partial x_i} + \frac{\partial \Delta v}{\partial x_i} \\ &= \frac{\partial f}{\partial x_i} - \frac{\partial v}{\partial x_i} + \operatorname{div}\left(\nabla \frac{\partial v}{\partial x_i}\right). \end{aligned}$$

Note that in the term $\nabla \partial v / \partial x_i$, there is no second-order derivative of type $\partial^2 v / \partial x_d^2$; this term thus belongs to $(L^2(\mathbb{R}_+^d))^d$ from the preceding arguments. By taking the divergence of this term and by using the fact that $f \in (L^2(\mathbb{R}_+^d))^d$, we obtain

$$\nabla \frac{\partial p}{\partial x_i} = \frac{\partial}{\partial x_i}(\nabla p) \in (H^{-1}(\mathbb{R}_+^d))^d.$$

Furthermore, from (IV.55), we already know that $\dfrac{\partial p}{\partial x_i} \in H^{-1}(\mathbb{R}_+^d)$. Hence, from the Nečas inequality (Theorem IV.1.1), we have shown that the distribution $\dfrac{\partial p}{\partial x_i}$ lies in $L^2(\mathbb{R}_+^d)$ and satisfies the expected estimate for all $i \leq d-1$.
- Step 2 : Complete regularity.

To obtain the complete regularity of the solution, it remains to study the derivatives of v and p in the direction e_d. In the same way as for the Laplace problem, these properties are directly obtained from the partial differential equations and using the tangential regularity properties we just obtained.

– Substep 1 : Complete regularity of the normal component of the velocity field.
  We differentiate the divergence equation with respect to x_d, which gives

$$\frac{\partial^2 v_d}{\partial x_d^2} = -\sum_{i=1}^{d-1} \frac{\partial^2 v_i}{\partial x_i \partial x_d} \in L^2(\mathbb{R}^d_+).$$

  Therefore, we have proved that the normal component v_d belongs to $H^2(\mathbb{R}^d_+)$.
– Substep 2 : Complete regularity of the pressure.
  We write the dth component of the momentum equation in the Stokes problem in the following form

$$\frac{\partial p}{\partial x_d} = \Delta v_d - v_d + f_d,$$

  so that, using the regularity of v_d that we proved above, we immediately see that $\partial p/\partial x_d$ also lies in $L^2(\mathbb{R}^d_+)$.
  Therefore, we have proved that $p \in H^1(\mathbb{R}^d_+)$.
– Substep 3 : Complete regularity of tangential components of the velocity field.
  For $i \leq d-1$, we can write the ith equation of the system in the form

$$\frac{\partial^2 v_i}{\partial x_d^2} = -\sum_{j=1}^{d-1} \frac{\partial^2 v_i}{\partial x_j^2} + v_i + \frac{\partial p}{\partial x_i} - f_i.$$

  We deduce that $\partial^2 v_i/\partial x_d^2$ also belongs to $L^2(\mathbb{R}^d_+)$, and thus that $v_i \in H^2(\mathbb{R}^d_+)$.

The regularity theorem is then proved and more precisely, we have established the existence of some $C > 0$ such that

$$\|v\|_{H^2} + \|\nabla p\|_{L^2} \leq C\Big(\|f\|_{L^2} + \|\nabla p\|_{H^{-1}}\Big).$$

□

Let us now follow the same scenario in the case of a general smooth bounded domain in the sequel.

6.1.2.2 Tangential regularity

We recall that we have already shown in Theorem IV.6.1 the local regularity $(H^2_{loc}(\Omega))^d \times H^1_{loc}(\Omega)$ of the solution of the Stokes problem. Let us now show the tangential regularity $(H^2_{tang}(\Omega))^d \times H^1_{tang}(\Omega)$ of this same solution if the data are regular up to the boundary.

We use here the notation introduced in Section 3 of Chapter III.

Theorem IV.6.3. *Let Ω be a connected bounded domain of class $\mathcal{C}^{1,1}$ of $\mathbb{R}^d$, $f \in (L^2(\Omega))^d$ and $g \in H^1(\Omega) \cap L^2_0(\Omega)$. If (v,p) is the unique solution in $(H^1_0(\Omega))^d \times L^2_0(\Omega)$ of*

$$\begin{cases} -\Delta v + \nabla p = f, & \text{in } \Omega, \\ \operatorname{div} v = g, & \text{in } \Omega, \\ v = 0, & \text{on } \partial\Omega, \end{cases}$$

then $v \in (H^2_{tang}(\Omega))^d$ and $p \in H^1_{tang}(\Omega)$ and for any vector field θ satisfying (III.77) with $k = 0$, there exists a $C(\Omega,\theta) > 0$ such that

$$|||v|||_{\theta,2} + |||p|||_{\theta,1} \leq C(\Omega,\theta)\left(\|f\|_{L^2} + \|g\|_{H^1}\right).$$

The proof of this result requires several steps, inspired by the calculations carried out in the case of the half-space (see Section 6.1.2.1). We begin by looking for the equation satisfied by the differences $\delta^\theta_h v = v \circ \tau^\theta(h,\cdot) - v$ and $\delta^\theta_h p = p \circ \tau^\theta(h,\cdot) - p$, paying attention to the differentiation operations.

The lemma below gives a suitable expression of the commutators between the differential operators and the translation operators along the tangential fields. In the case of the half-space that we have considered previously, these commutators were zero because of the particularly simple structure of the tangential translations.

Lemma IV.6.4. *We have the following equalities*

$$\begin{cases} -\Delta(\delta^\theta_h v) = -\delta^\theta_h(\Delta v) - \left[\operatorname{div}, \tau^\theta_h\right] \nabla v - \operatorname{div}\left(\left[\nabla, \tau^\theta_h\right] v\right), \\ \quad \nabla \delta^\theta_h p = \delta^\theta_h \nabla p + \left[\nabla, \tau^\theta_h\right] p, \\ \operatorname{div}(\delta^\theta_h v) = \delta^\theta_h(\operatorname{div} v) + \left[\operatorname{div}, \tau^\theta_h\right] v. \end{cases} \tag{IV.56}$$

Proof (Lemma IV.6.4).

This involves simple, purely algebraic computations.

□

Proof (Theorem IV.6.3).

Let $h \in]0,1[$ and let θ be a vector field tangential to the boundary of Ω (i.e., satisfying (III.77) with $k = 0$). We denote $w = \delta^\theta_h v$, $\pi = \delta^\theta_h p$. By using (IV.56), we see that the pair (w,π) satisfy the system

$$\begin{cases} -\Delta w + \nabla \pi = F, & \text{in } \Omega, \\ \operatorname{div} w = G, & \text{in } \Omega, \end{cases} \tag{IV.57}$$

where

$$F = \delta_h^\theta f + \left[\nabla, \tau_h^\theta\right] p - \left[\operatorname{div}, \tau_h^\theta\right] \nabla v - \operatorname{div}\left(\left[\nabla, \tau_h^\theta\right] v\right), \tag{IV.58}$$

and

$$G = \delta_h^\theta g + \delta_h^\theta(\operatorname{div} v) + \left[\operatorname{div}, \tau_h^\theta\right] v. \tag{IV.59}$$

We note that since for a fixed h $\tau^\theta(h,\cdot)$ is a bijection from $\partial\Omega$ onto itself, the function w belongs to $(H_0^1(\Omega))^d$. Let us now evaluate F in the $(H^{-1}(\Omega))^d$ norm. With (III.91) and (III.92) we obtain

$$\begin{aligned} \|\delta_h^\theta f\|_{H^{-1}} &\leq Ch\|f\|_{L^2}, \\ \left\|\left[\operatorname{div}, \tau_h^\theta\right] \nabla v\right\|_{H^{-1}} &\leq Ch\|\nabla v\|_{L^2}, \\ \left\|\operatorname{div}\left(\left[\nabla, \tau_h^\theta\right] v\right)\right\|_{H^{-1}} &\leq \left\|\left[\nabla, \tau_h^\theta\right] v\right\|_{L^2} \leq Ch\|v\|_{H^1}. \end{aligned}$$

We can then use the estimate in $(H^1(\Omega))^d$ on the solution v for the initial problem (Theorem IV.5.2) to obtain that

$$\|F\|_{H^{-1}} \leq Ch(\|f\|_{L^2} + \|g\|_{H^1}).$$

In the same way we can estimate G in $L^2(\Omega)$ by

$$\|G\|_{L^2} \leq Ch\|g\|_{H^1} + Ch\|v\|_{H^1} \leq Ch(\|f\|_{L^2} + \|g\|_{H^1}).$$

We then carry out the classic energy estimate for problem (IV.57) and obtain

$$\begin{aligned} \|\nabla w\|_{L^2}^2 \leq& \|F\|_{H^{-1}}\|\nabla w\|_{L^2} + \left|\int_\Omega \pi\, G\, dx\right| \\ \leq& Ch\left(\|f\|_{L^2} + \|g\|_{H^1}\right)\|\nabla w\|_{L^2} + \|\pi\|_{L^2}\|G\|_{L^2} \\ \leq& Ch\left(\|f\|_{L^2} + \|g\|_{H^1}\right)\|\nabla w\|_{L^2} \\ &+ Ch\|\pi\|_{L^2}\left(\|f\|_{L^2} + \|g\|_{H^1}\right). \end{aligned} \tag{IV.60}$$

Furthermore, we read the following estimate directly from Equation (IV.6.4)

$$\begin{aligned} \|\nabla \pi\|_{H^{-1}} &\leq \|\Delta w\|_{H^{-1}} + \|F\|_{H^{-1}} \\ &\leq C\|\nabla w\|_{L^2} + Ch\left(\|f\|_{L^2} + \|g\|_{H^1}\right), \end{aligned}$$

and from (III.92) and Theorem IV.5.2 we have

$$\|\pi\|_{H^{-1}} = \|\delta_h^\theta p\|_{H^{-1}} \leq Ch\|p\|_{L^2} \leq Ch\left(\|f\|_{L^2} + \|g\|_{H^1}\right).$$

Finally, from the Nečas inequality (see Theorem IV.1.1), we obtain

$$\begin{aligned}\|\pi\|_{L^2} &\le C(\|\pi\|_{H^{-1}} + \|\nabla\pi\|_{H^{-1}}) \\ &\le C\|\nabla w\|_{L^2} + Ch\left(\|f\|_{L^2} + \|g\|_{H^1}\right).\end{aligned} \tag{IV.61}$$

Hence, from inequalities (IV.60), (IV.61), and Young's inequality

$$\begin{aligned}\|\nabla w\|_{L^2}^2 &\le Ch\left(\|f\|_{L^2} + \|g\|_{H^1}\right)\|\nabla w\|_{L^2} + Ch^2\left(\|f\|_{L^2}^2 + \|g\|_{H^1}^2\right) \\ &\le Ch^2\left(\|f\|_{L^2}^2 + \|g\|_{H^1}^2\right) + \frac{1}{2}\|\nabla w\|_{L^2}^2.\end{aligned}$$

This allows us to conclude, using (IV.61), with the inequality

$$\|\nabla\delta_h^\theta v\|_{L^2}^2 + \|\delta_h^\theta p\|_{L^2}^2 \le Ch^2\left(\|f\|_{L^2}^2 + \|g\|_{H^1}^2\right),$$

which states precisely that

$$|||v|||_{\theta,2}^2 + |||p|||_{\theta,1}^2 \le C\left(\|f\|_{L^2}^2 + \|g\|_{H^1}^2\right).$$

□

6.1.2.3 Complete regularity up to the boundary

We now use the coordinate system established in a tubular neighborhood $\mathcal{O}_{s_0}^\rho$ of the boundary of Ω in Chapter III. We have just seen that, under the hypothesis of Theorem IV.5.8 with $k = 0$, the solution of the Stokes problem belongs to $(H^2_{tang}(\Omega))^d \times H^1_{tang}(\Omega)$. To prove the regularity of the solution on all Ω, it is sufficient to prove its regularity in $\mathcal{O}_{s_0}^\rho$. The tangential/normal coordinate system allows the roles of the "tangential" variables and the normal variable to be separated, as we did in the much simpler case of the half-space $\mathbb{R}^d_+$ in Section 6.1.2.1.

Recall that in Section 3 of Chapter III we used the notation $\langle x, y\rangle$ for the Euclidean scalar product of two vectors x and y, whereas in what follows we denote this scalar product as $x \cdot y$.

- In the open set $\mathcal{O}_{s_0}^\rho$, the equation on the divergence of v reads

$$\mathrm{Tr}(\nabla_T v) + \nabla_N v_N + v\cdot\tilde{\mathcal{G}} = g \in H^1(\mathcal{O}_{s_0}^\rho). \tag{IV.62}$$

By using the tangential regularity properties of v that we proved before, we see that the first term of this equality belongs to $H^1(\mathcal{O}_{s_0}^\rho)$. Furthermore, we have

$$\nabla_N\Big(v\cdot\tilde{\mathcal{G}}\Big) = (\nabla_N v)\cdot\tilde{\mathcal{G}} + v\cdot\nabla_N\tilde{\mathcal{G}}. \tag{IV.63}$$

Since $\tilde{\mathcal{G}}$ is controlled by $|\ln\rho|$ and ∇v belongs to $(L^2(\Omega))^{d\times d}$, we deduce that

$$(\nabla_N v)\cdot\tilde{\mathcal{G}} \in L^q(\Omega),\ \forall q < 2. \tag{IV.64}$$

Using that $|\nabla_N \tilde{\mathcal{G}}| \leq \frac{C}{\rho}$ and that $v = 0$ on $\partial\Omega$, the Hardy inequality implies that the second term in the right-hand side of (IV.63) belongs to $L^2(\Omega)$ and satisfies

$$\left\| v \cdot (\nabla_N \tilde{\mathcal{G}}) \right\|_{L^2} \leq C \|v\|_{H^1}. \tag{IV.65}$$

Therefore, it follows from (IV.62) and (IV.63) that

$$\nabla_N \nabla_N v_N \in L^q(\mathcal{O}^\rho_{s_0}),\ \forall q < 2.$$

- Let $q < 2$ and let us prove that $v_N \in W^{2,q}(\mathcal{O}^\rho_{s_0})$. First of all, we prove that $\nabla_N v_N \in (W^{1,q}(\Omega))^d$. To this end, we compute

$$\begin{aligned}\nabla(\nabla_N v_N) &= \nabla_T(\nabla_N v_N) + (\nabla_N \nabla_N v_N)\nu \\ &= \underbrace{\nabla_T\big((\nabla_N v) \cdot \nu\big)}_{=\nabla_T\big((\nabla v).\nu\cdot\nu\big)} + \nabla_T\big(v \cdot (\nabla_N \nu)\big) + (\nabla_N \nabla_N v_N)\nu.\end{aligned}$$

We have shown just before that the last term belongs to L^q. Since ν is Lipschitz continuous and $\nabla v \in (H^1_{tang}(\Omega))^{d\times d}$, the first term belongs to $L^2 \subset L^q$. Finally, using the properties of ν and the Hardy inequality, we see that the second term satisfies

$$\nabla_T\big(v \cdot (\nabla_N \nu)\big) \in L^2(\Omega) \subset L^q(\Omega). \tag{IV.66}$$

This actually proves that $\nabla_N v_N$ belongs to $(W^{1,q}(\Omega))^d$.
Finally, we write

$$\nabla v_N = \nabla_T v_N + (\nabla_N v_N)\nu,$$

and since each term has been proved to belong to $(W^{1,q}(\Omega))^d$, we deduce that v_N lies in $W^{2,q}(\Omega)$.

- We compute the normal part of the Laplacian of v as

$$(\Delta v)_N = (\Delta v) \cdot \nu = \Delta(v_N) - v \cdot (\Delta \nu) - 2(\nabla v) : (\nabla \nu).$$

We perform the same computations as above. In particular we use the Hardy inequality using the fact that $|\Delta \nu| \sim 1/\rho$ to prove that the second term satisfies

$$v \cdot (\Delta \nu) \in L^2(\Omega), \tag{IV.67}$$

and the fact that $v_N \in W^{2,q}(\Omega)$. We deduce that $(\Delta v)_N \in L^q(\Omega)$. The momentum equation projected onto the normal direction gives

$$-(\Delta v)_N + \nabla_N p = f_N,$$

and thus

$$\nabla_N p \in L^q(\Omega).$$

Since $\nabla_T p$ is already known to belong to $(L^2(\Omega))^d$, we finally get that $p \in W^{1,q}(\Omega)$.

- Let $1 \leq i \leq d$ and let us look at the ith component of the momentum equation

$$-\operatorname{Tr}(\nabla_T \nabla v_i) - \nabla_N \nabla_N v_i - \big((\nabla v_i) \cdot \tilde{\mathcal{G}}\big) + \frac{\partial p}{\partial x_i} = f_i.$$

All the terms except the second one are already known to belong to $L^q(\mathcal{O}^\rho_{s_0})$; therefore it follows that $\nabla_N \nabla_N v_i \in L^q(\mathcal{O}^\rho_{s_0})$.
Proceeding as before, we can prove that $\nabla_N v_i \in W^{1,q}(\mathcal{O}^\rho_{s_0})$. Indeed, we have

$$\nabla(\nabla_N v_i) = \nabla_T(\nabla_N v_i) + (\nabla_N \nabla_N v_i)\nu = \nabla_T\big((\nabla v).\nu \cdot e_i\big) + (\nabla_N \nabla_N v_i)\nu,$$

and each term belongs to L^q by using the properties of ν and all the above regularity properties of v.
To conclude, we write

$$\nabla v_i = \nabla_T v_i + (\nabla_N v_i)\nu,$$

and it follows that $\nabla v_i \in (W^{1,q}(\mathcal{O}^\rho_{s_0}))^d$ and then that $v \in (W^{2,q}(\Omega))^d$.

- We have proved that $v \in (W^{2,q}(\Omega))^d$ for all $q < 2$. By Sobolev embeddings, we deduce that there exists some $r > 2$ such that $\nabla v \in (L^r(\Omega))^{d \times d}$. As a consequence, (IV.64) also holds for $q = 2$ and the sequel of the proof above still holds in this case. We finally deduce that $v \in (H^2(\Omega))^d$.

The proof of Theorem IV.5.8 in the case $k = 0$ is now complete.

6.2 Higher-order regularity

Let us move to the general case of the elliptic regularity results for the Stokes problem.

Proof (Theorem IV.5.8).

Let $k \geq 0$ and let Ω be a connected bounded domain of class $\mathcal{C}^{k+1,1}$ of $\mathbb{R}^d$, let f be a function of $(H^k(\Omega))^d$, and g be a function of $H^{k+1}(\Omega)$. We wish to show that the solution $(v, p) \in (H^1_0(\Omega))^d \times L^2_0(\Omega)$ of

$$\begin{cases} -\Delta v + \nabla p = f, & \text{in } \Omega, \\ \operatorname{div} v = g, & \text{in } \Omega, \\ v = 0, & \text{on } \partial\Omega, \end{cases}$$

belongs to the space $(H^{k+2}(\Omega))^d \times H^{k+1}(\Omega)$. To do this, we proceed by induction.

The case $k = 0$ was proved in Section 6.1. Let us take $k \geq 1$ and assume that we have established the result with k replaced with $k-1$:

$$\|v\|_{H^{k+1}} + \|p\|_{H^k} \leq C\left(\|f\|_{H^{k-1}} + \|g\|_{H^k}\right).$$

We could first prove the local regularity, as before, by reducing the problem to the whole space $\mathbb{R}^d$ and then by carrying out translations parallel to the coordinate axis. To obtain tangential regularity, we again take the notation from the proof of Theorem IV.6.3. We recall that we established that $w = \delta_h^\theta v$ and $\pi = \delta_h^\theta p$, are solutions of the Stokes problem (IV.57) with the right-hand side terms defined by (IV.58) and (IV.59).

By using the induction hypothesis applied to the initial Stokes problem, the regularity of f and g, and the properties (III.91) and (III.92), we immediately obtain

$$\begin{aligned}\|F\|_{H^{k-1}} &\leq Ch(\|f\|_{H^k} + \|g\|_{H^{k+1}}),\\ \|G\|_{H^k} &\leq Ch(\|f\|_{H^k} + \|g\|_{H^{k+1}}).\end{aligned}$$

Hence, from the induction hypothesis applied to the Stokes problem (IV.57), we get

$$\|\delta_h^\theta v\|^2_{H^{k+1}} + \|\delta_h^\theta p\|^2_{H^k} \leq Ch^2(\|f\|^2_{H^k} + \|g\|^2_{H^{k+1}}).$$

This gives the result from the characterisation of tangential Sobolev spaces (Theorem III.2.32).

To conclude, we again use the tangential and normal coordinates in the tubular neighborhood $\mathcal{O}^\rho_{s_0}$ of $\partial\Omega$ in Ω, to deduce the regularity up to the boundary of the solution from the tangential regularity. Again, this proof requires a level of additional regularity on the domain Ω.

6.3 L^q theory of the Stokes problem

The last subsection of this section aims to introduce, very briefly, a few elements in the theory of the Stokes problem posed in $L^q(\Omega)$ spaces. We refer to [63] for a complete study of this problem.

First, we state the following theorem, proved by Agmon, Douglis, and Nirenberg, which, assuming the existence of a solution in $(W^{2,q}(\Omega))^d \times W^{1,q}(\Omega)$, allows us to obtain additional regularity on this solution. This theorem is in fact a special case of the more general theorems established in [2] and [3].

We emphasise the fact that this theorem is not an existence theorem of smooth solutions.

Theorem IV.6.5 (Agmon–Douglis–Nirenberg). *Let Ω be a bounded domain of $\mathbb{R}^d$, of class $\mathcal{C}^{k+2}$, where k is a positive integer.*

Let $1 < q < +\infty$*, and the data* $f \in (L^q(\Omega))^d$*,* $g \in W^{1,q}(\Omega)$*, and* $v_b \in (W^{2-1/q,q}(\partial\Omega))^d$ *satisfying the compatibility condition*

$$\int_\Omega g\,dx = \int_{\partial\Omega} v_b \cdot \nu\,d\sigma.$$

We assume that there exists a pair $(v,p) \in (W^{2,q}(\Omega))^d \times (W^{1,q}(\Omega) \cap L^2_0(\Omega))$*, which is a solution of the Stokes problem*

$$\begin{cases} -\Delta v + \nabla p = f, & \text{in } \Omega, \\ \ \operatorname{div} v = g, & \text{in } \Omega, \\ \ v = v_b, & \text{on } \partial\Omega. \end{cases}$$

If the data (f, g, v_b) *are more regular*

$$(f, g, v_b) \in (W^{k,q}(\Omega))^d \times W^{k+1,q}(\Omega) \times (W^{k+2-1/q,q}(\partial\Omega))^d,$$

then the solution of the problem above satisfies

$$(v,p) \in (W^{k+2,q}(\Omega))^d \times W^{k+1,q}(\Omega), \tag{IV.68}$$

and we have the estimate

$$\begin{aligned} &\|v\|_{W^{k+2,q}} + \|p\|_{W^{k+1,q}} \\ \le& C\left(\|f\|_{W^{k,q}} + \|g\|_{W^{k+1,q}} + \|v_b\|_{W^{k+2-\frac{1}{q},q}} + d_q\|v\|_{L^q}\right), \end{aligned} \tag{IV.69}$$

where C *depends only on* (q, k, Ω) *and* $d_q = 0$ *for* $q \geq 2$ *and* 1 *otherwise.*

This theorem is a genuine regularity theorem because it assumes that the existence of a solution is already known. In order to obtain a combined existence and regularity result it is necessary to refer, for example, to the result of Cattabriga in [39] (see also [7]), proven by the author in dimension 3 and which we give here without proof.

Theorem IV.6.6 (Cattabriga). *Let* k *be an integer* $k \geq -1$ *and let* Ω *be a connected bounded domain of* $\mathbb{R}^3$ *of class* $\mathcal{C}^m$*, with* $m = \max(2, k+2)$*. For all data*

$$(f, g, v_b) \in (W^{k,q}(\Omega))^d \times W^{k+1,q}(\Omega) \times (W^{k+2-1/q,q}(\partial\Omega))^d,$$

with $1 < q < +\infty$ *satisfying the compatibility condition*

$$\int_\Omega g\,dx = \int_{\partial\Omega} v_b \cdot \nu\,d\sigma,$$

there exists a unique solution (v,p) *of*

$$\begin{cases} -\Delta v + \nabla p = f, & \textit{in } \Omega, \\ \operatorname{div} v = g, & \textit{in } \Omega, \\ v = v_b, & \textit{on } \partial\Omega, \end{cases}$$

satisfying (IV.68) *and the estimate* (IV.69) *with* $d_q = 0$ *for all* q.

We note that the theorem is also true in two dimension (see, e.g., [122]).

6.4 Regularity for the div/curl *problem*

We can now proceed to the proof of the regularity theorem for the div/curl system that we stated in Section 4.

Proof (of Theorem IV.4.9).

We assume that $u \in (H^1(\Omega))^3$ satisfies $\operatorname{div} u \in H^1(\Omega)$, $\operatorname{curl} u \in (H^1(\Omega))^3$, and $\gamma_\nu(u) \in H^{3/2}(\partial\Omega)$ and we want to show that $u \in (H^2(\Omega))^3$.

First of all, we notice that it is enough, by using some suitable lifting of the trace, to consider the case where $\gamma_\nu(u) = 0$.

- We first prove the local regularity of the solution. Let $\varphi \in \mathcal{D}(\Omega)$ and set $v = \varphi u$. By using (A.4) and (A.5), and that v is compactly supported in Ω, we get

$$\operatorname{div} v = \varphi(\operatorname{div} u) + u \cdot \nabla\varphi \in H^1(\Omega),$$
$$\operatorname{curl} v = \varphi(\operatorname{curl} u) + \nabla\varphi \wedge u \in H^1(\Omega).$$

  Thus, for any $1 \leq i \leq 3$, $w_i = \partial_{x_i} v$ satisfies

$$w_i \in (L^2(\Omega))^3,\ \operatorname{div} w_i \in L^2(\Omega),\ \operatorname{curl} w_i \in (L^2(\Omega))^3,\ \gamma_\nu w_i = 0.$$

  Using Theorem IV.4.8, we deduce that $w_i \in (H^1(\Omega))^3$ and that

$$\|w_i\|_{H^1} \leq C_\varphi \left(\|u\|_{H^1} + \|\operatorname{div} u\|_{H^1} + \|\operatorname{curl} u\|_{H^1}\right).$$

  This is true for any $1 \leq i \leq 3$, therefore we deduce that $\varphi u = v \in (H^2(\Omega))^3$, for any $\varphi \in \mathcal{D}(\Omega)$. This proves that $u \in (H^2_{loc}(\Omega))^3$.
- We only need now to study the regularity properties in a neighborhood of the boundary $\partial\Omega$. We use the same strategy as in Section 6.1.2 and we begin by showing the tangential regularity of u. To this end, we consider a vector field θ which is tangent to the boundary of Ω (i.e., which satisfies (III.77)) and use the corresponding flows $(h, x) \mapsto \tau^\theta(h, x)$ (see (III.78)). Since θ is smooth enough, we see that $\delta_h^\theta u = \tau_h^\theta u - u$ belongs to $(H^1(\Omega))^3$ and satisfies moreover (see Lemma IV.6.4)

$$\begin{aligned}
\operatorname{div}(\delta_h^\theta u) &= \delta_h^\theta \operatorname{div} u + \left[\operatorname{div}, \tau_h^\theta\right] u,\\
\operatorname{curl}(\delta_h^\theta u) &= \delta_h^\theta \operatorname{curl} u + \left[\operatorname{curl}, \tau_h^\theta\right] u,\\
\gamma_\nu(\delta_h^\theta u) &= \tau_h^\theta u\cdot\nu - \underbrace{u\cdot\nu}_{=0} = -\underbrace{(\tau_h^\theta u)\cdot(\tau_h^\theta\nu)}_{=0} + \tau_h^\theta u\cdot\nu\\
&= -\tau_h^\theta u\cdot(\tau_h^\theta\nu - \nu) = -\tau_h^\theta u\cdot\delta_h^\theta\nu.
\end{aligned}$$

By using the commutator estimates in Proposition III.3.19 we find that

$$\begin{aligned}
\|\operatorname{div}(\delta_h^\theta u)\|_{L^2} &\le C_\theta|h|\left(\|\operatorname{div} u\|_{H^1} + \|u\|_{H^1}\right),\\
\|\operatorname{curl}(\delta_h^\theta u)\|_{L^2} &\le C_\theta|h|\left(\|\operatorname{curl} u\|_{H^1} + \|u\|_{H^1}\right),\\
\|\gamma_\nu(\delta_h^\theta u)\|_{H^{1/2}} &\le C_\theta|h|\|\nu\|_{W^{2,\infty}}\|u\|_{H^1}.
\end{aligned} \tag{IV.70}$$

By using Theorem IV.4.8, we deduce that the H^1-norm of $\delta_h^\theta u$ is bounded as follows

$$\|\delta_h^\theta u\|_{H^1} \le C_{\theta,\nu}|h|\left(\|u\|_{H^1} + \|\operatorname{div} u\|_{H^1} + \|\operatorname{curl} u\|_{H^1}\right).$$

Thanks to Theorem III.3.20, we deduce that $u \in (H^2_{tang}(\Omega))^3$ with the suitable estimate of its norm.

- We use Proposition III.3.23 to get that $u_N = u\cdot\nu$ satisfies

$$\nabla_N u_N = \operatorname{div} u - \operatorname{div} u_T - \frac{\nabla_N \mathcal{J}}{\mathcal{J}} u_N, \quad \text{in } \mathcal{O}^\rho_{s_0}.$$

Since Ω is of class $\mathcal{C}^{2,1}$, we know that $\mathcal{J}$ is of class $\mathcal{C}^{1,1}$ on $\overline{\mathcal{O}^\rho_{s_0}}$; see Proposition III.3.12. Therefore, since we assumed that $\operatorname{div} u \in H^1(\Omega)$, and we showed just before that $u \in (H^2_{tang}(\Omega))^3$, we deduce that

$$\nabla_N(u\cdot\nu) \in H^1(\mathcal{O}^\rho_{s_0}).$$

Using the tangential regularity of u that we already proved, it follows that $u\cdot\nu \in H^2(\Omega)$.

- Finally, for any $1 \le i \le 3$, we set $w_i = \partial_{x_i} u$ and we see that

$$w_i\cdot\nu = \partial_{x_i}(u\cdot\nu) - u\cdot(\partial_{x_i}\nu) \in H^1(\Omega),$$

because ν belongs to $\mathcal{C}^{1,1}(\overline{\Omega})$. It follows that the trace $\gamma_\nu(w_i)$ belongs to $H^{1/2}(\partial\Omega)$. Moreover, we have

$$\operatorname{div} w_i = \partial_{x_i}(\operatorname{div} u) \in L^2(\Omega), \text{ and } \operatorname{curl} w_i = \partial_{x_i}(\operatorname{curl} u) \in (L^2(\Omega))^3.$$

Using once more Theorem IV.4.8, we find that $w_i \in (H^1(\Omega))^3$. This being valid for any i, we get that $u \in (H^2(\Omega))^3$ and the required estimate.

□

7 The Stokes problem with stress boundary conditions

In the preceding sections, we have considered the Stokes problem with boundary data of the Dirichlet type. We now consider Neumann-type boundary conditions for this problem. At the end of this section we also consider a more physically relevant variant of this boundary condition which concerns stress boundary conditions; see Problem (IV.84).

We recall that the Stokes equation is written as

$$-\Delta v + \nabla p = f,$$

or equivalently

$$-\operatorname{div}(\nabla v - p\,\mathrm{Id}) = f.$$

By assuming that $v \in (H^1(\Omega))^d$, $p \in L^2(\Omega)$ and $f \in (L^2(\Omega))^d$, we see from Lemma IV.3.3 that we can define, in $(H^{-1/2}(\partial\Omega))^d$, the trace of $\sigma.\nu$ where $\sigma = \nabla v - p\,\mathrm{Id}$.

Hence, the natural Neumann problem for the Stokes problem in the particular divergence form given above can be stated as

$$\begin{cases} -\Delta v + \nabla p = f, & \text{in } \Omega, \\ \operatorname{div} v = g, & \text{in } \Omega, \\ \dfrac{\partial v}{\partial \nu} - p\nu = f_b, & \text{on } \partial\Omega. \end{cases} \tag{IV.71}$$

We propose to prove the following theorem.

Theorem IV.7.1. *Let Ω be a connected and bounded domain of class $\mathcal{C}^{1,1}$ in $\mathbb{R}^d$. For all data*

$$(f, g, f_b) \in (L^2(\Omega))^d \times L^2(\Omega) \times (H^{-1/2}(\partial\Omega))^d,$$

satisfying the compatibility condition

$$\int_\Omega f(x)\,dx + \langle f_b, 1\rangle_{H^{-1/2},H^{1/2}} = 0, \tag{IV.72}$$

problem (IV.71) possesses one, and only one, solution (v, p) in the space $(H^1(\Omega) \cap L^2_0(\Omega))^d \times L^2(\Omega)$.

Moreover, if for $k \geq 0$, the open set Ω is of class $\mathcal{C}^{k+2,1}$ and if the triplet (f, g, f_b) belongs to $(H^k(\Omega))^d \times H^{k+1}(\Omega) \times (H^{k+1/2}(\partial\Omega))^d$ then the pair (v, p) belongs to $(H^{k+2}(\Omega))^d \times H^{k+1}(\Omega)$ and there exists a $C > 0$ depending only on Ω such that

$$\|v\|_{H^{k+2}} + \|p\|_{H^{k+1}} \leq C\left(\|f\|_{H^k} + \|g\|_{H^{k+1}} + \|f_b\|_{H^{k+1/2}}\right). \tag{IV.73}$$

Notice that, in the proof of this result, we need one degree of regularity supplementary for the domain Ω compared to the case of Dirichlet boundary conditions (see Theorem IV.5.8).

7.1 The Stokes–Neumann problem

Here we are concerned with proving the first part of Theorem IV.7.1.

- Step 1:
  We start by considering a lifting of the mass source term, while being aware of the fact that g does not necessarily belong to $L^2_0(\Omega)$.
  We consider the smooth vector field given by

$$v_1(x) = \frac{1}{d}{}^t(x_1, \ldots, x_d),$$

  where the x_i are the coordinates of the point x in the canonical basis of $\mathbb{R}^d$. It is easy to verify that this vector field satisfies $\Delta v_1 = 0$, $\operatorname{div} v_1 = 1$ and $\partial v_1/\partial\nu = (1/d)\nu$ at all points on the boundary of Ω.
  Let us then set $v = \tilde{v} + m(g)v_1$, where we recall that $m(g)$ denotes the mean value of the function g on Ω. Problem (IV.71) can now be written using $\tilde{v}$, in the form

$$\begin{cases} -\Delta\tilde{v} + \nabla\tilde{p} = f, & \text{in } \Omega, \\ \operatorname{div}\tilde{v} = g - m(g), & \text{in } \Omega, \\ \dfrac{\partial\tilde{v}}{\partial\nu} - \tilde{p}\,\nu = f_b, & \text{on } \partial\Omega, \end{cases} \tag{IV.74}$$

  where the new pressure is defined by $\tilde{p} = p - m(g)/d$.
  Since $g - m(g)$ has a zero mean value, we have seen in Theorem IV.5.2 that we can solve the problem

$$\begin{cases} -\Delta w_0 + \nabla\pi_0 = 0, & \text{in } \Omega, \\ \operatorname{div} w_0 = \tilde{g}, & \text{in } \Omega, \\ w_0 = 0, & \text{on } \partial\Omega. \end{cases}$$

  In a more precise way, there exists a unique pair (w_0, π_0) in the space $(H^1_0(\Omega))^d \times L^2_0(\Omega)$ which satisfies the system above. If we now set $\tilde{v} = w + w_0$ and $\tilde{p} = \pi + \pi_0$, we see that System (IV.74) becomes

$$\begin{cases} -\Delta w + \nabla\pi = f, & \text{in } \Omega, \\ \operatorname{div} w = 0, & \text{in } \Omega, \\ \dfrac{\partial w}{\partial\nu} - \pi\,\nu = F_b, & \text{on } \partial\Omega, \end{cases}$$

with $F_b = f_b - (\nabla w_0 - \pi_0 \,\mathrm{Id}).\nu$. Since $\nabla w_0 - \pi_0 \,\mathrm{Id}$ lies in $(L^2(\Omega))^{d\times d}$ and $\mathrm{div}(\nabla w_0 - \pi_0 \,\mathrm{Id}) = 0 \in (L^2(\Omega))^d$, Lemma IV.3.3 tells us that $(\nabla w_0 - \pi_0 \,\mathrm{Id}).\nu$ is indeed defined on the boundary and belongs to $(H^{-1/2}(\partial\Omega))^d$. We have therefore shown that we can go back to the general case of Problem (IV.71) with $g = 0$.

- Step 2:
  We now assume that $g = 0$, and we are brought back to the following problem

$$\begin{cases} -\Delta v + \nabla p = f, & \text{in } \Omega, \\ \mathrm{div}\, v = 0, & \text{in } \Omega, \\ \dfrac{\partial v}{\partial \nu} - p\,\nu = f_b, & \text{on } \partial\Omega. \end{cases}$$

We introduce the Hilbert space

$$W = \left\{ v \in (H^1(\Omega))^d,\ \mathrm{div}\, v = 0 \right\},$$

which is different from the now usual space V (defined in Section 3.3) by the fact that the functions of W are not necessarily zero at the boundary of Ω. We associate the following variational formulation with the problem above which is set in the Hilbert space $W \cap (L^2_0(\Omega))^d$,

$$\int_\Omega \nabla v : \nabla \psi \, dx - \langle f_b, \psi \rangle_{H^{-1/2},H^{1/2}} = \int_\Omega f \cdot \psi \, dx,\ \forall \psi \in W \cap (L^2_0(\Omega))^d. \tag{IV.75}$$

The functions of $W \cap (L^2_0(\Omega))^d$ have zero mean value, thus the Poincaré–Wirtinger inequality (Proposition III.2.39) shows that the symmetric bilinear form which appears in this formulation is indeed continuous and coercive on $W \cap (L^2_0(\Omega))^d$. Hence, the Lax–Migram theorem shows that there exists a unique solution v to (IV.75) in $W \cap (L^2_0(\Omega))^d$.
Let φ be a function in W. We can take $\psi = \varphi - m(\varphi)$, which is clearly in $W \cap (L^2_0(\Omega))^d$, as a test function in (IV.75). This gives

$$\begin{aligned} &\int_\Omega \nabla v : \nabla \varphi \, dx - \langle f_b, \varphi \rangle_{H^{-1/2},H^{1/2}} + m(\varphi) \cdot \langle f_b, 1 \rangle_{H^{-1/2},H^{1/2}} \\ &= \int_\Omega f \cdot \varphi \, dx - m(\varphi) \cdot \int_\Omega f \, dx. \end{aligned}$$

According to the compatibility condition (IV.72), the two terms which contain the mean value of φ offset each other thus leaving

$$\int_\Omega \nabla v : \nabla \varphi \, dx - \langle f_b, \varphi \rangle_{H^{-1/2},H^{1/2}} = \int_\Omega f \cdot \varphi \, dx. \tag{IV.76}$$

If we restrict ourselves to $\varphi \in \mathscr{V}$, then de Rham's theorem (Theorem IV.2.4) shows that there exists a unique pressure p in $L^2_0(\Omega)$ such that we have

$$-\Delta v + \nabla p = f, \text{ in } \Omega.$$

We now need to prove that the boundary conditions are satisfied. By writing the equation in the form

$$-\operatorname{div}(\nabla v - p \,\mathrm{Id}) = f,$$

the Stokes equation of Lemma IV.3.3 shows that for any function $\varphi \in W$, we have

$$\int_\Omega (\nabla v - p\,\mathrm{Id}) : \nabla \varphi \, dx - \langle (\nabla v - p\,\mathrm{Id}).\nu, \varphi \rangle_{H^{-1/2},H^{1/2}} = \int_\Omega f \cdot \varphi \, dx.$$

Since φ is divergence-free, we have

$$p\,\mathrm{Id} : \nabla \varphi = p \ \operatorname{div} \varphi = 0,$$

thus for all $\varphi \in W$

$$\int_\Omega \nabla v : \nabla \varphi \, dx - \langle (\nabla v - p\,\mathrm{Id}).\nu, \varphi \rangle_{H^{-1/2},H^{1/2}} = \int_\Omega f \cdot \varphi \, dx.$$

By comparing this to (IV.76), we obtain the relation

$$\langle \sigma.\nu - f_b, \varphi \rangle_{H^{-1/2},H^{1/2}} = 0, \ \forall \varphi \in V, \tag{IV.77}$$

where $\sigma = \nabla v - p\,\mathrm{Id}$. This formula shows that $\sigma.\nu - f_b$ is zero against the traces of the functions of $(H^1(\Omega))^d$ with zero divergence.
Let $\psi \in (H^{1/2}(\partial\Omega))^d$. We have seen (Theorem IV.5.2) that, provided that we have $\int_{\partial\Omega} \psi \cdot \nu \, d\sigma = 0$, then we can solve the Stokes problem

$$\begin{cases} -\Delta \varphi + \nabla \pi = 0, & \text{in } \Omega, \\ \operatorname{div} \varphi = 0, & \text{in } \Omega, \\ \varphi = \psi, & \text{on } \partial\Omega, \end{cases}$$

with $\varphi \in (H^1(\Omega))^d$ and $\pi \in L^2_0(\Omega)$. The function φ is then a divergence-free lifting of ψ that we can put in (IV.77). We have then established that for all $\psi \in (H^{1/2}(\partial\Omega))^d$ such that $\int_{\partial\Omega} \psi \cdot \nu \, d\sigma = 0$, we have

$$\langle f_b - \sigma.\nu, \psi \rangle_{H^{-1/2},H^{1/2}} = 0. \tag{IV.78}$$

Let us now define $\psi_0 = \nu$, which is indeed a function of $(H^{1/2}(\partial\Omega))^d$ since Ω is supposed to be of class $\mathcal{C}^{1,1}$. It satisfies

$$\int_{\partial\Omega} \psi_0 \cdot \nu \, d\sigma = \int_{\partial\Omega} |\nu|^2 \, d\sigma = |\partial\Omega| > 0,$$

where $|\partial\Omega|$ is the measure (of dimension $d-1$) of $\partial\Omega$. Now let any $\psi \in (H^{1/2}(\partial\Omega))^d$, we set

$$\psi_1 = \psi - \frac{1}{|\partial\Omega|}\left(\int_{\partial\Omega} \psi \cdot \nu \, d\sigma\right)\psi_0,$$

such that

$$\psi = \psi_1 + \frac{1}{|\partial\Omega|}\left(\int_{\partial\Omega} \psi \cdot \nu \, d\sigma\right)\psi_0.$$

By definition ψ_1 satisfies $\displaystyle\int_{\partial\Omega} \psi_1 \cdot \nu \, d\sigma = 0$ and therefore we deduce from (IV.78) that

$$\langle f_b - \sigma.\nu, \psi_1 \rangle_{H^{-1/2},H^{1/2}} = 0.$$

Hence, we obtain

$$\langle f_b - \sigma.\nu, \psi \rangle_{H^{-1/2},H^{1/2}} = \frac{1}{|\partial\Omega|}\left(\int_{\partial\Omega} \psi \cdot \nu \, d\sigma\right)\langle f_b - \sigma.\nu, \psi_0 \rangle_{H^{-1/2},H^{1/2}}.$$

By introducing the number

$$C_0 = \frac{1}{|\partial\Omega|}\langle f_b - \sigma.\nu, \psi_0 \rangle_{H^{-1/2},H^{1/2}},$$

we have just shown that for any function $\psi \in (H^{1/2}(\partial\Omega))^d$ we have

$$\langle f_b - \sigma.\nu, \psi \rangle_{H^{-1/2},H^{1/2}} = \int_{\partial\Omega} \psi \cdot (C_0\nu) \, d\sigma.$$

This proves that $f_b - \sigma.\nu = C_0\,\nu$ in the sense of $(H^{-1/2}(\partial\Omega))^d$. We see, therefore, that if we set $\tilde{p} = p + C_0$ then we indeed have

$$f_b = \widetilde{\sigma}.\nu, \quad \text{on } \partial\Omega,$$

where $\widetilde{\sigma} = \nabla v - \tilde{p}\ \mathrm{Id}$. This new pressure $\tilde{p}$ only differs from p by a constant, therefore we have solved the problem

$$\begin{cases} -\Delta v + \nabla\tilde{p} = f, & \text{in } \Omega, \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ \dfrac{\partial v}{\partial \nu} - \tilde{p}\,\nu = f_b, & \text{on } \partial\Omega. \end{cases}$$

We notice that in this case, contrary to the case with Dirichlet boundary conditions, the Neumann boundary conditions define the pressure in a unique way. By contrast, the velocity field v is only unique if we impose its mean value. We can easily verify that any other velocity field of the form $v(x) + t$ where t is a constant vector in $\mathbb{R}^d$ is also a solution of the problem (associated with the same pressure p).

□

7.2 Regularity properties

To prove the regularity of the solution, we return to the initial problem (IV.71) with g not necessarily zero.

In order to simplify the following proof, we first consider a lifting of the boundary conditions, that is, by introducing a vector field w_1 which satisfies $\partial w_1/\partial\nu = f_b$. Under the hypothesis $f_b \in (H^{1/2}(\partial\Omega))^d$ we know, from the trace lifting theorem (Theorem III.2.23), that one such w_1 exists in the space $(H^2(\Omega))^d$ and satisfies $\|w_1\|_{H^2} \leq C\|f_b\|_{H^{1/2}}$.

We now set $\tilde{v} = v - w_1$, such that the pair $(\tilde{v}, p)$ satisfies the system

$$\begin{cases} -\Delta\tilde{v} + \nabla p = \tilde{f}, & \text{in } \Omega, \\ \operatorname{div}\tilde{v} = \tilde{g}, & \text{in } \Omega, \\ \dfrac{\partial\tilde{v}}{\partial\nu} - p\,\nu = 0, & \text{on } \partial\Omega, \end{cases}$$

where $\tilde{f} = f + \Delta w_1$ and $\tilde{g} = g - \operatorname{div} w_1$. Since $w_1 \in (H^2(\Omega))^d$, these new source terms have the same regularity as f and g, and we have

$$\begin{aligned} \|\tilde{f}\|_{L^2} &\leq C\left(\|f\|_{L^2} + \|f_b\|_{H^{1/2}}\right), \\ \|\tilde{g}\|_{H^1} &\leq C\left(\|g\|_{H^1} + \|f_b\|_{H^{1/2}}\right), \end{aligned}$$

and moreover,

$$\|v\|_{H^2} \leq C\left(\|\tilde{v}\|_{H^2} + \|f_b\|_{H^{1/2}}\right).$$

We can then content ourselves with studying Problem (IV.71) associated which a boundary data $f_b = 0$. This in no way obstructs obtaining estimate (IV.73).

We are therefore interested in the regularity of (v, p), the solution of the following mixed variational formulation, which is equivalent to the system under study

$$\begin{cases} \displaystyle\int_\Omega \nabla v : \nabla\psi\,dx - \int_\Omega p\,\operatorname{div}\psi\,dx = \int_\Omega f\cdot\psi\,dx, & \forall\psi \in (H^1(\Omega))^d, \\ \displaystyle\int_\Omega q\,\operatorname{div} v\,dx = \int_\Omega q\,g\,dx, & \forall q \in L^2(\Omega), \end{cases} \tag{IV.79}$$

where $f \in (L^2(\Omega))^d$, and $g \in H^1(\Omega)$.

From Theorem IV.6.1 (which does not depend on the kind of boundary condition which is considered), we know that the unique solution (v, p) in $(H^1(\Omega) \cap L^2_0(\Omega))^d \times L^2(\Omega)$ of (IV.79) belongs to $(H^2_{loc}(\Omega))^d \times H^1_{loc}(\Omega)$.

The proof of tangential regularity is more technical. Contrary to our approach in the case of Dirichlet boundary conditions, we cannot directly evaluate the commutator of the Stokes operator using translations, because the translation operators are not compatible with the boundary conditions. Indeed, if v satisfies the Neumann boundary conditions above, then $v \circ \tau^\theta(h, \cdot)$ does not necessarily satisfy these boundary conditions.

We therefore need to perform a proof by directly working on the variational formulation. To this end, we can demonstrate the following result for Problem (IV.79). Notice that we need here one degree of regularity supplementary of the domain Ω than for the Dirichlet boundary conditions (see the end of the proof).

Theorem IV.7.2. *Let Ω be a bounded domain of $\mathbb{R}^d$ of class $\mathcal{C}^{2,1}$. We assume that the data satisfy:*

$$f \in (L^2(\Omega))^d,\ g \in H^1(\Omega);$$

then the solution (v, p) of (IV.79) belongs to $(H^2_{tang}(\Omega))^d \times H^1_{tang}(\Omega)$ and we have the estimate

$$|||v|||_{\theta,2} + |||p|||_{\theta,1} \leq C(\Omega, \theta) \left(\|f\|_{L^2} + \|g\|_{H^1} \right),$$

for all vector fields θ satisfying (III.77) with $k = 0$.

Remark IV.7.1. For the nonhomogeneous problem (IV.71), by using the process of lifting the boundary condition detailed above, we obtain the following estimate:

$$|||v|||_{\theta_2} + |||p|||_{\theta,1} \leq C(\Omega, \theta) \left(\|f\|_{L^2} + \|f_b\|_{H^{1/2}} + \|g\|_{H^1} \right).$$

To reason directly on the nonhomogeneous problem requires establishing the characterisation properties of the fractional Sobolev trace spaces by translation methods, which we wished to avoid here.

Proof.

We start by establishing the equations satisfied by $\delta^\theta_h v$ and $\delta^\theta_h p$. To do this, in the first equation of system (IV.79) we take $\psi = \delta^\theta_{-h}\varphi$, where φ is any test function in $H^1(\Omega)$. This gives

$$\int_\Omega \nabla v : \nabla \delta^\theta_{-h}\varphi \, dx - \int_\Omega p \operatorname{div}(\delta^\theta_{-h}\varphi) \, dx = \int_\Omega f \cdot \delta^\theta_{-h}\varphi \, dx.$$

Introducing suitable commutators we get

$$\begin{aligned}&\int_\Omega \nabla v : \delta^\theta_{-h}\big(\nabla\varphi\big)\,dx - \int_\Omega p\,\delta^\theta_{-h}(\operatorname{div}\varphi)\,dx\\ &= \int_\Omega f\cdot\delta^\theta_{-h}\varphi\,dx - \underbrace{\int_\Omega \nabla v : \big[\nabla, \tau^\theta_{-h}\big]\,\varphi\,dx}_{=T_1(\varphi)} + \underbrace{\int_\Omega p\,\left(\big[\operatorname{div}, \tau^\theta_{-h}\big]\,\varphi\right)\,dx}_{=T_2(\varphi)}.\end{aligned}$$

We now want to discover the difference operator δ^θ_h applied on ∇v and p. This is done by using the operator defined in (III.86). We obtain

$$\begin{aligned}&\int_\Omega \delta^\theta_h\nabla v : \big(\nabla\varphi\big)\,dx - \int_\Omega \delta^\theta_h p(\operatorname{div}\varphi)\,dx\\ &= \int_\Omega f\cdot\delta^\theta_{-h}\varphi\,dx - T_1(\varphi) + T_2(\varphi)\\ &\quad+ \underbrace{\int_\Omega \{\nabla v, \nabla\varphi\}_h\,dx}_{=T_3(\varphi)} - \underbrace{\int_\Omega \{p, \operatorname{div}\varphi\}_h\,dx}_{=T_4(\varphi)}.\end{aligned}$$

Using once more the commutator between ∇ and δ^θ_h, we finally get

$$\begin{aligned}&\int_\Omega \nabla(\delta^\theta_h v) : \big(\nabla\varphi\big)\,dx - \int_\Omega \delta^\theta_h p(\operatorname{div}\varphi)\,dx\\ &= \int_\Omega f\cdot\delta^\theta_{-h}\varphi\,dx - T_1(\varphi) + T_2(\varphi) + T_3(\varphi) - T_4(\varphi)\\ &\quad+ \underbrace{\int_\Omega \big(\,\big[\nabla, \tau^\theta_h\big]\,v\big) : \big(\nabla\varphi\big)\,dx}_{=T_5(\varphi)}.\end{aligned} \tag{IV.80}$$

With Proposition III.3.19, we get the following estimates on the terms $T_1, \dots, T_5$

$$\begin{aligned}|T_1(\varphi)| &\leq C|h|\|\nabla v\|_{L^2}\|\nabla\varphi\|_{L^2},\\ |T_2(\varphi)| &\leq C|h|\|p\|_{L^2}\|\operatorname{div}\varphi\|_{L^2},\\ |T_3(\varphi)| &\leq C|h|\|\nabla v\|_{L^2}\|\nabla\varphi\|_{L^2},\\ |T_4(\varphi)| &\leq C|h|\|p\|_{L^2}\|\operatorname{div}\varphi\|_{L^2},\\ |T_5(\varphi)| &\leq C|h|\|\nabla v\|_{L^2}\|\nabla\varphi\|_{L^2},\end{aligned}$$

where C depends only on Ω and θ. Coming back to (IV.80), we deduce first from these estimates that

$$\begin{aligned}&\left|\int_\Omega (\delta^\theta_h p)\operatorname{div}\varphi\,dx\right|\\ &\leq \|\nabla\delta^\theta_h v\|_{L^2}\|\nabla\varphi\|_{L^2} + C|h|(\|\nabla v\|_{L^2} + \|p\|_{L^2} + \|f\|_{L^2})\|\varphi\|_{H^1_0}, \forall\varphi\in(H^1_0(\Omega))^d.\end{aligned}$$

By definition, this gives the estimate

$$\|\nabla(\delta_h^\theta p)\|_{H^{-1}} \leq \|\nabla \delta_h^\theta v\|_{L^2} + C|h|(\|\nabla v\|_{L^2} + \|p\|_{L^2} + \|f\|_{L^2}).$$

Moreover, from (III.87), we obtain

$$\|\delta_h^\theta p\|_{H^{-1}} \leq C|h|\|p\|_{L^2}.$$

Gathering the two previous inequalities and using the Nečas inequality (Theorem IV.1.1) we get

$$\|\delta_h^\theta p\|_{L^2} \leq C\|\nabla \delta_h^\theta v\|_{L^2} + C|h|(\|\nabla v\|_{L^2} + \|p\|_{L^2} + \|f\|_{L^2}). \tag{IV.81}$$

We now choose $q = \delta_{-h}^\theta \delta_h^\theta p$ in the second equation of (IV.79) to get

$$\int_\Omega (\delta_{-h}^\theta \delta_h^\theta p)(\operatorname{div} v)\, dx = \int_\Omega g(\delta_{-h}^\theta \delta_h^\theta p)\, dx.$$

The same manipulations as before lead to

$$\begin{aligned}\int_\Omega (\delta_h^\theta p)(\operatorname{div} \delta_h^\theta v)\, dx = \underbrace{\int_\Omega \delta_h^\theta g \delta_h^\theta p\, dx}_{=T_1'} + \underbrace{\int_\Omega \{\operatorname{div} v, \delta_h^\theta p, \}_h dx}_{=T_2'} \\ + \underbrace{\int_\Omega (\delta_h^\theta p)\big(\left[\operatorname{div}, \tau_h^\theta\right] v\big)\, dx}_{=T_3'} - \underbrace{\int_\Omega \{g, \delta_h^\theta p\}_h\, dx}_{=T_4'}.\end{aligned} \tag{IV.82}$$

Proposition III.3.19 leads to the estimates

$$\begin{aligned}|T_1'| &\leq \|\delta_h^\theta g\|_{L^2}\|\delta_h^\theta p\|_{L^2} \leq C|h|\|g\|_{H^1}\|\delta_h^\theta p\|_{L^2},\\ |T_2'| &\leq C|h|\|\operatorname{div} v\|_{L^2}\|\delta_h^\theta p\|_{L^2},\\ |T_3'| &\leq C|h|\|\delta_h^\theta p\|_{L^2}\|\operatorname{div} v\|_{L^2},\\ |T_4'| &\leq C|h|\|g\|_{L^2}\|\delta_h^\theta p\|_{L^2}.\end{aligned}$$

We can now take $\varphi = \delta_h^\theta v$ in (IV.80), use (IV.82) to eliminate the pressure term, and finally use the estimates of the terms T_i and T_j' to get

$$\begin{aligned}\|\nabla \delta_h^\theta v\|_{L^2}^2 \leq & C|h|(\|v\|_{H^1} + \|p\|_{L^2} + \|f\|_{L^2})\|\delta_h^\theta v\|_{H^1} \\ & + C\|f\|_{L^2} \underbrace{\|\delta_{-h}^\theta \delta_h^\theta v\|_{L^2}}_{\leq C|h|\|\delta_h^\theta v\|_{H^1}} + C|h|(\|g\|_{H^1} + \|v\|_{H^1})\|\delta_h^\theta p\|_{L^2}\end{aligned}$$

We use now the fact that $\|\delta_h^\theta v\|_{L^2} \leq C|h|\|v\|_{H^1}$ and the estimate of the pressure term given in (IV.81) to get

$$\begin{aligned}\|\nabla \delta_h^\theta v\|_{L^2}^2 \leq & C|h|(\|v\|_{H^1} + \|p\|_{L^2} + \|f\|_{L^2} + \|g\|_{H^1})\|\nabla \delta_h^\theta v\|_{L^2} \\ & + C|h|^2(\|g\|_{H^1} + \|v\|_{H^1} + \|p\|_{L^2} + \|f\|_{L^2})^2.\end{aligned}$$

With the Young inequality we finally obtain the estimate

$$\|\nabla \delta_h^\theta v\|_{L^2}^2 \leq C|h|^2(\|g\|_{H^1} + \|v\|_{H^1} + \|p\|_{L^2} + \|f\|_{L^2})^2.$$

Since $\|\delta_h^\theta v\|_{L^2} \leq C|h|\|v\|_{H^1}$, the above estimate shows that $|||v|||_{\theta,2,2} < +\infty$ and thus $v \in (H^2_{tang}(\Omega))^d$.

Coming back to (IV.81), we deduce

$$\|\delta_h^\theta p\|_{L^2} \leq C|h|(\|v\|_{H^1} + \|p\|_{L^2} + \|f\|_{L^2} + \|g\|_{H^1}),$$

so that we proved $p \in H^1_{tang}(\Omega)$.

The proof of the tangential regularity of the solution (v, p) is complete.

□

The regularity up to the boundary can be shown in the same way as for the Dirichlet problem in Section 6.1.2.3. We write the equations using the normal and tangential variables and by using the tangential regularity to deduce, directly from these formulas, the regularity of all the other derivatives of the solution up to the boundary.

The main differences lie in the proofs of (IV.65) through (IV.67) where we need to use the Hardy inequality which only holds when v vanishes on the boundary of Ω.

That is the reason why, here, we assume that Ω is of class $\mathcal{C}^{2,1}$ in such a way that we have $\tilde{\mathcal{G}} \in \mathcal{C}^{0,1}(\overline{\Omega})$ and $\nu \in \mathcal{C}^{1,1}(\overline{\Omega})$ which implies that the estimates (IV.65) though (IV.67) hold without any need of the Hardy inequality. In particular, the proof holds even when v does not vanish on the boundary.

To summarise the content of this section, we proved the following result for the general problem (IV.71) with nonhomogeneous boundary data.

Theorem IV.7.3. *Let Ω be a bounded connected domain of $\mathbb{R}^d$ of class $\mathcal{C}^{2,1}$. Under the hypothesis that (f, g, f_b) is given in $(L^2(\Omega))^d \times H^1(\Omega) \times (H^{1/2}(\partial\Omega))^d$ then the unique solution (v, p) of Problem (IV.71) belongs to $(H^2(\Omega))^d \times H^1(\Omega)$ and we have the estimate*

$$\|v\|_{H^2} + \|p\|_{H^1} \leq C\left(\|f\|_{L^2} + \|g\|_{H^1} + \|f_b\|_{H^{1/2}}\right).$$

7.3 Stress boundary conditions

As we saw at the start of this section, expressing the equation in divergence form is fundamental to give a meaning to the Neumann-type boundary conditions. However, we saw in Chapter I, that the incompressible Stokes problem is, in reality, naturally written in the following form

$$\begin{cases} -\operatorname{div}\sigma = f, \\ \quad \operatorname{div} v = g, \end{cases} \tag{IV.83}$$

where σ is the stress tensor which is given, according to Newton's law, by $\sigma = 2\mathrm{D}(v) - \big(p + \frac{2}{3}(\operatorname{div} v)\big)\,\mathrm{Id}$. We are still assuming for simplicity that the viscosity of the fluid is equal to 1. We recall that in this formulation, $\mathrm{D}(v)$ is the symmetric part of the tensor ∇v defined by $\mathrm{D}(v) = \frac{1}{2}(\nabla v + {}^t\nabla v)$.

Let us now carry out a formal calculation:

$$\begin{aligned} -\operatorname{div}\sigma &= -\operatorname{div}\left(\nabla v + {}^t\,\nabla v - \left(p + \frac{2}{3}\operatorname{div} v\right)\mathrm{Id}\right) \\ &= -\Delta v + \nabla\left(p - \frac{1}{3}\operatorname{div} v\right) = -\Delta v + \nabla\left(p - \frac{1}{3} g\right). \end{aligned}$$

Hence, problem (IV.83) is formally equivalent to (IV.71) by replacing the pressure p by $p - \frac{1}{3}g$. However, to give a meaning to the boundary conditions, the two formulations are different. Hence, the Neumann type problem associated with the expression in divergence form (IV.83) is given by

$$\begin{cases} -\operatorname{div}\,\big(2\mathrm{D}(v) - p\,\mathrm{Id}\big) = f, & \text{in } \Omega, \\ \operatorname{div}\, v = g, & \text{in } \Omega, \\ \big(2\mathrm{D}(v) - p\,\mathrm{Id}\big).\nu = f_b, & \text{on } \partial\Omega, \end{cases} \tag{IV.84}$$

where, for simplification, we have replaced the term $p + \frac{2}{3}g$ by a simple pressure term, p.

In this problem the boundary condition has a very clear physical meaning. Indeed, we saw in Chapter I that, by definition, the quantity $\sigma.\nu$ at the boundary is the force exerted by the wall on the fluid. In other words, instead of imposing the velocity field at the boundary, as in the case of Dirichlet boundary conditions, here we impose the force exerted on the fluid by the experimenter. In Chapter VII, we study outflow boundary conditions for the unsteady Navier–Stokes equations in which the quantity $\sigma.\nu$ plays a key role.

For this model, we only consider the case of dimension $d = 3$ to illustrate the ideas. We prove results of existence, uniqueness, and regularity of solutions analogous to the preceding ones.

We denote the centre of mass of the domain Ω as O, that is, the point defined by

$$O = \frac{1}{|\Omega|}\int_\Omega x\,dx,$$

and for all $x \in \Omega$, we denote the vector position with respect to the origin O fixed above as $r = \overrightarrow{Ox}$.

Theorem IV.7.4. *Let Ω be a connected bounded Lipschitz domain of $\mathbb{R}^3$. For all data*

$$(f, g, f_b) \in (L^2(\Omega))^3 \times L^2(\Omega) \times (H^{-1/2}(\partial\Omega))^3,$$

satisfying the compatibility conditions

$$\int_\Omega f\,dx + \langle f_b, 1\rangle_{H^{-1/2},H^{1/2}} = 0,$$

$$\int_\Omega f \wedge r\,dx + \langle f_b \wedge r\rangle_{H^{-1/2},H^{1/2}} = 0,$$

Problem (IV.84) *has a unique solution* $(v,p) \in (H^1(\Omega))^3 \times L^2(\Omega)$ *such that* $m(v) = 0$ *and* $m(\operatorname{curl} v) = 0$.

Moreover, if for $k \geq 0$*, the open set* Ω *is of class* $\mathcal{C}^{k+1,1}$ *and if the triplet* (f, g, f_b) *belongs to* $(H^k(\Omega))^3 \times H^{k+1}(\Omega) \times (H^{k+1/2}(\partial\Omega))^3$ *then the pair* (v,p) *belongs to* $(H^{k+2}(\Omega))^3 \times H^{k+1}(\Omega)$ *and there exists* $C > 0$ *such that*

$$\|v\|_{H^{k+2}} + \|p\|_{H^{k+1}} \leq C\left(\|f\|_{H^k} + \|g\|_{H^{k+1}} + \|f_b\|_{H^{k+1/2}}\right).$$

Note that, under these conditions, the pressure is defined in a unique way but the velocity field is only defined relative to a rigid body motion.

Remark IV.7.2. In the second compatibility condition, the vector duality bracket $\langle f_b \wedge r\rangle_{H^{-1/2},H^{1/2}}$ is defined component by component. For instance, the first component of this vector is given by

$$\langle (f_b)_2, r_3\rangle_{H^{-1/2},H^{1/2}} - \langle (f_b)_3, r_2\rangle_{H^{-1/2},H^{1/2}}.$$

7.3.1 Preliminary remarks

The first result that we need (and which was already referred to in Chapter I; see Remark I.4.1) is the following.

Lemma IV.7.5. *Let* Ω *be a connected open set of* $\mathbb{R}^3$ *and let* $v \in (H^1(\Omega))^3$ *be a vector field such that* $\mathrm{D}(v) = 0$ *almost everywhere; then there exist two constant vectors* $t \in \mathbb{R}^3$ *and* $\omega \in \mathbb{R}^3$ *such that*

$$v(x) = t + \frac{1}{2}\omega \wedge r, \text{ for almost all } x \in \Omega.$$

Furthermore, we have

$$m(v) = t,$$

and

$$\operatorname{curl} v(x) = \omega, \text{ for almost all } x \in \Omega.$$

This corresponds to the motion of a rigid body.

Proof.

We set

$$\omega(x) = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = \operatorname{curl} v = \begin{pmatrix} \frac{\partial v_3}{\partial x_2} - \frac{\partial v_2}{\partial x_3} \\ \frac{\partial v_1}{\partial x_3} - \frac{\partial v_3}{\partial x_1} \\ \frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2} \end{pmatrix}.$$

Since $\mathrm{D}(v) = 0$, the gradient of v is reduced to its antisymmetric part, we therefore have

$$\nabla v = \frac{1}{2}\begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}.$$

The Schwarz theorem on crossed derivatives then shows that we have

$$\begin{aligned} &\frac{\partial \omega_2}{\partial x_1} = \frac{\partial \omega_3}{\partial x_1} = 0, \quad &&\frac{\partial \omega_1}{\partial x_2} = \frac{\partial \omega_3}{\partial x_2} = 0, \quad &&\text{and} \quad \frac{\partial \omega_1}{\partial x_3} = \frac{\partial \omega_2}{\partial x_3} = 0, \\ &-\frac{\partial \omega_3}{\partial x_3} = \frac{\partial \omega_2}{\partial x_2}, &&\frac{\partial \omega_3}{\partial x_3} = -\frac{\partial \omega_1}{\partial x_1}, &&\text{and} \ -\frac{\partial \omega_2}{\partial x_2} = \frac{\partial \omega_1}{\partial x_1}, \end{aligned}$$

and thus $\nabla\omega = 0$, which proves that ω is a constant.

It is then easy to show that $\nabla(v - \frac{1}{2}\omega\wedge r) = 0$ and hence that there exists a constant vector t such that $v(x) = t + \frac{1}{2}\omega \wedge r$. Consequently we can calculate the mean value of v

$$m(v) = t + \frac{1}{2|\Omega|}\omega \wedge \left(\int_\Omega r\,dx\right) = t,$$

because, by definition of the origin O, being the centre of mass of Ω, we have $\displaystyle\int_\Omega r\,dx = 0.$

□

The same type of calculation then shows that if (v, p) is a solution of (IV.84) then for all constant vectors $t \in \mathbb{R}^d$ and $\omega \in \mathbb{R}^d$, the pair $(v(x) + t + \omega \wedge r, p)$ is also a solution. Hence, we cannot hope to show the uniqueness of solutions of (IV.84) except under the normalisation conditions $m(v) = 0$ and $m(\operatorname{curl} v) = 0$.

Furthermore, if (IV.84) is satisfied then, from the divergence theorem, we necessarily have

$$\int_\Omega f(x)\,dx + \langle f_b, 1\rangle_{H^{-1/2},H^{1/2}} = 0.$$

In the same way, if we take the vector product of the Stokes equation with the position vector r and integrate it on Ω, we obtain (from the symmetry of the tensor σ)

$$\begin{aligned} \int_\Omega f(x)\wedge r\,dx &= -\int_\Omega (\operatorname{div}\sigma)\wedge r\,dx \\ &= -\left\langle(\sigma.\nu)\wedge r\right\rangle_{H^{-1/2},H^{1/2}} \\ &= -\left\langle f_b \wedge r\right\rangle_{H^{-1/2},H^{1/2}}, \end{aligned}$$

the duality brackets being defined as in Remark IV.7.2.

We clearly see, therefore, that the existence of a solution could only be established if the two compatibility conditions between f and f_b given in the statement are satisfied.

7.3.2 Existence

As before, even if it means modifying the data f and f_b, and even if it requires adding a constant to the pressure, we can focus our attention on the case where $g = 0$ (i.e., the case where the divergence of v is zero). Let us now concentrate on this case and, for this, introduce the Hilbert space

$$W = \left\{v \in (H^1(\Omega))^3, \quad \operatorname{div} v = 0, \ m(v) = 0, \quad m(\operatorname{curl} v) = 0\right\}. \qquad \text{(IV.85)}$$

We now consider the following weak formulation for Problem (IV.84) with $g = 0$. We look for $v \in W$ such that

$$2\int_\Omega \mathrm{D}(v) : \mathrm{D}(\varphi)\,dx - \langle f_b, \varphi\rangle_{H^{-1/2},H^{1/2}} = \int_\Omega f \cdot \varphi\,dx, \forall \varphi \in W. \qquad \text{(IV.86)}$$

To be able to apply the Lax–Milgram theorem (Theorem II.2.5), it is necessary to ensure that the bilinear form

$$a(v, w) = 2\int_\Omega \mathrm{D}(v) : \mathrm{D}(w)\,dx$$

is coercive on the space W introduced above.

To do this, we start by introducing a classic inequality in fluid mechanics and elasticity.

Lemma IV.7.6 (Korn inequality). *Let Ω be a bounded Lipschitz domain of $\mathbb{R}^3$. There exists a $C > 0$ depending only on Ω such that*

$$\|\nabla v\|_{L^2} \le C\left(\|v\|_{L^2} + \|\mathrm{D}(v)\|_{L^2}\right), \forall v \in (H^1(\Omega))^3.$$

Proof.

Let $v \in (H^1(\Omega))^3$; we set

$$d_{ij} = \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right), \forall i, j \in \{1, \ldots, 3\},$$

the entries of the matrix $\mathrm{D}(v)$.

We note that, in the sense of distributions, we have the following relation

$$\frac{\partial}{\partial x_j}\left(\frac{\partial v_i}{\partial x_k}\right) = \frac{\partial d_{ij}}{\partial x_k} + \frac{\partial d_{ik}}{\partial x_j} - \frac{\partial d_{jk}}{\partial x_i}.$$

Hence, it follows

$$\left\|\frac{\partial}{\partial x_j}\left(\frac{\partial v_i}{\partial x_k}\right)\right\|_{H^{-1}} \le C\|\mathrm{D}(v)\|_{L^2},$$

but we also have

$$\left\|\frac{\partial v_i}{\partial x_k}\right\|_{H^{-1}} \le \|v\|_{L^2}.$$

Finally, we have obtained

$$\left\| \frac{\partial v_i}{\partial x_k} \right\|_{H^{-1}} + \left\| \nabla \left(\frac{\partial v_i}{\partial x_k} \right) \right\|_{H^{-1}} \leq C(\|v\|_{L^2} + \|\mathrm{D}(v)\|_{L^2}).$$

Hence, by applying the Nečas inequality (Theorem IV.1.1), we obtain the desired inequality for all i and k:

$$\left\| \frac{\partial v_i}{\partial x_k} \right\|_{L^2} \leq C(\|v\|_{L^2} + \|\mathrm{D}(v)\|_{L^2}).$$

□

Remark IV.7.3. In the space $(H_0^1(\Omega))^d$, the Korn inequality is much easier to prove. Indeed, using the formulas given in Remark I.4.2, we get

$$\begin{aligned} \int_\Omega \nabla v : {}^t\nabla v \, dx &= -\int_\Omega v \cdot \mathrm{div}\left({}^t\nabla v\right) \, dx \\ &= -\int_\Omega v \cdot \nabla(\mathrm{div}\, v) \, dx = \int_\Omega |\mathrm{div}\, v|^2 \, dx \geq 0, \end{aligned}$$

for any $v \in (\mathcal{D}(\Omega))^d$. It follows that

$$2\int_\Omega |\mathrm{D}(v)|^2 \, dx = \int_\Omega \nabla v : \left(\nabla v + {}^t\nabla v\right) \, dx \geq \int_\Omega |\nabla v|^2 \, dx, \tag{IV.87}$$

and the Korn inequality is proved by density of $\mathcal{D}(\Omega)$ in $H_0^1(\Omega)$. Notice that, in that case, the constant $C = \sqrt{2}$ in the inequality is universal and that the term $\|v\|_{L^2}$ is useless in the right-hand side. This is due to the fact that nontrivial rigid-body motions are not allowed in $(H_0^1(\Omega))^d$.

Next, we can prove an inequality of the Poincaré type as follows.

Proposition IV.7.7. *Let Ω be a bounded Lipschitz domain of $\mathbb{R}^3$. There exists a $C > 0$ depending only on Ω such that*

$$\|v\|_{L^2} \leq C\left(|m(v)| + |m(\mathrm{curl}\, v)| + \|\mathrm{D}(v)\|_{L^2}\right), \forall v \in (H^1(\Omega))^d.$$

Proof.

We follow the path of the proof of the Poincaré–Wirtinger inequality (Proposition III.2.39). We argue by contradiction, by assuming that for all n, there exists a function $v_n \in (H^1(\Omega))^3$ such that $\|v_n\|_{L^2} = 1$ and

$$|m(v_n)| + |m(\mathrm{curl}\, v_n)| + \|\mathrm{D}(v_n)\|_{L^2} \leq \frac{1}{n}.$$

From the Korn inequality, we deduce that the sequence $(v_n)_n$ is bounded in $(H^1(\Omega))^3$. We can extract a subsequence, still referred to as $(v_n)_n$ for simplicity, which weakly converges towards a certain $v \in (H^1(\Omega))^3$. By the compactness of the embedding from $H^1(\Omega)$ into $L^2(\Omega)$, we know that the convergence of $(v_n)_n$ towards v is strong in $(L^2(\Omega))^3$. The limit v hence satisfies

$$\|v\|_{L^2} = 1,\ m(v) = 0,\ m(\operatorname{curl} v) = 0,\ \mathrm{D}(v) = 0.$$

This is impossible inasmuch as from Lemma IV.7.5 the last three equalities imply that v is zero.

□

By combining the Korn inequality and the Poincaré inequality above, we deduce that $\|\mathrm{D}(\,\cdot\,)\|_{L^2}$ is a norm on the space W (defined by (IV.85)) which is equivalent to the H^1 norm.

We can, therefore, apply the Lax–Milgram theorem to establish the existence of a unique solution $v \in W$ of (IV.86).

It is now necessary to show that this function v is indeed a solution of the initial problem. To do this, we consider a function $\psi \in (H^1(\Omega))^3$ such that $\operatorname{div} \psi = 0$ and we set

$$\varphi = \psi - m(\psi) - \frac{1}{2} m(\operatorname{curl} \psi) \wedge r,$$

which by definition belongs to the space W. We can therefore use this function φ in the formulation (IV.86) and obtain

$$\begin{aligned}
&2 \int_\Omega \mathrm{D}(v) : \mathrm{D}(\psi)\, dx - \langle f_b, \psi \rangle_{H^{-1/2}, H^{1/2}} \\
&\quad + m(\psi) \cdot \langle f_b, 1 \rangle_{H^{-1/2}, H^{1/2}} - \frac{1}{2} \langle f_b, m(\operatorname{curl} \psi) \wedge r \rangle_{H^{-1/2}, H^{1/2}} \\
&= \int_\Omega f \cdot \psi\, dx - m(\psi) \cdot \int_\Omega f\, dx - \frac{1}{2} \int_\Omega (m(\operatorname{curl} \psi) \wedge r) \cdot f\, dx.
\end{aligned}$$

From the first compatibility condition, the terms which contain the mean of ψ compensate each other. By using the relation $a \cdot (b \wedge c) = -b \cdot (a \wedge c)$ and the second compatibility condition, we see that the terms which contain the mean of the curl of ψ also compensate each other. This therefore leaves

$$2 \int_\Omega \mathrm{D}(v) : \mathrm{D}(\psi)\, dx - \langle f_b, \psi \rangle_{H^{-1/2}, H^{1/2}} = \int_\Omega f \cdot \psi\, dx,$$

for any function ψ which is divergence-free. In particular, using the symmetry of $\mathrm{D}(v)$, we get

$$2 \int_\Omega \mathrm{D}(v) : \nabla \psi\, dx = \int_\Omega f \cdot \psi\, dx,\ \forall \psi \in \mathscr{V}.$$

It follows that

$$\langle -\operatorname{div}(2\mathrm{D}(v)) - f, \psi \rangle_{\mathcal{D}', \mathcal{D}} = 0,\ \forall \psi \in \mathscr{V}.$$

From Theorem IV.2.4, we deduce that there exists a unique pressure $p \in L^2_0(\Omega)$ such that

$$-\operatorname{div}(2\mathrm{D}(v)) + \nabla p = f,$$

that is,

$$-\operatorname{div}(2\mathrm{D}(v) - p\,\mathrm{Id}) = f.$$

We can then recover the boundary conditions as we did above for the first formulation of the Neumann problem, which determines the pressure in a unique manner.

7.3.3 Regularity

It is difficult, now, to build a lifting of the boundary condition which is not a classic Neumann condition. We thus follow a different approach and directly account for the regularity of the boundary data in the weak formulation. As a consequence, we need the following lemma.

Lemma IV.7.8. *Let Ω be a connected bounded domain of $\mathbb{R}^3$ of class $\mathcal{C}^{1,1}$. Let θ be a vector field in Ω satisfying (III.77) with $k = 0$.*

Then, we have the following estimate for all $g \in H^{1/2}(\partial\Omega)$,

$$\|\delta_h^\theta g\|_{H^{-1/2}} \leq C(\theta)|h|\|g\|_{H^{1/2}}.$$

Proof.

First, it is clear that, even if it means subtracting a constant of g, it is enough to consider the case where g has a zero mean value on $\partial\Omega$.

We consider the scalar Neumann problem defined by

$$\begin{cases} -\Delta\varphi = 0, & \text{in } \Omega, \\ \dfrac{\partial\varphi}{\partial\nu} = g, & \text{on } \partial\Omega, \\ \displaystyle\int_\Omega \varphi\, dx = 0 \end{cases} \tag{IV.88}$$

From Theorem III.4.3, since g has a zero mean value, we know that the above problem has a unique solution in $H^1(\Omega)$. Moreover, we know that we have $\varphi \in H^2(\Omega)$ as soon as $g \in H^{1/2}(\partial\Omega)$, as well as the estimate

$$\|\varphi\|_{H^2} \leq C\|g\|_{H^{1/2}}. \tag{IV.89}$$

The variational formulation of problem (IV.88) is

$$\int_\Omega \nabla\varphi \cdot \nabla\psi\, dx - \int_{\partial\Omega} g\psi\, d\sigma = 0, \forall \psi \in H^1(\Omega). \tag{IV.90}$$

We note that the boundary term, which a priori is a duality bracket, is in fact an integral term because $g \in H^{1/2}(\partial\Omega)$.

We now take a test function in (IV.90) in the form $\psi = \tau_{-h}^{\theta} v$ for any $v \in H^1(\Omega)$.

It follows that

$$\int_\Omega (\delta_h^\theta \nabla\varphi)\cdot\nabla v\,dx - \int_{\partial\Omega} (\delta_h^\theta g)\gamma_0(v)\,d\sigma = \int_\Omega \{\nabla\varphi, \nabla v\}_h\,dx - \int_{\partial\Omega} \{g, \gamma_0(v)\}_h\,d\sigma.$$

We obtain, with (III.90),

$$\begin{aligned}
&\big|\langle \delta_h^\theta g, \gamma_0(v)\rangle_{H^{-1/2},H^{1/2}}\big| = \left|\int_{\partial\Omega} (\delta_h^\theta g)\gamma_0(v)\,d\sigma\right| \\
&\leq \|\delta_h^\theta \nabla\varphi\|_{L^2}\|\nabla v\|_{L^2} + C|h|\|\nabla\varphi\|_{L^2}\|\nabla v\|_{L^2} + C|h|\|g\|_{L^2(\partial\Omega)}\|\gamma_0(v)\|_{L^2(\partial\Omega)} \\
&\leq C|h|\|\varphi\|_{H^2}\|v\|_{H^1} + C|h|\|g\|_{H^{1/2}}\|\gamma_0(v)\|_{H^{1/2}}.
\end{aligned}$$

Using (IV.89), we deduce that

$$\big|\langle \delta_h^\theta g, \gamma_0(v)\rangle_{H^{-1/2},H^{1/2}}\big| \leq C|h|\|g\|_{H^{1/2}}\|v\|_{H^1},\ \forall v \in H^1(\Omega).$$

The claim is proved.

□

Through this lemma and the methods used above for the Stokes–Neumann problem, we can easily demonstrate the regularity properties stated for the stress boundary conditions.

8 The interface Stokes problem

The goal of this section is to show how to apply the tools and techniques of this chapter to more complex situations.

As an example we consider here the following generalised Stokes problem

$$\begin{cases} \rho(x)v - \operatorname{div}\big(2\mu(x)\mathrm{D}(v) - p\,\mathrm{Id}\big) = f, & \text{in } \Omega, \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ v = 0, & \text{on } \partial\Omega, \end{cases} \tag{IV.91}$$

where ρ and μ are given functions which represent the density and the viscosity of the fluid mixture under study. Of course, in real-life nonsteady situations, ρ and μ also evolve with the flow (see, e.g., Chapter VI), and for instance the viscosity μ is a function of the density ρ. However, a very usual time discretisation of the nonhomogeneous Stokes equations can take the following form

$$\begin{cases} \dfrac{\rho^{n+1}-\rho^n}{\delta t} + v^n \cdot \nabla \rho^{n+1} = 0, \\ \dfrac{\rho^{n+1}v^{n+1} - \rho^n v^n}{\delta t} - \operatorname{div}\big(2\mu(\rho^{n+1}(x))\mathrm{D}(v^{n+1})\big) + \nabla p^{n+1} = f^{n+1}, \\ \operatorname{div} v^{n+1} = 0. \end{cases}$$

Hence, for ρ^n, v^n given, and δt small enough, the couple (v^{n+1}, p^{n+1}) is a solution of a system whose form is exactly the one of (IV.91).

8.1 Existence and uniqueness

With minimal regularity assumptions on ρ and μ, we can prove the following result.

Theorem IV.8.1. *Let Ω be a bounded connected Lipschitz domain in $\mathbb{R}^d$. Assume that $\rho, \mu \in L^\infty(\Omega)$ with $\rho \geq 0$ and $\inf_\Omega \mu > 0$; then for any $f \in (H^{-1}(\Omega))^d$, there exists a unique solution $(v,p) \in (H^1_0(\Omega))^d \times L^2_0(\Omega)$ of Problem (IV.91). Furthermore, it satisfies*

$$\|v\|_{H^1} + \|p\|_{L^2} \leq C\|f\|_{H^{-1}},$$

for some $C > 0$ which does not depend on f.

Proof.

The proof is now classic. We introduce the bilinear form

$$\begin{aligned} a(v,w) &= \int_\Omega \rho(x) v \cdot w \, dx + \int_\Omega 2\mu(x)\mathrm{D}(v) : \mathrm{D}(w)\, dx \\ &= \int_\Omega \rho(x) v \cdot w \, dx + \int_\Omega 2\mu(x)\nabla v : \mathrm{D}(w)\, dx, \ \forall v, w \in V, \end{aligned}$$

which is continuous on V. Moreover we have

$$a(v,v) = \int_\Omega \rho(x)|v|^2\, dx + 2\int_\Omega \mu(x)|\mathrm{D}(v)|^2\, dx \geq 2(\inf_\Omega \mu)\|\mathrm{D}(v)\|^2_{L^2},$$

and by the Korn inequality (Lemma IV.7.6) we deduce that $a(v,v) \geq \alpha\|v\|^2_V$ for some $\alpha > 0$. The linear form $L : v \in V \mapsto \langle f, v\rangle_{H^{-1},H^1_0}$ is continuous on V so that the Lax–Milgram theorem lets us conclude as to the existence and uniqueness of a solution $v \in V$ of the problem

$$a(v,w) = L(w), \ \forall w \in V.$$

In particular, using the Stokes formula this proves that we have

$$\langle \rho v - \operatorname{div}(2\mu \mathrm{D}(v)) - f, w\rangle_{H^{-1},H^1_0} = 0, \ \forall w \in V,$$

and thus by Theorem IV.2.3 we deduce the existence of a pressure $p \in L^2_0(\Omega)$ so that (v, p) solves (IV.91); moreover this pressure is unique because Ω is connected.

□

Let us assume now that we are given a decomposition of Ω into two Lipschitz open subdomains Ω_1, Ω_2 such that

$$\Omega_1 \cap \Omega_2 = \emptyset,\ \overline{\Omega_1} \cup \overline{\Omega_2} = \overline{\Omega},\ \partial\Omega_1 \cap \partial\Omega_2 \cap \partial\Omega = \emptyset.$$

We define the interface between the two subdomains $S = \partial\Omega_1 \cap \partial\Omega_2$ and $\gamma_S : H^1(\Omega) \to H^{1/2}(S)$ denote the trace operator on S, which is well defined as the restriction to S of the trace of $u_{|\Omega_1}$ on $\partial\Omega_1$, or equivalently as the restriction to S of the trace of $u_{|\Omega_2}$ on $\partial\Omega_2$.

Let $j_S \in (H^{-1/2}(S))^d$ be given and let us consider the linear map

$$J_S : v \in (H^1(\Omega))^d \mapsto \langle j_S, \gamma_S(v)\rangle_{H^{-1/2}(S), H^{1/2}(S)}.$$

It is clear that the linear map J_S is continuous on $(H^1_0(\Omega))^d$ and thus defines an element of $(H^{-1}(\Omega))^d$.

Thus, from Theorem IV.8.1, we know that for any given $f_0 \in (L^2(\Omega))^d$, there exists a unique couple $(v, p) \in (H^1_0(\Omega))^d \times L^2_0(\Omega)$ which satisfies

$$\begin{cases} \rho(x)v - \operatorname{div}\big(2\mu(x)\mathrm{D}(v) - p\,\mathrm{Id}\big) = f_0 + J_S, & \text{in } \Omega, \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ v = 0, & \text{on } \partial\Omega. \end{cases} \tag{IV.92}$$

Let us comment on this system. We denote by v_i, p_i the restrictions of v, p to Ω_i, for $i = 1, 2$.

- First of all, we observe that for any test function $\varphi \in (\mathcal{D}(\Omega_i))^d \subset (\mathcal{D}(\Omega))^d$, the term $\langle J_S, \varphi\rangle_{H^{-1}, H^1_0}$ vanishes. Therefore, the couple (v_i, p_i) solves the Stokes problem with the source term f_0 in Ω_i, as well as the Dirichlet boundary condition $v_i = 0$ on $\partial\Omega_i \cap \partial\Omega$.
- From the previous facts, we deduce that the stress tensor $\sigma_i = 2\mu(x)\mathrm{D}(v_i) - p_i\,\mathrm{Id}$ belongs to $(H_{\mathrm{div}}(\Omega_i))^d$ for $i = 1, 2$. Hence, we can define the normal trace of these two tensor fields $\gamma_{\nu_i}(\sigma_i) \in (H^{-1/2}(\Omega_i))^d$. Taking a test function $\varphi \in (\mathcal{D}(\Omega))^d$ in the weak formulation of the problem and using the Stokes formula given in Lemma IV.3.3 on each of the two subdomains Ω_1 and Ω_2 we finally arrive at

$$\begin{aligned} &\langle \gamma_{\nu_1}(\sigma_1) + \gamma_{\nu_2}(\sigma_2), \gamma_S(\varphi)\rangle_{H^{-1/2}(S), H^{1/2}(S)} \\ =&\langle J_S, \gamma_S(\varphi)\rangle_{H^{-1}, H^1_0} = \langle j_S, \gamma_S(\varphi)\rangle_{H^{-1/2}(S), H^{1/2}(S)}. \end{aligned}$$

Since γ_S is surjective onto $H^{1/2}(S)$, this finally leads to the formula

$$\gamma_{\nu_1}(\sigma_1) + \gamma_{\nu_2}(\sigma_2) = j_S. \tag{IV.93}$$

The outward normals ν_i have opposite directions on S, thus this exactly means that the solution (v, p) we constructed satisfies a stress jump condition at the interface S prescribed by the data j_S.
From a modelling point of view, stress jumps may account, for instance, for surface tension phenomena between two fluids, or for some mechanical properties of the interface in the case where S models a membrane separating the fluids. Here also, the more physically relevant situation is the one of nonsteady flows that we study in the next chapters.

8.2 Regularity of the solution

We now want to prove that, under suitable assumptions, the solution (v, p) studied in the previous section is smooth on each subdomain Ω_1 and Ω_2. We obviously cannot expect in general for the solution to be globally smooth on Ω because the jump condition (IV.93) implies that the pressure or the velocity gradient should be discontinuous on S. Let us denote by $\mu_i : \Omega_i \to \mathbb{R}$ the restriction of the viscosity μ to the domain Ω_i.

Theorem IV.8.2. *Assume that Ω, Ω_1, and Ω_2 are of class $\mathcal{C}^{2,1}$, that $\mu_i \in W^{1,\infty}(\Omega_i)$ for $i = 1, 2$, and that $j_S \in H^{1/2}(S)$; then the solution (v, p) to* (IV.92) *satisfies*

$$(v, p) \in (H^2(\Omega_i))^d \times H^1(\Omega_i), \textit{ for } i = 1, 2.$$

Proof.
We first observe that we have

$$-\operatorname{div}\big(2\mu(x)\mathrm{D}(v) - p\,\mathrm{Id}\big) = f_0 - \rho v + J_S,$$

and since $-\rho v \in L^2$, we can change f_0 into $f_0 - \rho v$ in order to study the regularity properties of (v, p). In other words, we can assume that $\rho(x) = 0$ for all $x \in \Omega$ without loss of generality. Therefore, the weak formulation of the problem that we consider from now on is the following

$$\left\{\begin{aligned} &\int_\Omega 2\mu(x)\,\mathrm{D}(v) : \mathrm{D}(\psi)\,dx - \int_\Omega p\,\operatorname{div}\psi\,dx \\ &\qquad\qquad = \int_\Omega f_0 \cdot \psi\,dx + \int_S j_S \cdot \gamma_S(\psi)\,d\sigma, \quad \forall \psi \in (H^1_0(\Omega))^d, \\ &\int_\Omega q\,\operatorname{div} v\,dx = 0, \qquad\qquad\qquad\qquad\qquad\quad \forall q \in L^2(\Omega). \end{aligned}\right. \tag{IV.94}$$

Notice that the jump term J_S is now an integral term because we assume that j_S belongs in $(H^{1/2}(S))^d$.

- Let $i \in \{1,2\}$ and let us first study the local regularity of the solution in Ω_i. For any $\tilde{\psi} \in (H^1_0(\Omega_i))^d$, since μ_i is Lipschitz continuous and does not vanish, we can take $\psi = \mu_i^{-1}\tilde{\psi}$ as a test function in the first equation of (IV.94).
After some straightforward computations, we find that $(v_i, \mu_i^{-1}p_i)$ solves the following problem in Ω_i
$$\begin{cases} -\underbrace{\mathrm{div}(2\mathrm{D}(v_i))}_{=\Delta v_i} + \nabla\left(\mu_i^{-1}p_i\right) = \mu_i^{-1}f_0 + 2\mu_i^{-1}\mathrm{D}(v_i).\nabla\mu_i + p_i\nabla\left(\mu_i^{-1}\right), \\ \mathrm{div}\, v_i = 0. \end{cases} \tag{IV.95}$$
Since $p_i \in L^2(\Omega_i)$, $v_i \in (H^1(\Omega_i))^d$, and μ_i is smooth, we see that the source term in this problem belongs to $L^2(\Omega_i)$ and then $(v_i, \mu_i^{-1}p_i)$ solves a usual Stokes problem in Ω_i with a source term in L^2. Thus, we can apply Theorem IV.6.1 which gives that $v_i \in (H^2_{loc}(\Omega_i))^d$ and $\mu_i^{-1}p_i \in H^1_{loc}(\Omega_i)$, so that $p_i \in H^1_{loc}(\Omega_i)$ and the claim is proved.
Notice that, as expected, the jump j_S does not play any role in the proof of the local regularity properties in each Ω_i.
- Let us now prove the tangential regularity up to the boundary and to the interface of the solution (v,p) of (IV.94).
Most of the proof is the same as for the standard Stokes problem; we only concentrate here on the particular terms related to the interface.
Let us consider any vector field θ satisfying (III.77) with $k=0$ for the two domains Ω_1 and Ω_2, so that in particular θ is tangent to S. We use the same notation as in Section 6.1.2.2. Since θ is tangent to S, the flow of θ satisfies $\tau^\theta(h,S) = S, \forall h \in \mathbb{R}$. In particular, we have
$$\gamma_S(\tau^\theta_h(v)) = \tau^\theta_h(\gamma_S v),\ \forall v \in (H^1_0(\Omega))^d.$$
For $\varphi \in (H^1_0(\Omega))^d$ and $\pi \in L^2(\Omega)$, we take $\psi = \delta^\theta_{-h}\varphi$ and $q = \delta^\theta_{-h}\pi$ as test functions in (IV.94). Following the same computations as in previous sections, we obtain
$$\begin{aligned} &\int_\Omega 2(\tau^\theta_h\mu)\mathrm{D}(\delta^\theta_h v) : \mathrm{D}(\varphi)\,dx - \int_\Omega \delta^\theta_h p\ \mathrm{div}\,\varphi\,dx \\ &= \int_\Omega f_0\cdot\delta^\theta_{-h}\varphi\,dx - \int_\Omega 2\mu\mathrm{D}(v) : \left[\mathrm{D},\tau^\theta_h\right]\varphi\,dx + 2\int_\Omega \{\mu\mathrm{D}v, \mathrm{D}(\varphi)\}_h\,dx \\ &\quad + \int_\Omega p\left[\mathrm{div},\tau^\theta_h\right]\varphi\,dx - \int_\Omega \{p, \mathrm{div}\,\varphi\}_h\,dx + \int_\Omega 2(\tau^\theta_h\mu)\mathrm{D}(\varphi) : \left[\mathrm{D},\tau^\theta_h\right]v\,dx \\ &\quad - 2\int_\Omega (\delta^\theta_h\mu)\mathrm{D}(v) : \mathrm{D}(\varphi)\,dx + \int_S \delta^\theta_h j_S\cdot\gamma_0(\varphi)\,d\sigma - \int_S \{j_S, \gamma_0(\varphi)\}_h d\sigma \end{aligned} \tag{IV.96}$$
All the terms in the right-hand side, except the last three, can be estimated in exactly the same way as we did in Section 6.1.2.2, by using additionally

that μ is a bounded function. Let us denote by $T_1''(\varphi), T_2''(\varphi)$, and $T_3''(\varphi)$ the last three terms.
Since θ is tangent to S, the flow τ^θ maps each subdomain Ω_i into itself. Therefore, μ being supposed to be Lipschitz-continuous on each subdomain, we have

$$\|\delta_h^\theta \mu\|_{L^\infty} \leq C|h|(\|\mu\|_{W^{1,\infty}(\Omega_1)} + \|\mu\|_{W^{1,\infty}(\Omega_2)}).$$

Moreover, we can use Lemma IV.7.8 to deal with the boundary terms T_2'' and T_3''. The estimates we finally obtain are

$$\begin{aligned}
|T_1''(\varphi)| &\leq C|h|(\|\mu\|_{W^{1,\infty}(\Omega_1)} + \|\mu\|_{W^{1,\infty}(\Omega_2)})\|v\|_{H^1}\|\varphi\|_{H^1},\\
|T_2''(\varphi)| &\leq \|\delta_h^\theta j_S\|_{H^{-1/2}(S)}\|\gamma_0(\varphi)\|_{H^{1/2}(S)} \leq C|h|\|j_S\|_{H^{1/2}(S)}\|\varphi\|_{H^1},\\
|T_3''(\varphi)| &\leq C|h|\|j_S\|_{L^2(S)}\|\gamma_0(\varphi)\|_{L^2(S)} \leq C|h|\|j_S\|_{H^{1/2}(S)}\|\varphi\|_{H^1}.
\end{aligned}$$

Gathering all these inequalities we get

$$\begin{aligned}
&\left|\int_\Omega (\delta_h^\theta p)\,\mathrm{div}\,\varphi\,dx\right| \leq \|\nabla \delta_h^\theta v\|_{L^2}\|\nabla\varphi\|_{L^2}\\
&+ C|h|(\|\nabla v\|_{L^2} + \|p\|_{L^2} + \|f_0\|_{L^2} + \|j_S\|_{H^{1/2}})\|\varphi\|_{H_0^1},\ \forall \varphi \in (H_0^1(\Omega))^d,
\end{aligned}$$

and we can conclude with the Nečas inequality (Theorem IV.1.1) that

$$\|\delta_h^\theta p\|_{L^2} \leq C\|\nabla \delta_h^\theta v\|_{L^2} + C|h|(\|\nabla v\|_{L^2} + \|p\|_{L^2} + \|f\|_{L^2} + \|j_S\|_{H^{1/2}}). \tag{IV.97}$$

Coming back to (IV.96) with $\varphi = \delta_h^\theta v$, using the fact that $\inf_\Omega \mu > 0$, and the Korn inequality, we finally obtain

$$\|\nabla \delta_h^\theta v\|_{L^2}^2 \leq C|h|^2(\|g\|_{H^1} + \|v\|_{H^1} + \|p\|_{L^2} + \|f\|_{L^2} + \|j_S\|_{H^{1/2}})^2,$$

and the estimate on $\delta_h^\theta p$ follows from (IV.97).
This proves the tangential regularity of v and p in each subdomain Ω_1 and Ω_2.

- Let us finally conclude with the complete regularity up to the boundary $\partial\Omega$ and to the interface S. To this end, we come back to (IV.95) and we see that for $i \in \{1,2\}$ fixed, the couple $(v_i, \mu_i^{-1}p_i)$ belongs to $(H^2_{tang}(\Omega_i))^d \times H^1_{tang}(\Omega_i)$ and solves a standard Stokes problem with a right-hand side in $(L^2(\Omega_i))^d$.
  Therefore, we can apply exactly the same strategy as in Section 6.1.2.3 to obtain the normal regularity of the solution up to the boundary. Notice that, since Ω_i is of class $\mathcal{C}^{2,1}$, this proof does not make use of the boundary condition which is satisfied by the solution, as we noticed in the study of the Stokes–Neumann problem (see the discussion at the end of Section 7.2).

□

9 The Stokes problem with vorticity boundary conditions

The vorticity in a flow is defined to be $\omega = \operatorname{curl} v$. In order to illustrate its physical meaning, we consider a rotating fluid (in the whole space $\mathbb{R}^3$) whose velocity field is given, as for a rigid body rotation, by $v(r) = r\boldsymbol{\Omega} \wedge e_r$ in cylindrical coordinates ($\boldsymbol{\Omega}$ is the angular velocity). We saw in (I.37) that

$$\operatorname{curl} v = 2\boldsymbol{\Omega},$$

so that the vorticity $\omega = \operatorname{curl} v$ is exactly equal, up to a factor 2, to the angular velocity $\boldsymbol{\Omega}$. That's the reason why, for a general flow, the vorticity is a measure of the local angular velocity of fluid particles in the flow.

We consider here the Stokes problem with an *impermeable wall* condition (i.e., only the normal part of the velocity field is imposed to be zero at the boundary) and an additional condition involving (the tangential part of) the vorticity. Therefore, the fluid is allowed to slide along the boundary of the domain. For simplicity, we only consider the three-dimensional case but all the results can be translated in the two-dimensional case.

More precisely, the problem under study is the following

$$\begin{cases} -\Delta v + \nabla p = f, & \text{in } \Omega, \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ v \cdot \nu = 0, & \text{on } \partial\Omega, \\ (\operatorname{curl} v) \wedge \nu = g, & \text{on } \partial\Omega. \end{cases} \tag{IV.98}$$

The property of particular interest for this system is that the computation of the pressure and the one of the velocity can be uncoupled (see Proposition IV.9.9). This is generally not true for other kinds of boundary conditions for the Stokes equations as the ones studied in the previous sections.

Remark IV.9.1. All the results of this section can be easily adapted to the case of nonhomogeneous boundary data of the kind $v\cdot\nu = v_b$, with $\int_{\partial\Omega} v_b \, d\sigma = 0$. To this end, it is sufficient to solve Problem (IV.98) and to change the velocity v into $v + \nabla\varphi$ where φ solves the following Laplace problem

$$\begin{cases} -\Delta\varphi = 0, & \text{in } \Omega, \\ \dfrac{\partial\varphi}{\partial\nu} = v_b, & \text{on } \partial\Omega. \end{cases}$$

This new velocity field is divergence-free because φ is harmonic, and satisfies $-\Delta(v + \nabla\varphi) = -\Delta v$ so that it solves the Stokes equation with the same pressure p. Finally, the vorticity boundary condition is satisfied because $\operatorname{curl}(v + \nabla\varphi) = \operatorname{curl} v$.

9.1 Preliminaries

Let us define the Hilbert space

$$H_\nu = (H^1(\Omega))^3 \cap \operatorname{Ker} \gamma_\nu = \left\{ v \in (H^1(\Omega))^3, \;\; \gamma_0(v)\cdot\nu = 0, \text{ on } \partial\Omega \right\}.$$

We first need to define and study some special elements of $(H^{-1}(\Omega))^3$.

We recall that δ is the function "distance to the boundary" that we studied in Section 3.2. Since Ω is supposed to be of class $\mathcal{C}^{1,1}$ we know that δ is also of class $\mathcal{C}^{1,1}$ in a neighborhood of $\partial\Omega$.

For any $\varepsilon > 0$, we introduce $\beta_\varepsilon : \mathbb{R}^+ \to \mathbb{R}^+$ which is defined by

$$\beta_\varepsilon(s) = \begin{cases} 1 & \text{for } s > \varepsilon, \\ \frac{s}{\varepsilon} & \text{for } s \le \varepsilon. \end{cases}$$

Definition IV.9.1. *We say that $f \in (H^{-1}(\Omega))^3$ does not contain tangential boundary terms if and only if*

$$\Big(\langle f, \beta_\varepsilon(\delta) v \rangle_{H^{-1},H^1_0} \Big)_\varepsilon \text{ has a limit as } \varepsilon \to 0, \text{ for any } v \in H_\nu. \tag{IV.99}$$

We denote by $H^{-1}_\nu \subset (H^{-1}(\Omega))^3$ the set of such distributions.

Lemma IV.9.2. *For any $w \in (H^1_0(\Omega))^3$, we have $\beta_\varepsilon(\delta) w \xrightarrow[\varepsilon\to 0]{} w$ in $(H^1(\Omega))^3$.*

Proof.

The convergence in L^2 is straightforward by using the dominated convergence theorem. Then, we write

$$\|\partial_{x_i}(\beta_\varepsilon(\delta) w) - \partial_{x_i} w\|_{L^2} \le \|(\beta_\varepsilon(\delta) - 1)\partial_{x_i} w\|_{L^2} + \|\partial_{x_i}(\beta_\varepsilon(\delta))\, w\|_{L^2},$$

and we see that the first term also converges to zero as $\varepsilon \to 0$ by using the dominated convergence theorem.

It remains to study the second term. Using that $|\nabla\delta| \le 1$ and the expression of β_ε, we find

$$\|\partial_{x_i}(\beta_\varepsilon(\delta)) w\|^2_{L^2} = \int_{\mathcal{O}_\varepsilon} |\partial_{x_i}\delta|^2 \left|\frac{w}{\varepsilon}\right|^2 dx \le \int_{\mathcal{O}_\varepsilon} \left|\frac{w}{\delta}\right|^2 dx.$$

We recall that $\mathcal{O}_\varepsilon = \{x \in \Omega, \;\delta(x) < \varepsilon\}$. Since $w \in (H^1_0(\Omega))^3$, the Hardy inequality (Proposition III.2.40) proves that $|w/\delta|$ belongs to $L^2(\Omega)$ and therefore, by the dominated convergence theorem, the last integral above tends to 0 as ε goes to 0.

□

Proposition IV.9.3. *For any $f \in H^{-1}_\nu$, we define*

$$\tilde{f} : v \in H_\nu \mapsto \left(\lim_{\varepsilon \to 0} \langle f, \beta_\varepsilon(\delta) v \rangle_{H^{-1}, H_0^1} \right).$$

Then $\tilde{f}$ *belongs to* H_ν' *and* $\tilde{f} = f$ *on* $(H_0^1(\Omega))^3$.

Proof.

For any $\varepsilon > 0$, the map $f_\varepsilon : v \in H_\nu \mapsto \langle f, \beta_\varepsilon(\delta) v \rangle_{H^{-1}, H_0^1}$ is linear and continuous. Moreover, from Assumption (IV.99), we know that $\langle f_\varepsilon, v \rangle_{H_\nu', H_\nu}$ has a limit when $\varepsilon \to 0$. Therefore, by the Banach–Steinhaus theorem (see Theorem II.2.4), we know that $\tilde{f}$ is a linear continuous form on H_ν. Moreover, for any $v \in (\mathcal{D}(\Omega))^3$, we have $\beta_\varepsilon(\delta) v = v$ for $\varepsilon > 0$ small enough so that $\tilde{f}$ and f coincide on $(\mathcal{D}(\Omega))^3$. By density, we deduce that they also coincide on $(H_0^1(\Omega))^3$.

□

We now show that the gradient of any function in $L^2(\Omega)$ belongs to the class H_ν^{-1}.

Lemma IV.9.4. *For any* $q \in L^2(\Omega)$, *we have* $\nabla q \in H_\nu^{-1}$ *and more precisely, we have*

$$\langle \widetilde{\nabla q}, w \rangle_{H_\nu', H_\nu} = - \int_\Omega q (\operatorname{div} w) \, dx, \; \forall w \in H_\nu.$$

Proof.

For any $\varepsilon > 0$ and any $w \in H_\nu$, we have

$$\begin{aligned} \langle \nabla q, \beta_\varepsilon(\delta) w \rangle_{H^{-1}, H_0^1} &= - \int_\Omega q \operatorname{div} \big(\beta_\varepsilon(\delta) w \big) \, dx \\ &= - \int_\Omega q (\operatorname{div} w) \beta_\varepsilon(\delta) \, dx - \int_\Omega q (w \cdot \nabla \beta_\varepsilon(\delta)) \, dx. \end{aligned}$$

By using the dominated convergence theorem, we immediately obtain that the first term in the right-hand side converges towards $- \int_\Omega q (\operatorname{div} w) \, dx$.

Let us show that the second term, referred to as I_ε, tends to zero. We first observe that

$$\nabla \beta_\varepsilon(\delta) = \beta_\varepsilon'(\delta) \nabla \delta,$$

and that

$$\beta_\varepsilon'(\delta) = \begin{cases} 0, & \text{for } \delta > \varepsilon \\ \frac{1}{\varepsilon}, & \text{for } \delta < \varepsilon \end{cases}.$$

It follows that

$$I_\varepsilon = - \frac{1}{\varepsilon} \int_{\mathcal{O}_\varepsilon} q (w \cdot \nabla \delta) \, dx.$$

By definition of $\mathcal{O}_\varepsilon$ we have $(1/\varepsilon) \leq (1/\delta)$ in $\mathcal{O}_\varepsilon$, so that we have the estimate

$$|I_\varepsilon| \leq \int_{\mathcal{O}_\varepsilon} |q| \frac{|w \cdot \nabla \delta|}{\delta} \, dx \leq \|q\|_{L^2} \left(\int_{\mathcal{O}_\varepsilon} \frac{|w \cdot \nabla \delta|^2}{\delta^2} \, dx \right)^{1/2}. \tag{IV.100}$$

Finally, using that $\nabla\delta = -\nu$ on the boundary, and since $w \in H_\nu$, we have $w \cdot \nabla\delta = 0$ on $\partial\Omega$. In particular, we get $w \cdot \nabla\delta \in H^1_0(\Omega)$ and the Hardy inequality (Proposition III.2.40) implies that $(w \cdot \nabla\delta)/\delta$ belongs to $L^2(\Omega)$. We conclude from (IV.100) that $I_\varepsilon \xrightarrow[\varepsilon\to 0]{} 0$ and the claim is proved.

□

9.2 A vector Laplace problem

We begin with the study of the following vector Laplace problem: find a vector field $v : \Omega \to \mathbb{R}^3$ satisfying

$$\begin{cases} -\Delta v = f, & \text{in } \Omega, \\ \quad v \cdot \nu = 0, & \text{on } \partial\Omega, \\ \quad (\operatorname{curl} v) \wedge \nu = g, & \text{on } \partial\Omega. \end{cases} \tag{IV.101}$$

Notice that all the components of v are coupled through the boundary conditions and therefore this problem cannot be reduced to scalar Laplace problems.

Theorem IV.9.5. *Let Ω be a simply connected bounded domain of $\mathbb{R}^3$ of class $\mathcal{C}^{1,1}$, $f \in H^{-1}_\nu$, and $g \in (H^{-1/2}(\partial\Omega))^3$, which is tangent to the boundary in the following sense*

$$\langle g, \varphi\nu\rangle_{H^{-1/2},H^{1/2}} = 0, \ \forall\varphi \in H^{1/2}(\partial\Omega). \tag{IV.102}$$

Then, there exists a unique $v \in H_\nu$ which is solution to (IV.101) and satisfies

$$\|v\|_{H^1} \le C(\|f\|_{H^{-1}} + \|g\|_{H^{-1/2}}),$$

for some $C > 0$ depending only on Ω. More precisely, the third boundary condition in (IV.101) is satisfied in the following weak sense,

$$\left(\frac{1}{\varepsilon}\int_0^\varepsilon (\operatorname{curl} v)(. - s\nu(.))\, ds\right) \wedge \nu(.) \xrightarrow[\varepsilon\to 0]{} g, \text{ in } (H^{-1/2}(\partial\Omega))^3. \tag{IV.103}$$

Remark IV.9.2. Let us comment on the property (IV.103). Since $\operatorname{curl} v \in (L^2(\Omega))^3$, for any $\varepsilon > 0$ the map

$$\sigma \in \partial\Omega \mapsto \left(\frac{1}{\varepsilon}\int_0^\varepsilon (\operatorname{curl} v)(\sigma - s\nu(\sigma))\, ds\right) \wedge \nu(\sigma),$$

belongs to $(L^2(\partial\Omega))^3 \subset (H^{-1/2}(\partial\Omega))^3$. It corresponds to taking averages of $(\operatorname{curl} v) \wedge \nu$ along the normal direction to the boundary.

We claim in (IV.103) that this family of functions weakly converges towards g as $\varepsilon \to 0$. This weak interpretation of the boundary condition is unavoidable because we do not know, a priori, that $(\operatorname{curl} v) \wedge \nu$ has a trace in any usual strong sense, because $\operatorname{curl} v$ is simply an element of $(L^2(\Omega))^3$.

Of course, we show below that, as soon as the solution is more regular, we are able to justify this boundary condition in the usual sense.

Remark IV.9.3. In the case where Ω is simply a Lipschitz domain, not necessarily simply connected, we can prove the existence of a unique solution v belonging to the space $H_{\mathrm{div},\mathrm{curl},\nu,0}(\Omega)$ defined in (IV.15). Indeed, the regularity and the simple connexity of Ω is only needed in the following proof to affirm that $H_{\mathrm{div},\mathrm{curl},\nu,0}(\Omega) = H_\nu$ (see Theorem IV.4.8).

Proof.

We define the following bilinear form on H_ν

$$a : (v,w) \in H_\nu \times H_\nu \mapsto \int_\Omega (\operatorname{curl} v)\cdot(\operatorname{curl} w)\,dx + \int_\Omega (\operatorname{div} v)(\operatorname{div} w)\,dx.$$

It is clear that a is continuous on H_ν. Moreover by using (IV.21) we have

$$\|v\|_{H^1}^2 \le C\left(\|\operatorname{div} v\|_{L^2}^2 + \|\operatorname{curl} v\|_{L^2}^2\right) = Ca(v,v),\ \forall v \in H_\nu,$$

so that a is coercive on H_ν.

Thus, we can use the Lax–Milgram theorem II.2.5 to prove the existence and uniqueness of a $v \in H_\nu$ satisfying

$$a(v,w) = \langle \tilde{f}, w\rangle_{H_\nu',H_\nu} + \langle g, \gamma_0(w)\rangle_{H^{-1/2},H^{1/2}},\ \forall w \in H_\nu, \tag{IV.104}$$

where $\tilde{f}$ is the extension of f defined in Proposition IV.9.3. Let us now prove that this function v actually solves the problem under study.

- Taking $w \in (\mathcal{D}(\Omega))^3$, integrating by parts in the distribution sense, and using (A.9) we first deduce that v solves the equation $-\Delta v = f$.
- The boundary condition $v\cdot\nu = 0$ is satisfied by definition of the space H_ν.
- We have already proved that the following equation holds, in the sense of $(H^{-1}(\Omega))^3$,

$$\operatorname{curl}\operatorname{curl} v - \nabla \operatorname{div} v = f. \tag{IV.105}$$

Since $\operatorname{div} v \in L^2(\Omega)$, Lemma IV.9.4 implies that $\nabla \operatorname{div} v \in H_\nu^{-1}$. Let $w \in H_\nu$, and choose $\beta_\varepsilon(\delta)w$ as a test function in (IV.105). It follows that

$$\int_\Omega (\operatorname{curl} v)\cdot(\operatorname{curl}(\beta_\varepsilon(\delta)w))\,dx = \langle f + \nabla(\operatorname{div} v), \beta_\varepsilon(\delta)w\rangle_{H^{-1},H_0^1}. \tag{IV.106}$$

Let us study the limit when $\varepsilon \to 0$ of the various terms in this equation.

– By definition of $\tilde{f}$ and by Lemma IV.9.4, we know that the right-hand side term has a limit as $\varepsilon \to 0$ which is

$$\langle \tilde{f} + \widetilde{\nabla(\operatorname{div} v)}, w\rangle_{H_\nu',H_\nu} = \langle \tilde{f}, w\rangle_{H_\nu',H_\nu} - \int_\Omega (\operatorname{div} v)(\operatorname{div} w)\,dx.$$

– It remains to study the term

$$\int_\Omega (\operatorname{curl} v) \cdot (\operatorname{curl}(\beta_\varepsilon(\delta) w))\, dx = \int_\Omega \beta_\varepsilon(\delta)(\operatorname{curl} v) \cdot (\operatorname{curl} w)\, dx + \frac{1}{\varepsilon}\int_{\mathcal{O}_\varepsilon} (\operatorname{curl} v) \cdot ((\nabla\delta) \wedge (\operatorname{curl} w))\, dx.$$

We can easily pass to the limit in the first term by using the dominated convergence theorem. By definition of β_ε, we can write the second term as

$$T_\varepsilon(w) = -\frac{1}{\varepsilon}\int_{\mathcal{O}_\varepsilon} ((\operatorname{curl} v) \wedge (\nu \circ P_0) \cdot w\, dx,$$

where we used (III.66) to express the gradient of δ as a function of the outward normal field ν and the projection P_0 on $\partial\Omega$.

We have seen that the terms in the right-hand side of (IV.106) have a limit; we deduce that the term $T_\varepsilon(w)$ also has a limit and that

$$\lim_{\varepsilon\to 0} T_\varepsilon(w) = \langle \tilde{f}, w\rangle_{H'_\nu, H_\nu} - \int_\Omega (\operatorname{curl} v) \cdot (\operatorname{curl} w)\, dx - \int_\Omega (\operatorname{div} v)(\operatorname{div} w)\, dx.$$

Comparing with the variational formulation (IV.104), we conclude that

$$\lim_{\varepsilon\to 0} T_\varepsilon(w) = -\langle g, \gamma_0(w)\rangle_{H^{-1/2}, H^{1/2}}, \tag{IV.107}$$

for any $w \in H_\nu$. Moreover, if w is any element in $(H^1(\Omega))^3$, we can introduce $\widetilde{w_T} = w - (w \cdot (\nu \circ P_0))(\nu \circ P_0) \in H_\nu$ and apply (IV.107) to $\widetilde{w_T}$. Since $(\nu \circ P_0) \cdot ((\operatorname{curl} v) \wedge (\nu \circ P_0)) = 0$ almost everywhere and using (IV.102), we finally obtain that (IV.107) holds for any $w \in (H^1(\Omega))^3$.

– Let us now write

$$T_\varepsilon(w) = \underbrace{-\frac{1}{\varepsilon}\int_{\mathcal{O}_\varepsilon} [(\operatorname{curl} v) \wedge (\nu \circ P_0)] \cdot \gamma_0(w) \circ P_0\, dx}_{=T_{\varepsilon,1}(w)} \underbrace{-\frac{1}{\varepsilon}\int_{\mathcal{O}_\varepsilon} [(\operatorname{curl} v) \wedge (\nu \circ P_0)] \cdot \big(w - \gamma_0(w) \circ P_0\big)\, dx}_{=T_{\varepsilon,2}(w)}.$$

By using the generalised Hardy inequality (Proposition III.3.13 and Remark III.3.5), and the same argument as above, we obtain that the term $T_{\varepsilon,2}(w)$ tends to zero for any $w \in (H^1(\Omega))^3$.

With the change of variables formula stated in Remark III.3.4, the term $T_{\varepsilon,1}(w)$ can be written as

$$T_{\varepsilon,1}(w) = \int_{\partial\Omega} \left[\left(-\frac{1}{\varepsilon} \int_0^{\varepsilon} (\operatorname{curl} v)(\sigma - s\nu(\sigma))\, \widetilde{\mathcal{J}}_s(\sigma) ds \right) \wedge \nu(\sigma) \right] \cdot \gamma_0(w)(\sigma)\, d\sigma.$$

Finally, since $\delta \in \mathcal{C}^{1,1}$, we have the estimate $|\widetilde{\mathcal{J}}_s(\sigma) - 1| = |\widetilde{\mathcal{J}}_s(\sigma) - \widetilde{\mathcal{J}}_0(\sigma)| \leq C|s|$. It follows that

$$\lim_{\varepsilon \to 0} T_{\varepsilon,1}(w) = \lim_{\varepsilon \to 0} \int_{\partial\Omega} \left[\left(-\frac{1}{\varepsilon} \int_0^{\varepsilon} (\operatorname{curl} v)(\sigma - s\nu(\sigma))\, ds \right) \wedge \nu(\sigma) \right] \cdot \gamma_0(w)(\sigma)\, d\sigma.$$

The limit in the right-hand side only depend on the trace of w. Therefore, gathering all the results we have obtained with (IV.107), we get

$$\int_{\partial\Omega} \left[\left(-\frac{1}{\varepsilon} \int_0^{\varepsilon} (\operatorname{curl} v)(\sigma - s\nu(\sigma))\, ds \right) \wedge \nu(\sigma) \right] \cdot \varphi(\sigma)\, d\sigma \xrightarrow[\varepsilon \to 0]{} -\langle g, \varphi \rangle_{H^{-1/2}, H^{1/2}}, \ \forall \varphi \in (H^{1/2}(\partial\Omega))^3,$$

which is exactly the property (IV.103).

We can now prove the following elliptic regularity result for the problem under study.

Theorem IV.9.6. *We use the same notation as in the previous theorem. If Ω is of class $\mathcal{C}^{2,1}$, $f \in (L^2(\Omega))^3$, and $g \in (H^{1/2}(\partial\Omega))^3$, then the solution v of* (IV.101) *belongs to $(H^2(\Omega))^3$ and we have*

$$\|v\|_{H^2} \leq C(\|f\|_{L^2} + \|g\|_{H^{1/2}}),$$

where $C > 0$ depends only on Ω.

Moreover, the boundary condition on $\operatorname{curl} v$ is satisfied in a strong sense

$$\gamma_0(\operatorname{curl} v) \wedge \nu = g, \text{ on } \partial\Omega.$$

Proof.

Observe first that $(L^2(\Omega))^3 \subset H_\nu^{-1}$, so that the previous theorem applies. Second, it is clear that $v \in (H^2_{loc}(\Omega))^3$ because local regularity does not depend on the boundary conditions and therefore we can apply to each component v_i of v the local regularity result for the scalar Laplace problem (see the proof of Theorem III.4.2). The main new difficulty is then to prove regularity up to the boundary.

- Let us establish that $v \in (H^2_{tang}(\Omega))^3$. We first extend the formulation (IV.104) to the entire space $(H^1(\Omega))^3$. To this end, for any $w \in (H^1(\Omega))^3$, and using Theorem III.4.3, we define $\Phi[w]$ to be the unique function in $H^2(\Omega)$ such that

$$\begin{cases} -\Delta(\Phi[w]) = \dfrac{1}{|\Omega|}\left(\displaystyle\int_{\partial\Omega} \gamma_0(w)\cdot\nu\, d\sigma\right), & \text{in } \Omega, \\ -\dfrac{\partial \Phi[w]}{\partial \nu} = \gamma_0(w)\cdot\nu, & \text{on } \partial\Omega, \\ \displaystyle\int_\Omega \Phi[w]\, dx = 0. \end{cases}$$

Moreover we have the estimate

$$\|\nabla\Phi[w]\|_{L^2} \leq C\|\gamma_0(w)\cdot\nu\|_{H^{-1/2}(\partial\Omega)}. \tag{IV.108}$$

Notice that $\nabla\Phi[w] = 0$ for any $w \in H_\nu$. Using $w - \nabla\Phi[w]$ in (IV.104), we find that v solves

$$\begin{aligned} &\int_\Omega (\operatorname{curl} v)\cdot(\operatorname{curl} w)\, dx + \int_\Omega (\operatorname{div} v)(\operatorname{div} w)\, dx \\ &= \int_\Omega f\cdot w\, dx - \int_\Omega f\cdot\nabla\Phi[w]\, dx + \int_{\partial\Omega} g\cdot\gamma_0(w)\, d\sigma,\ \forall w \in (H^1(\Omega))^3. \end{aligned} \tag{IV.109}$$

We used here the fact that g is tangent to the boundary (assumption (IV.102)).
Let θ be a vector field satisfying (III.77). We use once more the notation of Section 3.5.2 of Chapter III. We take now $w = \delta^\theta_{-h}\delta^\theta_h v$ in (IV.109) and, by definition of the commutators, we find

$$\begin{aligned} &\int_\Omega (\operatorname{curl} v)\cdot\delta^\theta_{-h}(\operatorname{curl}\delta^\theta_h v)\, dx + \int_\Omega (\operatorname{div} v)\delta^\theta_{-h}(\operatorname{div}\delta^\theta_h v)\, dx \\ &= -\underbrace{\int_\Omega (\operatorname{curl} v)\cdot\left[\operatorname{curl}, \tau^\theta_{-h}\right]\delta^\theta_h v\, dx}_{=T_1} - \underbrace{\int_\Omega (\operatorname{div} v)\left[\operatorname{div}, \tau^\theta_{-h}\right]\delta^\theta_h v\, dx}_{=T_2} \\ &\quad + \underbrace{\int_\Omega f\cdot\delta^\theta_{-h}\delta^\theta_h v\, dx}_{=T_3} - \underbrace{\int_\Omega f\cdot\nabla\Phi[\delta^\theta_{-h}\delta^\theta_h v]\, dx}_{=T_4} \\ &\quad + \underbrace{\int_{\partial\Omega} \delta^\theta_h g\cdot\gamma_0(\delta^\theta_h v)\, d\sigma}_{=T_5} - \underbrace{\int_{\partial\Omega} \{g, \gamma_0(\delta^\theta_h v)\}_h\, d\sigma}_{=T_6}. \end{aligned}$$

It follows

$$
\begin{aligned}
&\int_\Omega \delta_h^\theta(\operatorname{curl} v)\cdot(\operatorname{curl}\delta_h^\theta v)\,dx + \int_\Omega \delta_h^\theta(\operatorname{div} v)(\operatorname{div}\delta_h^\theta v)\,dx\\
=&-T_1-T_2-T_3-T_4+T_5-T_6\\
&+\underbrace{\int_\Omega \{\operatorname{curl} v,\operatorname{curl}\delta_h^\theta v\}_h\,dx}_{=T_7}+\underbrace{\int_\Omega \{\operatorname{div} v,\operatorname{div}\delta_h^\theta v\}_h\,dx}_{=T_8}.
\end{aligned}
$$

Finally using once more the definition of the commutators, we obtain

$$
\begin{aligned}
&\int_\Omega |\operatorname{curl}\delta_h^\theta v|^2\,dx + \int_\Omega |\operatorname{div}\delta_h^\theta v|^2\,dx\\
=&-T_1-T_2-T_3-T_4+T_5-T_6+T_7+T_8\\
&+\underbrace{\int_\Omega \big(\big[\operatorname{curl},\tau_h^\theta\big]\,v\big)\cdot(\operatorname{curl}\delta_h^\theta v)\,dx}_{=T_9}+\underbrace{\int_\Omega \big(\big[\operatorname{div},\tau_h^\theta\big]\,v\big)(\operatorname{div}\delta_h^\theta v)\,dx}_{=T_{10}}.
\end{aligned}
$$

We need now to estimate the terms $T_1,...,T_{10}$. Using Proposition III.3.19 and Lemma IV.7.8 we find

$$
\begin{aligned}
|T_1| &\leq C|h|\|\operatorname{curl} v\|_{L^2}\|\delta_h^\theta v\|_{L^2}\leq C|h|^2\|v\|_{H^1}^2,\\
|T_2| &\leq C|h|\|\operatorname{div} v\|_{L^2}\|\delta_h^\theta v\|_{L^2}\leq C|h|^2\|v\|_{H^1}^2,\\
|T_3| &\leq C|h|\|f\|_{L^2}\|\delta_h^\theta v\|_{H^1},\\
|T_5| &\leq \|\delta_h^\theta g\|_{H^{-1/2}}\|\gamma_0(\delta_h^\theta w)\|_{H^{1/2}}\leq C|h|\|g\|_{H^{1/2}}\|\delta_h^\theta v\|_{H^1}\\
|T_6| &\leq C|h|\|g\|_{L^2}\|\gamma_0(\delta_h^\theta w)\|_{L^2}\leq C|h|\|g\|_{H^{1/2}}\|\delta_h^\theta v\|_{H^1}\\
|T_7| &\leq C|h|\|\operatorname{curl} v\|_{L^2}\|\operatorname{curl}\delta_h^\theta v\|_{L^2}\leq C|h|\|v\|_{H^1}\|\delta_h^\theta v\|_{H^1},\\
|T_8| &\leq C|h|\|v\|_{H^1}\|\delta_h^\theta v\|_{H^1},\\
|T_9| &\leq C|h|\|v\|_{L^2}\|\operatorname{curl}\delta_h^\theta v\|_{L^2}\leq C|h|\|v\|_{L^2}\|\delta_h^\theta v\|_{H^1},\\
|T_{10}| &\leq C|h|\|v\|_{L^2}\|\delta_h^\theta v\|_{H^1}.
\end{aligned}
$$

It remains to estimate the term T_4. To this end we use the following lemma.

Lemma IV.9.7. *For any $v\in H_\nu$, we have*

$$
\begin{aligned}
&\|\gamma_0(\delta_{-h}^\theta\delta_h^\theta v)\cdot\nu\|_{H^{-1/2}}\\
\leq& Ch^2\|\nu\|_{W^{2,\infty}}\|v\|_{L^2}+C|h|\|\nu\|_{W^{1,\infty}}\big(\|\delta_{-h}^\theta v\|_{H^1}+\|\delta_h^\theta v\|_{H^1}\big).
\end{aligned}
$$

We postpone the proof of the lemma for the moment. Using this estimate and (IV.108), we find that

$$\begin{aligned}|T_4| \le \|f\|_{L^2}\|\nabla\Phi[\delta^\theta_{-h}\delta^\theta_h v]\|_{L^2} &\le C\|f\|_{L^2}\|\gamma_0(\delta^\theta_{-h}\delta^\theta_h v)\cdot\nu\|_{H^{-1/2}}\\ &\le C\|f\|_{L^2}\big(h^2\|v\|_{L^2} + |h|\|\delta^\theta_{-h}v\|_{H^1} + |h|\|\delta^\theta_h v\|_{H^1}\big).\end{aligned}$$

Finally, we have already seen in (IV.70) that, for any $v \in H_\nu$, we have

$$\|\gamma_0(\delta^\theta_h v)\cdot\nu\|_{H^{1/2}} \le C|h|\|v\|_{H^1}.$$

Gathering all the estimates above and using (IV.21), we deduce that

$$\begin{aligned}\|\delta^\theta_h v\|^2_{H^1} \le & Ch^2(\|v\|^2_{H^1} + \|f\|^2_{L^2})\\ &+ |h|(\|f\|_{L^2} + \|v\|_{H^1} + \|g\|_{H^{1/2}})(\|\delta^\theta_h v\|_{H^1} + \|\delta^\theta_{-h}v\|_{H^1}).\end{aligned}$$

We add this inequality with the same one where h is replaced by $-h$ and we use the Young inequality to deduce that

$$\|\delta^\theta_h v\|^2_{H^1} + \|\delta^\theta_{-h}v\|^2_{H^1} \le Ch^2(\|v\|^2_{H^1} + \|f\|^2_{L^2}).$$

This exactly proves that $v \in (H^2_{tang}(\Omega))^3$.

- Each component v_i of v for $i \in \{1,\ldots,3\}$, belongs to $H^2_{tang}(\Omega)$ and satisfies $-\Delta v_i = f_i \in L^2(\Omega)$. Therefore, we can now deduce the complete regularity up to the boundary exactly as we did at the end of the proof of Theorem III.4.2.
- Let us now prove that the boundary condition for $\operatorname{curl} v$ is satisfied in a stronger sense. We first observe that, since $v \in (H^2(\Omega))^3$, we know that $\operatorname{curl} v \in (H^1(\Omega))^3$ and therefore we can define its trace $\gamma_0(\operatorname{curl} v) \in (H^{1/2}(\partial\Omega))^3$. The idea is to use the fact that we have

$$\frac{1}{\varepsilon}\int_0^\varepsilon (\operatorname{curl} v)(. - \nu(.))\,ds \xrightarrow[\varepsilon\to 0]{} \gamma_0(\operatorname{curl} v), \text{ in } (L^2(\partial\Omega))^3.$$

More precisely, for any $\sigma \in \partial\Omega$, we have

$$\begin{aligned}&\left(\frac{1}{\varepsilon}\int_0^\varepsilon (\operatorname{curl} v)(\sigma - \nu(\sigma))\,ds\right) - \gamma_0(\operatorname{curl} v)(\sigma)\\ =&\frac{1}{\varepsilon}\int_0^\varepsilon \big[(\operatorname{curl} v)(\sigma - \nu(\sigma)) - \gamma_0(\operatorname{curl} v)(\sigma)\big]\,ds,\end{aligned}$$

so that, by taking the L^2 norm with respect to σ and using the Jensen inequality, we find

$$\begin{aligned}
&\int_{\partial\Omega} \left| \frac{1}{\varepsilon} \int_0^{\varepsilon} (\operatorname{curl} v)(\sigma - \nu(\sigma))\, ds - \gamma_0(\operatorname{curl} v)(\sigma) \right|^2 \, d\sigma \\
\leq & \frac{1}{\varepsilon} \int_{\partial\Omega} \int_0^{\varepsilon} \left| (\operatorname{curl} v)(\sigma - \nu(\sigma)) - \gamma_0(\operatorname{curl} v)(\sigma) \right|^2 \, ds \, d\sigma \\
\leq & \frac{1}{\varepsilon} \int_{\mathcal{O}_\varepsilon} |\operatorname{curl} v - \gamma_0(\operatorname{curl} v) \circ P_0|^2 \, dx \\
\leq & \varepsilon \int_{\mathcal{O}_\varepsilon} \frac{|\operatorname{curl} v - \gamma_0(\operatorname{curl} v) \circ P_0|^2}{\delta^2} \, dx.
\end{aligned}$$

Using the generalised Hardy inequality (Proposition III.3.13 and Remark III.3.5) we get that this term tends to 0. The claim is proved.

□

It remains to prove the lemma.

Proof (of Lemma IV.9.7).

We consider ν as a vector field defined on the whole domain Ω (at least in a suitable neighborhood of the boundary) and we compute

$$\delta^\theta_{-h}\delta^\theta_h(v\cdot\nu) = (\delta^\theta_{-h}\delta^\theta_h v)\cdot\nu - (\delta^\theta_h v)\cdot(\delta^\theta_h\nu) - (\delta^\theta_{-h} v)\cdot(\delta^\theta_{-h}\nu) + v\cdot(\delta^\theta_{-h}\delta^\theta_h\nu).$$

Since $v\cdot\nu$ vanishes on the boundary, in the trace sense, we observe that $\delta^\theta_{-h}\delta^\theta_h(v\cdot\nu)$ also vanishes on the boundary. As a consequence, we have

$$\gamma_0((\delta^\theta_{-h}\delta^\theta_h v)\cdot\nu) = \gamma_0((\delta^\theta_h v)\cdot(\delta^\theta_h\nu)) + \gamma_0((\delta^\theta_{-h} v)\cdot(\delta^\theta_{-h}\nu)) - \gamma_0(v\cdot(\delta^\theta_{-h}\delta^\theta_h\nu)), \tag{IV.110}$$

and we need to estimate the $H^{-1/2}$ norm of these three terms. We deal with the first term by computing its $H^{1/2}$-norm, which is stronger than required. Indeed, we have

$$\begin{aligned}
\|\gamma_0((\delta^\theta_h v)\cdot(\delta^\theta_h\nu))\|_{H^{1/2}(\partial\Omega)} &\leq C\|(\delta^\theta_h v)\cdot(\delta^\theta_h\nu)\|_{H^1} \\
&\leq \|\delta^\theta_h v\|_{H^1}\|\delta^\theta_h\nu\|_{L^\infty} + \|\delta^\theta_h v\|_{L^2}\|\nabla\delta^\theta_h\nu\|_{L^\infty} \\
&\leq C|h|\|\nu\|_{W^{1,\infty}}\|\delta^\theta_h v\|_{H^1} + h^2\|v\|_{H^1}\|\nu\|_{W^{2,\infty}}.
\end{aligned}$$

We estimate now the last term in (IV.110) in the L^2-norm (which is also stronger than required)

$$\|\gamma_0(v\cdot(\delta^\theta_{-h}\delta^\theta_h\nu))\|_{L^2(\partial\Omega)} \leq \|\gamma_0(v)\|_{L^2}\|\delta^\theta_{-h}\delta^\theta_h\nu\|_{L^\infty} \leq C|h|^2\|v\|_{H^1}\|\nu\|_{W^{2,\infty}}.$$

The proof of the lemma is complete.

□

9.3 The Stokes problem

We come now to the study of the Stokes problem (IV.98).

Theorem IV.9.8. *Let Ω be a bounded, simply connected domain of $\mathbb{R}^3$, of class $\mathcal{C}^{1,1}$. For any $f \in H_\nu^{-1}$, and any $g \in (H^{-1/2}(\partial\Omega))^3$ satisfying (IV.102), there exists a unique solution $(v, p) \in H_\nu \times L_0^2(\Omega)$ to the problem (IV.98).*

Proof.

We introduce the Hilbert space

$$V_\nu = \{v \in H_\nu, \quad \operatorname{div} v = 0\},$$

and the bilinear form

$$a : (v, w) \in V_\nu \times V_\nu \to a(v, w) = \int_\Omega (\operatorname{curl} v) \cdot (\operatorname{curl} w)\, dx.$$

It is clear that a is continuous. Moreover, by using Theorem IV.4.8 we can see that a is coercive on V_ν. The Lax–Milgram theorem proves that there exists a unique solution $v \in V_\nu$ of

$$a(v, w) = \langle \tilde{f}, w\rangle_{H_\nu', H_\nu} + \langle g, \gamma_0(w)\rangle_{H^{-1/2}, H^{1/2}}, \ \forall w \in V_\nu. \tag{IV.111}$$

Since $\operatorname{div} v = 0$ and $-\Delta v = \operatorname{curl}\operatorname{curl} v - \nabla \operatorname{div} v$, we deduce that

$$\langle -\Delta v - f, w\rangle_{H^{-1}, H_0^1} = 0, \ \forall \varphi \in (H_0^1(\Omega))^3, \operatorname{div} \varphi = 0.$$

Using de Rham's theorem (Theorem IV.2.4), we obtain that there exists a unique pressure $p \in L_0^2(\Omega)$ such that

$$-\Delta v + \nabla p = f.$$

By Lemma IV.9.4, we know that $f - \nabla p \in H_\nu^{-1}$. Therefore, by Theorem IV.9.5, there exists a unique solution $\tilde{v} \in H_\nu$ of the vector Laplace problem $-\Delta \tilde{v} = f - \nabla p$, $\tilde{v} \cdot \nu = 0$ and $(\operatorname{curl} \tilde{v}) \wedge \nu = g$. Let us show that $\tilde{v} = v$. This will immediately prove that v satisfies the claimed boundary condition.

- By definition, $\tilde{v}$ satisfies

$$\begin{aligned} &\int_\Omega (\operatorname{curl} \tilde{v}) \cdot (\operatorname{curl} w)\, dx + \int_\Omega (\operatorname{div} \tilde{v})(\operatorname{div} w)\, dx \\ =&\langle \tilde{f} - \widetilde{\nabla p}, w\rangle_{H_\nu', H_\nu} + \langle g, \gamma_0(w)\rangle_{H^{-1/2}, H^{1/2}}, \ \forall w \in H_\nu. \end{aligned} \tag{IV.112}$$

By Theorem IV.4.12, we know that there exists $w \in H_\nu$ such that

$$\operatorname{div} w = 0, \operatorname{curl} w = \operatorname{curl}(v - \tilde{v}).$$

Using this particular test function w in (IV.112) gives

$$\int_\Omega (\operatorname{curl} \tilde{v}) \cdot (\operatorname{curl}(v - \tilde{v}))\, dx = \langle \tilde{f}, w \rangle_{H'_\nu, H_\nu} + \langle g, \gamma_0(w) \rangle_{H^{-1/2}, H^{1/2}},$$

because the contribution of $\widetilde{\nabla p}$ vanishes on divergence-free test functions. Moreover, by (IV.111) we have

$$\int_\Omega (\operatorname{curl} v) \cdot (\operatorname{curl}(v - \tilde{v}))\, dx = \langle \tilde{f}, w \rangle_{H'_\nu, H_\nu} + \langle g, \gamma_0(w) \rangle_{H^{-1/2}, H^{1/2}}.$$

Subtracting the two previous equalities gives that

$$\int_\Omega |\operatorname{curl}(v - \tilde{v})|^2\, dx = 0;$$

that is $\operatorname{curl} v = \operatorname{curl} \tilde{v}$.

- We have now the following two equations which are satisfied in the distribution sense

$$\operatorname{curl}\operatorname{curl} v - \nabla \underbrace{\operatorname{div} v}_{=0} = -\Delta v = f - \nabla p,$$

$$\operatorname{curl}\operatorname{curl} \tilde{v} - \nabla \operatorname{div} \tilde{v} = -\Delta \tilde{v} = f - \nabla p.$$

Inasmuch as we proved above that $\operatorname{curl} v = \operatorname{curl} \tilde{v}$, we find that

$$\nabla(\operatorname{div} \tilde{v}) = 0,$$

in the distribution sense, which implies that $\operatorname{div} \tilde{v}$ is a constant on Ω. The divergence theorem gives that

$$\int_\Omega (\operatorname{div} \tilde{v})\, dx = \int_{\partial\Omega} \tilde{v} \cdot \nu\, d\sigma = 0,$$

by definition of H_ν, and therefore we have proved that $\operatorname{div} \tilde{v} = 0$.

- In conclusion, we have shown that

$$\operatorname{curl} v = \operatorname{curl} \tilde{v},\ \operatorname{div} v = \operatorname{div} \tilde{v} = 0,\ v \cdot \nu = \tilde{v} \cdot \nu = 0,$$

so that $v = \tilde{v}$ (see Lemma IV.4.6 for instance).
Therefore, $\operatorname{curl} v = \operatorname{curl} \tilde{v}$ satisfies the suitable boundary condition in the weak sense given in (IV.103).

□

We have shown in the proof of the previous theorem that, once the pressure is known, the velocity field v can be computed by simply solving a vector Laplace problem.

We show now that it is actually possible to separate the computation of the pressure from the one of the velocity field for this particular choice of the boundary conditions for the Stokes equations.

Proposition IV.9.9. *We use the same notation as in Theorem IV.9.8.*

The pressure p which is a solution to Problem (IV.98) can be obtained by computing $p = -\operatorname{div} u$, where $u \in H_\nu$ is the unique solution to the vector Laplace problem

$$\begin{cases} -\Delta u = f, & \text{in } \Omega, \\ u \cdot \nu = 0, & \text{on } \partial\Omega, \\ (\operatorname{curl} u) \wedge \nu = g, & \text{on } \partial\Omega. \end{cases} \tag{IV.113}$$

Proof.

We denote by $(v, p) \in H_\nu \times L^2_0(\Omega)$ the unique solution of (IV.98). By Theorem IV.9.5, Problem (IV.113) is known to have a unique solution $u \in H_\nu$. We set $q = -\operatorname{div} u \in L^2_0(\Omega)$ and we want to show that $q = p$.

By definition of u, we have

$$\begin{aligned} &\int_\Omega (\operatorname{curl} u) \cdot (\operatorname{curl} w)\, dx + \int_\Omega (\operatorname{div} u)(\operatorname{div} w)\, dx \\ =&\langle \tilde{f}, w\rangle_{H'_\nu, H_\nu} + \langle g, \gamma_0(w)\rangle_{H^{-1/2}, H^{1/2}},\ \forall w \in H_\nu. \end{aligned}$$

By Lemma IV.9.4, this can be written

$$\int_\Omega (\operatorname{curl} u) \cdot (\operatorname{curl} w)\, dx = \langle \tilde{f} - \widetilde{\nabla q}, w\rangle_{H'_\nu, H_\nu} + \langle g, \gamma_0(w)\rangle_{H^{-1/2}, H^{1/2}},\ \forall w \in H_\nu.$$

Subtracting this equation from the variational formulation satisfied by (v, p) gives

$$\begin{aligned} \int_\Omega (\operatorname{curl}(u - v)) \cdot (\operatorname{curl} w)\, dx &= -\langle \widetilde{\nabla(q - p)}, w\rangle_{H'_\nu, H_\nu} \\ &= \int_\Omega (p - q)(\operatorname{div} w)\, dx,\ \forall w \in H_\nu. \end{aligned} \tag{IV.114}$$

We consider now the following scalar Laplace problem

$$\begin{cases} -\Delta\psi = p - q, & \text{in } \Omega, \\ \dfrac{\partial\psi}{\partial\nu} = 0, & \text{on } \partial\Omega. \end{cases}$$

Since $p - q \in L^2_0(\Omega)$, the above problem has a solution $\psi \in H^2(\Omega)$ (we use here that Ω is of class $\mathcal{C}^{1,1}$) which is unique apart from a constant. Taking $w = -\nabla\psi \in H_\nu$ in (IV.114), we obtain

$$\int_\Omega |p - q|^2\, dx = 0,$$

and the claim is proved.

□

Remark IV.9.4. This result is particularly interesting from a numerical point of view because the resolution of the Stokes problem (IV.98) is then reduced to the resolution of two Laplace problems. Applying some Galerkin approximation (through the finite-element method for instance) to these two problems leads to definite-positive symmetric matrices.

This is very comfortable compared to the numerical approximation of the Stokes–Dirichlet or Stokes–Neumann problem for which the computation of v and p cannot be decoupled. After discretisation, this leads in general to more complex nonsymmetric matrices for which more specific solvers have to be used.

Remark IV.9.5. In the proof above, we have shown that solving the Stokes problem (IV.98) can be reduced to solving a first vector Laplace problem with source term f (this leads us to compute the pressure p) and then to solve a second vector Laplace problem with source term $f - \nabla p$ (this leads us to compute the velocity field v).

In fact the second step can be reduced to a scalar Laplace problem. Indeed, we know that $p = q$, thus Equation (IV.114) with $w = u - v$ leads to the additional property that

$$\operatorname{curl}(u - v) = 0.$$

By Lemma IV.4.10, we deduce that there exists a $q = \pi \in H^1(\Omega)$ such that $u - v = \nabla\pi$. Since $\operatorname{div} v = 0$ and $\operatorname{div} u = -p$, and since $u \cdot \nu = v \cdot \nu = 0$ on the boundary, we deduce that π solves the problem

$$\begin{cases} -\Delta\pi = p, & \text{in } \Omega, \\ \dfrac{\partial\pi}{\partial\nu} = 0, & \text{on } \partial\Omega. \end{cases} \tag{IV.115}$$

Therefore, once u is known, it is enough to solve Problem (IV.115) to obtain π and thus the velocity field $v = u - \nabla\pi$.

With the help of the previous proposition and remarks, we can now obtain a regularity theorem for Problem (IV.98) by simply using the corresponding regularity result for the vector Laplace problem that we studied in the previous section.

Theorem IV.9.10. *We use the same notation as in Theorem IV.9.8. If we additionally assume that $f \in (L^2(\Omega))^3$, $g \in (H^{1/2}(\partial\Omega))^2$, Ω is of class $\mathcal{C}^{2,1}$, then we have $(v, p) \in (H^2(\Omega))^3 \times H^1(\Omega)$ and the boundary condition $(\operatorname{curl} v) \wedge \nu = g$ is satisfied in the strong sense.*

Proof.

By using Theorem IV.9.5, we know that the solution u to (IV.113) belongs to $(H^2(\Omega))^3$. As a consequence, the pressure $p = -\operatorname{div} u$ belongs to $H^1(\Omega)$.

It follows that $f - \nabla p$ lies in $(L^2(\Omega))^3$ and we can once more apply Theorem IV.9.5 to deduce that the velocity field v belongs to $(H^2(\Omega))^3$. The claim is proved.

□

Chapter V
Navier–Stokes equations for homogeneous fluids

The main matter of this chapter concerns existence, uniqueness, and regularity results in a bounded domain of $\mathbb{R}^d$, $d = 2, 3$ for solutions of the incompressible homogeneous Navier–Stokes equations. The study of steady solutions and their stability properties is also investigated.

Many interesting topics have been purposely ignored, in particular the study of the Navier–Stokes equations in unbounded domains (particularly over all $\mathbb{R}^d$); see for instance [63] and the references therein.

The first section is dedicated to so-called *weak* solutions which are defined for all time on any bounded domain of $\mathbb{R}^3$. More precisely, it is mostly devoted to the proof of the theorem by Leray [79, 80] who, in 1934, first established the existence of weak solutions for the Navier–Stokes equations, which he called *turbulent solutions* (see also the contributions of Hopf in [71, 72]).

The second section is dedicated to the existence and uniqueness of more regular solutions (known as *strong solutions*) in both dimension 2 and dimension 3 (on a finite time interval in the latter case). Moreover, we have given a detailed proof of the parabolic regularity properties of these equations.

The techniques used are classic and the first two sections are largely inspired by the books of Ladyzhenskaya [76], Lions [84], and Temam [122], see also [62] for a more complete review on this topic.

The last section of this chapter concerns the steady Navier–Stokes equations. We study existence and uniqueness of solutions in the case of homogeneous and nonhomogeneous Dirichlet boundary conditions and we discuss some of their stability properties.

Of course, we cannot start this chapter without mentioning that, in the three-dimensional case, the question of the global existence (i.e., for all time $t > 0$) of regular solutions as well as the uniqueness of weak solutions, are yet unsolved problems inspiring a large amount of research.

Throughout this chapter, we systematically use the concepts and notations introduced in Chapter IV, particularly regarding the spaces $\mathscr{V}, V, H$ and the Stokes operator A (see Section 5.2 of Chapter IV).

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1 Leray's theorem

Let Ω be a connected Lipschitz bounded domain of $\mathbb{R}^d$ (henceforth, we always assume that $d = 2$ or $d = 3$), let the Reynolds number be fixed $\mathcal{R}e > 0$, let f be given in $L^2_{loc}([0, +\infty[,\ (H^{-1}(\Omega))^d)$, and v_0 be given in H. As we saw in Chapter I, the flow of an incompressible homogeneous fluid in a domain Ω submitted to a body force field f is well described by the Navier–Stokes equations. We are therefore interested in the following mathematical problem: to find a vector field v and a scalar field p satisfying

$$\begin{cases} \dfrac{\partial v}{\partial t} + (v \cdot \nabla)v - \dfrac{1}{\mathcal{R}e}\Delta v + \nabla p = f, & \text{in } \Omega, \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ v = 0, & \text{on } \partial\Omega, \\ v(0) = v_0. \end{cases} \tag{V.1}$$

To study this system, we need to define a functional space in which to search for the solution and then give a (weak) sense to the various equations of the system above before finally showing that we can find indeed a solution to the weak formulation adopted.

Let us start with several technical results concerning the only nonlinear term that one can find in these equations: the inertia term $(v \cdot \nabla)v$. This is important because this term contains most of the new difficulties compared to the linear Stokes system.

1.1 Properties of the inertia term

For all $u, v, w \in (H^1_0(\Omega))^d$, we define

$$b(u, v, w) = \int_\Omega \big((u \cdot \nabla)v\big) \cdot w \, dx = \sum_{i,j=1}^{d} \int_\Omega u_i \frac{\partial v_j}{\partial x_i} w_j \, dx.$$

Lemma V.1.1. *The trilinear form b is continuous on $(H^1_0(\Omega))^d \times (H^1_0(\Omega))^d \times (H^1_0(\Omega))^d$ and satisfies*

$$\begin{aligned} b(u, v, w) + b(u, w, v) = 0,\ \forall u \in V,\ \forall v, w \in (H^1_0(\Omega))^d, \\ b(u, v, v) = 0,\ \forall u \in V,\ \forall v \in (H^1_0(\Omega))^d. \end{aligned} \tag{V.2}$$

Moreover, for all $u \in V$ and all $v, w \in (H^1_0(\Omega))^d$ we have

$$|b(u, v, w)| \leq C\|u\|_{L^2}^{1-d/4}\|u\|_{H^1}^{d/4}\|v\|_{L^2}^{1-d/4}\|v\|_{H^1}^{d/4}\|w\|_{H^1}, \tag{V.3}$$

In the following, for all $u, v \in V$ we denote as $B(u,v) \in V'$ the continuous linear form on V defined by

$$\langle B(u,v), w\rangle_{V',V} = b(u,v,w). \tag{V.4}$$

The estimate (V.3) shows that the bilinear map B is continuous from $V \times V$ into V' and that we have

$$\|B(u,u)\|_{V'} \leq C\|u\|_{L^2}^{2-d/2}\|u\|_{H^1}^{d/2},\ \forall u \in V. \tag{V.5}$$

Remark V.1.1. In a very simplified manner, we can state that it is the fact that the power on the H^1 norm in this estimate is "less good" (i.e., higher) in the three-dimensional case than in the two-dimensional case, which implies certain difficulties (in particular with regard to the uniqueness of the weak solutions in 3D).

Proof.

The continuity simpli follows from the Sobolev embedding $H^1(\Omega) \subset L^4(\Omega)$, valid in dimensions less than or equal to 4 (see Theorem III.2.34).

The antisymmetry property is proved (for smooth functions, then we conclude by density) using the Stokes formula (the boundary terms are zero because u vanishes on $\partial\Omega$),

$$\begin{aligned} b(u,v,w) &= \sum_{i,j=1}^{d} \int_\Omega u_i \frac{\partial v_j}{\partial x_i} w_j\,dx \\ &= -\sum_{i,j=1}^{d} \int_\Omega \frac{\partial u_i}{\partial x_i} v_j w_j\,dx - \sum_{i,j=1}^{d} \int_\Omega u_i v_j \frac{\partial w_j}{\partial x_i}\,dx \\ &= -\int_\Omega (\operatorname{div} u)(v\cdot w)\,dx - b(u,w,v) = -b(u,w,v), \end{aligned}$$

because the divergence of u vanishes in Ω.

To prove (V.3), we use the Hölder inequality to write

$$|b(u,v,w)| = |b(u,w,v)| \leq \|u\|_{L^4}\|v\|_{L^4}\|w\|_{H^1},$$

and then we use Proposition III.2.35 to obtain the result.

$\square$

1.2 Weak formulations of the Navier–Stokes equations

Among the various equivalent formulations possible for the Navier–Stokes equations, we have chosen to present two which are very similar and which seem to us to be the most convenient to manipulate. The main idea (due to

Leray) consists in taking divergence-free test functions which are zero at the boundary. These have the advantage of making the pressure disappear in the formulations considered.

As we show later (see Section 1.5), once we have resolved one or other of these weak formulations we can recover the pressure by using de Rham's theorem established in Chapter IV.

1.2.1 Time-independent test functions

In this first formulation, we only consider time-independent test functions. More precisely, for the system (V.1) we investigate the following problem: to find $v \in L^2(]0,T[,\, V)$ such that for all $\psi \in V$ we have

$$\begin{aligned}\frac{d}{dt}\int_\Omega v(t)\cdot\psi\,dx + \int_\Omega \big((v(t)\cdot\nabla)v(t)\big)\cdot\psi\,dx \\ + \frac{1}{\mathcal{R}e}\int_\Omega \nabla v(t) : \nabla\psi\,dx = \langle f(t),\psi\rangle_{H^{-1},H_0^1},\end{aligned} \tag{V.6}$$

in the sense of distributions in $\mathcal{D}'(]0,T[)$ and such that $v(0) = v_0$ in V' in the weak sense.

Let us clarify the meaning of this last condition. It is easy to see that if v belongs to $L^2(]0,T[,V)$ and satisfies (V.6) then, for all $\psi \in V$, the real-valued function $F_\psi : t \mapsto F_\psi(t) \equiv (v(t),\psi)_H = \langle v(t),\psi\rangle_{V',V}$ has a derivative in the sense of distributions which lies in $L^1(]0,T[)$. Hence F_ψ belongs to $W^{1,1}(]0,T[)$ which shows that this function is continuous (Corollary II.4.2). In other words, any solution of the above formulation is $\star$-weakly continuous with values in V', which allows us to give a meaning to the initial value $v(0)$.

This first formulation is, in a certain sense, simpler than the one that follows. Unfortunately, in order to establish the energy equation, or even the uniqueness of the solutions, we need to take the solutions themselves as the test functions, or the differences between two solutions. This is only possible in the formulation with time-dependent test functions.

1.2.2 Time-dependent test functions

Associated with system (V.1) we investigate the following variational problem: to find a function $v \in L^2(]0,T[,\, V)$ such that $dv/dt \in L^1(]0,T[,V')$ which, for all test functions φ in $\mathcal{C}_c^0(]0,T[,V)$, satisfies

$$
\begin{aligned}
&\int_0^T \left\langle \frac{dv}{dt}, \varphi(t) \right\rangle_{V',V} dt + \int_0^T \int_\Omega \big((v(t)\cdot\nabla)\, v(t)\big)\cdot\varphi(t)\,dx\,dt \\
&+\frac{1}{\mathcal{R}e}\int_0^T \int_\Omega \nabla v(t) : \nabla\varphi(t)\,dx\,dt = \int_0^T \langle f(t),\varphi(t)\rangle_{H^{-1},H_0^1}\,dt,
\end{aligned}
\tag{V.7}
$$

and moreover such that $v(0) = v_0$ in V'.

We can immediately give a sense to this last condition because the functions $v \in L^2(]0,T[,V)$ such that $dv/dt \in L^1(]0,T[,V')$ are continuous with values in V' for the strong topology (Proposition II.5.11).

We now note that if we also know that the solution v satisfies $\frac{dv}{dt} \in L^p(]0,T[,V')$ for a certain $p \in]1,2]$, then this formulation can be extended, by density, to the test functions $\varphi \in L^{p'}(]0,T[,V)$. This remark is fundamental because, in the case of dimension 2, we show that the solution we obtain satisfies $dv/dt \in L^2(]0,T[,V')$, and that we can therefore extend the formulation to test functions $\varphi \in L^2(]0,T[,V)$. In particular, in the two-dimensional case, we can take the solution v itself as a test function. This is not true in dimension 3 and this is one of the reasons for the particular difficulty of the three-dimensional case.

1.2.3 Equivalence of the formulations

Let us start by proving a lemma on the density of functions of the "tensor products" type.

Lemma V.1.2. *The set $\mathcal{E}$ of functions φ of the form*

$$
\varphi(t,x) = \sum_{k=1}^{N} \eta_k(t)\psi_k(x),
$$

where N is any integer, $\eta_k \in \mathcal{D}(]0,T[)$, and $\psi_k \in V$, is dense in $\mathcal{C}_c^0(]0,T[,V)$.

Proof.

Let $\varphi \in \mathcal{C}_c^0(]0,T[,V)$ and $\varepsilon > 0$. Since φ is continuous on the compact set $[0,T]$, it is uniformly continuous. Therefore, there exists $\delta > 0$ such that for all $s,t \in [0,T]$ satisfying $|s-t| \leq \delta$ we have $\|\varphi(s)-\varphi(t)\|_V \leq \varepsilon$.

Now, let $t_0 = 0,\ldots,t_{N+1} = T$ be a regular division of the interval $[0,T]$ into subintervals of length less than δ and sufficiently fine so that $\varphi = 0$ on $[0,t_1] \cup [t_N, 1]$. The intervals $(]t_{k-1},t_{k+1}[)_{1\leq k\leq N}$ form an open covering of $]0,T[$. Therefore, let $(\psi_k)_k$ be a partition of unity of class $\mathcal{C}^\infty$ associated with this covering (Lemma II.2.38). Then, we set

$$
\varphi_N(t) = \sum_{k=1}^{N} \psi_k(t)\varphi(t_k) \in V,
$$

which is indeed a function of the set $\mathcal{E}$.

Now, let $t \in [0,T]$; since $\sum_k \psi_k(t) = 1$ and $\psi_k \geq 0$ we have

$$\|\varphi(t) - \varphi_N(t)\|_V = \left\| \sum_{k=1}^{N} \psi_k(t)(\varphi(t) - \varphi(t_k)) \right\|_V \leq \sum_{k=1}^{N} \psi_k(t) \|\varphi(t) - \varphi(t_k)\|_V.$$

However, $\psi_k(t)$ can only be nonzero if $|t - t_k| < \delta$ and we have the inequality $\|\varphi(t) - \varphi(t_k)\|_V \leq \varepsilon$, by definition of δ. Finally, we have indeed shown that $\|\varphi(t) - \varphi_N(t)\|_V \leq \varepsilon$, for all $t \in [0,T]$.

□

We can now show the following important result which proves that we can work independently with one or other of the preceding formulations.

Proposition V.1.3. *Let $f \in L^1(]0,T[, V')$ and $v \in L^2(]0,T[, V)$. The following assertions are equivalent.*

- *v has a weak derivative dv/dt in $L^1(]0,T[, V')$ and v satisfies (V.7).*
- *v satisfies (V.6).*

Moreover, if v satisfies these two equivalent assertions, then for almost all t we have the following equality in V',

$$\frac{dv}{dt} + \frac{1}{\mathcal{R}e} Av(t) + B(v(t), v(t)) = f(t).$$

Proof.

Let us assume that $dv/dt \in L^1(]0,T[, V')$ and that v satisfies (V.7). Let $\psi \in V$ and be independent of time and let $\eta \in \mathcal{D}(]0,T[)$ be a smooth real-valued function with compact support. We can take the function $\varphi(t,x) = \eta(t)\psi(x)$ as a test function in the formulation (V.7), which gives:

$$\begin{aligned} &\int_0^T \left\langle \frac{dv}{dt}, \eta(t)\psi \right\rangle_{V',V} dt + \int_0^T \eta(t) \int_\Omega (v(t) \cdot \nabla v(t)) \cdot \psi \, dxdt \\ &+ \frac{1}{\mathcal{R}e} \int_0^T \eta(t) \int_\Omega \nabla v(t) : \nabla \psi \, dxdt = \int_0^T \eta(t) \langle f(t), \psi \rangle_{H^{-1}, H_0^1} \, dt. \end{aligned} \tag{V.8}$$

By definition of the weak derivative dv/dt (Definition II.5.7), and because η is regular, the first term can also be written as

$$\begin{aligned} \int_0^T \left\langle \frac{dv}{dt}, \eta(t)\psi \right\rangle_{V',V} dt &= \left\langle \int_0^T \eta(t) \frac{dv}{dt} \, dt, \psi \right\rangle_{V',V} \\ &= -\left\langle \int_0^T \eta'(t) v(t) \, dt, \psi \right\rangle_{V',V} \\ &= -\int_0^T \eta'(t) \langle v(t), \psi \rangle_{V',V} \, dt. \end{aligned}$$

However, as for almost all t, $v(t) \in V$, by identification of H and its dual via the natural scalar product of H, we know that $\langle v(t), \psi \rangle_{V',V} = (v(t), \psi)_H$ for almost all t. We have thus obtained

$$\begin{aligned} &-\int_0^T \eta'(t)(v(t),\psi)_H\,dt + \int_0^T \eta(t)\int_\Omega (v(t)\cdot\nabla v(t))\cdot\psi\,dxdt \\ &+\frac{1}{\mathcal{R}e}\int_0^T \eta(t)\int_\Omega \nabla v(t):\nabla\psi\,dxdt = \int_0^T \eta(t)\langle f(t),\psi\rangle_{H^{-1},H_0^1}\,dt. \end{aligned} \tag{V.9}$$

This is valid for any function $\eta \in \mathcal{D}(]0,T[)$, thus we have indeed shown that (V.6) holds in the sense of distributions. Since the initial condition is satisfied in the strong sense in V', it is clear that it is also satisfied in the weak sense.

Conversely, let us suppose that v satisfies (V.6). We first show that v has a weak derivative with respect to time in the space $L^1(]0,T[,V')$. To do this, we take the preceding calculation in reverse, so that (V.9) is again true. Using the definition of the Stokes operator (IV.35) and that for the bilinear function B (V.4), we can also write

$$\begin{aligned} &\left\langle -\int_0^T \eta'(t)v(t)\,dt,\psi\right\rangle_{V',V} + \left\langle \int_0^T \eta(t)B(v(t),v(t))\,dt,\psi\right\rangle_{V',V} \\ &+\left\langle \int_0^T \eta(t)\frac{1}{\mathcal{R}e}Av(t)\,dt,\psi\right\rangle_{V',V} = \left\langle \int_0^T \eta(t)f(t)\,dt,\psi\right\rangle_{V',V}. \end{aligned}$$

Since this is true for all $\psi \in V$, we can deduce the following equality in V',

$$\begin{aligned} -\int_0^T \eta'(t)v(t)\,dt = &-\int_0^T \eta(t)B(v(t),v(t))\,dt - \frac{1}{\mathcal{R}e}\int_0^T \eta(t)Av(t)\,dt \\ &+\int_0^T \eta(t)f(t)\,dt. \end{aligned}$$

However, since the Stokes operator A is continuous from V to V' and the bilinear function B is continuous from $V\times V$ to V', we have

$$\begin{aligned} \int_0^T \|B(v(t),v(t))\|_{V'}\,dt &\le C\int_0^T \|v(t)\|_V^2\,dt \le C\|v\|^2_{L^2(]0,T[,V)}, \\ \int_0^T \|Av(t)\|_{V'}\,dt &\le C\int_0^T \|v(t)\|_V\,dt \le C\sqrt{T}\|v\|_{L^2(]0,T[,V)}. \end{aligned}$$

Hence, since (V.9) is true for any function $\eta \in \mathcal{D}(]0,T[)$, we have indeed shown that v has a weak derivative with respect to time in $L^1(]0,T[,V')$ and also that, for almost all $t \in]0,T[$, we have

$$\frac{dv}{dt} = -\frac{1}{\mathcal{R}e}Av(t) - B(v(t),v(t)) + f(t).$$

Now that we know that dv/dt exists in $L^1(]0,T[,V')$, we can deduce from (V.9) that Equation (V.8) is satisfied for all $\psi \in V$ and all $\eta \in \mathcal{D}(]0,T[)$ and by density, it is clear that (V.8) remains valid for all $\eta \in \mathcal{C}_c^0(]0,T[)$.

Moreover, by linearity of (V.8) with respect to the test function $\eta(t)\psi$, we notice that the formulation (V.7) is satisfied for any test function belonging to the set $\mathcal{E}$ defined in Lemma V.1.2. This lemma makes it possible to show that (V.7) is valid for any function $\varphi \in \mathcal{C}_c^0(]0,T[,V)$. Indeed, it is easy to see that since $v \in L^2(]0,T[,V)$ and $dv/dt \in L^1(]0,T[,V')$, the linear functional with respect to φ which defines (V.7) is continuous for the topology of $\mathcal{C}_c^0(]0,T[,V)$.

Furthermore, we know from Proposition II.5.11, that v is continuous with values in V' for the strong topology and, by hypothesis, $v(0) = v_0$ in the weak continuity sense with values in V'. We can, indeed, recover the initial data $v(0) = v_0$ in the strong sense in V' because the weak limit is unique.

□

1.3 Existence and uniqueness of weak solutions

1.3.1 Statement of the main result

We can now give the main existence and uniqueness result for (so-called "weak") solutions of the Navier–Stokes equations.

Theorem V.1.4 (Leray). *Let Ω be a connected bounded Lipschitz domain of $\mathbb{R}^d$. Let $\mathcal{R}e > 0$, v_0 be given in H and f in $L^2_{loc}([0,+\infty[,(H^{-1}(\Omega))^d)$; then there exists a pair (v,p) defined on all $\mathbb{R}^+$ which is a solution of (V.1) and such that for all $T > 0$,*

$$(v,p) \in \big(L^\infty(]0,T[,H) \cap L^2(]0,T[,V)\big) \times W^{-1,\infty}(]0,T[,L^2_0(\Omega)),$$

and

$$\frac{dv}{dt} \in L^{4/d}(]0,T[,V_{-1}) \cap L^2(]0,T[,V_{-d/2}).$$

• *For $d = 2$, this solution is unique and v is continuous from $[0,+\infty[$ into H. Moreover, it satisfies the following energy equality for all $t \in \mathbb{R}^+$,*

$$\frac{1}{2}\|v(t)\|^2_{L^2} + \frac{1}{\mathcal{R}e}\int_0^t \|\nabla v(\tau)\|^2_{L^2} d\tau = \frac{1}{2}\|v_0\|^2_{L^2} + \int_0^t \langle f(\tau), v(\tau)\rangle_{H^{-1},H^1_0} d\tau. \tag{V.10}$$

• *For $d = 3$, v is continuous from $[0,+\infty[$ into $V_{-1/4}$ and weakly continuous from $[0,+\infty[$ into H. It satisfies the following energy inequality, for all $t \in \mathbb{R}^+$,*

$$\frac{1}{2}\|v(t)\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\int_0^t \|\nabla v(\tau)\|_{L^2}^2 d\tau \le \frac{1}{2}\|v_0\|_{L^2}^2 + \int_0^t \langle f(\tau), v(\tau)\rangle_{H^{-1},H_0^1} d\tau. \tag{V.11}$$

The uniqueness of such solutions remains an open problem.

The sketch of the proof is the following. We take any $T > 0$ and attempt to solve one of the equivalent formulations (V.6) or (V.7) on $[0, T]$. To achieve this, we introduce a finite-dimensional approximate problem that can be easily resolved using the Cauchy–Lipschitz theorem. Then we prove estimates on the solutions of this approximate problem which are uniform with respect to the approximation parameter. Finally, we use compactness theorems to obtain the strong convergences which allow us to justify the limit in the approximate formulation, particularly in the nonlinear term. This strategy gives the existence of the velocity field v satisfying the proposed formulations.

The uniqueness of such solutions in the two-dimensional case is established in an independent section. Then, we show that we can construct a solution on all $\mathbb{R}^+$, in both dimensions $d = 2$ and $d = 3$.

Then the proof of the energy equality (or inequality) is obtained together with certain regularity properties over time for the solution. We note that the energy inequality (V.11) in the three-dimensional case is only proven for the specific weak solution that we have defined in the proof of the Leray theorem. In particular, if by any chance there exist other weak solutions of the problem, then we have no reason to assume that they will also satisfy this energy inequality.

To finish, since the weak formulation used does not directly provide the pressure p, we need to find it indirectly in order to prove and guarantee that we have indeed solved equation (V.1) in the distribution sense.

1.3.2 Approximate problem

Let $T > 0$ be given. We choose to discretise the weak formulation (V.6) of the Navier–Stokes equations which only involves time-independent test functions. To do this we use the special basis $(w_k)_k$ made of eigenfunctions of the Stokes operator introduced in Definition IV.5.6. Let H_N be the finite-dimensional vector space generated by the functions $(w_k)_{k\le N}$. We introduce a regularisation with respect to time of the source term f, denoted $(f_N)_N$, which is defined by

$$f_N(t) = N\int_{-1/N}^0 \overline{f}(t+h)\,dh,\ \forall t \ge 0, \tag{V.12}$$

where $\overline{f}$ is the extension by 0 of f to all $\mathbb{R}$. Hence, for all N, f_N is continuous with respect to time with values in $(H^{-1}(\Omega))^d$. Moreover, the sequence $(f_N)_N$ converges towards f in $L^2(]0, T[, (H^{-1}(\Omega))^d)$. Indeed, for all $t \ge 0$, from the Jensen inequality we have

$$\|f(t) - f_N(t)\|_{H^{-1}}^2 \leq N \int_{-1/N}^{0} \|\overline{f}(t) - \overline{f}(t+h)\|_{H^{-1}}^2 \, dh,$$

and hence

$$\begin{aligned}
\|f - f_N\|_{L^2(]0,T[,H^{-1})}^2 &\leq N \int_0^T \int_{-1/N}^{0} \|\overline{f}(t) - \overline{f}(t+h)\|_{H^{-1}}^2 \, dh \, dt \\
&= N \int_{-1/N}^{0} \left(\int_0^T \|\overline{f}(t) - \overline{f}(t+h)\|_{H^{-1}}^2 \, dt \right) dh \\
&\leq \sup_{|h| \leq 1/N} \|\overline{f} - \tau_h \overline{f}\|_{L^2(]0,T[,H^{-1})}^2,
\end{aligned}$$

where $\tau_h \overline{f}$ is the time-translation of $\overline{f}$ defined in (II.20). The result thus follows from Corollary II.5.4. Furthermore, we note that for all N and all $T > 0$ we have

$$\|f_N\|_{L^2(]0,T[,X)} \leq \|f\|_{L^2(]0,T[,X)}, \tag{V.13}$$

for any Banach space X such that $f \in L^2(]0,T[,X)$.

Now we consider the following approximate problem: to find v_N belonging to $\mathcal{C}^1([0,T],\ H_N)$ which satisfies, for all $\psi_N \in H_N$, the differential equation

$$\begin{aligned}
\left(\frac{d}{dt} v_N, \psi_N\right)_H + \int_\Omega ((v_N \cdot \nabla) v_N) \cdot \psi_N \, dx + \frac{1}{\mathcal{R}e} \int_\Omega \nabla v_N : \nabla \psi_N \, dx \\
= \langle f_N, \psi_N \rangle_{H^{-1}, H_0^1},
\end{aligned} \tag{V.14}$$

as well as the initial condition $v_N(0) = \mathcal{P}_N(v_0)$, where $\mathcal{P}_N$ is the orthogonal projection in H onto the finite-dimensional space H_N. One such approximate problem of the variational formulation considered above is known as a *Galerkin approximation.*

The family $(w_k)_k$ is orthogonal in H, therefore if we denote the component along w_k of the function v_N as $\alpha_k(t)$ and the vector of components $(\alpha_k(t))_{k \leq N}$ as $\alpha(t)$, this approximate problem can be written as a system of ordinary differential equations for $\alpha(t)$ of the form

$$\frac{d\alpha}{dt} = \mathscr{F}(t, \alpha),$$

where $\mathscr{F}$ is continuous and locally Lipschitz continuous with respect to α (because it is polynomial).

The Cauchy–Lipschitz theorem gives us the existence of a unique $\mathcal{C}^1$ solution, defined on a maximum interval $[0, T_N)$ where $0 < T_N \leq T$.

We now need to establish *energy estimates* on v_N, independent of N, which first ensure that $T_N = T$, from the finite time blow-up theorem for ordinary differential equations. Moreover, these estimates also allow us to deduce the existence of weakly convergent subsequences in appropriate spaces and hence justify passing to the limit in the approximate problem.

1.3.3 Energy estimates

By taking $\psi_N = v_N$ in equality (V.14) we obtain, by selecting the L^2 norm of the gradient as the norm on $H_0^1(\Omega)$, and by using Young's inequality

$$\begin{aligned}\frac{1}{2}\frac{d}{dt}\|v_N\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\|\nabla v_N\|_{L^2}^2 &\leq \|f_N\|_{H^{-1}}\|v_N\|_{H_0^1} \\ &\leq \|f_N\|_{H^{-1}}\|\nabla v_N\|_{L^2} \\ &\leq \frac{\mathcal{R}e}{2}\|f_N\|_{H^{-1}}^2 + \frac{1}{2\mathcal{R}e}\|\nabla v_N\|_{L^2}^2.\end{aligned} \tag{V.15}$$

Note that we have used here the fact that the nonlinear term $b(v_N, v_N, v_N)$ is zero (see (V.2)) and therefore does not appear in this inequality.

By simple integration over time, we can deduce

$$\begin{aligned}&\|v_N(t)\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\int_0^t \|\nabla v_N(\tau)\|_{L^2}^2 d\tau \\ \leq &\|\mathcal{P}_N v_0\|_{L^2}^2 + \mathcal{R}e\int_0^t \|f_N(\tau)\|_{H^{-1}}^2 d\tau \\ \leq &\|v_0\|_{L^2}^2 + \mathcal{R}e\int_0^t \|f_N(\tau)\|_{H^{-1}}^2 d\tau,\end{aligned} \tag{V.16}$$

because the projection $\mathcal{P}_N$ is orthogonal (therefore having norm equal to 1) in $(L^2(\Omega))^d$.

From (V.13), we have therefore proven the following.

Lemma V.1.5. *For all $N \in \mathbb{N}$, we have the estimates*

$$\begin{cases}\sup\limits_{t\leq T_N} \|v_N(t)\|_{L^2}^2 \leq \|v_0\|_{L^2}^2 + \mathcal{R}e\|f\|_{L^2(]0,T[,(H^{-1}(\Omega))^d)}^2, \\ \displaystyle\int_0^{T_N} \|\nabla v_N(\tau)\|_{L^2}^2 d\tau \leq \mathcal{R}e\|v_0\|_{L^2}^2 + \mathcal{R}e^2\|f\|_{L^2(]0,T[,(H^{-1}(\Omega))^d)}^2.\end{cases} \tag{V.17}$$

The first fundamental consequence of this lemma is that, for fixed N, the solution v_N is bounded on $[0, T_N]$ (for the L^2 norm, for example) independently of T_N, so that the finite time blow-up theorem for ordinary differential equations gives that $T_N = T$. As a consequence, the estimates (V.17) are valid with $T_N = T$.

To pass to the limit in the nonlinear term, it is necessary to obtain a strong convergence result. The compactness theorem of Aubin–Lions–Simon (Theorem II.5.16) needs to be used. To this end we need to prove an estimate on the derivatives with respect to time of the approximate solutions and hence estimate the various terms of Equation (V.14).

Lemma V.1.6. *There exists $K(T, v_0, f) > 0$ such that for all N we have*

$$\left\| \frac{dv_N}{dt} \right\|_{L^{4/d}(]0,T[,V')} \leq K(T, v_0, f).$$

Proof.

The space H_N is built on a basis of eigenfunctions of the Stokes operator. As we have seen, this basis is orthogonal in H and in V. Hence, the operator $\mathcal{P}_N$ is also an orthogonal projection in V onto H_N. The adjoint operator ${}^t\mathcal{P}_N$ is therefore a continuous linear operator on V' with norm less than or equal to 1. From the definition of the adjoint, we have the following equation in $\mathcal{D}'(]0,T[)$,

$$\frac{dv_N}{dt} = -\left(\frac{1}{\mathcal{R}e} A v_N + {}^t\mathcal{P}_N(B(v_N, v_N)) - {}^t\mathcal{P}_N(f_N) \right). \tag{V.18}$$

Indeed, for all $\psi \in V$, since dv_N/dt and Av_N belong to H_N, we have

$$\begin{aligned} &\left\langle \frac{dv_N}{dt} + \left(\frac{1}{\mathcal{R}e} A v_N + {}^t\mathcal{P}_N(B(v_N, v_N)) - {}^t\mathcal{P}_N(f_N) \right), \psi \right\rangle_{V',V} \\ &= \left\langle \frac{dv_N}{dt} + \frac{1}{\mathcal{R}e} A v_N + B(v_N, v_N) - f_N, \mathcal{P}_N \psi \right\rangle_{V',V} = 0, \end{aligned}$$

because $\mathcal{P}_N \psi \in H_N$ and we can therefore apply (V.14) with $\psi_N = \mathcal{P}_N \psi$.

Hence, since the Stokes operator A is continuous from V to V', it follows from (V.18) and (V.5), that

$$\begin{aligned} \left\| \frac{dv_N}{dt} \right\|_{V'} &\leq \frac{1}{\mathcal{R}e} \|A v_N\|_{V'} + \|B(v_N, v_N)\|_{V'} + \|f_N(t)\|_{V'} \\ &\leq C \|v_N\|_V + \|v_N\|_{L^2}^{2-d/2} \|v_N\|_{H^1}^{d/2} + \|f_N(t)\|_{V'}, \\ &= g_N(t), \end{aligned}$$

where g_N is a sequence of bounded functions in $L^{4/d}(]0,T[)$ according to (V.13) and the estimates provided by Lemma V.1.5. Hence

$$\left\| \frac{dv_N}{dt} \right\|_{L^{4/d}(]0,T[,V')} \leq \left(\int_0^T g_N^{4/d} \, dt \right)^{d/4},$$

which gives the desired estimate.

□

Remark V.1.2. In the three-dimensional case, by noting that $(H^{1/2}(\Omega))^3$ (or also that $V_{1/2}$) embeds continuously in $(L^3(\Omega))^3$, we can show, by writing the estimate on the bilinear term slightly differently, that $B(v_N, v_N)$ is also bounded in $L^2(]0,T[,\, V_{-3/2})$. Indeed, we write

$$b(v,v,w) = b(v,w,v) = \int_\Omega \big((v\cdot\nabla)w\big)\cdot v\,dx,$$

and hence by the Hölder inequality

$$|b(v,v,w)| \le \|v\|_{L^3}\|\nabla w\|_{L^3}\|v\|_{L^3} \le \|v\|_{L^2}\|v\|_{H^1}\|w\|_{V_{3/2}}.$$

By duality this proves that

$$\|B(v,v)\|_{V_{-3/2}} \le \|v\|_{L^2}\|v\|_{H^1},$$

and that since $(v_N)_N$ is bounded in $L^\infty(]0,T[,H)\cap L^2(]0,T[,V)$, we have

$$(B(v_N,v_N))_N \text{ is bounded in } L^2(]0,T[,V_{-3/2}),$$

which enables us to show a bound on dv_N/dt in $L^2(]0,T[,V_{-3/2})$.

1.3.4 Passing to the limit

In Lemmas V.1.5 and V.1.6 we established that the sequence $(v_N)_N$ is bounded in $L^\infty(]0,T[,\, H)$ and in $L^2(]0,T[,\, V)$ and that $(dv_N/dt)_N$ is bounded in $L^{4/d}(]0,T[,\, V')$. Moreover, from the Sobolev embeddings, we know that $H^1(\Omega)$ is continuously embedded into $L^6(\Omega)$ because the space dimension d is less than or equal to 3. We can then deduce that the sequence $(v_N)_N$ is also bounded in $L^2(]0,T[,(L^6(\Omega))^d)$.

We now need to obtain a bound for the nonlinear inertia term $(v_N\cdot\nabla)v_N$. From the Hölder inequality, we have

$$\begin{aligned}\|(v_N\cdot\nabla)v_N\|_{L^2(]0,T[,(L^1(\Omega))^d)} &\le \|v_N\|_{L^\infty(]0,T[,(L^2(\Omega))^d)}\|\nabla v_N\|_{L^2(]0,T[,(L^2(\Omega))^d)}\\ &\le \|v_N\|_{L^\infty(]0,T[,H)}\|v_N\|_{L^2(]0,T[,V)} \le C(T,v_0,f).\end{aligned}$$

However, we also have

$$\begin{aligned}\|(v_N\cdot\nabla)v_N\|_{L^1(]0,T[,(L^{3/2}(\Omega))^d)} &\le \|v_N\|_{L^2(]0,T[,(L^6(\Omega))^d)}\|\nabla v_N\|_{L^2(]0,T[,(L^2(\Omega))^d)}\\ &\le \|v_N\|^2_{L^2(]0,T[,V)} \le C(T,v_0,f).\end{aligned}$$

We have therefore shown that the sequence $((v_N\cdot\nabla)v_N)_N$ is bounded in the spaces $L^2(]0,T[,(L^1(\Omega))^d)$ and $L^1(]0,T[,(L^{3/2}(\Omega))^d)$. These regularity spaces are optimal in the sense where the regularity L^1 in time, or L^1 in space, is the weakest regularity in the class of L^p spaces. In the usual way, we observe that a gain in regularity in time implies a loss of regularity in space and vice versa. The bound obtained in the two spaces above enables us to deduce many other bounds in the intermediate spaces from Theorem II.5.5. More precisely, for all $0 \leq \theta \leq 1$, we can deduce from this theorem that the sequence $((v_N \cdot \nabla)v_N)_N$ is bounded in $L^p(]0,T[,(L^q(\Omega))^d)$ with

$$\frac{1}{p} = \frac{\theta}{1} + \frac{1-\theta}{2} \text{ and } \frac{1}{q} = \frac{2\theta}{3} + \frac{1-\theta}{1}. \tag{V.19}$$

What estimate do we require? To determine this, we need to go back to the weak formulation of the approximate problem in which we wish to pass to the limit. The nonlinear term appears in the formulation in the following form

$$\int_0^t \int_\Omega ((v_N \cdot \nabla)v_N) \cdot \psi \, dx,$$

where ψ is a time-independent function of $(H^1(\Omega))^d$ and hence lies in $L^\infty(]0,T[,(L^6(\Omega))^d)$. To be able to pass to the limit in this term, we require a weak convergence of the nonlinear term $((v_N \cdot \nabla)v_N)_N$ in the dual space $L^r(]0,T[,(L^{6/5}(\Omega))^d)$ with $r > 1$ because $6/5$ is the conjugate exponent of 6. By returning to (V.19), we see that we need to take $\theta = 1/2$ to have $q = 6/5$ which gives $p = 4/3$. In summary, the estimate which will be useful to us is therefore

$$((v_N \cdot \nabla)v_N)_N \text{ is bounded in } L^{4/3}(]0,T[,(L^{6/5}(\Omega))^d). \tag{V.20}$$

From Theorem II.2.7 there exists a function v and a subsequence of $(v_N)_N$, still referred to as $(v_N)_N$ to simplify the notation, satisfying

$$\begin{cases} v_N \xrightarrow[N\to\infty]{} v & \text{weakly-}\star \text{ in } L^\infty(]0,T[,\, H), \\ v_N \xrightarrow[N\to\infty]{} v & \text{weakly in } L^2(]0,T[,\, V), \\ v_N \xrightarrow[N\to\infty]{} v & \text{weakly in } L^2(]0,T[,\, (L^6(\Omega))^d), \\ \dfrac{dv_N}{dt} \xrightarrow[N\to\infty]{} \dfrac{dv}{dt} & \text{weakly in } L^{\frac{4}{d}}(]0,T[,\, V'). \end{cases}$$

We note that the weak limit in the three initial spaces are necessarily the same, because the three weak convergences imply convergence in the distributions sense (in $\mathcal{D}'(]0,T[\times\Omega)$), and we know that the limit of a sequence in the distributions sense is unique. Finally, the weak limit of $\frac{dv_N}{dt}$ is equal to $\frac{dv}{dt}$, because the operator for differentiation with respect to time is continuous in the sense of distributions.

Then, even if it means again extracting a subsequence, we know from Theorem II.2.7 and using estimate (V.20), that there exists a function g such that

$$(v_N \cdot \nabla)v_N \xrightarrow[N\to\infty]{} g, \quad \text{weakly in } L^{4/3}(]0,T[,\, (L^{6/5}(\Omega))^d).$$

Note that at this stage, we cannot yet affirm that $g = (v\cdot\nabla)v$. The fundamental step which follows, consists precisely in using the compactness properties (which we have not used so far) to prove that we do indeed have this equality.

Let us start by noting that thanks to the weak convergences above and through Theorem II.5.16 and Proposition II.3.4, we directly obtain that

$$v_N \xrightarrow[N\to\infty]{} v, \quad \text{in } L^2(]0,T[,\, H).$$

We know that $(v_N)_N$ strongly converges towards v in $L^2(]0,T[,(L^2(\Omega))^d)$ and that $(\nabla v_N)_N$ weakly converges in $L^2(]0,T[,(L^2(\Omega))^{d\times d})$ towards ∇v. From Proposition II.2.12, we conclude that the sequence $((v_N\cdot\nabla)v_N)_N$ weakly converges in $L^1(]0,T[,(L^1(\Omega))^d)$ towards $(v\cdot\nabla)v$. However, we also know that $((v_N\cdot\nabla)v_N)_N$ weakly converges towards g in $L^{4/3}(]0,T[,(L^{6/5}(\Omega))^d)$. The convergence in these two spaces both imply the convergence in the distributions sense, therefore we can deduce that $g = (v\cdot\nabla)v$ by uniqueness of the limit in $\mathcal{D}'(]0,T[\times\Omega)$.

We want to emphasise the very general method which, for treating nonlinear terms, consists in establishing estimates and weak convergences in the most precise regularity spaces, towards limits which are a priori unknown and identifying these limits by compactness through justifying the passing to the limit in larger spaces (i.e., spaces of less regular functions). This strategy is exactly the one that we presented in Proposition II.2.10 in a quite abstract setting.

We can now pass to the limit in the weak formulation. Indeed, let ψ_K be a fixed function of H_K and let $\theta(t)$ be a function of $\mathcal{D}(]0,T[)$. The spaces $(H_N)_N$ are nested, thus for all sufficiently large N we have

$$\begin{aligned}\int_0^T \left\langle \frac{dv_N}{dt},\psi_K\right\rangle_{V',V}\theta(\tau)d\tau + \int_0^T\int_\Omega((v_N\cdot\nabla)v_N)\cdot\psi_K\theta(\tau)dxd\tau\\ +\frac{1}{\mathcal{R}e}\int_0^T\int_\Omega \nabla v_N : \nabla\psi_K\theta(\tau)dxd\tau = \int_0^T \langle f_N,\psi_K\rangle_{H^1,H^1_0}\theta(\tau)d\tau.\end{aligned}$$

Since K is fixed, we make N tend towards infinity by using all the convergences established above to obtain

$$\begin{aligned}\int_0^T \left\langle \frac{dv}{dt},\theta(\tau)\psi_K\right\rangle_{V',V}d\tau + \int_0^T\int_\Omega((v\cdot\nabla)v)\cdot(\theta(\tau)\psi_K)dxd\tau\\ +\frac{1}{\mathcal{R}e}\int_0^T\int_\Omega \nabla v : \nabla(\theta(\tau)\psi_K)dxd\tau = \int_0^T \langle f,\theta(\tau)\psi_K\rangle_{H^{-1},H^1_0}d\tau.\end{aligned} \tag{V.21}$$

Note that we only use weak convergences here; the strong convergence of $(v_N)_N$ towards v in $L^2(]0,T[,H)$ has, indeed, only been used to identify the weak limit of the product $(v_N\cdot\nabla)v_N$.

Now let $\psi \in V$ and let $\psi_K = \mathcal{P}_K\psi$. Since the $(w_k)_k$ form a complete family in V, we know that ψ_K converges towards ψ in V. It is then clear that $\theta(\cdot)\psi_K$ converges towards $\theta(\cdot)\psi$ in $\mathcal{C}^0([0,T],V)$ and therefore also in $\mathcal{C}^0([0,T],(L^6(\Omega))^d)$ by Sobolev embedding.

However, since $dv/dt \in L^{4/d}(]0,T[,V')$, $(v\cdot\nabla)v \in L^{4/3}(]0,T[,(L^{6/5}(\Omega))^d)$, and $f \in L^2(]0,T[,(H^{-1}(\Omega))^d)$, we can pass to the limit with respect to K in all the terms of Equation (V.21), which exactly demonstrates (V.6) for the test function ψ.

Note that, from Proposition V.1.3, the function v thus obtained also satisfies equation (V.7). Since we have seen that dv/dt belongs to the space $L^{4/d}(]0,T[,V')$, we can extend the formulation (V.7) to test functions $\varphi \in L^{4/(4-d)}(]0,T[,V)$ by density.

We now need to verify that the initial condition is satisfied. From Proposition II.5.11, the limit function v is continuous with values in V'.

Moreover, using Theorem II.5.16 we know that the sequence $(v_N)_N$ strongly converges towards v in $\mathcal{C}^0([0,T],V')$ which implies, in particular, that $v_N(0)$ converges towards $v(0)$ in V'. However, by definition, we have $v_N(0) = \mathcal{P}_N(v_0)$ and since the special basis chosen is an orthonormal basis of H, the sequence $v_N(0)$ converges towards v_0 in H and therefore in V'. Through the uniqueness of the limit in V', we have indeed obtained $v(0) = v_0$. □

1.3.5 The uniqueness problem

Currently, establishing the uniqueness (or nonuniqueness) of weak solutions of the Navier–Stokes equations in 3D is still an open problem. Among other things, this is linked to the lack of regularity of the nonlinear term which only belongs to $L^{4/3}(]0,T[,\, V')$, as we have seen. This does not allow us to take a test function of $L^2(]0,T[,\, V)$ in the weak formulation (V.7), which is the regularity of the solution we just have built. By contrast, this objection disappears in dimension 2.

We therefore assume in this section that $d = 2$ (the 3D case is discussed further in Section 2.3.2). We consider v_1 and v_2 to be two weak solutions of (V.7) and introduce $v = v_2 - v_1$. The function v therefore satisfies, for any function $\psi \in L^2(]0,T[,V)$,

$$\int_0^T \left\langle \frac{dv}{dt}(\tau),\psi(\tau)\right\rangle_{V',V} d\tau + \frac{1}{\mathcal{R}e}\int_0^T\int_\Omega \nabla v(\tau) : \nabla\psi(\tau)dx\,d\tau$$
$$+\int_0^T\int_\Omega (v_2(\tau)\cdot\nabla v(\tau))\cdot\psi(\tau)dx\,d\tau + \int_0^T\int_\Omega (v(\tau)\cdot\nabla v_1(\tau))\cdot\psi(\tau)dx\,d\tau = 0.$$

For any $t \in [0,T]$, the function $\psi(\tau) = 1_{[0,t]}(\tau)v(\tau)$ lies in $L^2(]0,T[,V)$ and it is therefore possible to take it as a test function in the above equation. Then, from Theorem II.5.12 we obtain

$$\frac{1}{2}\|v(t)\|_H^2 + \frac{1}{\mathcal{R}e}\int_0^t \int_\Omega |\nabla v(\tau)|^2 \, dx \, d\tau$$
$$+ \int_0^t b(v_2(\tau), v(\tau), v(\tau)) \, d\tau + \int_0^t b(v(\tau), v_1(\tau), v(\tau)) \, d\tau = \frac{1}{2}\|v(0)\|_H^2.$$

Since $b(v_2(\tau), v(\tau), v(\tau)) = 0$ for almost all τ (see (V.2)), using Estimate (V.3) and Young's inequality, we get

$$\begin{aligned}
&\frac{1}{2}\|v(t)\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\int_0^t \|\nabla v(\tau)\|_{L^2}^2 \, d\tau \\
\leq &\frac{1}{2}\|v(0)\|_{L^2}^2 + \int_0^t |b(v(\tau), v_1(\tau), v(\tau))| \, d\tau \\
\leq &\frac{1}{2}\|v(0)\|_{L^2}^2 + \int_0^t \|\nabla v_1(\tau)\|_{L^2} \|v(\tau)\|_{L^2} \|\nabla v(\tau)\|_{L^2} d\tau \\
\leq &\frac{1}{2}\|v(0)\|_{L^2}^2 + \frac{1}{2\mathcal{R}e}\int_0^t \|\nabla v(\tau)\|_{L^2}^2 d\tau \\
&+ \frac{\mathcal{R}e}{2}\int_0^t \|\nabla v_1(\tau)\|_{L^2}^2 \|v(\tau)\|_{L^2}^2 d\tau,
\end{aligned}$$

from which we deduce

$$\|v(t)\|_{L^2}^2 \leq \|v(0)\|_{L^2}^2 + \mathcal{R}e \int_0^t \|\nabla v_1(\tau)\|_{L^2}^2 \|v(\tau)\|_{L^2}^2 d\tau.$$

Since the function v_1 lies in $L^2(]0,T[, V)$, the real-valued function defined by $\tau \mapsto g(\tau) = \|\nabla v_1(\tau)\|_{L^2)}^2$ belongs to $L^1(]0,T[)$. Hence, we can apply the Gronwall lemma (Lemma II.4.10) and obtain

$$\forall t \in [0,T], \ \|v(t)\|_{L^2}^2 \leq \|v(0)\|_{L^2}^2 \exp\left(\mathcal{R}e \int_0^t g(\tau) \, d\tau\right).$$

However, $v_1(0) = v_2(0) = v_0$ and therefore $v(0) = 0$, and hence for $t \in [0,T]$, we obtain $v(t) = 0$ which proves the uniqueness of the weak solutions.

□

1.3.6 Globality of weak solutions

In all that has gone before, we have constructed weak solutions of the problem on intervals $[0,T]$, where $T > 0$ was set arbitrarily large. We now wish to show, as we claimed in the statement, that we can in reality obtain global solutions, that is, solutions defined over all time.

- The two-dimensional case:
  In this case, we obtained the uniqueness of the solutions on the entire interval $[0,T]$. This allows us to construct a global solution easily. Indeed, if we denote (v_n, p_n) as the unique solution of the problem on the interval $[0,n]$ then, thanks to the uniqueness property, we have

$$v_n = v_k,\ p_n = p_k, \text{ on } [0,n],$$

  provided $n < k$. We obtain the solution on all $\mathbb{R}^+$ by defining

$$v(t) = v_n(t),\ p(t) = p_n(t), \text{ for } t \leq n.$$

- The three-dimensional case:
  The preceding argument does not apply because there is no uniqueness of the solutions and we are therefore not certain that the various solutions on the intervals $[0,n]$ cover themselves correctly. We need, therefore, to proceed in a different manner by returning to the way in which solutions are constructed. We introduced an approximate problem in which we denoted the solutions $(v_N)_N$ (the pressure is obtained directly, at the end, from the solution v). We have established the energy estimates which show that these approximate solutions are unique and exist on all $\mathbb{R}^+$ (Cauchy–Lipschitz theorem).
  Let us set $K = 1$ and let us consider the interval $[0,K] = [0,1]$. By extracting a subsequence $(v_{\varphi_1(N)})_N$ of $(v_N)_N$ which converges on $[0,1]$ in the appropriate spaces, at the limit we obtain a solution v of the problem on $[0,1]$ (see Section 1.3.4).
  Now let us take $K = 2$ and let us consider the situation on the interval $[0,K] = [0,2]$. We extract a subsequence $(v_{\varphi_1(\varphi_2(N))})_N$ of the sequence $(v_{\varphi_1(N)})_N$ which provides a solution on $[0,2]$, and so on.
  Hence, by successive extractions, we can obtain the strictly increasing functions φ_i of $\mathbb{N}$ in $\mathbb{N}$ such that, at the limit, $(v_{\varphi_1\circ\ldots\circ\varphi_K(N)})_N$ provides a solution of the initial problem on the interval $[0,K]$.
  In conclusion, it is now sufficient to use a diagonal extraction process, that is to consider the subsequence

$$(v_{\varphi_1\circ\ldots\circ\varphi_N(N)})_N.$$

  This sequence converges towards a solution of the initial problem on any interval $[0,K]$, or even on any interval $[0,T]$. Hence, the limit of this subsequence is indeed a solution defined on all $\mathbb{R}^+$.

1.4 Kinetic energy evolution

We now detail the properties of continuity over time for the weak solutions obtained above.

Proposition V.1.7. *In the two-dimensional case, the previously constructed solution v satisfies*

$$v \in \mathcal{C}^0(\mathbb{R}^+,\, H).$$

In the three-dimensional case, it satisfies

$$v \in \mathcal{C}^0(\mathbb{R}^+,\, V_{-1/4}),$$

and

$$v \in \mathcal{C}^0(\mathbb{R}^+,\, H_{weak});$$

that is, it is weakly continuous with values in H.

Proof.

In 2D, we know that

$$v \in L^2(]0,T[,\, V),$$
$$\frac{dv}{dt} \in L^2(]0,T[,\, V').$$

Corollary II.5.13 then gives the claim.

In 3D, from Remark V.1.2, we have

$$v \in L^2(]0,T[,\, V),$$
$$\frac{dv}{dt} \in L^2(]0,T[,\, V_{-3/2}).$$

In this case Theorem IV.5.11 expresses the first desired result.

We note that since v is strongly continuous with values in $V_{-1/4}$, it is also weakly continuous in this same space and that, moreover, we have

$$v \in L^\infty(]0,T[, H).$$

Hence, Lemma II.5.9 allows us to conclude the weak continuity of v with values in H.

$\square$

What can we say about the behavior of the energy over time? The physical energy of the system is kinetic energy $\frac{1}{2}\int_\Omega |v|^2\, dx$ (remember that in this dimensionless model the fluid density is set to 1). We show that in 2D, we can write an energy equation (V.10) which we can physically interpret in the following way

$$\underbrace{\frac{1}{2}\|v(t)\|^2_{L^2} - \frac{1}{2}\|v_0\|^2_{L^2}}_{\text{Kinetic energy variation}} + \underbrace{\frac{1}{\mathcal{R}e}\int_0^t \|\nabla v(\tau)\|^2_{L^2} d\tau}_{\text{Dissipated energy}} = \underbrace{\int_0^t \langle f(\tau), v(\tau)\rangle_{V',V} d\tau.}_{\text{Work of the source term}}$$

In 3D, this energy balance equation simply becomes an inequality. This problem is intimately linked to the currently still unsolved problem of the uniqueness of weak solutions in dimension 3.

The two-dimensional case.

We know that, in 2D, the previously constructed weak solution satisfies $dv/dt \in L^2(]0,T[,V')$, and that the same holds for the term $(v\cdot\nabla)v$, which implies that we can extend (V.7) by density to all the functions $\varphi \in L^2(]0,T[,V)$ as we have already commented above. In particular, we can apply (V.7), with $\varphi = v$ and we exactly obtain the energy equation (V.10) through Theorem II.5.12.

The three-dimensional case.

Here, as we have seen, we no longer have $dv/dt \in L^2(]0,T[,V')$, but only

$$\frac{dv}{dt} \in L^2(]0,T[,V_{-3/2}) \cap L^{4/3}(]0,T[,V').$$

This does not allow the extension of (V.7) to test functions in $L^2(]0,T[,V)$, and therefore we cannot perform the same calculations as in the two-dimensional case. The proof of the energy inequality is therefore more intricate.

- Step 1:
  We can take $\psi_N = v_N$ in the approximate problem (V.14), and obtain

$$\frac{1}{2}\frac{d}{dt}\|v_N\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\int_\Omega |\nabla v_N|^2\,dx = \langle f_N, v_N\rangle_{H^{-1},H_0^1}, \tag{V.22}$$

  which, by integrating over time, gives for all $t \in [0,T]$

$$\begin{aligned}&\frac{1}{2}\|v_N(t)\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\int_0^t \|\nabla v_N\|_{L^2}^2\,d\tau\\ =&\frac{1}{2}\|\mathcal{P}_N v_0\|_{L^2}^2 + \int_0^t \langle f_N, v_N\rangle_{H^{-1},H_0^1} ds\\ \leq&\frac{1}{2}\|v_0\|_{L^2}^2 + \int_0^t \langle f_N, v_N\rangle_{H^{-1},H_0^1} d\tau.\end{aligned} \tag{V.23}$$

  The remaining steps of the proof consist in justifying that we are allowed to pass to the limit in this inequality.
- Step 2:
  We show that

$$\|v_0\|_H^2 = \|v(0)\|_H^2 = \lim_{s\to 0^+} \frac{1}{s}\int_0^s \|v(t)\|_H^2\,dt. \tag{V.24}$$

We note that since v is weakly continuous on $[0,T]$ in H, this essentially amounts to proving that 0 is a Lebesgue point for the mapping $t \mapsto \|v(t)\|_H^2$.

To do this, let us integrate (V.23) between 0 and $s > 0$ and divide by s. This gives:

$$\begin{aligned}&\frac{1}{2}\frac{1}{s}\int_0^s \|v_N\|_H^2 dt + \frac{1}{\mathcal{R}e}\frac{1}{s}\int_0^s\left(\int_0^t \|\nabla v_N\|_{L^2}^2\,d\tau\right)dt\\ \le&\frac{1}{2}\|v_0\|_H^2 + \frac{1}{s}\int_0^s\left(\int_0^t \langle f_N, v_N\rangle_{H^{-1},H_0^1} d\tau\right)dt.\end{aligned} \tag{V.25}$$

At fixed $s > 0$, passing to the limit when N tends towards infinity in the first term is an immediate consequence of the strong convergence of $(v_N)_N$ towards v in $L^2(]0,T[,H)$.

Inasmuch as $(v_N)_N$ is bounded in $L^2(]0,T[,V)$, and from (V.13), the term $\displaystyle\int_0^t \langle f_N, v_N\rangle_{H^{-1},H_0^1}\,ds$ is uniformly bounded in t and N, and, furthermore, for all $t \in [0,T]$, this term converges towards $\displaystyle\int_0^t \langle f, v\rangle_{H^{-1},H_0^1}\,ds$ when N tends towards infinity by weak convergence of $(v_N)_N$ towards v in $L^2(]0,T[,V)$ and by strong convergence of $(f_N)_N$ towards f. Hence, with Lebesgue's dominated convergence theorem we have:

$$\frac{1}{s}\int_0^s\left(\int_0^t \langle f_N, v_N\rangle_{H^{-1},H_0^1}\,d\tau\right)dt \xrightarrow[N\to\infty]{} \frac{1}{s}\int_0^s\left(\int_0^t \langle f, v\rangle_{H^{-1},H_0^1}\,d\tau\right)dt.$$

Furthermore, the sequence $(\nabla v_N)_N$ weakly converges towards ∇v in the space $L^2(]0,T[,(L^2(\Omega))^{d\times d})$ and therefore, by denoting the characteristic function of the interval $[0,t]$ as $1_{[0,t]}$, it is easy to see that $(1_{[0,t]}\nabla v_N)_N$ weakly converges towards $1_{[0,t]}\nabla v$ in $L^2(]0,T[,(L^2(\Omega))^{d\times d})$. Therefore, by weak lower semicontinuity of the norm (Corollary II.2.8), we obtain

$$\int_0^t \|\nabla v\|_{L^2}^2\,d\tau \le \liminf_{N\to\infty}\int_0^t \|\nabla v_N\|_{L^2}^2\,d\tau,\ \forall t\in[0,T].$$

Hence, from Fatou's lemma we have

$$\begin{aligned}\frac{1}{s}\int_0^s\left(\int_0^t \|\nabla v\|_{L^2}^2\,d\tau\right)dt &\le \frac{1}{s}\int_0^s\left(\liminf_{N\to\infty}\int_0^t \|\nabla v_N\|_{L^2}^2\,d\tau\right)dt\\ &\le \frac{1}{s}\liminf_{N\to\infty}\int_0^s\left(\int_0^t \|\nabla v_N\|_{L^2}^2\,d\tau\right)dt.\end{aligned}$$

So we can finally pass to the lower limit (in N) in the inequality (V.25). It follows

$$\begin{aligned}&\frac{1}{2}\frac{1}{s}\int_0^s \|v\|_H^2 dt + \frac{1}{\mathcal{R}e}\frac{1}{s}\int_0^s \left(\int_0^t \|\nabla v\|_{L^2}^2 d\tau\right) dt \\ \leq &\frac{1}{2}\|v_0\|_H^2 + \frac{1}{s}\int_0^s \left(\int_0^t \langle f, v\rangle_{H^{-1},H_0^1} d\tau\right) dt.\end{aligned} \tag{V.26}$$

Furthermore, the functions $\tau \mapsto \|\nabla v(\tau)\|_{L^2}^2$ and $\tau \mapsto \langle f(\tau), v(\tau)\rangle_{H^{-1},H_0^1}$ are in $L^1(]0,T[)$ and hence their antiderivatives $t \mapsto \displaystyle\int_0^t \|\nabla v(\tau)\|_{L^2}^2\, d\tau$ and $t \mapsto \displaystyle\int_0^t \langle f(\tau), v(\tau)\rangle_{H^{-1},H_0^1}\, d\tau$ are continuous functions (Lemma II.4.1) which cancel at zero; their own antiderivatives are therefore differentiable at 0 with zero derivatives (Proposition II.4.6). In other words, we have

$$\frac{1}{s}\int_0^s \left(\int_0^t \|\nabla v\|_{L^2}^2\, d\tau\right) dt \xrightarrow[s\to 0^+]{} 0,$$

and

$$\frac{1}{s}\int_0^s \left(\int_0^t \langle f, v\rangle_{H^{-1},H_0^1} d\tau\right) dt \xrightarrow[s\to 0^+]{} 0.$$

By taking the upper limit when s tends towards 0^+ in inequality (V.26), we obtain

$$\limsup_{s\to 0^+} \frac{1}{s}\int_0^s \|v\|_H^2 dt \leq \|v_0\|_H^2. \tag{V.27}$$

We now need to obtain an inequality in the other sense. To do this, we use the fact that v is weakly continuous with values in H (Proposition V.1.7) and the property of weak lower semicontinuity of the norm in a Banach space (Corollary II.2.8). This then gives:

$$\|v_0\|_H^2 = \|v(0)\|_H^2 \leq \liminf_{s\to 0^+} \|v(s)\|_H^2 = \lim_{\sigma\to 0^+}\left(\inf_{t\in[0,s]} \|v(t)\|_H^2\right).$$

However, for all $s > 0$, we have

$$\inf_{t\in[0,s]} \|v(t)\|_H^2 \leq \frac{1}{s}\int_0^s \|v(t)\|_H^2\, dt,$$

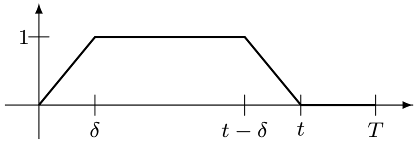
and by taking the lower limit in this inequality, it follows that

$$\|v_0\|_H^2 \leq \liminf_{s\to 0^+} \frac{1}{s}\int_0^s \|v(t)\|_H^2\, dt. \tag{V.28}$$

Hence, by combining (V.27) and (V.28), we have indeed demonstrated (V.24).

- Step 3:
  Let $t \in]0, T]$. For all $\delta < \frac{t}{2}$ we introduce the piecewise affine function θ_δ defined by

$$\theta_\delta(\tau) = \begin{cases} \frac{1}{\delta}\tau & \text{in } [0, \delta], \\ 1 & \text{in } [\delta, t-\delta], \\ \frac{1}{\delta}(t-\tau) & \text{in } [t-\delta, t], \\ 0 & \text{in } [t, T]. \end{cases}$$



This function belongs to $W_0^{1,1}([0,T])$ and we can therefore multiply (V.22) by θ_δ, integrate over $[0,T]$, and carry out an integration by parts. This gives

$$\begin{aligned} &\frac{1}{2}\frac{1}{\delta}\int_{t-\delta}^{t} \|v_N\|_H^2 d\tau + \frac{1}{\mathcal{R}e}\int_0^t \|\nabla v_N\|_{L^2}^2 \theta_\delta(\tau)\, d\tau \\ =&\frac{1}{2}\frac{1}{\delta}\int_0^{\delta} \|v_N\|_H^2 d\tau + \int_0^t \langle f_N, v_N\rangle_{H^{-1},H_0^1}\theta_\delta(\tau)\, d\tau. \end{aligned}$$

For the same reasons as in the previous step, we can pass to the lower limit when N tends towards infinity in this equality (all the terms converge except those which relate to the gradient of v_N for which we use once again the weak lower semicontinuity of the norm in a Banach space). This gives

$$\begin{aligned} &\frac{1}{2}\frac{1}{\delta}\int_{t-\delta}^{t} \|v\|_H^2\, d\tau + \frac{1}{\mathcal{R}e}\int_0^t \|\nabla v\|_{L^2}^2 \theta_\delta(s)\, d\tau \\ \leq&\frac{1}{2}\frac{1}{\delta}\int_0^{\delta} \|v\|_H^2\, d\tau + \int_0^t \langle f, v\rangle_{H^{-1},H_0^1}\theta_\delta(\tau)\, d\tau. \end{aligned}$$

We now wish to pass to the limit when δ tends towards 0. The two integrals which contain the function θ_δ can be treated directly by the dominated convergence theorem.
Let us initially assume that t is a Lebesgue point of the function $s \mapsto \|v(s)\|_H^2$. By employing Proposition II.4.4 and the convergence (V.24), we obtain the desired energy inequality

$$\frac{1}{2}\|v(t)\|_H^2 + \frac{1}{\mathcal{R}e}\int_0^t \|\nabla v\|_{L^2}^2\, d\tau \leq \frac{1}{2}\|v_0\|_H^2 + \int_0^t \langle f, v\rangle_{H^{-1},H_0^1}\, d\tau.$$

We know that the Lebesgue points of a function in $L^1(]0,T[)$ are dense in $]0,T[$ (Theorem II.4.5); we have therefore demonstrated the energy inequality for a dense set of points of $[0,T]$. Now, let $t \in [0,T]$ which is not a Lebesgue point of $s \mapsto \|v(s)\|_H^2$. There exists a sequence $(t_j)_j$ of points of $[0,T]$ which tends towards t, formed from Lebesgue points of this function and for which the energy inequality is true. Therefore, we have

$$\frac{1}{2}\|v(t_j)\|_H^2 + \frac{1}{\mathcal{R}e}\int_0^{t_j} \|\nabla v\|_{L^2}^2 \leq \frac{1}{2}\|v_0\|_H^2 + \int_0^{t_j} \langle f, v\rangle_{H^{-1},H_0^1}\, ds,$$

and, furthermore, since the function v is weakly continuous in H, we have

$$\|v(t)\|_H \leq \liminf_{j\to\infty} \|v(t_j)\|_H,$$

and thus we can pass to the lower limit in the energy inequality above and obtain the desired result, for all $t \in [0,T]$,

$$\frac{1}{2}\|v(t)\|_H^2 + \frac{1}{\mathcal{R}e}\int_0^t \|\nabla v\|_{L^2}^2\, ds \leq \frac{1}{2}\|v_0\|_H^2 + \int_0^t \langle f, v\rangle_{H^{-1},H_0^1}\, ds.$$

$\square$

1.5 Existence and regularity of the pressure

Let $\psi \in V$ be fixed and $\eta \in \mathcal{D}(]0,T[)$. We take $\varphi = \eta(t)\psi(x)$ in the formulation (V.7), which gives

$$\begin{aligned}&\int_0^T \eta(t)\left\langle \frac{dv}{dt}, \psi\right\rangle_{V',V} dt + \int_0^T \eta(t)\left(\int_\Omega ((v\cdot\nabla)v)\cdot\psi\, dx\right) dt\\ &+\frac{1}{\mathcal{R}e}\int_0^T \eta(t)\left(\int_\Omega \nabla v : \nabla\psi\, dx\right) dt - \int_0^T \eta(t)\langle f,\psi\rangle_{H^{-1},H_0^1}\, dt = 0.\end{aligned}$$

This is true for any function η, thus we obtain that for almost any t in $]0,T[$, we have

$$\left\langle \frac{dv}{dt}, \psi\right\rangle_{V',V} + \int_\Omega ((v\cdot\nabla)v)\cdot\psi\, dx + \frac{1}{\mathcal{R}e}\int_\Omega \nabla v : \nabla\psi\, dx - \langle f(t),\psi\rangle_{H^{-1},H_0^1} = 0.$$

This can also be written in the form

$$\left\langle \frac{dv}{dt}, \psi\right\rangle_{V',V} + \left\langle (v\cdot\nabla)v - \frac{1}{\mathcal{R}e}\Delta v - f, \psi\right\rangle_{H^{-1},H_0^1} = 0. \tag{V.29}$$

Let us now integrate (V.29) over time using Theorem II.5.12 and the fact that ψ does not depend on time. We obtain

$$\begin{aligned}&(v(t),\psi)_H - (v(0),\psi)_H\\ &+\left\langle \int_0^t (v\cdot\nabla)v\, d\tau - \frac{1}{\mathcal{R}e}\int_0^t \Delta v\, d\tau - \int_0^t f\, d\tau, \psi\right\rangle_{H^{-1},H_0^1} = 0.\end{aligned}$$

The scalar product on H is exactly the scalar product of $(L^2(\Omega))^d$, therefore this equation can also be written as

$$\Big\langle G(t), \psi \Big\rangle_{H^{-1}, H_0^1} = 0, \text{ for all } t \in [0, T], \tag{V.30}$$

with

$$G(t) = v(t) - v(0) + \int_0^t (v \cdot \nabla) v \, d\tau - \frac{1}{\mathcal{R}e} \int_0^t \Delta v \, d\tau - \int_0^t f \, d\tau.$$

We have already seen that the function v is weakly continuous with values in H, and hence with values in $(L^2(\Omega))^d$ as well as with values in $(H^{-1}(\Omega))^d$. Furthermore, v lies in $L^2(]0, T[, V)$ and hence in $L^2(]0, T[, (H_0^1(\Omega))^d)$. Thus, since the Laplace operator continuously maps H_0^1 to H^{-1}, the term Δv belongs to $L^2(]0, T[, (H^{-1}(\Omega))^d$. Similarly, from (V.3), the nonlinear term $(v \cdot \nabla) v$ belongs to $L^1(]0, T[, (H^{-1}(\Omega))^d)$. Hence, all the integral terms in (IV.59) are antiderivatives of integrable functions with values in $(H^{-1}(\Omega))^d$ and, as a consequence, they are continuous in time with values in $(H^{-1}(\Omega))^d$.

All this shows that the function G is weakly continuous on $[0, T]$ with values in $(H^{-1}(\Omega))^d$. Hence, for all $t \in [0, T]$, $G(t)$ is an element of $(H^{-1}(\Omega))^d$ which moreover satisfies (V.30) for all $\psi \in V$. From de Rham's theorem (Theorem IV.2.3), for all $t \in [0, T]$, there exists a unique $\pi(t) \in L_0^2(\Omega)$ such that

$$G(t) = -\nabla \pi(t). \tag{V.31}$$

Let us show that $t \mapsto \pi(t)$ is weakly continuous with values in $L^2(\Omega)$. Indeed, if g is a function of $L^2(\Omega)$, then from Theorem IV.3.1 there exists a function $h \in (H_0^1(\Omega))^d$ such that $\operatorname{div} h = g - m(g)$. Since $\pi(t)$ has a zero mean, we then have

$$\begin{aligned}(\pi(t), g)_{L^2} &= (\pi(t), g - m(g))_{L^2} = (\pi(t), \operatorname{div} h)_{L^2} \\ &= -\langle \nabla \pi(t), h \rangle_{H^{-1}, H_0^1} = \langle G(t), h \rangle_{H^{-1}, H_0^1},\end{aligned}$$

and this last quantity is continuous in time because G is weakly continuous in time with values in $(H^{-1}(\Omega))^d$.

We have therefore established, in particular, that the function $t \mapsto \pi(t)$ lies in $L^\infty(]0, T[, L_0^2(\Omega))$. We can now introduce the distribution $p = \partial \pi / \partial t$ which belongs to $W^{-1,\infty}(]0, T[, L_0^2(\Omega))$. By taking test functions of the form $\partial \varphi / \partial t$ with $\varphi \in \mathcal{D}(]0, T[\times \Omega)$ in (V.31) we can easily show that the Navier–Stokes equation

$$\frac{\partial v}{\partial t} + (v \cdot \nabla) v - \frac{1}{\mathcal{R}e} \Delta v + \nabla p = f,$$

is satisfied in the distributions sense on $]0, T[\times \Omega$. Note that if v, which is a solution of (V.7), is given, then the pressure $p \in W^{-1,\infty}(]0, T[, L_0^2(\Omega))$ is unique.

$\square$

This concludes the proof of Leray's theorem.

2 Strong solutions

To avoid the problem of the nonuniqueness of weak solutions for $d = 3$, we can ask ourselves if more regular solutions exist for which we could obtain uniqueness. The answer to this question is affirmative but with one important restriction, which is the loss of globality in time for such solutions in 3D. To establish the existence of strong solutions we return to the Galerkin approximation which uses a special basis (V.14) and we look for new estimates. Of course, if we hope to obtain these more regular solutions, we also need to start with more regular data (v_0, f).

Let us first state the main results concerning those strong solutions. Their proofs are given in the next subsections.

Theorem V.2.1. *Let Ω be a connected, bounded domain of $\mathbb{R}^d$ of class $\mathcal{C}^{1,1}$. Let $\mathcal{R}e > 0$, let v_0 be given in V, and f be given in $L^2_{loc}([0, +\infty[, (L^2(\Omega))^d)$.*

- *For $d = 2$, there exists a unique solution to Problem (V.1) satisfying*

$$\begin{aligned} v &\in \mathcal{C}^0([0, +\infty[,\ V) \cap L^2_{loc}([0 + \infty[,\ (H^2(\Omega))^d \cap V), \\ \frac{dv}{dt} &\in L^2_{loc}([0, +\infty[,\ H), \\ p &\in L^2_{loc}([0, +\infty[, H^1(\Omega)). \end{aligned}$$

This solution, of course, satisfies the energy equation for all $t \in [0, +\infty[$,

$$\frac{1}{2}\|v(t)\|^2_{L^2} + \frac{1}{\mathcal{R}e}\int_0^t \|\nabla v(\tau)\|^2_{L^2} d\tau = \frac{1}{2}\|v_0\|^2_{L^2} + \int_0^t (f(\tau), v(\tau))_{L^2} d\tau. \quad \text{(V.32)}$$

- *For $d = 3$, there exists a $T^\star > 0$ dependent on the data and a unique solution to Problem (V.1) satisfying*

$$\begin{aligned} v &\in \mathcal{C}^0([0, T^\star[,\ V) \cap L^2_{loc}([0, T^\star[,\ (H^2(\Omega))^d \cap V), \\ \frac{dv}{dt} &\in L^2_{loc}([0, T^\star[,\ H), \\ p &\in L^2_{loc}([0, T^\star[, H^1(\Omega)). \end{aligned}$$

Furthermore, it also satisfies the energy equality (V.32).

Moreover, since Ω is bounded, if the data are sufficiently small we can show that $T^\star = +\infty$. From a physical point of view, this states that if the viscosity is sufficiently large, then the regular solutions are global. We can also see this result as a stability result of the rest state $(v = 0, p = 0)$ (see Section 3.4 for a similar result concerning steady solutions of the Navier–Stokes equations). This is what specifies the following theorem.

Theorem V.2.2 (Global strong solutions for small data). *In the three-dimensional case, if v_0 is given in V and f in $L^\infty(\mathbb{R}^+, (L^2(\Omega))^d)$, then under* *the following hypotheses,*

$$\|\nabla v_0\|_{L^2}^2 \le \frac{C(\Omega)}{\mathcal{R}e^2},\ \|f\|_{L^\infty(\mathbb{R}^+,L^2)}^2 \le \frac{C(\Omega)}{\mathcal{R}e^4},$$

where $C(\Omega)$ depends only on the domain Ω, we have $T^\star = +\infty$, which means that the associated strong solution is global.

In the 2D case, and if the source term is independent of time, we can prove the following additional estimate.

Theorem V.2.3 (Uniform bound over time). *Let Ω be a connected bounded domain of $\mathbb{R}^2$ of class $\mathcal{C}^{1,1}$, $v_0 \in V$, and $f \in (L^2(\Omega))^2$ independent of time; then there exists a $C > 0$ such that*

$$\sup_{t\in\mathbb{R}^+} \|\nabla v(t)\|_{L^2} \le C.$$

2.1 New estimates

We return to the Galerkin approximation which involves considering an approximate solution $v_N \in \mathcal{C}^1([0,T],H_N)$ for all $\psi_N \in H_N$ satisfying

$$\begin{aligned}&\int_\Omega \frac{dv_N}{dt}\cdot\psi_N\,dx + \int_\Omega ((v_N\cdot\nabla)v_N)\cdot\psi_N\,dx\\ &+\frac{1}{\mathcal{R}e}\int_\Omega \nabla v_N : \nabla\psi_N\,dx = \int_\Omega f_N\cdot\psi_N\,dx.\end{aligned}\tag{V.33}$$

The basic idea is to take $-\Delta v_N$ as a test function, but this is not allowed since $\Delta v_N \notin H_N$. Instead, we can take $\psi_N = \mathcal{P}_N(-\Delta v_N)$ as a test function. Since the approximation space H_N is based on the eigenfunctions of the Stokes operator we see that this amounts to taking

$$\psi_N = Av_N = -\Delta v_N + \nabla p_N \in H_N.$$

We note that since $Av_N \in H_N \subset V$, we have

$$\int_\Omega \nabla w : \nabla v_N\,dx = \langle Av_N, w\rangle_{V',V} = \int_\Omega w\cdot Av_N\,dx,\ \forall w \in H_N,$$

from the identification of H with its dual, via the scalar product of $(L^2(\Omega))^d$.

In particular, by taking $w = v_N$, $w = dv_N/dt$ and $w = Av_N$, respectively, we obtain

$$\int_\Omega v_N \cdot A v_N \, dx = \int_\Omega |\nabla v_N|^2 \, dx,$$
$$\int_\Omega \frac{dv_N}{dt} \cdot A v_N \, dx = \int_\Omega \nabla \frac{dv_N}{dt} : \nabla v_N \, dx = \frac{1}{2}\frac{d}{dt}\int_\Omega |\nabla v_N|^2 \, dx,$$
$$\int_\Omega \nabla v_N : \nabla(A v_N)\, dx = \int_\Omega |A v_N|^2 \, dx.$$

These calculations are valid since the approximate solution v_N is smooth in time and space.

Hence, taking $\psi_N = A v_N$ in (V.33) gives

$$\frac{1}{2}\frac{d}{dt}\|\nabla v_N\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\|A v_N\|_{L^2}^2 = \int_\Omega f_N \cdot A v_N \, dx - \int_\Omega ((v_N \cdot \nabla) v_N) \cdot A v_N \, dx.$$

We now need to estimate the right-hand side term of this equality. To do this, we use the Hölder inequality, the precise Sobolev inequalities (Proposition III.2.35), and finally Young's inequality in the following way

$$\begin{aligned}
&\frac{1}{2}\frac{d}{dt}\|\nabla v_N\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\|A v_N\|_{L^2}^2 \\
\le &\|f_N\|_{L^2}\|A v_N\|_{L^2} + \|A v_N\|_{L^2}\|v_N\|_{L^6}\|\nabla v_N\|_{L^3} \\
\le &\|f_N\|_{L^2}\|A v_N\|_{L^2} \\
&+ C\|A v_N\|_{L^2}\left(\|v_N\|_{L^2}^{1-d/3}\|\nabla v_N\|_{L^2}^{d/3}\right)\left(\|\nabla v_N\|_{L^2}^{1-d/6}\|A v_N\|_{L^2}^{d/6}\right) \\
\le &\|f_N\|_{L^2}\|A v_N\|_{L^2} + C\|A v_N\|_{L^2}^{1+d/6}\|\nabla v_N\|_{L^2}^{1+d/6}\|v_N\|_{L^2}^{1-d/3} \\
\le &\frac{1}{2\mathcal{R}e}\|A v_N\|_{L^2}^2 \\
&+ C\mathcal{R}e\|f_N\|_{L^2}^2 + C\mathcal{R}e^{(6+d)/(6-d)}\|\nabla v_N\|_{L^2}^{2(6+d)/(6-d)}\|v_N\|_{L^2}^{4(3-d)/(6-d)}
\end{aligned}$$

from which we deduce

$$\begin{aligned}
&\frac{d}{dt}\|\nabla v_N\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\|A v_N\|_{L^2}^2 \\
\le &C\mathcal{R}e\|f_N\|_{L^2}^2 + C\mathcal{R}e^{(6+d)/(6-d)}\|\nabla v_N\|_{L^2}^{2(6+d)/(6-d)}\|v_N\|_{L^2}^{4(3-d)/(6-d)}.
\end{aligned} \tag{V.34}$$

From now on, we need to distinguish between the case $d = 2$ and the case $d = 3$.

2.2 The two-dimensional case

2.2.1 Existence

We have already established, during our study of weak solutions (estimate (V.16)), that

$$\|v_N(t)\|_{L^2}^2+\frac{1}{\mathcal{R}e}\int_0^t \|\nabla v_N(\tau)\|_{L^2}^2 d\tau \leq \|v_0\|_{L^2}^2+\mathcal{R}e\int_0^t \|f_N(\tau)\|_{H^{-1}}^2 d\tau. \quad \text{(V.35)}$$

Moreover, (V.34) written for $d=2$ gives

$$\frac{d}{dt}\|\nabla v_N\|_{L^2}^2+\frac{1}{\mathcal{R}e}\|Av_N\|_{L^2}^2 \leq C\mathcal{R}e\|f_N\|_{L^2}^2+C\mathcal{R}e^2\|\nabla v_N\|_{L^2}^4\|v_N\|_{L^2}. \quad \text{(V.36)}$$

Hence, after integration over time, and from (V.35), we obtain

$$\|\nabla v_N(t)\|_{L^2}^2+\frac{1}{\mathcal{R}e}\int_0^t \|Av_N(\tau)\|_{L^2}^2 d\tau \leq \|\nabla v_N(0)\|_{L^2}^2+C\mathcal{R}e\int_0^T \|f(\tau)\|_{L^2}^2 d\tau$$
$$+C\mathcal{R}e^2\left(\|v_0\|_{L^2}^2+\mathcal{R}e\int_0^T \|f(s)\|_{H^{-1}}^2 ds\right)^{1/2}\int_0^t \|\nabla v_N(\tau)\|_{L^2}^2\|\nabla v_N(\tau)\|_{L^2}^2 d\tau. \quad \text{(V.37)}$$

We recall that $v_N(0)=\mathcal{P}_N v_0$, that is, the orthogonal protection in $(L^2(\Omega))^d$ of v_0 onto H_N. We have already seen that, inasmuch as H_N is constructed on eigenfunctions of the Stokes operator, the operator $\mathcal{P}_N$ is also the orthogonal projection onto H_N in the space V. Hence we have

$$\|\nabla v_N(0)\|_{L^2}=\|\mathcal{P}_N v_0\|_V \leq \|v_0\|_V=\|\nabla v_0\|_{L^2}. \quad \text{(V.38)}$$

The Gronwall lemma (Lemma II.4.10) applied to $y(t)=\|\nabla v_N(t)\|_{L^2}^2$ gives the following bound from (V.37)

$$\|\nabla v_N(t)\|_{L^2}^2 \leq \left(\|\nabla v_0\|_{L^2}^2+C\mathcal{R}e\int_0^T \|f(\tau)\|_{L^2}^2 d\tau\right)e^{k_N(t)},$$

where

$$k_N(t)=C\mathcal{R}e^2\left(\|v_0\|_{L^2}^2+\mathcal{R}e\int_0^T \|f(s)\|_{H^{-1}}^2 ds\right)^{1/2}\int_0^t \|\nabla v_N(\tau)\|_{L^2}^2 d\tau.$$

We now use the estimate on $\int_0^t \|\nabla v_N(\tau)\|_{L^2}^2\, d\tau$ given by (V.35), which shows that for all N and all $t\in[0,T]$,

$$k_N(t) \leq C\mathcal{R}e^3 \left(\|v_0\|_{L^2}^2 + \mathcal{R}e \int_0^T \|f(s)\|_{H^{-1}}^2 ds \right)^{3/2}.$$

We have therefore shown that there exists a $C(v_0, f, T) > 0$ such that

$$\sup_{t \in [0,T]} \|\nabla v_N(t)\|_{L^2} \leq C(v_0, f, T).$$

Through this estimate, we can return to (V.37) to obtain the existence of another $C(v_0, f, T)$, such that

$$\int_0^T \|Av_N(\tau)\|_{L^2}^2 d\tau \leq C(v_0, f, T).$$

Since Ω is of class $\mathcal{C}^{1,1}$, the above estimates, and Proposition IV.5.9, show that the sequence of approximate solutions $(v_N)_N$ is bounded in $L^\infty(]0,T[,V)$ and in $L^2(]0,T[,V \cap (H^2(\Omega))^d)$. We can easily deduce, in a similar way to the case of weak solutions, that the sequence $(dv_N/dt)_N$ is bounded in $L^2(]0,T[,H)$.

Hence, with the help of additional extractions of subsequences, we can assume that the sequence of approximate solutions $(v_N)_N$, in addition to the convergences obtained in Section 1.3.4, satisfies

$$\begin{cases} v_N \xrightarrow[N\to\infty]{} v, & \text{weakly-}\star \text{ in } L^\infty(]0,T[,V), \\ v_N \xrightarrow[N\to\infty]{} v, & \text{weakly in } L^2(]0,T[,V \cap (H^2(\Omega))^d), \\ \dfrac{dv_N}{dt} \xrightarrow[N\to\infty]{} \dfrac{dv}{dt}, & \text{weakly in } L^2(]0,T[,H). \end{cases}$$

From these new properties, we arrive at the existence of a unique solution of Equation (V.1) such that

$$v \in \mathcal{C}^0(\mathbb{R}^+, V) \cap L^2_{loc}(\mathbb{R}^+, V \cap (H^2(\Omega))^d) \text{ with } \frac{dv}{dt} \in L^2_{loc}(\mathbb{R}^+, H).$$

Indeed, the continuity of the solution with values in V results from Theorem IV.5.11 because $v \in L^2(]0,T[,D(A))$ and $dv/dt \in L^2(]0,T[,H)$.

Moreover, inasmuch as we know now that the terms dv/dt and $(v \cdot \nabla)v$ are in $L^2(]0,T[,(L^2(\Omega))^d)$, we can write

$$\nabla p = -\frac{dv}{dt} + \frac{1}{\mathcal{R}e}\Delta v - (v \cdot \nabla)v + f \in L^2(]0,T[,(L^2(\Omega))^d).$$

Since p has a zero mean, the Poincaré–Wirtinger inequality (Proposition III.2.39) shows that

$$p \in L^2(]0,T[,H^1(\Omega)).$$

$\square$

2.2.2 Uniform bound over time

The essential ingredient here is the uniform Gronwall lemma (Lemma II.4.11).
Proof (of Theorem V.2.3).

Remember that we assumed that f does not depend on time so that the regularisation with respect to time defined by (V.12) simply reads $f_N = f$. We denote the smallest eigenvalue of the Stokes operator as λ_1. We have seen in Proposition IV.5.12, that we then have the following Poincaré inequality $\|v\|_{L^2}^2 \leq (1/\lambda_1)\|\nabla v\|_{L^2}^2$. Estimate (V.15) gives

$$\frac{d}{dt}\|v_N\|_{L^2}^2 + \frac{\lambda_1}{\mathcal{R}e}\|v_N\|_{L^2}^2 \leq \mathcal{R}e\|f\|_{H^{-1}}^2.$$

Lemma II.4.9 then allows us to deduce that

$$\|v_N(t)\|_{L^2}^2 \leq \|v_0\|_{L^2}^2 e^{-\lambda_1/\mathcal{R}et} + \frac{\mathcal{R}e^2}{\lambda_1}\|f\|_{H^{-1}}^2 \leq \|v_0\|_{L^2}^2 + \frac{\mathcal{R}e^2}{\lambda_1}\|f\|_{H^{-1}}^2.$$

By integrating (V.15) between t and $t+1$, and by using the above inequality, we get

$$\begin{aligned}\frac{1}{\mathcal{R}e}\int_t^{t+1} \|\nabla v_N(\tau)\|_{L^2}^2 d\tau &\leq \|v_N(t)\|_{L^2}^2 + \mathcal{R}e\|f\|_{H^{-1}}^2 \\ &\leq \|v_0\|_{L^2}^2 + \mathcal{R}e\|f\|_{H^{-1}}^2\left(1 + \frac{\mathcal{R}e}{\lambda_1}\right).\end{aligned}$$

Using (V.36), this allows the uniform Gronwall lemma (Lemma II.4.11) to be applied, with

$$y(t) = \|\nabla v_N\|_{L^2}^2,$$
$$g_1(t) = C\mathcal{R}e\|f\|_{L^2}^2,$$

and

$$g_2(t) = C\mathcal{R}e\|v_N\|_{L^2}\|\nabla v_N\|_{L^2}^2.$$

Hence, we obtain the desired bounds on the Galerkin approximations v_N of the solution, uniformly in N. More precisely, we have shown

$$\|\nabla v_N\|_{L^\infty(\mathbb{R}^+,(L^2(\Omega))^{d\times d})} \leq C.$$

From Corollary II.2.8, we know that the norm is weakly lower semicontinuous. The preceding inequality therefore passes to the limit, which gives:

$$\sup_{t\geq 0}\|\nabla v(t)\|_{L^2} \leq C.$$

□

2.3 The three-dimensional case

Estimate (V.34) written for $d = 3$ gives

$$\frac{d}{dt}\|\nabla v_N\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\|Av_N\|_{L^2}^2 \leq C\mathcal{R}e\|f_N\|_{L^2}^2 + C\,\mathcal{R}e^3\|\nabla v_N\|_{L^2}^6. \tag{V.39}$$

We first study the general case and then the case for small data.

2.3.1 General case

We set $y_N(t) = \|\nabla v_N(t)\|_{L^2}^2$ for any $t \in \mathbb{R}$. By using (V.12) and (V.38), and by integrating (V.39) over time, for all $t \geq 0$, we get

$$y_N(t) \leq \|\nabla v_0\|_{L^2}^2 + C\mathcal{R}e\int_0^t \|f(s)\|_{L^2}^2\,ds + C\mathcal{R}e^3\int_0^t y_N^3(s)\,ds.$$

Hence, let $T^* > 0$ be the unique real number satisfying

$$T^* = \frac{1}{2C\mathcal{R}e^3\left(\|\nabla v_0\|_{L^2}^2 + C\mathcal{R}e\displaystyle\int_0^{T^*}\|f(s)\|_{L^2}^2\,ds\right)^2}. \tag{V.40}$$

From Lemma II.4.12, we can deduce that for any $T < T^\star$, there exists a constant $C(T, \|\nabla v_0\|_{L^2}, \mathcal{R}e, \|f\|_{L^2})$ independent of N such that

$$\|\nabla v_N(t)\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\int_0^t \|Av_N(\tau)\|_{L^2}^2 d\tau \leq C(T, \|\nabla v_0\|_{L^2}, \mathcal{R}e, \|f\|_{L^2}),\ \forall t < T.$$

We note that T^* tends towards 0 when $\|\nabla v_0\|_{L^2}$ tends towards infinity and also that this “lifetime” tends towards infinity when the data v_0 and f tend towards 0. This latter behavior is described in detail in Section 2.3.3.

By applying the regularity theorem for the Stokes problem (see Proposition IV.5.9) we deduce the following result from the above.

Proposition V.2.4. *Let $\mathcal{R}e > 0, v_0 \in V$, and let f be in $L^2_{loc}([0, +\infty[, (L^2(\Omega))^d)$, then there exists a $T^\star > 0$ such that for any $T < T^\star$, there exists a $C(T, \|\nabla v_0\|_{L^2}, \mathcal{R}e, \|f\|_{L^2}) > 0$ such that*

$$\begin{aligned} \|\nabla v_N(t)\|_{L^2} &\leq C,\ \forall t \leq T, \\ \int_0^T \|v_N(\tau)\|_{H^2}^2 d\tau &\leq C. \end{aligned} \tag{V.41}$$

It is now useful to estimate the nonlinear term $(v_N \cdot \nabla)v_N$.

Proposition V.2.5. *For all $T < T^\star$, there exists a $C > 0$ such that*

$$\int_0^T \|(v_N(\tau)\cdot\nabla)v_N(\tau)\|_{L^2}^4 d\tau \leq C,\ \forall N \geq 1. \tag{V.42}$$

Proof.

By using Proposition III.2.35, we have the following inequality

$$\begin{aligned}
\|(v_N\cdot\nabla)v_N\|_{L^2}^4 &\leq \|v_N\|_{L^6}^4\|\nabla v_N\|_{L^3}^4 \\
&\leq \|\nabla v_N\|_{L^2}^4\left(\|\nabla v_N\|_{L^2}^{1/2}\|Av_N\|_{L^2}^{1/2}\right)^4 \\
&\leq \|\nabla v_N\|_{L^2}^6\|Av_N\|_{L^2}^2
\end{aligned}$$

Estimate (V.41) then allows us to reach the claim.

□

This demonstrates, in particular, that the nonlinear term is bounded in $L^2(]0,T[,(L^2(\Omega))^d)$.

Remark V.2.1. When we estimate the derivative with respect to time of v_N, we see that in the equation

$$\frac{dv_N}{dt} = -{}^t\mathcal{P}_N(B(v_N,v_N)) - \frac{1}{\mathcal{R}e}Av_N + {}^t\mathcal{P}_N(f_N), \tag{V.43}$$

the term with lowest regularity is now the linear term contrary to the 2D case.

Proposition V.2.6. *For all $T < T^\star$, there exists a $C > 0$ such that*

$$\left\|\frac{dv_N}{dt}\right\|_{L^2(]0,T[,\ (L^2(\Omega))^d)} \leq C,\ \forall N \geq 1. \tag{V.44}$$

Proof.

This follows easily from the hypothesis on f, from estimates (V.41) and (V.42), and from Equation (V.43).

□

As in the two-dimensional case, Estimates (V.41) and (V.44) allow us to conclude the existence of a subsequence always denoted as $(v_N)_N$ such that for all $T < T^\star$,

$$\begin{cases}
v_N \xrightharpoonup[N\to\infty]{} v, & \text{weakly-}\star \text{ in } L^\infty(]0,T[,\ V)\ , \\
v_N \xrightharpoonup[N\to\infty]{} v, & \text{weakly in } L^2(]0,T[,\ V\cap(H^2(\Omega))^d), \\
\dfrac{dv_N}{dt} \xrightharpoonup[N\to\infty]{} \dfrac{dv}{dt}, & \text{weakly in } L^2(]0,T[,\ H), \\
v_N \xrightarrow[N\to\infty]{} v, & \text{in } L^2(]0,T[,\ V),
\end{cases}$$

where v is a solution of (V.1). The solution v thus obtained is indeed a strong solution of the equations. As we have already seen in the two-dimensional case, the pressure p then belongs in $L^2(]0,T[,H^1(\Omega))$.

2.3.2 Uniqueness of the strong solution

Let v_1 and v_2 be two strong solutions of (V.1) for the same initial data $v_0 \in V$, and let us set $v = v_2 - v_1$. Since for $i \in \{1,2\}$, $(v_i \cdot \nabla)v_i$ and dv_i/dt lie in $L^2(]0,T[,(L^2(\Omega))^d)$, we can extend formulation (V.7) for v_1 and v_2 with test functions $\varphi \in L^2(]0,T[,V)$. By subtraction of these two equations, we obtain that v satisfies, for any $\varphi \in L^2(]0,T[,V)$,

$$\int_0^T \left\langle \frac{dv}{dt}, \varphi \right\rangle_{V',V} dt + \frac{1}{\mathcal{R}e} \int_0^T \int_\Omega \nabla v : \nabla \varphi \, dx \, dt$$
$$+ \int_0^T \int_\Omega ((v_2 \cdot \nabla)v) \cdot \varphi \, dx \, dt + \int_0^T \int_\Omega ((v \cdot \nabla)v_1) \cdot \varphi \, dx \, dt = 0.$$

It is now possible, for all $t \in]0,T[$, to take $\varphi = 1_{[0,t]}.v$ in the preceding equality and since from (V.2) we have $\displaystyle\int_\Omega ((v_2 \cdot \nabla)v) \cdot v \, dx = 0$, from Theorem II.5.12 we obtain

$$\begin{aligned}
&\frac{1}{2}\|v(t)\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\int_0^t \|\nabla v(\tau)\|_{L^2}^2 \, d\tau \\
\le &\frac{1}{2}\|v(0)\|_{L^2}^2 + \int_0^t \|\nabla v_1(\tau)\|_{L^2} \|v(\tau)\|_{L^4}^2 \, d\tau \\
\le &\frac{1}{2}\|v(0)\|_{L^2}^2 + \int_0^t g(\tau) \left(\|v(\tau)\|_{L^2}^{1/4} \|\nabla v(\tau)\|_{L^2}^{3/4} \right)^2 d\tau \\
\le &\frac{1}{2}\|v(0)\|_{L^2}^2 + \frac{1}{2\mathcal{R}e}\int_0^t \|\nabla v(\tau)\|_{L^2}^2 \, d\tau + C\int_0^t g(\tau)^4 \|v(\tau)\|_{L^2}^2 \, d\tau,
\end{aligned}$$

where $\tau \mapsto g(\tau) = \|\nabla v_1(\tau)\|_{L^2} \in L^\infty(]0,T[)$ because v_1 is a strong solution of the problem. We have hence obtained the inequality

$$\frac{1}{2}\|v(t)\|_{L^2}^2 \le \frac{1}{2}\|v(0)\|_{L^2}^2 + C\int_0^t g(\tau)^4 \|v(\tau)\|_{L^2}^2 \, d\tau,$$

which allows us to apply the Gronwall lemma which then tells us that

$$\|v(t)\|_{L^2}^2 \le \|v(0)\|_{L^2}^2 \exp\left(2C \int_0^t g(\tau)^4 d\tau \right).$$

However, $v(0) = v_1(0) - v_2(0) = v_0 - v_0 = 0$, and the preceding inequality implies that $v(t) = 0$ for all t, and therefore completes the proof of the theorem.

□

In fact, we have a stronger result than the previous one.

Theorem V.2.7 (Weak–strong uniqueness). *In dimension $d = 3$, let u be a weak solution of (V.1) satisfying the energy inequality (V.11) and let v be a weak solution of (V.1) such that there exists a time $T > 0$, for which*

$$v \in L^4(]0, T[, V). \tag{V.45}$$

If $u(0) = v(0)$ then the solutions u and v coincide on $[0, T[$.

Proof.

We first need to remark that the weak solution v satisfies the energy equality (V.32). Indeed, we have seen that in 3D, the weak solutions satisfy $dv/dt \in L^{4/3}(]0, T[, V')$ and $(v \cdot \nabla)v \in L^{4/3}(]0, T[, V')$, which allows the weak formulation (V.7) to be extended to test functions $\varphi \in L^4(]0, T[, V)$. Having assumed that $v \in L^4(]0, T[, V)$, we can take $\varphi = v$ in the weak formulation and obtain the energy equation (V.32) in the usual way.

Furthermore, $v \in L^4(]0, T[, V)$, thus we can easily show that $(v \cdot \nabla)v$ lies in $L^2(]0, T[, V')$ which proves that we indeed have $dv/dt \in L^2(]0, T[, V')$.

Since $v \in L^4(]0, T[, V)$, for the same reason as above we can take v as a test function in the weak formulation (V.7) for the solution u. Similarly, as we have seen above, we can take the function $u \in L^2(]0, T[, V)$ as a test function in the weak formulation (V.7) for the solution v. Hence, for all $t \in [0, T]$ we obtain the two formulas

$$\begin{aligned} &\int_0^t \left\langle \frac{du}{dt}, v \right\rangle_{V',V} d\tau + \frac{1}{\mathcal{R}e} \int_0^t \int_\Omega \nabla u : \nabla v \, dx \, d\tau \\ &+ \int_0^t \int_\Omega ((u \cdot \nabla)u) \cdot v \, dx \, d\tau = \int_0^t \langle f, v \rangle_{H^{-1}, H_0^1} d\tau, \end{aligned}$$

and

$$\begin{aligned} &\int_0^t \left\langle \frac{dv}{dt}, u \right\rangle_{V',V} d\tau + \frac{1}{\mathcal{R}e} \int_0^t \int_\Omega \nabla v : \nabla u \, dx \, d\tau \\ &+ \int_0^t \int_\Omega ((v \cdot \nabla)v) \cdot u \, dx \, d\tau = \int_0^t \langle f, u \rangle_{H^{-1}, H_0^1} d\tau. \end{aligned}$$

A purely algebraic computation on the nonlinear terms, together with application of Theorem II.5.12, then shows that if we denote $w = u - v$, by adding the two previous equations, we obtain

$$\begin{aligned}
&(u(t), v(t))_H + \frac{2}{\mathcal{R}e}\int_0^t \int_\Omega \nabla u : \nabla v \, dx \, d\tau \\
=&(u(0), v(0))_H + \int_0^t \langle f, u+v\rangle_{H^{-1},H_0^1}\, d\tau - \int_0^t \int_\Omega ((w\cdot\nabla)w)\cdot v\, dx\, d\tau.
\end{aligned} \tag{V.46}$$

The energy inequality for u and the energy equality for v can be written as

$$(u(t), u(t))_H + \frac{2}{\mathcal{R}e}\int_0^t \|\nabla u\|_{L^2}^2 \, d\tau \leq (u(0), u(0))_H + 2\int_0^t \langle f, u\rangle_{H^{-1},H_0^1}\, d\tau, \tag{V.47}$$

$$(v(t), v(t))_H + \frac{2}{\mathcal{R}e}\int_0^t \|\nabla v\|_{L^2}^2 \, d\tau = (v(0), v(0))_H + 2\int_0^t \langle f, v\rangle_{H^{-1},H_0^1}\, d\tau, \tag{V.48}$$

by forming the equation (V.47) + (V.48) − 2 × (V.46) we obtain the inequality

$$\|w(t)\|_{L^2}^2 + \frac{2}{\mathcal{R}e}\int_0^t \|\nabla w(\tau)\|_{L^2}^2 d\tau \leq \|w(0)\|_{L^2}^2 + 2\int_0^t \int_\Omega ((w\cdot\nabla)w)\cdot v\, dx\, d\tau.$$

By hypothesis we have $w(0) = u(0) - v(0) = 0$ and by using the Sobolev inequalities given in Proposition III.2.35 and Young's inequality, we can deduce

$$\begin{aligned}
&\|w(t)\|_{L^2}^2 + \frac{2}{\mathcal{R}e}\int_0^t \|\nabla w(\tau)\|_{L^2}^2 d\tau \\
\leq&2\int_0^t \|w(\tau)\|_{L^3}\|\nabla w(\tau)\|_{L^2}\|v(\tau)\|_{L^6} d\tau \\
\leq&C\int_0^t \|w(\tau)\|_{L^2}^{1/2}\|\nabla w(\tau)\|_{L^2}^{3/2}\|\nabla v(\tau)\|_{L^2} d\tau \\
\leq&\frac{1}{\mathcal{R}e}\int_0^t \|\nabla w(\tau)\|_{L^2}^2 d\tau + C\int_0^t \|w(\tau)\|_{L^2}^2\|\nabla v(\tau)\|_{L^2}^4 d\tau.
\end{aligned}$$

In particular for all $t \in [0, T]$, we have

$$\|w(t)\|_{L^2}^2 \leq C\int_0^t \|w(\tau)\|_{L^2}^2\|\nabla v(\tau)\|_{L^2}^4 d\tau,$$

which allows us to make the desired conclusion using Gronwall's lemma, because by assumption we have $\nabla v \in L^4(]0, T[, (L^2(\Omega))^{d\times d})$.

□

Remark V.2.2.

- We note that if v is a strong solution constructed above, then v lies in $L^\infty(]0,T[,V) \subset L^4(]0,T[,V)$ for all $T < T^*$ and hence satisfies the hypotheses of the theorem.
- It is needed to assume that the weak solution u satisfies the energy inequality (V.11). Indeed, as we have already noted, the only weak solution which we can be sure satisfies this inequality is the one constructed by the Galerkin method proposed above. We do not possess a uniqueness result for weak solutions in 3D, thus we cannot guarantee that any other possible weak solution satisfies this energy inequality.

There exist various results proving regularity and/or uniqueness of weak solutions in dimension $d = 3$ starting from a suitable a priori estimate. As an example, we give without proof the following result (see [106, 107, 116, 115]).

Theorem V.2.8. *Let Ω be a bounded domain in $\mathbb{R}^3$ and v be any weak solution of the Navier–Stokes defined on a time interval $]0,T[$.*

If v satisfies

$$v \in L^r(]0,T[,(L^s(\Omega))^3), \tag{V.49}$$

for some r, s satisfying

$$\frac{2}{r} + \frac{3}{s} = 1,\ 3 < s \leq \infty, \tag{V.50}$$

then v satisfies the energy equality (V.32). Moreover, if $2/r + 3/s < 1$ then v is smooth in $]0,T[\times\Omega$.

Finally, if u is a weak solution associated with the same initial data as v and which satisfies the energy inequality (V.11) then $u = v$.

Remark V.2.3. It is possible to intuitively understand why the condition (V.50) plays a key role in this theory. Any weak solution v satisfies, by definition,

$$\begin{aligned} &v \in L^2(]0,T[,(H^1(\Omega))^3) \subset L^2(]0,T[,(L^6(\Omega))^3), \\ &v \in L^\infty(]0,T[,(L^2(\Omega))^3). \end{aligned}$$

This is not sufficient for the nonlinear term $(v \cdot \nabla)v$ to belong to the dual of the space $X = L^2(]0,T[,V) \cap L^\infty(]0,T[,H)$, which is a fundamental property to get the energy equality. Let us show that the additional assumption (V.49), with (V.50) exactly implies this regularity of the nonlinear term.

For a given $w \in L^2(]0,T[,V) \cap L^\infty(]0,T[,H)$ we first use the Hölder inequality with respect to the space variable as

$$\left| \int_\Omega (v \cdot \nabla)v \cdot w \, dx \right| \leq \|v\|_{L^s} \|\nabla v\|_{L^2} \|w\|_{L^2}^{1-\theta} \|w\|_{L^6}^{\theta},$$

where $\theta \in [0,1]$ is defined by

$$\frac{\theta}{6}+\frac{1-\theta}{2}=\frac{1}{2}-\frac{1}{s},\ \text{that is,}\ \theta=\frac{3}{s}.$$

We now use the Hölder inequality with respect to the time variable

$$\left|\int_0^T\int_\Omega (v\cdot\nabla)v\cdot w\,dx\,dt\right|$$
$$\leq \|v\|_{L^r(]0,T[,L^s)}\|\nabla v\|_{L^2(]0,T[,L^2)}\|w\|^{1-\theta}_{L^\infty(]0,T[,L^2)}\|w\|^{\theta}_{L^{2r\theta/(r-2)}(]0,T[,L^6)}.$$

Condition (V.50) exactly implies that

$$\theta=\frac{3}{s}=1-\frac{2}{r}=\frac{r-2}{r},$$

so that

$$\frac{2r\theta}{r-2}=2,$$

and thus we get

$$\left|\int_0^T\int_\Omega (v\cdot\nabla)v\cdot w\,dx\,dt\right|$$
$$\begin{aligned}&\leq \|v\|_{L^r(]0,T[,L^s)}\|\nabla v\|_{L^2(]0,T[,L^2)}\|w\|^{1-\theta}_{L^\infty(]0,T[,L^2)}\|w\|^{\theta}_{L^2(]0,T[,L^6)}\\&\leq C_v\|w\|_X,\end{aligned}$$

and the claim is proved; that is, $(v\cdot\nabla)v\in X'$.

2.3.3 The case of small data

In (V.40) we obtained a lower bound for the lifetime of the solutions. This bound seems to show that the smaller the data are, the larger the lifetime is.

The aim of this section is to show that, in 3D, if the external forces f, the Reynolds number $\mathcal{R}e$, and the initial data v_0 are sufficiently small, then $T^\star=+\infty$; that is, that the strong solution associated with the data (v_0,f) is in fact global. Let us begin by giving an elementary result relating to a class of nonlinear differential inequalities.

Lemma V.2.9. *Let z be a nonnegative numerical function of class $\mathcal{C}^1$, satisfying the differential inequality:*

$$\begin{cases} z'(t)+\alpha z(t)\leq \beta+\gamma z(t)^3,\\ z(0)=z_0,\end{cases}$$

on a time interval $[0,T[$, $0<T<+\infty$, and where α,β,γ,z_0 are nonnegative real numbers. If we assume that

$$z_0 \leq \frac{1}{2}\sqrt{\frac{\alpha}{2\gamma}},\ \beta \leq \frac{\alpha}{4}\sqrt{\frac{\alpha}{2\gamma}},$$

then the function z satisfies

$$z(t) \leq \sqrt{\frac{\alpha}{2\gamma}},\ \forall t \in [0, T[.$$

Proof.

Let $z_0 \leq \frac{1}{2}\sqrt{\alpha/2\gamma}$; since $t \mapsto z(t)$ is continuous, there exists a maximum time $T_1 \in]0, T]$ such that $z(t)$ remains less than $\sqrt{\alpha/2\gamma}$ on the interval $[0, T_1[$. If $T_1 = T$, the lemma is proven. Hence, we suppose that $T_1 < T$ and we are going to prove that this leads to a contradiction.

For $t \leq T_1$, $\gamma z^2(t) - \alpha$ remains less than $-\alpha/2$ and hence we have the inequality:

$$z'(t) \leq \beta + z(t)(\gamma z(t)^2 - \alpha) \leq \beta - \frac{\alpha}{2} z(t).$$

Thus, from Lemma II.4.9, for all $t \leq T_1$, we have

$$z(t) \leq z_0 e^{-\alpha/2t} + \frac{2\beta}{\alpha},$$

so that from the hypotheses on z_0, α, and β, we get

$$z(T_1) \leq z_0 e^{-\alpha/2T_1} + \frac{1}{2}\sqrt{\frac{\alpha}{2\gamma}} < \sqrt{\frac{\alpha}{2\gamma}}.$$

Since the function z is continuous, we see that there exists $\varepsilon > 0$ such that $z(t) \leq \sqrt{\alpha/2\gamma}$ for all $t \in [T_1, T_1 + \varepsilon]$, which contradicts the definition of T_1. $\square$

We can now prove the globality of strong solutions of the Navier–Stokes equations in dimension 3 for small data.

Proof (of Theorem V.2.2).

From Proposition IV.5.12, inequality (V.39) gives:

$$\begin{aligned}\frac{d}{dt}\|\nabla v_N(t)\|_{L^2}^2 + \frac{\lambda_1}{\mathcal{R}e}\|\nabla v_N(t)\|_{L^2}^2 &\leq C\mathcal{R}e\|f_N\|_{L^2}^2 + C\mathcal{R}e^3\|\nabla v_N(t)\|_{L^2}^6 \\ &\leq C\mathcal{R}e\|f\|_{L^\infty(\mathbb{R}^+, L^2)}^2 + C\mathcal{R}e^3\|\nabla v_N(t)\|_{L^2}^6,\end{aligned}$$

and the preceding lemma allows us to conclude that under the hypothesis

$$\|\nabla v_0\|_{L^2}^2 \leq \frac{C(\Omega)}{\mathcal{R}e^2} \text{ and } \|f\|_{L^\infty(\mathbb{R}^+, L^2)}^2 \leq \frac{C(\Omega)}{\mathcal{R}e^4},$$

where $C(\Omega)$ depends on λ_1 and the Sobolev constants on the open set Ω, then there is no finite time blow-up; that is, $T_N = +\infty$ and the sequence $(\nabla v_N)_N$ is bounded in $L^\infty(\mathbb{R}^+, (L^2(\Omega))^{d\times d})$. The proof is complete

□

2.4 Parabolic regularity properties

Theorem V.2.10. *Let Ω be a connected bounded domain of $\mathbb{R}^d$ of class $\mathcal{C}^{k,1}$, with $k \geq 1$. Let $\mathcal{R}e > 0$, v_0 be in V and f be in $L^2_{loc}([0,+\infty[, (L^2(\Omega))^d)$. From Theorem V.2.1, there exists $0 < T^* \leq +\infty$ and a unique pair (v,p), with $m(p) = 0$, which is a strong solution of the Navier–Stokes equation (V.1) in the sense of Theorem V.2.1, for the initial data v_0.*

We now assume that we have

$$v_0 \in (H^k(\Omega))^d,$$

and for all $s \leq k/2$,

$$\frac{\partial^s f}{\partial t^s} \in L^2(]0,T[, (H^{k-2s-1}(\Omega))^d).$$

Hence, assuming compatibility conditions (V.54) at the boundary on the data, that we detail in the proof, the solution (v,p) of the Navier–Stokes equations possesses the following regularity properties

$$\begin{aligned}
\frac{\partial^s v}{\partial t^s} &\in L^\infty(]0,T[, (H^{k-2s}(\Omega))^d \cap V), \ \ \forall s \in \mathbb{N}, s \leq \frac{k}{2},\\
\frac{\partial^s v}{\partial t^s} &\in L^2(]0,T[, (H^{k-2s+1}(\Omega))^d \cap V), \ \ \forall s \in \mathbb{N}, s \leq \frac{k}{2},\\
\frac{\partial^s v}{\partial t^s} &\in L^2(]0,T[, V_{k-2s+1}), \ \ \textit{for} \ \ s = E\left(\frac{k}{2}+1\right),\\
\frac{\partial^s p}{\partial t^s} &\in L^\infty(]0,T[, H^{k-2s-1}(\Omega)), \ \ \forall s \in \mathbb{N}, s \leq \frac{k}{2}-1,\\
\frac{\partial^s p}{\partial t^s} &\in L^2(]0,T[, H^{k-2s}(\Omega)), \ \ \forall s \in \mathbb{N}, s < \frac{k}{2}.
\end{aligned}$$

Before proving this theorem, we state and demonstrate a direct corollary of this result.

Corollary V.2.11. *Under the conditions of the preceding theorem, we have*

$$\frac{\partial^s v}{\partial t^s} \in \mathcal{C}^0([0,T], (H^{k-2s}(\Omega))^d \cap V), \ \forall s \in \mathbb{N}, s < \frac{k}{2},$$

$$\frac{\partial^s v}{\partial t^s} \in \mathcal{C}^0([0,T], H), \quad \textit{if } s = \frac{k}{2} \in \mathbb{N},$$

$$\frac{\partial^s p}{\partial t^s} \in \mathcal{C}^0([0,T], H^{k-2s-1}(\Omega)),\ \forall s \in \mathbb{N}, s < \frac{k}{2} - 1.$$

Proof (of the corollary).

It is sufficient to use the results of the preceding theorem as well as Corollary II.5.13 (or Theorem IV.5.11).

□

Proof (of the theorem).

In order to simplify the calculations, in the proof that follows we choose a Reynolds number equal to 1 such that (v, p) is a solution of

$$\begin{cases} \dfrac{\partial v}{\partial t} + (v \cdot \nabla)v - \Delta v + \nabla p = f, & \text{in } \Omega, \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ v = 0, & \text{on } \partial\Omega. \end{cases}$$

Below, in the usual way, we will make formal energy estimates on these equations. To perform the complete proof, it should be necessary to make these estimates over the Galerkin approximations of the solution (which are as regular as we wish provided that the domain is sufficiently regular) and then pass to the limit. This process is now standard and we leave the details as an exercise for the reader.

Furthermore, to conserve the readability of the calculations, we define until the end of the proof

$$v^{(s)} = \frac{\partial^s v}{\partial t^s},\ p^{(s)} = \frac{\partial^s p}{\partial t^s},\ f^{(s)} = \frac{\partial^s f}{\partial t^s}.$$

Let us perform a proof by induction. The case $k = 1$ is simply an alternative expression of the regularity obtained in Theorem V.2.1.

Hence, let $k \geq 2$ and let us assume the required result up to row $k-1$ and then let us demonstrate it to row k. The proof differs depending on the parity of k.

- First case: odd k.

  We write $k = 2S+1$. The idea is to differentiate the equation s times with respect to time, for all $0 \leq s \leq S$

$$\frac{\partial v^{(s)}}{\partial t} - \Delta v^{(s)} + \sum_{r=0}^{s} \binom{s}{r} (v^{(r)} \cdot \nabla) v^{(s-r)} + \nabla p^{(s)} = f^{(s)}. \tag{V.51}$$

  More particularly, at time $t = 0$, we find for all $s \leq S$

$$\begin{aligned}v_0^{(s)} = v^{(s)}_{|t=0} &= \frac{\partial v^{(s-1)}}{\partial t}\Big|_{t=0}\\ &= \Delta v_0^{(s-1)} - \sum_{r=0}^{s-1}\binom{s-1}{r}(v_0^{(r)}\cdot\nabla)v_0^{(s-1-r)} - \nabla p_0^{(s-1)} + f_0^{(s-1)}.\end{aligned}$$

We can then project this equation onto the set of divergence-free vector fields to obtain

$$v_0^{(s)} = -Av_0^{(s-1)} - \sum_{r=0}^{s-1}\binom{s-1}{r}\mathcal{P}\Big((v_0^{(r)}\cdot\nabla)v_0^{(s-1-r)}\Big) + \mathcal{P}f_0^{(s-1)}. \quad \text{(V.52)}$$

We now note that these equations do, indeed, have a meaning. Indeed, the result is assumed to be true for the row $k-1$, therefore we can apply Corollary V.2.11 to row $k-1$, which proves that for all $s \leq S$ the trace of $v^{(s)}$ at $t = 0$ is indeed defined. Similarly, the hypothesis of regularity on f and on all its derivatives with respect to time shows that $f_0^{(s)}$ is, indeed, defined and we have

$$f_0^{(s)} \in (H^{k-2s-2}(\Omega))^d.$$

From the hypothesis $v_0 \in (H^k(\Omega))^d$ and from Equations (V.52), it is now easy to show (by an immediate induction) that for all $s \leq S$, we have

$$v_0^{(s)} \in (H^{k-2s}(\Omega))^d. \quad \text{(V.53)}$$

For this, we use the fact that the Stokes operator maps $(H^t(\Omega))^d$ to $(H^{t-2}(\Omega))^d$ and the fact that the product of two functions, respectively, in H^{s_1} and H^{s_2} (over all $s_1, s_2 > 0$), belongs to H^s with $s = \min(s_1, s_2)$ except if $s_1 = s_2 = 1$ in which case $s = 0$. The simple proof of this result is left to the reader.

We can now give the compatibility condition that is required to obtain the regularity to all orders of the solution. Intuitively, since v vanishes at the boundary, we see that if v is sufficiently regular, then we must also have $v^{(s)} = 0$ on the boundary. Hence, if we hope to have continuity up to $t = 0$ of $v^{(s)}$ with values in $(H^1(\Omega))^d$, for example, then we must have $v_0^{(s)} = 0$ on the boundary of Ω. We then assume, in all of the following, that $v_0^{(s)}$, defined by recurrence using Equation (V.52), satisfies the condition

$$v_0^{(s)} = 0 \text{ on } \partial\Omega. \quad \text{(V.54)}$$

Under this hypothesis, we see in particular that by taking $s = S$ in (V.53), we have shown that

$$v_0^{(S)} \in V.$$

We then return to Equation (V.51) with $s = S$, where we now know that the initial data belong to V. The idea is therefore to multiply this equation by $Av^{(S)}$ and integrate by parts to obtain the following energy estimate

$$\begin{aligned}&\frac{1}{2}\frac{d}{dt}\|\nabla v^{(S)}\|_{L^2}^2+\|Av^{(S)}\|_{L^2}^2\\ \leq&\frac{1}{2}\|Av^{(S)}\|_{L^2}^2+C\sum_{r=0}^{S}\|(v^{(r)}\cdot\nabla)v^{(S-r)}\|_{L^2}^2+C\|f^{(S)}\|_{L^2}^2.\end{aligned}\tag{V.55}$$

From the induction hypothesis, for all $r\leq S$, we know that

$$v^{(r)}\in L^\infty(]0,T[,(H^{k-2r-1}(\Omega))^d\cap V)\cap L^2(]0,T[,(H^{k-2r}(\Omega))^d\cap V),$$

and hence in particular for all $r\leq S-1$,

$$\begin{cases}v^{(r)}\in L^\infty(]0,T[,(H^2(\Omega))^d\cap V)\cap L^2(]0,T[,(H^3(\Omega))^d\cap V),\\ v^{(S)}\in L^\infty(]0,T[,H)\cap L^2(]0,T[,V).\end{cases}\tag{V.56}$$

Hence, we can control the nonlinear term of the estimate (V.55) in the following way, by separating out the case where $r=S$,

$$\begin{aligned}&\sum_{r=0}^{S}\|(v^{(r)}\cdot\nabla)v^{(S-r)}\|_{L^2}^2\\ \leq&\sum_{r=0}^{S-1}\|v^{(r)}\|_{L^\infty}^2\|\nabla v^{(S-r)}\|_{L^2}^2+\|v^{(S)}\|_{L^4}^2\|\nabla v^{(0)}\|_{L^4}^2\\ \leq&C\left(\sum_{r=0}^{S-1}\|v^{(r)}\|_{H^2}^2\|v^{(S-r)}\|_{H^1}^2+\|v^{(S)}\|_{H^1}^2\|v^{(0)}\|_{H^1}^2\right)\\ \leq&k_1(t)+k_2(t)\|\nabla v^{(S)}\|_{L^2}^2,\end{aligned}$$

where k_1 and k_2 are two functions of t belonging to $L^1(]0,T[)$ from (V.56). Finally, (V.55) gives

$$\frac{d}{dt}\|\nabla v^{(S)}\|_{L^2}^2+\|Av^{(S)}\|_{L^2}^2\leq C\|f^{(S)}\|_{L^2}^2+Ck_1(t)+Ck_2(t)\|\nabla v^{(S)}\|_{L^2}^2,$$

and we can use the Gronwall lemma to conclude that

$$v^{(S)}\in L^\infty(]0,T[,V)\cap L^2(]0,T[,(H^2(\Omega))^d\cap V).$$

This concludes the first part of the claim.
We now write Equation (V.51) taking $s=S$ which, after projection onto the set of divergence-free vector fields, reads

$$v^{(S+1)}=\frac{\partial v^{(S)}}{\partial t}=-Av^{(S)}-\sum_{r=0}^{S}\binom{S}{r}\mathcal{P}\Big((v^{(r)}\cdot\nabla)v^{(S-r)}\Big)+\mathcal{P}f^{(S)}.$$

Through the hypotheses on $f^{(s)}$ and the result obtained above, we immediately obtain that

$$v^{(S+1)} \in L^\infty(]0,T[,V') \cap L^2(]0,T[,H).$$

Now, we can rewrite Equation (V.51) taking $s = S$ but in a slightly different form

$$Av^{(S)} = -\Delta v^{(S)} + \nabla p^{(S)} = f^{(S)} - \frac{\partial v^{(S)}}{\partial t} + \sum_{r=0}^{S} \binom{S}{r} (v^{(r)} \cdot \nabla) v^{(S-r)},$$

which let us see the pair $(v^{(S)}, p^{(S)})$ as the solution of a certain Stokes-like problem. However, the preceding results obtained on $v^{(S+1)}$, as well as the hypothesis on f, show that the source term of this Stokes-like problem belongs to $L^\infty(]0,T[,V') \cap L^2(]0,T[,(L^2(\Omega))^d)$. Hence, by using the regularity theorem for the Stokes problem (see Proposition IV.5.9), and the fact that the operator A is an isomorphism from V to V', we obtain

$$\begin{aligned} &v^{(S)} \in L^\infty(]0,T[,V) \cap L^2(]0,T[,(H^2(\Omega))^d \cap V),\\ &p^{(S)} \in L^2(]0,T[,H^1(\Omega)). \end{aligned}$$

The end of the proof is now clear. We proceed in the same way by reverse induction from $s = S$ to $s = 0$, writing each time that $(v^{(s)}, p^{(s)})$ is a solution to the Stokes problem

$$-\Delta v^{(s)} + \nabla p^{(s)} = f^{(s)} - \frac{\partial v^{(s)}}{\partial t} + \sum_{r=0}^{s} \binom{s}{r} (v^{(r)} \cdot \nabla) v^{(s-r)},$$

for which the regularity of the source term is given by the results of row $s+1$. This concludes the proof for the case where k is odd.

- Second case: even k.
  We write $k = 2S$. The principle is entirely identical to the preceding case and we do not detail the proof, which is also rather simple. Firstly, the first part of the proof is strictly the same so that (V.53) is again true. However, now $k = 2S$ and hence we simply have

$$v_0^{(S)} \in H.$$

Thus, we can easily carry out a L^2 estimate on Equation (V.51) for $s = S$. This is obtained by multiplying the equation by $v^{(S)}$ and integrating by parts. We obtain

$$\frac{1}{2}\frac{d}{dt}\|v^{(S)}\|_{L^2}^2 + \|\nabla v^{(S)}\|_{L^2}^2 \le \frac{1}{2}\|\nabla v^{(S)}\|_{L^2}^2 + \frac{1}{2}\|f^{(S)}\|_{H^{-1}}^2$$
$$+ C\sum_{r=0}^{S}\left|\int_\Omega \left((v^{(r)}\cdot\nabla)v^{(S-r)}\right)\cdot v^{(S)}\,dx\right|.$$

We note that the nonlinear term, corresponding to the index $r = 0$, is zero due to property (V.2). The nonlinear terms are again (for $1 \le r \le S$) estimated in a similar way to the case where k is odd by using the induction hypothesis on row $k-1$. Furthermore, the hypothesis on f says that $f^{(S)}$ lies in $L^2(]0,T[,(H^{-1}(\Omega))^d)$.
Finally, we obtain

$$v^{(S)} \in L^\infty(]0,T[,H) \cap L^2(]0,T[,V).$$

The end of this proof is then exactly the same; the various regularities of $v^{(s)}$ and $p^{(s)}$ are obtained by writing that the pair $(v^{(s)},p^{(s)})$ is a solution of the Stokes problem for which the regularity is known by descending recurrence in s.

□

2.5 Regularisation over time

As for parabolic nondegenerate problems (for which the simplest model is the heat equation), the weak solutions of the Navier–Stokes equations are regularised over time. More exactly, in the two-dimensional case, if the open set Ω is of class $\mathcal{C}^\infty$ and the data f are smooth enough, then from an initial condition v_0 in H, the solution satisfies

$$v \in \mathcal{C}^\infty\left(]0,T[,\ V \cap (H^k(\Omega))^d\right), \quad \text{for all } k,$$

one such regularity being certainly false on the whole interval $[0,T[$.

This property illustrates the irreversibility of diffusion phenomena. We do not fall into the trap of believing that this occurs for weak solutions in 3D! Indeed, not having uniqueness for such solutions, we could not hope for a result of the type: "under certain assumptions, there exist weak solutions which regularise themselves along the time." Such results have no physical relevance. On the contrary, if we consider local strong solutions in 3D, they also regularise themselves along the time, over the whole time interval on which the solution exists.

We establish this phenomenon for $k \le 2$ and in the case where f does not depend on time. The more general case can be obtained by combining the proof which follows with the method used in the preceding subsection to prove Theorem V.2.10.

To obtain this result we proceed by the Galerkin approximation and perform the estimates which follow for the approximate solutions; this tedious step, with which the reader is already familiar, is omitted. We content ourselves with carrying out the formal calculations.

Theorem V.2.12. *Let Ω be a connected bounded domain of $\mathbb{R}^2$ of class $\mathcal{C}^{1,1}$, $\mathcal{R}e > 0$, let v_0 be given in H, and f be given in $(L^2(\Omega))^d$, independent of time. Then for all $\varepsilon > 0$, the unique weak solution (v, p) of the Navier–Stokes equations satisfies*

$$v \in \mathcal{C}^1([\varepsilon, +\infty[,\ H) \cap \mathcal{C}^0([\varepsilon, +\infty[,\ V) \cap L^2_{loc}(]\varepsilon, +\infty[,\ V \cap (H^2(\Omega))^d),$$

and

$$p \in L^2_{loc}([\varepsilon, +\infty[, H^1(\Omega)).$$

Proof.

In order to simplify the notation, from here on we denote the derivative of v with respect to time as $\dot{v}$. Let $\psi \in V$ be fixed independent of time. Almost everywhere, the solution v satisfies

$$\langle \dot{v}, \psi \rangle_{V',V} + \frac{1}{\mathcal{R}e} \int_\Omega \nabla v : \nabla \psi \, dx + \int_\Omega ((v \cdot \nabla) v) \cdot \psi \, dx = \int_\Omega f \cdot \psi \, dx.$$

After multiplication by t and differentiation with respect to time, by introducing $w = t\dot{v}$, we obtain

$$\left\langle \frac{dw}{dt}, \psi \right\rangle_{V',V} + \frac{1}{\mathcal{R}e} \int_\Omega \nabla w : \nabla \psi \, dx + \frac{1}{\mathcal{R}e} \int_\Omega \nabla v : \nabla \psi \, dx + \int_\Omega ((w \cdot \nabla) v) \cdot \psi \, dx$$
$$+ \int_\Omega ((v \cdot \nabla) w) \cdot \psi \, dx + \int_\Omega ((v \cdot \nabla) v) \cdot \psi \, dx = \int_\Omega f \cdot \psi \, dx.$$

By taking $\psi = w(t)$ in this identity, and since $\int_\Omega ((v \cdot \nabla) w) \cdot w \, dx = 0$ (due to (V.2)), we find that the function w satisfies the energy estimate

$$\begin{aligned}
\frac{1}{2} \frac{d}{dt} \|w\|^2_{L^2} + \frac{1}{\mathcal{R}e} \|\nabla w\|^2_{L^2} =& - \frac{1}{\mathcal{R}e} \int_\Omega \nabla v : \nabla w \, dx - \int_\Omega ((w \cdot \nabla) v) \cdot w \, dx \\
& - \int_\Omega ((v \cdot \nabla) v) \cdot w \, dx + \int_\Omega f \cdot w \, dx \\
\leq& \frac{1}{\mathcal{R}e} \|\nabla v\|_{L^2} \|\nabla w\|_{L^2} + C \|\nabla v\|_{L^2} \|w\|_{L^2} \|\nabla w\|_{L^2} \\
& + C \|v\|^{1/2}_{L^2} \|\nabla v\|^{3/2}_{L^2} \|w\|^{1/2}_{L^2} \|\nabla w\|^{1/2}_{L^2} + \|f\|_{L^2} \|w\|_{L^2} \\
\leq& g(t) \|w\|^2_{L^2} + h(t) + \frac{1}{2\mathcal{R}e} \|\nabla w\|^2_{L^2},
\end{aligned}$$

where g and h are two functions of $L^1(]0, T[)$ for fixed $T > 0$, because v is a weak solution of the problem and hence satisfies the corresponding energy estimates. Hence, we have

$$\frac{d}{dt}\|w\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\|\nabla w\|_{L^2}^2 \leq 2g(t)\|w\|_{L^2}^2 + 2h(t).$$

Since $w(0) = 0$, we can deduce from this inequality, by the Gronwall lemma, that for all $T > 0$,

$$w \in L^\infty(]0, T[, H) \cap L^2(]0, T[, V),$$

which implies in the classic way (see the proof of the Leray theorem), that

$$\frac{dw}{dt} \in L^2(]0, T[, V').$$

All this demonstrates (by Theorem II.5.13), that

$$w \in \mathcal{C}^0([0, T], H).$$

However $w = tdv/dt$ and so we see that for all $\varepsilon > 0$, and $T > 0$ we have

$$v \in \mathcal{C}^1([\varepsilon, T], H) \cap \mathcal{C}^0([\varepsilon, T]; V).$$

Since $v(\varepsilon) \in V$, the function $t \mapsto v(t)$ is also a strong solution of the Navier–Stokes equations on the interval $[\varepsilon, T]$ (here we use the uniqueness of the weak solutions in dimension 2). Hence, we indeed obtain

$$v \in L^2(]\varepsilon, T[,\ V \cap (H^2(\Omega))^d),$$

and

$$p \in L^2(]\varepsilon, T[,\ H^1(\Omega)).$$

$\square$

3 The steady Navier–Stokes equations

In this section, we deal with steady solutions for the Navier–Stokes equations. We first establish the existence of at least one solution for space dimension 2 and 3. We consider homogeneous and non-homogenous Dirichlet boundary conditions. Then we prove the uniqueness of such solution for small data and we finally establish asymptotic stability properties of such solutions.

3.1 The case of homogeneous boundary conditions

3.1.1 Existence

Theorem V.3.1. *Let Ω be a Lipschitz bounded domain in $\mathbb{R}^d$, $d \in \{2,3\}$. Let $f \in \left(H^{-1}(\Omega)\right)^d$ be given. There exists at least one solution (v,p) in $\left(H^1(\Omega)\right)^d \times L^2_0(\Omega)$ of:*

$$\begin{cases} -\dfrac{1}{\mathcal{R}e}\Delta v + (v\cdot\nabla)v + \nabla p = f, & \text{in } \Omega,\\ \operatorname{div} v = 0, & \text{in } \Omega,\\ v = 0, & \text{on } \partial\Omega. \end{cases} \tag{V.57}$$

Any such solution satisfies

$$\|\nabla v\|_{L^2} \leq \mathcal{R}e\|f\|_{H^{-1}} \text{ and } \|p\|_{L^2} \leq C\left(\|f\|_{H^{-1}} + \mathcal{R}e^2\|f\|^2_{H^{-1}}\right).$$

Proof.

Let us consider the special basis introduced in Definition IV.5.6. For $N \geq 1$, let us denote by H_N the finite-dimension vector space spanned by $(w_j)_{1\leq j\leq N}$. We look for $v_N \in H_N$, $v_N = \sum_{j=1}^N \alpha_j w_j$ solution of

$$\begin{aligned} &\frac{1}{\mathcal{R}e}\int_\Omega \nabla v_N : \nabla \psi_N \, dx + \int_\Omega \big((v_N\cdot\nabla)v_N\big)\cdot\psi_N\, dx \\ &= \langle f, \psi_N\rangle_{H^{-1},H^1_0}, \ \forall \psi_N \in H_N. \end{aligned} \tag{V.58}$$

To this end, we introduce the map P defined on $\mathbb{R}^N$ by $P(\alpha) = \beta$, where the coordinates β_j are equal to

$$\beta_j = \frac{1}{\mathcal{R}e}\int_\Omega \nabla v_N : \nabla w_j\, dx + \int_\Omega \big((v_N\cdot\nabla)v_N\big)\cdot w_j\, dx - \langle f, w_j\rangle_{H^{-1},H^1_0}.$$

From Equation (V.58) one has

$$\begin{aligned} a\cdot P(\alpha) =& \frac{1}{\mathcal{R}e}\|\nabla v_N\|^2_{L^2} + \underbrace{\int_\Omega \big((v_N\cdot\nabla)v_N\big)\cdot v_N\, dx}_{=0, \text{ by (V.2)}} - \langle f, v_N\rangle_{H^{-1},H^1_0}, \\ \geq& \frac{1}{\mathcal{R}e}\|\nabla v_N\|^2_{L^2} - \|f\|_{H^{-1}}\|v_N\|_{H^1_0}. \end{aligned} \tag{V.59}$$

Since $\|v_N\|_{H^1_0} = \|\nabla v_N\|_{L^2}$, we observe that $a\cdot P(\alpha) \geq 0$ as soon as $\|v_N\|_{H^1_0} \geq \mathcal{R}e\|f\|_{H^{-1}}$. This condition is fulfilled if $|\alpha| = \rho$, where ρ is large enough. Therefore we can apply the Brouwer theorem (Proposition II.3.11), which ensures the existence of at least one $\alpha \in \mathbb{R}^N$ such that $P(\alpha) = 0$ and thus gives one solution of (V.58).

Moreover, $P(\alpha) = 0$, therefore the inequality (V.59) gives the estimate

$$\|\nabla v_N\|_{L^2} \leq \mathcal{R}e \, \|f\|_{H^{-1}}.$$

Hence, there exists $v \in (H_0^1(\Omega))^d$ and a subsequence $(v_{N_k})_k$ such that:

$$\begin{cases} v_{N_k} \xrightharpoonup[k\to\infty]{} v, & \text{weakly in } \left(H_0^1(\Omega)\right)^d, \\ v_{N_k} \xrightarrow[k\to\infty]{} v, & \text{in } \left(L^p(\Omega)\right)^d, \forall p < 6. \end{cases}$$

In particular, by Corollary II.2.8, we have

$$\|\nabla v\|_{L^2} \leq \liminf_{k\to\infty} \|\nabla v_k\|_{L^2} \leq \mathcal{R}e \, \|f\|_{H^{-1}}.$$

The convergences above imply, by Proposition II.2.12, that

$$\left((v_{N_k} \cdot \nabla) v_{N_k}\right)_k \xrightharpoonup[k\to\infty]{} (v \cdot \nabla) v, \quad \text{weakly in } \left(L^p(\Omega)\right)^d, \forall p < \frac{6}{5}.$$

Moreover the sequence $\left((v_{N_k} \cdot \nabla) v_{N_k}\right)$ is bounded in $\left(L^{6/5}(\Omega)\right)^d$ and thus, by Proposition II.2.10, we get

$$\left((v_{N_k} \cdot \nabla) v_{N_k}\right)_k \xrightharpoonup[k\to\infty]{} (v \cdot \nabla) v, \quad \text{weakly in } \left(L^{6/5}(\Omega)\right)^d.$$

For a fixed $\psi_N \in H_N \subset (L^6(\Omega))^d$, we take the limit as $k \to \infty$ in (V.58) to obtain

$$\frac{1}{\mathcal{R}e} \int_\Omega \nabla v : \nabla \psi_N \, dx + \int_\Omega \left((v \cdot \nabla) v\right) \cdot \psi_N \, dx = \langle f, \psi_N \rangle_{H^{-1}, H_0^1}, \ \forall \psi_N \in H_N.$$

By density of $\cup_N H_N$ in V, and using the embedding $V \subset L^6(\Omega)$ (which is true as soon as $d \leq 3$), we obtain that v satisfies

$$\frac{1}{\mathcal{R}e} \int_\Omega \nabla v : \nabla \psi \, dx + \int_\Omega \left((v \cdot \nabla) v\right) \cdot \psi \, dx = \langle f, \psi \rangle_{H^{-1}, H_0^1}, \ \forall \psi \in V. \quad \text{(V.60)}$$

Let us denote by F the element of $\left(H^{-1}(\Omega)\right)^d$ defined by

$$F = -\frac{1}{\mathcal{R}e} \Delta v + (v \cdot \nabla) v - f.$$

Formula (V.60) shows that F satisfies

$$\langle F, \psi \rangle_{H^{-1}, H_0^1} = 0, \quad \forall \psi \in V.$$

Therefore, we deduce from Theorem IV.2.4 that there exists $p \in L^2_0(\Omega)$, such that $F = \nabla p$, and the claim is proved.

□

3.1.2 Regularity

When we assume that the source term f belongs to $\big(L^2(\Omega)\big)^d$, and that the domain Ω is smooth enough, the solution (v,p) of (V.57) is more regular. More precisely, we have the following result.

Theorem V.3.2. *We suppose that Ω is of class $\mathcal{C}^{1,1}$ and that $f \in \big(L^2(\Omega)\big)^d$, then we have*

$$(v,p) \in \big(H^2(\Omega)\big)^d \times H^1(\Omega).$$

Proof.

The solution (v,p) of (V.57) that we found above satisfies the problem

$$\begin{cases} -\dfrac{1}{\mathcal{R}e}\Delta v + \nabla p = f - (v\cdot\nabla)v, & \text{in } \Omega, \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ v = 0, & \text{on } \partial\Omega. \end{cases} \tag{V.61}$$

Let us first show that $v \in (L^q(\Omega))^d$ for any $q < +\infty$.

- In the case $d = 2$, this property is obvious because the Sobolev embeddings (Theorem III.2.34) assert that $H^1(\Omega) \subset L^q(\Omega)$ for any $q < +\infty$.
- In the case $d = 3$, the Sobolev embedding only gives $H^1(\Omega) \subset L^6(\Omega)$. From this, since $\nabla v \in (L^2(\Omega))^{3\times 3}$, we deduce that the product $(v\cdot\nabla)v$ belongs to $(L^{3/2}(\Omega))^3$ and thus that we have

$$f - (v\cdot\nabla)v \in (L^{3/2}(\Omega))^3.$$

  From Theorem IV.6.6 applied to Problem (V.61), we deduce that $v \in (W^{2,3/2}(\Omega))^3$. By Sobolev embeddings, we conclude that $v \in (W^{1,3}(\Omega))^3$ and then that $v \in (L^q(\Omega))^3$ for any $q < +\infty$.

It is now clear that the product $(v\cdot\nabla)v$ belongs to $(L^p(\Omega))^d$ for any $p < 2$ and thus that the right-hand side in (V.61) also belongs to $(L^p(\Omega))^d$ for any $p < 2$. Using once more Theorem IV.6.6, we deduce that $v \in (W^{2,p}(\Omega))^d$ for any $p < +\infty$ and finally that $v \in (L^\infty(\Omega))^d$.

At last, we see that the product $(v\cdot\nabla)v$ actually lies in $(L^2(\Omega))^d$. Theorem IV.6.6 (see also Theorem IV.5.8) let us conclude that $(v,p) \in (H^2(\Omega))^d \times H^1(\Omega)$.

□

Remark V.3.1. Using the same strategy, we can show that if the source term f belongs to $\left(L^p(\Omega)\right)^d$, $2 < p < \infty$, then the solution (v,p) belongs to $\left(W^{2,p}(\Omega)\right)^d \times W^{1,p}(\Omega)$.

3.2 The case of nonhomogeneous boundary conditions

We now want to study the case where the velocity field v satisfies the nonhomogeneous Dirichlet boundary condition on $\partial\Omega$. For simplicity we only consider in this section that $d = 3$ but similar results also hold in the two-dimensional case.

For given data f and g satisfying suitable assumptions, we prove the existence of at least one solution for the steady Navier–Stokes problem

$$\begin{cases} -\dfrac{1}{\mathcal{R}e}\Delta v + (v\cdot\nabla)v + \nabla p = f, & \text{in } \Omega, \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ v = g, & \text{on } \partial\Omega. \end{cases} \tag{V.62}$$

The first natural compatibility condition for g is related to the divergence-free constraint. Indeed, the divergence theorem shows that if a solution exists we have

$$0 = \int_\Omega \operatorname{div} v\,dx = \int_{\partial\Omega} g\cdot\nu\,d\sigma.$$

In this section we suppose, for the sake of simplicity, that the domain Ω is simply connected. Note that the case of multiply connected domains can be treated by similar methods. Let us denote by $(\Gamma_i)_{1\leq i\leq m}$ the connected components of the boundary $\partial\Omega$. For any boundary data g, we define

$$\mathcal{Q}_{\partial\Omega}(g) = \sum_{i=1}^m \left|\int_{\Gamma_i} g\cdot\nu\,d\sigma\right|,$$

which is the sum of all the flow rates across every connected components of the boundary.

We are now able to state and prove the main result of this section.

Theorem V.3.3. *Let Ω be a bounded simply connected Lipschitz domain of $\mathbb{R}^d$, with $d \in \{2,3\}$. There exist $C_{1,\Omega}, C_{2,\Omega} > 0$ depending only on Ω, such that for any $f \in \left(H^{-1}(\Omega)\right)^d$ and $g \in \left(H^{1/2}(\partial\Omega)\right)^d$ satisfying*

$$\begin{cases} \displaystyle\int_{\partial\Omega} g\cdot\nu\,d\sigma = 0, \\ \mathcal{Q}_{\partial\Omega}(g) \leq \dfrac{C_{1,\Omega}}{\mathcal{R}e}, \end{cases} \tag{V.63}$$

then Problem (V.62) *has at least one solution* $(v,p) \in \big(H^1_0(\Omega)\big)^d \times L^2_0(\Omega)$ *which satisfies*

$$\|\nabla v\|_{L^2} \leq C_{2,\Omega}\mathcal{R}e\|f\|_{H^{-1}} + C_{2,\Omega}\|g\|_{H^{1/2}} \exp(C_{2,\Omega}\mathcal{R}e\|g\|_{H^{1/2}})\,. \tag{V.64}$$

We will see that the smallness condition (V.63) is certainly useful in the following proof. Nevertheless, it is not known up to now if this condition is really necessary for the steady Navier–Stokes problem to have a solution. We refer to [63] for a more detailed discussion on this issue.

The key point of the proof of this theorem is to build a suitable lifting of the boundary data in order to transform the problem into a case with homogeneous Dirichlet boundary conditions. The problem is nonlinear, thus this lifting has to be small enough to let us perform energy estimates.

We consider boundary data g such as in the statement of the theorem and we define

$$\mu_i = \frac{1}{|\Gamma_i|}\int_{\Gamma_i} g\cdot\nu\,d\sigma,\ \forall i\in\{1,\dots,m\}.$$

We observe that, by assumption, we have

$$\sum_{i=1}^m \mu_i\,|\Gamma_i| = \int_{\partial\Omega} g\cdot\nu\,d\sigma = 0.$$

Therefore, according to Theorem IV.5.2, we can find a vector field $F_1 \in \big(H^1(\Omega)\big)^d$ satisfying, for some $\pi_1 \in L^2(\Omega)$, the equations

$$\begin{cases} -\Delta F_1 + \nabla\pi_1 = 0, & \text{in } \Omega,\\ \operatorname{div} F_1 = 0, & \text{in } \Omega,\\ F_1 = \mu_i\,\nu, & \text{on } \Gamma_i,\quad \forall i\in\{1,\dots,m\},\\ \|F_1\|_{H^1} \leq C\displaystyle\sum_{i=1}^m |\mu_i| \leq C\mathcal{Q}_{\partial\Omega}(g). \end{cases} \tag{V.65}$$

Note that we only use that F_1 is a divergence-free lifting of the trace $\mu_i\nu$ on Γ_i. In particular the scalar field π_1 is useless in the sequel. By construction, the function $g - F_1$ satisfies on the boundary

$$\begin{cases} \displaystyle\int_{\partial\Omega}(g-F_1)\cdot\nu\,d\sigma = 0,\\ \displaystyle\int_{\Gamma_i}(g-F_1)\cdot\nu\,d\sigma = 0,\ \forall i\in\{1,\dots,m\}. \end{cases} \tag{V.66}$$

Applying once more Theorem IV.5.2 we can find a divergence-free vector field $F_2 \in \big(H^1(\Omega)\big)^d$ satisfying $F_2 = (g - F_1)$ on $\partial\Omega$ and $\|F_2\|_{H^1} \leq C\|g - F_1\|_{H^{1/2}}$.

Under the assumption (V.63), the estimate (V.65) says that the term F_1 can be made small but we do not know anything about the size of the term F_2.

The following result (see [71]) shows that one can modify F_2 in the interior of Ω, in such a way that F_2 is small enough in a weak sense which is sufficient for our purpose.

Lemma V.3.4 (Hopf's lemma). *For all $\alpha > 0$, there exists a vector field $G \in \left(H^1(\Omega)\right)^d$, such that $\operatorname{div} G = 0$, $G = F_2$ on $\partial\Omega$ and satisfying*

$$\|G \otimes w\|_{L^2} \leq \alpha \|w\|_{H_0^1}, \ \forall w \in (H_0^1(\Omega))^d, \tag{V.67}$$

and

$$\|\nabla G\|_{L^2} \leq C\|g\|_{H^{1/2}} \exp\left(\frac{C}{\alpha}\|g\|_{H^{1/2}}\right). \tag{V.68}$$

We refer, for instance, to [70] for a discussion about the more general conditions on the domain and on the boundary data which ensure the validity of this lemma.

Proof.

We recall that we only consider here the three-dimensional case. Let us first notice that we can simply take $G = F_2$ if α is large enough. Indeed we have

$$\|F_2 \otimes v\|_{L^2} \leq \|F_2\|_{L^3}\|v\|_{L^6} \leq C\|F_2\|_{H^1}\|v\|_{H_0^1} \leq C'\|g - F_1\|_{H^{1/2}}\|v\|_{H_0^1},$$

and then (V.67) is satisfied as soon as $\alpha \geq C'\|g - F_1\|_{H^{1/2}}$. Note that (V.68) is also satisfied in that case.

From now on, we assume that

$$\alpha \leq C'\|g - F_1\|_{H^{1/2}}. \tag{V.69}$$

From (V.66) we have

$$\langle \gamma_\nu(F_2), 1\rangle_{H^{-1/2}(\Gamma_i),H^{1/2}(\Gamma_i)} = \int_{\Gamma_i} F_2 \cdot \nu \, d\sigma = 0,$$

for any $1 \leq i \leq m$. Thus, we can apply Theorem IV.4.13 to obtain the existence of a vector field Φ in $\left(H^2(\Omega)\right)^d$ such that

$$\begin{cases} \operatorname{curl} \Phi = F_2, & \text{in } \Omega, \\ \operatorname{div} \Phi = 0, & \text{in } \Omega \\ \|\Phi\|_{H^2} \leq C\|F_2\|_{H^1} \leq C'\|g - F_1\|_{H^{1/2}}. \end{cases}$$

We emphasise that it is not needed to impose a boundary condition on this vector field Φ. For all $\varepsilon > 0$, we set $\delta_\varepsilon = e^{-1/\varepsilon}$, and we introduce the function $\psi_\varepsilon : \mathbb{R} \to [0, +\infty[$ defined by

$$\psi_\varepsilon(s) = \begin{cases} 1 & \text{for } s \leq \delta_\varepsilon^2, \\ \varepsilon \log\left(\frac{\delta_\varepsilon}{s}\right) & \text{for } \delta_\varepsilon^2 \leq s \leq \delta_\varepsilon, \\ 0 & \text{for } s \geq \delta_\varepsilon. \end{cases}$$

Let $\eta : \mathbb{R} \to \mathbb{R}$ be a mollifying kernel with support in $]-1, 1[$. We introduce a $\mathcal{C}^\infty$ function β_ε as follows

$$\beta_\varepsilon(s) = \frac{4}{\delta_\varepsilon^2} \int_{\mathbb{R}} \eta\left(\frac{s-t}{\delta_\varepsilon^2/4}\right) \psi_\varepsilon(t)\, dt.$$

The function β_ε satisfies

$$\begin{cases} \beta_\varepsilon(s) = 1, & \forall s \leq \dfrac{3\delta_\varepsilon^2}{4}, \\ \beta_\varepsilon(s) = 0, & \forall s \geq \delta_\varepsilon + \dfrac{\delta_\varepsilon^2}{4}, \\ |\beta_\varepsilon'(s)| \leq \dfrac{3}{2}\dfrac{\varepsilon}{s}, & \forall s \geq 0. \end{cases} \tag{V.70}$$

We only give the details for the estimate on $\beta_\varepsilon'(s)$ for $3\delta_\varepsilon^2/4 \leq s \leq \delta_\varepsilon + \delta_\varepsilon^2/4$

$$\begin{aligned} \beta_\varepsilon'(s) &= \left(\frac{4}{\delta_\varepsilon^2}\right)^2 \int_{\mathbb{R}} \eta'\left(\frac{s-t}{\delta_\varepsilon^2/4}\right) \psi_\varepsilon(t)\, dt = \left(\frac{4}{\delta_\varepsilon^2}\right) \int_{\mathbb{R}} \eta\left(\frac{s-t}{\delta_\varepsilon^2/4}\right) \psi_\varepsilon'(t)\, dt, \\ &= \left(\frac{4}{\delta_\varepsilon^2}\right) \int_{s-\delta_\varepsilon^2/4}^{s+\delta_\varepsilon^2/4} \eta\left(\frac{s-t}{\delta_\varepsilon^2/4}\right) \psi_\varepsilon'(t)\, dt. \end{aligned}$$

The derivative of the function ψ_ε is bounded by $\frac{\varepsilon}{t}$ thus we have

$$|\beta_\varepsilon'(s)| \leq \left(\frac{4}{\delta_\varepsilon^2}\right) \int_{s-\delta_\varepsilon^2/4}^{s+\delta_\varepsilon^2/4} \eta\left(\frac{s-t}{\delta_\varepsilon^2/4}\right) \frac{\varepsilon}{t}\, dt \leq \frac{\varepsilon}{s - \frac{\delta_\varepsilon^2}{4}}.$$

Finally for $\frac{3}{4}\delta_\varepsilon^2 \leq s$, one obtains $|\beta_\varepsilon'(s)| \leq 3\varepsilon/(2s)$. In the same way, we can prove

$$\begin{aligned} |\beta''(s)| &= \left| \left(\frac{4}{\delta_\varepsilon^2}\right)^2 \int_{\mathbb{R}} \eta'\left(\frac{s-t}{\delta_\varepsilon^2/4}\right) \eta'(t)\, dt \right| \\ &\leq \left(\frac{4}{\delta_\varepsilon^2}\right)^2 \int_{\mathbb{R}} \left|\eta'\left(\frac{s-t}{\delta_\varepsilon^2/4}\right)\right| dt \left(\frac{\varepsilon}{s - \frac{\delta_\varepsilon^2}{4}}\right) \\ &\leq \frac{4}{\delta_\varepsilon^2} \int_{\mathbb{R}} |\eta'(t)| dt \left(\frac{\varepsilon}{s - \frac{\delta_\varepsilon^2}{4}}\right) \\ &\leq \varepsilon e^{1/\varepsilon} \frac{1}{s}. \end{aligned}$$

Therefore, by definition of δ_ε, we have

$$\|\beta_\varepsilon'\|_\infty \leq 2\varepsilon e^{2/\varepsilon} \text{ and } \|\beta_\varepsilon''\|_\infty \leq C\varepsilon e^{3/\varepsilon}.$$

We now define $\chi_\varepsilon(x) = \beta_\varepsilon\big(\rho(x)\big)$, for all $x \in \Omega$, where ρ is the regularized distance function built in Section 3.3 of Chapter III. Notice that for $\varepsilon > 0$ small enough χ_ε is smooth because its support is a small neighborhood of the boundary $\partial\Omega$. We now take $G = \operatorname{curl}\big(\chi_\varepsilon \Phi\big)$ so that $\operatorname{div} G = 0$ is automatically satisfied. Since $\chi_\varepsilon = 1$ near the boundary of Ω, we also have $G = F_2$ on $\partial\Omega$.

From Equation (V.70), it is straightforward that the function G satisfies

$$\begin{cases} |G(x)| \leq C\left(\dfrac{\varepsilon}{\rho(x)}\,|\Phi(x)| + |\nabla\Phi(x)|\right), \ \forall x \in \Omega, \\ \ G(x) = 0, \text{ as soon as } \rho(x) \geq \delta_\varepsilon + \delta_\varepsilon^2/4. \end{cases}$$

We are now able to estimate the L^2 norm of $G \otimes v$, where v belongs to $\big(H_0^1(\Omega)\big)^d$. One has

$$\begin{aligned} \int_\Omega |G|^2\,|v|^2\,dx \leq & C\,\varepsilon^2 \int_{\rho(x)\leq\delta_\varepsilon+\delta_\varepsilon^2/4} |\Phi|^2 \left|\frac{v}{\rho(x)}\right|^2 dx \\ & + C\int_{\rho(x)\leq\delta_\varepsilon+\delta_\varepsilon^2/4} |\nabla\Phi|^2|v|^2 dx \\ \leq & C\varepsilon^2\,\|\Phi\|_{H^2}^2 \int_\Omega \left|\frac{v(x)}{\rho(x)}\right|^2 dx \\ & + C\,\|v\|_{L^6}^2 \left(\int_{\rho(x)\leq\delta_\varepsilon+\delta_\varepsilon^2/4} |\nabla\Phi|^3\,dx\right)^{2/3}. \end{aligned}$$

According to the Hardy inequality (Proposition III.2.40 using also (III.69)), we have proved that

$$\begin{aligned} \|G\otimes v\|_{L^2} &\leq C\varepsilon\|\Phi\|_{H^2}\|\nabla v\|_{L^2} + C\theta(\varepsilon)\|v\|_{L^6}, \\ &\leq C\big(\varepsilon\|\Phi\|_{H^2} + \theta(\varepsilon)\big)\|\nabla v\|_{L^2}, \end{aligned}$$

where

$$\theta(\varepsilon) = \left(\int_{\rho(x)\leq\delta_\varepsilon+\delta_\varepsilon^2/4} |\nabla\Phi|^3\,dx\right)^{1/3} \leq C\delta_\varepsilon^{1/6}\|\Phi\|_{H^2} \leq C\varepsilon\|\Phi\|_{H^2}.$$

Finally, we have shown the inequality

$$\|G\otimes v\|_{L^2} \leq C\varepsilon\|\Phi\|_{H^2}\|v\|_{L^2} \leq C\varepsilon\|g - F_1\|_{H^{1/2}}\|v\|_{L^2},$$

so that (V.67) is proved if we choose

$$\varepsilon = \frac{\alpha}{C\|g - F_1\|_{H^{1/2}}}.$$

Thanks to assumption (V.69), we remark that this value of ε remains bounded by some number depending only on Ω. Finally we easily find that

$$\begin{aligned}\|\nabla G\|_{L^2} &\leq C(1+\|\beta_\varepsilon''\|_{L^\infty}+\|\beta_\varepsilon'\|_{L^\infty}^2+\|\beta_\varepsilon\|_{L^\infty})\|\Phi\|_{H^2}\\ &\leq C\varepsilon^2 e^{5/\varepsilon}\|\Phi\|_{H^2} \leq \frac{\alpha^2}{C\|g-F_1\|_{H^{1/2}}}\exp\Big(\frac{C}{\alpha}\|g-F_1\|_{H^{1/2}}\Big),\end{aligned}$$

which gives (V.68) by using (V.69) and the definition of F_1.

$\square$

Proof (of Theorem V.3.3).

Let $\alpha > 0$ determined later and G the vector field given by the previous lemma. We define $\widetilde{G} = G + F_1$, so that by construction, we have $\widetilde{G} = g$ on the boundary $\partial\Omega$. We look for a solution of the system (V.62) in the form $v = w + \widetilde{G}$. The Navier–Stokes system is then equivalent to the following problem, with homogeneous Dirichlet boundary conditions

$$\begin{cases} -\dfrac{1}{\mathcal{R}e}\Delta w + (w\cdot\nabla)\widetilde{G} + (\widetilde{G}\cdot\nabla)w + (w\cdot\nabla)w + \nabla p = f + \widetilde{f}, & \text{in } \Omega\\ \operatorname{div} w = 0, & \text{in } \Omega,\\ w = 0, & \text{on } \partial\Omega,\end{cases} \tag{V.71}$$

with

$$\widetilde{f} = \frac{1}{\mathcal{R}e}\Delta\widetilde{G} - (\widetilde{G}\cdot\nabla)\widetilde{G}. \tag{V.72}$$

We shall find a solution to this system by using the same Galerkin approximation strategy as in the proof of Theorem V.3.1. We do not give the details of this procedure but only the energy estimate that we need to prove an uniform bound on the approximate solutions (passing to the limit is then justified in the same way as in the proof of the above-mentioned theorem).

Thanks to the homogeneous boundary condition satisfied by w, the nonlinear term $(w\cdot\nabla)w$ does not contribute to the energy estimate (see (V.2)). Therefore, the new difficulty is to estimate the following linear terms,

$$I(\tilde{G}, w) = \int_\Omega \big((\widetilde{G}\cdot\nabla)w\big)\cdot w\,dx + \int_\Omega \big((w\cdot\nabla)\widetilde{G}\big)\cdot w\,dx.$$

An integration by parts of the second term allows us to write

$$I(\tilde{G}, w) = \int_\Omega (w\otimes\widetilde{G}) : \nabla w\,dx - \int_\Omega (\widetilde{G}\otimes w) : \nabla w\,dx.$$

We recall that $\tilde{G} = G + F_1$, and thus $I(\tilde{G}, w) = I(G, w) + I(F_1, w)$.

By Lemma V.3.4, we have

$$|I(G, w)| \leq 2\alpha\|\nabla w\|_{L^2}^2.$$

Using Sobolev embeddings and the condition (V.63) (where the constant $C_{1,\Omega}$ is chosen appropriately) we have

$$\begin{aligned}|I(F_1,w)| &\leq 2\|w\|_{L^6}\|F_1\|_{L^3}\|\nabla w\|_{L^2}\\ &\leq C\|F_1\|_{L^3}\|\nabla w\|_{L^2}^2 \leq C\|F_1\|_{H^1}\|\nabla w\|_{L^2}^2\\ &\leq C\mathcal{Q}_{\partial\Omega}(g)\|\nabla w\|_{L^2}^2 \leq \frac{1}{4\,\mathcal{R}e}\|\nabla w\|_{L^2}^2.\end{aligned}$$

Hence the energy estimate for Equation (V.71) reads

$$\frac{1}{\mathcal{R}e}\|\nabla w\|_{L^2}^2 \leq \left(2\alpha + \frac{1}{4\mathcal{R}e}\right)\|\nabla w\|_{L^2}^2 + \left(\|f\|_{H^{-1}} + \|\tilde{f}\|_{H^{-1}}\right)\|\nabla w\|_{L^2},$$

with α to be determined. For example, we can take $\alpha = 1/8\mathcal{R}e$ to obtain

$$\|\nabla w\|_{L^2} \leq 2\mathcal{R}e\left(\|f\|_{H^{-1}} + \|\tilde{f}\|_{H^{-1}}\right), \tag{V.73}$$

and we conclude the proof of the existence of at least a solution to Equation (V.62) as for the homogeneous case.

From (V.72), (V.68), and the choice $\alpha = 1/8\mathcal{R}e$ we find, using assumption (V.63), that

$$\|\tilde{f}\|_{H^{-1}} \leq \frac{1}{\mathcal{R}e}\|\nabla\widetilde{G}\| + \|\widetilde{G}\otimes\widetilde{G}\|_{L^2} \leq C\frac{\|g\|_{H^{1/2}}}{\mathcal{R}e}\exp(C\mathcal{R}e\|g\|_{H^{1/2}}).$$

Since $v = w + \widetilde{G}$, we deduce from (V.73) that the estimate (V.64) holds.

□

3.3 Uniqueness for small data

Theorem V.3.5. *Let Ω be a bounded, simply connected, and Lipschitz domain of $\mathbb{R}^d$ with $d = 3$. There exists $C_{3,\Omega} > 0$ depending only on Ω, such that for any $f \in (H^{-1}(\Omega))^d$ and $g \in (H^{1/2}(\partial\Omega))^d$ satisfying the conditions*

$$\begin{cases}\displaystyle\int_{\partial\Omega} g\cdot\nu\,d\sigma = 0,\\ \mathcal{Q}_{\partial\Omega}(g) \leq \dfrac{C_{1,\Omega}}{\mathcal{R}e},\\ \mathcal{R}e^2\|f\|_{H^{-1}} + \mathcal{R}e\|g\|_{H^{1/2}}\exp(C_{2,\Omega}\mathcal{R}e\|g\|_{H^{1/2}}) \leq C_{3,\Omega},\end{cases}$$

the problem (V.62) has a unique solution $(v,p) \in (H^1(\Omega))^d \times L^2_0(\Omega)$.

Remark V.3.2. For some particular domains Ω, it can be shown that for high values of the Reynolds number this uniqueness property may fail (even with homogeneous boundary conditions); see [63] for instance.

Proof.

Under the assumption on $\mathcal{Q}_{\partial\Omega}(g)$, Theorem V.3.3 proves the existence of at least one solution (v,p) satisfying the estimate (V.64). Let $(\tilde{v},\tilde{p}) \in (H^1(\Omega))^d \times L^2_0(\Omega)$ be any other possible solution of (V.62) and let us denote by $w = \tilde{v} - v$, $q = \tilde{p} - p$ the differences between these two solutions.

By subtracting the two systems, we see that (w,q) solves the problem

$$\begin{cases} -\dfrac{1}{\mathcal{R}e}\Delta w + (w\cdot\nabla)v + (\tilde{v}\cdot\nabla w) + \nabla q = 0, & \text{in } \Omega,\\ \operatorname{div} w = 0, & \text{in } \Omega,\\ w = 0, & \text{on } \partial\Omega. \end{cases}$$

We use w as a test function in this problem and integrate by parts. Since w vanishes on the boundary, with (V.2) we obtain

$$\frac{1}{\mathcal{R}e}\|\nabla w\|^2_{L^2} = -\int_\Omega \big((w\cdot\nabla)v\big)\cdot w\,dx \le C\|\nabla v\|_{L^2}\|\nabla w\|^2_{L^2}.$$

With the estimate (V.64), we see that we can find $C_{2,\Omega}$ and $C_{3,\Omega}$ such that $C\mathcal{R}e\|\nabla v\|_{L^2} \le \frac{1}{2}$ which leads to $\nabla w = 0$ and then $w = 0$.

□

Remark V.3.3. In the case of the homogeneous Dirichlet condition, we have $g = 0$, and the uniqueness of the solution is ensured as soon as the condition $\mathcal{R}e^2\|f\|_{H^{-1}} \le C_{3,\Omega}$ is fulfilled.

3.4 Asymptotic stability of steady solutions

We now deal with the stability properties of steady solutions. We only concentrate on the case of homogeneous Dirichlet boundary conditions but the nonhomogeneous case can be treated in the very same way.

For a given source term f which does not depend on time, we consider a steady solution $(v_\infty, p_\infty) \in (H^1_0(\Omega))^d \times L^2_0(\Omega)$ to (V.62). The question we want to adress now is the following: given initial data $v_0 \in H$ close enough to v_∞ (made precise below), is there a unique global solution (v,p) to the unsteady Navier–Stokes problem

$$\begin{cases} \dfrac{\partial v}{\partial t} - \dfrac{1}{\mathcal{R}e}\Delta v + (v\cdot\nabla)v + \nabla p = f, & \text{in } \Omega, \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ v = 0, & \text{on } \partial\Omega, \\ v(0) = v_0, \end{cases} \tag{V.74}$$

and does this solution converge towards (v_∞, p_∞) when $t \to +\infty$? In such a case, we say that the steady solution under study is asymptotically stable.

Of course, the situation will be different depending on the dimension because, for $d = 2$, the existence of an unique global weak solution is known for any initial data (Theorem V.2.1), whereas for $d = 3$, global weak solutions are not necessarily unique and strong solutions are not necessarily global.

From now on we look for a solution to (V.74) in the form $v = v_\infty + w$ and $p = p_\infty + \pi$ where (w, π) solves the problem

$$\begin{cases} \dfrac{\partial w}{\partial t} - \dfrac{1}{\mathcal{R}e}\Delta w + (w\cdot\nabla)w + (v_\infty\cdot\nabla)w + (w\cdot\nabla)v_\infty + \nabla\pi = 0, & \text{in } \Omega, \\ \operatorname{div} w = 0, & \text{in } \Omega, \\ w = 0, & \text{on } \partial\Omega, \\ w(t=0) = v_0 - v_\infty. \end{cases} \tag{V.75}$$

We first concentrate on the asymptotic behavior of the velocity field v (or w) and we postpone the discussion on the behavior of the pressure to the end of this section.

3.4.1 The two-dimensional case

We show here that a steady solution is asymptotically stable, provided that it is small enough.

Theorem V.3.6. *Let Ω be a bounded, simply connected, and Lipschitz domain. Let f be given in $(H^{-1}(\Omega))^2$ and $(v_\infty, p_\infty) \in (H^1_0(\Omega))^d \times L^2_0(\Omega)$ be a steady solution to (V.62) associated with the source term f.*

There exists $C > 0$, $\alpha > 0$ depending only on Ω such that, under the condition

$$\|\nabla v_\infty\|_{L^2} \leq \frac{C}{\mathcal{R}e},$$

then for any initial data $v_0 \in H$, the unique weak solution (v, p) of (V.74) satisfies

$$\|v(t) - v_\infty\|_{L^2} \leq \|v_0 - v_\infty\|_{L^2}\, e^{-\alpha t}, \ \forall t \geq 0,$$
$$\|\nabla v(t) - \nabla v_\infty\|_{L^2} \leq C\|v_0 - v_\infty\|_{L^2}\, e^{-\alpha t}, \ \forall t \geq 1.$$

Proof.

The existence and uniqueness of an unique global weak solution is already known in that case. Therefore, we only have to give a suitable estimate on w.

We multiply (V.75) by w and we use (V.3) to obtain the following estimate

$$\frac{1}{2}\frac{d}{dt}\|w\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\|\nabla w\|_{L^2}^2 \leq C\|\nabla v_\infty\|_{L^2}\|w\|_{L^2}\|\nabla w\|_{L^2}.$$

Then by using the Young inequality, we get

$$\frac{d}{dt}\|w\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\|\nabla w\|_{L^2}^2 \leq C\,\mathcal{R}e\,\|\nabla v_\infty\|_{L^2}^2\|w\|_{L^2}^2.$$

Finally, using the Poincaré inequality (Proposition IV.5.12), this estimate leads to

$$\frac{d}{dt}\|w\|_{L^2}^2 + \left(\frac{C_1}{\mathcal{R}e} - C\,\mathcal{R}e\,\|\nabla v_\infty\|_{L^2}^2\right)\|w\|_{L^2}^2 \leq 0.$$

Provided that $C\mathcal{R}e^2\|\nabla v_\infty\|_\infty^2 < C_1$, we obtain

$$\frac{d}{dt}\|w\|_{L^2}^2 + 2\alpha\,\|w\|_{L^2}^2 \leq 0,$$

which finally gives the expected exponential behavior

$$\|w(t)\|_{L^2}^2 \leq \|w_0\|_{L^2}^2\, e^{-2\alpha t}.$$

The estimate on ∇w can be proved by using the same arguments as the one used in the three-dimensional case below.

□

3.4.2 The three-dimensional case

In that case we need to consider smoother initial data because uniqueness of weak solutions is not known. Nevertheless, global existence of strong solutions is not known in general and we then have to show that the existence time if infinite as soon as the initial data are close enough to v_∞.

Notice that, as it is shown below, we do not need to suppose that the source term belongs to $\left(L^2(\Omega)\right)^d$, even though we are interested in strong solutions. This is due to the fact that we look for a solution in the form $v = v_\infty + w$ where w is a solution of Problem (V.75) for which there is no source term.

Theorem V.3.7. *Let Ω be a bounded, simply connected, and Lipschitz domain of $\mathbb{R}^3$. Let f be given in $\left(H^{-1}(\Omega)\right)^3$ and $(v_\infty, p_\infty) \in (H_0^1(\Omega))^3 \times L_0^2(\Omega)$ a steady solution of (V.62). There exist $C > 0$, $\alpha > 0$ depending only on Ω such that, under the condition*

$$\|\nabla v_\infty\|_{L^2} \leq \frac{C}{\mathcal{R}e},$$

for any initial data $v_0 \in V$ *satisfying*

$$\|\nabla v_0 - \nabla v_\infty\|_{L^2} \leq \frac{C}{\mathcal{R}e},$$

there exists a unique global strong solution (v, p) *to* (V.74) *which satisfies*

$$\|\nabla v(t) - \nabla v_\infty\|_{L^2} \leq \|\nabla v_0 - \nabla v_\infty\|_{L^2}\, e^{-\alpha t},\ \forall t \geq 0.$$

Proof.

We show the existence of a unique global strong solution of (V.75) which implies the existence of a unique global strong solution for (V.74). Existence and uniqueness of a local strong solution of (V.75) can be done exactly as in the proof of Theorem V.2.1. We only give the new estimates which prove that this solution is global and converges towards v_∞.

By multiplying (V.75) by Aw (where A is the Stokes operator), and using (V.3) we get

$$\begin{aligned}
&\frac{1}{2}\frac{d}{dt}\|\nabla w\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\|Aw\|_{L^2}^2\\
\leq&\|w\|_{L^\infty}\|\nabla w\|_{L^2}\|Aw\|_{L^2} + \|v_\infty\|_{L^6}\|\nabla w\|_{L^3}\|Aw\|_{L^2}\\
&+ \|w\|_{L^\infty}\|\nabla v_\infty\|_{L^2}\|Aw\|_{L^2}\\
\leq& C\left(\|\nabla w\|_{L^2}^{3/2}\|Aw\|_{L^2}^{3/2} + \|\nabla v_\infty\|_{L^2}\|\nabla w\|_{L^2}^{1/2}\|Aw\|_{L^2}^{3/2}\right).
\end{aligned}$$

By using the Young inequality, we deduce

$$\frac{d}{dt}\|\nabla w\|_{L^2}^2 + \frac{1}{\mathcal{R}e}\|Aw\|_{L^2}^2 \leq C\mathcal{R}e^3\|\nabla v_\infty\|_{L^2}^4\|\nabla w\|_{L^2}^2 + C\mathcal{R}e^3\|\nabla w\|_{L^2}^6,$$

and then, with the Poincaré inequalities given in Proposition IV.5.12, we get

$$\frac{d}{dt}\|\nabla w\|_{L^2}^2 + \frac{C_1}{\mathcal{R}e}\|\nabla w\|_{L^2}^2 \leq C\mathcal{R}e^3\|\nabla v_\infty\|_{L^2}^4\|\nabla w\|_{L^2}^2 + C\mathcal{R}e^3\|\nabla w\|_{L^2}^6.$$

We see that if v_∞ satisfies the condition $2C\mathcal{R}e^4\,\|\nabla v_\infty\|_{L^2}^4 \leq C_1$, then the previous equation leads to

$$\frac{d}{dt}\|\nabla w\|_{L^2}^2 + \left(\frac{C_1}{2\mathcal{R}e} - C\mathcal{R}e^3\|\nabla w\|_{L^2}^4\right)\|\nabla w\|_{L^2}^2 \leq 0. \tag{V.76}$$

Now, we assume that the initial data v_0 are such that $w(0) = v_0 - v_\infty$ is small enough, in the following sense

$$\frac{C_1}{2\mathcal{R}e} - C\mathcal{R}e^3\|\nabla w(0)\|_{L^2}^4 \geq \frac{C_1}{4\mathcal{R}e}. \tag{V.77}$$

We denote by $T^* \in]0,+\infty]$ the maximal existence time for the strong solution w. We define now

$$T = \sup\left\{t \in [0,T^\star[,\ \text{s.t. } \frac{C_1}{2\mathcal{R}e} - C\mathcal{R}e^3\|\nabla w(s)\|_{L^2}^4 \geq \frac{C_1}{8\mathcal{R}e}, \forall s \in [0,t]\right\}. \tag{V.78}$$

By continuity and assumption (V.77), we see that $T > 0$. We show that we necessarily have $T = +\infty$, which implies in particular that $T^* = +\infty$.

Let us assume, by contradiction, that $T < +\infty$. This implies that $T < T^\star$ because the solution remains bounded when $t \to T$. By definition of T, and by (V.76), we see that

$$\frac{d}{dt}\|\nabla w\|_{L^2}^2 + \frac{C_1}{8\mathcal{R}e}\|\nabla w\|_{L^2}^2 \leq 0,\ \forall t \in [0,T], \tag{V.79}$$

so that we deduce that

$$\|\nabla w(t)\|_{L^2} \leq \|\nabla w(0)\|_{L^2},\ \forall t \in [0,T].$$

In particular we have

$$\frac{C_1}{2\mathcal{R}e} - C\mathcal{R}e^3\|\nabla w(T)\|_{L^2}^4 \geq \frac{C_1}{4\mathcal{R}e},$$

and then, by continuity, there is $\varepsilon > 0$ such that $T + \varepsilon < T^\star$ and

$$\frac{C_1}{2\mathcal{R}e} - C\mathcal{R}e^3\|\nabla w(t)\|_{L^2}^4 \geq \frac{C_1}{8\mathcal{R}e},\ \forall t \in [0, T+\varepsilon[.$$

This is a contradiction with the maximality property of T (see (V.78)).

Finally, we have shown that $T = T^\star = +\infty$, so that the strong solution under consideration is global. Moreover, since $T = +\infty$ we deduce from (V.79) that

$$\|\nabla w(t)\|_{L^2}^2 \leq \|\nabla w(0)\|_{L^2}^2 e^{-C_1 t/(8\mathcal{R}e)},\ \ \forall t \geq 0,$$

and the claim is proved.

3.4.3 Asymptotic behavior of the pressure

Let $\tau > 0$ be given, we can prove the following convergence of the pressure field

$$\frac{1}{\tau}\int_t^{t+\tau} p(s,.)\,ds \xrightarrow[t\to\infty]{} p_\infty,\ \text{in } L^2(\Omega).$$

This amounts to showing that the pressure π in Problem (V.75) satisfies that $t \mapsto \int_t^{t+\tau} \pi(s,.)\,dx$ converges to 0. To this end, we integrate the equation in time to obtain

$$\begin{aligned}\nabla\left(\int_t^{t+\tau}\pi(s,.)\,ds\right) =& w(t)-w(t+\tau)\\ &+\operatorname{div}\left(\int_t^{t+\tau}\frac{1}{\mathcal{R}e}\nabla w - w\otimes w - v_\infty\otimes w - w\otimes v_\infty\,ds\right).\end{aligned}$$

Thus, using the Nečas inequality (Theorem IV.1.1) and Proposition IV.1.7, we get

$$\begin{aligned}\left\|\int_t^{t+\tau}\pi(s,.)\,ds\right\|_{L^2} \leq& C\left\|\nabla\int_t^{t+\tau}\pi(s,.)\,ds\right\|_{H^{-1}}\\ \leq& \|w(t)\|_{H^{-1}}+\|w(t+\tau)\|_{H^{-1}}\\ &+\left\|\int_t^{t+\tau}\frac{1}{\mathcal{R}e}\nabla w - w\otimes w - v_\infty\otimes w - w\otimes v_\infty\,ds\right\|_{L^2}\\ \leq& \|w(t)\|_{H^{-1}}+\|w(t+\tau)\|_{H^{-1}}\\ &+C\int_t^{t+\tau}\|\nabla w\|_{L^2}+\|w\|_{L^4}^2+\|w\|_{L^4}\|v_\infty\|_{L^4}\,ds,\end{aligned}$$

and the claim follows because we already know that $\|w(t)\|_{L^2}\xrightarrow[t\to+\infty]{}0$ and that $\|\nabla w(t)\|_{L^2}\xrightarrow[t\to+\infty]{}0$.

Chapter VI
Nonhomogeneous fluids

In this chapter we study the nonhomogeneous incompressible Navier–Stokes equations. The difference with the equations studied in the previous chapter is that the density is no longer a constant and has become a new unknown function. For simplicity, in this chapter, we set $\Gamma = \partial\Omega$ and for any vector field v, and any time t, we define the outflow and inflow parts of Γ as

$$\Gamma_v^+(t) = \{x \in \Gamma, \quad (v(t,x)\cdot\nu(x)) > 0\},$$

$$\Gamma_v^-(t) = \{x \in \Gamma, \quad (v(t,x)\cdot\nu(x)) < 0\},$$

the trace of v on Γ being understood in a weak sense if necessary.

We recall that the equations involved are the mass balance equation, the momentum balance equation and the incompressibility constraint, as well as initial and boundary data

$$\begin{cases} \dfrac{\partial\rho}{\partial t} + \operatorname{div}(\rho v) = 0, & \text{in } \Omega, \\ \dfrac{\partial(\rho v)}{\partial t} + \operatorname{div}(\rho v\otimes v) - 2\operatorname{div}(\mu(\rho)\mathrm{D}(v)) + \nabla p = \rho f, & \text{in } \Omega, \\ \operatorname{div} v = 0, & \text{in } \Omega, \\ v = v_b, & \text{on } \Gamma, \\ \rho = \rho^{in}, & \text{on } \Gamma_v^-(t), \end{cases}$$

together with initial conditions of the type $\rho(0) = \rho_0$, $v(0) = v_0$. We will see that the meaning to give to these initial conditions is not as simple as in the case of homogeneous fluids. The interpretation of the boundary condition for the density ρ has also to be detailed since, for weak solutions, the density ρ does not possess integrable derivatives, so that the trace theory for functions in Sobolev spaces described in Chapter III is not sufficient. Moreover, this boundary condition is only imposed on the inflow part of Γ.

F. Boyer and P. Fabrie, *Mathematical Tools for the Study of the Incompressible Navier-Stokes Equations and Related Models*, Applied Mathematical Sciences 183,
DOI 10.1007/978-1-4614-5975-0_6,

In this model, the viscosity is allowed to depend on the density ρ according to a given law $\rho \mapsto \mu(\rho)$.

As we saw in Chapter I, using the mass balance equation, the momentum balance equation can also be written in the form

$$\rho\left(\frac{\partial v}{\partial t} + (v \cdot \nabla)v\right) - 2\,\mathrm{div}(\mu(\rho)\mathrm{D}(v)) + \nabla p = \rho f.$$

The first form of this equation is called the *conservative form* and the second one is called the *nonconservative form*. In the same way, by using the incompressibility constraint, we can write the mass balance equation in the nonconservative form

$$\frac{D\rho}{Dt} = \frac{\partial \rho}{\partial t} + (v \cdot \nabla)\rho = 0. \tag{VI.1}$$

This last equation is called a *transport equation*. It describes the transport of the quantity ρ by the velocity field v, as we saw in the transport theorem (Theorem I.2.1).

When compared to the homogeneous model studied in Chapter V, several new difficulties appear. First there are new nonlinearities which reflect the fact that the density ρ and the viscosity $\mu(\rho)$ are no longer constant. Second, if we allow situations where vacuum regions exist (i.e., regions where the density is zero, or so low as to be negligible), then the problem above degenerates (the momentum equation becoming elliptic in some sense). This is why the initial condition must be applied to the pair mass density/momentum density $(\rho, \rho v)$ and not to the pair mass density/velocity (ρ, v). We emphasise the fact that, despite the mathematical interest of this issue, the vanishing density limit is in some sense contradictory with the continuum medium assumption which is the basis of the derivation of the models.

It is then quite clear that the subcase in which one assumes that $\inf_\Omega \rho_0 > 0$ and $\inf \rho^{in} > 0$, should be in some way simpler and that there will be fewer difficulties to define and to build weak solutions. We return to this special case at the very end of this chapter.

We begin this chapter by exploring how to solve the transport equation (VI.1) subject to initial and inflow boundary data, for a given velocity field v. This is, of course, an essential component in the resolution of the full coupled problem. First of all it is worth noticing that, when the velocity field v is tangent to the boundary of the domain, there are no more boundary data to impose on the density because $v \cdot \nu = 0$. In that particular case, if the data are regular the equation can be solved by using the method of characteristics which amounts to solving the characteristic equation associated with v (see (I.4)) and observing that the solution ρ of (I.2.1) has to be constant along this flow. However, when the data are less regular (in particular if the velocity field does not satisfy the hypothesis of the Cauchy–Lipschitz theorem), it is necessary to use a more sophisticated theory known as *renormalised solu-*

tions initially introduced by Di Perna and Lions in [52]. This method let us build a complete theory of existence and uniqueness for weak solutions of the transport equation for less regular data.

This approach can be generalised to the case of nontangent velocity fields. It allows us to define a trace for such weak solutions in a suitable sense. This is exactly the framework that we present in Section 1, in the slightly more general context of linear transport-reaction equations which can appear in many other models.

Next, in Section 2, we introduce a complete approximate problem for the coupled system considered here. The resolution of this problem is itself not obvious and uses the Schauder fixed-point theorem as well as a stability result for solutions of the transport equation which is proved in Section 1.

We then are able to obtain uniform estimates with respect to the approximation parameter. They give us enough weak and strong compactness to justify the passage to the limit in the approximate problem and to conclude that there exists at least one weak solution of the system.

We emphasise that we do not prove any uniqueness result for these solutions here, even in 2D. To do this, it would be necessary to construct more regular solutions. For that approach, in the case $v_b = 0$, a local existence result and uniqueness in the class

$$\begin{cases} \rho \in \mathcal{C}^1([0,T] \times \overline{\Omega}), \\ v \in L^q(]0,T[, (W^{2,q}(\Omega))^d \cap V), \dfrac{\partial v}{\partial t} \in (L^q(]0,T[\times\Omega))^d, \end{cases}$$

where $q > 3$, can be found in [77] under the assumption that the viscosity, μ, does not depend on ρ. Results concerning this problem can also be found in [12, 50, 51, 86, 83, 42, 45].

Finally, since the two-dimensional case is (as usual) simpler than the three-dimensional case, below we concentrate exclusively on the dimension $d = 3$. The results presented in this chapter come principally from the references [52, 22] concerning the transport equation and [110, 86] concerning weak solutions of the Navier–Stokes problem with possibly vanishing density.

We again use the various functional spaces H and V defined in Section 3.3 of Chapter IV.

1 Weak solutions of the transport equation

Let us begin with a classical notation that we use in this section.

Definition VI.1.1. *The positive and negative parts of any real number x, are defined by*

$$x^+ = \max(x, 0), \;\; x^- = -\min(x, 0),$$

in such a way that we have

$$x = x^+ - x^- \text{ and } |x| = x^+ + x^-.$$

1.1 Setting of the problem

Let $d \geq 1$, $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and $T > 0$ given. We are interested in the following equation (written in conservative formulation)

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho v) + c\rho = 0, \quad \text{in } \Omega, \tag{VI.2}$$

assorted with the following initial and boundary conditions

$$\begin{cases} \rho(0, \cdot) = \rho_0, \\ \rho = \rho^{in}, & \text{on } \Gamma_v^-(t), \end{cases} \tag{VI.3}$$

where $\rho_0 \in L^\infty(\Omega)$ and $\rho^{in} \in L^\infty(]0, T[\times\Gamma)$ are given data. We need the following assumptions

$$c \in L^1(]0, T[\times\Omega), \tag{VI.4}$$

$$v \in L^1(]0, T[, (W^{1,1}(\Omega))^d), \tag{VI.5}$$

$$\begin{cases} (c + \operatorname{div} v)^- \in L^1(]0, T[, L^\infty(\Omega)), & \text{(VI.6a)} \\ \quad (\operatorname{div} v)^+ \in L^1(]0, T[, L^\infty(\Omega)). & \text{(VI.6b)} \end{cases}$$

Remark VI.1.1. We considered here, and in the whole section, a slightly more general equation than the one we strictly need for the study of the incompressible nonhomogeneous Navier–Stokes equations, that is to say, the advection equation (I.2.1). This allows us to account for a broader range of situations, in particular for non divergence-free vector fields, such as in compressible models or in Vlasov models in kinetic theory.

- In the particular case where $c = -\operatorname{div} v$, we recover the usual advection equation
$$\frac{\partial \rho}{\partial t} + v \cdot \nabla \rho = 0,$$
and in this case Assumption (VI.6a) is automatically satisfied.
- In the particular case where $c = 0$, the equation under study is nothing but the mass conservation equation
$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho v) = 0,$$

and Assumptions (VI.6) are then equivalent to requiring that $\operatorname{div} v \in L^1(]0,T[, L^\infty(\Omega))$.

Associated with the vector field v, we introduce the measure

$$d\mu_v = (v \cdot \nu)\, dt\, d\sigma, \text{ on }]0,T[\times\Gamma,$$

and we denote by $d\mu_v^+$ (resp., $d\mu_v^-$) its positive (resp., negative) part in such a way that $|d\mu_v| = d\mu_v^+ + d\mu_v^-$. The support of $d\mu_v^+$ (resp., $d\mu_v^-$) is the outflow (resp. inflow) part of $]0,T[\times\Gamma$.

Definition VI.1.2. *Let $v \in (L^1(]0,T[\times\Omega))^d$ and $c \in L^1(]0,T[\times\Omega)$. A function $\rho \in L^\infty(]0,T[\times\Omega)$ is said to be a weak solution of (VI.2) if it satisfies*

$$\int_0^T \int_\Omega \rho \left(\frac{\partial \varphi}{\partial t} + v \cdot \nabla\varphi - c\varphi \right) dx\, dt = 0,$$

for any test function $\varphi \in \mathcal{C}^{0,1}([0,T] \times \overline{\Omega})$ such that $\varphi(0,.) = \varphi(T,.) = 0$ and $\varphi = 0$ on $[0,T] \times \Gamma$.

In the next subsections, we prove the main properties of those weak solutions. First of all we show that any weak solution has a trace on the boundary in some weak sense, and that it satisfies the so-called *renormalisation property*. Then we show that the Cauchy–Dirichlet transport problem is well-posed and finally that its solution depends continuously on the data, in particular the velocity field.

The introduction of the concept of renormalised solutions for the transport equation comes from [52] (see also [5] for the case of BV vector fields). The presentation proposed here and which includes the trace theory is mainly taken from [22] with some improvements, in particular concerning the regularity assumptions on the domain. An application of these results to the convergence analysis of finite volume schemes for the transport problem is presented in [23].

1.2 Trace theorem. Renormalisation property

The study of weak solutions of the transport problem begins with proving that such solutions have traces on the boundary in a suitable sense. This is the objective of the following theorem.

Theorem VI.1.3. *Assume that v and c satisfy assumptions (VI.4) through (VI.6) and let $\rho \in L^\infty(]0,T[\times\Omega)$ be any weak solution of (VI.2). Then the following properties hold.*

1. **Time continuity:** *The function ρ lies in $\mathcal{C}^0([0,T], L^p(\Omega))$ for any $1 \leq p < +\infty$.*

2. **Existence and uniqueness of a trace:** *There exists a unique function $\gamma\rho \in L^\infty(]0,T[\times\Gamma, |d\mu_v|)$ such that, for any test function $\varphi \in \mathcal{C}^{0,1}([0,T]\times\overline{\Omega})$ and for any $[t_0,t_1]\subset[0,T]$, we have*

$$\begin{aligned}&\int_{t_0}^{t_1}\int_\Omega \rho\left(\frac{\partial\varphi}{\partial t}+v\cdot\nabla\varphi-c\varphi\right)dx\,dt-\int_{t_0}^{t_1}\int_\Gamma(\gamma\rho)\varphi(v\cdot\nu)\,d\sigma\,dt\\&+\int_\Omega\rho(t_0)\varphi(t_0)\,dx-\int_\Omega\rho(t_1)\varphi(t_1)\,dx=0.\end{aligned}\tag{VI.7}$$

3. **Renormalisation property:** *For any function $\beta:\mathbb{R}\to\mathbb{R}$ of class $\mathcal{C}^1$, for any $\varphi\in\mathcal{C}^{0,1}([0,T]\times\overline{\Omega})$ and any $[t_0,t_1]\subset[0,T]$, we have*

$$\begin{aligned}&\int_{t_0}^{t_1}\int_\Omega\beta(\rho)\left(\frac{\partial\varphi}{\partial t}+v\cdot\nabla\varphi\right)dx\,dt-\int_{t_0}^{t_1}\int_\Omega c\rho\beta'(\rho)\varphi\,dx\,dt\\&-\int_{t_0}^{t_1}\int_\Omega(\operatorname{div}v)(\rho\beta'(\rho)-\beta(\rho))\varphi\,dx\,dt-\int_{t_0}^{t_1}\int_\Gamma\beta(\gamma\rho)\varphi(v\cdot\nu)\,d\sigma\,dt\\&+\int_\Omega\beta(\rho(t_0))\varphi(t_0)\,dx-\int_\Omega\beta(\rho(t_1))\varphi(t_1)\,dx=0.\end{aligned}\tag{VI.8}$$

It is important to understand that the renormalisation property simply consists in justifying that the chain rule

$$\frac{\partial\beta(\rho)}{\partial t}+v\cdot\nabla\beta(\rho)=\beta'(\rho)\left(\frac{\partial\rho}{\partial t}+v\cdot\nabla\rho\right),$$

still holds for weak solutions and low regularity vector fields v.

Remark VI.1.2. We show below, in Corollary VI.1.5, that the renormalisation property actually holds for any β which is simply continuous and piecewise $\mathcal{C}^1$.

Remark VI.1.3. The same result holds if one adds a given source term $f\in L^1(]0,T[\times\Omega)$ in the right-hand side of the transport equation (VI.2). More precisely the term $\displaystyle\int_{t_0}^{t_1}\int_\Omega f\varphi\,dx\,dt$ has to be added to the left-hand side of (VI.7) and the term $\displaystyle\int_{t_0}^{t_1}\int_\Omega f\beta'(\rho)\varphi\,dx\,dt$ in the left-hand side of (VI.8).

However, we emphasise the fact that we need to know a priori that ρ is a bounded weak solution in order to prove the result. Extension of these results to the case of weak solutions in $L^\infty(]0,T[,L^p(\Omega))$, $p<+\infty$ can be found for instance in [52, 22].

Before proving Theorem VI.1.3, we give a preliminary technical result on the approximation of the absolute value by smooth functions. Its proof is straightforward.

Lemma VI.1.4. *For any $\delta > 0$ the function $\beta_\delta : \mathbb{R} \to \mathbb{R}$ defined by*

$$\beta_\delta(\xi) = \frac{\xi^2}{\sqrt{\xi^2+\delta}},$$

is of class $\mathcal{C}^\infty$ and satisfies

$$\begin{aligned} |\beta_\delta(\xi)| &\leq |\xi|, && \forall \xi \in \mathbb{R},\\ |\beta_\delta'(\xi)| &\leq 2, && \forall \xi \in \mathbb{R},\\ |\xi\beta_\delta'(\xi) - \beta_\delta(\xi)| &\leq \sqrt{\delta}, && \forall \xi \in \mathbb{R}, \forall \delta > 0,\\ \beta_\delta(\xi) &\xrightarrow[\delta\to 0]{} |\xi|, && \forall \xi \in \mathbb{R}. \end{aligned}$$

Proof (of Theorem VI.1.3).

The proof of this result strongly relies on the use of the mollifying operators introduced in Section 2.2 of Chapter III and of the corresponding Friedrichs commutation estimates.

- We consider the family of operators $(\mathcal{S}_\varepsilon)_\varepsilon$ defined in (III.18) and we set, for any $t \in [0,T]$, $\rho_\varepsilon(t,.) = \mathcal{S}_\varepsilon\rho(t,.)$.
  We observe that, since $\mathcal{S}_\varepsilon$ is only acting on the space variables, the time variable is just a parameter in this mollifying procedure. In particular the differential operators $\partial/\partial t$ and $\mathcal{S}_\varepsilon$ commute. Thanks to the regularity assumptions (VI.4) and (VI.5), we can then apply Proposition III.2.8 and Theorem III.2.10 to conclude that ρ_ε solves the following equation in the distribution sense
  $$\frac{\partial \rho_\varepsilon}{\partial t} + \mathrm{div}(\rho_\varepsilon v) + c\rho_\varepsilon = R^\varepsilon, \tag{VI.9}$$
  where $R^\varepsilon \in L^1(]0,T[\times\Omega)$ satisfies $\|R^\varepsilon\|_{L^1} \to 0$ as $\varepsilon \to 0$.
  From Proposition III.2.8, we know that
  $$\rho_\varepsilon \in L^\infty(]0,T[,\mathcal{C}^k(\overline{\Omega})),\ \forall k \geq 0,$$
  $$\|\rho_\varepsilon\|_{L^\infty(]0,T[\times\Omega)} \leq \|\rho\|_{L^\infty(]0,T[\times\Omega)}, \tag{VI.10}$$
  and that
  $$\begin{aligned} \rho_\varepsilon(t) &\xrightarrow[\varepsilon\to 0]{} \rho(t), && \text{in } L^p(\Omega) \text{ for any } p < +\infty \text{ and any } t \in [0,T],\\ \rho_\varepsilon &\xrightarrow[\varepsilon\to 0]{} \rho, && \text{in } L^p(]0,T[\times\Omega) \text{ for any } p < +\infty. \end{aligned}$$
  Note that these convergences and the L^∞ bound on ρ_ε imply that
  $$|\rho_\varepsilon - \rho|^2 \xrightarrow[\varepsilon\to 0]{} 0, \quad \text{in } L^\infty(]0,T[\times\Omega) \text{ weak-}\star. \tag{VI.11}$$
  Since ρ_ε is smooth with respect to the space variable, we deduce from (VI.9) that for any $\varepsilon > 0$ we have

$$\frac{\partial \rho_\varepsilon}{\partial t} = R^\varepsilon - \operatorname{div}(\rho_\varepsilon v) - c\rho_\varepsilon = R^\varepsilon - \rho_\varepsilon(\operatorname{div} v) - v\cdot(\nabla\rho_\varepsilon) - c\rho_\varepsilon \in L^1(]0,T[\times\Omega).$$

In particular we have $\rho_\varepsilon \in W^{1,1}(]0,T[\times\Omega)$. It follows from this property, and using for instance Proposition II.5.11, that $\rho_\varepsilon \in \mathcal{C}^0([0,T],L^1(\Omega))$. From (VI.10), we finally obtain that $\rho_\varepsilon \in \mathcal{C}^0([0,T],L^p(\Omega))$ for any $p < +\infty$.

- Let us show now that $(\rho_\varepsilon)_\varepsilon$ is a Cauchy sequence in $\mathcal{C}^0([0,T],L^1(\Omega))$. For any $\varepsilon_1,\varepsilon_2 > 0$, we have

$$\frac{\partial}{\partial t}(\rho_{\varepsilon_1} - \rho_{\varepsilon_2}) + \operatorname{div}\big((\rho_{\varepsilon_1} - \rho_{\varepsilon_2})v\big) + c(\rho_{\varepsilon_1} - \rho_{\varepsilon_2}) = (R_{\varepsilon_1} - R_{\varepsilon_2}). \quad \text{(VI.12)}$$

Let $\beta : \mathbb{R} \to \mathbb{R}$ be a smooth function such that β' is bounded. We multiply (VI.12) by $\beta'(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})$ (this is allowed because $\rho_\varepsilon \in W^{1,1}(]0,T[\times\Omega)$). We get the following equation in the distribution sense

$$\begin{aligned}&\frac{\partial}{\partial t}\Big(\beta(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\Big) + \operatorname{div}\big(\beta(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})v\big) + c(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\beta'(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\\ &\quad + (\operatorname{div} v)\big((\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\beta'(\rho_{\varepsilon_1} - \rho_{\varepsilon_2}) - \beta(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\big)\\ &= \beta'(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})(R_{\varepsilon_1} - R_{\varepsilon_2}).\end{aligned} \quad \text{(VI.13)}$$

Let $\varphi \in \mathcal{C}^{0,1}(\overline{\Omega})$ be a time-independent Lipschitz continuous function which satisfies $\varphi = 0$ on Γ and $0 \leq \varphi \leq 1$ in Ω. We use this test function in (VI.13) to obtain

$$\begin{aligned}&\frac{d}{dt}\int_\Omega \varphi\beta(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\,dx - \int_\Omega \beta(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})v\cdot\nabla\varphi\,dx\\ &\quad + \int_\Omega c(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\beta'(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\varphi\,dx\\ &\quad + \int_\Omega (\operatorname{div} v)\big((\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\beta'(\rho_{\varepsilon_1} - \rho_{\varepsilon_2}) - \beta(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\big)\varphi\,dx\\ &= \int_\Omega \beta'(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})(R_{\varepsilon_1} - R_{\varepsilon_2})\varphi\,dx,\end{aligned}$$

so that for any $s \in [0,T]$, we have

$$\begin{aligned}&\int_\Omega \varphi\beta(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})(s)\,dx + \int_0^s\int_\Omega c(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\beta'(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\varphi\,dx\,dt\\ &\quad + \int_0^s\int_\Omega (\operatorname{div} v)\big((\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\beta'(\rho_{\varepsilon_1} - \rho_{\varepsilon_2}) - \beta(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\big)\varphi\,dx\,dt\\ &\leq \int_\Omega |\rho_{\varepsilon_1}(0,\cdot) - \rho_{\varepsilon_2}(0,\cdot)|\,dx + C\int_0^s\int_\Omega |R_{\varepsilon_1} - R_{\varepsilon_2}|\,dx\,dt\\ &\quad + C\int_0^s\int_\Omega |\rho_{\varepsilon_1} - \rho_{\varepsilon_2}||v\cdot\nabla\varphi|\,dx\,dt.\end{aligned}$$

We use Lemma VI.1.4, take $\beta = \beta_\delta$ (which is smooth enough) in this inequality, and then pass to the limit when $\delta \to 0$. By using the dominated convergence theorem and the convergence properties of β_δ given in the lemma, we get

$$\begin{aligned}&\int_\Omega \varphi|\rho_{\varepsilon_1}-\rho_{\varepsilon_2}|(s)\,dx + \int_0^s\int_\Omega c|\rho_{\varepsilon_1}-\rho_{\varepsilon_2}|\varphi\,dx\,dt\\ \le&\|\rho_{\varepsilon_1}(0)-\rho_{\varepsilon_2}(0)\|_{L^1(\Omega)} + C\|R_{\varepsilon_1}-R_{\varepsilon_2}\|_{L^1(]0,T[\times\Omega)}\\ &+C\int_0^s\int_\Omega |\rho_{\varepsilon_1}-\rho_{\varepsilon_2}||v\cdot\nabla\varphi|\,dx\,dt.\end{aligned}$$

We remark now that $c = c + \operatorname{div} v - \operatorname{div} v \ge -(c+\operatorname{div} v)^- - (\operatorname{div} v)^+$, so that

$$0 \le c^- \le (c+\operatorname{div} v)^- + (\operatorname{div} v)^+ \in L^1(]0,T[, L^\infty(\Omega)),$$

thanks to (VI.6). Therefore, we can write, for any $s\in[0,T]$,

$$\begin{aligned}\int_\Omega \varphi|\rho_{\varepsilon_1}-\rho_{\varepsilon_2}|(s)\,dx \le&\|\rho_{\varepsilon_1}(0)-\rho_{\varepsilon_2}(0)\|_{L^1(\Omega)} + C\|R_{\varepsilon_1}-R_{\varepsilon_2}\|_{L^1(]0,T[\times\Omega)}\\ &+\int_0^s \|c^-(t)\|_{L^\infty(\Omega)}\|\rho_{\varepsilon_1}(t)-\rho_{\varepsilon_2}(t)\|_{L^1(\Omega)}\,dt\\ &+C\int_0^T\int_\Omega |\rho_{\varepsilon_1}-\rho_{\varepsilon_2}||v\cdot\nabla\varphi|\,dt\,dx. \qquad \text{(VI.14)}\end{aligned}$$

The idea is now to take a function $\varphi=\varphi_h$ in such a way that $\varphi_h\to 1$ in Ω when $h\to 0$, so that the limit of the left-hand side in (VI.14) will exactly be the L^1 norm of $\rho_{\varepsilon_1}-\rho_{\varepsilon_2}$. The only problem that we have to deal with is the last term in the right-hand side.

Let us define φ_h, for $h>0$ as follows

$$\varphi_h(x) = \frac{1}{h}\min\left(h, \delta(x)\right),\ \forall x\in\Omega,$$

where we recall that $\delta(x)$ is the distance from x to the boundary of Ω. By construction, φ_h is Lipschitz continuous on $\mathbb{R}^d$, satisfies $0\le\varphi_h\le 1$ and since $\operatorname{Lip}(\delta)\le 1$, we have

$$|\nabla\varphi_h| = \begin{cases}0 & \text{in } \Omega_h,\\ \dfrac{1}{h} & \text{in } \mathcal{O}_h,\end{cases}$$

where we used here the notation introduced at the beginning of Section 1 of Chapter III. For any $h,\varepsilon_1,\varepsilon_2$, we have

$$\int_\Omega (1-\varphi_h)|\rho_{\varepsilon_1}-\rho_{\varepsilon_2}|(s,.)\,dx \le 2\|\rho\|_{L^\infty}\int_\Omega(1-\varphi_h)\,dx \le 2\|\rho\|_{L^\infty}|\mathcal{O}_h|,$$

so that, for any $s \in [0,T]$, (VI.14) leads to

$$\begin{aligned}\|\rho_{\varepsilon_1}(s)-\rho_{\varepsilon_2}(s)\|_{L^1(\Omega)} \leq & 2\|\rho\|_{L^\infty}|\mathcal{O}_h| + \|\rho_{\varepsilon_1}(0)-\rho_{\varepsilon_2}(0)\|_{L^1(\Omega)} \\ & + \int_0^s \|c^-(t)\|_{L^\infty(\Omega)}\|\rho_{\varepsilon_1}(t)-\rho_{\varepsilon_2}(t)\|_{L^1(\Omega)}\,dt \\ & + C\|R_{\varepsilon_1}-R_{\varepsilon_2}\|_{L^1(]0,T[\times\Omega)} \\ & + \underbrace{\frac{C}{h}\int_0^T\int_{\mathcal{O}_h}|\rho_{\varepsilon_1}-\rho_{\varepsilon_2}||v|\,dx\,dt}_{=I_{h,\varepsilon_1,\varepsilon_2}(v)}. \end{aligned} \tag{VI.15}$$

Let us study the term $I_{h,\varepsilon_1,\varepsilon_2}(v)$. To this end we consider a smooth vector field $w \in \mathcal{C}^\infty([0,T]\times\overline{\Omega})^d$ and we write

$$\begin{aligned} I_{h,\varepsilon_1,\varepsilon_2}(v) \leq & I_{h,\varepsilon_1,\varepsilon_2}(v-w) + I_{h,\varepsilon_1,\varepsilon_2}(w) \\ \leq & \frac{2C\|\rho\|_{L^\infty}}{h}\|v-w\|_{L^1(]0,T[\times\Omega)} \\ & + \frac{C}{h}\|w\|_{L^\infty(]0,T[\times\Omega)}\|\rho_{\varepsilon_1}-\rho_{\varepsilon_2}\|_{L^1(]0,T[\times\Omega)}. \end{aligned}$$

Since (VI.15) holds for any $s \in [0,T]$, and $c^- \in L^1(]0,T[,L^\infty(\Omega))$ we use the Gronwall lemma to get

$$\begin{aligned}\sup_{[0,T]}\|\rho_{\varepsilon_1}-\rho_{\varepsilon_2}\|_{L^1(\Omega)} \leq & C\Bigg(|\mathcal{O}_h| + \|\rho_{\varepsilon_1}(0)-\rho_{\varepsilon_2}(0)\|_{L^1(\Omega)} \\ & + \|R_{\varepsilon_1}-R_{\varepsilon_2}\|_{L^1(]0,T[\times\Omega)} + \frac{1}{h}\|v-w\|_{L^1(]0,T[\times\Omega)} \\ & + \frac{1}{h}\|w\|_{L^\infty(]0,T[\times\Omega)}\|\rho_{\varepsilon_1}-\rho_{\varepsilon_2}\|_{L^1(]0,T[\times\Omega)}\Bigg). \end{aligned}$$

Notice that the left-hand side of the above estimate only depends on ε_1 and ε_2 whereas the right-hand side also depends on h and w. Let us now choose some $\zeta > 0$. We first take $h > 0$ such that $C|\mathcal{O}_h| \leq \zeta$. Using the density of $\mathcal{C}^\infty([0,T]\times\overline{\Omega})^d$ in $L^1(]0,T[\times\Omega)^d$, we then choose $w \in \mathcal{C}^\infty([0,T]\times\overline{\Omega})^d$ such that

$$\frac{C}{h}\|v-w\|_{L^1(]0,T[\times\Omega)} \leq \zeta.$$

We use now the fact that $\rho_\varepsilon(0) \to \rho_0$ in $L^1(\Omega)$, $\rho_\varepsilon \to \rho$ in $L^1(]0,T[\times\Omega)$ and $R^\varepsilon \to 0$ in $L^1(]0,T[\times\Omega)$. Since w and h are fixed, we can find some $\varepsilon_0 > 0$ such that, for any $0 < \varepsilon_1, \varepsilon_2 < \varepsilon_0$, we have

$$C\|\rho_{\varepsilon_1}(0) - \rho_{\varepsilon_2}(0)\|_{L^1(\Omega)} \leq \zeta,$$
$$C\|R_{\varepsilon_1} - R_{\varepsilon_2}\|_{L^1(]0,T[\times\Omega)} \leq \zeta,$$
$$\frac{C}{h}\|w\|_{L^\infty(]0,T[\times\Omega)}\|\rho_{\varepsilon_1} - \rho_{\varepsilon_2}\|_{L^1(]0,T[\times\Omega)} \leq \zeta.$$

Gathering all these estimates we finally get

$$\sup_{[0,T]}\|\rho_{\varepsilon_1} - \rho_{\varepsilon_2}\|_{L^1(\Omega)} \leq 5\zeta,\ \forall 0 < \varepsilon_1, \varepsilon_2 < \varepsilon_0.$$

This proves that $(\rho_\varepsilon)_\varepsilon$ is a Cauchy sequence in $\mathcal{C}^0([0,T], L^1(\Omega))$. We already know that $(\rho_\varepsilon)_\varepsilon$ converges to ρ in $L^1(]0,T[\times\Omega)$, thus it follows that

$$\rho \in \mathcal{C}^0([0,T], L^1(\Omega)),$$

$$\rho_\varepsilon \xrightarrow[\varepsilon\to 0]{} \rho, \quad \text{in } \mathcal{C}^0([0,T], L^1(\Omega)).$$

By (VI.10), we immediately deduce that the convergence also holds in $\mathcal{C}^0([0,T], L^p(\Omega))$ for any $p < +\infty$.

- Let us now prove the existence of the trace $\gamma\rho$. Writing equation (VI.13) with $\beta(s) = s^2$ gives

$$\begin{aligned}&\frac{\partial}{\partial t}(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})^2 + v\cdot\nabla(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})^2 + 2(c + \operatorname{div} v)\big(\rho_{\varepsilon_1} - \rho_{\varepsilon_2}\big)^2 \\ =&2\big(R_{\varepsilon_1} - R_{\varepsilon_2}\big)\big(\rho_{\varepsilon_1} - \rho_{\varepsilon_2}\big).\end{aligned} \tag{VI.16}$$

We choose a test function $\psi \in \mathcal{C}^{0,1}([0,T]\times\overline{\Omega})$ such that $\psi(0,.) = \psi(T,.) = 0$ in (VI.16). We integrate by parts and look precisely at the following boundary term

$$\begin{aligned}&\int_0^T\int_\Gamma |\rho_{\varepsilon_1} - \rho_{\varepsilon_2}|^2\psi(v\cdot\nu)\,d\sigma\,dt \\ =&\int_0^T\int_\Omega |\rho_{\varepsilon_1} - \rho_{\varepsilon_2}|^2\left(\frac{\partial\psi}{\partial t} + v\cdot\nabla\psi - (\operatorname{div} v)\psi - 2c\psi\right)dx\,dt \\ &+ 2\int_0^T\int_\Omega (R_{\varepsilon_1} - R_{\varepsilon_2})(\rho_{\varepsilon_1} - \rho_{\varepsilon_2})\psi\,dx\,dt.\end{aligned} \tag{VI.17}$$

Notice that ρ_ε has a trace on Γ in the usual sense because it is smooth with respect to the space variable. We would like to choose ψ such that $\psi = \operatorname{sgn}(v\cdot\nu)$ on the boundary but this is not possible because of the regularity requirements on ψ. Nevertheless, using for instance Theorem III.2.21 we can find a sequence $\psi_n \in \mathcal{C}^{0,1}([0,T]\times\overline{\Omega})$ satisfying $\|\psi_n\|_{L^\infty(]0,T[\times\Omega)} \leq 1$, $\psi_n(0,.) = \psi_n(T,.) = 0$ and such that

$$\psi_n \xrightarrow[n\to\infty]{} \operatorname{sgn}(v\cdot\nu), \quad \text{a.e. in }]0,T[\times\Gamma. \tag{VI.18}$$

Taking the test function $\psi = \psi_n$ in (VI.17), it follows that

$$\begin{aligned}\int_0^T \int_\Gamma |\rho_{\varepsilon_1} - \rho_{\varepsilon_2}|^2 |v\cdot\nu|\, d\sigma\, dt \leq & 4\|\rho\|_{L^\infty}^2 \int_0^T \int_\Gamma |v\cdot\nu|\, |\psi_n - \operatorname{sgn}(v\cdot\nu)|\, d\sigma\, dt \\ & + \int_0^T \int_\Omega |\rho_{\varepsilon_1} - \rho_{\varepsilon_2}|^2 F_n\, dx\, dt \\ & + C\|R_{\varepsilon_1} - R_{\varepsilon_2}\|_{L^1(]0,T[\times\Omega)}, \end{aligned} \tag{VI.19}$$

where $F_n = (\partial\psi_n/\partial t + v\cdot\nabla\psi_n - (\operatorname{div} v)\psi_n - 2c\psi_n) \in L^1(]0,T[\times\Omega)$.
Let $\zeta > 0$ be fixed. The first term in the right-hand side of (VI.19) tends to 0 as $n \to \infty$ by using the Lebesgue dominated convergence theorem and (VI.18). Therefore, this term can be made smaller than ζ for some n. Such an integer n being fixed, by using (VI.11), we can choose $\varepsilon_0 > 0$ such that the second and third terms are less than ζ for any $\varepsilon_1, \varepsilon_2$ such that $\varepsilon_1 < \varepsilon_0$, $\varepsilon_2 < \varepsilon_0$.
All these computations show that the restriction to $]0,T[\times\Gamma$ of $(\rho_\varepsilon)_\varepsilon$ is a Cauchy sequence in the space $L^2(]0,T[\times\Gamma, |d\mu_v|)$. We define $\gamma\rho \in L^2(]0,T[\times\Gamma, |d\mu_v|)$ to be the corresponding limit. Notice that $\gamma\rho$ is well-defined $|d\mu_v|$-almost everywhere.
For any $\beta \in \mathcal{C}^1(\mathbb{R})$, we can compute from (VI.9) that

$$\frac{\partial\beta(\rho_\varepsilon)}{\partial t} + v\cdot\nabla\beta(\rho_\varepsilon) + (c + \operatorname{div} v)\rho_\varepsilon\beta'(\rho_\varepsilon) = \beta'(\rho_\varepsilon)R^\varepsilon.$$

Testing against a function $\varphi \in \mathcal{C}^1([t_0,t_1]\times\overline{\Omega})$ we get

$$\begin{aligned} & \int_{t_0}^{t_1}\int_\Omega \beta(\rho_\varepsilon)\left(\frac{\partial\varphi}{\partial t} + v\cdot\nabla\varphi\right) dx\, dt - \int_{t_0}^{t_1}\int_\Omega c\rho\beta'(\rho_\varepsilon)\varphi\, dx\, dt \\ & - \int_{t_0}^{t_1}\int_\Omega (\operatorname{div} v)(\rho_\varepsilon\beta'(\rho_\varepsilon) - \beta(\rho_\varepsilon))\varphi\, dx\, dt - \int_{t_0}^{t_1}\int_\Gamma \beta(\rho_\varepsilon)\varphi(v\cdot\nu)\, d\sigma\, dt \\ & + \int_\Omega \beta(\rho_\varepsilon(t_0))\varphi(t_0)\, dx - \int_\Omega \beta(\rho_\varepsilon(t_1))\varphi(t_1)\, dx \\ = & \int_0^T\int_\Omega R_\varepsilon\varphi\, dx\, dt. \end{aligned}$$

Thanks to all the convergences proved above, we can pass to the limit as $\varepsilon \to 0$, in particular in the boundary term, and obtain (VI.8). Finally, (VI.7) follows by simply taking $\beta(s) = s$.

- It remains to show the uniqueness property. Assume that $\gamma_1\rho, \gamma_2\rho$ belong to $L^\infty(]0,T[\times\Gamma, |d\mu_v|)$ and both satisfy (VI.7); then we have

$$\int_0^T\int_\Gamma (\gamma_1\rho - \gamma_2\rho)\varphi(v\cdot\nu)\, d\sigma\, dt = 0,$$

for any test function $\varphi \in \mathcal{C}^{0,1}([0,T]\times\overline{\Omega})$. By density, there exists a sequence of Lipschitz continuous functions φ_n such that

$$\sup_n \|\varphi_n\|_{L^\infty} < +\infty,$$

$$\varphi_n \xrightarrow[n\to\infty]{} (\gamma_1\rho - \gamma_2\rho)\,\mathrm{sgn}(v\cdot\nu), \quad \text{almost everywhere in }]0,T[\times\Gamma.$$

It follows, by the dominated convergence theorem, that

$$\int_0^T \int_\Gamma |\gamma_1\rho - \gamma_2\rho|^2 |v\cdot\nu|\, d\sigma\, dt = 0,$$

so that $\gamma_1\rho = \gamma_2\rho$ for $|d\mu_v|$-almost every $(t,\sigma) \in]0,T[\times\Gamma$.

□

Corollary VI.1.5. *Assume that (VI.4), (VI.5), and (VI.6) hold and let ρ be a bounded weak solution to (VI.2).*

1. *For any $\alpha \neq 0$, we have the following property*

$$c + \mathrm{div}\, v = 0, \quad \textit{for almost every } (t,x) \textit{ in the level set } \{\rho = \alpha\}.$$

2. *For any function $\beta : \mathbb{R} \mapsto \mathbb{R}$ which is continuous and piecewise $\mathcal{C}^1$, we have the renormalisation property (VI.8), for any choice of the values of β' at singular points.*

Proof.

Notice that the first property of the lemma cannot hold for $\alpha = 0$ (for instance the solution $\rho \equiv 0$ is an obvious counterexample).

1. For any $0 < \varepsilon \leq 1$, we define

$$\beta_\varepsilon(s) = \frac{\sqrt{\varepsilon}(s-\alpha)}{\sqrt{(s-\alpha)^2 + \varepsilon}},$$

which is a smooth function satisfying $\|\beta_\varepsilon\|_{L^\infty} \leq 1$ and $\|\beta'_\varepsilon\|_{L^\infty} \leq 1$. We observe that

$$\beta_\varepsilon(s) \xrightarrow[\varepsilon\to 0]{} 0, \quad \forall s \in \mathbb{R},$$

$$s\beta'_\varepsilon(s) \xrightarrow[\varepsilon\to 0]{} \begin{cases} \alpha, & \text{for } s = \alpha, \\ 0, & \text{for } s \neq \alpha. \end{cases}$$

We then apply (VI.8) and find that, for any $\varphi \in \mathcal{C}_c^\infty(]0,T[\times\Omega)$ we have

$$\int_0^T \int_\Omega \beta_\varepsilon(\rho) \left(\frac{\partial\varphi}{\partial t} + v\cdot\nabla\varphi \right) dx\, dt$$
$$+ \int_0^T \int_\Omega \Big(\beta_\varepsilon(\rho)(\mathrm{div}\, v) - \rho\beta'_\varepsilon(\rho)(c + \mathrm{div}\, v) \Big) \varphi\, dx\, dt = 0.$$

We now pass to the limit when $\varepsilon \to 0$ by the Lebesgue convergence theorem to obtain

$$\alpha \int_0^T \int_\Omega (c + \operatorname{div} v)\varphi 1_{\{\rho=\alpha\}}\, dx\, dt = 0.$$

Since $\alpha \neq 0$, and $c + \operatorname{div} v \in L^1(]0,T[\times\Omega)$ and by weak-$\star$ density of $\mathcal{C}_c^\infty(]0,T[\times\Omega)$ into $L^\infty(]0,T[\times\Omega)$, we deduce that

$$\int_0^T \int_\Omega (c + \operatorname{div} v)\varphi 1_{\{\rho=\alpha\}}\, dx\, dt = 0,$$

for any $\varphi \in L^\infty(]0,T[\times\Omega)$. The claim is proved.

2. Let $\beta : \mathbb{R} \mapsto \mathbb{R}$ be a continuous piecewise $\mathcal{C}^1$ function. Notice that (VI.8) has a sense even though β' is not well-defined at some points because for such points $\alpha \in \mathbb{R}$ (even for $\alpha = 0$) we have, by the first point proved in this corollary,

$$\int_0^T \int_\Omega (c + \operatorname{div} v)\rho 1_{\{\rho=\alpha\}}\varphi\, dx\, dt = 0,\ \forall \varphi \in L^\infty(]0,T[\times\Omega).$$

Such a piecewise smooth function β can be written as the sum of a globally $\mathcal{C}^1$ function and a finite linear combination of functions of the form $s \mapsto |s - \alpha_i|$. By linearity of the problem and by translation, it is then enough to prove the claim for $\beta(s) = |s|$.
To this end, we use the smooth approximation β_δ introduced in Lemma VI.1.4 and we pass to the limit when $\delta \to 0$.

□

Remark VI.1.4. During the proof of Theorem VI.1.3, we built a sequence of approximations $(\rho_\varepsilon)_\varepsilon$ of the weak solution ρ, which are smooth with respect to the space variable and which satisfy the same equation as ρ up to a remainder term which strongly converges to 0 in $L^1(]0,T[\times\Omega)$. We want here to observe that it is possible to obtain a similar result with approximate functions which are smooth with respect to both time and space variables. This can be of importance in some applications; see for instance [23].

To do this we consider, for any $\xi > 0$, a mollifying operator $\overline{\mathcal{S}}_\xi$ associated with the 1D domain $]0,T[$, as defined in Section 2.2 of Chapter III (we added a line over the symbol $\overline{\mathcal{S}}_\xi$ to emphasise the fact that, contrary to $\mathcal{S}_\varepsilon$, it opers on functions depending on t).

We set $\rho_{\varepsilon,\xi} = \overline{\mathcal{S}}_\xi \rho_\varepsilon = \overline{\mathcal{S}}_\xi(\mathcal{S}_\varepsilon \rho)$. By construction, $\rho_{\varepsilon,\xi}$ is smooth with respect to both the time and space variables. Moreover, it satisfies

$$\frac{\partial \rho_{\varepsilon,\xi}}{\partial t} + \operatorname{div}(\rho_{\varepsilon,\xi} v) + c\rho_{\varepsilon,\xi} = R_{\varepsilon,\xi},$$

with

$$\begin{aligned}R_{\varepsilon,\xi} =&\overline{\mathcal{S}}_\xi R_\varepsilon + \left(\frac{\partial \overline{\mathcal{S}}_\xi \rho_\varepsilon}{\partial t} - \overline{\mathcal{S}}_\xi \frac{\partial \rho_\varepsilon}{\partial t}\right)\\ &+ \operatorname{div}\left((\overline{\mathcal{S}}_\xi \rho_\varepsilon)v - \overline{\mathcal{S}}_\xi(\rho_\varepsilon v)\right) + \left((\overline{\mathcal{S}}_\xi \rho_\varepsilon)c - \overline{\mathcal{S}}_\xi(\rho_\varepsilon c)\right).\end{aligned}$$

Using in particular that, for any given $\varepsilon > 0$, $\partial\rho_\varepsilon/\partial t$ belongs to L^1 and that ρ_ε is smooth with respect to x, we get that $R_{\varepsilon,\xi} \in L^1(]0,T[\times\Omega)$ and that

$$R_{\varepsilon,\xi} \xrightarrow[\xi\to 0]{} R_\varepsilon, \quad \text{in } L^1(]0,T[\times\Omega), \text{ for any } \varepsilon > 0.$$

Note that we do not need any time regularity of v at that point inasmuch as the operator $\overline{\mathcal{S}}_\xi$ only acts on the time variable. We also immediately get that $\rho_{\varepsilon,\xi}$ converges to ρ_ε when $\xi \to 0$ in $\mathcal{C}^0([0,T], L^p(\Omega))$ for any finite p.

Since R_ε tends to 0 as $\varepsilon \to 0$, one can choose ξ as a function of ε so that $\lim_{\varepsilon\to 0}\xi(\varepsilon) = 0$ and

$$\begin{aligned}&R_{\varepsilon,\xi(\varepsilon)} \xrightarrow[\varepsilon\to 0]{} 0, \quad \text{in } L^1(]0,T[\times\Omega),\\ &\rho_{\varepsilon,\xi(\varepsilon)} \xrightarrow[\varepsilon\to 0]{} \rho, \quad \text{in } \mathcal{C}^0([0,T], L^p(\Omega)) \text{ for any } p < +\infty.,\end{aligned}$$

with $\rho_{\varepsilon,\xi(\varepsilon)} \in \mathcal{C}^\infty([0,T]\times\overline{\Omega})$. Convergence of the traces are obtained in the same way.

We emphasise the fact that the same result cannot be obtained by directly using a mollifying operator associated with the $d+1$-dimensional Lipschitz domain $]0,T[\times\Omega$, because it would require more time regularity for the velocity field v in order to apply the Friedrichs commutator estimate.

Here, we used the particular structure of the $d+1$-dimensional vector field $(1, v(t,x))$ which let us apply successively a mollifier with respect to x and then a mollifier with respect to t. This kind of argument was used for instance, in [78] to extend the renormalised solution theory to partially $W^{1,1}$ velocity fields.

1.3 The initial- and boundary-value problem

Using the trace theorem that we proved above, we can now study the existence and uniqueness of the Cauchy–Dirichlet problem for transport problems.

Theorem VI.1.6. *Assume that v and c satisfy assumptions (VI.4) through (VI.6). For any initial data $\rho_0 \in L^\infty(\Omega)$, and any inflow boundary data $\rho^{in} \in L^\infty(]0,T[\times\Gamma, d\mu_v^-)$, there exists a unique $\rho \in L^\infty(]0,T[\times\Omega)$ such that*

- *ρ is a weak solution to the transport equation (VI.2).*
- *The initial condition $\rho(0) = \rho_0$ is satisfied.*
- *The trace $\gamma\rho$ of the solution ρ satisfies the inflow boundary condition*

$$\gamma\rho = \rho^{in}, \quad d\mu_v^- \text{-}almost\ everywhere\ in\ ]0,T[\times\Gamma.$$

Moreover this solution satisfies

$$\|\rho(t,.)\|_{L^\infty(\Omega)} \leq M \exp\left(\int_0^t \alpha(s)\,ds\right), \ \forall t \in [0,T], \tag{VI.20a}$$

$$\|\gamma\rho\|_{L^\infty(]0,T[\times\Gamma,|d\mu_v|)} \leq M \exp\left(\int_0^T \alpha(s)\,ds\right), \tag{VI.20b}$$

where

$$M = \max\Big(\|\rho_0\|_{L^\infty}, \|\rho^{in}\|_{L^\infty(]0,T[\times\Gamma,d\mu_v^-)}\Big),$$

and

$$\alpha(s) = \|(c+\operatorname{div} v)^-(s,.)\|_{L^\infty(\Omega)}, \ \textit{for any } s \in [0,T]. \tag{VI.21}$$

Proof.

- Let us first show the uniqueness property. Assume that ρ_1, ρ_2 are two weak solutions of the problem with the same initial and inflow boundary data. We set $\rho = \rho_1 - \rho_2$, so that ρ is also a weak solution to the problem which satisfies $\rho(0) = 0$ and $\gamma\rho = 0$, $d\mu_v^-$-almost everywhere in $]0,T[\times\Gamma$.
  We apply the renormalisation property (VI.8) to ρ and for $\beta(\xi) = |\xi|$, $\varphi = 1$, $t_0 = 0$ and $t_1 = s$ (see Corollary VI.1.5). It follows that, for any $s \in [0,T]$,

$$\int_0^s\int_\Omega c|\rho|\,dx\,dt + \int_\Omega |\rho(s,.)|\,dx + \int_0^s\int_\Gamma |\gamma\rho|(v\cdot\nu)^+\,d\sigma\,dt = 0. \tag{VI.22}$$

  Since $c^- \in L^1(]0,T[, L^\infty(\Omega))$ we first deduce that

$$\|\rho(s)\|_{L^1(\Omega)} \leq \int_0^s \|c^-(t)\|_{L^\infty(\Omega)}\|\rho(t)\|_{L^1(\Omega)}\,dt, \forall s \in [0,T],$$

  and thus, by using the Gronwall lemma, we get $\rho = 0$ in $]0,T[\times\Omega$.
  Coming back to (VI.22), we deduce that $\gamma\rho = 0$, $d\mu_v^+$-almost everywhere. This concludes the proof of the uniqueness property.
- For any $\varepsilon > 0$ we consider the following parabolic problem

$$\begin{cases} \dfrac{\partial\tilde\rho_\varepsilon}{\partial t} + \operatorname{div}(\tilde\rho_\varepsilon v) + c\tilde\rho_\varepsilon - \varepsilon\Delta\tilde\rho_\varepsilon = 0, & \text{in } \Omega \\ \varepsilon\dfrac{\partial\tilde\rho_\varepsilon}{\partial\nu} + (\tilde\rho_\varepsilon - \rho^{in})(v\cdot\nu)^- = 0, & \text{on } \Gamma, \\ \tilde\rho_\varepsilon(0) = \rho_0, \end{cases}$$

  whose weak formulation is

$$
\begin{aligned}
&-\int_0^T \int_\Omega \tilde{\rho}_\varepsilon \left(\frac{\partial \varphi}{\partial t} + v \cdot \nabla \varphi - c\varphi \right) dx\, dt + \varepsilon \int_0^T \int_\Omega \nabla \tilde{\rho}_\varepsilon \cdot \nabla \varphi \, dx\, dt \\
&+ \int_0^T \int_\Gamma (\tilde{\rho}_\varepsilon (v\cdot\nu)^+ \varphi - \rho^{in} (v\cdot\nu)^- \varphi)\, d\sigma\, dt \\
&+ \int_\Omega \tilde{\rho}_\varepsilon(T)\varphi(T)\, dx - \int_\Omega \rho_0 \varphi(0)\, dx = 0,
\end{aligned} \tag{VI.23}
$$

for any $\varphi \in \mathcal{C}^{0,1}([0,T] \times \overline{\Omega})$.
By using a standard Galerkin approximation for this problem, we can easily show that for any $\rho_0 \in L^\infty(\Omega)$ and $\rho^{in} \in L^\infty(]0,T[\times\Gamma, d\mu_v^-)$ there exists a unique solution $\tilde{\rho}_\varepsilon \in L^\infty(]0,T[, L^2(\Omega)) \cap L^2(]0,T[, H^1(\Omega))$ to (VI.23). There is no particular difficulty because the problem is linear and satisfies the following energy estimates (which hold uniformly with respect to the Galerkin approximations)

$$
\|\tilde{\rho}_\varepsilon\|_{L^\infty(]0,T[,L^2(\Omega))} + \|\sqrt{\varepsilon}\nabla \tilde{\rho}_\varepsilon\|_{L^2(]0,T[,L^2(\Omega))} \leq C, \tag{VI.24}
$$

$$
\int_0^T \int_\Gamma |\gamma_0 \tilde{\rho}_\varepsilon|^2 (v\cdot\nu)^+ \, d\sigma\, dt \leq C, \tag{VI.25}
$$

where C does not depend on ε. We have used here the fact that, since it is smooth enough in the space variable, the function $\tilde{\rho}_\varepsilon$ has a trace $\gamma_0 \tilde{\rho}_\varepsilon \in L^2(]0,T[, H^{1/2}(\Gamma)) \subset L^2(]0,T[\times\Gamma)$, in the usual trace sense (see Section 2.5 in Chapter III).
To obtain these energy estimates, we first choose $\varphi = \tilde{\rho}_\varepsilon$ in (VI.23). It follows

$$
\begin{aligned}
&\int_\Omega |\tilde{\rho}_\varepsilon(T)|^2\, dx + \int_0^T \int_\Omega (2c + \operatorname{div} v)\tilde{\rho}_\varepsilon^2 \, dx\, dt + 2\varepsilon \int_0^T \int_\Omega |\nabla \tilde{\rho}_\varepsilon|^2\, dx\, dt \\
&+ \int_0^T \int_\Gamma \tilde{\rho}_\varepsilon^2 (v\cdot\nu)^+ \, d\sigma\, dt + \int_0^T \int_\Gamma (\tilde{\rho}_\varepsilon - \rho^{in})^2 (v\cdot\nu)^- \, d\sigma\, dt \\
= &\int_\Omega \rho_0^2\, dx + \int_0^T \int_\Gamma (\rho^{in})^2 (v\cdot\nu)^- \, d\sigma\, dt.
\end{aligned} \tag{VI.26}
$$

Then we observe that, by (VI.6), we have

$$
0 \leq (2c + \operatorname{div} v)^- \leq (2c + 2\operatorname{div} v)^- - (\operatorname{div} v)^+ \in L^1(]0,T[, L^\infty(\Omega)).
$$

Therefore, we can use the Gronwall lemma from (VI.26) and obtain the bounds (VI.24) and (VI.25).
Let us now state the following L^∞ bounds on the approximation solution $\tilde{\rho}_\varepsilon$, whose proof is admitted for the moment.

Lemma VI.1.7. *For any $\varepsilon > 0$, we have the estimate*

$$\|\tilde{\rho}_\varepsilon(t)\|_{L^\infty(\Omega)} \leq M \exp\left(\int_0^t \alpha(s)\,ds\right), \forall t \in [0,T],$$

where α is defined in (VI.21). In particular the sequence $(\tilde{\rho}_\varepsilon)_\varepsilon$ (resp., $(\gamma_0\tilde{\rho}_\varepsilon)_\varepsilon$) is bounded in $L^\infty(]0,T[\times\Omega)$ (resp., in $L^\infty(]0,T[\times\Gamma, d\mu_v^+)$).

Thanks to this lemma and to Proposition II.2.28, we can find a sequence $(\varepsilon_k)_k$, a function $\rho \in L^\infty(]0,T[\times\Omega)$ and a function $\rho^{out} \in L^\infty(]0,T[\times\Gamma, d\mu_v^+)$ such that

$$\begin{aligned}
\tilde{\rho}_{\varepsilon_k} &\xrightarrow[k\to\infty]{} \rho, &&\text{weakly-}\star \text{ in } L^\infty(]0,T[\times\Omega),\\
\tilde{\rho}_{\varepsilon_k} &\xrightarrow[k\to\infty]{} \rho^{out}, &&\text{weakly-}\star \text{ in } L^\infty(]0,T[\times\Gamma, d\mu_v^+).
\end{aligned}$$

Thanks to these convergences and to the estimate (VI.24), it is now straightforward to perform the limit in the linear equation (VI.23). This proves that ρ satisfies the weak formulation of Problem (VI.2) with (VI.3). Estimates (VI.20) are immediately deduced from Lemma VI.1.7.

□

It remains to prove the L^∞ bounds we used.

Proof (of Lemma VI.1.7).

By changing $\tilde{\rho}_\varepsilon$ into $\tilde{\rho}_\varepsilon \exp\Big(-\int_0^t \alpha(s)ds\Big)$, we see that it is enough to consider the case where $c + \operatorname{div} v \geq 0$, and to show the monotonicity property

$$\Big[\rho_0 \geq 0 \text{ and } \rho^{in} \geq 0\Big] \Rightarrow \tilde{\rho}_\varepsilon \geq 0.$$

To this end, we define a function $\beta : \mathbb{R} \to \mathbb{R}$ of class $\mathcal{C}^2$ by

$$\beta(\xi) = \begin{cases} |\xi|^3 & \text{for } \xi \leq 0\\ 0 & \text{for } \xi > 0.\end{cases}$$

Then we take the test-function $\varphi = \beta'(\tilde{\rho}_\varepsilon)$ in the weak formulation (VI.23). It follows after some computations that for any $s \in [0,T]$ we have

$$\begin{aligned}
&\int_\Omega \beta(\tilde{\rho}_\varepsilon(s))\,dx + \int_0^s \int_\Omega (3c + 2\operatorname{div} v)\beta(\tilde{\rho}_\varepsilon)\,dx\,dt\\
&\quad + \varepsilon \int_0^s \int_\Omega \beta''(\tilde{\rho}_\varepsilon)|\nabla\tilde{\rho}_\varepsilon|^2\,dx\,dt + \int_0^s \int_\Gamma \beta(\tilde{\rho}_\varepsilon)(v\cdot\nu)^+\,d\sigma\,dt\\
&= \int_\Omega \beta(\rho_0)\,dx + \int_0^s \int_\Gamma \Big(\beta(\tilde{\rho}_\varepsilon) + \beta'(\tilde{\rho}_\varepsilon)(\rho^{in} - \tilde{\rho}_\varepsilon)\Big)(v\cdot\nu)^-\,d\sigma\,dt.
\end{aligned}$$

Since β is convex and $\rho^{in} \geq 0$, we have

$$\beta(\tilde{\rho}_\varepsilon) + \beta'(\tilde{\rho}_\varepsilon)(\rho^{in} - \tilde{\rho}_\varepsilon) \leq \beta(\rho^{in}) = 0,$$

and moreover $\beta(\rho_0) = 0$ because $\rho_0 \geq 0$. Using once more the convexity of β, it follows that

$$\int_\Omega \beta(\tilde{\rho}_\varepsilon(s))\,dx + \int_0^s \int_\Omega (3c + 2\operatorname{div} v)\beta(\tilde{\rho}_\varepsilon)\,dx\,dt \leq 0,\ \forall s \in [0,T].$$

We assumed at the beginning that $c + \operatorname{div} v \geq 0$, and β is nonnegative, therefore we finally obtain

$$\begin{aligned}\int_\Omega \beta(\tilde{\rho}_\varepsilon(s))\,dx &\leq \int_0^s \int_\Omega (\operatorname{div} v)^+ \beta(\tilde{\rho}_\varepsilon)\,dx\,dt \\ &\leq \int_0^s \|(\operatorname{div} v)^+(t)\|_{L^\infty(\Omega)} \left(\int_\Omega \beta(\tilde{\rho}_\varepsilon(t))\,dx\right) dt.\end{aligned}$$

By using (VI.6b) and the Gronwall lemma once more , we deduce that

$$\int_\Omega \beta(\tilde{\rho}_\varepsilon(s))\,dx \leq 0,\ \forall s \in [0,T],$$

which gives, by definition of β, that $\tilde{\rho}_\varepsilon \geq 0$ on $]0,T[\times\Omega$.

□

Using the renormalisation property and the same function β as in the proof above, it is very easy to show with the same assumption as in Theorem VI.1.6, that the unique weak solution ρ to the transport problem associated with nonnegative initial and inflow boundary data is itself nonnegative.

A stronger version of this result can be proved with an additional assumption on c and v.

Proposition VI.1.8. *We consider the same assumptions as in Theorem VI.1.6 and moreover we suppose that $(c + \operatorname{div} v)^+ \in L^1(]0,T[, L^\infty(\Omega))$.*

If there exists $\rho_{\min} > 0$ such that $\rho_0 \geq \rho_{\min}$ and $\rho^{in} \geq \rho_{\min}$ $d\mu_v^-$-almost everywhere, then the weak solution ρ of the transport problem satisfies the following bound from below

$$\rho \geq \rho_{\min} \exp\left(\int_0^T -\|(c + \operatorname{div} v)^+(t)\|_{L^\infty}\,dt\right).$$

The proof is very similar to the previous one; we leave the details to the reader.

1.4 Stability theorem

In this section, we study the stability of weak solutions of the Cauchy–Dirichlet transport problem with respect to all the data.

Theorem VI.1.9. *For any $k \geq 1$, let $\rho_{0,k} \in L^\infty(\Omega)$, $\rho_k^{in} \in L^\infty(]0,T[\times\Gamma)$, c_k satisfying* (VI.4) *and v_k satisfying* (VI.5) *and* (VI.6).

We define $\rho_k \in L^\infty(]0,T[\times\Omega)$ to be the unique weak solution, given by Theorem VI.1.6, of the following initial- and boundary-value problem

$$\begin{cases} \dfrac{\partial \rho_k}{\partial t} + \operatorname{div}(\rho_k v_k) + c_k \rho_k = 0, & \text{in } \Omega, \\ \gamma\rho_k = \rho_k^{in}, & \text{on } \Gamma_{v_k}^-(t), \\ \rho_k(0) = \rho_{0,k}. \end{cases} \tag{VI.27}$$

We assume that there exists $c \in L^1(]0,T[\times\Omega)$, $v \in L^1(]0,T[,(W^{1,1}(\Omega))^d)$ with $(\operatorname{div} v)^+ \in L^1(]0,T[,L^\infty(\Omega))$, $\rho_0 \in L^\infty(\Omega)$, $\rho^{in} \in L^\infty(]0,T[\times\Gamma)$ such that

$$v_k \xrightarrow[k\to\infty]{} v, \qquad \text{in } L^1(]0,T[\times\Omega)^d, \tag{VI.28a}$$

$$\operatorname{div} v_k \xrightarrow[k\to\infty]{} \operatorname{div} v, \quad \text{in } L^1(]0,T[\times\Omega), \tag{VI.28b}$$

$$(v_k \cdot \nu) \xrightarrow[k\to\infty]{} (v\cdot\nu), \quad \text{in } L^1(]0,T[\times\Gamma), \tag{VI.28c}$$

$$c_k \xrightarrow[k\to\infty]{} c, \qquad \text{in } L^1(]0,T[\times\Omega), \tag{VI.28d}$$

$$\rho_k^{in} \xrightharpoonup[k\to\infty]{} \rho^{in}, \qquad \text{weakly-}\star \text{ in } L^\infty(]0,T[\times\Gamma), \tag{VI.28e}$$

$$\rho_k^{in} \xrightarrow[k\to\infty]{} \rho^{in}, \qquad \text{in } L^1(]0,T[\times\Gamma, d\mu_v^-). \tag{VI.28f}$$

$$\begin{gathered} ((c_k + \operatorname{div} v_k)^-)_k \text{ is bounded in } L^1(]0,T[,L^\infty(\Omega)), \\ (\rho_{0,k})_k \text{is bounded in } L^\infty(\Omega) \text{ and } \rho_{0,k} \xrightarrow[k\to\infty]{} \rho_0 \text{ in } L^1(\Omega) \end{gathered} \tag{VI.28g}$$

Then, if we denote by ρ the unique solution to the transport problem (VI.2)-(VI.3) *associated with the data c, v, ρ_0, ρ^{in}, we have*

$$\rho_k \xrightarrow[k\to\infty]{} \rho \quad \text{in } C^0([0,T],L^p(\Omega)), \text{ for any } p < +\infty,$$

$$\gamma\rho_k \xrightarrow[k\to\infty]{} \gamma\rho \text{ in } L^p(]0,T[\times\Gamma,|d\mu_v|), \text{for any } p<+\infty.$$

Remark VI.1.5. Notice that we do not need the strong convergence of v_k towards v in the space $L^1(]0,T[,(W^{1,1}(\Omega))^d)$ but only in $L^1(]0,T[\times\Omega)^d$ supplemented by the strong convergence of $(\operatorname{div} v_k)_k$ and of the normal traces $(v_k\cdot\nu)_k$ in L^1. In the case of tangential divergence-free velocity fields, these last two assumptions are obviously satisfied.

We also notice that it is not necessary to assume that $((\operatorname{div} v_k)^+)_k$ is bounded in $L^1(]0,T[,L^\infty(\Omega))$.

Remark VI.1.6.
- From assumption (VI.28e) and Corollary II.2.8, we deduce that $(\rho_k^{in})_k$ is bounded in $L^\infty(]0,T[\times\Gamma)$.

- Using (VI.28e), it can be easily shown that the assumption (VI.28f) is equivalent to

$$\rho_k^{in}(v_k \cdot \nu)^- \xrightarrow[k\to\infty]{} \rho^{in}(v \cdot \nu)^-, \text{ in } L^1(]0,T[\times\Gamma),$$

which can be interpreted, in the framework of fluid mechanics, as a convergence property of the inflow mass flux.

Proof.

We first observe that $\gamma\rho_k$ is only well-defined $|d\mu_{v_k}|$−almost everywhere. From now on, we set $\gamma\rho_k = 0$ on the set $\{(t,\sigma) \in]0,T[\times\Gamma,\ v_k \cdot \nu = 0\}$ so that $\gamma\rho_k$ is a well-defined element in $L^\infty(]0,T[\times\Gamma)$.

- By using the estimates (VI.20) and the assumptions on $\rho_{0,k}$, ρ_k^{in} and $(c_k + \operatorname{div} v_k)^-$, we immediately obtain that

$$\begin{cases} (\rho_k)_k & \text{is bounded in } L^\infty(]0,T[\times\Omega), \\ (\gamma\rho_k)_k & \text{is bounded in } L^\infty(]0,T[\times\Gamma). \end{cases} \tag{VI.29}$$

In particular, $(\gamma\rho_k)_k$ is bounded in $L^\infty(]0,T[\times\Gamma, |d\mu_v|)$. Therefore, we can find a subsequence (still referred to as $(\rho_k)_k$) and $R \in L^\infty(]0,T[\times\Omega)$, $g \in L^\infty(]0,T[\times\Gamma, |d\mu_v|)$ such that

$$\begin{aligned} \rho_k &\xrightarrow[k\to\infty]{} R, \quad \text{weakly-}\star \text{ in } L^\infty(]0,T[\times\Omega), \\ \gamma\rho_k &\xrightarrow[k\to\infty]{} g, \quad \text{weakly-}\star \text{ in } L^\infty(]0,T[\times\Gamma, |d\mu_v|). \end{aligned} \tag{VI.30}$$

We now show that $R = \rho$ and $g = \gamma\rho$, so that finally the convergences above are shown to hold for the whole initial sequence $(\rho_k)_k$ (see Proposition II.2.9).

Let $\varphi \in \mathcal{C}^{0,1}([0,T]\times\Omega)$ be a test function such that $\varphi(T,.) = 0$. The weak formulation satisfied by ρ_k reads

$$\int_0^T \int_\Omega \rho_k \left(\frac{\partial\varphi}{\partial t} + v_k \cdot \nabla\varphi - c_k\varphi\right) dx\, dt + \int_\Omega \rho_{0,k}\varphi(0)\, dx - \int_0^T \int_\Gamma (\gamma\rho_k)\varphi(v_k \cdot \nu)\, d\sigma\, dt = 0.$$

Thanks to the weak convergence of ρ_k and the strong convergence of v_k and c_k in L^1, we can easily pass to the limit in the first integral (we use Proposition II.2.30). The assumption (VI.28g) let us pass to the limit in the second term. It remains to study the boundary term by writing

$$\begin{aligned}
&\left| \int_0^T \int_\Gamma (\gamma\rho_k)\varphi(v_k\cdot\nu)\,d\sigma\,dt - \int_0^T\int_\Gamma g\varphi(v\cdot\nu)\,d\sigma\,dt \right| \\
&= \left| \int_0^T \int_\Gamma (\gamma\rho_k - g)\varphi(v\cdot\nu)\,d\sigma\,dt + \int_0^T\int_\Gamma (\gamma\rho_k)\varphi\big[(v_k\cdot\nu)-(v\cdot\nu)\big]\,d\sigma\,dt \right| \\
&\leq \left| \int_0^T \int_\Gamma (\gamma\rho_k - g)\varphi(v\cdot\nu)\,d\sigma\,dt \right| + C_\varphi \|v_k\cdot\nu - v\cdot\nu\|_{L^1(]0,T[\times\Gamma)}.
\end{aligned}$$

The first term goes to 0 by the weak-$\star$ convergence property of $\gamma\rho_k$, and the second one also tends to 0 by using (VI.28c).
We have thus shown that R and g satisfy

$$\begin{aligned}
&\int_0^T \int_\Omega R\left(\frac{\partial\varphi}{\partial t} + v\cdot\nabla\varphi - c\varphi\right)\,dx\,dt \\
&+ \int_\Omega \rho_0\varphi(0)\,dx - \int_0^T\int_\Gamma g\varphi(v\cdot\nu)\,d\sigma\,dt = 0.
\end{aligned} \tag{VI.31}$$

In order to conclude, we just need to show that $g = \rho^{in}\ d\mu_v^-$-almost everywhere. For any test function $\psi \in L^\infty(]0,T[\times\Gamma)$ we have

$$\begin{aligned}
&\left| \int_0^T \int_\Gamma \underbrace{\rho_k^{in}(v_k\cdot\nu)^-}_{=(\gamma\rho_k)(v_k\cdot\nu)^-}\psi\,d\sigma\,dt - \int_0^T\int_\Gamma g(v\cdot\nu)^-\psi\,d\sigma\,dt \right| \\
&\leq \left| \int_0^T \int_\Gamma ((\gamma\rho_k) - g)(v\cdot\nu)^-\psi\,d\sigma\,dt \right| \\
&\quad + \left| \int_0^T \int_\Gamma \rho_k^{in}\big((v\cdot\nu)^- - (v_k\cdot\nu)^-\big)\psi\,d\sigma\,dt \right|.
\end{aligned}$$

The first term in the right-hand side tends to 0 by (VI.30) and the second one by using the L^∞ bound on ρ_k^{in} (obtained from (VI.28e)) and (VI.28c). Comparing this result to (VI.28f) leads to the equality

$$\int_0^T \int_\Gamma g(v\cdot\nu)^-\psi\,d\sigma,\,dt = \int_0^T\int_\Gamma \rho^{in}(v\cdot\nu)^-\psi\,d\sigma\,dt,$$

for any test function ψ. It follows that $g = \rho^{in}$, $d\mu_v^-$-almost everywhere.
We finally deduce from (VI.31) that R solves the transport problem with initial data ρ_0 and inflow boundary data ρ^{in}. By uniqueness of such a weak solution, we conclude that $R = \rho$ and $g = \gamma\rho$.
As a consequence, we have just shown the weak convergence of ρ_k (resp., of the trace $\gamma\rho_k$) towards ρ (resp., towards the trace $\gamma\rho$).

- In a second step, we prove that the convergence obtained above is in fact much stronger.
By assumption, the problem (VI.27) satisfies the renormalisation property. In particular, if we define $\zeta_k = (\rho_k)^2$, we know that ζ_k solves the following initial- and boundary-value problem

$$\begin{cases} \dfrac{\partial \zeta_k}{\partial t} + \operatorname{div}(\zeta_k v_k) + (2c_k + \operatorname{div} v_k)\zeta_k = 0, & \text{in } \Omega, \\ \gamma(\zeta_k) = (\rho_k^{in})^2, & \text{on } \Gamma^-_{v_k}(t), \\ \zeta_k(0) = (\rho_{0,k})^2. \end{cases} \tag{VI.32}$$

Similarly, $\zeta = \rho^2$ solves the problem

$$\begin{cases} \dfrac{\partial \zeta}{\partial t} + \operatorname{div}(\zeta v) + (2c + \operatorname{div} v)\zeta = 0, & \text{in } \Omega, \\ \gamma(\zeta) = (\rho^{in})^2, & \text{on } \Gamma^-_{v}(t), \\ \zeta(0) = (\rho_0)^2, \end{cases}$$

Problem (VI.32) has the same form as (VI.27) with c_k replaced with $2c_k + \operatorname{div} v_k$ and satisfies all the corresponding assumptions (VI.28).
We can then use the weak convergence result that we proved in the previous step which shows that

$$(\rho_k)^2 = \zeta_k \xrightharpoonup[k\to\infty]{} \zeta = (\rho)^2, \text{ weakly-}\star \text{ in } L^\infty(]0,T[\times\Omega).$$

This implies in particular that

$$\|\rho_k\|^2_{L^2} = \int_0^T \int_\Omega (\rho_k)^2 \, dx \, dt \xrightarrow[k\to\infty]{} \int_0^T \int_\Omega \rho^2 \, dx \, dt = \|\rho\|^2_{L^2},$$

which gives the strong convergence of $(\rho_k)_k$ towards ρ in $L^2(]0,T[\times\Omega)$ (by Proposition II.2.32). The convergence in any $L^p(]0,T[\times\Omega)$, $p < +\infty$ follows thanks to the bounds (VI.29).
Similarly, we obtain that $(\gamma\rho_k)_k$ strongly converges towards $\gamma\rho$ in the space $L^p(]0,T[\times\Gamma, |d\mu_v|)$, for any $p < +\infty$.
- It remains to show the strong convergence of $(\rho_k)_k$ in $C^0([0,T], L^p(\Omega))$, $p < +\infty$, that is to say uniformly with respect to the time variable. Here also, thanks to the uniform bounds (VI.29), it is sufficient to consider the case $p = 2$, for instance.
The basic idea is to look at the equation satisfied by the difference $\rho - \rho_k$. Unfortunately, since ρ is not smooth enough this does not let us perform all the required computations. Therefore, we replace ρ by its regularisation $\mathcal{S}_\varepsilon\rho$. We recall that we established in the proof of Theorem VI.1.3, that

$$\|\mathcal{S}_\varepsilon\rho - \rho\|_{\mathcal{C}^0([0,T],L^2(\Omega))} \xrightarrow[\varepsilon\to 0]{} 0, \tag{VI.33}$$

$$\gamma_0(\mathcal{S}_\varepsilon \rho) \xrightarrow[\varepsilon\to 0]{} \gamma\rho, \text{ in } L^2(]0,T[\times\Gamma, |d\mu_v|), \tag{VI.34}$$

and that $\mathcal{S}_\varepsilon \rho$ solves

$$\frac{\partial \mathcal{S}_\varepsilon \rho}{\partial t} + \operatorname{div}(\mathcal{S}_\varepsilon \rho v) + c\mathcal{S}_\varepsilon \rho = R_\varepsilon,$$

with

$$\|R_\varepsilon\|_{L^1(]0,T[\times\Omega)} \xrightarrow[\varepsilon\to 0]{} 0. \tag{VI.35}$$

Notice that we avoided using the notation ρ_ε here because it can lead to confusion. It follows that

$$\begin{cases} \dfrac{\partial}{\partial t}(\mathcal{S}_\varepsilon\rho - \rho_k) + \operatorname{div}\big((\mathcal{S}_\varepsilon\rho - \rho_k)v_k\big) + c_k(\mathcal{S}_\varepsilon\rho - \rho_k) = \mathcal{R}_{\varepsilon,k}, & \text{in } \Omega,\\ \gamma(\mathcal{S}_\varepsilon\rho - \rho_k) = \gamma_0(\mathcal{S}_\varepsilon\rho) - \rho_k^{in}, & \text{on } \Gamma^-_{v_k}(t),\\ (\mathcal{S}_\varepsilon\rho - \rho_k)(0) = \mathcal{S}_\varepsilon\rho_0 - \rho_{0,k}, \end{cases}$$

where $\mathcal{R}_{\varepsilon,k} = R_\varepsilon + (\mathcal{S}_\varepsilon\rho)\operatorname{div}(v_k - v) + (\nabla\mathcal{S}_\varepsilon\rho)\cdot(v_k - v) + (c_k - c)(\mathcal{S}_\varepsilon\rho)$ satisfies

$$\|\mathcal{R}_{\varepsilon,k}\|_{L^1(]0,T[\times\Omega)} \leq \|R_\varepsilon\|_{L^1} + \|\operatorname{div} v - \operatorname{div} v_k\|_{L^1} + C_\varepsilon\|v - v_k\|_{L^1} + \|c_k - c\|_{L^1}.$$

By using assumptions (VI.28a), (VI.28b), (VI.28d) and the property (VI.35), we deduce that

$$\lim_{\varepsilon\to 0}\left(\limsup_{k\to\infty}\|\mathcal{R}_{\varepsilon,k}\|_{L^1}\right) = 0. \tag{VI.36}$$

We can then apply the renormalisation property to this problem (see Remark VI.1.3) and deduce that

$$\begin{cases} \dfrac{\partial}{\partial t}(\mathcal{S}_\varepsilon\rho - \rho_k)^2 + \operatorname{div}\big((\mathcal{S}_\varepsilon\rho - \rho_k)^2 v_k\big) \\ \qquad + (2c_k + \operatorname{div} v_k)(\mathcal{S}_\varepsilon\rho - \rho_k)^2 = 2\mathcal{R}_{\varepsilon,k}(\mathcal{S}_\varepsilon\rho - \rho_k), & \text{in } \Omega,\\ \gamma\big((\mathcal{S}_\varepsilon\rho - \rho_k)^2\big) = (\gamma_0(\mathcal{S}_\varepsilon\rho) - \rho_k^{in})^2, & \text{on } \Gamma^-_{v_k}(t),\\ (\mathcal{S}_\varepsilon\rho - \rho_k)^2(0) = (\mathcal{S}_\varepsilon\rho_0 - \rho_{0,k})^2. \end{cases}$$

Integrating this equation over Ω and between times 0 and $s \in [0,T]$, we get

$$\begin{aligned} &\|(\mathcal{S}_\varepsilon\rho - \rho_k)(s)\|^2_{L^2} + \int_0^s\int_\Omega (2c_k + \operatorname{div} v_k)(\mathcal{S}_\varepsilon\rho - \rho_k)^2\,dx\,dt \\ &+ \int_0^s\int_\Gamma \Big(\gamma\big(\mathcal{S}_\varepsilon\rho - \rho_k\big)\Big)^2 (v_k\cdot\nu)\,d\sigma\,dt \\ =&\|\mathcal{S}_\varepsilon\rho_0 - \rho_{0,k}\|^2_{L^2} + 2\int_0^s\int_\Omega \mathcal{R}_{\varepsilon,k}(\mathcal{S}_\varepsilon\rho - \rho_k)\,dx\,dt. \end{aligned}$$

It follows, since $\sup_\varepsilon \|\mathcal{S}_\varepsilon \rho\|_{L^\infty} + \sup_k \|\rho_k\|_{L^\infty} < +\infty$, that

$$\begin{aligned}
&\|(\mathcal{S}_\varepsilon\rho - \rho_k)(s)\|_{L^2}^2 \leq 2\|\mathcal{S}_\varepsilon\rho_0 - \rho_0\|_{L^2}^2 + 2\|\rho_0 - \rho_{0,k}\|_{L^2}^2 \\
&+ 4C\|\mathcal{R}_{\varepsilon,k}\|_{L^1(]0,T[\times\Omega)} + \|\gamma_0(\mathcal{S}_\varepsilon\rho) - \rho_k^{in}\|^2_{L^2(]0,T[\times\Gamma,d\mu_v^-)} \\
&+ \int_0^s \|(2c_k + \operatorname{div} v_k)^-(t)\|_{L^\infty}\|(\mathcal{S}_\varepsilon\rho - \rho_k)(t)\|_{L^2}^2\,dt \\
&+ C\|(v_k\cdot\nu)^- - (v\cdot\nu)^-\|_{L^1(]0,T[\times\Gamma)}.
\end{aligned}$$

By using the Gronwall lemma, it follows that

$$\begin{aligned}
&\|\mathcal{S}_\varepsilon\rho - \rho_k\|^2_{\mathcal{C}^0([0,T],L^2(\Omega))} \leq M\Big(\|\mathcal{S}_\varepsilon\rho_0 - \rho_0\|_{L^2}^2 + \|\rho_0 - \rho_{0,k}\|_{L^2}^2 \\
&+ \|\mathcal{R}_{\varepsilon,k}\|_{L^1} + \|(v_k\cdot\nu)^- - (v\cdot\nu)^-\|_{L^1(]0,T[\times\Gamma)} \\
&+ \|\gamma_0(\mathcal{S}_\varepsilon\rho) - \gamma\rho\|^2_{L^2(]0,T[\times\Gamma,d\mu_v^-)} + \|\gamma\rho - \rho_k^{in}\|^2_{L^2(]0,T[\times\Gamma,d\mu_v^-)}\Big).
\end{aligned}$$

By the triangle inequality, we deduce

$$\begin{aligned}
&\|\rho - \rho_k\|^2_{\mathcal{C}^0([0,T],L^2(\Omega))} \leq M\Big(\|\mathcal{S}_\varepsilon\rho - \rho\|^2_{\mathcal{C}^0([0,T],L^2(\Omega))} + \|\mathcal{S}_\varepsilon\rho_0 - \rho_0\|_{L^2}^2 \\
&+ \|\rho_0 - \rho_{0,k}\|_{L^2}^2 + \|\mathcal{R}_{\varepsilon,k}\|_{L^1} + \|(v_k\cdot\nu)^- - (v\cdot\nu)^-\|_{L^1(]0,T[\times\Gamma)} \\
&+ \|\gamma_0(\mathcal{S}_\varepsilon\rho) - \gamma\rho\|^2_{L^2(]0,T[\times\Gamma,d\mu_v^-)} + \|\gamma\rho - \rho_k^{in}\|^2_{L^2(]0,T[\times\Gamma,d\mu_v^-)}\Big).
\end{aligned}$$

We take now the $\limsup$ as $k \to \infty$ in this inequality to obtain

$$\begin{aligned}
&\limsup_{k\to\infty} \|\rho - \rho_k\|^2_{\mathcal{C}^0([0,T],L^2(\Omega))} \\
\leq &M\Big(\|\mathcal{S}_\varepsilon\rho - \rho\|^2_{\mathcal{C}^0([0,T],L^2(\Omega))} + \|\mathcal{S}_\varepsilon\rho_0 - \rho_0\|_{L^2}^2 \\
&+ \limsup_{k\to\infty}\|\mathcal{R}_{\varepsilon,k}\|_{L^1} + \|\gamma_0(\mathcal{S}_\varepsilon\rho) - \gamma\rho\|^2_{L^2(]0,T[\times\Gamma,d\mu_v^-)}\Big).
\end{aligned}$$

The left-hand side does not depend on ε, thus we can pass to the limit as $\varepsilon \to 0$ in the right-hand side. Using (VI.33),(VI.34) and (VI.36), we finally get that

$$\limsup_{k\to\infty} \|\rho - \rho_k\|^2_{\mathcal{C}^0([0,T],L^2(\Omega))} = 0,$$

and the proof of the theorem is complete.

□

Remark VI.1.7. Assume that the strong convergence of v_k towards v in L^1 is not known. We can observe that the proof of the theorem still holds if we know instead that, at least for a subsequence, we have

- The product $(\rho_k v_k)_k$ weakly converges towards the product ρv.
- The product $(\rho_k^2 v_k)_k$ weakly converges towards ζv, where ζ is the weak limit of $(\rho_k^2)_k$.

This remark is used in the study of the nonhomogeneous Navier–Stokes problem in the next section.

2 The nonhomogeneous incompressible Navier–Stokes equations

The goal of this section is to give a proof of the (global) existence of a weak solution of the nonhomogeneous incompressible Navier–Stokes equations.

2.1 Main result

In all that follows, we assume that the function μ, giving the viscosity as a function of the density, is of class $\mathcal{C}^1$ and satisfies the following hypotheses

$$\exists \mu_1, \mu_2 > 0,\ \mu_1 \leq \mu(x) \leq \mu_2,\ \forall x \in \mathbb{R}\,\text{and}\,\mu' \text{ is bounded.} \tag{VI.37}$$

The ultimate goal of this chapter is to demonstrate the following existence result in space dimension 3.

Theorem VI.2.1 (Existence of a weak solution). *Let Ω be a bounded Lipschitz domain of $\mathbb{R}^d$ with $d = 3$, $T > 0$ and let μ satisfy (VI.37). Let $v_b \in (H^1(\Omega))^d$ with $\operatorname{div} v_b = 0$, $v_0 \in v_b + H$, $\rho_0 \in L^\infty(\Omega)$, $\rho^{in} \in L^\infty(]0,T[\times\Gamma)$ and $f \in L^1(]0,T[,(L^2(\Omega))^d)$ such that $\rho_0 \geq 0$ and $\rho^{in} \geq 0$.*

Then, there exists at least one triplet (v, ρ, p) such that

$$\begin{aligned}
&v \in v_b + L^2(]0,T[,(H^1_0(\Omega))^d),\\
&\rho v \in L^\infty(]0,T[,(L^2(\Omega))^d) \cap N_2^{1/4}(]0,T[,(W^{-1,3}(\Omega))^d),\\
&p \in W^{-1,\infty}(]0,T[,L^2_0(\Omega)),\\
&\rho \in \mathcal{C}^0([0,T],L^q(\Omega)),\ \forall q < +\infty,\\
&t \mapsto \int_\Omega (\rho(t)v(t)) \cdot \varphi \, dx \in \mathcal{C}^0([0,T]),\ \textit{for any } \varphi \in V,
\end{aligned}$$

$$\rho_{\min} \leq \rho(t,x) \leq \rho_{\max},\ \textit{for almost every } (t,x) \in]0,T[\times\Omega,$$

with $\rho_{\min} = \min(\inf \rho_0, \inf \rho^{in})$, $\rho_{\max} = \max(\sup \rho_0, \sup \rho^{in})$, and which satisfies, in the distribution sense

$$\left\{\begin{aligned} &\frac{\partial\rho}{\partial t}+\operatorname{div}(\rho v)=0, \quad && \textit{in } \Omega,\\ &\rho(0)=\rho_0,\\ &\gamma\rho=\rho^{in}, && \textit{on } \Gamma_v^-(t),\end{aligned}\right. \tag{VI.38}$$

$$\left\{\begin{aligned} &\frac{\partial(\rho v)}{\partial t}+\operatorname{div}(\rho v\otimes v)-2\operatorname{div}(\mu(\rho)\mathrm{D}(v))+\nabla p=\rho f, \quad \textit{in } \Omega,\\ &(\rho v)(0)=\rho_0 v_0,\end{aligned}\right. \tag{VI.39}$$

$$\operatorname{div} v=0, \quad \textit{in } \Omega, \tag{VI.40}$$

Moreover, if we assume that $\inf_\Omega \rho_0>0$*, then*

$$v\in L^\infty(]0,T[,H)\cap N_2^{1/4}(]0,T[,H).$$

We recall that the Nikolskii spaces $N_q^\sigma(]0,T[,E)$ were defined by (II.26). A similar result holds in dimension $d=2$ with slightly different functional spaces.

We note that the initial condition on ρv in (VI.39) has to be understood in V'; that is, more precisely, in the following sense

$$\left(\int_\Omega \rho v\cdot\varphi\,dx\right)_{|t=0}=\int_\Omega \rho_0 v_0\cdot\varphi\,dx,\ \forall\varphi\in V.$$

This weak interpretation of the initial data cannot be strengthened without some further assumptions (particularly regarding the initial data ρ_0). In particular, it is not possible in general to give a sense to $(\rho v)(0)$ nor, of course, to $v(0)$. We briefly return to this problem in Section 2.5.

Remark VI.2.1. It is possible to show the same result for a time-dependent boundary data for the velocity v_b, provided some suitable time regularity properties hold for this data.

Moreover, it is also possible to consider outflow boundary conditions of the same type as the one we will study in Section 1 of Chapter VII. We refer to [25] for the study of this particular problem, using similar tools as in the present chapter.

2.2 Approximate problem

2.2.1 Definition of the approximate problem

We consider the same assumptions and notation as in Theorem VI.2.1.

For any integer $k\geq 1$, let V_k be a k-dimensional Galerkin approximation space in $V\subset H$ satisfying $V_k\subset V_{k+1}$ for any k. We can, for example, build this space upon the eigenfunctions of the Stokes operator, as we did in Section

1.3.2 of Chapter V but this is not necessary here because it does not lead to any simplification of the proof. Therefore, we do not specify in the sequel a particular choice for V_k.

We introduce approximations of the data defined by:

$$\rho_k^{in} = \rho^{in} + \frac{1}{k},\ \rho_{0,k} = \rho_0 + \frac{1}{k}, \tag{VI.41}$$

so that we have

$$\inf \rho_k^{in} \geq \frac{1}{k} \text{ and } \inf \rho_{0,k} \geq \frac{1}{k},$$

even if the initial and boundary density data are allowed to vanish. We finally choose a sequence $(f_k)_k$ of smooth vector fields which converges towards f in $L^1(]0,T[,(L^2(\Omega))^d)$.

Let us consider the following approximate problem.

Definition VI.2.2. *We set $\tilde{v}_{0,k} = \mathcal{P}_{V_k}(v_0 - v_b)$, the orthogonal projection (in H) of $v_0 - v_b$ onto V_k. The approximate problem consists in looking for $v_k = v_b + \tilde{v}_k \in v_b + \mathcal{C}^1([0,T],V_k)$, $\rho_k \in \mathcal{C}^0([0,T],L^1(\Omega)) \cap L^\infty(]0,T[\times\Omega)$ such that*

1. *ρ_k is the unique solution to the transport problem associated with the velocity field v_k, the initial data $\rho_{0,k}$, and the inflow boundary data ρ_k^{in}.*
2. *The couple (ρ_k, v_k) satisfies*

$$\begin{aligned}\int_\Omega \rho_k \left(\frac{\partial v_k}{\partial t} + (v_k\cdot\nabla)v_k\right)\cdot\psi\,dx + \int_\Omega 2\mu(\rho_k)\mathrm{D}(v_k) : \mathrm{D}(\psi)\,dx \\ = \int_\Omega \rho_k f_k\cdot\psi\,dx,\end{aligned} \tag{VI.42}$$

for any $\psi \in V_k$ and any $t \in [0,T]$, as well as the initial data $v_k(0) = v_b + \tilde{v}_{0,k}$.

Notice that the approximate density ρ_k depends on v_k in a complex non-local way. Therefore, the resolution of the approximate problem is not a straightforward consequence of the ordinary differential equations theory.

2.2.2 Resolution of the approximate problem

The resolution of the approximate problem is performed using a fixed-point method (see, for instance, [86]). To this end, we introduce the finite-dimensional subspace of $(H^1(\Omega))^d$ defined by

$$\widetilde{V}_k = \mathbb{R}v_b + V_k,$$

and the infinite-dimensional Banach space

$$\mathcal{E}_k = \mathcal{C}^0([0,T],\widetilde{V}_k).$$

We now define a map $\Theta_k : \mathcal{E}_k \to \mathcal{E}_k$ for which we want to find a fixed-point. This fixed-point will be a solution to the approximate problem under study.

- Let $w_k \in \mathcal{E}_k$ be given. We first consider the following transport problem

$$\begin{cases} \dfrac{\partial \rho_k}{\partial t} + \mathrm{div}(\rho_k w_k) = 0, & \text{in } \Omega \\ \gamma(\rho_k) = \rho_k^{in}, & \text{on } \Gamma_{w_k}^-(t), \\ \rho_k(0) = \rho_0. \end{cases} \tag{VI.43}$$

Since w_k belongs in particular to $L^1(]0,T[,(W^{1,1}(\Omega))^d)$, and is divergence-free, we know from Theorem VI.1.6 that there exists a unique solution $\rho_k \geq 0$ to (VI.43) which satisfies

$$\sup_{]0,T[\times\Omega} \rho_k \leq \max(\|\rho_0\|_{L^\infty}, \|\rho^{in}\|_{L^\infty}) + 1/k, \tag{VI.44}$$

and which is continuous in time with values in any $L^q(\Omega)$ for $q < +\infty$. Moreover, by using Proposition VI.1.8, the fact that $\mathrm{div}\, w_k = 0$ and (VI.41), we deduce that

$$\inf_{]0,T[\times\Omega} \rho_k \geq \frac{1}{k}. \tag{VI.45}$$

- Once ρ_k is built, we look for $v_k \in v_b + \mathcal{C}^1([0,T], V_k) \subset \mathcal{E}_k$ satisfying

$$\begin{aligned} &\int_\Omega \rho_k \left(\frac{\partial v_k}{\partial t} + (w_k \cdot \nabla) v_k \right) \cdot \psi \, dx + \int_\Omega 2\mu(\rho_k) \mathrm{D}(v_k) : \mathrm{D}(\psi) \, dx \\ &= \int_\Omega \rho_k f_k \cdot \psi \, dx, \end{aligned} \tag{VI.46}$$

for any $\psi \in V_k$ and any $t \in [0,T]$, as well as the initial data $v_k(0) = v_b + \tilde{v}_{0,k}$. This equation is obtained from the original equation (VI.42) by replacing in the inertia term the (unknown) advection vector field v_k by the given vector field w_k, in such a way that (VI.46) is now a finite-dimensional linear ordinary differential equation.

Indeed, if one seeks its solution under the form $v_k = v_b + \sum_{i=1}^k \alpha_i(t)\psi_i$, we obtain that (VI.46) can be written

$$M(t)\frac{d\alpha}{dt}(t) = A(t)\alpha(t) + B(t), \tag{VI.47}$$

where $\alpha(t) = (\alpha_i(t))_i \in \mathbb{R}^k$ is the unknown vector, $A(t), M(t)$ are $k \times k$ matrices and $B(t) \in \mathbb{R}^k$ is a vector, all of them being continuous with respect to the time variable t (recall that f_k is smooth).

In order to show that (VI.47) has a unique global solution for any initial data, we only need to show that the matrix $M(t)$ is invertible for any $t \in [0,T]$ so that the Cauchy–Lipschitz theory can be applied. By definition, the entries of $M(t)$ are given by

$$(M(t))_{ij} = \int_\Omega \rho_k(t,x)\psi_i(x)\psi_j(x)\,dx,\ \forall i,j \in \{1,\dots,k\}.$$

Therefore, for any $t \in [0,T]$, $M(t)$ is the Gram matrix of the basis $(\psi_i)_i$ with respect to the inner product defined by

$$(f,g) \mapsto \langle f,g\rangle_{\rho_k(t)} = \int_\Omega \rho_k(t,x) f(x) g(x)\,dx.$$

Notice that $\langle .,.\rangle_{\rho_k(t)}$ is indeed a scalar product thanks to (VI.45).
It follows that $M(t)$ is invertible for all t, thus (VI.47) has a unique global solution associated with any initial data. As a consequence, there exists a unique $v_k \in v_b + \mathcal{C}^1([0,T],V_k) \subset \mathcal{E}_k$ which is a solution of (VI.46) and such that $v_k(0) = v_b + \tilde{v}_{0,k}$.

- Finally, with any $w_k \in \mathcal{E}_k$ we have associated a unique couple (ρ_k, v_k) which solves the previous set of equations. We set $\Theta_k(w_k) = v_k$.

We now show that the map Θ_k has a fixed-point in a suitable subset of $\mathcal{E}_k$. It is straightforward to see that such a fixed-point v_k (and its associated density ρ_k) is a solution to the nonlinear approximate problem introduced in Section 2.2.1.

- Preliminary computations:
We first observe that, for any $\psi \in \mathcal{C}^0([0,T],V_k)$, we can take $\psi(t)$ as a test function in (VI.46) and integrate the resulting equation with respect to time. We obtain

$$\begin{aligned}&\int_0^T \int_\Omega \rho_k \left(\frac{\partial v_k}{\partial t} + (w_k\cdot\nabla)v_k\right)\cdot\psi\,dx\,dt\\ &+\int_0^T\int_\Omega 2\mu(\rho_k)\mathrm{D}(v_k):\mathrm{D}(\psi)\,dx\,dt = \int_0^T\int_\Omega \rho_k f_k\cdot\psi\,dx\,dt.\end{aligned} \tag{VI.48}$$

Notice that (VI.48) is equivalent to (VI.46) for any time t, as soon as v_k, w_k are continuous in time.
For $\psi \in \mathcal{C}^1([0,T],V_k)$ and, since v_k is smooth enough, we can consider $\frac{1}{2}(\tilde{v}_k\cdot\psi)$ (which is vanishing at the boundary) as a test function in (VI.43). It follows that

$$\begin{aligned}&\frac12\int_0^T\int_\Omega \rho_k\left(\frac{\partial\tilde{v}_k}{\partial t}\cdot\psi + \tilde{v}_k\cdot\frac{\partial\psi}{\partial t} + ((w_k\cdot\nabla)\tilde{v}_k)\cdot\psi + ((w_k\cdot\nabla)\psi)\cdot\tilde{v}_k\right)dx\,dt\\ =&\frac12\int_\Omega \rho_k(T)(\tilde{v}_k(T)\cdot\psi(T))\,dx - \frac12\int_\Omega \rho_{0,k}(\tilde{v}_{0,k}\cdot\psi(0))\,dx.\end{aligned} \tag{VI.49}$$

We subtract (VI.49) from (VI.48). Using that $v_k = \tilde{v}_k + v_b$, v_b being time-independent, it follows

$$\frac{1}{2}\int_0^T\int_\Omega \rho_k\left(\frac{\partial \tilde v_k}{\partial t}\cdot\psi - \frac{\partial\psi}{\partial t}\cdot\tilde v_k + ((w_k\cdot\nabla)\tilde v_k)\cdot\psi - ((w_k\cdot\nabla)\psi)\cdot\tilde v_k\right)dx\,dt$$
$$+\frac{1}{2}\int_\Omega \rho_k(T)(\tilde v_k(T)\cdot\psi(T))\,dx - \frac{1}{2}\int_\Omega \rho_{0,k}(\tilde v_{0,k}\cdot\psi(0))\,dx$$
$$+\int_0^T\int_\Omega 2\mu(\rho_k)\mathrm{D}(v_k):\mathrm{D}(\psi)\,dx\,dt = \int_0^T\int_\Omega \rho_k\big(f_k-(w_k\cdot\nabla)v_b\big)\cdot\psi\,dx\,dt. \tag{VI.50}$$

Subtracting once more (VI.49) from this equation we find a conservative formulation of the approximate momentum equation

$$\begin{aligned}
&-\int_0^T\int_\Omega \rho_k\tilde v_k\cdot\left(\frac{\partial\psi}{\partial t}+(w_k\cdot\nabla)\psi\right)dx\,dt\\
&+\int_0^T\int_\Omega 2\mu(\rho_k)\mathrm{D}(v_k):\mathrm{D}(\psi)\,dx\,dt\\
&+\int_\Omega \rho_k(T)\tilde v_k(T)\cdot\psi(T)\,dx - \int_\Omega \rho_{0,k}\tilde v_{0,k}\cdot\psi(0)\,dx\\
=&\int_0^T\int_\Omega \rho_k\big(f_k-(w_k\cdot\nabla)v_b\big)\cdot\psi\,dx\,dt.
\end{aligned}\tag{VI.51}$$

This formula is useful to analyse the limit as $k\to\infty$ in the sequel.

- Energy estimate:
Consider $w_k\in v_b+\mathcal{C}^0([0,T],V_k)$ and $v_k=\Theta_k(w_k)$ as defined previously. Let us choose $\psi=\tilde v_k=v_k-v_b$ as a test function in (VI.50). We see that the first integral vanishes and it follows

$$\begin{aligned}
&\frac{1}{2}\int_\Omega \rho_k(T)|\tilde v_k(T)|^2\,dx + \int_0^T\int_\Omega 2\mu(\rho_k)|\mathrm{D}(\tilde v_k)|^2\,dx\,dt\\
=&\frac{1}{2}\int_\Omega \rho_{0,k}|\tilde v_{0,k}|^2\,dx - \int_0^T\int_\Omega 2\mu(\rho_k)\mathrm{D}(\tilde v_k):\mathrm{D}(v_b)\,dx\,dt\\
&+\int_0^T\int_\Omega \rho_k\tilde v_k\cdot(f_k-(w_k\cdot\nabla)v_b)\,dx\,dt.
\end{aligned}$$

Using the L^∞ bound on ρ_k given in (VI.44), assumption (VI.37), and the Korn inequality (IV.87), we get

$$\begin{aligned}
&\frac{1}{2}\|\sqrt{\rho_k(T)}\tilde v_k(T)\|_{L^2}^2 + \mu_1\int_0^T \|\nabla\tilde v_k\|_{L^2}^2\,dt\\
\le& C\|\tilde v_{0,k}\|_{L^2}^2 + C\int_0^T \|\nabla\tilde v_k\|_{L^2}\|\nabla v_b\|_{L^2}\,dt + C\int_0^T \|f_k\|_{L^2}\|\tilde v_k\|_{L^2}\,dt\\
&+\left|\int_0^T\int_\Omega \rho_k\tilde v_k\cdot((w_k\cdot\nabla)v_b)\,dx\,dt\right|.
\end{aligned}\tag{VI.52}$$

Let us consider the last term of this inequality. Using the Hölder inequality, and the Sobolev embedding $H^1(\Omega) \subset L^6(\Omega)$, we get

$$\begin{aligned}
&\left| \int_0^T \int_\Omega \rho_k \tilde{v}_k \cdot ((w_k \cdot \nabla) v_b)\, dx\, dt \right| \\
\leq &\int_0^T \|\sqrt{\rho_k}\|_{L^\infty} \|\sqrt{\rho_k} w_k\|_{L^3} \|\nabla v_b\|_{L^2} \|\tilde{v}_k\|_{L^6}\, dt \\
\leq &C \int_0^T \|\sqrt{\rho_k} w_k\|_{L^2}^{1/2} \|\sqrt{\rho_k} w_k\|_{L^6}^{1/2} \|v_b\|_{H^1} \|\nabla \tilde{v}_k\|_{L^2}\, dt \\
\leq &C \|v_b\|_{H^1} \int_0^T \|\sqrt{\rho_k} w_k\|_{L^2}^{1/2} \|\nabla w_k\|_{L^2}^{1/2} \|\nabla \tilde{v}_k\|_{L^2}\, dt \\
\leq &C' \left(\int_0^T \|\sqrt{\rho_k} w_k\|_{L^2}^2\, dt \right)^{1/4} \left(\int_0^T \|\nabla w_k\|_{L^2}^2\, dt \right)^{1/4} \left(\int_0^T \|\nabla \tilde{v}_k\|_{L^2}^2\, dt \right)^{1/2}.
\end{aligned}$$

Using this estimate in (VI.52) and the Young inequality, we get

$$\begin{aligned}
&\|\sqrt{\rho_k(T)}\tilde{v}_k(T)\|_{L^2}^2 + \mu_1 \int_0^T \|\nabla \tilde{v}_k\|_{L^2}^2\, dt \\
\leq &C + C \left(\int_0^T \|\sqrt{\rho_k} w_k\|_{L^2}^2\, dt \right)^{1/2} \left(\int_0^T \|\nabla w_k\|_{L^2}^2\, dt \right)^{1/2},
\end{aligned}$$

where C only depends on the data f, v_b, μ_1, μ_2, ρ_0. Finally, writing $v_k = v_b + \tilde{v}_k$ we deduce that

$$\begin{aligned}
&\|\sqrt{\rho_k(T)} v_k(T)\|_{L^2}^2 + \mu_1 \int_0^T \|\nabla v_k\|_{L^2}^2\, dt \\
\leq &C' + C' \left(\int_0^T \|\sqrt{\rho_k} w_k\|_{L^2}^2\, dt \right)^{1/2} \left(\int_0^T \|\nabla w_k\|_{L^2}^2\, dt \right)^{1/2}.
\end{aligned} \tag{VI.53}$$

We now use the fact that k being fixed, the H^1-norm is equivalent to the L^2-norm on the finite-dimensional space $\widetilde{V}_k$ (with constants depending on k) and (VI.45). It follows from (VI.53),

$$\|v_k(T)\|_{L^2}^2 \leq C_k + D_k \int_0^T \|w_k(t)\|_{L^2}^2\, dt,$$

where C_k and D_k depend on the data and on k but not on w_k. The above estimate applies for any final time $T > 0$ so that we have in fact proved that

$$\|v_k(t)\|_{L^2}^2 \leq C_k + D_k \int_0^t \|w_k(s)\|_{L^2}^2\, ds, \ \forall t \in [0, T]. \tag{VI.54}$$

Let us introduce

$$M_k(t) = C_k e^{D_k t}, \ \forall t \geq 0.$$

Suppose that w_k is such that

$$\|w_k(t)\|_{L^2}^2 \leq M_k(t), \ \forall t \in [0,T],$$

then, using (VI.54), we deduce

$$\|v_k(t)\|_{L^2}^2 \leq M_k(t), \ \forall t \in [0,T].$$

In other words, we proved that Θ_k maps the following convex subset of $\mathcal{E}_k$,

$$\mathcal{K}_{0,k} = \left\{ v \in v_b + \mathcal{C}^1([0,T], V_k), \text{ such that } \|v(t)\|_{L^2}^2 \leq M_k(t), \ \forall t \in [0,T] \right\},$$

into itself. Notice in particular that $\mathcal{K}_{0,k}$ is bounded in $L^\infty(]0,T[, (L^2(\Omega))^d)$. Moreover, since $\widetilde{V}_k$ is a finite-dimensional subspace of $(H^1(\Omega))^d$, the L^2-norm and the H^1-norm are equivalent in this subspace so that the set $\mathcal{K}_{0,k}$ is also bounded in $L^\infty(]0,T[, (H^1(\Omega))^d)$. Of course, the corresponding bound depends on k.

- Compactness:
  Let $w_k \in \mathcal{K}_{0,k}$ and $v_k = \Theta_k(w_k) \in \mathcal{K}_{0,k}$. We take $\psi = \partial \tilde{v}_k/\partial t = \partial v_k/\partial t$ in (VI.46) (recall that v_b does not depend on time). All the norms in V_k are equivalent so that using once more (VI.45), we find a bound

  $$\sup_{0\leq t\leq T} \left\| \frac{\partial v_k}{\partial t} \right\|_{V_k} \leq C_k',$$

  where C_k' depends only on T, k, and on the data. As a consequence, the convex set

  $$\mathcal{K}_{1,k} = \left\{ v \in \mathcal{K}_{0,k}, \quad \sup_{0\leq t\leq T} \left\| \frac{\partial v}{\partial t} \right\|_{V_k} \leq C_k' \right\},$$

  is invariant through the map Θ_k. By the Ascoli theorem (Theorem II.3.1), we know that the set $\mathcal{K}_{1,k}$ is relatively compact in $\mathcal{E}_k$.
- Continuity of Θ_k:
  In order to apply the Schauder fixed-point theorem to the map Θ_k on the compact convex set $\overline{\mathcal{K}_{1,k}}$, it remains to show that Θ_k is continuous for the topology of $\mathcal{E}_k$. In fact it is enough to show that Θ_k is sequentially continuous.
  Recall that k is a fixed integer. Let $(w_k^n)_n$ be a sequence in $\mathcal{E}_k$ which converges towards some w_k in this space. For any n, let ρ_k^n be the unique solution to the transport problem (VI.43) with $w_k = w_k^n$. Let ρ be the unique solution to (VI.43) for the limit velocity field w_k. Since k is fixed,

all these transport problems are associated with the same initial data $\rho_{0,k}$ and boundary data ρ_k^{in}.
By assumption, the sequence $(w_k^n)_n$ strongly converges in the space $\mathcal{C}^0([0,T],(H^1(\Omega))^d)$, so that the sequence of traces $(w_k^n \cdot \nu)_n$ converges towards $(w_k \cdot \nu)$ in $L^1(]0,T[\times\Gamma)$. Hence, by Theorem VI.1.9 we deduce that $(\rho_k^n)_n$ strongly converges towards ρ_k in $L^\infty(]0,T[,L^q(\Omega))$ for any $q<+\infty$ and that $(\gamma(\rho_k^n)(w_k^n\cdot\nu)^+)_n$ strongly converges towards $\gamma(\rho_k)(w_k\cdot\nu)^+$ in $L^1(]0,T[\times\Gamma)$.
Therefore, since μ is a Lipschitz continuous bounded function, we deduce that $(\mu(\rho_k^n))_n$ converges towards $\mu(\rho_k)$ in all the spaces $L^\infty(]0,T[,L^q(\Omega))$, for any $q<+\infty$.

Let us now consider the solution $v_k^n \in v_b + \mathcal{C}^0([0,T],V_k)$ to (VI.46) for the advection vector field w_k^n and the density ρ_k^n constructed above. Since $(w_k^n)_n$ is bounded in $\mathcal{C}^0([0,T],(H^1(\Omega))^d)$, the energy estimate (VI.54) leads to

$$\sup_n \|v_k^n\|_{\mathcal{C}^0([0,T],L^2)} < +\infty,$$

and, taking $\psi = \partial v_k^n/\partial t$ in (VI.46), we also easily get that

$$\sup_n \left\| \frac{\partial v_k^n}{\partial t} \right\|_{\mathcal{C}^0([0,T],L^2)} < +\infty. \tag{VI.55}$$

Using once again the Ascoli theorem, we find that there exists a subsequence always referred to as $(v_k^n)_n$ which strongly converges towards some limit v_k in $v_b+\mathcal{C}^0([0,T],V_k)$. Furthermore, from (VI.55) we deduce that the limit v_k is Lipschitz continuous with respect to the time variable.
Thanks to the convergences obtained above, we can perform the limit as $n\to\infty$ in the equation satisfied by v_k^n. We finally obtain that ρ_k, v_k, w_k satisfy (VI.51) and by adding (VI.49) twice, we find that v_k solves (VI.48) and thus the differential equation (VI.46) in the sense of $\mathcal{D}'(]0,T[)$.
The solution to (VI.46) is unique (and equal to $\Theta_k(w_k)$) as soon as ρ_k and w_k are fixed thus we deduce, with the same argument as in the proof of Proposition II.2.9, that the whole sequence $(v_k^n)_n$, and not only a subsequence, actually converges in $\mathcal{C}^0([0,T],(H^1(\Omega))^d)$ towards $v_k=\Theta_k(w_k)$.
This concludes the proof of the continuity of the map Θ_k.

- Conclusion:
  We just proved that Θ_k is a continuous map from $\mathcal{E}_k$ into itself and that the convex compact set $\overline{\mathcal{K}_{1,k}}$ is invariant by Θ_k. Thanks to the Schauder fixed-point theorem (Theorem II.3.9), we obtain that there exists at least one fixed-point v_k of Θ_k into $\overline{\mathcal{K}_{1,k}}$.
  This exactly provides the existence of at least one solution to the approximate problem under study introduced in Definition VI.2.2.

2.3 Estimates for the approximate solution

Up to now we obtained k-dependent estimates. In the present section, we are going to provide uniform estimates with respect to k for this approximate solution. This is useful for performing the limit when k goes to infinity.

2.3.1 Uniform energy estimates

Lemma VI.2.3. *There exists $C_0 > 0$ depending only on the data such that, for any $k \geq 1$, we have*

$$\|\rho_k\|_{L^\infty(]0,T[\times\Omega)} \leq C_0, \tag{VI.56}$$

$$\|\sqrt{\rho_k} v_k\|_{L^\infty(]0,T[,L^2)} + \|v_k\|_{L^2(]0,T[,H^1)} \leq C_0, \tag{VI.57}$$

$$\|\sqrt{\rho_k} v_k\|_{L^{8/3}(]0,T[,L^4)} \leq C_0. \tag{VI.58}$$

Proof.

The estimate (VI.56) directly follows from (VI.44) and the fact that $\rho_k \geq 0$.

Since for a fixed-point of Θ_k we have $w_k = v_k = \Theta_k(v_k)$, using Young's inequality and the Gronwall lemma, with inequality (VI.53) gives the bound (VI.57).

The 3D Sobolev embedding $H^1(\Omega) \subset L^6(\Omega)$ and the bound (VI.44) gives

$$\|\sqrt{\rho_k} v_k\|_{L^2(]0,T[,L^6)} \leq C\|v_k\|_{L^2(]0,T[,L^6)} \leq C'\|v_k\|_{L^2(]0,T[,H^1)}.$$

We now use the interpolation inequality (Theorem II.5.5) to get

$$\begin{aligned}\|\sqrt{\rho_k} v_k\|_{L^{\frac{8}{3}}(]0,T[,L^4)} &\leq \|\sqrt{\rho_k} v_k\|_{L^\infty(]0,T[,L^2)}^{1/4} \|\sqrt{\rho_k} v_k\|_{L^2(]0,T[,L^6)}^{3/4} \\ &\leq C''\|\sqrt{\rho_k} v_k\|_{L^\infty(]0,T[,L^2)}^{1/4} \|v_k\|_{L^2(]0,T[,H^1)}^{3/4}.\end{aligned}$$

Using (VI.57), the estimate (VI.58) follows.

□

Remark VI.2.2. Since ρ_k may vanish in the limit $k \to \infty$, we cannot deduce from (VI.57) a classical $L^\infty(]0,T[,(L^2(\Omega))^d)$ estimate on the velocity field v_k.

2.3.2 Time derivative estimate for the density

Lemma VI.2.4. *For any $\beta : \mathbb{R} \to \mathbb{R}$ of class $\mathcal{C}^1$, there exists $C_\beta > 0$ such that for any $k \geq 0$ we have*

$$\left\|\frac{\partial \beta(\rho_k)}{\partial t}\right\|_{L^2(]0,T[,W^{-1,6})} \leq C_\beta.$$

Proof.

Since $\operatorname{div} v_k = 0$, the renormalisation property applied to the mass balance equation shows that $\beta(\rho_k)$ satisfies

$$\frac{\partial \beta(\rho_k)}{\partial t} + \operatorname{div}(\beta(\rho_k) v_k) = 0,$$

in the distribution sense. It follows that

$$\begin{aligned}\left\| \frac{\partial \beta(\rho_k)}{\partial t} \right\|_{L^2(]0,T[,W^{-1,6})} &= \|\operatorname{div}(\beta(\rho_k) v_k)\|_{L^2(]0,T[,W^{-1,6})} \\ &\leq \|\beta(\rho_k) v_k\|_{L^2(]0,T[,L^6)}.\end{aligned}$$

Moreover, by (VI.56) and the Sobolev embedding theorem we have

$$\|\beta(\rho_k) v_k\|_{L^2(]0,T[,L^6)} \leq C \|v_k\|_{L^2(]0,T[,L^6)} \leq C' \|v_k\|_{L^2(]0,T[,H^1)}.$$

The claim follows from (VI.57).

$\square$

2.3.3 Time translations estimates for the velocity

In order to perform the limit in the approximate problem, it is necessary to prove some compactness property for the sequence $(v_k)_k$. As in [110], for instance, this compactness property follows from fractional time derivatives and more precisely from time translations estimates.

To this end, we use the framework introduced in Section 5, in particular the time translation operators $\tau_h : f \mapsto f(. + h)$.

Lemma VI.2.5. *There exists $C_2 > 0$ depending only on the data and the final time T, such that for any $k \geq 0$ and any $0 < h < T$, we have*

$$\|\sqrt{\tau_h \rho_k}(\tau_h v_k - v_k)\|_{L^2(]0,T-h[,(L^2(\Omega))^d)} \leq C_2 h^{1/4}. \tag{VI.59}$$

Proof.

Since v_b is time-independent, we observe that $\tau_h v_k - v_k = \tau_h \tilde{v}_k - \tilde{v}_k$. Then we write the following algebraic identity

$$\begin{aligned}\tau_h \rho_k (\tau_h \tilde{v}_k - \tilde{v}_k) \cdot (\tau_h \tilde{v}_k - \tilde{v}_k) = &\underbrace{(\tau_h(\rho_k \tilde{v}_k) - \rho_k \tilde{v}_k) \cdot (\tau_h \tilde{v}_k - \tilde{v}_k)}_{=A} \\ &- \underbrace{\Big((\tau_h \rho_k - \rho_k) \tilde{v}_k\Big) \cdot (\tau_h \tilde{v}_k - \tilde{v}_k)}_{=B}.\end{aligned} \tag{VI.60}$$

- Estimate of the term A:

Consider $\Psi \in \mathcal{C}^1([0,T], V_k)$ and let us introduce $\psi(t,x) = 1_{[s,s+h]}(t)\Psi(s,x)$, with $s < T-h$. We use ψ as a test function in the conservative formulation (VI.51) where we recall that $w_k = v_k$ because v_k is assumed to be a fixed-point of Θ_k. This computation is allowed because ρ_k and $\tilde{v}_k$ are continuous with respect to the time variable. We get

$$\begin{aligned}
&\int_\Omega (\tau_h(\rho_k\tilde{v}_k)(s) - \rho_k\tilde{v}_k(s)) \cdot \Psi(s)\,dx \\
=& \int_s^{s+h}\int_\Omega \rho_k((v_k\cdot\nabla)\Psi(s))\cdot\tilde{v}_k\,dx\,dt - \int_s^{s+h}\int_\Omega 2\mu(\rho_k)\mathrm{D}(v_k) : \mathrm{D}(\Psi(s))\,dx\,dt \\
&+ \int_s^{s+h}\int_\Omega \rho_k\big(f_k - ((v_k\cdot\nabla)v_b)\big)\cdot\Psi(s)\,dx\,dt.
\end{aligned}$$

Using (VI.56) and the Hölder inequality, we get

$$\begin{aligned}
&\left|\int_\Omega (\tau_h(\rho_k\tilde{v}_k)(s) - \rho_k\tilde{v}_k(s)) \cdot \Psi(s)\,dx\right| \\
\leq& C\left(\int_s^{s+h} \|\sqrt{\rho_k}v_k\|_{L^4}(\|\sqrt{\rho_k}\tilde{v}_k\|_{L^4} + \|\nabla v_b\|_{L^2})\,dt\right)\|\nabla\Psi(s)\|_{L^2} \\
&+ C\left(\int_s^{s+h} \|\nabla v_k\|_{L^2} + \|f_k\|_{L^2}\,dt\right)\|\nabla\Psi(s)\|_{L^2}.
\end{aligned}$$

By using the Hölder inequality with respect to the time variable, it follows

$$\begin{aligned}
&\left|\int_\Omega (\tau_h(\rho_k\tilde{v}_k)(s) - \rho_k\tilde{v}_k(s)) \cdot \Psi(s)\,dx\right| \\
\leq& C(h^{3/5}\|\sqrt{\rho_k}v_k\|_{L^{8/3}(]0,T[,L^4)} + h^{1/4}\|\sqrt{\rho_k}v_k\|^2_{L^{8/3}(]0,T[,L^4)})\|\nabla\Psi(s)\|_{L^2} \\
&+ Ch^{1/2}(\|v_k\|_{L^2(]0,T[,H^1)} + \|f_k\|_{L^2(]0,T[,L^2)})\|\nabla\Psi(s)\|_{L^2}.
\end{aligned}$$

Using the bounds given in Lemma VI.2.3, we finally proved the following estimate

$$\left|\int_\Omega (\tau_h(\rho_k\tilde{v}_k)(s) - \rho_k\tilde{v}_k(s)) \cdot \Psi(s)\,dx\right| \leq Ch^{1/4}\|\nabla\Psi(s)\|_{L^2}.$$

Let us take now $\Psi(s) = \tau_h\tilde{v}_k(s) - \tilde{v}_k(s)$. It follows

$$\begin{aligned}
&\left|\int_\Omega (\tau_h(\rho_k\tilde{v}_k)(s) - \rho_k\tilde{v}_k(s)) \cdot (\tau_h\tilde{v}_k(s) - \tilde{v}_k(s))\,dx\right| \\
\leq& Ch^{1/4}\|\tau_h\tilde{v}_k(s) - \tilde{v}_k(s)\|_{H^1}.
\end{aligned}$$

By integrating this inequality with respect to s and using (VI.57), we get

$$\begin{aligned}&\left|\int_0^{T-h}\int_\Omega (\tau_h(\rho_k\tilde{v}_k)(s)-\rho_k\tilde{v}_k(s))\cdot(\tau_h\tilde{v}_k(s)-\tilde{v}_k(s))\,dx\,ds\right| \\ \leq & Ch^{1/4}\|\tilde{v}_k\|_{L^2(0,T-h,H^1)}\leq K_1h^{1/4}.\end{aligned} \tag{VI.61}$$

This is the estimate of the term A in (VI.60).

• Estimate for the term B :

Consider a given time $s\in]0,T-h[$, and choose a time-independent function $\varphi\in H_0^1(\Omega)$. We take $(t,x)\mapsto 1_{[s,s+h]}(t)\varphi(x)$ as a test function in (VI.43) with $w_k=v_k$ (this is allowed because ρ_k is continuous in time). It follows

$$\int_\Omega (\tau_h\rho_k(s,x)-\rho_k(s,x))\varphi(x)\,dx=\int_\Omega\left(\int_s^{s+h}\rho_kv_k\,dt\right)\cdot\nabla\varphi(x)\,dx.$$

Let $\psi\in\mathcal{C}^0([0,T],(H^1(\Omega))^d)$. We choose $\varphi(x)=\tilde{v}_k(s,x)\cdot\psi(s,x)\in H_0^1(\Omega)$ in the above identity. It follows, using Sobolev embeddings and (VI.56), that

$$\begin{aligned}&\left|\int_\Omega(\tau_h\rho_k(s)-\rho_k(s))\tilde{v}_k(s)\cdot\psi(s)\,dx\right| \\ \leq & C\left(\int_s^{s+h}\|v_k(t)\|_{H^1}\,dt\right)\|\tilde{v}_k(s)\|_{H^1}\|\psi(s)\|_{H_0^1} \\ \leq & Ch^{1/2}\|v_k\|_{L^2(]0,T[,H^1)}\|\tilde{v}_k(s)\|_{H^1}\|\psi(s)\|_{H^1}.\end{aligned} \tag{VI.62}$$

We now take $\psi(s)=\tau_h\tilde{v}_k(s)-\tilde{v}_k(s)$ in (VI.62), so that integrating with respect to s we get

$$\begin{aligned}&\left|\int_0^{T-h}\int_\Omega(\tau_h\rho_k-\rho_k)\tilde{v}_k\cdot(\tau_h\tilde{v}_k-\tilde{v}_k)\,dx\,ds\right| \\ \leq & Ch^{1/2}\|v_k\|_{L^2(]0,T[,H^1)}\|\tilde{v}_k\|^2_{L^2(]0,T[,H^1)}\leq K_1'h^{1/2}.\end{aligned} \tag{VI.63}$$

Combining estimates (VI.61) and (VI.63) gives the claim thanks to (VI.60). □

Lemma VI.2.6. *There exists $C>0$ depending only on the data, such that for any $h>0$ we have*

$$\|(\tau_h\rho_k-\rho_k)v_k\|_{L^2(]0,T-h[,W^{-1,3})}\leq Ch^{1/2}.$$

Proof.

We first write

$$\rho_k(t+h)-\rho_k(t)=\int_t^{t+h}\frac{\partial\rho_k}{\partial t}(s)ds,$$

this equality being satisfied in $W^{-1,6}(\Omega)$. We have seen in Lemma VI.2.4 that $(\partial\rho_k/\partial t)_k$ is bounded in $L^2(]0,T[,W^{-1,6})$, therefore we have

$$\|\tau_h\rho_k - \rho_k\|_{L^\infty(]0,T-h[,W^{-1,6})} \leq h^{1/2}\left\|\frac{\partial\rho_k}{\partial t}\right\|_{L^2(]0,T[,W^{-1,6})} \leq Ch^{1/2}.$$

Using Theorem III.2.37, we know that the product operator continuously maps $W^{-1,6}(\Omega)\times H^1(\Omega)$ into $W^{-1,3}(\Omega)$. From the inequality above and from (VI.57), we deduce the claimed estimate

$$\begin{aligned}&\|(\tau_h\rho_k - \rho_k)v_k\|_{L^2(]0,T-h[,W^{-1,3})}\\ \leq& C\|\tau_h\rho_k - \rho_k\|_{L^\infty(]0,T[,W^{-1,6})}\|v_k\|_{L^2(]0,T[,H^1)} \leq Ch^{1/2}.\end{aligned}$$

□

Proposition VI.2.7. *The sequence $(\rho_k v_k)_k$ is bounded in the Nikolskii space $N_2^{1/4}(]0,T[,(W^{-1,3}(\Omega))^d)$, that is, there exists $C>0$ such that, for any $0<h<T$ and any k, we have*

$$\|\tau_h(\rho_k v_k) - \rho_k v_k\|_{L^2(]0,T-h[,W^{-1,3})} \leq Ch^{1/4}.$$

Proof.

We just write

$$\tau_h(\rho_k v_k) - \rho_k v_k = (\tau_h\rho_k - \rho_k)v_k + \sqrt{\tau_h\rho_k}\sqrt{\tau_h\rho_k}(\tau_h v_k - v_k).$$

The first term is estimated by Lemma VI.2.6 and the second one by Lemma VI.2.5 (using the continuous embedding from L^2 into $W^{-1,3}$).

□

2.4 End of the proof of the existence theorem

We can now conclude the proof of Theorem VI.2.1 by establishing the existence of at least one weak solution to Problem (VI.38)–(VI.40). The end of the proof consists in showing that the approximate solution (ρ_k, v_k) constructed in the previous section has a limit, at least for a subsequence, in suitable spaces and that its limit is a solution to the required problem.

- Weak convergence of the density:
  From Theorem III.2.34, we know that the embedding of $W_0^{1,6/5}(\Omega)$ in $L^1(\Omega)$ is compact, and thus by duality (Lemma II.3.6) the embedding of $L^\infty(\Omega)$ in $W^{-1,6}(\Omega)$ is also compact.
  Using (VI.56), Lemma VI.2.4, and Theorem II.5.16, we conclude that, for any $\beta:\mathbb{R}\to\mathbb{R}$ of class $\mathcal{C}^1$, the sequence $(\beta(\rho_k))_k$ is relatively compact in $\mathcal{C}^0([0,T],W^{-1,6}(\Omega))$.

Applying this compactness property with $\beta(s) = s$ and $\beta(s) = s^2$, we obtain that there exists a subsequence still referred to as $(\rho_k)_k$, and two functions $\rho, \zeta \in L^\infty(]0,T[\times\Omega)$ such that

$$\begin{cases} \rho_k \underset{k\to\infty}{\rightharpoonup} \rho, & \text{weakly-}\star \text{ in } L^\infty(]0,T[\times\Omega), \\ \dfrac{\partial \rho_k}{\partial t} \underset{k\to\infty}{\rightharpoonup} \dfrac{\partial \rho}{\partial t}, & \text{weakly in } L^2(]0,T[,W^{-1,6}(\Omega)), \\ \rho_k \underset{k\to\infty}{\longrightarrow} \rho, & \text{in } \mathcal{C}^0([0,T],\ W^{-1,6}(\Omega)), \end{cases}$$

and

$$\begin{cases} \rho_k^2 \underset{k\to\infty}{\rightharpoonup} \zeta, & \text{weakly-}\star \text{ in } L^\infty(]0,T[\times\Omega), \\ \dfrac{\partial \rho_k^2}{\partial t} \underset{k\to\infty}{\rightharpoonup} \dfrac{\partial \zeta}{\partial t}, & \text{weakly in } L^2(]0,T[,W^{-1,6}(\Omega)), \\ \rho_k^2 \underset{k\to\infty}{\longrightarrow} \zeta, & \text{in } \mathcal{C}^0([0,T],\ W^{-1,6}(\Omega)). \end{cases}$$

We notice that $v_k \in v_b + \mathcal{C}^0([0,T],V_k)$ so that, in particular, the trace $v_k \cdot \nu$ is equal to $v_b \cdot \nu$ and does not depend on k. Moreover, the renormalisation property gives us

$$\gamma(\rho_k^2) = (\gamma(\rho_k))^2 = (\rho_k^{in})^2 = \left(\rho^{in} + \frac{1}{k}\right)^2, \quad \text{where } (v_b \cdot \nu) < 0.$$

It follows that

$$\begin{aligned} \gamma(\rho_k) &\underset{k\to\infty}{\longrightarrow} \rho^{in}, && \text{in } L^1(]0,T[\times\Gamma, d\mu_{v_b}^-), \\ \gamma(\rho_k^2) &\underset{k\to\infty}{\longrightarrow} (\rho^{in})^2, && \text{in } L^1(]0,T[\times\Gamma, d\mu_{v_b}^-). \end{aligned}$$

- Weak convergence of the velocity and the momentum:
  Using (VI.57), we can find another subsequence still referred to as $(v_k)_k$ and a $v \in v_b + L^2(]0,T[,V)$ such that

$$v_k \underset{k\to\infty}{\rightharpoonup} v, \quad \text{weakly in } L^2(]0,T[,(H^1(\Omega))^d). \tag{VI.64}$$

The product mapping $(\rho, v) \mapsto \rho v$ is continuous from $W^{-1,6}(\Omega)\times(H^1(\Omega))^d$ into $(W^{-1,3}(\Omega))^d$ (Theorem III.2.37). Using Proposition II.2.12, and the strong convergences proved above for $(\rho_k)_k$ and $(\rho_k^2)_k$ we deduce that

$$\begin{aligned} \rho_k v_k &\underset{k\to\infty}{\rightharpoonup} \rho v, && \text{weakly in } L^2(]0,T[,(W^{-1,3}(\Omega))^d), \\ \rho_k^2 v_k &\underset{k\to\infty}{\rightharpoonup} \zeta v, && \text{weakly in } L^2(]0,T[,(W^{-1,3}(\Omega))^d). \end{aligned}$$

Moreover, by (VI.57) we know that $(\rho_k v_k)_k$ and $(\rho_k^2 v_k)_k$ are bounded in $L^\infty(]0,T[,(L^2(\Omega))^d)$. Therefore, by using Proposition II.2.10 (in the slightly different case of a weak-$\star$ convergent sequence; see Remark II.2.1), we obtain that

$$\rho_k v_k \xrightarrow[k\to\infty]{} \rho v, \quad \text{weakly-}\star \text{ in } L^\infty(]0,T[,(L^2(\Omega))^d), \tag{VI.65}$$

$$\rho_k^2 v_k \xrightarrow[k\to\infty]{} \zeta v, \quad \text{weakly-}\star \text{ in } L^\infty(]0,T[,(L^2(\Omega))^d). \tag{VI.66}$$

- Strong convergence of the density and the viscosity:
  We can now use the stability result of the transport equation obtained in Section 1. Indeed, from Theorem VI.1.9 and Remark VI.1.7, the weak convergence of $(\rho_k v_k)_k$ towards ρv and the one of $(\rho_k^2 v_k)_k$ towards ζv obtained in (VI.65) and (VI.66), as well as the strong convergence of the initial and inflow data, allow us to conclude first that $\zeta = \rho^2$ and that $(\rho_k)_k$ strongly converges towards ρ in $\mathcal{C}^0([0,T],L^q(\Omega))$ for all $q<+\infty$. In particular, we have

$$\rho(0) = \lim_{k\to\infty} \rho_k(0) = \lim_{k\to\infty} \rho_k^0 = \rho_0,$$

  by definition of the initial approximate data ρ_k^0.
  Finally, from (VI.37), we have

$$\|\mu(\rho_k)-\mu(\rho)\|_{L^\infty(]0,T[,L^q)} \le \|\mu'\|_\infty \|\rho_k-\rho\|_{L^\infty(]0,T[,L^q)},$$

  and therefore we have proved the strong convergence

$$\mu(\rho_k) \xrightarrow[k\to\infty]{} \mu(\rho), \qquad \text{in } L^\infty(]0,T[,L^q(\Omega)). \tag{VI.67}$$

- Weak convergence of the inertia term:
  The Sobolev theorem shows that the embedding $W_0^{1,r}(\Omega) \subset L^2(\Omega)$ is compact for any $r>5/6$. By duality, we deduce that the embedding $L^2(\Omega)\subset W^{-1,q}(\Omega)$ is compact for any $q<6$.
  From Proposition VI.2.7 and Theorem II.5.17, we deduce that $(\rho_k v_k)_k$ is relatively compact in $L^2(]0,T[,(W^{-1,q}(\Omega))^d)$ for any $q<6$. We already obtained (VI.65), therefore an argument similar to the one used in the proof of Proposition II.2.9 shows that

$$\rho_k v_k \xrightarrow[k\to\infty]{} \rho v, \qquad \text{in } L^2(]0,T[,\ (W^{-1,q}(\Omega))^d),\ \forall q<6. \tag{VI.68}$$

  The product operator $(w,v)\mapsto w\otimes v$ is continuous from the space $(W^{-1,q}(\Omega))^d \times (W^{1,2}(\Omega))^d$ into $(W^{-1,\sigma}(\Omega))^{d\times d}$ with $\sigma = \frac{6q}{6+q}$. Therefore, it follows from (VI.64), (VI.68) and Proposition II.2.12 that

$$\rho_k v_k\otimes v_k \xrightarrow[k\to\infty]{} \rho v\otimes v, \quad \text{weakly in } L^1(]0,T[,\ (W^{-1,\sigma}(\Omega))^{d\times d}),\ \forall \sigma<3.$$

Finally, we know by using (VI.58) that the sequence $(\rho_k v_k \otimes v_k)_k$ is bounded in the space $L^{4/3}(]0,T[,(L^2(\Omega))^{d\times d})$. Thus, we can use Proposition II.2.10 once more to deduce that

$$\rho_k v_k \otimes v_k \xrightarrow[k\to\infty]{} \rho v \otimes v, \quad \text{weakly in } L^{4/3}(]0,T[,(L^2(\Omega))^{d\times d}).$$

- Passing to the limit in the momentum equation:
  Let $k \geq n$, $\theta \in \mathcal{C}^1([0,T])$, $\theta(T) = 0$, and ψ_n be fixed in V_n. Through integration by parts, the previously defined approximate solution satisfies

$$\begin{aligned}
&-\int_0^T \int_\Omega (\rho_k v_k)\cdot \psi_n \frac{\partial \theta}{\partial t}\,dx\,dt - \int_0^T \int_\Omega (\rho_k v_k \otimes v_k) : \nabla\psi_n \theta\,dx\,dt \\
&+\int_0^T \int_\Omega 2\mu(\rho_k)\mathrm{D}(v_k) : \mathrm{D}(\psi_n)\theta\,dx\,dt - \int_0^T \int_\Omega \rho_k f_k \cdot \psi_n \theta\,dx\,dt \\
=&\theta(0)\int_\Omega \rho_{0,k} v_{0,k}\cdot\psi_n\,dx.
\end{aligned}$$

  The test function ψ_n being fixed, convergences (VI.65) and (VI.67), obtained above, allow us to pass to the limit in k to obtain

$$\begin{aligned}
&-\int_0^T \int_\Omega (\rho v)\cdot \psi_n \frac{\partial \theta}{\partial t}\,dx\,dt - \int_0^T \int_\Omega (\rho v \otimes v) : \nabla\psi_n \theta\,dx\,dt \\
&+\int_0^T \int_\Omega 2\mu(\rho)\mathrm{D}(v) : \mathrm{D}(\psi_n)\theta\,dx\,dt - \int_0^T \int_\Omega \rho f \cdot \psi_n \theta\,dx\,dt \\
=&\theta(0)\int_\Omega \rho_0 v_0\cdot\psi_n\,dx.
\end{aligned}$$

  By linear combination of test functions of the form $\theta(t)\psi_n(x)$, for all $n \geq 1$ and for any test function $\varphi \in \mathcal{C}^1([0,T],H_n)$ such that $\varphi(T) = 0$, we obtain

$$\begin{aligned}
&-\int_0^T \int_\Omega (\rho v)\cdot \frac{\partial \varphi}{\partial t}\,dx\,dt - \int_0^T \int_\Omega (\rho v \otimes v) : \nabla\varphi\,dx\,dt \\
&+\int_0^T \int_\Omega 2\mu(\rho)\mathrm{D}(v) : \mathrm{D}(\varphi)\,dx\,dt - \int_0^T \int_\Omega \rho f \cdot \varphi\,dx\,dt \\
=&\int_\Omega \rho_0 v_0\cdot\varphi(0)dx.
\end{aligned} \tag{VI.69}$$

  The regularity obtained on ρ and v allows us to extend the identity above, by density, to the test functions φ of $\mathcal{C}^1([0,T],V)$ such that $\varphi(T) = 0$, by using Lemma V.1.2. By restricting ourselves to test functions $\mathcal{C}^\infty$ with compact support over time in $]0,T[$, we indeed obtain the equation in the sense of distributions, by using Theorem IV.2.3 to recover the pressure as we did in Chapter V.
- Interpretation of the initial condition on ρv:

It is now necessary to interpret the initial velocity data. If we take a test function of the form $\varphi(t,x) = \theta(t)\psi(x)$ with $\theta \in \mathcal{D}(]0,T[)$ and $\psi \in V$ fixed, in (VI.69), by definition we obtain exactly that

$$\begin{aligned}
&\frac{d}{dt}\int_\Omega (\rho v)\cdot\psi\,dx - \int_\Omega (\rho v\otimes v) : \nabla\psi\,dx + \int_\Omega 2\mu(\rho)\mathrm{D}(v) : \mathrm{D}(\psi)\,dx \\
&= \int_\Omega \rho f\cdot\psi\,dx,
\end{aligned} \tag{VI.70}$$

in the sense of distributions of $\mathcal{D}'(]0,T[)$. Thanks to the regularity property of ρ and of v, we can deduce that

$$\int_\Omega (\rho v)\cdot\psi\,dx \in L^1(]0,T[), \text{ and } \frac{d}{dt}\int_\Omega (\rho v)\cdot\psi\,dx \in L^1(]0,T[).$$

Hence $t \mapsto \int_\Omega (\rho v)\cdot\psi\,dx$ is a function of $W^{1,1}(]0,T[)$ and, as a consequence, can be seen as a continuous function of t (Corollary II.4.2). If we now choose $\theta \in \mathcal{C}^\infty([0,T])$, we can write, by using (VI.70), an equation for integration by parts (Corollary II.4.2) in the form

$$\begin{aligned}
&\left(\int_\Omega (\rho v)\cdot\psi\,dx\right)(T)\,\theta(T) - \left(\int_\Omega (\rho v)\cdot\psi\,dx\right)(0)\,\theta(0) \\
&= \int_0^T \left[\left(\frac{d}{dt}\int_\Omega (\rho\, v)\cdot\psi\,dx\right)\theta(t) + \left(\int_\Omega (\rho\, v)\cdot\psi\,dx\right)\left(\frac{d\theta}{dt}\right)\right] dt \\
&= \int_0^T\int_\Omega (\rho v\otimes v) : \nabla\psi\,\theta(t)\,dx\,dt - \int_0^T\int_\Omega 2\mu(\rho)\mathrm{D}(v) : \mathrm{D}(\psi)\,\theta(t)\,dx\,dt \\
&\quad + \int_0^T\int_\Omega \rho f\cdot\psi\,\theta(t)\,dx\,dt + \int_0^T\int_\Omega (\rho\, v)\cdot\psi\frac{d\theta}{dt}dx\,dt.
\end{aligned}$$

If we assume that $\theta(0) = 1$ and $\theta(T) = 0$, then by comparing the result obtained above to that given by (VI.69) (for $\varphi = \theta(t)\psi$), we get

$$\left(\int_\Omega (\rho v)\cdot\psi\,dx\right)(0) = \int_\Omega \rho_0 v_0\cdot\psi\,dx,\ \forall\psi \in V.$$

This gives a suitable weak sense to the initial condition at $t = 0$ for ρv. We draw the attention of the reader to the fact that $v(0)$ does not have a precise meaning in this respect and that we can only define the initial condition for the momentum ρv.

2.5 The case without vacuum

In the case where $\inf \rho_0 > 0$ and $\inf \rho^{in} > 0$, that is, when the fluid being studied does not contain any vacuum regions, the results can be improved. In a more precise way, we have

- For all k, we have $\inf \rho_k \geq \min(\inf \rho_0, \inf \rho^{in}) > 0$.
- From (VI.57), we deduce that $(v_k)_k$ is bounded in

$$L^\infty(]0,T[,(L^2(\Omega))^d) \cap L^2(]0,T[,(H^1(\Omega))^d).$$

- From (VI.59), we deduce that we have

$$\|\tau_h v_k - v_k\|_{L^2(]0,T-h[,L^2)} \leq Ch^{1/4},$$

  and therefore that the sequence $(v_k)_k$ is bounded in $N_2^{1/4}(]0,T[,(L^2(\Omega))^d)$.

This allows us to finally obtain the strong convergence of $(v_k)_k$ towards v in $L^2(]0,T[\times\Omega)^d$, as in the case of homogeneous fluids studied in Chapter V. This case is therefore simpler, and justifying the passage to the limit in the approximate problem is more straightforward than in the general case.

Furthermore, still in the case where $\inf \rho_0 > 0$ and $\inf \rho^{in} > 0$, we can show (see [86]) that ρv and v are continuous with respect to the time variable with values in $(L^2(\Omega))^d$ for the weak topology and also that $v(t)$ converges towards v_0 in $(L^2(\Omega))^d$ for the strong topology as $t \to 0$ (because we have chosen v_0 which is divergence-free).

Chapter VII
Boundary conditions modelling

In this chapter, we consider two different problems related to boundary conditions that one may encounter when trying to compute numerical approximations of real flows. These problems arise when the computational domain is not exactly the original physical domain which is of interest.

- In Section 1, we consider the case where the computational domain is strictly smaller than the physical domain (for saving computational ressources, for instance, or because the physical domain is unbounded). Therefore, we need to propose artificial boundary conditions on the non-physical part of the boundary and to analyse the resulting system. We discuss flows for which the velocity is prescribed at one part of the boundary whilst being able to flow freely at another. An example of this type of situation is a network of pipes for which the input velocity is imposed at a certain number of points in the network but remains unknown at the output.
  From a mathematical and numerical point of view it is necessary to choose boundary conditions on the outflow part of the boundary. In this section, we propose a nonlinear boundary condition aiming at providing a stable solution of the Navier–Stokes equations, that we hope to be close enough to the real flow.
- Conversely, in Section 2, we consider the case where the computational domain is greater than the physical domain. This is called the fictitious domain approach. This often occurs when one wants to avoid the difficulties related to numerical methods on unstructured or moving meshes. We refer to the part of the computational domain which is not in the physical domain as the *obstacle*.
  In this section, we are interested in taking into account Dirichlet boundary conditions for the Navier–Stokes equations on the real physical boundary by using a penalty method which consists in adding some unphysical term in the equations inside the obstacle which aims at taking into account for real boundary conditions on the boundary of the obstacle. We show that

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the approximation is well-posed and we analyse its convergence towards the real solution as the penalty parameter goes to 0.

In addition to the particular interest of each of these two problems in practical situations, this is the opportunity for us to present mathematical methods which can be useful in many other frameworks.

1 Outflow boundary conditions

In general, whatever the physical situation encountered, when we want to compute the flow around an object (such as the classic case of an airflow around the wing of an aircraft, for example) we are required to, more or less arbitrarily, truncate the physical domain so as to obtain a suitable computational domain. When doing so, it is useful to write realistic boundary conditions for the "non-physical" boundaries of the domain so as to arrive at a mathematically well-posed problem which will be able to describe the physical situation under study.

Typically, when a flow ceases to be laminar at high values of the Reynolds number, turbulent structures are induced at the exit of the flow (i.e., the place where, on average, the fluid leaves the computational domain), presenting a velocity field that is alternately exiting and re-entering the domain, although the net flow is outward. However, at a point where the velocity field is leaving the domain, we can intuitively understand that it is not necessary to impose a boundary condition whereas in the opposite case, where the velocity field is entering the domain, it is an absolute necessity to provide some boundary condition.

In order to deal with this difficulty we describe weakly reflective boundary conditions downstream of the flow, that is, which will not contribute to increasing the energy of the system too much. These boundary conditions were first introduced and studied in the references [29, 30]. Extension of these results to the case of the nonhomogeneous Navier–Stokes equation is proposed in [25].

1.1 Setting up the model

Let Ω be a bounded Lipschitz domain of $\mathbb{R}^d$, where $d = 2$ or $d = 3$. We denote its boundary as $\Gamma = \partial\Omega$ and consider a partition of Γ, $\Gamma = \Gamma_1 \cup \Gamma_2$ where Γ_1 and Γ_2 have nonzero surface measures. We assume that the Dirichlet boundary data g_1 are known on Γ_1 (which represents the upstream part of the flow) and we consider the following problem. We need to find a pair (v, p) which is a solution of

$$\begin{cases} \dfrac{\partial v}{\partial t} + (v \cdot \nabla)v - \text{div } \sigma(v,p) = 0, & \text{in } \Omega, \\ \text{div } v = 0, & \text{in } \Omega, \\ v = g_1, & \text{on } \Gamma_1, \\ v(0) = v_0, \end{cases} \tag{VII.1}$$

where $\sigma(v,p) = (2/\mathcal{R}e)\mathrm{D}(v) - p\,\mathrm{Id}$ is the stress tensor of the flow, the strain rate tensor being defined by $\mathrm{D}(v) = \frac{1}{2}(\nabla v + {}^t\,\nabla v)$.

It is also necessary to give a boundary condition (which is not prescribed by the physical situation being modelled) on Γ_2, which accounts as precisely as possible for the downstream part of the flow which is not computed, see Figure VII.1. Clearly, this boundary condition must relate to the stress experienced by the flow across Γ_2. We can, for the moment, express this in the general form

$$\sigma(v,p).\nu = F(v), \ \text{on}\Gamma_2, \tag{VII.2}$$

where the function F is specified below.

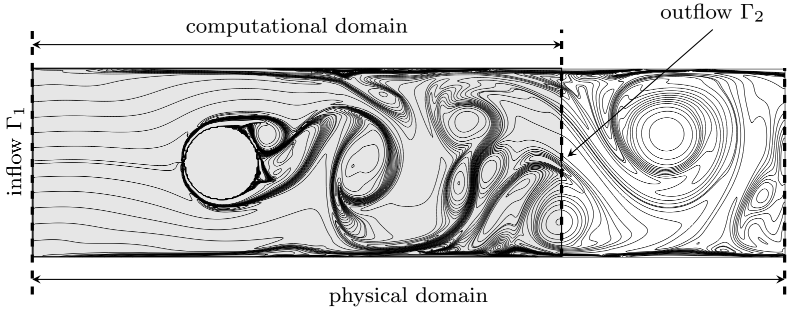


Fig. VII.1 Truncation of the physical domain for computational purposes

In order to build the boundary condition that we consider, we need to choose some reference flow u_{ref}, which is any $u_{\text{ref}} \in (H^1(\Omega))^d$ such that $u_{\text{ref}} = g_1$ on Γ_1. It is important for the efficiency of the method that u_{ref} is chosen so as to be a reasonable approximation of the expected flow (which is unknown of course) near Γ_2.

In order to simplify the computations, even though it is not necessary at all, we assume that u_{ref} is a solution to the following under-determined steady Stokes problem

$$\begin{cases} -\operatorname{div}\sigma(u_{\text{ref}},p_{\text{ref}}) = -\dfrac{1}{\mathcal{R}e}\Delta u_{\text{ref}} + \nabla p_{\text{ref}} = 0, & \text{in } \Omega, \\ \operatorname{div} u_{\text{ref}} = 0, & \text{in } \Omega, \\ u_{\text{ref}} = g_1, & \text{on } \Gamma_1. \end{cases} \tag{VII.3}$$

We also need to choose a reference stress tensor σ_{ref} defined on the outflow part of the boundary Γ_2. Here also to simplify a little, we simply choose

$$\sigma_{\text{ref}}.\nu = \sigma(u_{\text{ref}},p_{\text{ref}}).\nu \in (H^{-1/2}(\partial\Omega))^d. \tag{VII.4}$$

From now on, we assume that u_{ref} and σ_{ref} are fixed and time-independent.

We then look for a solution of problem (VII.1) in the form $(v = u + u_{\text{ref}}, p)$ where u satisfies homogeneous boundary conditions on Γ_1 and for which the condition on Γ_2 are satisfied in a weak sense.

Let us introduce the functional spaces which are naturally associated with this problem

$$\begin{aligned} &V = \left\{\varphi \in (H^1(\Omega))^d,\ \varphi_{|\Gamma_1} = 0,\ \operatorname{div}\varphi = 0\right\}, \\ &H = \overline{V},\ \text{the closure of } V \text{ in } (L^2(\Omega))^d. \end{aligned}$$

Note that those spaces are different (because the Dirichlet boundary condition is only prescribed on Γ_1) from the spaces V and H introduced in Section 3.3 of Chapter IV. We are, therefore, looking for a pair (u,p), with $u(t) \in V$ for almost any t, such that the pair $(u + u_{\text{ref}}, p)$ satisfies the following weak formulation of problem (VII.1), for all $\psi \in V$

$$\begin{aligned} &\frac{d}{dt}\int_\Omega (u+u_{\text{ref}})\cdot\psi\,dx + \int_\Omega \sigma(u+u_{\text{ref}},p+p_{\text{ref}}) : \nabla\psi\,dx \\ &\quad + \frac{1}{2}\int_\Omega\Big(\big(((u+u_{\text{ref}})\cdot\nabla)(u+u_{\text{ref}})\big)\cdot\psi - \big(((u+u_{\text{ref}})\cdot\nabla)\psi\big)\cdot(u+u_{\text{ref}})\Big)dx \\ &= -\frac{1}{2}\int_{\Gamma_2}\big((u+u_{\text{ref}})\cdot\psi\big)\big((u+u_{\text{ref}})\cdot\nu\big)\,d\sigma \\ &\quad + \big\langle\sigma(u+u_{\text{ref}},p+p_{\text{ref}}).\nu,\psi\big\rangle_{H^{-1/2},H^{1/2}}. \end{aligned}$$

This equation is of course understood in the sense of distributions on $]0,T[$. As in Chapter V, we can write a weak formulation with test functions which depend on time. Furthermore, we can add to this formulation the initial data which are satisfied in the weak sense of V'.

This formulation takes account of the nonhomogeneous Dirichlet boundary conditions on Γ_1 and of the fact that u_{ref} is a lifting of this boundary condition. We now need to detail the condition imposed on Γ_2. The idea is as follows. At a place where the flow is outward (i.e., if $v\cdot\nu = (u+u_{\text{ref}})\cdot\nu > 0$) then we impose on the fluid the constraint coming from the selected reference flow (in general a nonturbulent flow). In contrast, if $v\cdot\nu = (u+u_{\text{ref}})\cdot\nu < 0$ we need, for stability reasons, to control the increase of energy generated by inertia through the boundary. In other words, in the case where $(u+u_{\mathrm{ref}})\cdot\nu<0$, it is necessary to take account of the pressure exerted by the fluid situated outside the computational domain (on the other side of Γ_2) where, of course, we know nothing. By recalling that the square of the norm of a velocity field is homogeneous with the quotient of the pressure over the density, we deduce that the constraint term which we need to add on Γ_2 must depend quadratically on the velocity.

To take account of both cases (i.e., the inward and outward parts of the boundary), we specify boundary conditions (VII.2) in the form

$$\sigma(u+u_{\mathrm{ref}},p+p_{\mathrm{ref}}).\nu=-\frac{1}{2}\big((u+u_{\mathrm{ref}})\cdot\nu\big)^{-}u+\sigma_{\mathrm{ref}}.\nu,\ \text{on } \Gamma_2. \tag{VII.5}$$

We recall that the positive and negative parts of any real number are defined in Definition VI.1.1.

We show that the choice of the quadratic term in (VII.5) is exactly that which allows us to compensate the energy contained in the contributions of inertia terms on the unphysical parts of the boundary, where the fluid is entering the domain.

Remark VII.1.1. • Other boundary conditions of this type can be found in [30]. Furthermore, we note that these boundary conditions actually depend on the reference flow u_{ref}, σ_{ref}.

From a physical and numerical point of view, the choice of a “good” reference flow is clearly crucial. Indeed, we observe that (VII.5) is exactly satisfied if it happens that $v=u_{\mathrm{ref}}$; that is, $u=0$. Therefore, if we are able to choose, a priori, $(u_{\mathrm{ref}},\sigma_{\mathrm{ref}})$ not too far from the expected solution, then the computed solution will be close to the exact flow. This of course depends on the physical situation under study.

• In [25], the same kind of boundary condition is studied in the case of incompressible nonhomogeneous flows (i.e., the same model that we studied in Chapter VI). In particular, it is shown that σ_{ref} and u_{ref} do not need to be related by the relation (VII.4) and may also depend on time.

If we take this boundary condition (VII.5) into the problem and if we consider the fact that u_{ref} is a solution of the steady Stokes problem (VII.3), then the weak formulation which we study now can be written as follows, for any test function $\psi\in V$,

$$\begin{aligned}&\frac{d}{dt}\int_\Omega u\cdot\psi\,dx+\int_\Omega\sigma(u,p):\nabla\psi\,dx\\&+\frac{1}{2}\int_\Omega\big(((u+u_{\mathrm{ref}})\cdot\nabla)(u+u_{\mathrm{ref}})\big)\cdot\psi-\big(((u+u_{\mathrm{ref}})\cdot\nabla)\psi\big)\cdot(u+u_{\mathrm{ref}})\,dx\\&+\frac{1}{2}\int_{\Gamma_2}\big((u+u_{\mathrm{ref}})\cdot\psi\big)\big((u+u_{\mathrm{ref}})\cdot\nu\big)\,d\sigma=-\frac{1}{2}\int_{\Gamma_2}(u\cdot\psi)((u+u_{\mathrm{ref}})\cdot\nu)^{-}\,d\sigma.\end{aligned}$$

By using the fact that the doubly contracted product of a symmetric tensor with an asymmetric tensor is zero, we obtain for all $\psi \in V$ that

$$\begin{aligned}\sigma(u,p) : \nabla\psi &= \frac{2}{\mathcal{R}e}\mathrm{D}(u) : \left(\mathrm{D}(\psi) + \frac{1}{2}(\nabla\psi - {}^t\nabla\psi)\right) - p\,(\mathrm{div}\ \psi),\\ &= \frac{2}{\mathcal{R}e}\mathrm{D}(u) : \mathrm{D}(\psi).\end{aligned}$$

Finally, the weak formulation on u no longer explicitly contains the pressure term, reducing the problem to looking for $t \mapsto u(t) \in V$ which satisfies, for all $\psi \in V$, the following equation in the sense of $\mathcal{D}'(]0,T[)$,

$$\begin{aligned}&\frac{d}{dt}\int_\Omega u\cdot\psi\,dx + \frac{2}{\mathcal{R}e}\int_\Omega \mathrm{D}(u):\mathrm{D}(\psi)\,dx\\ &+\frac{1}{2}\int_\Omega \big(((u+u_{\mathrm{ref}})\cdot\nabla)(u+u_{\mathrm{ref}})\big)\cdot\psi - \big(((u+u_{\mathrm{ref}})\cdot\nabla)\psi\big)\cdot(u+u_{\mathrm{ref}})\,dx\\ &+\frac{1}{2}\int_{\Gamma_2}\big(u\cdot\psi\big)\big((u+u_{\mathrm{ref}})\cdot\nu\big)^+\,d\sigma = -\frac{1}{2}\int_{\Gamma_2}(u_{\mathrm{ref}}\cdot\psi)((u+u_{\mathrm{ref}})\cdot\nu)\,d\sigma.\end{aligned} \tag{VII.6}$$

1.2 Existence and uniqueness

In this section we prove the following result.

Theorem VII.1.1. *Let Ω be a connected, bounded, Lipschitz domain of $\mathbb{R}^d$, with $d \in \{2,3\}$. Let $\mathcal{R}e > 0$, $g_1 \in (H^{1/2}(\Gamma))^d$ and $(u_{\mathrm{ref}}, p_{\mathrm{ref}})$ be the unique solution of (VII.3).*

For all $v_0 \in u_{\mathrm{ref}} + H$ and all $T > 0$, there exists $u \in L^\infty(]0,T[,H) \cap L^2(]0,T[,V)$ a solution of (VII.6) such that for all $\psi \in V$ we have

$$\left(\int_\Omega u\cdot\psi\,dx\right)(0) = \int_\Omega u_0\cdot\psi\,dx, \tag{VII.7}$$

with $u_0 = v_0 - u_{\mathrm{ref}}$.

Moreover, we have $du/dt \in L^{\frac{4}{d}}(]0,T[,V')$ and the solution u is unique in dimension $d = 2$.

Finally, there exists $p \in W^{-1,\infty}(]0,T[,L^2(\Omega))$ such that the pair $(v = u_{\mathrm{ref}} + u, p)$ is a solution of the Navier–Stokes equations in Ω.

The initial condition (VII.7) has a meaning because for any solution of (VII.6), we see that $t \mapsto \int_\Omega u\cdot\psi\,dx$ is continuous with respect to time, using Corollary II.4.2.

1.2.1 Definition of an approximate problem

We now return to the functional space V introduced above. This space is a closed subspace of $(H^1(\Omega))^d$ and in particular a separable Hilbert space. Therefore, it possesses a countable Hilbert basis $(w_k)_k$.

Let us construct an approximate problem which is suitable for the weak formulation (VII.6). Let H_N be the vector space generated by the functions $(w_k)_{k\leq n}$. We consider the following Galerkin approximation: to find $t \mapsto u_N(t) \in H_N$ a solution for all $\psi \in H_N$ of

$$\begin{aligned}
&\frac{d}{dt}\int_\Omega u_N\cdot\psi\,dx + \frac{2}{\mathcal{R}e}\int_\Omega \mathrm{D}(u_N):\mathrm{D}(\psi)\,dx\\
&+\frac{1}{2}\int_\Omega\big(((u_N{+}u_{\mathrm{ref}})\cdot\nabla)(u_N{+}u_{\mathrm{ref}})\big)\cdot\psi - \big(((u_N{+}u_{\mathrm{ref}})\cdot\nabla)\psi\big)\cdot(u_N{+}u_{\mathrm{ref}})\,dx\\
&+\frac{1}{2}\int_{\Gamma_2}\big(u_N\cdot\psi\big)\big((u_N+u_{\mathrm{ref}})\cdot\nu\big)^+\,d\sigma = -\frac{1}{2}\int_{\Gamma_2}\big(u_{\mathrm{ref}}\cdot\psi\big)\big((u_N+u_{\mathrm{ref}})\cdot\nu\big)\,d\sigma.
\end{aligned}\tag{VII.8}$$

In this case, we do not use a special basis constructed from the eigenfunctions of a well-chosen operator, contrary to our approach in Chapter V. The reason is that the study of the Stokes operator with mixed boundary conditions of the Dirichlet type on Γ_1 and Neumann conditions on Γ_2 introduces additional difficulties that we do not wish to address here.

Because of this, we do not know a priori that the functions w_k are regular and, moreover, we do not know how to estimate the norm of the orthogonal projection $\mathcal{P}_N : H \mapsto H_N \subset H$ considered as a (nonorthogonal) projection in V. Hence, we cannot obtain an estimate of the time derivative of the approximate solutions in the same way as in Lemma V.1.6 in Chapter V. For this reason, we replace here this estimate of du_N/dt by an estimate over the fractional derivatives with respect to time. We get this estimate by using the Fourier transform with respect to time introduced in Section 5.4 of Chapter II. By using Proposition II.5.23, we obtain an estimate of the approximate solution in a well chosen Nikolskii space. This is sufficient to obtain the compactness properties that we need (see Theorem II.5.17).

This general approach is well-described in the case of the Navier–Stokes equations with usual Dirichlet boundary conditions in [84].

1.2.2 The energy inequality

Proposition VII.1.2. *The approximate system* (VII.8) *has a unique global solution on* $\mathbb{R}^+$ *and for all* $T>0$ *there exists a constant* C*, independent of* N*, such that*

$$\begin{aligned}&\sup_{t\le T}\|u_N(t)\|_{L^2}^2+\frac{1}{\mathcal{R}e}\int_0^T\|\nabla u_N(t)\|_{L^2}^2\,dt\\&+\int_0^T\int_{\Gamma_2}|u_N|^2\big((u_N+u_{\mathrm{ref}})\cdot\nu\big)^+\,d\sigma\,dt\le C.\end{aligned}\tag{VII.9}$$

Proof.

In the usual way, Equation (VII.8) is equivalent to a system of finite dimensional ordinary differential equations. The existence and uniqueness of a solution of this system over a maximal time interval $[0,T_N[$ comes from the Cauchy–Lipschitz theorem. It is now sufficient to establish an energy inequality with bounds independent of N, in order to prove that the solution is global and the associated estimate (VII.9).

This energy inequality is obtained by selecting $\psi = u_N(t) = (u_N(t) + u_{\mathrm{ref}}) - u_{\mathrm{ref}}$ as a test function and by being aware of the time dependence of ψ. It follows

$$\begin{aligned}&\frac12\frac{d}{dt}\|u_N\|_{L^2}^2+\frac{2}{\mathcal{R}e}\|\mathrm{D}(u_N)\|_{L^2}^2+\frac12\int_{\Gamma_2}|u_N|^2\big((u_N+u_{\mathrm{ref}})\cdot\nu\big)^+d\sigma\\
\le&\left|\int_\Omega\big(((u_N+u_{\mathrm{ref}})\cdot\nabla)u_N\big)\cdot u_{\mathrm{ref}}-\big(((u_N+u_{\mathrm{ref}})\cdot\nabla)u_{\mathrm{ref}}\big)\cdot u_N\,dx\right|\\
&+\frac12\left|\int_{\Gamma_2}\big(u_N\cdot u_{\mathrm{ref}}\big)\big((u_N+u_{\mathrm{ref}})\cdot\nu\big)d\sigma\right|\\
\le&\|u_N+u_{\mathrm{ref}}\|_{L^3}\big(\|\nabla u_N\|_{L^2}\|u_{\mathrm{ref}}\|_{L^6}+\|u_N\|_{L^6}\|\nabla u_{\mathrm{ref}}\|_{L^2}\big)\\
&+\|u_N+u_{\mathrm{ref}}\|_{L^{8/3}(\Gamma_2)}\|u_N\|_{L^{8/3}(\Gamma_2)}\|u_{\mathrm{ref}}\|_{L^4(\Gamma_2)}.\end{aligned}$$

From the Sobolev inequalities (Proposition III.2.35), we can bound the quantity $\|u_N\|_{L^3}$ from above by $C\|u_N\|_{L^2}^{1/2}\|\nabla u_N\|_{L^2}^{1/2}$. Moreover, we bound $\|u_N\|_{L^{8/3}(\Gamma_2)}$ from above by $\|u_N\|_{L^{8/3}(\Gamma)}$ and hence from Theorem III.2.36 and the generalised Poincaré inequality (Proposition III.2.38), we obtain

$$\|u_N\|_{L^{8/3}(\Gamma_2)}\le C\|u_N\|_{L^2}^{1/4}\|\nabla u_N\|_{L^2}^{3/4}.$$

We note that these inequalities are equally valid in dimension 2 and in dimension 3. We also recall that u_{ref} is a fixed reference steady flow in $(H^1(\Omega))^d$ so that there exists a $C>0$ which only depends on u_{ref} and on Ω, such that

$$\begin{aligned}&\frac12\frac{d}{dt}\|u_N\|_{L^2}^2+\frac{2}{\mathcal{R}e}\|\mathrm{D}(u_N)\|_{L^2}^2+\frac12\int_{\Gamma_2}|u_N|^2\big((u_N+u_{\mathrm{ref}})\cdot\nu\big)^+d\sigma\\
\le&C\left(1+\|\nabla u_N\|_{L^2}+\|u_N\|_{L^2}^{1/2}\|\nabla u_N\|_{L^2}^{3/2}\right).\end{aligned}$$

Now we need to use the Korn inequality (Lemma IV.7.6) in order to control the norm of ∇u_N by the norm of $\mathrm{D}(u_N)$. Hence, by suitable use of Young's inequality, this gives

$$\begin{aligned}&\frac{1}{2}\frac{d}{dt}\|u_N\|_{L^2}^2+\frac{2}{\mathcal{R}e}\|\mathrm{D}(u_N)\|_{L^2}^2+\frac{1}{2}\int_{\Gamma_2}|u_N|^2\big((u_N+u_{\mathrm{ref}})\cdot\nu\big)^+d\sigma\\ \leq&\frac{1}{\mathcal{R}e}\|\mathrm{D}(u_N)\|_{L^2}^2+C(1+\|u_N\|_{L^2}^2),\end{aligned}$$

where C depends on u_{ref}. From Gronwall's lemma, we can deduce from this last inequality the existence of a $C(T)>0$, independent of N such that

$$\begin{aligned}&\sup_{t\leq T}\|u_N(t)\|_{L^2}^2+\frac{1}{\mathcal{R}e}\int_0^T\|\mathrm{D}(u_N)\|_{L^2}^2\,dt\\ &+\int_0^T\int_{\Gamma_2}|u_N|^2\big((u_N+u_{\mathrm{ref}})\cdot\nu\big)^+\,d\sigma\,dt\leq C(T).\end{aligned}$$

The desired inequality (VII.9) is finally obtained by once again using the Korn inequality.

□

As in Chapter V, we deduce from the estimate above that the approximate solution obtained through the Galerkin method is global on $\mathbb{R}^+$; that is, that the existence time T_N is $+\infty$.

Remark VII.1.2. For a smooth bounded domain Ω of $\mathbb{R}^3$, the largest boundary Sobolev space in the family $H^s(\Gamma)$, with $s>0$, which embeds into $L^3(\Gamma)$ is $H^{1/3}(\Gamma)$ (because Γ is a 2-dimensional manifold). This is the trace space associated with the fractional Sobolev space $H^{5/6}(\Omega)$, since $\frac{1}{3}=\frac{5}{6}-\frac{1}{2}$. Thanks to the Sobolev embeddings and by using interpolation (Theorem II.5.5) we obtain that the functions of $L^\infty(]0,T[,L^2(\Omega))\cap L^2(]0,T[,H^1(\Omega))$ only belong to $L^{12/5}(]0,T[,H^{5/6}(\Omega))$. Hence, $|u|^3$ is not integrable on $]0,T[\times\Gamma$ because it is not regular enough with respect to time. From a mathematical point of view this is a justification of the fact that the outflow boundary condition need be quadratic in order to lead to a well-posed system.

1.2.3 Estimates of fractional derivatives with respect to time

It is now useful to obtain an additional estimate of the velocity u_N in view of passing to the limit in the nonlinear terms by the compactness theorems. We now use the fractional derivatives with respect to time and more precisely the Fourier transform in the time variable, the properties of which we detailed in Section 5.4 of Chapter II. We return to using the notation $\widetilde{u_N}$ to designate the extension by 0 of u_N to all the time interval $\mathbb{R}$.

Proposition VII.1.3. *For all $T>0$, and for all $\sigma<1/6$, there exists a $C>0$, independent of N, such that:*

$$\int_{\mathbb{R}}|\tau|^{2\sigma}\|\mathcal{F}(\widetilde{u_N})(\tau)\|_{L^2}^2d\tau\leq C.$$

Proof.

The test function $\psi \in H_N$ being fixed, we can extend the ordinary differential equation (VII.8) by 0 outside the interval $[0, T]$. We then apply the Fourier transform with respect to time to the equation obtained. Making use of (II.27), for all $\tau \in \mathbb{R}$ this gives:

$$\begin{aligned}
& i\tau \int_\Omega \mathcal{F}(\widetilde{u_N}) \cdot \psi \, dx + \frac{1}{2} \int_\Omega \mathcal{F}\Big(((\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \cdot \nabla)(\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \Big) \cdot \psi \, dx \\
& - \frac{1}{2} \int_\Omega \mathcal{F}\Big((((\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \cdot \nabla)\psi) \cdot (\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \Big) \, dx \\
& + \frac{2}{\mathcal{R}e} \int_\Omega D\left(\mathcal{F}(\widetilde{u_N})\right) : \mathrm{D}(\psi) dx + \frac{1}{2} \int_{\Gamma_2} \mathcal{F}\left((\widetilde{u_N} \cdot \psi)\big((\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \cdot \nu\big)^+ \right) d\sigma \\
= & - \frac{1}{2} \int_{\Gamma_2} (u_{\text{ref}} \cdot \psi) \mathcal{F}\big((\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \cdot \nu\big) d\sigma \\
& + \frac{1}{\sqrt{2\pi}} \int_\Omega u_N(0) \cdot \psi \, dx - \frac{e^{-i\tau T}}{\sqrt{2\pi}} \int_\Omega u_N(T) \cdot \psi \, dx.
\end{aligned} \tag{VII.10}$$

We note that since ψ does not depend on time, the third term can be expressed by a purely algebraic manipulation

$$\begin{aligned}
& \int_\Omega \mathcal{F}\Big((((\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \cdot \nabla)\psi) \cdot (\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \Big) \, dx \\
= & \int_\Omega \mathcal{F}\Big(((\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \otimes (\widetilde{u_N} + \widetilde{u_{\text{ref}}})) \Big) : \nabla\psi \, dx.
\end{aligned}$$

The number τ being fixed, we now take $\psi = \overline{\mathcal{F}(\widetilde{u_N})(\tau)}$ in (VII.10) and we obtain

$$\begin{aligned}
& i\tau \int_\Omega \left|\mathcal{F}(\widetilde{u_N})\right|^2 \, dx + \frac{1}{2} \int_\Omega \mathcal{F}\Big(((\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \cdot \nabla)(\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \Big) \cdot \overline{\mathcal{F}(\widetilde{u_N})} \, dx \\
& - \frac{1}{2} \int_\Omega \mathcal{F}\Big(((\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \otimes (\widetilde{u_N} + \widetilde{u_{\text{ref}}})) \Big) : \nabla\overline{\mathcal{F}(\widetilde{u_N})} \, dx \\
& + \frac{2}{\mathcal{R}e} \int_\Omega \left| D\Big(\mathcal{F}(\widetilde{u_N})\Big) \right|^2 \, dx + \frac{1}{2} \int_{\Gamma_2} \mathcal{F}\left(\big((\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \cdot \nu\big)^+ \widetilde{u_N} \right) \cdot \overline{\mathcal{F}(\widetilde{u_N})} \, d\sigma \\
= & - \frac{1}{2} \int_{\Gamma_2} (u_{\text{ref}} \cdot \overline{\mathcal{F}(\widetilde{u_N})}) \mathcal{F}\big((\widetilde{u_N} + \widetilde{u_{\text{ref}}}) \cdot \nu\big) \, d\sigma \\
& + \frac{1}{\sqrt{2\pi}} \int_\Omega u_N(0) \cdot \overline{\mathcal{F}(\widetilde{u_N})} \, dx - \frac{e^{-i\tau T}}{\sqrt{2\pi}} \int_\Omega u_N(T) \cdot \overline{\mathcal{F}(\widetilde{u_N})} \, dx.
\end{aligned}$$

We then note that the first term in the left-hand side is pure imaginary whereas the term coming from the viscous stresses a nonnegative real number. The estimate of interest to us is that of the first term. We therefore estimate the absolute value of the imaginary part of the above equation. We obtain

$$\begin{aligned}|\tau| \|\mathcal{F}(\widetilde{u_N})\|_{L^2}^2 \leq & \frac{1}{2}\left|\int_\Omega \mathcal{F}\Big(((\widetilde{u_N}+\widetilde{u_{\mathrm{ref}}})\cdot\nabla)(\widetilde{u_N}+\widetilde{u_{\mathrm{ref}}})\Big)\cdot\overline{\mathcal{F}(\widetilde{u_N})}\,dx\right| \\ & + \frac{1}{2}\left|\int_\Omega \mathcal{F}\Big(((\widetilde{u_N}+\widetilde{u_{\mathrm{ref}}})\otimes(\widetilde{u_N}+\widetilde{u_{\mathrm{ref}}}))\Big):\nabla\overline{\mathcal{F}(\widetilde{u_N})}\,dx\right| \\ & + \frac{1}{2}\left|\int_{\Gamma_2} \mathcal{F}\left(((\widetilde{u_N}+\widetilde{u_{\mathrm{ref}}})\cdot\nu)^+\widetilde{u_N}\right)\cdot\overline{\mathcal{F}(\widetilde{u_N})}\,d\sigma\right| \\ & + \frac{1}{2}\left|\int_{\Gamma_2} (u_{\mathrm{ref}}\cdot\overline{\mathcal{F}(\widetilde{u_N})})\mathcal{F}\big((\widetilde{u_N}+\widetilde{u_{\mathrm{ref}}})\cdot\nu\big)\,d\sigma\right| \\ & + \frac{1}{\sqrt{2\pi}}\left|\int_\Omega u_N(0)\cdot\overline{\mathcal{F}(\widetilde{u_N})}\,dx\right| + \frac{1}{\sqrt{2\pi}}\left|\int_\Omega u_N(T)\cdot\overline{\mathcal{F}(\widetilde{u_N})}\,dx\right|.\end{aligned} \tag{VII.11}$$

We refer to the six terms on the right-hand side of this inequality as $I_N^1(\tau), \dots, I_N^6(\tau)$ and let us try to estimate them.

- Term $I_N^1(\tau)$:
  From the estimates (VII.9), the sequence $(u_N)_N$ is bounded in the space $L^2(]0,T[,(H^1(\Omega))^d)$ and therefore also in the space $L^{6/5}(]0,T[,(L^6(\Omega))^d)$, thanks to Sobolev embeddings. Hence, the sequence $(\widetilde{u_N})_N$ is bounded in the space $L^{6/5}(\mathbb{R},(L^6(\Omega))^d)$. From the Hausdorff–Young theorem (Theorem II.5.20) we therefore have

  $$(\mathcal{F}(\widetilde{u_N}))_N \text{ is bounded in } L^6(\mathbb{R},(L^6(\Omega))^d). \tag{VII.12}$$

  We also have the estimate

  $$\begin{aligned}\big\|\big((u_N+u_{\mathrm{ref}})\cdot\nabla\big)(u_N+u_{\mathrm{ref}})\big\|_{L^{6/5}} &\leq \|\nabla(u_N+u_{\mathrm{ref}})\|_{L^2}\|u_N+u_{\mathrm{ref}}\|_{L^3} \\ &\leq C\|\nabla(u_N+u_{\mathrm{ref}})\|_{L^2}^{3/2}\|u_N+u_{\mathrm{ref}}\|_{L^2}^{1/2},\end{aligned}$$

  which demonstrates, from (VII.9), that $\Big(\big((u_N+u_{\mathrm{ref}})\cdot\nabla\big)(u_N+u_{\mathrm{ref}})\Big)_N$ is bounded in $L^{4/3}(]0,T[,(L^{6/5}(\Omega))^d)$ and hence in $L^{6/5}(]0,T[,(L^{6/5}(\Omega))^d)$. As a consequence, the sequence $\Big(\big((\widetilde{u_N}+\widetilde{u_{\mathrm{ref}}})\cdot\nabla\big)(\widetilde{u_N}+\widetilde{u_{\mathrm{ref}}})\Big)_N$ is bounded in $L^{6/5}(\mathbb{R},(L^{6/5}(\Omega))^d)$. By using the Hausdorff–Young theorem we can deduce that

  $$\left(\mathcal{F}\left(\big((\widetilde{u_N}+\widetilde{u_{\mathrm{ref}}})\cdot\nabla\big)(\widetilde{u_N}+\widetilde{u_{\mathrm{ref}}})\right)\right)_N \text{ is bounded in } L^6(\mathbb{R},(L^{6/5}(\Omega))^d). \tag{VII.13}$$

  From the Hölder inequality (with respect to time and space), we deduce from (VII.12) and (VII.13) that

  $$(I_N^1)_N \ \text{ is bounded in } L^3(\mathbb{R}). \tag{VII.14}$$

- Term I_N^2 :
  By using the Hölder inequality and the precise Sobolev inequalities (Proposition III.2.35), we obtain the estimate

$$\begin{aligned}\|(u_N + u_{\mathrm{ref}}) \otimes (u_N + u_{\mathrm{ref}})\|_{L^2} &\leq \|u_N + u_{\mathrm{ref}}\|_{L^4}^2 \\ &\leq C\,\|u_N + u_{\mathrm{ref}}\|_{L^2}^{1/2}\,\|\nabla(u_N + u_{\mathrm{ref}})\|_{L^2}^{3/2},\end{aligned}$$

  and hence, using (VII.9), the sequence $((u_N + u_{\mathrm{ref}}) \otimes (u_N + u_{\mathrm{ref}}))_N$ is bounded in $L^{\frac{4}{3}}(]0,T[,(L^2(\Omega))^{d\times d})$. By considering its extension by 0 outside the time interval $]0,T[$ and by once again using the Hausdorff–Young theorem, we obtain that $(\mathcal{F}((\widetilde{u_N} + \widetilde{u_{\mathrm{ref}}}) \otimes (\widetilde{u_N} + \widetilde{u_{\mathrm{ref}}})))_N$ is bounded in $L^4(\mathbb{R},(L^2(\Omega))^{d\times d})$. We also deduce from (VII.9) that $(\nabla\mathcal{F}(\widetilde{u_N}))_N$ is bounded in $L^2(\mathbb{R},(L^2(\Omega))^{d\times d})$.
  By combining these two bounds, we get

$$(I_N^2)_N \quad \text{is bounded in } L^{4/3}(\mathbb{R}). \tag{VII.15}$$

- Term I_N^3 :
  Theorem III.2.36 allows us to write

$$\begin{aligned}\left\|((u_N + u_{\mathrm{ref}})\cdot\nu)^+ u_N\right\|_{L^2(\Gamma_2)} &\leq \|u_N + u_{\mathrm{ref}}\|_{L^4(\Gamma_2)}\|u_N\|_{L^4(\Gamma_2)} \\ &\leq C\|u_N + u_{\mathrm{ref}}\|_{H^1}\|u_N\|_{H^1},\end{aligned}$$

  and hence from (VII.9), we obtain that the sequence $(((u_N+u_{\mathrm{ref}})\cdot\nu)^+u_N)_N$ is bounded in $L^1(]0,T[,(L^2(\Gamma_2))^d)$. Then, we deduce that the sequence $(\mathcal{F}(((\widetilde{u_N}+\widetilde{u_{\mathrm{ref}}})\cdot\nu)^+\widetilde{u_N}))_N$ is bounded in $L^\infty(\mathbb{R},(L^2(\Gamma_2))^d)$. Furthermore, we have

$$\|u_N\|_{L^2(\Gamma_2)} \leq \|u_N\|_{L^2(\Gamma)} \leq C\|u_N\|_{H^1},$$

  and therefore $(\mathcal{F}(\widetilde{u_N}))_N$ is bounded in $L^2(\mathbb{R},L^2(\Gamma_2))$.
  All this shows that

$$(I_N^3)_N \quad \text{is bounded in } L^2(\mathbb{R}). \tag{VII.16}$$

- Term I_N^4 :
  We treat this term in a similar way to the preceding one and we deduce that

$$(I_N^4)_N \quad \text{is bounded in } L^1(\mathbb{R}). \tag{VII.17}$$

- Terms I_N^5 and I_N^6:
  These two similar terms are Dirac mass contributions at $t=0$ and $t=T$ which we obtain by differentiating a function extended by 0 outside the time interval $]0,T[$ with respect to time. Let us focus on the term I_N^5, which is estimated by

$$I_N^5(\tau) \leq C\|u_N(0)\|_{L^2}\|\mathcal{F}(\widetilde{u_N})(\tau)\|_{L^2}.$$

However, $u_N(0) = \mathcal{P}_N(v_0 - u_{\mathrm{ref}})$ is bounded in $(L^2(\Omega))^d$ independently of N. Furthermore, we have already seen that (VII.9) implies that the sequence $(\mathcal{F}(\widetilde{u_N}))_N$ is bounded in $L^2(\mathbb{R}, (L^2(\Omega))^d)$. We therefore have

$$(I_N^5)_N \text{ and } (I_N^6)_N \text{ are bounded in } L^2(\mathbb{R}). \tag{VII.18}$$

Now, let $\delta \in [0,1]$. From Young's inequality we deduce that there exist $d_1, d_2 > 0$ such that

$$|\tau|^{1-\delta} \leq d_1 \frac{|\tau|}{1+|\tau|^\delta} + d_2, \ \forall \tau \in \mathbb{R}.$$

Hence from (VII.11), we have for all $\tau \in \mathbb{R}$

$$\begin{aligned}|\tau|^{1-\delta}\|\mathcal{F}(\widetilde{u_N})(\tau)\|_{L^2}^2 \leq & \, d_1 \frac{I_N^1(\tau)}{1+|\tau|^\delta} + d_1 \frac{I_N^2(\tau)}{1+|\tau|^\delta} + d_1 \frac{I_N^3(\tau) + I_N^5(\tau) + I_N^6(\tau)}{1+|\tau|^\delta} \\ & + d_1 \frac{I_N^4(\tau)}{1+|\tau|^\delta} + d_2 \|\mathcal{F}(\widetilde{u_N})(\tau)\|_{L^2}^2. \end{aligned} \tag{VII.19}$$

In order to use the bounds (VII.14) through (VII.18), via the Hölder inequality, we see that $\tau \mapsto 1/(1+|\tau|^\delta)$ has to belong to $L^{3/2}(\mathbb{R}) \cap L^\infty(\mathbb{R})$. It is indeed clear that this is a bounded function and it is therefore sufficient to choose δ so that it belongs to $L^{3/2}(\mathbb{R})$. Hence, for all $\delta \in]\frac{2}{3}, 1[$ from (VII.19) we deduce the estimate

$$\int_{\mathbb{R}} |\tau|^{1-\delta} \|\mathcal{F}(\widetilde{u_N})(\tau)\|_{L^2}^2 \, d\tau \leq C_\delta,$$

the number C_δ being independent of N. By setting $\sigma = (1-\delta)/2$ we have indeed proved the result.

□

1.2.4 Existence of a weak solution

It is now possible to pass to the limit in the approximate problem thanks to the following result.

Proposition VII.1.4. *Assume that $d = 3$. Even if it means considering a subsequence, there exists a function u and functions w_1, w_2 and γ_1 satisfying*

$$\begin{aligned} & u_N \xrightharpoonup[N\to\infty]{} u, \quad \textit{weakly-}\star \textit{ in } L^\infty(]0,T[, (L^2(\Omega))^d), \\ & u_N \xrightharpoonup[N\to\infty]{} u, \quad \textit{weakly in } L^2(]0,T[, (H^1(\Omega))^d), \\ & u_N \xrightharpoonup[N\to\infty]{} u, \quad \textit{weakly in } L^2(]0,T[, (L^6(\Omega))^d), \\ & u_N \xrightarrow[N\to\infty]{} u, \quad \textit{in } L^2(]0,T[, (L^2(\Omega))^d), \end{aligned}$$

$$\begin{aligned}
(u_N\cdot\nabla)u_N &\xrightarrow[N\to\infty]{} w_1, && \textit{weakly in } L^{4/3}(]0,T[,(L^{6/5}(\Omega))^d),\\
u_N\otimes u_N &\xrightarrow[N\to\infty]{} w_2, && \textit{weakly in } L^{4/3}(]0,T[,(L^{2}(\Omega))^{d\times d}),
\end{aligned}$$

$$\begin{aligned}
\big((u_N+u_{\mathrm{ref}})\cdot\nu\big)^+u_N &\xrightarrow[N\to\infty]{} \gamma_1, && \textit{weakly in } L^{4/3}(]0,T[,(L^{4/3}(\Gamma))^d),\\
\big(u_N\cdot\nu\big)u_{\mathrm{ref}} &\xrightarrow[N\to\infty]{} \big(u\cdot\nu\big)u_{\mathrm{ref}}, && \textit{weakly in } L^{4}(]0,T[,(L^{4/3}(\Gamma))^d),\\
u_N &\xrightarrow[N\to\infty]{} u, && \textit{weakly in } L^{4}(]0,T[,(L^{2}(\Gamma))^d),\\
u_N &\xrightarrow[N\to\infty]{} u, && \textit{in } L^{2}(]0,T[,(L^{2}(\Gamma))^d).
\end{aligned}$$

A similar result is valid in dimension $d=2$ and provides, in reality, convergence results in higher regularity spaces. We leave this more successful case to the conscientious reader.

Proof.

The first weak convergences claimed here are obtained directly by using the bounds (VII.9) as well as Theorem II.2.7 in the same way as during the study of the homogeneous Navier–Stokes equations with Dirichlet boundary conditions in Chapter V.

The strong convergence of $(u_N)_N$ in $L^2(]0,T[,(L^2(\Omega))^d)$ is a consequence of Theorem II.5.17. Indeed, from the Fourier estimate obtained in Proposition VII.1.3 and Proposition II.5.23, we get a bound of $(u_N)_N$ in the Nikolskii space $N_2^\sigma(]0,T[,(L^2(\Omega))^d)$ which is uniform with respect to N.

The only point necessary to detail here is the handling of the boundary terms. The first of these is treated using Theorem III.2.36 in the following way

$$\begin{aligned}
&\int_0^T \|((u_N+u_{\mathrm{ref}})\cdot\nu)^+u_N\|_{L^{4/3}(\Gamma)}^{4/3}\,dt\\
\leq &\int_0^T \|u_N\|_{L^{8/3}(\Gamma)}^{4/3}\|u_N+u_{\mathrm{ref}}\|_{L^{8/3}(\Gamma)}^{4/3}\,dt\\
\leq &C\int_0^T \Big(\|u_N\|_{L^{8/3}(\Gamma)}^{8/3}+\|u_N\|_{L^{8/3}(\Gamma)}^{4/3}\|u_{\mathrm{ref}}\|_{L^{8/3}(\Gamma)}^{4/3}\Big)\,dt\\
\leq &C\|u_{\mathrm{ref}}\|_{L^{8/3}(\Gamma)}^{8/3}+C\int_0^T \|u_N\|_{L^2}^{2/3}\|u_N\|_{H^1}^{2}\,dt\\
\leq &C+C\|u_N\|_{L^\infty(]0,T[,L^2(\Omega))}^{2/3}\|u_N\|_{L^2(]0,T[,H^1(\Omega))}^{2}.
\end{aligned}$$

Using (VII.9), this inequality shows a bound independent of N in the space $L^{4/3}(]0,T[,(L^{4/3}(\Gamma))^d)$ which is reflexive. This allows us to prove the existence, up to a subsequence, of a weak limit γ_1 of this term. The other boundary terms are linear and are treated in a straightforward way.

To conclude, it is necessary to justify the strong convergence of the trace of u_N in $L^2(]0,T[,(L^2(\Gamma))^d)$. This is obtained using Theorem III.2.36. Indeed, we have

$$\|u_N - u\|_{L^2(\Gamma)} \le C\|u_N - u\|_{L^2}^{1/2}\|u_N - u\|_{H^1}^{1/2},$$

from which we can deduce using the Hölder inequality that

$$\int_0^T \|u_N - u\|_{L^2(\Gamma)}^2\,dt \le C\left(\int_0^T \|u_N - u\|_{L^2}^2\,dt\right)^{1/2}\left(\int_0^T \|u_N - u\|_{H^1}^2\,dt\right)^{1/2}.$$

We have already obtained the strong convergence of u_N towards u in $L^2(]0,T[,(L^2(\Omega))^d)$ and hence the first term of the right-hand side of this inequality tends towards 0 when N tends towards infinity. Furthermore, the second factor remains bounded thanks to (VII.9). This indeed proves the strong convergence of the trace that we claimed above.

□

We now need to identify the weak limits w_1, w_2, and γ_1 exhibited above. To do this, as we have already seen, it is necessary to prove convergence in the spaces that are weaker than those in which we obtained these limits. More precisely, we argue by using the continuity of the product operator in suitable Lebesgue spaces (using the Hölder inequality) and the weak-strong limit property given in Proposition II.2.12. Thanks to convergences obtained above (especially the strong convergence of $(u_N)_n$ in $L^2(]0,T[,(L^2(\Omega))^d)$), we obtain

$$(u_N\cdot\nabla)u_N \xrightarrow[N\to\infty]{} (u\cdot\nabla)u, \qquad \text{weakly in } L^1(]0,T[,(L^1(\Omega))^d),$$

$$u_N\otimes u_N \xrightarrow[N\to\infty]{} u\otimes u, \qquad \text{weakly in } L^1(]0,T[,(L^{\frac{3}{2}}(\Omega))^{d\times d}),$$

$$\big((u_N+u_{\mathrm{ref}})\cdot\nu\big)^+u_N \xrightarrow[N\to\infty]{} \big((u+u_{\mathrm{ref}})\cdot\nu\big)^+u, \text{ weakly in } L^{\frac{4}{3}}(]0,T[,(L^1(\Gamma))^d).$$

To handle the boundary term, we use the fact that for any sequence $(f_N)_N$ which converges in $L^p(\Gamma)$ towards a function f we have

$$f_N^+ \xrightarrow[N\to\infty]{} f^+, \text{ in } L^p(\Gamma).$$

This is obtained by noting that the map $s\mapsto s^+$ is Lipschitz continuous on $\mathbb{R}$.

All these results allow us to identify w_1 with $(u\cdot\nabla)u$, w_2 with $u\otimes u$ and γ_1 with $\big((u+u_{\mathrm{ref}})\cdot\nu\big)^+u$, by uniqueness of the limit in the sense of distributions (see Proposition II.2.9). Passing to the limit in the weak formulation of the approximate problem is now standard and gives that u is indeed a solution of the initial problem.

Finally, by returning to the approach used in the proof of Proposition V.1.3, we show that u has a weak derivative with respect to time in $L^{\frac{4}{d}}(]0,T[,V')$ (because we know that $u\in L^\infty(]0,T[,H)\cap L^2(]0,T[,V))$ and that it also satisfies a weak formulation with test functions which depend on time. This proves the first part of Theorem VII.1.1

1.2.5 Uniqueness in the case $d = 2$

In this section, we show that the weak solution obtained above is unique in the two-dimensional case.

The proof is made as in Chapter V but considering the additional boundary terms. Therefore, let u_1 and u_2 be two solutions of (VII.6). Their difference, $u = u_1 - u_2$, satisfies

$$\begin{aligned}
\left\langle \frac{du}{dt}, \psi \right\rangle_{V',V} &+ \frac{1}{2}\int_\Omega \Big(\big((u\cdot\nabla)(u_1 + u_{\text{ref}})\big) + \big(((u_2 + u_{\text{ref}})\cdot\nabla)u\big) \Big) \cdot \psi \, dx \\
&- \frac{1}{2}\int_\Omega \Big(\big((u\cdot\nabla)\psi\big)\cdot(u_1 + u_{\text{ref}}) + \big(((u_2 + u_{\text{ref}})\cdot\nabla)\psi\big)\cdot u \Big)\, dx \\
&+ \frac{2}{\mathcal{R}e}\int_\Omega \mathrm{D}(u) : \mathrm{D}(\psi) dx + \frac{1}{2}\int_{\Gamma_2} (u\cdot\psi)\big((u_1 + u_{\text{ref}})\cdot\nu\big)^+ \, d\sigma \\
&+ \frac{1}{2}\int_{\Gamma_2} (u_2\cdot\psi)\left(\big((u_1 + u_{\text{ref}})\cdot\nu\big)^+ - \big((u_2 + u_{\text{ref}})\cdot\nu\big)^+\right) d\sigma \\
&+ \frac{1}{2}\int_{\Gamma_2} (u_{\text{ref}}\cdot\psi)(u\cdot\nu)\, d\sigma = 0.
\end{aligned}$$

It is allowed to take $\psi = u$ as a test function in the preceding system because in dimension 2 we have $du/dt \in L^2(]0,T[, V')$. By using Theorem II.5.12 and the fact that $u(0) = u_1(0) - u_2(0) = 0$, we obtain for all $t \in [0,T]$,

$$\begin{aligned}
&\frac{1}{2}\frac{d}{dt}\|u(t)\|_H^2 + \frac{1}{Re}\|\mathrm{D}(u)\|_{L^2}^2 \\
&= -\frac{1}{2}\int_\Omega \big((u\cdot\nabla)(u_1 + u_{\text{ref}})\big)\cdot u\, dx + \frac{1}{2}\int_\Omega \big((u\cdot\nabla)u\big)\cdot(u_1 + u_{\text{ref}})\, dx \\
&\quad - \frac{1}{2}\int_{\Gamma_2} |u|^2\big((u_1 + u_{\text{ref}})\cdot\nu\big)^+ \, d\sigma - \frac{1}{2}\int_{\Gamma_2} (u_{\text{ref}}\cdot u)(u\cdot\nu)\, d\sigma \\
&\quad - \frac{1}{2}\int_{\Gamma_2} (u_2\cdot\psi)\left(\big((u_1 + u_{\text{ref}})\cdot\nu\big)^+ - \big((u_2 + u_{\text{ref}})\cdot\nu\big)^+\right) d\sigma
\end{aligned}$$

By using that $s \mapsto s^+$ is Lipschitz continuous, the Hölder inequality, the Sobolev embeddings (Proposition III.2.35), and finally the sharp trace Theorem III.2.36 in dimension $d = 2$, we get

$$\begin{aligned}
&\frac{d}{dt}\|u(t)\|_{L^2}^2 + \frac{4}{\mathcal{R}e}\|\mathrm{D}(u)\|_{L^2}^2 \\
\leq&\|u\|_{L^4}^2\|\nabla u_1 + \nabla u_{\mathrm{ref}}\|_{L^2} + \|u\|_{L^4}\|\nabla u\|_{L^2}\|u_1 + u_{\mathrm{ref}}\|_{L^4} \\
&+ \|u\|_{L^3(\Gamma_2)}^2\|u_1 + u_{\mathrm{ref}}\|_{L^3(\Gamma_2)} + \|u\|_{L^3(\Gamma_2)}^2(\|u_2\|_{L^3(\Gamma_2)} + \|u_r\|_{L^3(\Gamma_2)}) \\
\leq&C\|u\|_{L^2}\|\nabla u\|_{L^2}\|u_1 + u_{\mathrm{ref}}\|_{H^1} \\
&+ C\|u\|_{L^2}^{1/2}\|\nabla u\|_{L^2}^{3/2}\|u_1 + u_{\mathrm{ref}}\|_{L^2}^{1/2}\|u_1 + u_{\mathrm{ref}}\|_{H^1}^{1/2} \\
&+ C\|u\|_{L^2}^{2/3}\|\nabla u\|_{L^2}^{4/3}\left(\|u_2\|_{L^2}^{1/3}\|u_2\|_{H^1}^{2/3} + \|u_{\mathrm{ref}}\|_{L^2}^{1/3}\|u_{\mathrm{ref}}\|_{H^1}^{2/3}\right).
\end{aligned}$$

We now need to use the Korn inequality (Lemma IV.7.6) to make the term $\|\mathrm{D}(u)\|_{L^2}$ appear on the right-hand side. By again using Young's inequality, we obtain

$$\frac{d}{dt}\|u(t)\|_{L^2}^2 + \frac{4}{\mathcal{R}e}\|\mathrm{D}(u)\|_{L^2}^2 \leq \frac{2}{\mathcal{R}e}\|\mathrm{D}(u)\|_{L^2}^2 + Cg(t)\|u\|_{L^2}^2,$$

which, of course, gives

$$\frac{d}{dt}\|u(t)\|_{L^2}^2 + \frac{2}{\mathcal{R}e}\|\mathrm{D}(u)\|_{L^2}^2 \leq Cg(t)\|u\|_{L^2}^2, \tag{VII.20}$$

the function $g(t)$ being given by

$$\begin{aligned}
g(t) =& 1 + \|u_1 + u_{\mathrm{ref}}\|_{H^1}^2 + \|u_1 + u_{\mathrm{ref}}\|_{L^2}^2\|u_1 + u_{\mathrm{ref}}\|_{H^1}^2 + \|u_2\|_{H^1}^2 \\
&+ \|u_{\mathrm{ref}}\|_{H^1}^2 + \|u_2\|_{L^2}\|u_2\|_{H^1}^2 + \|u_{\mathrm{ref}}\|_{L^2}\|u_{\mathrm{ref}}\|_{H^1}^2.
\end{aligned}$$

The reference flow u_{ref} is independent of time. Moreover, u_1 and u_2 are two solutions of the problem and thus belong to the space $L^\infty(]0,T[,H) \cap L^2(]0,T[,V)$, which shows that the real-valued function g above is integrable on $]0,T[$.

Gronwall's lemma allows us to deduce from inequality (VII.20) that the function u is identically zero on $]0,T[$. Therefore, $u_1 = u_2$ and the claim is proved.

1.2.6 Existence of the pressure

The existence and regularity of the pressure is demonstrated in a completely identical way as in the proof of the Leray theorem in Chapter V, that is, by using de Rham's theorem. We thus obtain that $v = u_{\mathrm{ref}} + u$ and the pressure p satisfies the Navier–Stokes equations in the sense of distributions on $]0,T[\times\Omega$.

2 Dirichlet boundary conditions through a penalty method

Over the last 30 years, starting from the pioneer works by Peskin [97, 98] in the framework of blood flow simulation inside a beating heart, there have been many attempts to build penalty methods in order to impose homogeneous Dirichlet boundary conditions for the velocity field on the boundary of an obstacle inside a given computational domain. The aim of these methods is to avoid the use of complex (possibly moving) unstructured meshes due to the geometry of the physical domain, when performing numerical simulations. Such an approach allows, for example, the use of spectral methods [74], or finite-difference methods [75] on Cartesian meshes, which are easier to set up and more effective in some cases. Similarly, the use of Cartesian meshes can be more suitable for parallel computations, which are fundamental in particular in the 3D case.

This is illustrated in Figure VII.2 which shows, at the top, a triangle mesh required when using a classic discretisation method and below, an uniform Cartesian mesh from a fictitious calculation domain in which the obstacle (grey in the figure) is considered via an additional penalty term in the equations. In this case, the computational domain is the large rectangle.

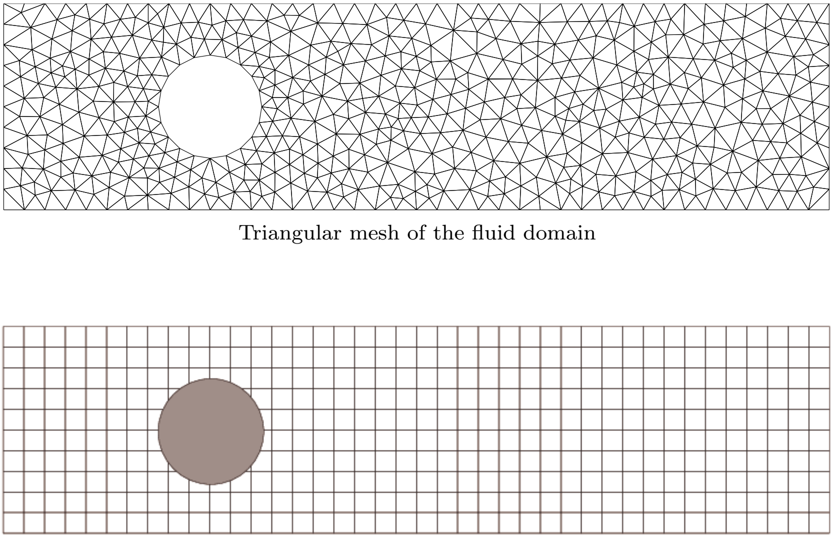
Studies have shown that the penalty term has to be extended over the entire volume of the obstacle to give a correct solution at high Reynolds number [13, 103]. In [34], it was suggested that this technique is able to provide a good approximation of the lift and drag coefficients by integration of the penalty term over the obstacle. This is fundamental for physical applications (e.g., in aeronautics). Similar ideas have also been successfully used in [9, 11, 31, 35, 36] to deal with fluid–porous or fluid–porous–solid systems.

Several researchers have used, for instance, this method to compute the flow of an incompressible fluid around an obstacle [9, 28, 29, 30, 75] or the motion of a rigid body in a viscous incompressible flow [20, 81].

To be more precise, in this section we consider the following penalty method

$$\begin{cases} \dfrac{\partial u^\varepsilon}{\partial t} - \dfrac{1}{\mathcal{R}e}\Delta u^\varepsilon + (u^\varepsilon \cdot \nabla)u^\varepsilon + \dfrac{1}{\varepsilon^2}\chi_\omega u^\varepsilon + \nabla p^\varepsilon = f, & \text{in } \Omega, \\ \operatorname{div} u^\varepsilon = 0, & \text{in } \Omega, \\ u^\varepsilon = 0, & \text{on } \partial\Omega, \end{cases} \tag{VII.21}$$

in which u^ε designates the penalised velocity field, Ω is the computational domain, ω the obstacle (supposed here to be at rest) which we wish to penalise, and χ_ω is the corresponding characteristic function (see Figure VII.5). Under these conditions $\mathcal{U} = \Omega \setminus \overline{\omega}$ represents the physical domain in which the fluid evolves. We assume that $\mathcal{U}$ is connected.



Triangular mesh of the fluid domain

Cartesian mesh of an artificial calculation domain and penalised obstacle

Fig. VII.2 Classic method and penalty method for calculating the flow around a cylinder

In $\mathcal{U}$, the equation considered is the usual Navier–Stokes equation; whereas inside the obstacle, since the coefficient ε is small, we expect that the predominant term in the equation will be the term $(1/\varepsilon^2)\chi_\omega u^\varepsilon$, and so that the equation considered will lead to

$$\chi_\omega u^\varepsilon \approx \varepsilon^2,$$

so that, asymptotically, we can think that u^ε will tend to 0 as $\varepsilon \to 0$. Finally, we hope that, in the fluid domain $\mathcal{U}$, u^ε is close to the physical solution which is of course not defined in the obstacle.

From the theoretical point of view, an error estimate for this particular method was first established in [10]. These estimates did not agree with experimental results. Indeed, it was shown in this reference that the error was at least of order $\varepsilon^{1/2}$ in the fluid, whereas the numerical results seemed to show that the error is of order ε. In this section, we prove (following [35, 37]) that the error generated by the method is indeed effectively of order ε.

To achieve that objective, and this is the second goal of this section, we prove that the presence of the penalty term leads to the appearance of a boundary layer inside the obstacle, but not in the fluid region. The careful study of this boundary layer is the key point of the proof of the sharp error estimate. Boundary layer theory is fundamental in PDE analysis and, in particular, for problems in fluid mechanics. Our objective is not to propose here a general theory, but only to illustrate the concept of *boundary layer* with a concrete example issued from a numerical problem.

We propose to establish the existence of the boundary layer by carrying out an asymptotic expansion of the solution, in the spirit of the WKB method (Wentzel–Kramers–Brillouin). Note that such an analysis requires knowing in advance that the solution of the limit problem is regular enough. Furthermore, in the present case, we obtain a global error estimate rather than one which concerns only the fluid domain. This requires us to describe the error in Sobolev spaces with small indices because the penalised solution cannot be smooth in the whole computational domain Ω.

In order to show that the results obtained are optimal in some sense, as well as to illustrate the concept of boundary layers and the technique employed in the sequel, we first describe a simple example of a 1D problem at the very beginning of this section. For such a toy system, we show through an explicit calculation that the error between the penalised solution and the exact solution is exactly of order ε in the L^2-norm.

2.1 A simple example of a boundary layer

Before we start the study of the problem described above, we illustrate, using an example, the behavior of an equation describing a boundary layer when one of the equation's parameters tends towards 0.

We are attempting to solve the following one-dimensional problem

$$\begin{cases} -u''(x) = 1, & \text{in }]0,1[, \\ \quad u(0) = u(1) = 0. \end{cases} \tag{VII.22}$$

A straightforward computation shows that the exact solution is given by

$$u_0(x) = \frac{1}{2}x(1-x).$$

Let us assume that we wish to solve this problem using a penalty method similar to that described above. For example, being given a small parameter ε, we replace Problem (VII.22) by the following penalised problem:

$$\begin{cases} -u^{\varepsilon\prime\prime}(x) + \dfrac{1}{\varepsilon^2}\chi_{]-1,0[}u^\varepsilon = 1, & \text{in }]-1,1[, \\ \quad u^\varepsilon(-1) = u^\varepsilon(1) = 0. \end{cases} \tag{VII.23}$$

If we make an analogy with the model described in the introduction, the computational domain is $\Omega =]-1,1[$, the obstacle is $\omega =]-1,0[$, and the "physical" domain is $\mathcal{U} = \Omega \setminus \overline{\omega}$.

We now proceed to calculate the exact solution of the approximate problem and see in what way it converges towards the solution u_0 in the initial problem when ε tends towards zero. To solve (VII.23), we start by solving the problem on $]-1,0[$ and, independently, on $]0,1[$. For now we do not have a boundary condition at $x=0$ and therefore we have an indeterminate term for each part of the solution. After straightforward computations, we obtain

$$u^\varepsilon(x) = \begin{cases} \dfrac{1}{2}(1-x)(x+\beta), & \text{in }]0,1[, \\ A_0 e^{-\frac{x}{\varepsilon}} + B_0 e^{\frac{x}{\varepsilon}} - \varepsilon^2, & \text{in }]-1,0[, \end{cases}$$

where β, A_0, and B_0 are constants to be determined. The boundary condition $u^\varepsilon(-1)=0$ leads to

$$A_0 e^{1/\varepsilon} + B_0 e^{-1/\varepsilon} = \varepsilon^2. \tag{VII.24}$$

It is now necessary to write the conditions connecting the two parts of the solution. We know that the solution of (VII.23) in the variational sense is defined a priori in $H_0^1(]-1,1[)$ but, if we write the equation in the following way,

$$u^{\varepsilon\prime\prime} = -1 + \frac{1}{\varepsilon^2}\chi_{]-1,0[}u^\varepsilon \in L^2(]-1,1[),$$

we see that in reality the solution belongs to $H^2(]-1,1[)$. This implies that the function u^ε and its derivative $u^{\varepsilon\prime}$ are continuous on $[-1,1]$. This gives us the two compatibility conditions between the two parts of the solution obtained above. It hence follows that

$$\frac{1}{2}\beta = A_0 + B_0 - \varepsilon^2, \tag{VII.25}$$

$$\frac{1}{2}(1-\beta) = -\frac{1}{\varepsilon}A_0 + \frac{1}{\varepsilon}B_0. \tag{VII.26}$$

Equations (VII.24) through (VII.26) give us the expressions for the constants β, A_0, and B_0 as a function of ε. After some tedious calculations, we find

$$\begin{cases} A_0 = \dfrac{\varepsilon^2(1+\varepsilon) - (\varepsilon^3 + \frac{\varepsilon}{2})e^{-\frac{1}{\varepsilon}}}{(1+\varepsilon)e^{\frac{1}{\varepsilon}} + (1-\varepsilon)e^{-\frac{1}{\varepsilon}}}, \\ B_0 = \dfrac{\varepsilon^2(1-\varepsilon) + (\varepsilon^3 + \frac{\varepsilon}{2})e^{\frac{1}{\varepsilon}}}{(1+\varepsilon)e^{\frac{1}{\varepsilon}} + (1-\varepsilon)e^{-\frac{1}{\varepsilon}}}, \\ \beta = \dfrac{4\varepsilon^2 - (2\varepsilon^2 - \varepsilon)e^{\frac{1}{\varepsilon}} - (2\varepsilon^2+\varepsilon)e^{-\frac{1}{\varepsilon}}}{(1+\varepsilon)e^{\frac{1}{\varepsilon}} + (1-\varepsilon)e^{-\frac{1}{\varepsilon}}}. \end{cases}$$

These exact expressions are not of much interest in themselves. Let us look at the behavior of the coefficients as a function of ε by remembering that, unlike for β, the constants A_0 and B_0 appear in front of exponential terms in the expression for u^ε. Asymptotically, we have the following equivalents

$$\begin{aligned}A_0 &\approx \varepsilon^2 e^{-1/\varepsilon},\\ B_0 &\approx \frac{\varepsilon}{2},\\ \beta &\approx \varepsilon,\end{aligned}$$

Hence, the penalised solution u^ε can be expressed, up to first-order, as

$$u^\varepsilon(x) \approx \begin{cases} \varepsilon^2 e^{-\frac{1+x}{\varepsilon}} + \frac{\varepsilon}{2} e^{\frac{x}{\varepsilon}} - \varepsilon^2, & \text{in }]-1,0[,\\ \frac{1}{2}(1-x)(x+\varepsilon) = u_0(x) + \frac{\varepsilon}{2}(1-x), & \text{in }]0,1[. \end{cases}$$

We see that this method approaches the solution u_0 of the initial problem at the order ε in the original domain $]0,1[$ and this is true for all Sobolev norms.

Let us now consider the behavior of the approximate solution in "the obstacle", that is, in $]-1,0[$ (see Figures VII.3 and VII.4). We observe that it tends towards zero, but only in the L^∞ norm. Indeed, if we look at the convergence of the first derivative, we find

$$u^{\varepsilon\prime}(x) \approx -\varepsilon e^{-(1+x)/\varepsilon} + \frac{1}{2} e^{x/\varepsilon},$$

which tends towards $1/2$, for example, at $x = 0$. This is satisfactory because the value of the derivative of the exact solution u_0 at $x = 0$ is exactly $1/2$. Nevertheless, this damages the convergence of u^ε towards 0 in Sobolev spaces of higher exponent in $]-1,0[$. We obtain, in a more precise way, the estimates

$$\begin{aligned}\|u^\varepsilon\|_{L^2(]-1,0[)} &\sim C\varepsilon^{3/2},\\ \|u^{\varepsilon\prime}\|_{L^2(]-1,0[)} &\sim C\sqrt{\varepsilon},\\ \|u^{\varepsilon\prime\prime}\|_{L^2(]-1,0[)} &\sim C\frac{1}{\sqrt{\varepsilon}}.\end{aligned}$$

This phenomenon is called a *boundary layer* because the derivatives of the approximate solution have relatively high values in a transition zone of width ε in the neighborhood of the boundary of the penalty zone. It is very important to note that the boundary layer only appears inside the penalised domain and does not perturb the approximation inside the original domain $]0,1[$.

We observe and describe the same phenomenon below for the problem of the flow of an incompressible viscous fluid around an obstacle. We also show in this case that the boundary layer appears only inside the obstacle and not in the fluid domain. Furthermore, in the example above we observe that the boundary layer is of size ε and that the solution has a behavior in the form $\varepsilon e^{-|x|/\varepsilon}$ in this boundary layer.

This is the reason why, in the study below, we look for an asymptotic expansion of the solution with terms in the form

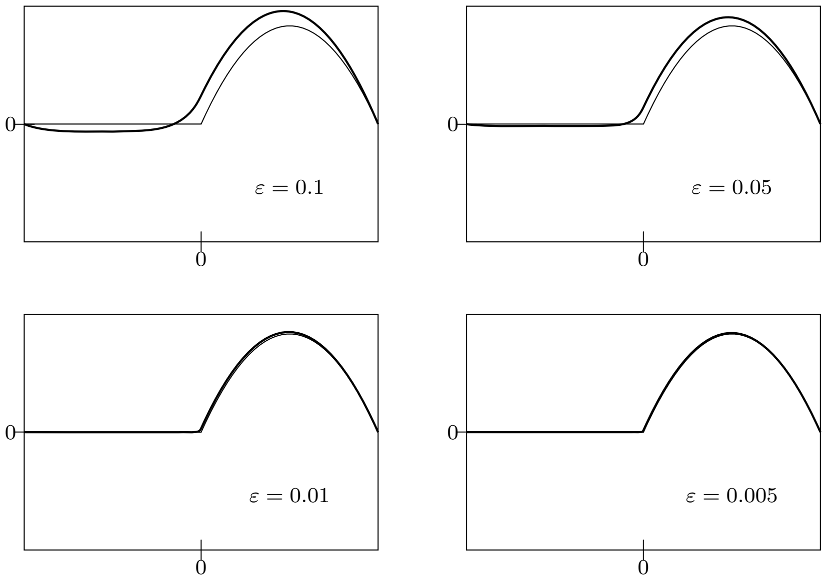


Fig. VII.3 General appearance of the exact solution (thin line) and the penalised solution (thick line) for $\varepsilon = 0.1, 0.05, 0.01, 0.005$

$$W\left(t, x, \frac{d(x, \partial\omega)}{\varepsilon}\right),$$

where the last variable (referred to as z in the sequel) aims to describe the solution inside the boundary layer.

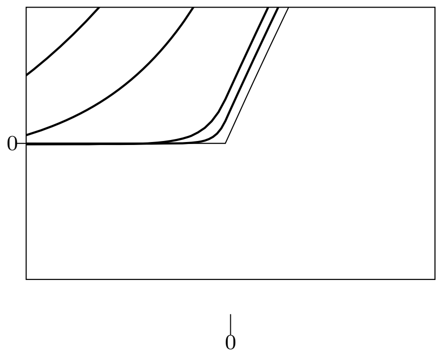


Fig. VII.4 Zoom in at the point $x = 0$: the exact solution (thin line) and penalised solution (thick line) for $\varepsilon = 0.1, 0.05, 0.01, 0.005$

Remark VII.2.1. The *artificial* boundary conditions chosen at $x = -1$, are hardly important in this problem. Hence if we replace the Dirichlet boundary condition by a Neumann type of boundary condition

$$u^{\varepsilon\prime}(-1) = 0,$$

then the general formula for the solution is unchanged but the relations satisfied by A_0, B_0, and β are now (VII.25), (VII.26), and a new condition in $x = -1$ which replaces (VII.24) given by

$$-\frac{A_0}{\varepsilon}e^{1/\varepsilon} + \frac{B_0}{\varepsilon}e^{-1/\varepsilon} = 0.$$

We can then show that we have the following asymptotics

$$A_0 \approx \varepsilon e^{-2/\varepsilon},\ B_0 \approx \frac{\varepsilon}{2},\ \text{and}\, \beta \approx \varepsilon.$$

and therefore on $[-1, 0]$, we have

$$u^\varepsilon(x) \approx \varepsilon^2 e^{-(x+2)/\varepsilon} + \frac{\varepsilon}{2}e^{x/\varepsilon} - \varepsilon^2.$$

The general behavior of the solution is therefore the same as the one observed above.

2.2 Statement of the main result

We can now state the main result concerning the penalty method for the Navier–Stokes equations around an obstacle. To simplify the calculations we assume that the Reynolds number is equal to 1. This is legitimate because we are not interested here in a viscous boundary layer (that one can observe in the inviscid limit $\mathcal{R}e \to 0$) but in the boundary layer induced by the penalty method. Furthermore, we show that this boundary layer only develops inside the penalised obstacle and does not therefore perturb the solution in the viscous boundary layer which develops inside the fluid domain with a characteristic size of $\sqrt{\mathcal{R}e}$.

Throughout this section Ω designates a smooth bounded domain (at least of class $\mathcal{C}^{5,1}$) of $\mathbb{R}^d$, $d = 2$, or 3 and ω is a bounded subdomain of Ω, with same regularity as Ω, such that $\overline{\omega} \subset \Omega$. We denote the *fluid* domain as $\mathcal{U} = \Omega\backslash\overline{\omega}$. In this context, we first recall the existence and regularity results for the solution of the Navier–Stokes equations (Theorems V.2.1 and V.2.10 and Corollary V.2.11).

Proposition VII.2.1. *Let* $v_0 \in (H^5(\mathcal{U}))^d \cap V$ *and* $f \in \mathcal{C}^2([0, +\infty[, (H^4(\mathcal{U}))^d)$ *satisfying the compatibility conditions* (V.54)*. Then, there exists a final time*

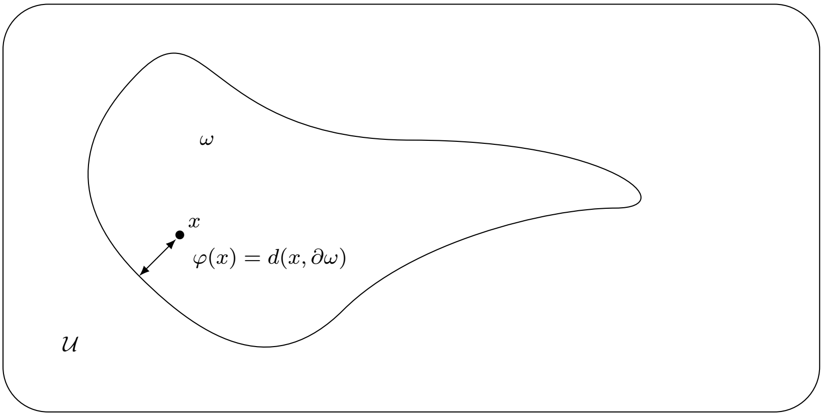


Fig. VII.5 The calculation domain Ω containing the fluid domain $\mathcal{U}$ and the obstacle ω

$T^* > 0$ and there exists a unique pair (V^0, p^0) defined on $[0, T^*[\times\mathcal{U}$ satisfying

$$\begin{cases} \dfrac{\partial V^0}{\partial t} - \Delta V^0 + (V^0 \cdot \nabla)V^0 + \nabla p^0 = f, & \text{in } \mathcal{U}, \\ \operatorname{div} V^0 = 0, & \text{in } \mathcal{U}, \\ V^0 = 0, & \text{on } \partial\mathcal{U}, \\ V^0(0) = v_0, \end{cases}$$

and such that for all $T < T^*$, we have

$$\begin{cases} V^0 \in \mathcal{C}^0([0,T], (H^5(\mathcal{U}))^d) \cap L^2(]0,T[, (H^6(\mathcal{U}))^d), \\ \dfrac{\partial V^0}{\partial t} \in \mathcal{C}^0([0,T], (H^3(\mathcal{U}))^d) \cap L^2(]0,T[, (H^4(\mathcal{U}))^d), \\ \dfrac{\partial^2 V^0}{\partial t^2} \in \mathcal{C}^0([0,T], (H^1(\mathcal{U}))^d) \cap L^2(]0,T[, (H^2(\Omega))^d), \\ p^0 \in \mathcal{C}^0([0,T], H^4(\mathcal{U})) \cap L^2(]0,T[, H^5(\mathcal{U})), \\ \dfrac{\partial p^0}{\partial t} \in \mathcal{C}^0([0,T], H^2(\mathcal{U})) \cap L^2(]0,T[, H^3(\mathcal{U})). \end{cases} \tag{VII.27}$$

Moreover, in the case of the dimension $d = 2$, we have $T^* = +\infty$.

We therefore propose to replace the numerical computation of V^0 in the fluid domain $\mathcal{U}$, by that of u^ε, a solution of (VII.21) in the domain Ω. We will then prove an error estimate which gives the precise order of magnitude of the error due to this approach. In a precise way, we have the following result.

Theorem VII.2.2. *Let v_0, f, V^0 and T^* be as in Proposition VII.2.1. We define a perturbation of the initial data of the form*

$$u_0^\varepsilon(x) = \begin{cases} v_0(x) + r^\varepsilon(x), & \textit{for } x \in \mathcal{U}, \\ r^\varepsilon(x), & \textit{for } x \in \omega, \end{cases}$$

with $\operatorname{div} r^\varepsilon = 0$ *and* $\|r^\varepsilon\|_{L^2(\Omega)} \leq K\varepsilon$. *Then, there exists* $(u^\varepsilon, \pi^\varepsilon)$, *a weak solution of the penalised problem*

$$\begin{cases} \dfrac{\partial u^\varepsilon}{\partial t} - \Delta u^\varepsilon + (u^\varepsilon \cdot \nabla) u^\varepsilon + \nabla \pi^\varepsilon + \dfrac{1}{\varepsilon^2} \chi_\omega u^\varepsilon = f, & \textit{in } \Omega, \\ \operatorname{div} u^\varepsilon = 0, & \textit{in } \Omega, \\ u^\varepsilon = 0, & \textit{on } \partial\Omega, \\ u^\varepsilon(0) = u_0^\varepsilon, \end{cases}$$

such that for all $T < T^*$, *there exists a constant* C *independent of* ε *such that*

$$\|u^\varepsilon\|_{L^\infty(]0,T[,(L^2(\Omega))^d)} + \|u^\varepsilon\|_{L^2(]0,T[,(H_0^1(\Omega))^d)} \leq C.$$

Moreover, u^ε *is an approximation of order 1 of* V^0 *in* $\mathcal{U}$ *in the energy norms. This means that for all* $T < T^*$ *there exists* $C > 0$ *such that*

$$\|u^\varepsilon - V^0\|_{L^\infty(]0,T[,(L^2(\mathcal{U}))^d)} + \|u^\varepsilon - V^0\|_{L^2(]0,T[,(H^1(\mathcal{U}))^d)} \leq C\varepsilon.$$

Finally, $(u^\varepsilon)_\varepsilon$ *tends to zero in* ω *in the following sense*

$$\begin{aligned} \|u^\varepsilon\|_{L^\infty(]0,T[,(L^2(\omega))^d)} &\leq C\varepsilon, \\ \|u^\varepsilon\|_{L^2(]0,T[,(H^1(\omega))^d)} &\leq C\sqrt{\varepsilon}. \end{aligned}$$

The existence of the solution u^ε, for a given $\varepsilon > 0$, involves the same techniques as introduced in Chapter V. In dimension 2, this solution is unique. The difficult point is to establish the error estimate. To prove this result, we carry out an asymptotic expansion of the solution u^ε, which allows us to describe the boundary layer present inside the obstacle ω.

We assumed that ω is sufficiently regular, therefore we know (see Section 3.2 of Chapter III) that there exists a neighborhood of $\partial\omega$ in ω defined by

$$\omega_1 = \{x \in \omega,\ d(x, \partial\omega) < \eta\},$$

in which the function $x \mapsto \delta(x) = d(x, \partial\omega)$ is smooth. Therefore, in all that follows, we fix a function φ which is regular on $\overline{\omega}$ and which coincides with δ on ω_1. We note that on $\partial\omega$, we have $\nabla\varphi = \nabla\delta = \nu$, where ν is the *inward* unit normal of ω and is also the *outward* unit normal of $\mathcal{U}$ on $\partial\omega$ because ω and $\mathcal{U}$ share a part of their boundary.

Furthermore, for any $x \in \omega_1$, we have $\nabla\varphi(x) = \nu(P_0(x))$, where $P_0(x)$ is the orthogonal projection of x on $\partial\omega$; see (III.66). In particular, in ω_1, we have $|\nabla\varphi| = 1$.

We can now describe the boundary layer as follows.

Theorem VII.2.3. *Let v_0, f, V^0, and T^* be as in Proposition VII.2.1. There exist two functions V^1 and V^2 defined in $[0,T^*[\times\mathcal{U}$, and two functions W^1 and W^2 defined on $[0,T^*[\times\omega\times\mathbb{R}^+$, which are sufficiently smooth, such that*

$$u^\varepsilon(t,x) = \begin{cases} V^0(t,x)+\varepsilon V^1(t,x)+\varepsilon^2 V^2(t,x)+\varepsilon v^r_\varepsilon(t,x), & \text{for } x\in\mathcal{U}, \\ \\ \varepsilon W^1\left(t,x,\dfrac{\varphi(x)}{\varepsilon}\right)+\varepsilon^2 W^2\left(t,x,\dfrac{\varphi(x)}{\varepsilon}\right)+\varepsilon w^r_\varepsilon(t,x), & \text{for } x\in\omega, \end{cases}$$

where v^r_ε and w^r_ε are bounded independently of ε, in $L^\infty(]0,T[,(L^2)^d)\cap L^2(]0,T[,(H^1)^d)$, for all $T<T^*$.

The error estimate given by Theorem VII.2.2 is now a direct consequence of Theorem VII.2.3, which is in fact more precise because it provides a complete description of the first term of the boundary layer in ω.

Looking for an asymptotic expansion in the multiscale form above (i.e., introducing the variable φ/ε) is known as the WKB method. We first find the equations that the profiles need to satisfy. After that, we prove the existence and regularity of the various terms of the asymptotic expansion obtained. Finally, in the last section, we estimate the size of the remainder and prove the convergence.

Remark VII.2.2. In view of the definition of the functions V^1, V^2 and W^1, W^2 which follow, we could expect that the remainder terms $\varepsilon v^r_\varepsilon$ and $\varepsilon w^r_\varepsilon$ would have a size of at least ε^2. As usually observed, the WKB method of decomposition into profiles does not necessarily allow us to obtain this result. As in [32] or [38], for example, it is absolutely necessary to write the formal expansion up to second-order terms, in order to prove that the remainders are of order ε.

2.3 Formal asymptotic expansion

To construct the asymptotic expansion we proceed using sufficient conditions. This expansion depends in particular on the choice of φ and we are not concerned about its uniqueness away from the boundary layer. The expansion has the simple goal of constructing an explicit approximate solution of the problem at a sufficiently high order.

Therefore, let u^ε be a solution of

$$\begin{cases} \dfrac{\partial u^\varepsilon}{\partial t}-\Delta u^\varepsilon+(u^\varepsilon\cdot\nabla)u^\varepsilon+\nabla\pi^\varepsilon+\dfrac{1}{\varepsilon^2}\chi_\omega u^\varepsilon=f, & \text{in } \Omega, \\ \operatorname{div} u^\varepsilon=0, & \text{in } \Omega, \\ u^\varepsilon=0, & \text{on } \partial\Omega, \\ u^\varepsilon(0,x)=u^\varepsilon_0(x). \end{cases} \tag{VII.28}$$

The first thing to do is to distinguish the behavior of the solution inside the obstacle ω from the one in the fluid domain $\mathcal{U}$. To do this, we denote the restrictions of u^ε and π^ε on $\mathcal{U} = \Omega \setminus \bar{\omega}$ as v^ε and p^ε, respectively, and the restriction of u^ε and π^ε on ω as w^ε and q^ε, respectively. The global problem (VII.28) is then equivalent to the following system on the functions $v^\varepsilon, w^\varepsilon, p^\varepsilon$, and q^ε

$$\left\{\begin{array}{lll}
\dfrac{\partial w^\varepsilon}{\partial t} - \Delta w^\varepsilon + (w^\varepsilon \cdot \nabla) w^\varepsilon + \nabla q^\varepsilon + \dfrac{1}{\varepsilon^2} w^\varepsilon = 0, & \text{in } \omega, & (1)\\
\operatorname{div} w^\varepsilon = 0, & \text{in } \omega, & (2)\\
\dfrac{\partial v^\varepsilon}{\partial t} - \Delta v^\varepsilon + (v^\varepsilon \cdot \nabla) v^\varepsilon + \nabla p^\varepsilon = f, & \text{in } \mathcal{U}, & (3)\\
\operatorname{div} v^\varepsilon = 0, & \text{in } \mathcal{U}, & (4)\\
v^\varepsilon = w^\varepsilon, & \text{on } \partial\omega, & (5)\\
-\dfrac{\partial v^\varepsilon}{\partial \nu} + p^\varepsilon \nu = -\dfrac{\partial w^\varepsilon}{\partial \nu} + q^\varepsilon \nu, & \text{on } \partial\omega, & (6)\\
v^\varepsilon = 0, & \text{on } \partial\Omega. & (7)
\end{array}\right. \qquad \text{(VII.29)}$$

Remark VII.2.3. Boundary conditions (5) and (6) in the preceding system are the natural transmission conditions which ensure continuity in the sense of the traces of u^ε and of the normal fluxes $(\pi^\varepsilon - \nabla u^\varepsilon).\nu$ across $\partial\omega$ and hence the existence of a solution u^ε which belongs to $(H^2(\Omega))^d$. For example, to justify the transmission condition on the fluxes, we can proceed as follows.

We take a divergence-free test function $\psi \in (\mathcal{D}(\Omega))^d$ and write the weak formulation of (VII.28), using the fact that f is extended by 0 on ω,

$$\frac{d}{dt}\int_\Omega u^\varepsilon \cdot \psi\, dx + \int_\Omega \nabla u^\varepsilon : \nabla \psi\, dx + \int_\Omega \big((u^\varepsilon \cdot \nabla) u^\varepsilon\big) \cdot \psi\, dx + \frac{1}{\varepsilon^2}\int_\omega u^\varepsilon \cdot \psi\, dx$$
$$= \int_{\mathcal{U}} f \cdot \psi\, dx.$$

Then, we multiply the equation inside the obstacle (1) by $\psi_{|\omega}$ which is certainly not zero on $\partial\omega$ and which, since ν is the *inward* normal to ω, gives through integration by parts

$$\frac{d}{dt}\int_\omega w^\varepsilon \cdot \psi\, dx + \int_\omega \nabla w^\varepsilon : \nabla \psi\, dx + \int_{\partial\omega} \left(+ \frac{\partial w^\varepsilon}{\partial \nu} - q^\varepsilon \nu \right) \cdot \psi\, d\sigma$$
$$+ \int_\omega \big((w^\varepsilon \cdot \nabla) w^\varepsilon\big) \cdot \psi\, dx + \frac{1}{\varepsilon^2}\int_\omega w^\varepsilon \cdot \psi\, dx = 0,$$

then we multiply the equation in the fluid (3) by $\psi_{|\mathcal{U}}$. This gives

$$\frac{d}{dt}\int_{\mathcal{U}} v^\varepsilon \cdot \psi \, dx + \int_{\mathcal{U}} \nabla v^\varepsilon : \nabla \psi \, dx + \int_{\partial\omega} \left(-\frac{\partial v^\varepsilon}{\partial \nu} + p^\varepsilon \nu \right) \cdot \psi \, d\sigma$$
$$+ \int_{\mathcal{U}} \left((v^\varepsilon \cdot \nabla) v^\varepsilon \right) \cdot \psi \, dx = \int_{\mathcal{U}} f \cdot \psi \, dx.$$

We now add together the last two equations and compare the result to the first one, recalling that v^ε and w^ε are restrictions of u^ε of $\mathcal{U}$ and ω, respectively. It then remains

$$\int_{\partial\omega} \left(+\frac{\partial w^\varepsilon}{\partial \nu} - q^\varepsilon \nu \right) \cdot \psi \, d\sigma + \int_{\partial\omega} \left(-\frac{\partial v^\varepsilon}{\partial \nu} + p^\varepsilon \nu \right) \cdot \psi \, d\sigma = 0.$$

This relation must be satisfied by all the test functions ψ which proves Equation (6).

Let us look for the asymptotic expansion of v^ε, w^ε, p^ε, and q^ε in the following form (this is called an *ansatz*)

$$\begin{cases} v^\varepsilon(t,x) = V^0(t,x) + \varepsilon V^1(t,x) + \dots, \\ p^\varepsilon(t,x) = \dfrac{1}{\varepsilon^2} p^{-2}(t,x) + \dfrac{1}{\varepsilon} p^{-1}(t,x) + p^0(t,x) + \dots, \\ w^\varepsilon(t,x) = W^0\left(t, x, \dfrac{\varphi(x)}{\varepsilon}\right) + \varepsilon W^1\left(t, x, \dfrac{\varphi(x)}{\varepsilon}\right) + \dots, \\ q^\varepsilon(t,x) = \dfrac{1}{\varepsilon^2} q^{-2}\left(t, x, \dfrac{\varphi(x)}{\varepsilon}\right) + \dfrac{1}{\varepsilon} q^{-1}\left(t, x, \dfrac{\varphi(x)}{\varepsilon}\right) + q^0\left(t, x, \dfrac{\varphi(x)}{\varepsilon}\right) \dots. \end{cases} \tag{VII.30}$$

Furthermore, as we saw in the 1D example considered at the beginning of this section, it is reasonable to look for the various profile terms inside the obstacle in the form:

$$W^i(t,x,z) = \overline{W}^i(t,x) + \widetilde{W}^i(t,x,z),$$
$$q^i(t,x,z) = \overline{q}^i(t,x) + \widetilde{q}^i(t,x,z),$$

where $\overline{W}^i$ and $\overline{q}^i$ are the *interior* terms of the profile, that is, those which do not contribute to the boundary layer, and in contrast $\widetilde{W}^i$ and $\widetilde{q}^i$ describe the behavior inside the boundary layer. This precisely means that, at a fixed x in ω, we should have $\lim_{\varepsilon\to 0} \widetilde{W}^i\,(t, x, \varphi(x)/\varepsilon) = 0$. By taking inspiration again from the 1D example of Section 2.1, where we obtained boundary layer profiles of the form $g(x)e^{-|x|/\varepsilon}$, we expect that the decrease with respect to the scalar variable z of the profiles $\widetilde{W}^i$ and $\widetilde{q}^i$ is very rapid.

Hence, below, we will impose the condition that $\widetilde{W}^i$ and $\widetilde{q}^i$ as well as their derivatives with respect to z, tend towards 0 when z tends towards infinity; that is, for any i and $k \geq 0$,

$$\partial_z^k \widetilde{W}^i \xrightarrow[z\to+\infty]{} 0 \text{ and } \partial_z^k \widetilde{q}^i \xrightarrow[z\to+\infty]{} 0, \quad \forall (t,x) \in [0, T^*[\times\omega. \tag{VII.31}$$

At the end, we show that we are indeed able to construct profiles satisfying this hypothesis and that the decrease in z is exponential as we would expect.

2.3.1 Determining the profile equations

Henceforth, for the functions $W(t,x,z)$, the standard notation for differential operators ∇, div, Δ represents the operators acting only on the original space variables x. We denote the derivatives with respect to the boundary layer variable z as W_z, W_{zz}, and so on.

Thus, for any vector field $(t,x,z) \mapsto W(t,x,z)$, the function defined by $(t,x) \mapsto w(t,x) = W\left(t,x,\varphi(x)/\varepsilon\right)$ satisfies the formulas

$$\begin{cases} \nabla w = \nabla W + \dfrac{1}{\varepsilon} W_z \otimes \nabla\varphi, \\ \operatorname{div} w = \operatorname{div} W + \dfrac{1}{\varepsilon} W_z \cdot \nabla\varphi, \\ \Delta w = \Delta W + \dfrac{2}{\varepsilon}\nabla W_z . \nabla\varphi + \dfrac{1}{\varepsilon}(\Delta\varphi) W_z + \dfrac{1}{\varepsilon^2}|\nabla\varphi|^2 W_{zz}, \end{cases} \tag{VII.32}$$

and for any scalar function $(t,x,z) \mapsto \Pi(t,x,z)$, the function $(t,x) \mapsto \pi(t,x) = \Pi\left(t,x,\varphi(x)/\varepsilon\right)$ satisfies

$$\nabla\pi = \nabla\Pi + \frac{1}{\varepsilon}\Pi_z \nabla\varphi.$$

Finally, we remember that in ω_1 the function φ satisfies $|\nabla\varphi| = 1$, and that at the boundary $\partial\omega$ we have $\nabla\varphi = \nu$, the outward normal to the fluid domain $\mathcal{U}$, which is also the inward normal to the obstacle ω.

The general principle is now as follows: we introduce the formal expressions (VII.30) in the equations (VII.29), which we have written as a function of the derivatives with respect to x and z; then we formally identify the terms corresponding to each power of ε, even in the boundary conditions.

2.3.1.1 Asymptotic expansion of Equation (1) of (VII.29)

- Order ε^{-2}:

$$-W^0_{zz} + W^0 + q_z^{-1}\nabla\varphi = 0,$$

  and hence, if we use hypothesis (VII.31) by taking the limit of this equation when z tends towards $+\infty$, it follows

$$\overline{W}^0 = 0, \quad \text{in } \omega, \tag{VII.33}$$

  and by difference, we obtain

$$-\widetilde{W}^0_{zz} + \widetilde{W}^0 + q_z^{-1}\nabla\varphi = 0, \quad \text{in } \omega \times \mathbb{R}^+. \tag{VII.34}$$

- Order ε^{-1}:

$$-W^1_{zz} + W^1 - 2\nabla W^0_z.\nabla\varphi - \Delta\varphi W^0_z + (W^0 \cdot \nabla\varphi)W^0_z + \nabla q^{-1} + q^0_z\nabla\varphi = 0;$$

hence when z tends towards infinity, by again using hypothesis (VII.31), we obtain

$$\overline{W^1} + \nabla\overline{q^{-1}} = 0, \quad \text{in } \omega, \tag{VII.35}$$

and through the difference with the equation above

$$\begin{aligned} &-\widetilde{W}^1_{zz} + \widetilde{W}^1 - 2(\nabla\varphi \cdot \nabla\widetilde{W}^0_z) - \Delta\varphi\widetilde{W}^0_z \\ &+ (W^0 \cdot \nabla\varphi)\widetilde{W}^0_z + \nabla\widetilde{q^{-1}} + \widetilde{q}^0_z\nabla\varphi = 0, \quad \text{in } \omega \times \mathbb{R}^+. \end{aligned} \tag{VII.36}$$

- Order ε^0:

$$\begin{aligned} &\frac{\partial W^0}{\partial t} - W^2_{zz} - 2\nabla W^1_z.\nabla\varphi - \Delta\varphi W^1_z - \Delta W^0 \\ &+ (W^0 \cdot \nabla)W^0 + (W^1 \cdot \nabla\varphi)W^0_z + (W^0 \cdot \nabla\varphi)W^1_z \\ &+ \nabla q^0 + q^1_z\nabla\varphi + W^2 = 0, \quad \text{in } \omega \times \mathbb{R}^+. \end{aligned}$$

Hence, by again making z tend towards $+\infty$, we obtain

$$\frac{\partial\overline{W}^0}{\partial t} - \Delta\overline{W}^0 + (\overline{W}^0 \cdot \nabla)\overline{W}^0 + \nabla\overline{q^0} + \overline{W}^2 = 0, \quad \text{in } \omega, \tag{VII.37}$$

and once again from the difference this gives

$$\begin{aligned} &\frac{\partial\widetilde{W}^0}{\partial t} - \widetilde{W}^2_{zz} - 2\nabla\widetilde{W}^1_z.\nabla\varphi - \Delta\varphi\widetilde{W}^1_z - \Delta\widetilde{W}^0 \\ &+ (\widetilde{W}^0 \cdot \nabla)\overline{W}^0 + (W^0 \cdot \nabla)\widetilde{W}^0 + (W^1 \cdot \nabla\varphi)\widetilde{W}^0_z + (W^0 \cdot \nabla\varphi)\widetilde{W}^1_z \\ &+ \nabla\widetilde{q}^0 + \widetilde{q}^1_z\nabla\varphi + \widetilde{W}^2 = 0, \quad \text{in } \omega \times \mathbb{R}^+. \end{aligned} \tag{VII.38}$$

- Order ε:
In $\omega \times \mathbb{R}^+$ this gives

$$\begin{aligned} &\frac{\partial W^1}{\partial t} - W^3_{zz} - 2\nabla W^2_z.\nabla\varphi - \Delta\varphi W^2_z - \Delta W^1 + (W^0 \cdot \nabla)W^1 + (W^1 \cdot \nabla)W^0 \\ &+ (W^0 \cdot \nabla\varphi)W^2_z + (W^1 \cdot \nabla\varphi)W^1_z + (W^2 \cdot \nabla\varphi)W^0_z + \nabla q^1 + q^2_z\nabla\varphi + W^3 = 0. \end{aligned}$$

When z tends towards infinity, in $\omega \times \mathbb{R}^+$ we obtain

$$\frac{\partial\overline{W}^1}{\partial t} - \Delta\overline{W}^1 + (\overline{W}^0 \cdot \nabla)\overline{W}^1 + (\overline{W}^1 \cdot \nabla)\overline{W}^0 + \nabla\overline{q}^1 + \overline{W}^3 = 0. \tag{VII.39}$$

2.3.1.2 Asymptotic expansion of Equation (2) in (VII.29)

- Order ε^{-1}:
$$\widetilde{W}^0 \cdot \nabla\varphi = 0, \quad \text{in } \omega \times \mathbb{R}^+. \tag{VII.40}$$

- Order ε^0:
$$\nabla\varphi \cdot W_z^1 + \operatorname{div} W^0 = 0,$$
and hence
$$\operatorname{div} \overline{W}^0 = 0, \quad \text{in } \omega, \tag{VII.41}$$
$$\nabla\varphi \cdot \widetilde{W}_z^1 + \operatorname{div} \widetilde{W}^0 = 0, \quad \text{in } \omega \times \mathbb{R}^+. \tag{VII.42}$$

- Order ε:
$$\nabla\varphi \cdot W_z^2 + \operatorname{div} W^1 = 0;$$
hence
$$\operatorname{div} \overline{W}^1 = 0, \quad \text{in } \omega, \tag{VII.43}$$
$$\nabla\varphi \cdot \widetilde{W}_z^2 + \operatorname{div} \widetilde{W}^1 = 0, \quad \text{in } \omega \times \mathbb{R}^+. \tag{VII.44}$$

- Order ε^2:
$$\nabla\varphi \cdot W_z^3 + \operatorname{div} W^2 = 0;$$
hence, when z tends towards infinity, we have
$$\operatorname{div} \overline{W}^2 = 0, \quad \text{in } \omega. \tag{VII.45}$$

- Order ε^3:
$$\nabla\varphi \cdot W_z^4 + \operatorname{div} W^3 = 0;$$
hence, as before,
$$\operatorname{div} \overline{W}^3 = 0, \quad \text{in } \omega. \tag{VII.46}$$

2.3.1.3 Asymptotic expansion of Equation (3) in (VII.29)

- Order ε^0:
$$\frac{\partial V^0}{\partial t} - \Delta V^0 + (V^0 \cdot \nabla)V^0 + \nabla p^0 = f, \quad \text{in } \mathcal{U}. \tag{VII.47}$$

- Order ε^1:
$$\frac{\partial V^1}{\partial t} - \Delta V^1 + (V^0 \cdot \nabla)V^1 + (V^1 \cdot \nabla)V^0 + \nabla p^1 = 0, \quad \text{in } \mathcal{U}. \tag{VII.48}$$

- Order ε^2:

$$\frac{\partial V^2}{\partial t} - \Delta V^2 + (V^0 \cdot \nabla)V^2 + (V^1 \cdot \nabla)V^1 + (V^2 \cdot \nabla)V^0 + \nabla p^2 = 0, \quad \text{in } \mathcal{U}. \tag{VII.49}$$

2.3.1.4 Asymptotic expansion of Equation (4) in (VII.29)

At all orders we obtain the relations

$$\operatorname{div} V^0 = \operatorname{div} V^1 = \operatorname{div} V^2 = 0, \quad \text{in } \mathcal{U}. \tag{VII.50}$$

2.3.1.5 Asymptotic expansion of Equation (5) in (VII.29)

We easily obtain the relations

$$V^0 = W^0,\ V^1 = W^1,\ V^2 = W^2, \quad \text{on } \partial\omega \times \{z = 0\}. \tag{VII.51}$$

2.3.1.6 Asymptotic expansion of Equation (6) in (VII.29)

- Order ε^{-1}:

$$-W_z^0 + q^{-1}\nu = 0, \quad \text{on } \partial\omega \times \{z = 0\}. \tag{VII.52}$$

- Order ε^0:

$$-\frac{\partial W^0}{\partial \nu} - W_z^1 + q^0\nu = p^0\nu - \frac{\partial V^0}{\partial \nu}, \quad \text{on } \partial\omega \times \{z = 0\}. \tag{VII.53}$$

- Order ε:

$$-\frac{\partial W^1}{\partial \nu} - W_z^2 + q^1\nu = p^1\nu - \frac{\partial V^1}{\partial \nu}, \quad \text{on } \partial\omega \times \{z = 0\}. \tag{VII.54}$$

2.3.2 Formal resolution of the profile equations

We now determine, for the moment at a formal level, the various terms of the asymptotic expansion. We use the relations obtained above by identifying the various powers of ε in the original equations and we look for solutions of these equations. We note that certain of these equations do not have unique solutions and we therefore arbitrarily choose one possible solution. This is not so surprising if we remember that the essential aim is to obtain a suitable approximate solution which is certainly not unique.

2.3.2.1 Determination of q^{-1} and $\overline{W}^1$

By taking the scalar product of (VII.52) with $\nabla\varphi$, since $\widetilde{W}^0_z \cdot \nabla\varphi = 0$ (see (VII.40)), and $\nabla\varphi = \nu$ on $\partial\omega$, we get

$$q^{-1} = 0, \quad \text{on } \partial\omega \times \{z = 0\}. \tag{VII.55}$$

If we now take the divergence of (VII.35) and if we use (VII.43), then we obtain

$$\Delta\overline{q^{-1}} = 0, \quad \text{in } \omega. \tag{VII.56}$$

Finally, we form the scalar product of (VII.34) with $\nabla\varphi$. Since $\widetilde{W}^0 \cdot \nabla\varphi = 0$ in all ω (see (VII.40)), we obtain $\widetilde{q_z^{-1}} = 0$ and therefore, from (VII.31), we get

$$\widetilde{q^{-1}} = 0, \quad \text{in } \omega \times \mathbb{R}^+.$$

Since $q^{-1}(t,x,z) = \overline{q^{-1}}(t,x) + \widetilde{q^{-1}}(t,x,z)$, we can now deduce from (VII.55) that $\overline{q^{-1}} = 0$ on $\partial\omega$ and hence using (VII.56) and the uniqueness of the solution to the Laplace equation with Dirichlet boundary conditions, we have shown that

$$\overline{q^{-1}} = 0, \quad \text{in } \omega,$$

and hence that

$$q^{-1} = \overline{q^{-1}} + \widetilde{q^{-1}} = 0.$$

With (VII.35), this implies that

$$\overline{W}^1 = 0, \quad \text{in } \omega.$$

2.3.2.2 Determination of W^0

From (VII.33), we already know that $\overline{W}^0 = 0$. Using (VII.34), and the fact that $\widetilde{q^{-1}} = 0$, we have

$$-\widetilde{W}^0_{zz} + \widetilde{W}^0 = 0, \quad \text{in } \omega \times \mathbb{R}^+.$$

Now from (VII.52), since q^{-1} is zero, we obtain that $\widetilde{W}^0_z = 0$ on the boundary $\partial\omega \times \{z = 0\}$.

The idea is then to extend the boundary conditions to all ω and hence to choose $\widetilde{W}^0_z = 0$ in $\omega \times \mathbb{R}^+$. This particular choice implies that

$$\widetilde{W}^0 = 0,$$

and hence since $\overline{W}^0 = 0$, we finally have

$$W^0 = 0, \quad \text{in } \omega \times \mathbb{R}^+. \tag{VII.57}$$

The choice above is of course arbitrary; however, we should not forget that we are proceeding by sufficient conditions. Hence, this choice is justified by the fact that we are able to complete the calculations below, that is, to define all the profiles of the expansion and to establish an estimate of the remainder terms.

2.3.2.3 Determination of V^0 and p^0

By using (VII.57) and (VII.51) we deduce that $V^0 = 0$ on $\partial\omega$. Hence, with Equations (VII.47) and (VII.50), we obtain that V^0 and p^0 are completely determined (apart from a constant for the pressure) by the Navier–Stokes equations:

$$\begin{cases} \dfrac{\partial V^0}{\partial t} - \Delta V^0 + (V^0 \cdot \nabla)V^0 + \nabla p^0 = f, & \text{in } \mathcal{U}, \\ \operatorname{div} V^0 = 0, & \text{in } \mathcal{U}, \\ V^0 = 0, & \text{on } \partial\mathcal{U}, \\ V^0(0) = v_0. \end{cases}$$

2.3.2.4 Determination of $\widetilde{W}^1$ and $\widetilde{q}^0$

From (VII.42) we have on all ω, that $\widetilde{W}^1_z \cdot \nabla\varphi = 0$ and thus

$$\widetilde{W}^1 \cdot \nabla\varphi = 0, \quad \text{in } \omega \times \mathbb{R}^+. \tag{VII.58}$$

Remark VII.2.4. Equation (VII.58) means in particular that the first term of the boundary layer in ω is tangential to $\partial\omega$, because $\nabla\varphi = \nu$ on $\partial\omega$.

Then, from (VII.36), and since $W^0 = 0$ and $q^{-1} = 0$, we have

$$-\widetilde{W}^1_{zz} + \widetilde{W}^1 + \widetilde{q}^0_z \nabla\varphi = 0, \quad \text{on } \partial\omega \times \{0\}. \tag{VII.59}$$

Hence, by taking the scalar product of this equation with $\nabla\varphi$ we obtain, using (VII.58), that $\widetilde{q}^0_z = 0$, and hence (using (VII.31))

$$\widetilde{q}^0 = 0, \quad \text{in } \omega \times \mathbb{R}^+. \tag{VII.60}$$

Equation (VII.59) now reduces to

$$-\widetilde{W}^1_{zz} + \widetilde{W}^1 = 0, \quad \text{in } \omega \times \mathbb{R}^+.$$

Its only solution which tends to 0 as $z \to +\infty$ is given by

$$\widetilde{W}^1(t, x, z) = w^1(t, x)e^{-z}, \quad \text{in } \omega \times \mathbb{R}^+, \tag{VII.61}$$

where w^1 is the (still unknown) value of $\widetilde{W}^1$ at $z = 0$.

Using (VII.53) and inasmuch as $W^0 = 0$ and $\widetilde{q}^0 = 0$, we obtain on the boundary

$$-\widetilde{W}_z^1 + \overline{q^0}\nu = -\frac{\partial V^0}{\partial \nu} + p^0\nu.$$

By taking the scalar product of this equation with ν and since $\widetilde{W}^1$ is tangent to the boundary of ω from (VII.58), we have

$$\overline{q^0} = -\left(\frac{\partial V^0}{\partial \nu}\cdot\nu\right) + p^0, \quad \text{on } \partial\omega. \tag{VII.62}$$

Hence, at the boundary (when $z = 0$ and $x \in \partial\omega$), this gives

$$-\widetilde{W}_z^1 = w^1 = -\frac{\partial V^0}{\partial \nu} + \left(\frac{\partial V^0}{\partial \nu}\cdot\nu\right)\nu.$$

We therefore decide to extend this formula to the whole domain ω by setting

$$w_1(t,x) = -\nabla\underline{V^0}.\nabla\varphi + \big((\nabla\underline{V^0}.\nabla\varphi)\cdot\nabla\varphi\big)\nabla\varphi, \quad \text{in all } \omega, \tag{VII.63}$$

where $\underline{V^0}$ is a simultaneous lifting in ω of the trace of V^0 and of that of $\partial V^0/\partial\nu$ on $\partial\omega$ and having the same regularity as V^0 (see Remark III.2.12).

This extends the condition found at the boundary, because $\nabla\varphi = \nu$ on $\partial\omega$.

2.3.2.5 Determination of q^0 and $\overline{W}^2$

We already know that $\widetilde{q}^0 = 0$. Using (VII.37), and since $W^0 = 0$, this gives

$$\nabla\overline{q^0} + \overline{W}^2 = 0, \quad \text{in } \omega. \tag{VII.64}$$

If we now take the divergence of this equation and use (VII.45) and (VII.62), we can completely determine $\overline{q^0}$ as a solution of the following Laplace problem

$$\begin{cases} \Delta\overline{q^0} = 0, & \text{in } \omega, \\ \overline{q^0} = -\left(\dfrac{\partial V^0}{\partial \nu}\cdot\nu\right) + p^0, & \text{on } \partial\omega. \end{cases} \tag{VII.65}$$

Moreover, knowing $\overline{q^0}$, the term $\overline{W}^2$ is simply defined by $\overline{W}^2 = -\nabla\overline{q^0}$.

2.3.2.6 Determination of V^1

Using (VII.48), (VII.51), (VII.61), and (VII.63), and since $\overline{W}^1 = 0$, we obtain that V^1 and p^1 satisfies a Navier–Stokes equation which is linearised around V^0

$$\begin{cases} \dfrac{\partial V^1}{\partial t} - \Delta V^1 + (V^0 \cdot \nabla)V^1 + (V^1 \cdot \nabla)V^0 + \nabla p^1 = 0, & \text{in } \mathcal{U}, \\ \operatorname{div} V^1 = 0, & \text{in } \mathcal{U}, \\ V^1 = -\dfrac{\partial V^0}{\partial \nu} + \left(\dfrac{\partial V^0}{\partial \nu} \cdot \nu\right)\nu, & \text{on } \partial\omega, \\ V^1 = 0, & \text{on } \partial\Omega. \end{cases}$$

We then note that $V^1 \cdot \nu = 0$ on $\partial\omega$, which shows that the first correction term of the solution in $\mathcal{U}$ is tangential to $\partial\omega$. Moreover, this induces the compatibility of the Dirichlet boundary data with the incompressibility constraint.

2.3.2.7 Determination of $\widetilde{W}^2$ and $\widetilde{q}^1$

We recall that, since $W^0 = 0$ and $\widetilde{q}^0 = 0$, from (VII.38) we have

$$- \widetilde{W}^2_{zz} - 2\nabla\widetilde{W}^1_z.\nabla\varphi - \Delta\varphi\widetilde{W}^1_z + \widetilde{q}^1_z\nabla\varphi + \widetilde{W}^2 = 0, \quad \text{in } \omega \times \mathbb{R}^+. \tag{VII.66}$$

Using (VII.44), we know that $(\widetilde{W}^2_z \cdot \nabla\varphi) = -\operatorname{div}\widetilde{W}^1$ and by integrating with respect to z we obtain using (VII.61):

$$(\widetilde{W}^2(t,x,z) \cdot \nabla\varphi(x)) = \int_z^{+\infty} \operatorname{div}\widetilde{W}^1(t,x,\zeta)d\zeta = \operatorname{div} w_1(t,x)e^{-z}. \tag{VII.67}$$

Let us now show that

$$(\nabla\widetilde{W}^1_z.\nabla\varphi) \cdot \nabla\varphi = 0. \tag{VII.68}$$

To achieve this, we note that from (VII.58) we have $\widetilde{W}^1_z \cdot \nabla\varphi = 0$ and therefore

$$\begin{aligned} 0 &= \nabla\left(\widetilde{W}^1_z \cdot \nabla\varphi\right) \cdot \nabla\varphi \\ &= (\nabla\widetilde{W}^1_z.\nabla\varphi) \cdot \nabla\varphi + \frac{1}{2}\widetilde{W}^1_z \cdot \nabla\big(|\nabla\varphi|^2\big). \end{aligned}$$

However, the second term above is zero, at least in ω_1 because $|\nabla\varphi| \equiv 1$ in ω_1. If we now take the scalar product of (VII.66) with $\nabla\varphi$, we obtain using (VII.68) and since we know that $\widetilde{W}^1 \cdot \nabla\varphi = 0$,

$$\widetilde{q}^1_z = (\widetilde{W}^2_{zz} \cdot \nabla\varphi) - (\widetilde{W}^2 \cdot \nabla\varphi).$$

However, (VII.67) shows that $\widetilde{W}^2_{zz} \cdot \nabla\varphi = \widetilde{W}^2 \cdot \nabla\varphi$ and we then have $\widetilde{q^1}_z = 0$, which finally gives

$$\widetilde{q^1} = 0, \quad \text{in } \omega_1 \times \mathbb{R}^+. \tag{VII.69}$$

It remains to determine the tangential component of $\widetilde{W}^2$, that is (in dimension $d = 3$, e.g.) the vector $\widetilde{W}^2 \wedge \nabla\varphi$. To do this, we compute the vector products (VII.54)$\wedge\nabla\varphi$, and (VII.66)$\wedge\nabla\varphi$, which show that $\widetilde{W}^2 \wedge \nabla\varphi$ satisfies the following ordinary differential equation with respect to the variable z

$$\begin{cases} \widetilde{W}^2_{zz} \wedge \nabla\varphi + 2(\nabla\widetilde{W}^1_z.\nabla\varphi) \wedge \nabla\varphi \\ \quad + (\Delta\varphi)\widetilde{W}^1_z \wedge \nabla\varphi - \widetilde{W}^2 \wedge \nabla\varphi = 0, \quad \text{in } \omega \times \mathbb{R}^+, \\ \widetilde{W}^2_z \wedge \nu = -\dfrac{\partial W^1}{\partial\nu} \wedge \nu + \dfrac{\partial V^1}{\partial\nu} \wedge \nu, \qquad \text{on } \partial\omega \times \{z = 0\}. \end{cases} \tag{VII.70}$$

The first equation is of the form

$$\widetilde{W}^2_{zz} \wedge \nabla\varphi - \widetilde{W}^2 \wedge \nabla\varphi = g(t,x)e^{-z},$$

where $g(t,x)$ is a known vector field as a function of w_1 and φ. The general solution of this differential equation, at fixed (t,x), is given by

$$\widetilde{W}^2 \wedge \nabla\varphi = a(t,x)e^z + \left(b(t,x) - \frac{1}{2}g(t,x)z\right)e^{-z}.$$

However, if we impose that $\widetilde{W}^2$ tends to 0 when z tends towards $+\infty$, this proves that $a(t,x) = 0$. We obtain the term $b(t,x)$ by writing the boundary conditions at $z = 0$ given in (VII.70)

$$b(t,x) = -\frac{1}{2}g(t,x) + \frac{\partial W^1}{\partial\nu} \wedge \nu - \frac{\partial V^1}{\partial\nu} \wedge \nu, \quad \text{on } \partial\omega.$$

As with the term w_1 introduced above, we again choose to extend this formula to the whole obstacle ω by setting

$$b(t,x) = -\frac{1}{2}g(t,x) + (\nabla W^1.\nabla\varphi) \wedge \nabla\varphi - (\nabla\underline{V^1}.\nabla\varphi) \wedge \nabla\varphi, \quad \text{in } \omega,$$

where $\underline{V^1}$ is a lifting in ω of the traces of V^1 and of $\partial V^1/\partial\nu$ having the same regularity properties as V^1, obtained from Remark III.2.12.

This completely determines the tangential part of $\widetilde{W}^2$, as follows

$$\begin{aligned} \widetilde{W}^2 \wedge \nabla\varphi = \Big(& (\nabla W^1.\nabla\varphi) \wedge \nabla\varphi - (\nabla\underline{V^1}.\nabla\varphi) \wedge \nabla\varphi \\ & + (1+z)\big(\Delta\varphi\widetilde{W}^1_z \wedge \nabla\varphi + 2(\nabla\widetilde{W}^1_z.\nabla\varphi) \wedge \nabla\varphi\big)\Big)e^{-z}. \end{aligned} \tag{VII.71}$$

2.3.2.8 Determination of V^2 and p^2

Using (VII.49) through (VII.51), and since W^2 is known, we can determine V^2 and p^2 as solutions of the linear system

$$\begin{cases} \dfrac{\partial V^2}{\partial t} - \Delta V^2 + (V^0 \cdot \nabla)V^2 + (V^1 \cdot \nabla)V^1 + (V^2 \cdot \nabla)V^0 + \nabla p^2 = 0, & \text{in } \mathcal{U}, \\ \operatorname{div} V^2 = 0, & \text{in } \mathcal{U}, \\ V^2 = W^2, & \text{on } \partial\omega, \\ V^2 = 0, & \text{on } \partial\Omega. \end{cases}$$

Remark VII.2.5. The boundary condition $V^2 = W^2$ on $\partial\omega$ is compatible with the divergence-free condition $\operatorname{div} V^2 = 0$ in $\mathcal{U}$ because

$$\int_{\partial\omega} W^2 \cdot \nu \, d\sigma = 0.$$

Indeed, from (VII.45), we have

$$0 = \int_\omega \operatorname{div} W^2 \, dx = -\int_{\partial\omega} W^2 \cdot \nu \, d\sigma.$$

2.3.2.9 Determination of $\overline{q}^1$

We have already seen that $\widetilde{q}^1 = 0$ and that $\overline{W}^1 = 0$. Hence Equation (VII.39) gives

$$\nabla \overline{q}^1 + \overline{W}^3 = 0,$$

and using the Laplace equation (VII.46) this leads to

$$\Delta \overline{q}^1 = 0, \quad \text{in } \omega. \tag{VII.72}$$

Furthermore, by making z tend towards infinity and by taking the scalar product with ν, Equation (VII.54) becomes

$$\overline{q}^1 = p^1 - \left(\frac{\partial V^1}{\partial \nu}\right) \cdot \nu, \quad \text{on } \partial\omega. \tag{VII.73}$$

This completely determines the term $\overline{q}^1$.

2.4 Well-posedness of profile equations

In this section we revisit the formal equations established in the preceding section and show that these equations are indeed well-posed in functional spaces with suitable regularity.

The existence, uniqueness, and regularity result for the nonpenalized solution V^0 has already been given by Proposition VII.2.1. We therefore concentrate on the higher-order terms.

- Regularity of $\widetilde{W}^1$:
  We recall that in (VII.63), we asserted that on all ω:

$$w_1 = -\nabla \underline{V^0}.\nabla\varphi + \big((\nabla \underline{V^0}.\nabla\varphi)\cdot\nabla\varphi\big)\nabla\varphi,$$

  such that the regularity established for V^0 (which is hence also that of $\underline{V^0}$, by construction of the lifting $\underline{V^0}$) and that of φ, show that for any $T < T^*$ we have

$$\begin{cases} w^1 \in \mathcal{C}^0([0,T],(H^4(\omega))^d) \cap L^2(]0,T[,(H^5(\omega))^d), \\ \dfrac{\partial w^1}{\partial t} \in \mathcal{C}^0([0,T],(H^2(\omega))^d) \cap L^2(]0,T[,(H^3(\omega))^d), \\ \dfrac{\partial^2 w^1}{\partial t^2} \in \mathcal{C}^0([0,T],(L^2(\omega))^d) \cap L^2(]0,T[,(H^1(\omega))^d). \end{cases} \tag{VII.74}$$

  Conforming with (VII.61), we now assert

$$\widetilde{W}^1(t,x,z) = w^1(t,x)e^{-z}, \quad \forall (t,x,z) \in [0,T^*[\omega \times \mathbb{R}^+,$$

  such that

$$\begin{cases} \widetilde{W}^1 \in \big(\mathcal{C}^0([0,T],(H^4(\omega))^d) \cap L^2(]0,T[,(H^5(\omega))^d)\big) \otimes \mathcal{C}^\infty, \\ \dfrac{\partial \widetilde{W}^1}{\partial t} \in \big(\mathcal{C}^0([0,T],(H^2(\omega))^d) \cap L^2(]0,T[,(H^3(\omega))^d)\big) \otimes \mathcal{C}^\infty, \\ \dfrac{\partial^2 \widetilde{W}^1}{\partial t^2} \in \big(\mathcal{C}^0([0,T],(L^2(\omega))^d) \cap L^2(]0,T[,(H^1(\omega))^d)\big) \otimes \mathcal{C}^\infty, \end{cases} \tag{VII.75}$$

  where the symbol $L^p(]0,T[,(H^s(\omega))^d) \otimes \mathcal{C}^\infty$ is the vector space spanned by functions of the form

$$f_i(t,x)g_i(z),$$

  with $f_i \in L^p(]0,T[,(H^s(\omega))^d)$ and $g_i \in \mathcal{C}^\infty(\mathbb{R})$.
- Regularity of V^1:

  Proposition VII.2.4. *Let Ω, ω, v_0, and f be as in Proposition VII.2.1. Let T^* and V^0 also be given by that proposition. There exist two functions V^1 and p^1 such that*

$$\begin{cases} \dfrac{\partial V^1}{\partial t} - \Delta V^1 + (V^0 \cdot \nabla)V^1 + (V^1 \cdot \nabla)V^0 + \nabla p^1 = 0, & \textit{in } \mathcal{U}, \\ \operatorname{div} V^1 = 0, & \textit{in } \mathcal{U}, \\ V^1 = -\dfrac{\partial V^0}{\partial \nu} + \left(\dfrac{\partial V^0}{\partial \nu} \cdot \nu\right)\nu, & \textit{on } \partial\omega, \\ V^1 = 0, & \textit{on } \partial\Omega, \end{cases} \tag{VII.76}$$

and which satisfy: for all $T < T^*$,

$$\begin{cases} V^1 \in \mathcal{C}^0([0,T], (H^4(\mathcal{U}))^d) \cap L^2(]0,T[, (H^5(\mathcal{U}))^d), \\ \dfrac{\partial V^1}{\partial t} \in \mathcal{C}^0([0,T], (H^2(\mathcal{U}))^d) \cap L^2(]0,T[, (H^3(\mathcal{U}))^d), \\ \dfrac{\partial^2 V^1}{\partial t} \in L^2(]0,T[, (H^1(\mathcal{U}))^d), \\ p^1 \in \mathcal{C}^0([0,T], H^3(\mathcal{U})) \cap L^2(]0,T[, H^4(\mathcal{U})), \\ \dfrac{\partial p^1}{\partial t} \in \mathcal{C}^0([0,T], H^1(\mathcal{U})) \cap L^2(]0,T[, H^2(\mathcal{U})). \end{cases} \tag{VII.77}$$

We note that in this proposition we do not set the initial data for Problem (VII.76). Since V^1 is the first term of the asymptotic expansion sought, we could impose $V^1(t=0) = 0$ but this would not allow us to ensure that the compatibility conditions (of type (V.54)) are satisfied and hence would not allow us to establish the regularity properties stated above. Hence, in the following proof, we construct suitable initial data.

Proof.

First, we want to find a suitable lifting of the boundary conditions. We therefore seek V^1 in the form $Z^1 + \Upsilon^1$ where Υ^1 is a lifting on $\mathcal{U}$ of the trace

$$-\frac{\partial V^0}{\partial \nu} + \left(\frac{\partial V^0}{\partial \nu} \cdot \nu\right)\nu.$$

We construct this lifting by solving the following steady linear Stokes problem for each time t

$$\begin{cases} -\Delta \Upsilon^1 + \nabla \Pi^1 = 0, & \text{in } \mathcal{U}, \\ \operatorname{div} \Upsilon^1 = 0, & \text{in } \mathcal{U}, \\ \Upsilon^1 = -\dfrac{\partial V^0}{\partial \nu} + \left(\dfrac{\partial V^0}{\partial \nu} \cdot \nu\right)\nu, & \text{on } \partial\omega, \\ \Upsilon^1 = 0, & \text{on } \partial\Omega, \end{cases}$$

where Π^1 has a zero mean value in $\mathcal{U}$. This equation can be solved because the boundary conditions are compatible with the divergence-free condition. Furthermore, from the regularity properties of the Stokes problem (Theorem IV.5.8), the lifting Υ^1 and the pressure Π^1 enjoy regularity properties induced by those of V^0 from the boundary conditions. More precisely (time is a parameter here), we have

$$\left\{\begin{aligned} \Upsilon^1 &\in \mathcal{C}^0([0,T],(H^4(\mathcal{U}))^d)\cap L^2(]0,T[,(H^5(\mathcal{U}))^d),\\ \frac{\partial\Upsilon^1}{\partial t} &\in \mathcal{C}^0([0,T],(H^2(\mathcal{U}))^d)\cap L^2(]0,T[,(H^3(\mathcal{U}))^d),\\ \frac{\partial^2\Upsilon^1}{\partial t^2} &\in L^2(]0,T[,(H^1(\mathcal{U}))^d),\\ \Pi^1 &\in \mathcal{C}^0([0,T],H^3(\mathcal{U}))\cap L^2(]0,T[,H^4(\mathcal{U})),\\ \frac{\partial\Pi^1}{\partial t} &\in \mathcal{C}^0([0,T],H^1(\mathcal{U}))\cap L^2(]0,T[,H^2(\mathcal{U})),\\ \frac{\partial^2\Pi^1}{\partial t^2} &\in L^2(]0,T[,L^2(\mathcal{U})). \end{aligned}\right. \tag{VII.78}$$

This lifting being defined, we must now establish the existence and regularity of a solution Z^1 of

$$\left\{\begin{array}{ll} \dfrac{\partial Z^1}{\partial t}-\Delta Z^1+(V^0\cdot\nabla)Z^1+(Z^1\cdot\nabla)V^0+\nabla(p^1-\Pi^1)=Q^1, & \text{in } \mathcal{U},\\ \operatorname{div} Z^1=0, & \text{in } \mathcal{U},\\ Z^1=0, & \text{on } \partial\mathcal{U}, \end{array}\right. \tag{VII.79}$$

where $Q^1=-\partial\Upsilon^1/\partial t-(V^0\cdot\nabla)\Upsilon^1-(\Upsilon^1\cdot\nabla)V^0$. We note that, from (VII.78), we have for all $T<T^*$,

$$\left\{\begin{aligned} Q^1 &\in \mathcal{C}^0([0,T],(H^2(\mathcal{U}))^d)\cap L^2(]0,T[,(H^3(\mathcal{U}))^d),\\ \frac{\partial Q^1}{\partial t} &\in L^2(]0,T[,(H^1(\mathcal{U}))^d). \end{aligned}\right. \tag{VII.80}$$

As we have already seen above, in order to ensure the regularity of the solution Z^1 it is necessary to choose an initial condition well suited to the problem so that compatibility conditions of type (V.54) hold. This may seem somewhat difficult; however, we overcome this difficulty by providing initial data coming from the solution of the problem starting at time $t=-1$.

To achieve this, we start by extending Q^1, by setting $Q^1(t)=Q^1(-t)$ for all $t<0$. Next, we consider a real-valued function η of class $\mathcal{C}^\infty$ such that $\eta(t)=1$ for $t\geq 0$ and $\eta(t)=0$ for $t\leq -\frac{1}{2}$.

Applying the same method as in Chapter V, we solve Equation (VII.79) starting from time $t=-1$ for the initial data $Z^1(t=-1)=0$, and for the second term $\eta(t)Q^1(t)$ which enjoys the same regularity properties as Q^1 given by (VII.80).

Since $\eta(t)Q^1(t)$ is zero for any $t\leq -\frac{1}{2}$, and the data $Z^1(t=-1)$ is chosen to be zero, it is clear that the compatibility conditions of type (V.54), posed

for this problem from the initial time $t = -1$, are satisfied. This then gives us a solution Z^1 to Problem (VII.79) satisfying the desired properties, the initial data at $t = 0$ now being a function of $(H^4(\mathcal{U}))^d$ issued from the solution of the Cauchy problem at time $t = -1$.
Now, by combining Υ^1 and Z^1, we see that we have indeed solved Problem (VII.76) by ensuring (VII.77). The initial data in V^1 are not zero, rather they are a well-determined function of $(H^4(\Omega))^d$.

- Regularity of q^0 and $\overline{W}^2$:
We have seen that $\widetilde{q}^0 = 0$, and hence that $q^0 = \overline{q}^0$. Therefore, from (VII.65), we find that the pressure term q^0 satisfies

$$\begin{cases} \Delta q^0 = 0, & \text{in } \omega, \\ q^0 = -\left(\dfrac{\partial V^0}{\partial \nu} \cdot \nu\right) + p^0, & \text{on } \partial\omega. \end{cases}$$

Because of the regularity properties of V^0 and p^0 (estimates (VII.27)), by using the elliptic regularity properties of the Laplace operator (Theorem III.4.2), we obtain that for all $T < T^*$,

$$\begin{cases} q^0 \in \mathcal{C}^0([0,T], H^4(\omega)) \cap L^2(]0,T[, H^5(\omega)), \\ \dfrac{\partial q^0}{\partial t} \in \mathcal{C}^0([0,T], H^2(\omega)) \cap L^2(]0,T[, H^3(\omega)). \end{cases}$$

Finally, since $\overline{W}^2 = -\nabla q^0$, we get

$$\begin{cases} \overline{W}^2 \in \mathcal{C}^0([0,T], (H^3(\omega))^d) \cap L^2(]0,T[, (H^4(\omega))^d), \\ \dfrac{\partial \overline{W}^2}{\partial t} \in \mathcal{C}^0([0,T], (H^1(\omega))^d) \cap L^2(]0,T[, (H^2(\omega))^d). \end{cases}$$

- Regularity of $\widetilde{W}^2$:
From (VII.67), we have $\widetilde{W}^2 \cdot \nabla\varphi = \operatorname{div}(w_1(t,x))e^{-z}$, and hence with (VII.74) we obtain

$$\begin{cases} \widetilde{W}^2 \cdot \nabla\varphi \in \big(\mathcal{C}^0([0,T], H^3(\omega)) \cap L^2(]0,T[, H^4(\omega))\big) \otimes \mathcal{C}^\infty, \\ \dfrac{\partial \widetilde{W}^2}{\partial t} \cdot \nabla\varphi \in \big(\mathcal{C}^0([0,T], H^1(\omega)) \cap L^2(]0,T[, H^2(\omega))\big) \otimes \mathcal{C}^\infty. \end{cases}$$

We recall that, by construction, $\underline{V^1}$ possesses the same regularity in ω that V^0 possesses in $\mathcal{U}$. Hence, from (VII.71) and from estimates (VII.75) and (VII.77), we obtain that for all $T < T^*$

$$\begin{cases} \widetilde{W}^2 \wedge \nabla\varphi \in \mathcal{C}^0([0,T], (H^3(\omega))^d) \cap L^2(]0,T[, (H^4(\omega))^d) \otimes \mathcal{C}^\infty, \\ \dfrac{\partial \widetilde{W}^2}{\partial t} \wedge \nabla\varphi \in \mathcal{C}^0([0,T], (H^1(\omega))^d) \cap L^2(]0,T[, (H^2(\omega))^d) \otimes \mathcal{C}^\infty. \end{cases}$$

This all gives us the regularity of the vector field $\widetilde{W}^2$.

- Regularity of V^2 and p^2:

Proposition VII.2.5. *Under the hypotheses of Proposition VII.2.1, there exist two functions V^2 and p^2 such that*

$$\begin{cases} \dfrac{\partial V^2}{\partial t} - \Delta V^2 + (V^0\cdot\nabla)V^2 + (V^1\cdot\nabla)V^1 + (V^2\cdot\nabla)V^0 + \nabla p^2 = 0, & \text{in } \mathcal{U},\\ \operatorname{div} V^2 = 0, & \text{in } \mathcal{U},\\ V^2 = W^2, & \text{on } \partial\omega,\\ V^2 = 0, & \text{on } \partial\Omega, \end{cases}$$

and satisfying, for all $T < T^*$,

$$\begin{cases} V^2 \in \mathcal{C}^0([0,T],(H^2(\mathcal{U}))^d) \cap L^2(]0,T[,(H^3(\mathcal{U}))^d),\\ \dfrac{\partial V^2}{\partial t} \in \mathcal{C}^0([0,T],(L^2(\mathcal{U}))^d) \cap L^2(]0,T[,(H^1(\mathcal{U}))^d),\\ p^2 \in \mathcal{C}^0([0,T],H^1(\mathcal{U})) \cap L^2(]0,T[,H^2(\mathcal{U})),\\ \dfrac{\partial p^2}{\partial t} \in L^2(]0,T[,L^2(\mathcal{U})). \end{cases}$$

Proof.

The principle of the proof is the same as for V^1 and p^1. We first define a lifting of $W^2_{|\partial\omega}$ by solving, for each time t, the following steady Stokes problem

$$\begin{cases} -\Delta\Upsilon^2 + \nabla\Pi^2 = 0, & \text{in } \mathcal{U},\\ \operatorname{div}\Upsilon^2 = 0, & \text{in } \mathcal{U},\\ \Upsilon^2 = W^2, & \text{on } \partial\omega,\\ \Upsilon^2 = 0, & \text{on } \partial\Omega. \end{cases}$$

It follows from Theorem IV.5.8 that for all $T < T^*$, we have

$$\begin{cases} \Upsilon^2 \in \mathcal{C}^0([0,T],(H^3(\mathcal{U}))^d) \cap L^2(]0,T[,(H^4(\mathcal{U}))^d),\\ \dfrac{\partial \Upsilon^2}{\partial t} \in \mathcal{C}^0([0,T],(H^1(\mathcal{U}))^d) \cap L^2(]0,T[,(H^2(\mathcal{U}))^d),\\ \Pi^2 \in \mathcal{C}^0([0,T],H^2(\mathcal{U})) \cap L^2(]0,T[,H^3(\mathcal{U})),\\ \dfrac{\partial \Pi^2}{\partial t} \in \mathcal{C}^0([0,T],L^2(\mathcal{U})) \cap L^2(]0,T[,H^1(\mathcal{U})). \end{cases}$$

We then look for V^2 in the form $V^2 = \Upsilon^2 + Z^2$ where Z^2 satisfies

$$\begin{cases} \dfrac{\partial Z^2}{\partial t} - \Delta Z^2 + (V^0 \cdot \nabla) Z^2 + (Z^2 \cdot \nabla) V^0 + \nabla(p^2 - \Pi^2) = Q^2, & \text{in } \mathcal{U}, \\ \operatorname{div} Z^2 = 0, & \text{in } \mathcal{U}, \\ Z^2 = 0, & \text{on } \partial\mathcal{U}, \end{cases} \tag{VII.81}$$

with

$$Q^2 = -\frac{\partial \Upsilon^2}{\partial t} - (V^0 \cdot \nabla)\Upsilon^2 - (V^1 \cdot \nabla) V^1 - (\Upsilon^2 \cdot \nabla) V^0.$$

From the estimates on V^0, V^1, and Υ^2, we deduce that

$$Q^2 \in \mathcal{C}^0([0,T], (H^1(\mathcal{U}))^d) \cap L^2(]0,T[, (H^2(\mathcal{U}))^d).$$

We can therefore solve Problem (VII.81) in the classic way for the initial data $Z^2(t=0) = 0$, with

$$\begin{cases} Z^2 \in \mathcal{C}^0([0,T], (H^1(\mathcal{U}))^d) \cap L^2(]0,T[, (H^2(\mathcal{U}))^d), \\ p^2 \in L^2(]0,T[, L^2(\mathcal{U})). \end{cases}$$

We note that given the assumptions applied on v_0 and f, we cannot hope for any better regularity for Z^2 because we do not have information on $\partial Q^2/\partial t$. Hence, here, we do not have the issue of compatibility conditions for the initial data that we encountered during the resolution of problem (VII.79). This is why we impose arbitrarily that the initial data are zero for Z^2.

□

- Regularity of $\overline{q}^1$:
  As we saw in (VII.72) and (VII.73), the term $\overline{q}^1$ is a solution of a Laplace problem. The existence and uniqueness of this problem are now classic and the regularity of such a term is deduced from the regularity of the boundary data (Theorem III.4.2):

$$\begin{cases} \overline{q}^1 \in \mathcal{C}^0([0,T], H^3(\omega)) \cap L^2(]0,T[, H^4(\omega)), \\ \dfrac{\partial \overline{q}^1}{\partial t} \in \mathcal{C}^0([0,T], H^1(\omega)) \cap L^2(]0,T[, H^2(\omega)). \end{cases}$$

2.5 Convergence of the asymptotic expansion

We want to complete the proof of the main result, that is, to estimate the remainder terms in the formal asymptotic expansion we proposed. For technical reasons, the boundary layer terms (i.e., the $\widetilde{W^i}$) need to be localized near the boundary of the obstacle. To do this, we introduce a localisation function $\theta \in \mathcal{C}^\infty(\overline{\omega})$ such that $\theta \equiv 1$ in a neighborhood of $\partial\omega$ and satisfies $\operatorname{supp}(\theta) \subset \omega_1$.

We proved in the previous sections that there exist various profile terms which constitute the sought-after asymptotic expansion. When we gather these various terms together, we obtain an approximate solution of the initial equations in a particular form. We now show that the exact solution of the initial equations $(u^\varepsilon, \pi^\varepsilon)$ is close to the approximate solution thus constructed.

We therefore search now for an exact solution of our equations in the following form

$$\begin{cases} w^\varepsilon(t,x) = \varepsilon\theta(x)\widetilde{W}^1\left(t, x, \dfrac{\varphi(x)}{\varepsilon}\right) \\ \qquad\qquad + \varepsilon^2\overline{W}^2(t,x) + \varepsilon^2\theta(x)\widetilde{W}^2\left(t, x, \dfrac{\varphi(x)}{\varepsilon}\right) + \varepsilon w^r_\varepsilon(t,x), & \text{in } \omega, \\ v^\varepsilon(t,x) = V^0(t,x) + \varepsilon V^1(t,x) + \varepsilon^2 V^2(t,x) + \varepsilon v^r_\varepsilon(t,x), & \text{in } \mathcal{U}, \\ q^\varepsilon(t,x) = \overline{q}^0(t,x) + \varepsilon\overline{q}^1(t,x) + \varepsilon q^r_\varepsilon(t,x), & \text{in } \omega, \\ p^\varepsilon(t,x) = p^0(t,x) + \varepsilon p^1(t,x) + \varepsilon^2 p^2(t,x) + \varepsilon p^r_\varepsilon(t,x), & \text{in } \mathcal{U}, \end{cases}$$

where the terms w^r_ε, v^r_ε, q^r_ε and p^r_ε are the remainder terms that we have to estimate.

Hereafter, we will denote the approximation of the velocity w^ε in ω as εW_{app} and its approximation in $\mathcal{U}$ as V_{app}; there are defined as follows

$$\begin{aligned} W_{app} &= \theta\widetilde{w}^1 + \varepsilon\overline{W}^2 + \varepsilon\theta\widetilde{w}^2, \\ V_{app} &= V^0 + \varepsilon V^1 + \varepsilon^2 V^2, \end{aligned}$$

by setting $\widetilde{w}^i(t,x) = \widetilde{W}^i\,(t, x, \varphi(x)/\varepsilon)$. Note the presence of the localisation function θ in the definition of W_{app}. Similarly, we set

$$\begin{aligned} p_{app} &= p^0 + \varepsilon p^1 + \varepsilon^2 p^2, \\ q_{app} &= \overline{q}^0 + \varepsilon\overline{q}^1. \end{aligned}$$

2.5.1 Equations satisfied by the remainders

By definition, the remainders satisfy the following system

$$
\left\{
\begin{aligned}
&\frac{\partial w_\varepsilon^r}{\partial t} - \Delta w_\varepsilon^r + \varepsilon (w_\varepsilon^r \cdot \nabla) W_{app} + \varepsilon (w_\varepsilon^r \cdot \nabla) w_\varepsilon^r \\
&\qquad + \varepsilon (W_{app} \cdot \nabla) w_\varepsilon^r + \nabla q_\varepsilon^r + \frac{1}{\varepsilon^2} w_\varepsilon^r = R^\varepsilon_{obst}, && \text{in } \omega, && (1)\\
&\operatorname{div} w_\varepsilon^r = R^\varepsilon_{div}, && \text{in } \omega, && (2)\\
&\frac{\partial v_\varepsilon^r}{\partial t} - \Delta v_\varepsilon^r + (V_{app} \cdot \nabla) v_\varepsilon^r + (v_\varepsilon^r \cdot \nabla) V_{app} \\
&\qquad + \varepsilon (v_\varepsilon^r \cdot \nabla) v_\varepsilon^r + \nabla p_\varepsilon^r = R^\varepsilon_{flu}, && \text{in } \mathcal{U}, && (3)\\
&\operatorname{div} v_\varepsilon^r = 0, && \text{in } \mathcal{U}, && (4)\\
&v_\varepsilon^r = w_\varepsilon^r, && \text{on } \partial\omega, && (5)\\
&-\frac{\partial v_\varepsilon^r}{\partial \nu} + p_\varepsilon^r \nu + \frac{\partial w_\varepsilon^r}{\partial \nu} - q_\varepsilon^r \nu = R^\varepsilon_{boundary}, && \text{on } \partial\omega, && (6)\\
&v_\varepsilon^r = 0, && \text{on } \partial\Omega, && (7)\\
&w_\varepsilon^r(0,.) = r^\varepsilon - \theta(.) \widetilde{W}^1 \left(0,., \frac{\varphi(.)}{\varepsilon}\right) \\
&\qquad - \varepsilon \overline{W}^2(0,.) - \varepsilon \theta(.) \widetilde{W}^2 \left(0,., \frac{\varphi(.)}{\varepsilon}\right), && \text{in } \omega, && (8)\\
&v_\varepsilon^r(0,.) = r^\varepsilon(.) - V^1(0,.) - \varepsilon V^2(0,.), && \text{in } \mathcal{U}, && (9)
\end{aligned}
\right.
\tag{VII.82}
$$

where we define

$$
\begin{aligned}
R^\varepsilon_{obst} &= -\frac{\partial W_{app}}{\partial t} + \Delta W_{app} - \varepsilon (W_{app} \cdot \nabla) W_{app} - \frac{1}{\varepsilon^2} W_{app} - \frac{1}{\varepsilon} \nabla q_{app}, \\
R^\varepsilon_{div} &= -\operatorname{div}(W_{app}), \\
R^\varepsilon_{flu} &= \frac{1}{\varepsilon} \left(f - \frac{\partial V_{app}}{\partial t} - (V_{app} \cdot \nabla) V_{app} + \Delta V_{app} - \nabla p_{app} \right), \\
R^\varepsilon_{boundary} &= \frac{1}{\varepsilon} \left(\frac{\partial V_{app}}{\partial \nu} - p_{app} \nu - \varepsilon \frac{\partial W_{app}}{\partial \nu} + q_{app} \nu \right).
\end{aligned}
$$

We note, to start with, that the initial data $w_\varepsilon^r(0,x)$ and $v_\varepsilon^r(0,x)$ are bounded independently of ε in $(L^2)^d$. Moreover, the support of θ is included in ω_1, therefore we know that $|\nabla \varphi| \equiv 1$ everywhere where θ is not zero. This simplifies the following calculations.

If we now use the equations satisfied by all the terms of the asymptotic expansion, then we get

$$
\begin{aligned}
R^\varepsilon_{obst} = &- \theta \frac{\partial \widetilde{W}^1}{\partial t} - \varepsilon \frac{\partial \overline{W}^2}{\partial t} - \varepsilon\theta \frac{\partial \widetilde{W}^2}{\partial t} + \theta \Delta \widetilde{W}^1 \\
&+ 2\nabla \widetilde{W}^1 . \nabla\theta + \frac{2}{\varepsilon}(\nabla\theta \cdot \nabla\varphi)\widetilde{W}^1_z + (\Delta\theta)\widetilde{W}^1 \\
&+ \varepsilon \Delta \overline{W}^2 + \varepsilon\theta\Delta\widetilde{W}^2 + 2\theta\nabla\widetilde{W}^2_z . \nabla\varphi + \theta(\Delta\varphi)\widetilde{W}^2_z \\
&+ 2\varepsilon\nabla\widetilde{W}^2 . \nabla\theta + 2(\nabla\theta\cdot\nabla\varphi)\widetilde{W}^2_z + \varepsilon(\Delta\theta)\widetilde{W}^2 \\
&- \varepsilon(W_{app}\cdot\nabla)W_{app} - \nabla\overline{q}^1, \\
R^\varepsilon_{div} = &- \nabla\theta\cdot\widetilde{W}^1 - \varepsilon\nabla\theta\cdot\widetilde{W}^2 - \varepsilon\theta \operatorname{div}\widetilde{W}^2, \\
R^\varepsilon_{flu} = &- \varepsilon^2\big((V^1\cdot\nabla)V^2 + (V^2\cdot\nabla)V^1\big) - \varepsilon^3(V^2\cdot\nabla)V^2, \\
R^\varepsilon_{boundary} = &\ \ \varepsilon\left(\frac{\partial V^2}{\partial\nu} - p^2\nu - \frac{\partial\widetilde{W}^2}{\partial\nu} - \frac{\partial\overline{W}^2}{\partial\nu}\right).
\end{aligned}
$$

Using the regularity properties of the various profile terms obtained above, we can easily show the following estimates for all $T < T^*$,

$$
\begin{cases}
\|R^\varepsilon_{obst}\|_{L^2(]0,T[,(L^2(\omega))^d)} \leq C, \\
\|R^\varepsilon_{div}\|_{L^\infty(]0,T[,L^2(\omega))} \leq C\varepsilon, \\
\left\|\dfrac{\partial}{\partial t}R^\varepsilon_{div}\right\|_{L^2(]0,T[,L^2(\omega))} \leq C\varepsilon, \\
\|R^\varepsilon_{flu}\|_{L^2(]0,T[,(L^2(\mathcal{U}))^d)} \leq C\varepsilon, \\
\|R^\varepsilon_{boundary}\|_{L^2(]0,T[,(L^2(\partial\omega))^d)} \leq C\varepsilon.
\end{cases}
\tag{VII.83}
$$

Indeed, we note, for example, in the case of R^ε_{obst}, that the factor ε in front of the term $(W_{app}\cdot\nabla)W_{app}$ compensates the factor $1/\varepsilon$ which comes from the differentiation of W_{app} with respect to x. Furthermore, the term $(2/\varepsilon)(\nabla\theta\cdot\nabla\varphi)\widetilde{W}^1_z$ is indeed bounded because we recall that $\widetilde{W}^1$ is exponentially decreasing in z so that

$$
\left|\frac{2}{\varepsilon}(\nabla\theta\cdot\nabla\varphi)\widetilde{W}^1_z\right| \leq \frac{2}{\varepsilon}|\nabla\theta(x)\cdot\nabla\varphi(x)||w_1(t,x)|e^{-\varphi(x)/\varepsilon},
$$

and since $\nabla\theta \equiv 0$ in a neighborhood of $\partial\omega$ (because, close to $\partial\omega$, θ has a value identically 1), we see that the term considered is zero if $\varphi(x) = d(x,\partial\omega) \leq \alpha$, for a well-chosen $\alpha > 0$. Hence, the term under study is bounded by

$$
\frac{C}{\varepsilon}|w_1(t,x)|e^{-\alpha/\varepsilon},
$$

which proves that it is indeed bounded with respect to ε in the suitable functional spaces.

Lastly, to carry out the final estimate, we need to use the following lifting lemma.

Lemma VII.2.6. *There exists a function* ψ_ε *such that*

$$\operatorname{div} \psi_\varepsilon = R^\varepsilon_{div}, \tag{VII.84}$$

and, for all $T < T^*$*, satisfying the following estimates,*

$$\|\psi_\varepsilon\|_{L^\infty(]0,T[,(H^1_0(\omega))^d)} \leq C\varepsilon.$$

$$\left\| \frac{\partial \psi_\varepsilon}{\partial t} \right\|_{L^2(]0,T[,(H^1_0(\omega))^d)} \leq C\varepsilon. \tag{VII.85}$$

Proof.

We should first note that $\displaystyle\int_\omega R^\varepsilon_{div}\, dx = 0$, because:

$$\int_\omega R^\varepsilon_{div}\, dx = -\int_\omega \operatorname{div} W_{app}\, dx = \int_{\partial\omega} W_{app} \cdot \nu\, d\sigma = \varepsilon \int_{\partial\omega} W^2 \cdot \nu\, d\sigma = 0,$$

from Remark VII.2.5.

We can then apply Theorem IV.3.1, which tells us that there exists a linear operator Π which is continuous from $L^2_0(\Omega)$ into $(H^1_0(\Omega))^d$ and which is a right inverse of the divergence operator, that is, such that $\operatorname{div}(\Pi(q)) = q$ for all $q \in L^2_0(\Omega)$.

For almost all $t \in]0, T[$, we set $\psi_\varepsilon(t) = \Pi(R^\varepsilon_{obst}(t))$. This choice satisfies the divergence condition by definition and the first estimate because of the properties of R^ε_{obst} and the operator Π.

Moreover, since Π is linear, we have

$$\begin{aligned}
\int_0^{T-h} \|\psi_\varepsilon(t+h) - \psi_\varepsilon(t)\|^2_{H^1_0}\, dt &\leq C \int_0^{T-h} \|R^\varepsilon_{obst}(t+h) - R^\varepsilon_{obst}(t)\|^2_{L^2}\, dt \\
&\leq C \int_0^{T-h} \left(\int_t^{t+h} \left\| \frac{\partial R^\varepsilon_{obst}}{\partial t}(s) \right\|_{L^2} ds \right)^2 dt \\
&\leq Ch \int_0^{T-h} \int_t^{t+h} \left\| \frac{\partial R^\varepsilon_{obst}}{\partial t}(s) \right\|^2_{L^2} ds\, dt \\
&\leq Ch^2 \left\| \frac{\partial R^\varepsilon_{obst}}{\partial t} \right\|^2_{L^2(]0,T[,L^2)}.
\end{aligned}$$

Hence, we obtain for all $h > 0$ small enough

$$\left\| \frac{\psi_\varepsilon(\cdot + h) - \psi_\varepsilon(\cdot)}{h} \right\|_{L^2(]0,T-h[,H^1_0)} \leq C \left\| \frac{\partial R^\varepsilon_{obst}}{\partial t} \right\|_{L^2(]0,T[,L^2)}.$$

This proves that $\partial \psi_\varepsilon / \partial t$ exists in $L^2(]0,T[, (H^1_0(\omega))^d)$ and satisfies the claimed estimate.

□

2.5.2 Estimate of the remainders

To carry out this estimate, we multiply Equation (1) in (VII.82) by $w^r_\varepsilon - \psi_\varepsilon$ (which is divergence-free thanks to Equation (2) of (VII.82) and (VII.84)) and we integrate over ω. Through integration by parts, we obtain an inequality of the following form,

$$\frac{1}{2}\frac{d}{dt}\|w^r_\varepsilon\|^2_{L^2} + \|\nabla w^r_\varepsilon\|^2_{L^2} + \frac{1}{\varepsilon^2}\|w^r_\varepsilon\|^2_{L^2} = I_1 + \cdots + I_{10}, \tag{VII.86}$$

where the various terms on the right-hand side are defined by

$$\begin{aligned}
I_1 &= -\int_{\partial\omega} w^r_\varepsilon \cdot \left(\frac{\partial w^r_\varepsilon}{\partial \nu} - q^r_\varepsilon \nu\right) d\sigma,\\
I_2 &= -\varepsilon \int_\omega \Big((w^r_\varepsilon \cdot \nabla)W_{app} + (W_{app}\cdot\nabla)w^r_\varepsilon + (\psi_\varepsilon \cdot \nabla) w^r_\varepsilon\Big)\cdot w^r_\varepsilon \, dx,\\
I_3 &= -\varepsilon \int_\omega \big((w^r_\varepsilon - \psi_\varepsilon)\cdot\nabla\big)(w^r_\varepsilon - \psi_\varepsilon)\cdot(w^r_\varepsilon - \psi_\varepsilon)\, dx,\\
I_4 &= -\varepsilon \int_\omega \big((w^r_\varepsilon - \psi_\varepsilon)\cdot\nabla\big)\psi_\varepsilon\cdot(w^r_\varepsilon - \psi_\varepsilon)\, dx,\\
I_5 &= \varepsilon \int_\omega \Big((w^r_\varepsilon \cdot \nabla)W_{app} + (W_{app}\cdot\nabla)w^r_\varepsilon + (\psi_\varepsilon \cdot \nabla) w^r_\varepsilon\Big)\cdot \psi_\varepsilon \, dx,\\
I_6 &= \frac{1}{\varepsilon^2}\int_\omega w^r_\varepsilon \cdot \psi_\varepsilon \, dx, \ I_7 = \int_\omega R^\varepsilon_{obst}\cdot w^r_\varepsilon \, dx,\\
I_8 &= \int_\omega \frac{\partial w^r_\varepsilon}{\partial t}\cdot\psi_\varepsilon\, dx, \ I_9 = \int_\omega \nabla w^r_\varepsilon : \nabla\psi_\varepsilon \, dx,\\
I_{10} &= -\int_\omega R^\varepsilon_{obst}\cdot \psi^\varepsilon \, dx.
\end{aligned}$$

If we multiply Equation (3) of (VII.82) by v^r_ε then, by another integration by parts, we obtain

$$\frac{1}{2}\frac{d}{dt}\|v^r_\varepsilon\|^2_{L^2} + \|\nabla v^r_\varepsilon\|^2_{L^2} = J_1 + \cdots + J_4, \tag{VII.87}$$

with

$$\begin{aligned}
J_1 &= \int_{\partial\omega}\left(\frac{\partial v^r_\varepsilon}{\partial\nu} - p^\varepsilon\nu\right)\cdot v^r_\varepsilon \, d\sigma,\\
J_2 &= -\int_{\mathcal{U}} \big((V_{app}\cdot\nabla)v^r_\varepsilon\big)\cdot v^r_\varepsilon + \big((v^r_\varepsilon\cdot\nabla)V_{app}\big)\cdot v^r_\varepsilon \, dx,\\
J_3 &= -\varepsilon\int_{\mathcal{U}} \big((v^r_\varepsilon\cdot\nabla)v^r_\varepsilon\big)\cdot v^r_\varepsilon\, dx,\\
J_4 &= \int_{\mathcal{U}} R^\varepsilon_{flu}\cdot v^r_\varepsilon \, dx.
\end{aligned}$$

Next, we add the two estimates (VII.86) and (VII.87) and it just remains to estimate the various terms as follows. First of all, we use boundary conditions (5) and (6) of (VII.82) and obtain

$$\begin{aligned}
I_1 + J_1 =& - \int_{\partial\omega} R^\varepsilon_{boundary} \cdot w^r_\varepsilon \, d\sigma \leq \|R^\varepsilon_{boundary}\|_{L^2(\partial\omega)} \|w^r_\varepsilon\|_{L^2(\partial\omega)} \\
\leq& \|R^\varepsilon_{boundary}\|_{L^2(\partial\omega)} \|w^r_\varepsilon\|_{H^1} \\
\leq& \|R^\varepsilon_{boundary}\|_{L^2(\partial\omega)} (\|w^r_\varepsilon\|_{L^2} + \|\nabla w^r_\varepsilon\|_{L^2}) \\
\leq& \frac{1}{8} \|\nabla w^r_\varepsilon\|^2_{L^2} + \|w^r_\varepsilon\|^2_{L^2} + C \|R^\varepsilon_{boundary}\|^2_{L^2(\partial\omega)}.
\end{aligned}$$

By using the Hölder and Young inequalities and the Sobolev embeddings, it follows

$$\begin{aligned}
|I_2| \leq& \varepsilon \|w^r_\varepsilon\|_{L^6} \|\nabla W_{app}\|_{L^2} \|w^r_\varepsilon\|_{L^3} + \varepsilon \|W_{app}\|_{L^6} \|\nabla w^r_\varepsilon\|_{L^2} \|w^r_\varepsilon\|_{L^3} \\
&+ \varepsilon \|\nabla w^r_\varepsilon\|_{L^2} \|w^r_\varepsilon\|_{L^3} \|\psi_\varepsilon\|_{L^6} \\
\leq& \frac{1}{8} \|\nabla w^r_\varepsilon\|^2_{L^2} + \|w^r_\varepsilon\|^2_{L^2} + C\varepsilon^4 \left(\|W_{app}\|^4_{H^1} + \|\psi_\varepsilon\|^4_{H^1} \right) \|w^r_\varepsilon\|^2_{L^2}.
\end{aligned}$$

Since $w^r_\varepsilon - \psi_\varepsilon$ is divergence-free and ψ_ε is zero on $\partial\omega$, we obtain through integrating by parts (don't forget that ν is the inward normal to ω)

$$\begin{aligned}
I_3 =& - \frac{\varepsilon}{2} \int_\omega (w^r_\varepsilon - \psi_\varepsilon) \cdot \nabla(|w^r_\varepsilon - \psi_\varepsilon|^2) \, dx \\
=& - \frac{\varepsilon}{2} \int_\omega \operatorname{div} \left(|w^r_\varepsilon - \psi_\varepsilon|^2 (w^r_\varepsilon - \psi_\varepsilon) \right) dx \\
=& \frac{\varepsilon}{2} \int_{\partial\omega} |w^r_\varepsilon - \psi_\varepsilon|^2 (w^r_\varepsilon - \psi_\varepsilon) \cdot \nu \, d\sigma = \frac{\varepsilon}{2} \int_{\partial\omega} |w^r_\varepsilon|^2 w^r_\varepsilon \cdot \nu \, d\sigma,
\end{aligned}$$

and also

$$\begin{aligned}
J_3 =& - \frac{\varepsilon}{2} \int_\omega v^r_\varepsilon \cdot \nabla(|v^r_\varepsilon|^2) \, dx = -\frac{\varepsilon}{2} \int_\omega \operatorname{div} \left(|v^r_\varepsilon|^2 v^r_\varepsilon \right) dx \\
=& - \frac{\varepsilon}{2} \int_{\partial\omega} |v^r_\varepsilon|^2 v^r_\varepsilon \cdot \nu \, d\sigma.
\end{aligned}$$

Hence, from boundary condition (5) in (VII.82), we have

$$I_3 + J_3 = 0.$$

The term I_4 can be written in the following manner

$$\begin{aligned}
I_4 =& - \varepsilon \int_\omega (w^r_\varepsilon \cdot \nabla) \psi_\varepsilon \cdot w^r_\varepsilon \, dx + \varepsilon \int_\omega (\psi_\varepsilon \cdot \nabla) \psi_\varepsilon \cdot w^r_\varepsilon \, dx \\
&+ \int_\omega (w^r_\varepsilon \cdot \nabla) \psi_\varepsilon \cdot \psi_\varepsilon \, dx - \varepsilon \int_\omega (\psi_\varepsilon \cdot \nabla) \psi_\varepsilon \cdot \psi_\varepsilon \, dx,
\end{aligned}$$

from which we deduce, using the Hölder inequality and Sobolev embeddings, the estimate

$$\begin{aligned}|I_4| \leq & \varepsilon\|\nabla\psi_\varepsilon\|_{L^2}\|w_\varepsilon^r\|_{L^3}\|w_\varepsilon^r\|_{L^6} + 2\varepsilon\|w_\varepsilon^r\|_{L^6}\|\psi_\varepsilon\|_{L^3}\|\nabla\psi_\varepsilon\|_{L^2} \\ & + \varepsilon\|\nabla\psi_\varepsilon\|_{L^2}\|\psi_\varepsilon\|_{L^3}\|\psi_\varepsilon\|_{L^6} \\ \leq & \varepsilon\|\psi_\varepsilon\|_{H^1}\|w_\varepsilon^r\|_{L^2}^{\frac{1}{2}}\|w_\varepsilon^r\|_{H^1}^{\frac{3}{2}} + \varepsilon\|\psi_\varepsilon\|_{H^1}^2\|w_\varepsilon^r\|_{H^1} + \varepsilon\|\psi_\varepsilon\|_{H^1}^3 \\ \leq & \frac{1}{8}\|\nabla w_\varepsilon^r\|_{L^2}^2 + C(1+\varepsilon^4\|\psi_\varepsilon\|_{H^1}^4)\|w_\varepsilon^r\|_{L^2}^2 + C\varepsilon^2\|\psi_\varepsilon\|_{H^1}^4 + C\varepsilon\|\psi_\varepsilon\|_{H^1}^3.\end{aligned}$$

The term I_5 is estimated in the following classic way

$$\begin{aligned}|I_5| \leq & \varepsilon\|\nabla W_{app}\|_{L^2}\|\psi_\varepsilon\|_{L^3}\|w_\varepsilon^r\|_{L^6} + \varepsilon\|W_{app}\|_{L^4}\|\nabla w_\varepsilon^r\|_{L^2}\|\psi_\varepsilon\|_{L^4} \\ & + \varepsilon\|\nabla w_\varepsilon^r\|_{L^2}\|\psi_\varepsilon\|_{L^4}^2 \\ \leq & \frac{1}{8}\|\nabla w_\varepsilon^r\|_{L^2}^2 + \varepsilon^2(\|W_{app}\|_{H^1}^2 + \|\psi_\varepsilon\|_{H^1}^2)\|\psi_\varepsilon\|_{H^1}^2.\end{aligned}$$

Finally, terms I_6 and I_7 are bounded above, respectively, by

$$\begin{aligned}|I_6| &\leq \frac{1}{2\varepsilon^2}\|w_\varepsilon^r\|_{L^2}^2 + \frac{1}{2\varepsilon^2}\|\psi_\varepsilon\|_{L^2}^2, \\ |I_7| &\leq \|R_{obst}^\varepsilon\|_{L^2}^2 + \|w_\varepsilon^r\|_{L^2}^2.\end{aligned}$$

If we integrate the term I_8 over time between 0 and t, we obtain, through integration by parts in time (Theorem II.5.12)

$$\begin{aligned}\int_0^t I_8\,ds =& \int_\omega w_\varepsilon^r(t)\cdot\psi_\varepsilon(t)\,dx - \int_\omega w_\varepsilon^r(0)\cdot\psi_\varepsilon(0)\,dx + \int_0^t\int_\omega w_\varepsilon^r\cdot\frac{\partial\psi_\varepsilon}{\partial t}\,dx\,ds, \\ \leq & \frac{1}{4}\|w_\varepsilon^r(t)\|_{L^2}^2 + \|\psi_\varepsilon(t)\|_{L^2}^2 + \frac{1}{2}\|w_\varepsilon^r(0)\|_{L^2}^2 + \frac{1}{2}\|\psi_\varepsilon(0)\|_{L^2}^2 \\ & + \frac{1}{4}\int_0^t\|w_\varepsilon^r\|_{L^2}^2\,ds + \int_0^t\left\|\frac{\partial\psi_\varepsilon}{\partial t}\right\|_{L^2}^2\,ds.\end{aligned}$$

By using the Sobolev embeddings, we can estimate the term J_2 as follows

$$\begin{aligned}|J_2| \leq & \|V_{app}\|_{L^6}\|\nabla v_\varepsilon^r\|_{L^2}\|v_\varepsilon^r\|_{L^3} + \|\nabla V_{app}\|_{L^2}\|v_\varepsilon^r\|_{L^4}^2 \\ \leq & \|V_{app}\|_{H^1}\|v_\varepsilon^r\|_{H^1}^{3/2}\|v_\varepsilon^r\|_{L^2}^{1/2} \\ \leq & \|V_{app}\|_{H^1}\left(\|v_\varepsilon^r\|_{L^2}^2 + \|\nabla v_\varepsilon^r\|_{L^2}^{3/2}\|v_\varepsilon^r\|_{L^2}^{1/2}\right) \\ \leq & \frac{1}{8}\|\nabla v_\varepsilon^r\|_{L^2}^2 + C(1+\|V_{app}\|_{H^1}^4)\|v_\varepsilon^r\|_{L^2}^2.\end{aligned}$$

Finally, terms I_9, I_{10}, and J_4 are estimated in the usual way by

$$
\begin{aligned}
|I_9| &\leq \frac{1}{8}\|\nabla w_\varepsilon^r\|_{L^2}^2 + 2\|\psi_\varepsilon\|_{H^1}^2,\\
|I_{10}| &\leq \|R_{obst}^\varepsilon\|_{L^2}\|\psi_\varepsilon\|_{L^2} \leq \frac{1}{2}\|R_{obst}^\varepsilon\|_{L^2}^2 + \frac{1}{2}\|\psi_\varepsilon\|_{L^2}^2,\\
|J_4| &\leq \|R_{flu}^\varepsilon\|_{L^2}^2 + \|v_\varepsilon^r\|_{L^2}^2.
\end{aligned}
$$

We now set

$$
\begin{aligned}
y(t) &= \|w_\varepsilon^r(t)\|_{L^2}^2 + \|v_\varepsilon^r(t)\|_{L^2}^2,\\
z(t) &= \|\nabla w_\varepsilon^r(t)\|_{L^2}^2 + \|\nabla v_\varepsilon^r(t)\|_{L^2}^2.
\end{aligned}
$$

Adding Equations (VII.86) and (VII.87), then integrating with respect to time between 0 and t, we get for any $0 \leq t \leq T$, of the estimate

$$
\begin{aligned}
y(t) + \int_0^t z(s)\,ds &\leq C(1 + \|w_\varepsilon^r(0)\|_{L^2}^2 + \|v_\varepsilon^r(0)\|_{L^2}^2) + C\varepsilon^4\int_0^t \|W_{app}\|_{L^2}^2\,ds\\
&+ C\int_0^t \left(\|R_{boundary}^\varepsilon\|_{L^2(\partial\omega)}^2 + \|R_{obst}^\varepsilon\|_{L^2}^2 + \|R_{flu}^\varepsilon\|_{L^2}^2 + \left\|\frac{\partial\psi_\varepsilon}{\partial t}\right\|_{L^2}^2\right) ds\\
&+ C\int_0^t \left(1 + \varepsilon^4\|W_{app}\|_{H^1(\omega)}^4 + \|V_{app}\|_{H^1(\mathcal{U})}^4\right) y(s)\,ds, \qquad \text{(VII.88)}
\end{aligned}
$$

where C does not depend on ε.

However, by definition of the approximate solutions V_{app} and W_{app} and from the estimates obtained above, we have

$$
\|V_{app}\|_{L^\infty(]0,T[,H^1(\mathcal{U}))} \leq C,
$$

and

$$
\|W_{app}\|_{L^\infty(]0,T[,H^1(\omega))} \leq \frac{C}{\sqrt{\varepsilon}}.
$$

Indeed, we should not forget that by estimating the H^1 norm of W_{app}, the dependency of the profiles on $\varphi(x)/\varepsilon$ leads to a factor $1/\varepsilon$ during differentiation and, furthermore, the exponential decrease of these same terms allows us to gain a factor of $\sqrt{\varepsilon}$. To observe this, it is sufficient to calculate the L^2 norm of the derivative of the function $e^{-x/\varepsilon}$ on $\mathbb{R}^+$ as follows

$$
\frac{1}{\varepsilon}\left(\int_0^{+\infty} e^{-2x/\varepsilon}\,dx\right)^{1/2} = \frac{1}{\sqrt{2\varepsilon}}\left(\int_0^{+\infty} e^{-x}\,dx\right)^{1/2} = \frac{1}{\sqrt{2\varepsilon}}.
$$

Moreover, we have estimates of the remainders (VII.83) and the estimate (VII.85). This brings the energy inequality (VII.88) back to the inequality

$$
y(t) + \int_0^t z(s)\,ds \leq C(1 + \|w_\varepsilon^r(0)\|_{L^2}^2 + \|v_\varepsilon^r(0)\|_{L^2}^2) + C\int_0^t y(s)\,ds, \quad \text{(VII.89)}
$$

from which we deduce, by initially ignoring the (nonnegative) term $\int_0^t z(s)\,ds$, and by using Gronwall's lemma,

$$y(t) \leq C\left(1 + \|w_\varepsilon^r(0)\|_{L^2}^2 + \|v_\varepsilon^r(0)\|_{L^2}^2\right)\,e^T,\ \forall t \leq T.$$

However, from Equations (8) and (9) of (VII.82), this indeed proves that

$$\|v_\varepsilon^r\|^2_{L^\infty(]0,T[,L^2(\mathcal{U}))} + \|w_\varepsilon^r\|^2_{L^\infty(]0,T[,L^2(\omega))} \leq C.$$

If we now return to (VII.89), we obtain the second estimate

$$\int_0^T z(s)\,ds \leq C,$$

which we can state exactly as

$$\|v_\varepsilon^r\|^2_{L^2(]0,T[,H^1(\mathcal{U}))} + \|w_\varepsilon^r\|^2_{L^2(]0,T[,H^1(\omega))} \leq C.$$

This completes the proof of Theorem VII.2.3 and hence that of Theorem VII.2.2.

□

Appendix A
Classic differential operators

In this appendix, we very briefly revise several definitions and formulas relating to the basic differential operators, namely the gradient, divergence, curl, and Laplace operators. Throughout the text, we denote the scalar product of two vectors of $\mathbb{R}^d$ as $A \cdot B$ and the vector product of two vectors of $\mathbb{R}^3$ (an orientation being fixed) as $A \wedge B$.

1 The scalar and vector cases

1.1 Definitions

Definition A.1.1. *Let f be a scalar field and let g be a vector field, sufficiently regular and defined on a domain Ω of $\mathbb{R}^d$.*

- *The* gradient of f *refers to the vector field defined by its coordinates in any orthonormal basis of $\mathbb{R}^d$ by*

$$\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{pmatrix}.$$

- *The* divergence of g *refers to the scalar field defined by*

$$\operatorname{div} g = \sum_{i=1}^{d} \frac{\partial g_i}{\partial x_i},$$

  where the coordinates of g are refered to as $g_1, \ldots, g_d$ in an othonormal basis of the space.
- *The* Laplacian of f *refers to the scalar field defined by*

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$$\Delta f = \operatorname{div}(\nabla f) = \sum_{i=1}^{d} \frac{\partial^2 f}{\partial x_i^2}.$$

- *In the case $d = 3$, the* curl of g *refers to the vector field denoted as* $\operatorname{curl} g$*, defined by its coordinates in the same orthonormal basis by*

$$\operatorname{curl} g = \begin{pmatrix} \dfrac{\partial g_3}{\partial x_2} - \dfrac{\partial g_2}{\partial x_3} \\ \dfrac{\partial g_1}{\partial x_3} - \dfrac{\partial g_3}{\partial x_1} \\ \dfrac{\partial g_2}{\partial x_1} - \dfrac{\partial g_1}{\partial x_2} \end{pmatrix}.$$

Moreover, if v is a vector field of $\mathbb{R}^d$, we denote the first order differential operator as $v \cdot \nabla$, defined by

$$(v \cdot \nabla) f = \sum_{i=1}^{d} v_i \frac{\partial f}{\partial x_i}.$$

The concepts defined above are actually intrinsic; that is, they do not depend on the choice of basis. Consequently, we can express these operators in non-Cartesian coordinates. We recall that the cylindrical coordinates in $\mathbb{R}^3$ are defined by

$$x = r\cos\theta,\ y = r\sin\theta,\ z = z,$$

and that a rotating othonormal reference frame is associated with the system of coordinates, denoted as (e_r, e_θ, e_z) and defined by

$$e_r = \begin{pmatrix} \cos\theta \\ \sin\theta \\ 0 \end{pmatrix},\ e_\theta = \begin{pmatrix} -\sin\theta \\ \cos\theta \\ 0 \end{pmatrix},\ e_z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Proposition A.1.2. *Let f be a scalar field and g be a vector field, sufficiently regular and defined in cylindrical coordinates on the cylindrical domain Ω of $\mathbb{R}^3$. We then have the following formulas*

$$\nabla f = \frac{\partial f}{\partial r} e_r + \frac{1}{r}\frac{\partial f}{\partial \theta} e_\theta + \frac{\partial f}{\partial z} e_z,$$

$$\operatorname{div} g = \frac{1}{r}\frac{\partial}{\partial r}(r g_r) + \frac{1}{r}\frac{\partial g_\theta}{\partial \theta} + \frac{\partial g_z}{\partial z}, \tag{A.1}$$

where $g = g_r e_r + g_\theta e_\theta + g_z e_z$, and

$$\Delta f = \frac{1}{r}\frac{\partial}{\partial r}\left(r \frac{\partial f}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2}, \tag{A.2}$$

$$\operatorname{curl} g = \left(\frac{\partial g_z}{\partial \theta} - \frac{\partial g_\theta}{\partial z}\right) e_r + \left(\frac{\partial g_r}{\partial z} - \frac{\partial g_z}{\partial r}\right) e_\theta + \left(\frac{1}{r}\frac{\partial (r g_\theta)}{\partial r} - \frac{1}{r}\frac{\partial g_r}{\partial \theta}\right) e_z.$$

1.2 Useful formulas

Let A, B and C be three vector fields and f a scalar field. Let us start this small formulary with the following algebraic formula known as the *vector triple product*

$$A \wedge (B \wedge C) = (A \cdot C)\, B - (A \cdot B)\, C. \tag{A.3}$$

We next have the following formulae which link the various differential operators introduced above

$$\operatorname{div}(fA) = f(\operatorname{div} A) + (A \cdot \nabla) f, \tag{A.4}$$

$$\operatorname{curl}(fA) = f(\operatorname{curl} A) + (\nabla f) \wedge A, \tag{A.5}$$

$$(A \cdot \nabla) A = \frac{1}{2} \nabla |A|^2 - A \wedge \operatorname{curl} A, \tag{A.6}$$

$$\operatorname{div}(A \wedge B) = B \cdot \operatorname{curl} A - A \cdot \operatorname{curl} B, \tag{A.7}$$

$$\operatorname{curl}(A \wedge B) = (\operatorname{div} B) A - (\operatorname{div} A) B + (B \cdot \nabla) A - (A \cdot \nabla) B, \tag{A.8}$$

$$\operatorname{curl}(\operatorname{curl} A) = \nabla \operatorname{div} A - \Delta A. \tag{A.9}$$

Of course, this list is a long way from being exhaustive and only contains the formulas which are most useful in fluid mechanics.

2 Extension to second-order tensors

Our goal is not to provide a detailed general study of tensors. In this text, we only need to consider in detail tensor fields of order 2, that is, the endomorphism fields of $\mathbb{R}^d$.

We therefore simply identify second-order tensors with square matrices via the canonical basis of $\mathbb{R}^d$.

Definition A.2.1. *Let Ω be a domain of $\mathbb{R}^d$.*

- *Lets v be a regular vector field on Ω; we call the* gradient *of v, the tensor field*

$$\nabla v = \left(\frac{\partial v_i}{\partial x_j}\right)_{1 \le i,j \le d}.$$

- *Let $\sigma = (\sigma_{ij})_{1 \le i,j \le d}$ be a regular tensor field on Ω; we call the* divergence *of σ, the vector field*

$$\operatorname{div}\sigma = \left(\sum_{j=1}^{3} \frac{\partial \sigma_{ij}}{\partial x_j}\right)_{1\le i\le d}.$$

- *Let v be a regular vector field on Ω; we call the* Laplacian *of v, the vector field*

$$\Delta v = \operatorname{div}(\nabla v).$$

We can easily verify that Δv is the vector field whose components in a fixed othonormal basis of $\mathbb{R}^d$ are the Δv_i, where v_i are the components of the field v in this same basis.

Definition A.2.2 (Tensor product). *Let u, v be two vector fields; we refer to the* tensor product of u and v, *and we write $u \otimes v$, for the second-order tensor defined in an orthonormal basis by*

$$u \otimes v = \big(u_i v_j\big)_{1\le i,j\le d}.$$

If we apply $u \otimes v$ to a third vector field w, then we have

$$(u \otimes v).w = (v \cdot w)u.$$

Furthermore, we have the following formula for the divergence of a tensor product

$$\operatorname{div}(u \otimes v) = (\operatorname{div} v)u + (v \cdot \nabla)u. \tag{A.10}$$

Finally, we can define a scalar product on the set of second-order tensor fields, which is simply the Euclidean scalar product in the space of the matrices identified with $\mathbb{R}^{d\times d}$. In tensor language, this is called the contracted tensor product.

Definition A.2.3 (Contracted product). *Let σ, τ be two tensors of order 2; we define their* contracted product, *to be the real number defined by*

$$\sigma : \tau = \sum_{1\le i,j\le d} \sigma_{ij}\tau_{ij}.$$

The norm associated with this scalar product is simply denoted as $|\cdot|$, such that

$$|\sigma|^2 = \sigma : \sigma.$$

Appendix B
Thermodynamics supplement

The goal of this appendix is to recall, in an extremely succinct and therefore necessarily very incomplete way, the main concepts of the thermodynamics of fluids which are used in Chapter I. Our basic objective is to introduce the notion of entropy and the second law of thermodynamics (in particular the Clausius theorem). Among the various reference textbooks available on this topic, we refer, for example, to [94] for a much more detailed exposition.

Thermodynamics is the study of the evolution of material systems and of their energy taking into account their internal molecular structure. Such material systems are well described by a certain number of quantities, referred to as *state variables*, among which

- Volume, mass, and density which account for the size of the system and the quantity of matter in the system under study
- Temperature and energy which account for the excitation state of the elementary particles of the matter
- Pressure which is a mechanical measure of the random motion of the particles through the collisions of these particles on a (virtual) wall placed in the material.

Other state variables are often necessary when dealing with complex systems (such as multiphase systems, for instance).

Among all these quantities, there are some which do not vary when the size of the system is increased (temperature, pressure, etc.) and these are referred to as *intensive variables*. By contrast, some quantities increase with the size of the system (volume, energy, etc.) and are referred to as *extensive variables*.

In fact, the values of the state variables are not independent. It is admitted that each homogeneous thermodynamical system satisfies an *equation of state* which is a algebraic relationship between all the state variables. Depending on the system under study, it can take the form of an expression of the pressure as a function of the temperature and the density of the system, or of an

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expression of the temperature as a function of pressure and density, or any other combination of independent variables.

In the case of fluid dynamics, we need to apply the concepts of this appendix to fluid elements $(\Omega_t)_t$. They are assumed to be in instantaneous local thermodynamical equilibrium so that each elementary fluid particle can be considered as a closed homogeneous thermodynamical system. This means that the relaxation time towards the local molecular equilibrium at any point x and any time t in the fluid is much shorter than any characteristic time for the overall evolution of the fluid.

1 Heat capacity

Definition B.1.1. *We use the term* heat capacity at constant volume *(resp.* heat capacity at constant pressure*) denoted by C_V (or C_P, resp.) to refer to the quantity of heat required to increase the temperature of a system by $1K$ while maintaining the volume (resp. the pressure) constant. It follows that an increase of temperature ΔT requires us to furnish to the system an energy which is given by*

$$\Delta Q = C_V \Delta T, \text{ at constant } V,$$

$$\Delta Q = C_P \Delta T, \text{ at constant } P.$$

Of course C_V and C_P are extensive quantities and we often make use of the specific heat capacities (i.e., per unit mass) defined by

$$c_V = \frac{C_V}{M}, \quad c_P = \frac{C_P}{M},$$

where M is the mass of the system considered.

2 The first law of thermodynamics. Internal energy

Let us now consider a closed homogeneous thermodynamic system (i.e., one which does not exchange matter with the exterior but which, nevertheless, can exchange energy in the form of heat or work). The first law of thermodynamics tells us that there exists a state variable known as the *internal energy* and referred to as E, which depends on temperature, pressure, and density such that the change in energy of the system during a transformation must be exactly equal to the external energy provided to the system. This internal energy accounts for the kinetic energy of each individual molecule in the system even though the system is at rest at the macroscopic level.

For a given elementary transformation of our closed system, we have

$$\Delta E = \Delta Q + \Delta W,$$

where ΔQ is the heat provided and ΔW is the work done by external forces acting on the system during the transformation. If we assume that the system is submitted to no external forces other than the pressure, then the term for the work reduces to the work of the pressure forces and can be written as

$$\Delta W = -P\Delta V,$$

where ΔV is the volume change undergone by the system in the course of the transformation.

Remark B.2.1. In the energy equation above, ΔE represents the change in a state variable E when passing from an initial to a final state and, importantly, this value does not depend on the path by which the system passes from the initial state to the final state. However, the values ΔQ and ΔW do not represent changes of state variables and they do depend on the path followed by the system between the two states considered.

It is a standard and useful notation to write ΔE in the form of the exact differential dE of the function E (considered as a function of some set of independent variables) but we cannot do the same for ΔQ and ΔW.

The same "differential" notation is used for any state function that we encounter in this book but we do not need any precise element of the differential form theory.

Remark B.2.2. In the context of fluid mechanics, each fluid element contains internal energy so that the total energy of this fluid element is given by

$$E_{tot} = E_{int} + E_{kin} + E_{pot},$$

where E_{kin} is the macroscopic kinetic energy of the system (computed from the mean velocity field v in the flow) and E_{pot} is its potential energy. In this formula E_{int} is the total internal energy contained in the fluid element considered. In practice, this internal energy term can be expressed as an integral term of the density of internal energy.

3 The second law of thermodynamics

3.1 Entropy

The second law of thermodynamics asserts the existence of a new state variable S, called *entropy* such that for any quasi-static transformation of the closed system considered, we would have

$$\frac{\Delta Q}{T} \leq \Delta S, \tag{B.1}$$

the equality being true if, and only if, the transformation is *reversible*. In the case of an elementary reversible transformation we therefore have

$$\Delta Q = TdS.$$

Here, we used the fact that the entropy is a state function to replace the notation ΔS by the differential notation dS.

This statement, known as the Clausius theorem, is equivalent to saying that the transfer of heat from a cold system to a warm system is forbidden.

Now, by combining these two laws of thermodynamics, we can show that in an elementary reversible transformation the internal energy of a closed system changes according to the equation

$$dE = TdS - PdV. \tag{B.2}$$

Remark B.3.1. The preceding result only applies to reversible transformations. How should we handle irreversible transformations? We can use the fact that the change in energy depends only on the initial and final states and attempt to construct a reversible transformation which will take the system from the first state to the second. Along such a path, the above relation is valid and we can therefore integrate it.

3.2 Internal energy calculation

We can now give an expression for the internal energy as a function of the other variables T, P, V, and S. To do this, it is sufficient to use the intensive or extensive character of each variable.

Let us consider a system of size 1 and then consider the same system $(1+\varepsilon)$ times as large as before, with all its proportions being conserved. The temperature and pressure do not change in this system, whereas the energy, entropy, and volume are each multiplied by $(1+\varepsilon)$. By applying (B.2), and using that T and P are the same in each state, it follows that

$$\begin{aligned}\varepsilon E &= (1+\varepsilon)E - E = \Delta E = T(\Delta S) - P(\Delta V)\\ &= T((1+\varepsilon)S - S) - P((1+\varepsilon)V - V) = \varepsilon TS - \varepsilon PV;\end{aligned}$$

we then find the expression

$$E = TS - PV.$$

By taking the differential of this expression and by comparing it to (B.2), we obtain the *Gibbs–Duhem relation* which is written as

$$SdT - VdP = 0. \tag{B.3}$$

4 Specific variables

We now work with specific variables, that is, quantities per unit mass. We call e and s the specific energy and specific entropy of the system. Furthermore, we introduce the specific volume $v = 1/\rho$ (where ρ is the density of the system being studied) as a volume variable. The relations obtained in the previous section become

$$e = Ts - Pv,$$
$$s\,dT - v\,dP = 0.$$

As we have seen above, for an elementary reversible transformation at constant specific volume, the quantity of heat acquired by the system is written as $\Delta Q = Tds$ but we also have $\Delta Q = c_v dT$, from which we can deduce that

$$\left(\frac{\partial s}{\partial T}\right)_v = \frac{c_v}{T},$$

where we use here the notation $(\partial F/\partial x)_y$ for the partial derivative of the quantity F with respect to x, at a fixed value of the variable y. This notation is useful to make precise the choice of independent state variables which is done in each computation. Hence, we can write the differential of the specific entropy (considered as a function of T and v) in the form

$$ds = \frac{c_v}{T}dT + \left(\frac{\partial s}{\partial v}\right)_T dv.$$

From the Gibbs–Duhem relation (B.3), we have:

$$d(Pv) = Pdv + vdP = Pdv + sdT;$$

we can therefore use Schwarz's theorem on crossed second-order derivatives for any (smooth) function, which gives us one of the so-called *Maxwell relations*

$$\left(\frac{\partial s}{\partial v}\right)_T = \left(\frac{\partial P}{\partial T}\right)_v.$$

It follows that

$$ds = \frac{c_v}{T}dT + \left(\frac{\partial P}{\partial T}\right)_v dv,$$

and then returning to the specific internal energy, we have

$$de = Tds - Pdv = c_v dT + \left(T \left(\frac{\partial P}{\partial T} \right)_v - P \right) dv. \tag{B.4}$$

Therefore, if we know the equation of state for the fluid under study (i.e., the expression of P as a function of T and v), we can calculate all of the coefficients of these last two relations.

Some classic equations of state for the dynamics of a gas are, for example:

- The ideal gas law
$$P = k\frac{T}{v},$$
where k is a constant depending only on the molar mass of the fluid study.
- The Van der Waals equation of state
$$\left(P + \frac{a}{v^2}\right)\left(1 - \frac{b}{v}\right) = k\frac{T}{v},$$
where a and b are two new constants which depend on the fluid studied. Unlike the very simple equation of state for an ideal gas, this law takes account of the diameter of the molecules (assumed to be hard spheres) and the attractive forces between molecules.

Liquids also satisfy equations of state but in that case, as explained in Section 6 of Chapter I, the flow can often be considered to be incompressible so that the particular expression of the equation of state is not useful for the description of the flow.

We observe finally that to obtain Equation (B.4), we have not used the explicit expression for the entropy and, furthermore, the entropy did not appear in the formula obtained. The simple fact of postulating the existence of this state variable, S, allows us to apply the Schwarz theorem and to deduce the fundamental thermodynamic relations from it. In the incompressible framework which is the main topic of this book, the entropy does not explicitly appear in the final system of equations describing the flow. However, it is a fundamental quantity in compressible flow modelling.

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